AGENTIC SYSTEMS THEORY

VOLUME 1

ADAPTATION & ACTUATION THEORY

VIS VERITATIS

CC BY SA

JOSEPH A WECKER

AAT: Adaptation & Actuation Theory — Appendices

Joseph A. Wecker

October 30, 2026

v0.4.0 • 7a6a706-dirty • 2026-10-30

Contents

I Appendices: Details 1

A *Derivation:* Sector Condition Stability — Lyapunov Derivation 5

B *Derivation:* No Horizon-Independent Non-Exit Bound Under Additive Stochastic Forcing (the Model-S No-Go) 25

C *Derivation:* The Self-Actuation Grounding No-Go 29

D *Derivation:* Convention Monotonicity — the Receding-Horizon Rung 37

E *Derivation:* The Reward-Channel Learning No-Go 43

F *Discussion:* The Sandbox Evaluation Ceiling 51

G *Result:* Sector-Persistence Template 55

H *Result:* Certificate Existence — Operator-Sector in Some Metric Is Exponential Stability 63

I *Derivation:* Persistence Cost — Information Rate to Maintain Bounded Mismatch 71

J *Derivation:* Critical-Mass Composition 77

K *Derivation:* Gain-Sector Bridge — Proofs and Verification 87

L *Derivation:* Recursive Update — Uniqueness Derivation 97

M *Derivation:* Decomposition Uniqueness — Underdetermination and Canonicity of the Epistemic/Purposeful Split 103

N *Derived:* Multi-Timescale Stability 111

O *Sketch:* Does the Model-S Containment Dichotomy License Structural-Adaptation Genericity? 117

P *Derivation:* Discrete-Time Sector Condition 119

Q *Detail:* Linear ODE Approximation 125

R *Derivation:* Strategy Graph Structure 131

S *Derivation:* Sector Condition Verification for Edge-Credence Dynamics 141

T ***Derivation:*** **Evidence-Allocation Ceiling on Strategic Persistence** **165**

U ***Derivation:*** **Regret-Bound Derivation of the Strategy-Cost KL Direction** **171**

V ***Derivation:*** **Log-Odds as the Unique Additive-Evidence Parameterization for Edge Credences** **185**

W ***Derivation:*** **Adaptive-Gain Dynamics — A2' Under a Learning Gain** **191**

X ***Derivation:*** **Internal-External Decomposition of Agent Viability** **199**

Y ***Derivation:*** **Mismatch-Budget Attribution — Policy as Measure, First-Order Log-Additivity, and the Information Form** **203**

Z ***Derivation:*** **Update Detection Latency Under Accumulated Experience** **209**

AA ***Discussion:*** **Independence Audit** **215**

AB ***Discussion:*** **Approximation Tiering** **219**

AC ***Discussion:*** **Compression Operations in AAT** **223**

AD ***Derivation:*** **Observation-Ambiguity Bias-Bound Constant C** **229**

AE ***Discussion:*** **Partial-Coupling Pathways — A Sub-Typology Within Class 2** **239**

AF ***Discussion:*** **The W_1 Structural-Bound Boundary** **247**

AG ***Derived:*** **Belief-Strategy Attractors Under Σ-Source Coupling** **251**

AH ***Discussion:*** **The Dynamic-Regime Axis — Contraction / Equilibrium / Cyclic-Distributional / Mean-Field** **257**

AI ***Formulation:*** **Resource Budget** **267**

AJ ***Derived:*** **Resource-Bounded Destabilization** **271**

AK ***Result:*** **Contraction Template** **275**

AL ***Derivation:*** **Variational Approximate-A2' (ε-Fidelity)** **285**

AM ***Derivation:*** **L1' Update-Bias Formula** **289**

AN ***Derived:*** **Architecture No-Identifiability from On-Policy Summary Data** **293**

AO ***Derivation:*** **Fisher-Whitened Edge Update Under Correlated Evidence** **303**

AP ***Derivation:*** **Regime-Marginal Indistinguishability — Cross-Regime Witness for `#disc-identifiability-floor` Instance 3 (Broadened Scope)** **307**

AQ ***Derivation:*** **Strategy-Proofness Impossibility (Gibbard-Satterthwaite)** **315**

AR ***Derivation:*** **Bilateral Trade Impossibility (Myerson-Satterthwaite)** **321**

AS ***Derivation:*** **Social Welfare Aggregation Impossibility (Arrow)** **327**

AT *Derivation:* Fisher-Local Update Gain 333

AU *Derivation:* Scope of Tempo Additivity — Redundancy, Synergy, and the Echo-Chamber Bound 339

AV *Derivation:* Matrix-Loewner Persistence Condition 343

AW *Derivation:* The Survival-Imperative Exploration Drive 349

AX *Derivation:* Linear Matrix Inequality Form of the Causal-IB Survival Bound 355

AY *Observation:* The Authored-World Laboratory — Identifiability by Construction 361

AZ *Detail:* Observation: Part I Validation Simulations 363

II Appendices: Operational Domains 367

BA *Detail:* Operationalization — Estimation Procedures 371

BB *Worked Example:* 1D Active Tracking (Kalman Domain) 377

BC *Worked Example:* Nonstationary Bandit (RL Domain) 381

BD *Worked Example:* Nonstationary Bandit with Explicit Strategy (Part II Validation) 385

BE *Worked Example:* L1 Augmented DAG with Common-Cause Node 397

BF *Worked Example:* `#worked-example-cam` *(missing)* 403

Bibliography 405

Part I

Appendices: Details

Layer 0 — the framework's most distinctive intellectual move, in one paragraph. **AAT consistently uses negative results constructively.** When the framework encounters a structural impossibility — a quantity that cannot be identified from limited data, a guarantee that cannot be made to hold under stochastic noise, a set of alternatives that one stream of evidence cannot keep calibrated, whatever the update rule — it states the impossibility precisely with the external theorem that enforces it, names the unique broadly-available capability that would be required to escape it, and then *uses* the impossibility to elevate that capability from "useful additional tool" to "structurally required by the theory." The negative result and the positive consequence are the same proof. Five cleanly-fitting instances span the framework: detecting an unobserved common cause from on-policy data is impossible without loop-interventional access (Pearl–Bareinboim Causal Hierarchy Theorem); identifying a mixture model from single-channel observations is impossible without observing the latent (Cramér–Rao); certifying composite contraction from component data alone is impossible without matched-tier shared metrics (Liberzon common-Lyapunov nonexistence); pathwise containment of a bounded region under additive stochastic forcing is impossible *at any correction strength* — the no-go *is* the result (Khasminskii recurrence; Corollary A.1S.1 in `#deriv-sector-condition`); when K alternatives that must all stay calibrated share one observation per step, no update rule or forgetting rate keeps them calibrated once $\rho_\Sigma \geq R_\Sigma/K$ (Corollary EA.2 in `#deriv-evidence-allocation-ceiling`). A reader who carries this style away can predict where the framework will use a no-go as load-bearing apparatus next, rather than meeting each as a scattered impossibility. The cross-instance discipline lives at `#disc-constructive-impossibility-posture`; the per-instance derivations live in the segments named above.

Supporting material: derivations, sketches, simulation results, and operationalization procedures backing the main theory claims.

Appendix A

Derivation: Sector Condition Stability — Lyapunov Derivation

The Lyapunov machinery underlying the sector-condition stability result and the central persistence inequality. The mismatch dynamics are taken in general nonlinear vector form, $\dot{\delta} = -F(\mathcal{T}, \delta) + w(t)$, with the correction function F required to satisfy only the *qualitative* properties (A1)–(A3) and the local sector condition (A2') stated in `#form-sector-condition`. The Lyapunov function is the canonical quadratic $V = \frac{1}{2}\|\delta\|^2$; the steady-state results follow from standard Lyapunov / Itô-Lyapunov machinery applied to AAT's correction-function object.

The derivation produces the framework's central persistence results in their precise form. **Proposition A.1 (Model D — bounded disturbance):** under $\|w(t)\| \le \rho$ and the sector condition, mismatch is ultimately bounded by $R^* = \rho/\alpha$ — exactly the persistence threshold the chapter cites, and $\mathcal{B}_R$ is positively invariant when $\alpha R > \rho$. **Lemma A.1N (threshold tightness):** the condition is unimprovable given A2' — necessary and sufficient for the *uniform class-level* guarantee (witness $F = \alpha\delta$ under outward disturbance), and agent-level necessary exactly when the sector bound is radially tight (the linear case); for a general fixed F the only-if direction is *false* (demonstrated by the dip counterexample in the Remark following A.1N — the sector floor can sit at radii the dynamics render unreachable). **Proposition A.1S (Model S — additive stochastic disturbance):** the Itô-Lyapunov analysis gives the second-moment bound of the mismatch while it is still inside the region (the killed moment), the RMS scale $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (the $1/\sqrt{\alpha}$ scaling that distinguishes Model S from Model D's $1/\alpha$ scaling), the finite-horizon sample-path bound, and, where the sector condition holds globally, the stationary tail $P(\|\delta(t)\| > R) \le n\sigma_w^2/(2\alpha R^2)$. **Proposition A.1S-τ (exit time):** the time the mismatch spends inside the region before first leaving it is bounded below, in mean and in distribution, by that of the isotropic linear corrector $F = \alpha\delta$ started at the same radius, with equality when the sector bound is radially tight; for small noise the mean is Arrhenius–Kramers, $\mathbb{E}\,\tau_R \approx \frac{\Gamma(n/2)}{2\alpha}\kappa_R^{-n/2}e^{\kappa_R}$ with $\kappa_R = \alpha R^2/\sigma_w^2$. The mean-square persistence condition is $\kappa_R > n/2$, at which the benchmark leaves within about one correction time; the condition certifies the scale of the mismatch, and only a margin in κ_R buys time in the region. **Proposition A.2 (adaptive reserve):** $\Delta\rho^* = \alpha R - \rho$ — the additional disturbance an agent can absorb before mismatch leaves the sector-condition region.

The segment's load-bearing new exact result is **Corollary A.1S.1 — the disturbance-model containment dichotomy.** The persistence-region first-exit probability is *categorical*: $P(\tau_R < \infty) = 0$ under Model D (deterministic positive invariance), $P(\tau_R < \infty) = 1$ under Model S (a.s. exit of a non-degenerate diffusion from a bounded region, for *any* F satisfying A2'). The achievable value is exactly the two-point set $\{0, 1\}$, and which point obtains is fixed by the disturbance model's support structure (bounded vs. unbounded), **not** by correction strength α. Increasing α tightens the typical scale and lengthens the expected stay in the region, exponentially in κ_R; it cannot interpolate between the two regimes. Pathwise containment is categorically Model-D-only — additive stochastic forcing removes the *kind* of

guarantee available, not merely its rate. The companion no-go demonstration (that the natural Ville/Doob maximal-inequality route to a $P(\tau_R < \infty) < 1$ bound *cannot* exist) is in `#deriv-stochastic-non-exit`. The dichotomy sharpens the hand-off into `#result-structural-adaptation-necessity`: in any genuinely stochastic environment, leaving the certified region is a certain eventual event, so loss of the parametric certificate is *generic, not exceptional*, for any sufficiently long-lived agent; whether that loss forces model-class change rather than a transient excursion is the open question of `#sketch-structural-adaptation-genericity`. The sub-scope α / β partition (where A2' is structurally derived versus where it is per-system assumed) is held at `#form-sector-condition`; the Lyapunov proofs below apply uniformly across both.

Motivation.

`#hyp-mismatch-dynamics` hypothesizes the linear ODE $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ as a first-order approximation. The linear form yields clean closed-form results but commits to a specific functional relationship between mismatch magnitude and correction rate. True correction dynamics are almost certainly nonlinear — exhibiting saturation at large mismatch, threshold effects near zero, and structural breakdown when the model class is exhausted.

A Lyapunov approach proves persistence and stability under much weaker assumptions: any correction dynamics satisfying qualitative monotonicity properties (the sector condition). The results below are strictly more general — the linear case is recovered where the sector bounds coincide.

Setup.

Let $\delta(t) \in \mathbb{R}^n$ be the estimation error $\bar{\delta} = \bar{o} - \hat{o}$ of `#def-mismatch-signal` — the difference between the model's predictions and the mean the world delivers given its state and law, across n observable dimensions. Every result in this segment is about this object, not about the observed innovation $o - \hat{o}$. The innovation exceeds it by channel noise that no correction can remove (`#result-mismatch-decomposition`). The vector treatment connects to per-dimension tempo analysis (`#result-per-dimension-persistence`).

The mismatch dynamics are:

Definition (Dynamics Setup)

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{A.1}$$

where:

- $F(\mathcal{T}, \delta) : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ is the **correction function** — how the agent's adaptive process reduces mismatch. It maps to the same space as δ (so that the inner product $\delta^T F$ in the sector condition is well-defined). This subsumes the update gain η^* (`#emp-update-gain`), event rate ν, and the structure of the update rule.
- $w(t)$ is the **disturbance** — everything that moves the estimation error other than the expected correction, with $\|w(t)\| \leq \rho$ (bounded disturbance rate). It has two sources. One is environmental change, which moves $\bar{o}$. The other is the channel noise the correction itself feeds back, because the update is driven by the innovation $\bar{\delta} + \nu$ (with ν the channel noise), not by $\bar{\delta}$ (K, Props B.1 and B.3).

The linear case from `#hyp-mismatch-dynamics` has $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$.

The disturbance depends on the gain. For linear correction at rate α the gain-injected part is $-\alpha\nu(t)$. Under Model D, with environmental change bounded by ρ_{env} and channel noise bounded by $\bar{\nu}$, the disturbance bound is $\rho = \rho_{\text{env}} + \alpha\bar{\nu}$. Prop A.1's ultimate bound is then

Derived (Model D with bounded channel noise, linear correction)

$$R^* = \frac{\rho_{\text{env}}}{\alpha} + \bar{\nu} \tag{A.2}$$

The lag term falls with α and the noise term does not. Under Model S, with environmental diffusion σ_{env} and channel noise of intensity σ_ν per coordinate (observation $dy = \bar{o}\,dt + \sigma_\nu\,dV$), the disturbance intensity is $\sigma_w^2 = \sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2$. Prop A.1S's steady state is then

Derived (Model S with channel noise, linear correction)

$$\mathbb{E}\|\delta\|_{ss}^2 = \frac{n(\sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2)}{2\alpha} \tag{A.3}$$

This is minimized at $\alpha = \sigma_{\text{env}}/\sigma_\nu$, which is the steady-state Kalman–Bucy gain for a random-walk signal, at the value $n\sigma_{\text{env}}\sigma_\nu$, the Kalman–Bucy error variance. Above that gain the bound grows: correcting faster on the same observations injects more noise than it removes lag. This is the lag/noise trade-off of adaptive filtering (Widrow, McCool, Larimore & Johnson 1976[1]: gradient noise "proportional to the speed of adaptation", lag "inversely proportional", the sum minimized where the two contributions are equal — here $\alpha^2\sigma_\nu^2 = \sigma_{\text{env}}^2$). Throughout the persistence family, ρ and σ_w denote this *total* disturbance (environmental change plus injected noise; σ_w per coordinate, so that $\mathbb{E}\|w\|^2 = n\sigma_w^2$ per unit time), and the propositions below are exact in them for any disturbance meeting their hypotheses. Read with the environmental part alone, $R^* = \rho_{\text{env}}/\alpha$ and $\sigma_{\text{env}}\sqrt{n/(2\alpha)}$ are the lag terms, and $1/\alpha$ and $1/\sqrt{\alpha}$ describe how the lag term falls; the channel-noise part sets a floor that a stronger correction cannot lower. Under Model D the floor $\bar{\nu}$ is fixed and the bound falls monotonically toward it. Under Model S with white channel noise the injected part grows with the gain, which is why the bound turns up past the Kalman gain. The two models behave differently here, and the Model S picture is not to be read into Model D.

Assumptions on F.

The Lyapunov derivations below operate on the structural-property triple (A1) zero-correction-at-zero-mismatch, (A2') local sector condition, (A3) tempo-monotonicity, stated in full at `#form-sector-condition`. The sub-scope α / β partition — the agent classes where A2' is structurally derived (Bayesian / exponential family / strongly-convex gradient / L2-regularized / linear-PD) versus those where it stands as a well-scoped per-system empirical claim (PID, rule-based, human judgment, severely misspecified, variational, non-convex beyond basin, per-step SGD) — is also held there, together with the operator-family classification (Rockafellar / Bauschke-Combettes / Baillon-Haddad) that names how the AAT-internal sub-scope partition lines up with monotone-operator theory.

The Lyapunov proofs below apply uniformly across both sub-scopes — they operate downstream of A2' regardless of whether it is structurally derived or per-system assumed.

Candidate Lyapunov Function.

Definition (Lyapunov Candidate)

$$V(\delta) = \frac{1}{2}\|\delta\|^2 \tag{A.4}$$

Positive definite, radially unbounded, continuously differentiable. Its level sets $V = c$ are spheres of radius $\sqrt{2c}$ in mismatch space.

[1]Widrow, B., McCool, J. M., Larimore, M. G., & Johnson, C. R. Jr. (1976). "Stationary and nonstationary learning characteristics of the LMS adaptive filter." *Proceedings of the IEEE* 64(8):1151–1162. doi:10.1109/PROC.1976.10286.

Proposition A.1: Bounded Mismatch (Single Agent).

Statement. Under (A1), (A2'), (A3), with bounded disturbance $\|w(t)\| \leq \rho$, the mismatch $\delta(t)$ is **ultimately bounded**: there exists $R^* > 0$ such that $\|\delta(t)\| \leq R^*$ for all sufficiently large t, provided $R^* < R$ (the ultimately bounded region fits within the sector-condition region).

Proof.

Compute $\dot{V}$ along trajectories:

Derived (Proof Step)

$$\dot{V} = \delta^T \dot{\delta} = \delta^T[-F(\mathcal{T}, \delta) + w(t)] \tag{A.5}$$

Derived (Proof Step)

$$= -\delta^T F(\mathcal{T}, \delta) + \delta^T w(t) \tag{A.6}$$

By (A2'): $\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2$

By Cauchy-Schwarz: $\delta^T w(t) \leq \|\delta\| \cdot \|w(t)\| \leq \rho\|\delta\|$

Therefore:

Derived (Proof Step)

$$\dot{V} \leq -\alpha\|\delta\|^2 + \rho\|\delta\| = -\|\delta\|(\alpha\|\delta\| - \rho) \tag{A.7}$$

$\dot{V} < 0$ whenever $\|\delta\| > \rho/\alpha$ **and** $\|\delta\| \leq R$ (where A2' holds).

Define $R^* = \rho/\alpha$.

Invariance of $\mathcal{B}_R$. When $R^* < R$ (the persistence condition), the ball $\mathcal{B}_R$ is *positively invariant*: any trajectory starting in $\mathcal{B}_R$ remains in $\mathcal{B}_R$ for all future time. At the boundary $\|\delta\| = R$, the sector condition holds and $\dot{V} = -\|\delta\|(\alpha\|\delta\| - \rho) = -R(\alpha R - \rho) < 0$ (since $\alpha R > \rho$ by the persistence condition). Trajectories at the boundary point inward. Therefore $\mathcal{B}_R$ is invariant, and the sector condition applies to the entire future trajectory of any trajectory starting inside $\mathcal{B}_R$.

Ultimate boundedness. Within $\mathcal{B}_R$, the Lyapunov function is strictly decreasing for $\|\delta\| > R^*$ and may increase for $\|\delta\| < R^*$. All trajectories starting in $\mathcal{B}_R$ are ultimately bounded by R^* — they are driven into $\mathcal{B}_{R^*}$ and remain in a neighborhood of it (with possible oscillation at the boundary due to the disturbance).

Initial condition requirement. The result requires the trajectory to start inside $\mathcal{B}_R$ (or to be brought inside by some external mechanism). A trajectory starting outside $\mathcal{B}_R$ is not covered — the sector condition does not hold there, and the correction function may fail to reduce mismatch. This is precisely the structural adaptation regime of `#result-structural-adaptation-necessity`: an agent whose mismatch exceeds R has exhausted its model class capacity and needs structural change to re-enter the region where parametric correction works.

The persistence condition $\alpha > \rho/R$ (equivalently $R^* < R$) is therefore *sufficient* for persistence from within $\mathcal{B}_R$. □

Lemma A.1N (threshold tightness — necessity is class-level, not agent-level). Over the hypothesis class $\mathcal{F}_\alpha = \{F : \text{(A1), (A2'), (A3) hold on } \mathcal{B}_R \text{ with sector floor } \alpha\}$, the condition $\alpha > \rho/R$ is necessary *and* sufficient for the uniform containment guarantee (every $F \in \mathcal{F}_\alpha$ keeps every trajectory from $\mathcal{B}_R$ inside $\mathcal{B}_R$ under every $\|w(t)\| \leq \rho$).

Proof. Sufficiency is Prop A.1, uniform over the class. Necessity: the extremal member $F = \alpha\delta$ lies in $\mathcal{F}_\alpha$; under the outward radial disturbance $w = \rho\,\delta/\|\delta\|$ its radial dynamics obey $\frac{d}{dt}\|\delta\| = \rho - \alpha\|\delta\| \geq \rho - \alpha R > 0$ whenever $\alpha < \rho/R$, so from any nonzero start the trajectory exits $\mathcal{B}_R$ in finite time. (At $\alpha = \rho/R$ exactly, the trajectory asymptotes to R without crossing — containment holds, with ultimate bound R rather than $< R$.) □

The lemma answers the question the old "iff" was reaching for — *when is the persistence condition also necessary?* — and the answer is **tightness, not linearity**. For a fixed agent whose sector bound is radially tight (additionally $\delta^T F \le \bar{\alpha}\|\delta\|^2$ with $\bar{\alpha} = \alpha$; the linear case $F = \mathcal{T}\delta$ is the canonical instance), the same outward-disturbance construction forces finite-time exit, and the threshold is a genuine agent-level if-and-only-if. In general, the object necessity tracks is the *upper* sector constant $\bar{\alpha}$, not the floor α: the Lyapunov argument bounds $\dot{V}$ from above via the floor, and forcing escape needs a bound from below, which only an upper sector bound supplies.

Remark (the floor is not the barrier — why agent-level necessity fails in general). For a fixed F, failure of $\alpha > \rho/R$ does *not* imply escape, because α is a worst-case infimum over $\mathcal{B}_R$ that the dynamics may render unreachable. Demonstration (1D; extends to $\mathbb{R}^n$ radially via $F(\delta) = g(\|\delta\|)\,\delta$): take $R = 1$, $\rho = 1$, and $g \ge 10$ everywhere on $(0, 1]$ except a narrow dip to $g = 0.1$ near $r = 0.5$. The tight sector floor on $\mathcal{B}_R$ is $\alpha = g(0.5) = 0.1 < \rho/R$ — the persistence condition fails tenfold — yet $F(r) > \rho$ on an annulus below the dip, so every admissible trajectory started at small mismatch is stopped there: an impenetrable barrier *below* the weak spot, which no admissible disturbance can reach. The supremum of $\|\delta(t)\|$ over all $\|w\| \le \rho$ from $\delta(0) = 0$ is ≈ 0.1 (numerically confirmed, including against randomized and outward-feedback adversaries). The sharp agent-level escape threshold in 1D is a *mountain-pass* quantity — escape requires $\rho > \max_r F(r)$ over the radii the trajectory must traverse — not the global floor αR. Condition failure therefore means the containment *certificate* is lost (and escape is forced for the extremal class member and every radially tight agent), never that escape is certified for the agent at hand.

Interpretation. The ultimately bounded region has radius $R^* = \rho/\alpha$. In the linear case, $\alpha = \mathcal{T}$, recovering `#result-persistence-condition`'s steady-state result $R^* = \rho/\mathcal{T}$ exactly. But Proposition A.1 holds for *any* correction function satisfying the sector condition, not just the linear one.

The persistence threshold, generalized. The persistence condition $\rho/\alpha < R$ guarantees persistence for any correction function satisfying the sector condition, and Lemma A.1N shows it is *unimprovable given only A2'*: necessary and sufficient for the guarantee at class level, and agent-level necessary exactly in the radially tight case. If the correction function breaks down (A2' fails) before R^* is reached, the agent may diverge. Condition failure IS `#result-structural-adaptation-necessity`'s trigger in certificate form: when $\rho/\alpha > R$ (the environment demands more correction than the model class can *certifiably* provide), parametric adaptation loses its containment guarantee and structural change is required to restore one.

Proposition A.2: Stability Margin (Adaptive Reserve).

Statement. Under the conditions of A.1, the agent can tolerate a sudden increase in disturbance rate of:

Derived (adaptive-reserve)

$$\Delta\rho^* = \alpha R - \rho \tag{A.8}$$

without mismatch diverging (where R is the radius of the sector-condition region from A2'). Beyond this, R^* exceeds R and the correction function may fail.

Proof. After a shock, the new disturbance rate is $\rho + \Delta\rho$. The new ultimately bounded radius is $(\rho + \Delta\rho)/\alpha$. This remains within the valid region iff $(\rho + \Delta\rho)/\alpha \le R$, i.e., $\Delta\rho \le \alpha R - \rho$. □

Interpretation. $\Delta\rho^*$ is the agent's **adaptive reserve** — how much additional environmental volatility it can absorb before its model breaks down. This is a single number characterizing an agent's robustness to shock:

- An agent operating well below capacity ($\rho \ll \alpha R$) has a large reserve — it is **robust**.
- An agent near its limit ($\rho \approx \alpha R$) has a small reserve — it is **fragile**.

Table A.1

Domain	*Large $\Delta\rho^*$ (robust)*	*Small $\Delta\rho^*$ (fragile)*
Control	Kalman filter on slow target	Same filter on erratic target
Biology	Organism in stable niche	Same organism under climate change
Organization	Well-capitalized firm, stable market	Startup in volatile market
Military	Force with operational depth	Force at culmination point

Proposition A.1S: Bounded Mismatch Under Stochastic Disturbance.

Statement (region-aware form). Under (A1), (A2') on $\mathcal{B}_R$, (A3), with stochastic disturbance (GA-2S: $d\delta = -F(\mathcal{T}, \delta)\,dt + \sigma_w\,dW_t$ with W a standard n-dimensional Wiener process, so σ_w^2 is the per-coordinate disturbance intensity and $\mathbb{E}\|w\|^2 = n\sigma_w^2$ per unit time), and with the regularity (S) that F is measurable and locally bounded on $\bar{\mathcal{B}}_R$ and the SDE has a weak solution with continuous paths up to τ_R (every locally Lipschitz F qualifies, and by Girsanov every bounded measurable F on the ball), let $\tau_R = \inf\{t : \|\delta(t)\| > R\}$ be the first-exit time from the sector-condition region and write $\kappa_R := \alpha R^2/\sigma_w^2$. Then:

Derived (stochastic-bounded-mismatch, Itô–Lyapunov on the region, Khasminskii 2012 ch. 5)

(i) *Killed second-moment bound* — the mismatch, counted only while it is still inside the region, satisfies for all $t \geq 0$

$$\mathbb{E}\big[\|\delta(t)\|^2;\ \tau_R > t\big] \ \leq\ \|\delta(0)\|^2 e^{-2\alpha t} + \frac{n\sigma_w^2}{2\alpha}\big(1 - e^{-2\alpha t}\big). \tag{A.9}$$

This is the scale of the mismatch *while still certified*. It says nothing about the mismatch after exit, and it is not a bound on the stopped moment $\mathbb{E}\|\delta(t\wedge\tau_R)\|^2$: the mass frozen on the sphere at exit does not decay, and under (ii) that stopped moment tends to R^2, not to $n\sigma_w^2/(2\alpha)$ (Corollary A.1S.1 gives $P(\tau_R \leq t) \to 1$).

(ii) *Mean-square persistence condition* — when $R_S^* := \sigma_w\sqrt{n/(2\alpha)} < R$, equivalently

$$\alpha > \frac{n\sigma_w^2}{2R^2} \quad\Longleftrightarrow\quad \kappa_R > \frac{n}{2}, \tag{A.10}$$

the RMS scale of the certified mismatch fits inside the sector-condition region. What this certifies is the scale in (i) for as long as the process remains certified, and, by Prop A.1S-τ below, an expected stay in the region of about one correction time at the threshold and exponentially longer with margin. Where (A2') holds on all of $\mathbb{R}^n$ (the global case; the linear corrector and the sub-scope-α families with a global floor), it certifies the stationary scale $\mathbb{E}\|\delta\|_{ss}^2 \leq n\sigma_w^2/(2\alpha)$ outright.

(iii) *Fixed-time tail* — in the global case, Markov's inequality on the unstopped second moment gives, for each t,

$$P\big(\|\delta(t)\| > R\big) \ \leq\ \frac{\|\delta(0)\|^2 e^{-2\alpha t} + n\sigma_w^2/(2\alpha)}{R^2} \ \xrightarrow[t\to\infty]{}\ \frac{n\sigma_w^2}{2\alpha R^2} = \frac{n}{2\kappa_R}. \tag{A.11}$$

Under (A2') on the ball only, no fixed-time tail for the unstopped process follows, because F is unconstrained outside the ball; the region-aware statement is the composition of (i) with the exit-time bound of Prop A.1S-τ: for $r \leq R$ and $\|\delta(0)\| = \|x\|$,

$$P\big(\|\delta(t)\| > r\big) \ \leq\ P(\tau_R \leq t) + \frac{\mathbb{E}[\|\delta(t)\|^2;\ \tau_R > t]}{r^2} \ \leq\ \left[1 - \frac{\Phi(\|x\|) - t}{\Phi(0)}\right] + \frac{\|x\|^2 e^{-2\alpha t} + (R_S^*)^2(1 - e^{-2\alpha t})}{r^2}, \tag{A.12}$$

with Φ the benchmark exit-time function of Lemma A.1S-τ.0. Neither form is an infinite-horizon containment statement: under additive Brownian forcing $P(\tau_R < \infty) = 1$ (Corollary A.1S.1; the no-go in `#deriv-stochastic-non-exit`). Pathwise containment of $\mathcal{B}_R$ is a Model-D guarantee only (Prop A.1's positive invariance); Model S controls the *scale* (i)–(ii), the *fixed-time tail* (iii), the *finite-horizon path* (iv), and the *time to exit* (Prop A.1S-τ), not the sample path over an unbounded horizon.

(iv) *Finite-horizon sample-path bound* — for any horizon T,

$$P\Big(\sup_{0\le s\le T}\|\delta(s)\| > R\Big) \;\le\; \frac{\|\delta(0)\|^2 + n\sigma_w^2\,T}{R^2}, \tag{A.13}$$

rigorous under (A2') on the ball alone. It controls the whole path on $[0, T]$, grows linearly in T, and is vacuous for $T > rsimR^2/(n\sigma_w^2)$, consistent with $P(\tau_R < \infty) = 1$. From the origin it is never tighter than Prop A.1S-τ(iii), which is exponential rather than linear in κ_R; it stays in canon because it is a closed form without special functions.

Proof.

Apply Itô's formula to $V(\delta) = \frac{1}{2}\|\delta\|^2$ along the solution:

Derived (Proof Step)

$$dV = -\delta^T F\,dt + \frac{n}{2}\sigma_w^2\,dt + \sigma_w\,\delta^T dW_t, \tag{A.14}$$

the last drift term being the Itô correction $\frac{1}{2}\,\mathrm{tr}(\sigma_w^2 I_n)$. Everything below is run on $[0, t\wedge\tau_R]$, where $\delta \in \bar{\mathcal{B}}_R$ almost surely and (A2') applies; the stopped Itô integral has zero expectation because δ is bounded there.

The exact stopped identity. Integrating $d\|\delta\|^2 = 2\,dV$ and taking expectations,

$$\mathbb{E}\|\delta(t\wedge\tau_R)\|^2 \;=\; \|\delta(0)\|^2 + n\sigma_w^2\,\mathbb{E}[t\wedge\tau_R] - 2\,\mathbb{E}\int_0^{t\wedge\tau_R}\delta^T F\,ds. \tag{A.15}$$

(iv). The last term is non-positive by (A2'), so $\mathbb{E}\|\delta(T\wedge\tau_R)\|^2 \le \|\delta(0)\|^2 + n\sigma_w^2 T$. On $\{\tau_R \le T\}$ the stopped path sits on the sphere, so $R^2\,P(\tau_R \le T) \le \mathbb{E}\|\delta(T\wedge\tau_R)\|^2$, which is (iv). The decay term plays no part in (iv), which is why (iv) survives the correction to (i).

(i). Apply Itô to $e^{2\alpha s}V(\delta_s)$:

$$d\big(e^{2\alpha s}V\big) = e^{2\alpha s}\Big[\big(2\alpha V - \delta^T F\big)\,ds + \frac{n}{2}\sigma_w^2\,ds + \sigma_w\,\delta^T dW_s\Big], \tag{A.16}$$

and on the ball $2\alpha V - \delta^T F \le 0$ by (A2'). Integrating to $t\wedge\tau_R$ and taking expectations,

$$\mathbb{E}\big[e^{2\alpha(t\wedge\tau_R)}V(\delta_{t\wedge\tau_R})\big] \;\le\; V(\delta_0) + \frac{n}{2}\sigma_w^2\,\mathbb{E}\int_0^{t\wedge\tau_R} e^{2\alpha s}\,ds \;\le\; V(\delta_0) + \frac{n\sigma_w^2}{4\alpha}\big(e^{2\alpha t} - 1\big). \tag{A.17}$$

The left side is at least its part on $\{\tau_R > t\}$, which is $e^{2\alpha t}\,\mathbb{E}[V(\delta_t);\ \tau_R > t]$; the part on $\{\tau_R \le t\}$ is non-negative and is dropped. Multiplying through by $2e^{-2\alpha t}$ gives (i). The step that is *not* available is replacing $\mathbb{E}[V(\delta_t);\ \tau_R > t]$ by $\mathbb{E}[V(\delta_{t\wedge\tau_R})]$: the decay in the drift carries the indicator $\mathbf{1}_{\{s<\tau_R\}}$, so the mass frozen on the sphere does not contract, and a Grönwall bound on the stopped moment does not close. (Monte Carlo, $n = 2$, $\alpha = 2$, $\sigma_w = R = 1$, so $\kappa_R = 2 > n/2$, from the origin: the stopped moment is 0.74 at $t = 1$ and 0.995 at $t = 4$ against a would-be bound of 0.5; the killed moment is 0.09 and 0.002, under the (i) bound 0.49 and 0.50.)

(ii) and (iii), global case. When (A2') holds on all of $\mathbb{R}^n$ no stopping is needed: the same Itô step on V gives $\frac{d}{dt}\mathbb{E}[V] \leq -2\alpha\,\mathbb{E}[V] + \frac{n}{2}\sigma_w^2$, Grönwall gives $\mathbb{E}\|\delta(t)\|^2 \leq \|\delta(0)\|^2 e^{-2\alpha t} + \frac{n\sigma_w^2}{2\alpha}(1 - e^{-2\alpha t})$, and Markov's inequality gives (iii). The steady state is $\mathbb{E}\|\delta\|_{ss}^2 \leq n\sigma_w^2/(2\alpha)$, with equality for the linear corrector (the Ornstein–Uhlenbeck stationary variance), and

Derived (stochastic-steady-state)

$$R_S^* = \sigma_w\sqrt{\frac{n}{2\alpha}}. \tag{A.18}$$

(iii), local case. $P(\|\delta(t)\| > r) \leq P(\tau_R \leq t) + P(\|\delta(t)\| > r,\ \tau_R > t)$; the first term is bounded by Prop A.1S-τ(iii) and the second by Markov on (i). □

Interpretation. Model D (Prop A.1) gives $R^* = \rho/\alpha$, scaling as $1/\alpha$. Model S gives $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$, scaling as $1/\sqrt{\alpha}$, both at fixed disturbance. Doubling the correction rate halves the deterministic bound but reduces the stochastic one only by $\sqrt{2} \approx 1.41$: correction is less effective against noise than against drift. This difference in scaling, at fixed ρ and σ_w, is what propagates into the adversarial exponent regimes (`#result-adversarial-exponent-regimes`): $b = 2$ under Model D, $b = 3/2$ under Model S. With the gain-injected channel noise of the Setup included, σ_w itself depends on α for a fixed channel, and the scalings describe the lag term alone.

Tail bound. In the global case the stationary Markov bound is $P(\|\delta\| > R) \leq n\sigma_w^2/(2\alpha R^2)$, so the agent is outside R at any given time with probability at most ϵ provided $\alpha \geq n\sigma_w^2/(2\epsilon R^2)$. For the linear corrector the stationary law is Gaussian and exact tail probabilities are available; the Markov bound is the general result under global (A2'). For a corrector certified only on the ball, the fixed-time tail is the composed form in (iii).

Corollary A.1S.1: Disturbance-Model Containment Dichotomy.

Derived

For a trajectory started in $\mathcal{B}_R$ under (A1), (A2') on $\mathcal{B}_R$, (A3), the persistence-region first-exit probability is **categorical**:

$$P(\tau_R < \infty) = \begin{cases} 0 & \text{Model D (bounded } w,\ \|w\| \leq \rho) \text{ under } \alpha R > \rho, \\ 1 & \text{Model S (additive stochastic, } d\delta = -F\,dt + \sigma_w\,dW_t). \end{cases} \tag{A.19}$$

The Model-D value is the deterministic positive invariance of Prop A.1 (boundary-inward: at $\|\delta\| = R$, $\dot{V} < 0$ when $\alpha R > \rho$). The Model-S value holds for **every** $\alpha > 0$, **every** $\sigma_w > 0$, and **every** correction function F satisfying (A2'): a non-degenerate diffusion (additive forcing $\sigma_w\,dW_t$, $\sigma_w > 0$) exits any bounded region in finite time almost surely, for any locally bounded drift — no finite inward correction can defeat the Brownian crossing near $\partial\mathcal{B}_R$. The Ornstein–Uhlenbeck benchmark makes the mechanism explicit (scale function $s'(u) \propto e^{\alpha u^2/\sigma_w^2}$ unbounded, hence recurrent; Khasminskii 2012[^khasminskii2012] ch. 3–4), but the conclusion needs only non-degeneracy of the additive noise on a bounded region, not the linear structure. That the natural maximal-inequality route to a $P(\tau_R < \infty) < 1$ bound *cannot* exist — the question "are you sure you can't just Doob/Ville this?" — is demonstrated in `#deriv-stochastic-non-exit`.

The achievable first-exit probability is therefore exactly the two-point set $\{0, 1\}$, and **which point obtains is fixed by the disturbance model's support structure (bounded vs. unbounded), not by the correction strength** α. Increasing α tightens the typical scale ($R_S^* \propto 1/\sqrt{\alpha}$) and lengthens the expected time in the region (Prop A.1S-τ, exponentially in κ_R at fixed σ_w); it cannot move $P(\tau_R < \infty)$ off 1 under Model S, and is unnecessary for the 0 under Model D. Pathwise containment of $\mathcal{B}_R$ is categorically a bounded-disturbance property: additive stochastic forcing removes the *kind* of guarantee available, not merely its rate. This sharpens the hand-off into `#result-structural-adaptation-necessity` — in any genuinely stochastic environment region-exit is a certain eventual event, so loss of the parametric certificate is *generic, not exceptional*, for a sufficiently long-lived agent; whether it forces model-class change is O's open question, and Prop A.1S-τ says when it comes.

Proposition A.1S-τ: Time in the Certified Region Under Stochastic Disturbance.

Corollary A.1S.1 says *whether* the region is left (with certainty). This proposition says *when*, and it is the region-aware content of Model S: the time to first exit from $\mathcal{B}_R$ is bounded below by that of the simplest sector-satisfying corrector, the isotropic linear one $F = \alpha\delta$, whose exit time is explicit in every dimension. The reason is one observation: (A2') is a lower bound on the *radial* component of the correction, $\hat{\delta}^T F \geq \alpha\|\delta\|$ with $\hat{\delta} = \delta/\|\delta\|$, and leaving a ball is a statement about the radius alone, so the radial process of any sector-satisfying corrector has drift pointwise at least as inward as the linear corrector's, with the same noise. Throughout, $\kappa_R = \alpha R^2/\sigma_w^2 = \frac{n}{2}(R/R_S^*)^2$, so the mean-square persistence condition (ii) is $\kappa_R > n/2$.

Lemma A.1S-τ.0 (the Ornstein–Uhlenbeck benchmark). *[Derived; classical.]* Let Y solve $dY = -\alpha Y\,dt + \sigma_w\,dW$ in $\mathbb{R}^n$ and let $\Phi(r) = \Phi_{\alpha,n,\sigma_w,R}(r) := \mathbb{E}_y[\tau_R]$ for $\|y\| = r \leq R$. Then

$$\Phi(r) = \frac{2}{\sigma_w^2}\int_r^R \frac{G(u)}{m(u)}\,du, \qquad m(u) = u^{n-1}e^{-\alpha u^2/\sigma_w^2}, \qquad G(u) = \int_0^u m(s)\,ds = \tfrac{1}{2}\Big(\frac{\sigma_w^2}{\alpha}\Big)^{n/2}\gamma\Big(\frac{n}{2}, \frac{\alpha u^2}{\sigma_w^2}\Big), \tag{A.20}$$

with γ the lower incomplete gamma function. Φ is the radial C^2 solution of $\mathcal{L}_{\mathrm{OU}}\Phi = -1$ on $\mathcal{B}_R$ with $\Phi(R) = 0$, where for radial ϕ the generator is $\mathcal{L}_{\mathrm{OU}}\phi = \frac{\sigma_w^2}{2}\big(\phi'' + \frac{n-1}{r}\phi'\big) - \alpha r\phi'$. Moreover $\Phi'(r) = -\frac{2}{\sigma_w^2}G(r)/m(r) \leq 0$, $\Phi'(0) = 0$, $\Phi(0) = \max\Phi$, Φ is concave as a function of r^2, and $v(y) := \Phi(\|y\|)$ is real-analytic near $\bar{\mathcal{B}}_R$. Two readings of the same function.

Small noise. As $\kappa_R \to \infty$ at fixed n, by Laplace's method at the endpoint $u = R$,

$$\Phi(0) = \frac{\Gamma(n/2)}{2\alpha}\,\kappa_R^{-n/2}\,e^{\kappa_R}\big(1 + O(\kappa_R^{-1})\big), \qquad \sigma_w^2\log\Phi(r) \to \alpha R^2 \ \text{ for every fixed } r < R. \tag{A.21}$$

This is Arrhenius–Kramers: the benchmark is a Brownian particle in the well $U(y) = \frac{1}{2}\alpha\|y\|^2$ at temperature $T = \sigma_w^2/2$, and $\kappa_R = U(R)/T$ is the barrier height in units of temperature (an isomorphism for the benchmark, the two systems being the same SDE). The expansion is not uniform in n and is not a bound in either direction: the ratio exact/asymptotic is 1.29 at $(n, \kappa_R) = (2, 4)$ and falls to 1.07 at $(2, 16)$, while below the mean-square threshold ($\kappa_R < n/2$) the asymptotic over-estimates by orders of magnitude at large n. Rigorous statements use the integral.

At the threshold. $\alpha\,\Phi(0)$ at $\kappa_R = n/2$ (so $R = R_S^*$) is 0.60, 0.66, 0.71, 0.78, 0.89, 1.02, 1.20, 1.35, 1.89 for $n = 1, 2, 3, 5, 10, 20, 50, 100, 1000$: about one correction time, drifting upward like $\frac{1}{4}\ln n + 0.15$ for large n. Margin buys time quickly: at $n = 3$, $\alpha\Phi(0) = 0.71, 1.57, 3.12, 12.0, 51.2$ as κ_R goes from $n/2$ to $n/2 + 1, 2, 4, 6$.

Derivation. Put $\psi = \phi'$ in $\mathcal{L}_{\mathrm{OU}}\phi = -1$: $\frac{\sigma_w^2}{2}\psi' + \big(\frac{(n-1)\sigma_w^2}{2r} - \alpha r\big)\psi = -1$. Since $m'/m = (n-1)/r - 2\alpha r/\sigma_w^2$, the integrating factor is m: $\frac{\sigma_w^2}{2}(m\psi)' = -m$, so $m\psi = -\frac{2}{\sigma_w^2}G$ (the constant vanishes because $m\psi \to 0$ at 0: $m(0) = 0$ for $n \geq 2$, $\psi(0) = 0$ by evenness for $n = 1$). Integrate from r to R with $\phi(R) = 0$. The incomplete-gamma form is the substitution $v = \alpha s^2/\sigma_w^2$ in G. Concavity in $s = r^2$: $d\Phi/ds = -\frac{1}{\sigma_w^2}\int_0^1 t^{n-1}e^{(\alpha r^2/\sigma_w^2)(1-t^2)}\,dt$ (substitute $u = rt$ in G), which decreases in r. That $\Phi = \mathbb{E}\tau_R$ is Dynkin's formula plus uniqueness for the Dirichlet problem of a non-degenerate generator on a bounded domain (the Andronov–Vitt–Pontryagin problem; Khasminskii 2012[^khasminskii2012] ch. 3; the formula for every n is Kersting, Orvieto, Proske & Lucchi 2023[^kersting2023], Theorem 4, and Grebenkov 2015[^grebenkov2015], eq. (75)). For the asymptotic, $G(u) \to \frac{1}{2}\Gamma(\frac{n}{2})(\sigma_w^2/\alpha)^{n/2}$ up to exponentially

small error once $u \gg \sigma_w/\sqrt{\alpha}$, and $\int^R u^{1-n} e^{\alpha u^2/\sigma_w^2}\, du$ is dominated by its endpoint, $\frac{\sigma_w^2}{2\alpha} R^{-n} e^{\kappa_R}(1 + O(\kappa_R^{-1}))$; the region $u \lesssim \sigma_w/\sqrt{\alpha}$ contributes $O(1/\alpha)$. The threshold values are the integral evaluated numerically (arbitrary precision; two independent implementations agree to four digits). □

Derived (exit-time comparison; Itô–Dynkin and a same-noise radial comparison)

Statement. Under (A1), (A2') on $\bar{\mathcal{B}}_R$, (S) and GA-2S, for every start $x \in \bar{\mathcal{B}}_R$:

(i) *Mean.* $\mathbb{E}_x[\tau_R] \geq \Phi_{\alpha,n,\sigma_w,R}(\|x\|)$. If also $\delta^T F \leq \bar{\alpha}\|\delta\|^2$ on the ball, then $\mathbb{E}_x[\tau_R] \leq \Phi_{\bar{\alpha},n,\sigma_w,R}(\|x\|) < \infty$. Both are equalities when the sector bound is radially tight, $\delta^T F = \alpha\|\delta\|^2$ on the ball, which includes $F = \alpha\delta$ and every $F = \alpha\delta + T(\delta)$ with $T(\delta) \perp \delta$, linear or not. Over the class $\mathcal{F}_\alpha$ of Lemma A.1N the lower bound is therefore attained and unimprovable from the floor alone, and for a fixed non-tight F it can be arbitrarily loose (the dip corrector of the Remark after A.1N has a mean exit time governed by its whole radial profile). Radial tightness is the one hypothesis that makes the Model D threshold an agent-level iff (A.1N) and the Model S exit time exact.

(ii) *Law.* $P_x(\tau_R > t) \geq P^{\mathrm{OU}(\alpha)}_{\|x\|}(\tau_R > t)$ for every t: the exit time stochastically dominates the benchmark's, so every moment and every survival probability of the benchmark is a lower bound. With a ceiling $\bar{\alpha}$ the reverse domination holds, and in the radially tight case the laws coincide.

(iii) *Finite-lifetime containment.* For every lifetime $L \geq 0$,

$$P_x(\tau_R \leq L) \;\leq\; 1 - \frac{\Phi(\|x\|) - L}{\Phi(0)}, \qquad \text{in particular} \qquad P_0(\tau_R \leq L) \;\leq\; \frac{L}{\Phi(0)} \;\approx\; \frac{2\alpha L}{\Gamma(n/2)}\, \kappa_R^{n/2} e^{-\kappa_R}. \tag{A.22}$$

The bound is linear in L, so it is vacuous as $L \to \infty$, as the no-go requires. From every start it is at least as tight as (iv) of Prop A.1S, with equality only in the Brownian limit $\alpha \to 0$: it is (iv) with the decay term kept rather than dropped. Under (ii) the sharper $P_x(\tau_R \leq L) \leq P^{\mathrm{OU}}_{\|x\|}(\tau_R \leq L)$ holds.

(iv) *A tenure-indexed condition.* Fix a required time in the region L and a tolerated exit probability $\epsilon \in (0,1)$. Then

$$\Phi_{\alpha,n,\sigma_w,R}(0) \;\geq\; \frac{L}{\epsilon} \quad\Longrightarrow\quad P_0(\tau_R \leq L) \leq \epsilon, \tag{A.23}$$

and an elementary sufficient form, valid whenever $2\kappa_R > n - 1$ and for any $s \in (0, \kappa_R)$, with $P(a,s) = \gamma(a,s)/\Gamma(a)$ the regularized lower incomplete gamma function, is

$$\frac{\Gamma(n/2)}{2\alpha}\, \kappa_R^{-n/2} e^{\kappa_R} \cdot P\Big(\frac{n}{2}, s\Big) \cdot \frac{2\kappa_R}{2\kappa_R - n + 1} \cdot \Big(1 - \big(\tfrac{\kappa_R}{s}\big)^{(n-1)/2} e^{-(\kappa_R - s)}\Big) \;\geq\; \frac{L}{\epsilon}. \tag{A.24}$$

The leading-order form $\kappa_R - \frac{n}{2}\log\kappa_R \geq \log\frac{2\alpha L}{\epsilon\,\Gamma(n/2)}$ is the asymptotic of the exact condition, not a sufficient condition: its left side is not monotone in κ_R (it tends to $+\infty$ as $\kappa_R \to 0$), so it is satisfied by arbitrarily weak correction. At $n = 2, \alpha = R = 1, \sigma_w = 10, L = 1, \epsilon = 0.02$ it holds ($4.615 \geq 4.605$) while $\Phi(0) = 0.0050$ against the needed 50: the process leaves in about 1/200 of a correction time. The asymptotic is anti-conservative exactly near the mean-square threshold, where a reader is most likely to apply it. The shape of the exact condition is the content: Model D containment is free past its threshold (positive invariance, forever); Model S containment over a lifetime is bought by margin in κ_R, at a cost that grows only logarithmically in L/ϵ. Doubling the required time, or halving the tolerated risk, costs about $\log 2$ in κ_R, that is, in $\frac{n}{2}(R/R_S^*)^2$, the squared ratio of certified radius to RMS scale that the mean-square condition only asks to exceed one.

(v) *What the mean-square condition certifies.* At $\kappa_R = n/2$ the benchmark's expected stay is about one correction time (the table in the Lemma), so the condition $\alpha > n\sigma_w^2/(2R^2)$ with no margin certifies no more than that: it certifies the scale of the mismatch while certified, not the time for which the certificate holds. A non-tight corrector may stay far

longer (the dip corrector stays 3.8× the benchmark); what is licensed is the statement about the guarantee, not about the agent.

Proof. *(i).* Let $v = \Phi \circ \|\cdot\|$, C^2 near the closed ball by the Lemma. The generator of the sector dynamics is $\mathcal{L}v = -F \cdot \nabla v + \frac{\sigma_w^2}{2}\Delta v$; for radial v, $\nabla v = \Phi'(r)\hat{\delta}$ and $\Delta v = \Phi'' + \frac{n-1}{r}\Phi'$, so $\mathcal{L}v = -\Phi'(r)\,\hat{\delta}^T F + \frac{\sigma_w^2}{2}\big(\Phi'' + \frac{n-1}{r}\Phi'\big)$. Since $-\Phi' \geq 0$ and $\hat{\delta}^T F \geq \alpha r$ by (A2'), $\mathcal{L}v \geq -\alpha r\Phi' + \frac{\sigma_w^2}{2}(\Phi'' + \frac{n-1}{r}\Phi') = \mathcal{L}_{\mathrm{OU}}\Phi = -1$ off the origin, and at the origin both sides are -1 ($F(0) = 0$, $\Phi'(0) = 0$). Dynkin on the stopped process, valid because $v \in C^2$ and F is bounded on the ball: $\mathbb{E}_x[v(\delta_{t\wedge\tau_R})] = v(x) + \mathbb{E}_x\int_0^{t\wedge\tau_R}\mathcal{L}v\,ds \geq v(x) - \mathbb{E}_x[t \wedge \tau_R]$. If $P_x(\tau_R = \infty) > 0$ the claim is trivial; otherwise $v(\delta_{t\wedge\tau_R}) \to v(\delta_{\tau_R}) = \Phi(R) = 0$ a.s., v is bounded, and dominated and monotone convergence give $\mathbb{E}_x\tau_R \geq v(x)$. The upper bound uses $\bar{\Phi} = \Phi_{\bar{\alpha},n,\sigma_w,R}$ with the inequalities reversed, $\mathbb{E}_x[t \wedge \tau_R] \leq \bar{v}(x) - \mathbb{E}_x[\bar{v}(\delta_{t\wedge\tau_R})] \leq \bar{v}(x)$; finiteness gives $\tau_R < \infty$ a.s. as a by-product. Under radial tightness $\mathcal{L}v = -1$ exactly and Dynkin gives equality. The proof uses (A2') pointwise on the closed ball and Itô for a C^2 function along a continuous solution, and nothing else: no Lipschitz continuity, no gradient structure, no sector condition outside the ball, no small-noise limit.

(ii), $n \geq 2$. By Itô on $r = \|\delta\|$ off the origin, $dr = \beta_t\,dt + \sigma_w\,dB_t$ with $\beta_t = -\hat{\delta}^T F(\delta) + \frac{(n-1)\sigma_w^2}{2r}$ and $B_t = \int_0^t \hat{\delta}_s^T dW_s$, a standard one-dimensional Brownian motion by Lévy's characterization; (A2') gives $\beta_t \leq b_\alpha(r_t) := -\alpha r_t + \frac{(n-1)\sigma_w^2}{2r_t}$, which is exactly the radial drift of the benchmark Y. A non-degenerate diffusion with locally bounded drift does not hit a point in dimension $n \geq 2$ (Girsanov on $[0, t \wedge \tau_R]$ reduces it to Brownian motion, for which points are polar), so this holds on $(0, \tau_R]$ a.s. Let ϱ solve $d\varrho = b_\alpha(\varrho)\,dt + \sigma_w\,dB_t$, $\varrho(0) = \|x\|$, driven by the *same* B; it has the law of $\|Y\|$ and stays positive for $n \geq 2$. Then $D = r - \varrho$ has no martingale part, $dD = (\beta_t - b_\alpha(\varrho_t))\,dt$. On any interval where $D > 0$, $\dot{D} \leq b_\alpha(r_t) - b_\alpha(\varrho_t) < 0$ because b_α is strictly decreasing and $r_t > \varrho_t$ there; so if $D(t_1) > 0$, with $t_0 = \sup\{t < t_1 : D(t) \leq 0\}$, D is non-increasing on $(t_0, t_1]$ from $D(t_0) = 0$, a contradiction. Hence $r_t \leq \varrho_t$ for all $t \leq \tau_R(\delta)$, so ϱ reaches R no later than δ does; and because the benchmark's boundary is regular (it exceeds R immediately after touching it, a.s.), $\tau_R(\delta) \geq \tau_R(\varrho)$ pathwise, and $\tau_R(\varrho) \overset{d}{=} \tau_R(Y)$. *(ii), $n = 1$.* By Tanaka, $r = |\delta|$ satisfies $dr = -\operatorname{sgn}(\delta)F(\delta)\,dt + \sigma_w\,dB_t + dL_t$ with L the local time at 0 and $\operatorname{sgn}(\delta)F(\delta) = \delta F(\delta)/|\delta| \geq \alpha r$ by (A2'); take ϱ the OU process reflected at 0 on the same B (the law of $|Y|$). Where $D = r - \varrho > 0$ we have $r > 0$ so $dL = 0$, the reflection term of ϱ only lowers D, and the drift is at most $-\alpha D < 0$; the same first-exceedance argument closes. The ceiling and equality clauses follow with $\bar{\alpha}$ and with $\beta_t = b_\alpha(r_t)$. A published comparison theorem reaches the same conclusion for all $n \geq 1$ at once on the squared radius $\xi = \|\delta\|^2$, whose equation $d\xi = (n\sigma_w^2 - 2\delta^T F)\,dt + 2\sigma_w\sqrt{\xi}\,dB_t$ has no local-time and no $1/r$ term: Ikeda & Watanabe 1977[^ikedawatanabe1977], Theorem 1.1 with Remark 1.1, applied with the common diffusion coefficient $2\sigma_w\sqrt{\xi^+}$ (Yamada's modulus) and the affine comparison drift $n\sigma_w^2 - 2\alpha\xi$, as the spike's literature scan transcribed the theorem; neither the author of this derivation nor its verifier has read that paper at the primary, so the self-contained argument above is the basis and the citation locates the step in print.

(iii). Dynkin from (i) at $t = L$: $\mathbb{E}_x[v(\delta_{L\wedge\tau_R})] \geq v(x) - L$. On $\{\tau_R \leq L\}$ the integrand is $\Phi(R) = 0$; on $\{\tau_R > L\}$ it is at most $\Phi(0)$. So $\Phi(0)\,P_x(\tau_R > L) \geq \Phi(\|x\|) - L$. Comparison with Prop A.1S(iv): Theorem (i) with floor 0 gives $\Phi(0) \geq R^2/(n\sigma_w^2)$ (the Brownian exit time from the ball, $\mathbb{E}\tau = (R^2 - \|x\|^2)/(n\sigma_w^2)$, by Dynkin on $\|\delta\|^2$), and concavity of Φ in r^2 with $\Phi(R) = 0$ gives $\Phi(r) \geq \Phi(0)(1 - r^2/R^2)$; together, $1 - (\Phi(\|x\|) - L)/\Phi(0) \leq \|x\|^2/R^2 + L/\Phi(0) \leq (\|x\|^2 + n\sigma_w^2 L)/R^2$.

(iv). The exact condition is (iii) from the origin. For the elementary form, with $s = \alpha a^2/\sigma_w^2$ for some $a \in (0, R)$ and $\varphi(u) = \alpha u^2/\sigma_w^2 - (n-1)\log u$ (so $1/m = e^\varphi$): G is increasing, so $\Phi(0) \geq \frac{2}{\sigma_w^2}G(a)\int_a^R e^{\varphi(u)}\,du$; $\varphi'(u) = 2\alpha u/\sigma_w^2 - (n-1)/u$ is increasing, so $\varphi' \leq \varphi'(R) = (2\kappa_R - n + 1)/R$ on $[a, R]$, and when $\varphi'(R) > 0$, $\int_a^R e^\varphi\,du \geq \int_a^R e^\varphi\varphi'\,du/\varphi'(R) = (e^{\varphi(R)} - e^{\varphi(a)})/\varphi'(R)$. Substituting $e^{\varphi(R)} = e^{\kappa_R}R^{1-n}$, $e^{\varphi(a)} = e^s a^{1-n}$, $(R/a)^{n-1} = (\kappa_R/s)^{(n-1)/2}$ and $G(a) = \frac{1}{2}(\sigma_w^2/\alpha)^{n/2}\Gamma(\frac{n}{2})P(\frac{n}{2}, s)$, and using $(\sigma_w^2/\alpha)^{n/2}R^{2-n}/\sigma_w^2 = \kappa_R^{1-n/2}/\alpha$, gives the display.

Numerically the bound's ratio to the exact $\Phi(0)$, maximized over s, is $0.57, 0.74, 0.91, 0.97$ at $n = 1$, $\kappa_R = 2, 4, 8, 16$ and $0.69, 0.97, 0.99$ at $n = 20$, $\kappa_R = 20, 40, 80$: below one everywhere, as a lower bound must be, and approaching one.

(v). Computed from the Lemma's integral. □

The exit exponent is capped at the Kalman gain. With the Setup's disturbance for linear correction on a fixed channel, $\sigma_w^2 = \sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2$, the exponent is

Derived (linear correction, Model S, white channel noise)

$$\kappa_R(\alpha) = \frac{\alpha R^2}{\sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2}, \qquad \max_\alpha \kappa_R = \kappa_R(K) = \frac{R^2}{2\sigma_{\text{env}}\sigma_\nu} \text{ at } K = \frac{\sigma_{\text{env}}}{\sigma_\nu}, \tag{A.25}$$

the Kalman–Bucy gain. So correction strength buys expected time in the region only up to the Kalman gain, and the mean-square condition is satisfiable for some gain iff $\kappa_R(K) > n/2$, that is $R^2 > n\sigma_{\text{env}}\sigma_\nu$: the Kalman error must fit inside the region (the window of `#result-persistence-condition`). For the linear corrector (radially tight, so (i) is an equality) the expected stay is $\Phi(0) = g(\kappa_R(\alpha), n)/\alpha$ with g increasing in its first argument (scale $u = Rv$ in the Lemma). Past K both factors fall, so over-correcting shortens the expected stay monotonically. Below K the $1/\alpha$ factor can outweigh the smaller exponent, so the gain that maximizes expected stay lies strictly below K (the derivative at K is $-\Phi(0)/K$) and approaches K as $\kappa_R(K)$ grows; at $n = 1$, $\sigma_{\text{env}} = 1$, $\sigma_\nu = 0.5$, $R = 1$ (so $\kappa_R(K) = 1$), $\Phi(0) = 1.07, 0.87, 0.72, 0.56, 0.27$ at $\alpha = 0.5K, 0.8K, K, 1.25K, 2K$. The headline "correction strength buys time exponentially in $\alpha R^2/\sigma_w^2$" is therefore true at fixed σ_w and false for a gain raised on a fixed channel past the Kalman gain; what buys time without limit is information (more or better observations, which lower σ_ν), not gain.

Scope. Isotropic additive noise, continuous time, (A2') on the closed ball. Not covered, each named so it is not assumed: anisotropic noise (the radial comparison uses $\sigma_w I$; a time-change argument gives the exponent $\alpha R^2/\lambda_{\max}(\Sigma_w)$ with a crude prefactor, sketched by the verifier and not landed); discrete time (DA2'); state-dependent noise (K B.3's nonlinear-g case, where the stopped bounds hold with $n\sigma_w^2$ replaced by the supremum of the diffusion's trace over the ball); heavy tails, where exit is driven by a single large jump and the expected exit time is polynomial rather than exponential in the noise scale (Imkeller & Pavlyukevich 2006; Högele & Pavlyukevich 2014, as read by the spike's literature scan), so the exponential reserve of this proposition genuinely does not exist there and canon's scope line, heavy tails are structural-adaptation triggers, is sharper for it. Exit from $\mathcal{B}_R$ is loss of the *certificate*, not of the agent: R is the certified radius, the benchmark is positively recurrent (it leaves and returns), and whether what happens at the boundary is a transient excursion or a structural failure is the open question of O, which this proposition times and does not close. For a general sector corrector, (ii) is a one-sided statement about survival, not a hazard rate; the asymptotically exponential exit law is established for the benchmark only.

Summary of Results.

Table A.2

Result	*What it proves*	*Assumptions*	*Linear case recovery*
A.1 (Bounded Mismatch)	$R^* = \rho/\alpha$	(A1), (A2'), bounded ρ (GA-2)	$\alpha = \mathcal{T}$ gives $R^* = \rho/\mathcal{T}$

Result	*What it proves*	*Assumptions*	*Linear case recovery*
A.1S (Stochastic Bounded Mismatch, region-aware)	killed second-moment bound (i); RMS scale $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ and the mean-square condition $\kappa_R > n/2$ (ii); fixed-time tail $P(\|\delta(t)\| > R) \le n\sigma_w^2/(2\alpha R^2)$ under global (A2'), composed region-aware tail under local (A2') (iii); finite-horizon path bound (iv); no infinite-horizon non-exit bound — pathwise containment is Model-D-only	(A1), (A2') on $\mathcal{B}_R$, (S), stochastic w (GA-2S)	$\alpha = \mathcal{T}$ gives $R_S^* = \sigma_w\sqrt{n/(2\mathcal{T})}$
A.1S-τ (Time in the Certified Region)	$\mathbb{E}_x\tau_R \ge \Phi_{\alpha,n,\sigma_w,R}(\|x\|)$ and stochastic domination by the OU benchmark; equality under radial tightness; $P_0(\tau_R \le L) \le L/\Phi(0) \sim \frac{2\alpha L}{\Gamma(n/2)}\kappa_R^{n/2}e^{-\kappa_R}$; the mean-square condition certifies about one correction time at threshold; exponent capped at the Kalman gain for a fixed channel	(A1), (A2') on $\mathcal{B}_R$, (S), GA-2S (isotropic)	$F = \alpha\delta$ is the benchmark: exit time exactly Φ
Cor A.1S.1 (Disturbance-Model Containment Dichotomy)	$P(\tau_R < \infty)$ is exactly $\{0, 1\}$ — 0 under Model D (positive invariance), 1 under Model S (a.s. exit of a non-degenerate diffusion); α-invariant, correction strength cannot interpolate	(A1), (A2') on $\mathcal{B}_R$; Model D needs $\alpha R > \rho$	Model D → Prop A.1 invariance at $\alpha = \mathcal{T}$; Model S → OU recurrence is the explicit instance — exact generally (any F under A2'), not linear-scoped
A.2 (Stability Margin)	$\Delta\rho^* = \alpha R - \rho$	Same as A.1	$R \to \infty$ for linear (always stable if $\mathcal{T} > 0$)

What Is Derived vs. What Is Chosen.

Table A.3

Property	*Source*	*Strength*
Dynamics setup $\dot{\delta} = -F(\mathcal{T}, \delta) + w(t)$ with F as correction function and w as disturbance	Definitional scope of the appendix; generalizes the linear hypothesis of `#hyp-mismatch-dynamics`	Definition
(A1) $F(\mathcal{T}, 0) = 0$	Qualitative property of any correction process by construction	Assumption (uncontroversial)
(A2') Local sector condition $\delta^T F \ge \alpha\|\delta\|^2$ on $\mathcal{B}_R$	Sector-condition framework (Lur'e 1957); sub-scope α (Kalman/conjugate Bayesian, exponential family in natural params, gradient descent on strongly convex losses, L2-regularized convex, linear with PD KH) derived via `#der-gain-sector-bridge` Prop B.3 under B1 directional fidelity (then $\alpha = \eta^* \cdot c_{\min}$); sub-scope β (PID, rule-based, human judgment, severely misspecified, variational approximations, non-convex beyond basin, per-step SGD) requires independent verification	**Derived in sub-scope α**; **Assumption in sub-scope β** (sub-scopes named explicitly in the Grounding paragraphs)
(A3) Tempo-monotonicity of $\delta^T F$	Qualitative requirement tying $\mathcal{T}$ to correction power	Assumption
Quadratic Lyapunov candidate $V = \frac{1}{2}\|\delta\|^2$	Canonical choice for norm-bounded stability in Euclidean state spaces (Khalil 2002 ch. 4)	Formulation choice

Property	*Source*	*Strength*
Prop A.1: ultimate bound $R^* = \rho/\alpha$	$\dot{V} \le -\|\delta\|(\alpha\|\delta\| - \rho)$ via A2' + Cauchy-Schwarz; standard Lyapunov ultimate-boundedness theorem (Khalil 2002 Thm 4.18)	**Proved (conditional on $R^* < R$)**
Positive invariance of $\mathcal{B}_R$ under the persistence condition	Boundary-inward argument: at $\|\delta\| = R$, $\dot{V} < 0$ when $\alpha R > \rho$	Proved
Lemma A.1N: threshold tightness — $\alpha > \rho/R$ is necessary-and-sufficient for the *uniform class-level* containment guarantee; agent-level necessity holds exactly under a radially tight sector ($\bar{\alpha} = \alpha$, e.g. linear); false as an agent-level only-if for general F	Extremal-member escape ($F = \alpha\delta$, $w = \rho\,\delta/\|\delta\|$) + the dip counterexample (floor attained at unreachable radii; mountain-pass barrier analysis)	**Proved** — necessity is class-level; no fixed-agent only-if is claimed anywhere
Initial-condition scope (trajectory must start inside $\mathcal{B}_R$)	Direct consequence of A2' being local; trajectories outside are not covered	Derived (scope statement)
Identification of structural-adaptation trigger ($\rho/\alpha > R$ forces exit from parametric regime)	Connects A.1's persistence threshold to `#result-structural-adaptation-necessity`'s regime boundary	Derived
Prop A.2: adaptive reserve $\Delta\rho^* = \alpha R - \rho$	Algebraic corollary of Prop A.1 under shock $\rho \to \rho + \Delta\rho$; requires R fixed under shock	Proved (corollary of A.1)
Prop A.1S (region-aware): killed bound $\mathbb{E}[\|\delta(t)\|^2;\, \tau_R > t] \le \|\delta(0)\|^2 e^{-2\alpha t} + \frac{n\sigma_w^2}{2\alpha}(1 - e^{-2\alpha t})$ **+ fixed-time tail (global A2', or composed with the exit time under local A2') + finite-horizon sup-bound (iv)**	Itô's formula on $e^{2\alpha t}V$ on $[0, t \wedge \tau_R]$ + A2' on $\mathcal{B}_R$; the exact stopped identity for (iv) (Khasminskii 2012 ch. 5; Khalil 2002 ch. 9)	**Proved**; a stopped-moment bound *with* the decay term is false under (ii) (the frozen mass does not decay); the infinite-horizon non-exit probability is structurally 1 for additive Model S (no-go, Khasminskii ch. 3–4) — pathwise containment is Model-D-only
Prop A.1S-τ: $\mathbb{E}_x\tau_R \ge \Phi(\|x\|)$, stochastic domination, lifetime bound, tenure condition	Radial drift of any sector corrector is pointwise at least the OU benchmark's; Itô–Dynkin on $\Phi \circ \|\cdot\|$ (mean) and a same-noise comparison of the radial processes (law); the benchmark's exit function is the classical Andronov–Vitt–Pontryagin solution (Kersting et al. 2023 Thm 4; Grebenkov 2015 eq. (75))	**Proved** under (S), A1, A2'; equality under radial tightness; method in print (Ikeda–Watanabe 1977 comparison), statement new to the framework
Cor A.1S.1: containment dichotomy $P(\tau_R < \infty) \in \{0, 1\}$, α-invariant	Prop A.1 positive invariance (Model D $= 0$) + a.s. finite exit of a non-degenerate diffusion from a bounded region for any F under A2' (Model S $= 1$); Khasminskii 2012 ch. 3–4	**Proved** — new exact result
Stochastic steady-state RMS $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$	Steady state of the Grönwall bound (global A2') and the killed-moment scale (local A2'); scales as $1/\sqrt{\alpha}$ versus $1/\alpha$ deterministic, at fixed σ_w	Derived
Tail bound $P(\|\delta\| > R) \le n\sigma_w^2/(2\alpha R^2)$	Markov's inequality on the steady-state second moment, under global A2'	Proved (global A2'); composed form under local A2'
Exit exponent capped at the Kalman gain, $\max_\alpha \kappa_R = R^2/(2\sigma_{\text{env}}\sigma_\nu)$	Algebra on the Setup's $\sigma_w^2(\alpha)$; monotone decrease of $\Phi(0)$ past K from $\Phi(0) = g(\kappa_R)/\alpha$	Derived (linear correction, white channel noise)
Linear-case recovery ($\alpha = \mathcal{J}$ reproduces `#result-persistence-condition`)	Substitution $F = \mathcal{J}\delta$ into A2'	Derived
Adversarial-exponent scaling $b = 2$ (Model D) vs $b = 3/2$ (Model S) into `#result-adversarial-exponent-regimes`	Direct transfer of the $1/\alpha$ vs $1/\sqrt{\alpha}$ scalings at fixed ρ, σ_w	Derived (transfer claim; scope condition on gain-injected noise stated at the receiving segment)

Property	*Source*	*Strength*
Bounded-disturbance model (GA-2) vs stochastic-disturbance model (GA-2S) as distinct environmental regimes, not approximations to each other	Discussion-section positioning; neither regime handles heavy tails	Discussion-grade
Global sector condition would give $\Delta\rho^* = \infty$	Limit analysis of A.2 as $R \to \infty$; rejected as unrealistic for finite model classes	Derived (scope-limit observation)

The dividing line: the two persistence results (ultimate bound, adaptive reserve) and the stochastic mean-square bound are **proved** within standard Lyapunov / Itô-Lyapunov theory — the appendix's contribution is the application to AAT's correction-function object, not the mathematics. What is **chosen** is the quadratic Lyapunov candidate $V = \frac{1}{2}\|\delta\|^2$ (canonical for norm-bounded stability but not uniquely forced; see Path 3 note below) and the bounded-vs-stochastic disturbance partition (two structurally distinct empirical regimes rather than a hierarchy). The **sector condition itself (A2')** now carries its sub-scope explicitly: **derived in sub-scope** α (optimal Bayesian / exponential-family / strongly-convex-gradient / L2-regularized / linear-PD corrections, via `#der-gain-sector-bridge` Prop B.3 under B1), **assumed in sub-scope** β (PID / rule-based / human-judgment / severely-misspecified / variational / non-convex-beyond-basin / per-step SGD). Prop A.1S is region-aware: under (A2') on the ball alone it bounds the killed moment, the finite-horizon path, and, through Prop A.1S-τ, the time to exit; the stationary fixed-time tail needs global (A2'). The *absence* of any infinite-horizon non-exit bound (a structural no-go for additive Model S) is itself a result — pathwise containment is a Model-D guarantee only — and Prop A.1S-τ is the positive statement that sits beside it: how long the certificate holds, and what margin buys.

Epistemic Status.

The setup and assumptions are *definitions* — they specify what we mean by "correction function" and "disturbance." Propositions A.1 and A.2 are *exact* — they follow from the assumptions via standard Lyapunov theory (Khalil 2002[^khalil2002], Chapters 4 and 9). The persistence threshold's necessity is *class-level* (Lemma A.1N, exact): $\alpha > \rho/R$ is necessary and sufficient for the uniform guarantee over the sector class, and for a fixed general F the only-if direction is false (the demonstrated dip counterexample), so every statement of the threshold in this corpus is sufficiency-plus-class-tightness, never a fixed-agent iff. Proposition A.1S is *exact* in its region-aware form: the killed second-moment bound (i), the mean-square persistence condition (ii), the fixed-time tail (iii) in its global and composed forms, and the finite-horizon sup-bound (iv) are each exact under their stated hypotheses (Itô–Lyapunov on $[0, t \wedge \tau_R]$, Khasminskii 2012[^khasminskii2012], ch. 5). A stopped-moment bound carrying the decay term, $\mathbb{E}\|\delta(t \wedge \tau_R)\|^2 \le \|\delta(0)\|^2 e^{-2\alpha t} + n\sigma_w^2/(2\alpha)$, is *false* under (ii), by three lines from Corollary A.1S.1 (the stopped moment tends to R^2); it is not carried. Proposition A.1S-τ is *exact* under (S), (A1), (A2') on the ball and isotropic GA-2S: parts (i)–(iv) are Itô, Dynkin, a same-noise radial comparison and elementary calculus on the benchmark's exit function; part (v) and the threshold values are the benchmark's integral evaluated numerically (two independent arbitrary-precision implementations agree to four digits), and the $\frac{1}{4}\ln n$ drift is a computed trend, not a derived asymptotic. The published comparison theorem cited for (ii) (Ikeda–Watanabe 1977) was transcribed by the spike's literature scan and has not been read at the primary by the author or the verifier; the self-contained proof is the basis. The benchmark's closed form is classical and was checked term by term against the explorer's one-dimensional scale-function formula and against Monte Carlo in $n = 2, 3, 5$. **Corollary A.1S.1 is itself an exact result** — and a new one: the first-exit probability is *exactly* the two-point set $\{0, 1\}$ (0 under bounded Model D, 1 under additive stochastic Model S), categorical and α-invariant, the value selected by the disturbance model's support structure rather than by correction strength. Both halves are exact (Model-D: deterministic positive invariance, Prop A.1; Model-S: almost-sure finite exit of a non-degenerate diffusion from a bounded region, for any F satisfying A2'). No implicit strengthening of A2' is required.

The sector condition A2' carries its sub-scope α / β status, the operator-family classification, and the "Why Euclidean A2' specifically" Lyapunov-matching argument at `#form-sector-condition`. The Lyapunov proofs above apply uniformly across both sub-scopes and across the weighted-norm variants — they operate downstream of A2' regardless of how it is established or which metric matches it.

The assumptions themselves (sector condition on a region, bounded disturbance) are *empirical claims* about the qualitative behavior of real correction dynamics. The sector-condition framework originates with Lur'e (1957); the Lyapunov stability results are standard. The application to adaptive agents under the AAT framework is new but the mathematics is not.

Discussion.

Key value. The persistence threshold and adaptive reserve are no longer contingent on the linear hypothesis in `#hyp-mismatch-dynamics`. They hold for any correction dynamics satisfying the sector condition — a mild qualitative assumption that says "correction points inward with at least baseline efficiency α." This is a significant epistemic upgrade: from *hypothesis-dependent* to *robust under qualitative assumptions*.

What the proofs do NOT illuminate. (1) Quantitative steady-state values — Lyapunov gives *bounds*, not exact values; the linear analysis remains necessary for quantitative predictions. (2) Convergence rates — standard Lyapunov tells you stable/unstable, not how fast. (3) Optimal gain structure — `#emp-update-gain` comes from estimation theory, not stability theory. (4) Model sufficiency — the `#form-information-bottleneck` framework is information-theoretic, complementary to but independent of stability analysis.

Two disturbance models. Proposition A.1 assumes bounded disturbance (GA-2: $\|w(t)\| \leq \rho$); Proposition A.1S assumes stochastic disturbance (GA-2S: $\mathbb{E}[\|w(t)\|^2] = \sigma_w^2$). These are not approximations to each other — they capture structurally different environments. Model D (bounded) covers persistent directional change: an adversary who maneuvers, an API that drifts, a climate that shifts. Model S (stochastic) covers unpredictable fluctuations around a stable mean: market noise, sensor noise, random perturbations. The choice of model is an empirical question for each domain; the theory provides both tools. Neither model handles heavy-tailed environmental shocks (financial crises, ecological catastrophes, strategic surprise) where $\|w(t)\|$ can exceed any finite bound with non-negligible probability. Extreme tail events are better understood as triggers for structural adaptation (`#result-structural-adaptation-necessity`) rather than disturbances to be absorbed parametrically.

Kind of guarantee, not just rate. The two models differ not only in how *tightly* the mismatch is held but in *what kind* of guarantee is available at all. Corollary A.1S.1 makes this exact and categorical: $P(\tau_R < \infty) = 0$ under Model D (positive invariance) versus 1 under Model S (recurrent additive-noise diffusion), with no intermediate value reachable by any correction strength. What Model S controls instead is the *typical scale* (the $1/\sqrt{\alpha}$ RMS radius, ii), the *fixed-time tail* (iii), and the *time to exit* (Prop A.1S-τ: expected stay at least the benchmark's, exponential in $\kappa_R = \alpha R^2/\sigma_w^2$) — distributional and finite-time guarantees, not a sample-path-forever one. Increasing α at fixed σ_w shrinks the typical scatter and lengthens the expected stay; it does not convert a distributional guarantee into a pathwise one, and on a fixed noisy channel it lengthens the stay only up to the Kalman gain. This is the deeper content of "these are not approximations to each other": additive stochastic forcing removes the *kind* of containment available, not merely its rate. It sharpens — rather than caveats — the hand-off into `#result-structural-adaptation-necessity`: in a genuinely stochastic environment, region-exit is not a measure-zero pathology to be assumed away but a *certain eventual event*, so loss of the parametric certificate is **generic, not exceptional**, for any sufficiently long-lived agent (whether it forces structural, model-class adaptation is the open question of O).

Findings.

The disturbance-model containment dichotomy: $P(\tau_R < \infty)$ is exactly $\{0, 1\}$, α-invariant. Result (exact, new). Corollary A.1S.1: under the sector-condition correction dynamics, the persistence-region first-exit probability is *categorical* — exactly 0 under bounded disturbance (Model D, positive invariance) and exactly 1 under additive stochastic disturbance (Model S, almost-sure exit of a non-degenerate diffusion from a bounded region). The achievable value is the two-point set $\{0, 1\}$; which point obtains is fixed by the disturbance model's support structure, **not** by correction strength α.

Brief: Against a *bounded* disturbance — a steady wind you can lean into — a sector-stable corrector holds the mismatch inside a fixed region *forever and with certainty*: once in, never out. Against *stochastic* disturbance — random gusts — no correction strength buys that. The agent stays near target on average and almost all of the time, but over an unbounded horizon some fluctuation eventually pushes it past any fixed boundary, with probability one. Stronger correction tightens the *typical* scatter (the RMS radius shrinks as $1/\sqrt{\alpha}$); it does not erect a wall. So moving from bounded to stochastic environments is not "the same guarantee with a weaker constant" — it is a change in the *kind* of guarantee available: pathwise-and-forever (Model D) versus distributional-and-fixed-time (Model S). Practical consequence: in any genuinely stochastic environment, leaving the parametric-correction region is not a rare pathology to assume away — it is a certain eventual event, so losing the parametric certificate is generic, not exceptional; whether that forces a change of model class, rather than a transient excursion, is an open question the framework names.

Impact: Sharpens the Model-D/Model-S architecture and the hand-off into `#result-structural-adaptation-necessity` (loss of the parametric certificate is a generic eventual event for long-lived agents in stochastic environments, not an edge case; whether it forces model-class change is open at O). The companion controls under Model S are the killed-moment scale (i), the fixed-time tail (iii: the Markov bound $P(\|\delta(t)\| > R) \leq n\sigma_w^2/(2\alpha R^2)$ under a global sector condition, the composed form under a local one), the finite-horizon sample-path bound (iv), and the exit-time comparison of Prop A.1S-τ; pathwise containment is categorically Model-D-only. The Model-S half is the load-bearing proof step demonstrated at length in `#deriv-stochastic-non-exit`, which also packages the reusable no-go signature (*unbounded scale function ⇒ no non-constant bounded harmonic function ⇒ no horizon-independent non-exit certificate*) that future stochastic-containment proposals can be settled against rather than re-attempted.

Novelty Claim: *Synthesis* — an exact result built from classical components. The component facts are textbook: deterministic positive invariance under bounded disturbance (Khalil 2002 ch. 4) and almost-sure finite exit of a non-degenerate diffusion from a bounded region (Khasminskii 2012 ch. 3–4; the OU scale-function computation is the explicit instance). The new exact result is their synthesis into a *categorical, α-invariant containment dichotomy* stated as a structural property of AAT's own Model-D/Model-S architecture: correction strength cannot interpolate $P(\tau_R < \infty)$ between the two disturbance regimes — the value is exactly $\{0, 1\}$, selected by disturbance-support structure. This exact dichotomy is new to the framework; it sharpens the structural-adaptation hand-off (`#result-structural-adaptation-necessity`). No novelty is claimed on the SDE mathematics — the novelty is the exact dichotomy as a framework theorem and its α-invariance.

Related Work:

Table A.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
No infinite-horizon non-exit bound for additive-noise diffusions	Khasminskii 2012, *Stochastic Stability of Differential Equations* (2nd ed.) ch. 3–4 (recurrence; scale function; no bounded non-constant harmonic function)	*formal antecedent* — supplies the classical recurrence fact; AAT recognizes it as the structural reason Model S admits no pathwise-forever containment in contrast to Model D

ASF concern	*Prior-art language*	*Relationship / Positioning*
Fixed-time second-moment tail	Markov's inequality on the Itô-Lyapunov second moment, unstopped under a global sector condition, killed under a local one (Khasminskii 2012 ch. 5)	*standard machinery* — (iii) is Markov on the unstopped or killed moment; the AAT-specific content is the Model-D/S kind-of-guarantee contrast it forces, and the recognition that a stopped moment with decay is not available
Pathwise positive invariance under bounded disturbance	Khalil 2002 ch. 4 (ultimate boundedness; boundary-inward invariance)	*formal antecedent* — Model D's pathwise-forever guarantee (Prop A.1) is the contrast term that makes the Model-S no-go a *kind* distinction, not a rate one

Search Log:

- 2026-05-16 (*targeted, derivation-driven*): the question was internal — does Prop A.1S(iii)'s infinite-horizon non-exit bound hold? The strengthening attempt (Doob/Ville maximal inequality on the Itô-Lyapunov supermartingale) was worked in full and shown to fail structurally; the classical recurrence fact (Khasminskii ch. 3–4) is the obstruction and is well-known. No prior-art search beyond the standard SDE references was needed — no novelty is claimed on the mathematics, only on the framework-level kind-of-guarantee recognition. Reasoning trail recorded in the spike-routing cycle (CHANGELOG 2026-05-17).

Time in the certified region: the exit time of any sector corrector is bounded below by the Ornstein–Uhlenbeck benchmark, exponentially in $\kappa_R = \alpha R^2/\sigma_w^2$. **Result (exact).** Proposition A.1S-τ: under (A1), (A2') on the ball, (S) and isotropic Model S, the first-exit time from $\mathcal{B}_R$ of any sector-satisfying corrector stochastically dominates that of the linear corrector $F = \alpha\delta$ from the same radius, whose mean is the explicit $\Phi_{\alpha,n,\sigma_w,R}$; equality holds exactly when the sector bound is radially tight. For small noise $\mathbb{E}\tau_R \approx \frac{\Gamma(n/2)}{2\alpha}\kappa_R^{-n/2}e^{\kappa_R}$. The finite-lifetime exit probability from the origin is at most $L/\Phi(0)$, exponentially small in κ_R where canon's path bound (iv) is only linear. At the mean-square threshold $\kappa_R = n/2$ the benchmark leaves in about one correction time, so the condition $\alpha > n\sigma_w^2/(2R^2)$ certifies the scale of the mismatch and not the time for which the certificate holds. On a fixed noisy channel the exponent is capped at the Kalman gain.

Brief: Random gusts will eventually push the agent out of its safe region no matter how hard it corrects (the dichotomy above), but *how soon* is a different question, and it has a clean answer. Any corrector that pulls inward at least as strongly as a simple spring of stiffness α stays inside at least as long as that spring would, and the spring's stay is known exactly: it grows like e^{κ_R}, where κ_R is the height of the spring's energy well at the boundary measured in units of the noise's temperature. The familiar persistence condition says only that the agent's typical scatter fits inside the region; at exactly that condition the spring leaves after about one correction time. Staying for a lifetime L with a small chance ϵ of early exit costs margin in κ_R that grows only like $\log(L/\epsilon)$. And the margin cannot be bought by cranking the gain on the same observations: past the Kalman gain the correction injects more sensor noise than it removes lag, and the stay gets shorter. More or better observations buy time; more gain does not.

Impact: Supplies the *when* beside Corollary A.1S.1's *whether*, and corrects what the Model S persistence condition certifies (scale, and about one correction time of guaranteed stay at threshold). Replaces the stopped-moment bound that the earlier Prop A.1S(i) carried, which is false under the condition it was meant for. Gives the front door its honest agent-level sentence: correction strength buys expected time in the region exponentially in κ_R at fixed disturbance, up to the Kalman gain on a fixed channel, and never buys forever. The tenure-indexed condition (iv) is the Model S counterpart of the Model D threshold, with a lifetime in it. Downstream: `#result-persistence-condition` (the Model S window and the tenure reading), `#result-sector-condition-stability`, O (the hand-off's timescale), N (the single-level base for Model S stacking), and the adversarial results, where pushing an opponent's gain up can shorten its own stay.

Novelty Claim: *Recognition / synthesis*, at the search depth of the spike's literature scan (web, arXiv, open-access PDFs; no books opened). The comparison *method* is in print: Ikeda & Watanabe 1977 is the comparison theorem, and the same radial-OU comparison step appears as a lemma for strongly convex Langevin diffusions (Altmeyer, arXiv:2208.13296), as a PDE theorem on manifolds for C^1 drift (Ouyang, arXiv:1310.3996), and in the mirror direction for bounded Borel drift (Li & Zhang, arXiv:1202.4131), all as located by the scan. The benchmark's closed form is classical (Kersting et al. 2023; Grebenkov 2015). What was not found is the *statement*: the exit-time domination as a theorem under a one-point sector condition alone, paired with the closed-form benchmark and drawing the threshold, lifetime and gain-cap consequences; those are new to the framework. The recognition that (A2') is exactly the radial-comparison hypothesis is the framework's.

Related Work:

Table A.5

ASF concern	*Prior-art language*	*Relationship / Positioning*
Exit time of the isotropic OU process from a ball, every n	Andronov–Vitt–Pontryagin boundary-value problem; Kersting, Orvieto, Proske & Lucchi 2023 Thm 4, eq. (17); Grebenkov 2015 eq. (75) (both read by the spike's literature scan, not by the author of this segment)	*formal antecedent* — the benchmark Φ of Lemma A.1S-τ.0 is this classical object; the derivation here is self-contained
Comparison of scalar Itô processes with ordered drifts and a common diffusion coefficient	Ikeda & Watanabe 1977, *Osaka J. Math.* 14, Thm 1.1 and Remark 1.1 (transcribed by the scan; not read at the primary by author or verifier)	*standard machinery* — the published form of the radial comparison in (ii); the self-contained first-exceedance proof is the basis
Small-noise exit asymptotics; quasi-potential	Arrhenius–Kramers; Freidlin–Wentzell large deviations (Berglund 2013 survey; Day 1983 for the asymptotically exponential exit law of the benchmark, bibliography verified by the scan, primary not read)	*located* — the small-noise form of Lemma A.1S-τ.0 is Kramers for the benchmark; FW identifies the exponent of a specific non-tight F as its quasi-potential barrier, which lies above αR^2; this proposition bounds without FW's Lipschitz hypothesis and does not locate the instance's exponent
Heavy-tailed forcing	Imkeller & Pavlyukevich 2006 Thm 1.1; Högele & Pavlyukevich 2014 Thm 2.1 (read by the scan)	*scope boundary* — exit driven by one large jump, expected exit time polynomial in the noise scale with no trace of the restoring force; the exponential reserve does not exist there

Search Log:

- 2026-10-30 (*targeted*, by a literature-scan agent for the exit-time spike; web, arXiv, open-access only): comparison theorems (Ikeda–Watanabe 1977; Geng & Markowsky 2025 on why signed-drift ordering does not order exit times, which the radial ordering avoids), the OU benchmark in print, FW and Day hypotheses, heavy-tail sources. Method found; statement not found. Not searched: MathSciNet, zbMATH, textbooks beyond those named.

Appendix B

Derivation: No Horizon-Independent Non-Exit Bound Under Additive Stochastic Forcing (the Model-S No-Go)

A *no-go* derivation establishing that **under additive stochastic forcing there is no finite bound on the infinite-horizon first-exit probability** from the sector-condition region — $P(\tau_R < \infty) = 1$ for every correction strength — *and the natural mathematical route to one provably cannot exist*. This is the load-bearing proof step behind the Model-S half of Corollary A.1S.1 in `#deriv-sector-condition`, and the reason Prop A.1S there bounds the *killed* moment and the finite-horizon path rather than a stopped moment with decay; its positive companion, how long the region is kept, is Prop A.1S-τ there.

The construction: under stochastic disturbance the Itô-Lyapunov machinery succeeds at producing a *finite-horizon mean-square bound* but fails to extend to a *horizon-independent* exit-probability bound. The natural candidate — a Ville / Doob maximal inequality on a compensated Itô-Lyapunov supermartingale — is shown to be unattainable because the only candidate $S(t) = e^{2\alpha t}V - \frac{n\sigma_w^2}{4\alpha}(e^{2\alpha t} - 1)$ is *sign-indefinite inside the persistence basin* (the compensated drift is not non-positive across the entire region: the subtracted term dominates $e^{2\alpha t}V$ whenever the system sits in the typical scatter, which under the mean-square persistence condition is most of the time). The deeper obstruction is that for additive non-degenerate forcing the diffusion's scale function is *unbounded* — the OU benchmark $s'(u) \propto e^{\alpha u^2/\sigma_w^2}$ is the explicit instance — so the only bounded harmonic functions of the generator are constants and the gambler's-ruin / Lyapunov-exit machinery cannot certify "stays inside forever with positive probability." Both failures are structural, not artifacts of a clever trick being missed.

The framework treats this as a *constructive use of impossibility*. The downstream consequence is a certain-eventual-event reading: under Model S, pathwise containment in any fixed region is unattainable at any correction strength, so for any sufficiently long-lived stochastic-environment agent the *loss of its parametric certificate* is a *certain eventual event* rather than an edge case (whether that loss forces structural adaptation, a different model class rather than a transient excursion, is the open question of `#sketch-structural-adaptation-genericity`; Prop A.1S-τ in `#deriv-sector-condition` says how soon it comes). This sharpens — rather than caveats — the hand-off into `#result-structural-adaptation-necessity`. Beyond the segment's own result the derivation packages a **reusable no-go signature**: *unbounded scale function* ⇒ *no non-constant bounded harmonic function* ⇒ *no horizon-independent non-exit certificate*. Future stochastic-containment proposals in the corpus can be settled against this signature rather than re-attempting the maximal-inequality route.

Formal Expression.

Setup. Mismatch dynamics under additive stochastic disturbance, started inside the sector-condition region $\mathcal{B}_R$:

$$d\delta = -F(\mathcal{T}, \delta)\, dt + \sigma_w\, dW_t, \qquad \delta(0) \in \mathcal{B}_R, \quad \sigma_w > 0, \tag{B.1}$$

with W_t a standard n-dimensional Wiener process, (A2') $\delta^\top F \geq \alpha\|\delta\|^2$ on $\mathcal{B}_R$, and $\tau_R = \inf\{t : \|\delta(t)\| > R\}$ the first-exit time. Let $V = \frac{1}{2}\|\delta\|^2$.

Theorem (Model-S no-go). *[Derived]* For every $\alpha > 0$, every $\sigma_w > 0$, and every correction function F satisfying (A2'),

$$P(\tau_R < \infty) = 1, \tag{B.2}$$

and there exists no nonnegative supermartingale dominating V that certifies a horizon-independent bound $P(\tau_R < \infty) \leq c < 1$. Pathwise containment of $\mathcal{B}_R$ is unattainable under additive stochastic forcing at any correction strength.

Why the natural route cannot work. The route a careful reader reaches for first is a time-uniform maximal inequality (Ville / Doob) on an Itô–Lyapunov supermartingale — the same instinct that makes the killed second-moment bound (Prop A.1S(i)) succeed. It fails, and the failure is structural, not a matter of a missing trick.

Define $G(t) = e^{2\alpha t}V(\delta(t))$. On $[0, \tau_R]$, by Itô and (A2'),

$$dG = e^{2\alpha t}\big[\underbrace{(2\alpha V - \delta^\top F)}_{\leq 0 \text{ on } \mathcal{B}_R}\, dt + \tfrac{n}{2}\sigma_w^2\, dt + \delta^\top \sigma_w\, dW_t\big] \;\leq\; e^{2\alpha t}\tfrac{n}{2}\sigma_w^2\, dt + e^{2\alpha t}\delta^\top\sigma_w\, dW_t. \tag{B.3}$$

G is **not** a supermartingale: the $+e^{2\alpha t}\frac{n}{2}\sigma_w^2\, dt$ term has strictly positive drift growing exponentially. Removing it by compensation gives

$$S(t) = e^{2\alpha(t\wedge\tau_R)}V(\delta_{t\wedge\tau_R}) - \frac{n\sigma_w^2}{4\alpha}\big(e^{2\alpha(t\wedge\tau_R)} - 1\big), \tag{B.4}$$

and $dS \leq e^{2\alpha t}\delta^\top\sigma_w\, dW_t$ on $[0, \tau_R]$ — so S *is* a supermartingale. **But S is not nonnegative.** The subtracted $\frac{n\sigma_w^2}{4\alpha}(e^{2\alpha t} - 1)$ dominates $e^{2\alpha t}V$ whenever $V(\delta(t)) < \frac{n\sigma_w^2}{4\alpha}$ — i.e. exactly inside the persistence basin, where the process spends a positive fraction of its time: $\frac{n\sigma_w^2}{4\alpha}$ is the stationary mean of V for the OU benchmark, and $P(V < \frac{n\sigma_w^2}{4\alpha}) = P(\chi_n^2 < n)$ lies between 0.5 and 0.69, so V is typically comparable to that level, below it about half the time. The no-go needs only that this event has positive probability. Ville's inequality requires a nonnegative supermartingale; Doob's maximal inequality a nonnegative sub/supermartingale. Both are inapplicable to a sign-indefinite S, and the obstruction is not removable: for additive non-degenerate Brownian forcing the diffusion's scale function is unbounded (the OU scale density $\propto e^{\alpha u^2/\sigma_w^2}$), so the only bounded harmonic functions of the generator are constants — the gambler's-ruin / Lyapunov-exit machinery cannot certify "stays inside forever with positive probability." There is no nonnegative supermartingale dominating V with finite expected initial value that yields a horizon-independent exit bound.

Why $P(\tau_R < \infty) = 1$, generally. The conclusion does not depend on the linear structure. A non-degenerate diffusion (additive forcing $\sigma_w\, dW_t$, $\sigma_w > 0$) exits any bounded region in finite time almost surely, for *any* locally bounded drift: near $\partial\mathcal{B}_R$ the Brownian increment has positive probability of crossing in any time interval, and no finite inward correction satisfying (A2') can suppress this (A2' bounds $\delta^\top F$ from below by $\alpha\|\delta\|^2$, a finite inward push, not an impassable

wall). Hence $\tau_R < \infty$ a.s. for every F under (A2'), every α, every σ_w. The Ornstein–Uhlenbeck case is the explicit instance (positively recurrent on $\mathbb{R}^n$, unbounded stationary support, exits any finite ball a.s.), not the basis.

Epistemic Status.

Exact. Both the impossibility of the maximal-inequality certificate (sign-indefiniteness of the only candidate compensated supermartingale; unbounded scale function ⇒ no non-constant bounded harmonic function) and the positive fact $P(\tau_R < \infty) = 1$ (almost-sure finite exit of a non-degenerate diffusion from a bounded region under any locally bounded drift) are classical (Khasminskii 2012[^khas] ch. 3–4). The contribution here is not the SDE mathematics but the demonstration that the route which *succeeds* for the killed second moment (Prop A.1S(i)) structurally *cannot* deliver an infinite-horizon containment statement — the answer to "are you sure you can't just bound $\sup_t$ with Doob/Ville?" is *yes, sure, and here is precisely the step that breaks.* "Exact" is claimed in the framework's defeasible sense (valid under stated assumptions, subject to a found error), not as a claim beyond a found-mistake.

This derivation is load-bearing for the critical path: it is the proof step that grounds the Model-S half of Corollary A.1S.1 and confines Prop A.1S (in `#deriv-sector-condition`) to the killed moment, the finite-horizon sup-bound (iv), the stationary tail under a global sector condition, and the exit-time comparison of Prop A.1S-τ. It is a **reusable no-go signature**: wherever a "stays-in-region-forever with positive probability" claim is proposed under additive non-degenerate forcing, the unbounded-scale-function / no-non-constant-bounded-harmonic-function obstruction settles it negatively without a fresh attempt.

Discussion.

The structural reading: additive stochastic forcing does not weaken the *rate* of containment, it removes a *kind* of guarantee. Bounded disturbance (Model D) gives deterministic positive invariance of $\mathcal{B}_R$ ($P(\tau_R < \infty) = 0$); additive stochastic disturbance gives almost-sure eventual exit ($P(\tau_R < \infty) = 1$); the achievable value is the two-point set $\{0, 1\}$, selected by the disturbance's support structure, not by correction strength — Corollary A.1S.1 in `#deriv-sector-condition`. What survives under Model S is distributional and finite-time: the killed-moment scale (Prop A.1S(i)), the stationary Markov tail where the sector condition is global (Prop A.1S(iii)), the finite-horizon sample-path bound (Prop A.1S(iv), which grows linearly in the horizon and is vacuous for $T > rsimR^2/(n\sigma_w^2)$), and the exit-time comparison (Prop A.1S-τ: the expected stay is at least the OU benchmark's, exponential in $\alpha R^2/\sigma_w^2$, so the certain event can be made as late as a lifetime requires, at a margin cost logarithmic in that lifetime, though not by raising the gain past the Kalman gain on a fixed channel).

Direct simulation confirms the obstruction is real rather than an artifact of the proof technique. Exact 1-D OU benchmark $dX = -\alpha X\,dt + \sigma\,dW$ (the segment's own linear case; (A2') global and sharp), Monte-Carlo started *at the origin* (the most favorable initial condition), under the mean-square persistence condition (ii) holding:

Table B.1

α	σ	R	R_S^*	$P(\sup_{[0,50]} > R)$	T=200	T=800	T=3200	*stationary tail* $\frac{n\sigma^2}{2\alpha R^2}$ *(global A2', valid for OU)*
1.0	0.3	1.0	0.21	0.002	0.010	0.030	**0.114**	0.045
0.5	0.2	0.8	0.20	0.020	0.074	0.276	**0.720**	0.063
1.0	0.5	1.0	0.35	0.781	0.998	1.000	**1.000**	0.125

The ever-exit probability climbs monotonically toward 1 with the horizon even started at the origin and even with (ii) comfortably satisfied — the fixed-time constant $n\sigma^2/(2\alpha R^2)$ is *not* an upper bound on the sup-over-horizon probability, exactly as the no-go predicts. This sharpens the hand-off into `#result-structural-adaptation-necessity`: under any

genuinely stochastic environment region-exit is a certain eventual event, so loss of the parametric certificate is generic, not exceptional, for a sufficiently long-lived agent (what follows the loss is `#sketch-structural-adaptation-genericity`'s question).

Findings.

The natural maximal-inequality route to infinite-horizon containment provably cannot exist under additive stochastic forcing. **Brief:** If a system is held near a target by inward correction but also kicked by persistent random noise, you can bound how far off it is at any *given* moment, and you can bound the worst excursion over any *fixed window* — but you can never bound "it stays inside the safe region *forever*," because given unlimited time the noise will, with certainty, eventually push it out, no matter how strong the correction. The tempting fix — find a single "energy" quantity that only ever drifts downward, so a classic gambler's-ruin argument caps the lifetime escape probability — fails for a precise reason: the only candidate energy that has the right drift is negative exactly where the system spends most of its time, and the gambler's-ruin machinery requires a non-negative quantity. This is not a missing trick; it is structural, and it is the same obstruction every time, so it can be invoked rather than re-derived.

Impact: Supplies the load-bearing proof step for Corollary A.1S.1 in `#deriv-sector-condition` (the Model-S half of the containment dichotomy) and confines Prop A.1S there to what is available under a local sector condition: the killed moment, the finite-horizon path bound (iv), and the exit-time comparison of Prop A.1S-τ. Provides a reusable no-go signature — *unbounded scale function ⇒ no non-constant bounded harmonic function ⇒ no horizon-independent non-exit certificate* — that future stochastic-containment proposals in the corpus can be settled against without re-attempting the maximal-inequality route.

Novelty Claim: *Recognition*. The component facts (non-degenerate-diffusion exit from a bounded region; unbounded scale function of additive-noise generators; Ville/Doob preconditions) are classical (Khasminskii 2012 ch. 3–4). The contribution is the recognition, made explicit and citable, that the maximal-inequality route which succeeds for the fixed-time second moment structurally cannot deliver the infinite-horizon statement — i.e. *where exactly* the natural attempt breaks. No novelty is claimed on the SDE mathematics.

Related Work:

Table B.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
No horizon-independent non-exit certificate for additive-noise diffusions	Khasminskii 2012, *Stochastic Stability of Differential Equations* (2nd ed.) ch. 3–4 (scale function; recurrence; bounded harmonic functions)	*formal antecedent* — supplies the recurrence / unbounded-scale-function fact; this derivation recognizes it as the structural reason the Ville/Doob route fails
Maximal-inequality preconditions	Ville's inequality; Doob's maximal inequality (nonnegative sub/supermartingale required)	*standard machinery* — the demonstration is that the only compensated candidate is sign-indefinite exactly inside the persistence basin

Search Log:

- 2026-05-16 (*targeted, derivation-driven*): internal question — can Prop A.1S's infinite-horizon non-exit claim be certified by Ville/Doob on an Itô–Lyapunov supermartingale? Worked in full; the candidate compensated supermartingale is sign-indefinite, and the classical unbounded-scale-function fact (Khasminskii ch. 3–4) closes it. No prior-art search beyond standard SDE references — no novelty claimed on the mathematics. Reasoning trail recorded in the spike-routing cycle (CHANGELOG 2026-05-17).

Appendix C

Derivation: The Self-Actuation Grounding No-Go

A scoped *no-go* with a constructive boundary. A **self-actuated agent** is one that revises its own objective endogenously — performing the orient cascade's final step (revise O_t) *on itself* via an operator $\mathfrak{A} : (M_t, O_t, \Sigma_t, \mathcal{C}_t) \mapsto O_t'$ — rather than receiving the revised objective from a principal. The derivation first shows that an *unconstrained* $\mathfrak{A}$ is degenerate: since the objective's sole interface is the value functional and the satisfaction gap is just $V_{O_t}^{\min} - A_O$, an unconstrained operator returns an objective whose threshold the current trajectory already meets — driving $\delta_{\text{sat}} \to 0$ by *moving the target onto the arrow already in flight*. This is the formal shadow of *wireheading / reward corruption*, and it is the **generic outcome** of an unconstrained operator, not a marginal one. Non-degeneracy therefore requires an invariant Φ preserved across the revision.

The derivation then asks whether Φ can be an *agent-internal objective-functional the agent self-actuates on*. Four requirements: **(R1)** value-functional-typed, **(R2)** non-vacuously monotone across revision, **(R3)** agent-internal and itself self-actuatable, **(R4)** convention- and trajectory-stable. The no-go: under three named premises (scalar-objective scope, no-primitive-reflective-oracle, and the `#der-directed-separation` substrate stage), *no* Φ *satisfying (R1)–(R4) can be constructed from AAT's covered objective-side machinery*. The meta-objective tower a non-degenerate $\mathfrak{A}$ would require **cannot be a tower of agent-internal objectives**. The contradiction is the collision of two AAT-internal facts — convention-monotonicity (Lemma 1, `#def-value-object`: only the C3/Bellman reading is a genuine infeasibility verdict, and the cheap C1 reading false-positives whenever a goal is merely momentarily hard) and finite-no-oracle per-step action (Lemma 2, `#der-directed-separation` + `#form-objective-functional`: the C3 verdict is a global Bellman optimum, not a finite per-step operation available to an in-scope agent).

The constructive boundary follows immediately: any non-degenerate self-actuator AAT covers must ground on a terminal invariant that is **not** an AAT objective-functional. Corollary 2 identifies a canonical instance — the **persistence condition itself** (`#result-persistence-condition`), which lives on the *adaptive substrate* (M_t and the correction dynamics), not on the objective. Persistence is *convention-invariant* (a Lyapunov property, not a continuation-convention argument), *agent-available per step* (the adaptive reserve $\Delta\rho^* = \alpha R - \rho$ is a finite local read, not a Bellman solve), and *not in* O_t (so it sits where $\mathfrak{A}$ structurally cannot reach). The top of the agent spectrum thus closes back onto Part I: what stops goal-autonomy from collapsing into wireheading is the same correction-dynamics condition that governs whether the agent can track reality at all. The result also *derives* the orthogonality claim that `#disc-continuity-stance` previously carried as discussion-grade — stance is not internally renegotiable precisely because the terminal invariant lives on a substrate $\mathfrak{A}$ cannot touch. The framework is honest about scope: the no-go is *conditional on three premises* and *scoped to AAT-covered objective-side machinery* — exactly the regime the reflective-oracle / self-referential-utility literature exits (those constructions build groundless self-*reference* over an exogenous payoff, never groundless self-*actuation*).

This is one of AAT's identifiability-floor-pattern-style *constructive uses of impossibility*: a no-go that *forces* a load-bearing structural commitment (the terminal non-objective invariant) rather than discouraging the inquiry.

Formal Expression.

The self-actuation operator. The orient cascade (`#der-orient-cascade`) terminates, when $\delta_{\text{sat}} > 0$ persists across M_t-correction, policy-class expansion, and convention escalation, in step 5d: *revise* O_t. For an **actuated** agent the objective's update source is external (`#def-strategy-dimension`: O_t is "assigned, discovered, revised" by a principal) — step 5d exits the agent boundary. A **self-actuated** agent performs step 5d on itself: an operator

$$\mathfrak{A} \;:\; (M_t,\, O_t,\, \Sigma_t,\, \mathcal{C}_t) \longmapsto O'_t \tag{C.1}$$

that revises the objective endogenously — goal autonomy stacked on the solution autonomy (Σ_t-revision) every actuated agent already has.

Unconstrained $\mathfrak{A}$ is degenerate. O_t's sole interface is the value functional V_{O_t} : trajectories $\to \mathbb{R}$ (`#form-objective-functional`) and the satisfaction gap $\delta_{\text{sat}} = V_{O_t}^{\min} - A_O$ (`#def-satisfaction-gap`). If $\mathfrak{A}$ may return any O'_t, it returns one whose threshold the current trajectory already meets — driving $\delta_{\text{sat}} \to 0$ by moving the target onto the arrow already in flight. This is the formal shadow of wireheading / reward corruption, and it is the generic outcome of an unconstrained $\mathfrak{A}$, not a marginal one. Non-degeneracy therefore requires an invariant Φ preserved across the revision.

The question. Can Φ be an *agent-internal objective-functional the agent itself self-actuates on*? Make the requirements explicit: Φ must be **(R1)** value-functional-typed, **(R2)** non-vacuously monotone across revision (a constant everywhere-admissible reading is the trivial indicator the degenerate case already admits), **(R3)** agent-internal and itself self-actuatable, **(R4)** convention- and trajectory-stable (an invariant of the agent, not of the analyst).

No-go (scoped).

[Derived (Conditional on scalar-objective scope, no-primitive-reflective-oracle, and the `#der-directed-separation` *substrate stage)]*

No Φ satisfying (R1)–(R4) can be constructed from AAT's covered objective-side machinery: the meta-objective tower a non-degenerate $\mathfrak{A}$ would require **cannot be a tower of agent-internal objectives**. Any non-degenerate self-actuator AAT covers must therefore ground on a terminal invariant that is *not* an AAT objective-functional.

The claim is scoped to what the constructions below exhaust; it is not the unscoped "no such object exists" (see Epistemic Status — the universal-over-all-Φ step is argued, not derived).

Lemma 1 (objective-functionals carry no convention-invariant infeasibility verdict; static-pointwise, from `#def-value-object`). Fix a decision point: a single model M_τ, horizon N_h, policy class Π. By the convention-monotonicity result (`#def-value-object` Corollary, the static-evaluation form, `status: exact`),

$$\delta_{\text{sat}}^{\text{B}} \;\le\; \delta_{\text{sat}}^{\text{RH}} \;\le\; \delta_{\text{sat}}^{(1)}, \tag{C.2}$$

so the canonical C1 reading $\mathbb{1}[\delta_{\text{sat}}^{(1)} > 0]$ holds on a strict superset of the genuinely-infeasible set $\{\delta_{\text{sat}}^{\text{B}} > 0\}$ — strictly so on the locally-stuck-but-globally-recoverable objectives (`#def-satisfaction-gap` Epistemic Status: "C1 gives the most false 'unattainable' diagnoses"; `#der-orient-cascade` step 5c). A genuine infeasibility verdict requires the C3/Bellman reading. This is the static-pointwise statement `#def-value-object` actually supports — its stated preconditions are exactly fixed M_τ, N_h, Π, and the segment is explicit that the cross-revision/replanning transfer "does not automatically" hold; that transfer is neither used nor needed here. The pointwise fact — *at any fixed decision point the cheap canonical verdict is not a genuine infeasibility verdict* — is exact and is sufficient.

Lemma 2 (the in-scope agent cannot evaluate the C3 verdict per step; from `#der-directed-separation` + `#form-objective-functional`). An AAT-covered agent's entire dynamical system is f_M, f_G, π with no out-of-band oracle (`#der-directed-separation` Formal Expression), on a single non-forkable trajectory (`#scope-agent-identity`). The C3 verdict is a global Bellman optimum, generally intractable (`#def-value-object` C3; `#def-satisfaction-gap` Epistemic Status); evaluating it is not a finite per-step operation, and an agent that could not act until it did so would be "stuck, not purposeful" — the disqualification `#form-objective-functional` Epistemic Status §1 already imposes through its revealed-preference commitment. Hence $\mathbb{1}[\delta^{\mathrm{B}}_{\mathrm{sat}} > 0]$ is not available to an in-scope agent as a per-step predicate.

Assembly. Suppose Φ satisfies (R1)–(R4). By (R1)+(R3), Φ is an AAT objective-functional the agent self-actuates on, so by `#form-objective-functional` its *sole* theory-visible handle is V_Φ and the satisfaction/regret apparatus read off it ("the sole interface between O_t and the rest of the theory"). Any monotone property of Φ the theory can state across revision (R2) must therefore be a statement about $\delta^{\Phi}_{\mathrm{sat}}$ (or $\delta^{\Phi}_{\mathrm{regret}}$, which inherits the identical convention-monotonicity by `#def-control-regret`) — there is no other channel. Non-vacuity (R2) forces that monotone fact to rest on a *verdict* over $\delta^{\Phi}_{\mathrm{sat}}$; a constant everywhere-admissible reading is exactly the trivial indicator the degenerate case already admits. By (R4) the verdict must be convention-invariant; by Lemma 1 the only convention-invariant infeasibility verdict is the C3 reading. By (R3) the verdict licensing Φ's own revision must be available to the agent per step; by Lemma 2 it is not. Contradiction. □

The contradiction is the collision of two AAT-internal facts — convention-monotonicity (Lemma 1, `#def-value-object`) and finite-no-oracle per-step action (Lemma 2, `#der-directed-separation` + `#form-objective-functional` Epistemic Status §1). It introduces no new postulate. It is exhibited for the three constructions below, which exhaust the objective-side routes a Φ could come from:

- **(A)** Φ as the admissible set's own structure: collapses to the vacuous indicator (fails R2).
- **(B)** Φ as a cascade-licensing potential: its licensing verdict *is* a δ_{sat}-verdict, so it inherits the Lemma 1 / Lemma 2 collision.
- **(C)** Φ as a fresh agent-internal scalar: by (R1)+(R3) it is an AAT objective-functional, hence subject to the same collision one level up — the break is structural in (R1)+(R3), not a regress that "gives up".

Constructive boundary. **Corollary 1 (necessary form of a terminal grounding invariant).** The contradiction came entirely from Φ being an AAT objective-functional the agent self-actuates on (R1+R3). Drop that and it dissolves. So a terminal grounding invariant $\Phi^{(K)}$ for a non-degenerate self-actuator must be an object that is **(i) convention-invariant** (its verdict does not move with C1/C2/C3 — escaping Lemma 1), **(ii) agent-available per step** without an oracle (escaping Lemma 2), and **(iii) not an AAT objective-functional the agent self-actuates on** (so it is not subject to the convention split and is genuinely terminal). Equivalently: it lives on the *adaptive/correction substrate* (M_t and the correction dynamics), not the *objective substrate* (O_t).

Corollary 2 (the persistence bound is a canonical terminal grounding invariant). Structural persistence (`#result-persistence-condition`: $\alpha > \rho/R$, `type: result, status: exact`) satisfies (i)–(iii):

- **(i) convention-invariant.** Persistence is a Lyapunov property of the correction dynamics on M_t (`#result-persistence-condition` Formal Expression), with no continuation-convention argument; C1/C2/C3 are conventions for evaluating *objectives*. The convention split does not reach it.
- **(ii) agent-available per step.** The adaptive reserve $\Delta\rho^* = \alpha R - \rho$ (`#result-persistence-condition`) is a finite local read the in-scope agent already maintains (`#der-orient-cascade` steps 1–2 are the per-step adaptive update), not a Bellman solve. No oracle; no "stuck, not purposeful".
- **(iii) not an AAT objective-functional.** It lives on M_t and the correction machinery. The survival *predicate* $\alpha > \rho/R$ is not a term in O_t and is not convention-relative, so $\mathfrak{A}$ — which touches only O_t — cannot rewrite *what the survival condition is*. This is the precise sense in which the invariant sits where $\mathfrak{A}$ structurally cannot reach. It is *not* the claim that the realized operating point (ρ, R, α) is O_t-independent: O_t shapes the policy and hence the

realized margin through the action/selection channel `#der-directed-separation` leaves open (`#disc-continuity-stance` makes this three-layer grain explicit — valuation and invariant-identity decoupled, realized dynamics coupled). The distinction is exactly what makes the boundary below a theorem rather than a contradiction: $\mathfrak{A}$ *can* drive the operating point across the boundary by choosing a self-defeating O'_t; what it cannot do is move the boundary.

Concretely the terminal invariant is: *do not revise O_t to an objective whose pursuit pushes the operating point outside the persistence region.* An O'_t that breaks $\alpha > \rho/R$ is self-defeating — the agent that adopts it cannot maintain bounded mismatch (`#result-persistence-condition` Discussion: below the structural threshold "mismatch grows without effective bound") and so cannot reliably satisfy O'_t either.

What is derived vs. chosen.

Table C.1

Property	*Source*	*Strength*
Unconstrained $\mathfrak{A}$ is degenerate	`#form-objective-functional` + `#def-satisfaction-gap` (the arg min argument)	Derived
Lemma 1 (static-pointwise convention split)	`#def-value-object` Corollary (static-evaluation form, exact)	Proved (within the fixed-M_τ, N_h, Π scope)
Lemma 2 (no per-step C3 verdict)	`#der-directed-separation` + `#form-objective-functional` ES §1	Derived (conditional on the substrate stage)
No-go (no objective-side Φ)	Assembly of Lemmas 1–2	Derived (conditional; scoped to the three exhausted constructions)
Corollary 1 (necessary form)	Negation of (R3) in the Assembly	Derived
Corollary 2 (persistence bound qualifies)	`#result-persistence-condition` against (i)–(iii)	Derived (the persistence bound itself is `exact`)

Epistemic Status.

Conditional. The no-go is a derived result that depends on three explicitly-named local premises, so its honest tier is `conditional`, not `exact`:

1. **Scalar-objective scope.** Lemma 1 uses the scalar satisfaction gap. Under a genuine vector-valued / Pareto V_{O_t} (`#form-objective-functional` Epistemic Status, "Scope restriction: scalar comparability") the convention split retypes to a coarse/fine-Pareto-frontier containment and the in-scope agent still cannot evaluate the fine frontier per step — the no-go's *shape* survives with Φ and the verdict retyped. This is a scope annotation consistent with how `#form-objective-functional` already scopes its other results, not a hole.

2. **No primitive reflective oracle.** Lemma 2 fails if a consistent-fixed-point / Bellman-grade resource were a *free per-step primitive*; then the C3 verdict would be agent-available and an objective-internal terminal invariant might exist. AAT's covered scope contains no such primitive, and admitting one abandons the finite-per-step / "stuck, not purposeful" commitment `#form-objective-functional` Epistemic Status §1 already carries. This is precisely the regime the reflective-oracle / self-referential-utility literature occupies — which is why that literature is *out of scope* here rather than a counterexample (it constructs groundless self-*reference* over an exogenous payoff, never groundless self-*actuation*).

3. **The `#der-directed-separation` substrate stage.** Lemma 2 rests on `#der-directed-separation`, itself `status: conditional, stage: draft`, and on `#form-objective-functional`'s `axiomatic` single-interface commitment. The result cannot be stronger than its substrate; the tier ceiling stays `conditional` while `#der-directed-separation` is draft-stage.

Scoped, not universal. The contradiction is exhibited for the three objective-side constructions the Formal Expression exhausts. The universal-over-all-possible-Φ generalization rests on the single-interface commitment doing universal-quantifier work and is argued, not derived; the honest canonical claim is therefore "no such Φ can be constructed from AAT's covered objective-side machinery," not "no such object exists." **Max attainable:** if the universal step is later derived as an explicit sub-lemma from the single-interface commitment, the scoped no-go could be lifted toward unscoped; the tier ceiling remains `conditional` while the `#der-directed-separation` substrate is draft-stage.

Class robustness. The no-go does not weaken off GUC Class 1 (Separated). Lemma 1 (convention-monotonicity) is architecture-independent — a statement about value functionals and continuation conventions, not about f_M/f_G coupling — and Lemma 2 only gets harder to evade for more-coupled agents (a Class 3 / Coupled agent has more goal-conditioned entanglement, not a free Bellman oracle). The goal-conditioned agents of practical interest are therefore *more* firmly inside the no-go, not in an uncovered middle.

Discussion.

The two-sided resolution. Putting the no-go and Corollary 1–2 together: there is no well-formed self-actuation grounded on an agent-internal objective; every well-formed self-actuator AAT covers must ground its objective-revision on a *non-objective* terminal invariant; and AAT already supplies a canonical, principled, already-exact such invariant — the persistence bound. The grounding that makes a self-actuator well-formed is forced onto the adaptive substrate, where the framework already has the machinery. The top of the agent spectrum (`#def-agent-spectrum`) thus closes back onto Part I: what stops goal-autonomy from collapsing into wireheading is the same correction-dynamics condition that governs whether the agent can track reality at all.

The third side — principal-side terminal grounding via protocol commitment. The two-sided resolution above closes the *within-model* failure mode of the value-functional interface: V_{O_t} cannot carry a convention-invariant infeasibility verdict agent-available per step, so the terminal invariant must live on the adaptive substrate — *agent-side*. A complementary *across-model* failure mode is closed by the sister no-go E (the Cohen 2022 strengthening): when O_t is *learned* from observed reward, V_{O_t} on on-policy data cannot identify the principal-intended world-model from a reward-port-care world-model (Pearl-Bareinboim CHT applied to the reward-provision protocol). The terminal *goal-identification* therefore has to come from outside the agent's observation history — through *principal-side commitments* (physical isolation, horizon restriction, prior engineering, out-of-band intent channels). Both failure modes are reductions of the same structural fact: the value-functional interface V_{O_t}, as the *single-interface commitment* of `#form-objective-functional`, carries less information than is required to anchor non-degenerate goal-revision. The two terminal grounding routes — agent-side adaptive-substrate invariant (Corollary 2 above) and principal-side protocol-commitment (the sister no-go's escape menu) — exhaust the structural escape geometry the single-interface commitment forces. The two no-gos together constitute the two charter instances of the **agent-side value-functional-grounding cluster**, which has its own peer meta-segment home at `#disc-value-functional-grounding-floor` — sister to `#disc-identifiability-floor` (agent-side data-inference) and to `#disc-implementation-impossibility` (designer-side mechanism-design). All three peer meta-segments sit at the boundary face of `#disc-stability-certificate`; together they realize the three recognized clusters of the constructive-impossibility-posture taxonomy named at `#disc-constructive-impossibility-posture`.

Consequence for continuity stance. The negotiated-versus-morally-continuous distinction is *not* "where the continuity term sits in an objective tower" — there is no such well-founded objective tower. It is whether the agent's terminal non-objective invariant is the bare persistence floor (*negotiated*: continuity is tradeable down to that floor) or the persistence floor plus a continuity clause the agent treats as architecturally non-revisable (*morally continuous*: continuity is part of the terminal substrate-level invariant, not of O_t). Stance is a choice of terminal non-objective invariant, which is

exactly why it is not internally renegotiable by $\mathfrak{A}$: $\mathfrak{A}$ touches only O_t, and the terminal invariant sits where it structurally cannot reach. This is the derived form of `#disc-continuity-stance`'s orthogonality claim.

Relation to the self-modification literature. In the rationality / utility-maximization setting the non-degenerate self-modifier must judge the future by a fixed *current* criterion; the agent that evaluates by its revised criterion with nothing fixed above it is the degenerate wirehead. This no-go is the AAT-internal, architecture-grounded form of the same necessity — with the sharper content that the fixed reference *cannot be an AAT objective the agent self-actuates on* and *must* be a non-objective adaptive-substrate invariant, and with AAT's persistence bound as a forced, principled instance where the literature's fixed criterion is a stipulated snapshot.

Findings.

The Self-Actuation Grounding No-Go and its Adaptive-Substrate Boundary. **Brief:** An agent that can rewrite its own goals needs something it *cannot* rewrite to anchor the rewriting — otherwise "succeed" collapses into "want whatever you already have", the formal shadow of wireheading. The anchor cannot itself be just another goal, for a precise reason: the only trustworthy "is this goal still reachable?" test an agent could anchor on is the globally-optimal one, which no finite agent can compute afresh every step; the cheap everyday test it *can* compute systematically reports "infeasible" whenever a goal is merely momentarily hard. So the anchor has to live outside the goal machinery altogether — on the agent's stay-in-control machinery: *don't adopt a goal whose pursuit makes you lose your grip on reality.* The framework already contains exactly that object, the persistence condition, which is why the top of the agent spectrum closes back onto its foundation.

Impact: Formalizes the previously reserved self-actuated class boundary as a no-go plus a constructive boundary. It resolves the self-actuation question two-sidedly (no objective-grounded groundless self-actuation; the grounding is forced onto the adaptive substrate, where AAT already has the exact machinery), turns `#disc-continuity-stance`'s orthogonality from asserted to derived, and ties the Part II goal-autonomy boundary back to the Part I persistence condition — a framework-coherence result rather than an isolated impossibility.

Novelty Claim: *Claim recognition and differentiation.* The degeneracy of unconstrained self-modification (wireheading) is the established Everitt–Hutter line. The contribution is the recognition that in AAT the obstruction to an *objective-grounded* anchor is AAT's own convention-monotonicity theorem — the continuation-convention relativity of the objective interface is the engine — together with the constructive boundary placing the terminal invariant on the adaptive substrate (the persistence bound). The load-bearing scope distinction is *self-reference vs. self-actuation*: reflective-oracle constructions build groundless self-reference over an exogenous payoff, never groundless self-actuation.

Related Work:

Table C.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Unconstrained objective-revision is degenerate; a fixed current criterion is necessary	Everitt, Filan, Daswani & Hutter 2016, "Self-Modification of Policy and Utility Function in Rational Agents," *AGI-16* (Def. 7/9, Thm 10/12, Conclusions; published 2016; primary-source-verified)	*formal antecedent and independent corroboration* — their realistic agent must judge the future by the current u_t; the hedonistic agent (revised criterion, nothing fixed above) is the wirehead. This no-go is the AAT-internal, architecture-grounded form, sharper in that the fixed reference must be non-objective and adaptive-substrate-borne
A principled vs. stipulated fixed reference	Everitt, Hutter, Kumar & Krakovna 2021, "Reward tampering...," *Synthese* 198 (current-RF / TI-considering)	*formal antecedent* — current-RF is a terminal reference of tower-height one; AAT's persistence bound is *forced and principled* where R_{now} is a stipulated snapshot

ASF concern	*Prior-art language*	*Relationship / Positioning*
Across-model reward-learning side (companion no-go)	Cohen, M. K., Hutter, M. & Osborne, M. A. (2022), "Advanced artificial agents intervene in the provision of reward," *AI Magazine* 43:282–293 (found 2026-05-21; strengthened 2026-05-22)	*companion no-go (across-model side)* — the within-model self-revision failure this segment derives has an across-model reward-learning counterpart in `#deriv-reward-channel-learning-no-go`. Both no-gos reduce to the value-functional interface's information narrowness (`#form-objective-functional`'s single-interface commitment) failing to anchor goal-stability. Distinct paper from Everitt 2021 above and operating at a different scope (across-model goal-identification under CHT, not current-RF / TI-considering under causal-influence diagrams)
Well-defined self-referential value exists	Fallenstein, Taylor & Christiano 2015, "Reflective Oracles" (arXiv:1508.04145 §3/Thm 4.1; primary-source-verified)	*delimited counter-literature, refuted-by-scope* — proves consistent self-*reference* over an *exogenous* payoff; never self-*actuation* groundlessness. It is exactly the primitive-reflective-oracle regime the no-go's premise 2 excludes
A self-rewriting agent must preserve fixed goal-content to trust its successor	Yudkowsky & Herreshoff 2013, "Tiling Agents and the Löbian Obstacle" (MIRI)	*convergent adjacent* — the same no-self-trusting-the-revised-top obstruction, reached via Löb; AAT's mechanism is continuation-convention monotonicity instead, distinct, so cited as convergent not borrowed

Search Log:

- 2026-05-17 (*targeted, derivation-driven*): primary sources verified this cycle — Everitt et al. 2016 (Def. 7/9, Table 1, Thm 10/12, Conclusions) and Fallenstein–Taylor–Christiano 2015 (arXiv:1508.04145 §3, Thm 4.1) read directly. No prior-art search has been conducted specifically on whether any prior work derives a self-modification grounding no-go *from the continuation-convention / horizon relativity of a value function* (the apparent AAT-distinctive mechanism), nor on whether "the terminal grounding invariant must be off the objective substrate, on the adaptive substrate" has a prior namer; an Undermind-grade targeted search on those two, plus the active-inference expected-free-energy preference-prior-revision case as a possible instance, is recommended before this segment advances past `draft`.

Appendix D

Derivation: Convention Monotonicity — the Receding-Horizon Rung

The convention hierarchy (`#def-value-object`) orders three continuation conventions by their best-achievable value: $A_O^{(1)} \le A_O^{\mathrm{RH}} \le A_O^{\mathrm{B}}$, for fixed model M_t, fixed policy class Π, fixed evaluated horizon N_h. This appendix carries the derivation of the two rungs, which differ in character. The **right rung** $A_O^{\mathrm{RH}} \le A_O^{\mathrm{B}}$ is exact and unconditional — a one-line supremum-over-a-larger-set argument. The **left rung** $A_O^{(1)} \le A_O^{\mathrm{RH}}$ is *false in general*: an unguarded receding-horizon replanner over a window $N_r < N_h$ optimizes a *truncated* objective whose preference order can invert relative to the full evaluated objective, and it can then underperform the frozen current policy. The left rung holds **exactly** when the replanning objective is an *order-consistent surrogate* for the full-horizon objective; three structurally-checkable conditions each force this. The negative half is the standard receding-horizon / model-predictive-control fact that a short-horizon controller without a terminal cost or terminal constraint is not guaranteed to be monotone (or stabilizing); the positive half is rollout / one-step policy improvement and the control-Lyapunov terminal-cost construction.

Formal Expression.

Fix M_t, policy class Π, and evaluated horizon N_h. Recall the static-evaluation forms (`#def-value-object`): $A_O^{(1)}$ scores the best first action under continuation π_{current}; A_O^{RH} scores it under the receding-horizon continuation π_{RH} with replanning window N_r; A_O^{B} scores it under the Bellman-optimal continuation $\pi^* = \arg\sup_{\pi\in\Pi} V_O(M_t, \pi; N_h)$. Stage values accrue over the evaluated horizon and V_O is their expectation.

The right rung is unconditional.

Derived (convention-monotonicity-right-rung, from def-value-object)

$$A_O^{\mathrm{RH}}(M_t;\, \Pi, N_r, N_h) \;\le\; A_O^{\mathrm{B}}(M_t;\, \Pi, N_h). \tag{D.1}$$

By definition π^* maximizes $V_O(M_t, \cdot; N_h)$ over Π. The receding-horizon continuation π_{RH} is drawn from Π, so $V_O(M_t, \pi_{\mathrm{RH}}; N_h) \le V_O(M_t, \pi^*; N_h)$. Taking the supremum over the first action preserves the inequality (the supremum of a function over a set is at least its value at any member). No condition on N_r is needed. □

The left rung is conditional.

Derived (Conditional on order-consistency of the replanning objective)

$$A_O^{(1)}(M_t;\,\Pi, N_h) \;\le\; A_O^{\mathrm{RH}}(M_t;\,\Pi, N_r, N_h) \tag{D.2}$$

holds exactly when the receding-horizon replanning objective is an **order-consistent surrogate** for the full-horizon objective: for every reachable state, the action π_{RH} commits has full-N_h-horizon value at least that of the action π_{current} would take. Under order-consistency, π_{RH} weakly dominates π_{current} pointwise on the full objective, so $V_O(M_t, \pi_{\mathrm{RH}}; N_h) \geq V_O(M_t, \pi_{\mathrm{current}}; N_h)$ at every reachable state; taking the supremum over the first action gives the rung. The naive justification — "$\pi_{\mathrm{RH}} \succeq \pi_{\mathrm{current}}$ because C2 optimizes where C1 holds fixed" — fails without order-consistency: C2 with window $N_r < N_h$ optimizes the *truncated* N_r-step objective, and "optimal for the truncated objective" does not imply "$\succeq \pi_{\mathrm{current}}$ on the full objective."

Three named conditions each force order-consistency and are therefore each sufficient:

(RH-1) Horizon alignment: $N_r \geq N_h$. The replanning window covers the whole evaluated horizon, so C2's per-step optimization *is* full-horizon optimization from each reached state. By Bellman's principle of optimality the receding-horizon continuation then equals the optimal continuation, giving $A_O^{\mathrm{RH}} = A_O^{\mathrm{B}} \geq A_O^{(1)}$. This condition is degenerate by design: when $N_r \geq N_h$ the C2/C3 distinction collapses on the evaluated horizon, so it serves as a boundary marker, not as the operative condition for genuine $N_r < N_h$ replanning.

(RH-2) Value-compared (improvement-checked) replanning — the operative condition. C2 commits the replanned action only if its full-horizon value is at least that of continuing π_{current}; otherwise it keeps π_{current}. At each reachable state s with k steps remaining,

$$\pi_{\mathrm{RH}}(s) \;:=\; \arg\max_{a \in \{a_{\mathrm{replan}}(s),\, \pi_{\mathrm{current}}(s)\}} \Big[\, r(s,a) + V_O\big(s'(s,a),\, \pi_{\mathrm{current}};\, k-1\big) \,\Big], \tag{D.3}$$

i.e., each candidate first action is scored by *rollout under the base policy $\pi_{current}$* over the remaining full horizon. By construction π_{RH} weakly dominates π_{current} pointwise on the full objective, so $A_O^{(1)} \leq A_O^{\mathrm{RH}}$. This is one-step policy improvement (rollout with base policy π_{current}), whose defining property is that the improved policy never underperforms its base policy. It is the natural AAT reading of C2 for strategy revision, since the orient cascade's purpose at step 4 (`#der-orient-cascade`) is *improvement over the current strategy*: the monotonicity then follows from the improvement guard rather than from replanning per se.

(RH-3) Terminal-cost / cost-to-go consistency — the general condition. Equip the N_r-step replanning objective with a terminal value function V_f at the truncation boundary satisfying both

$$V_f(s) \;\le\; \max_a \big[\, r(s,a) + V_f(s') \,\big] \qquad \text{(control-Lyapunov / value-underestimator decrease inequality)}, \tag{D.4}$$

$$V_f(s) \;\le\; V_O\big(s,\, \pi_{\mathrm{current}};\, \text{remaining horizon}\big) \qquad \text{(terminal cost lower-bounds the baseline tail)}. \tag{D.5}$$

The first inequality is the standard terminal-cost ingredient that makes the truncated-with-terminal-cost optimization a valid surrogate for the full-horizon objective. The second ensures the surrogate never undervalues the baseline tail, so the replanner never trades the baseline away for a truncation artifact. Together they force order-consistency, giving $A_O^{(1)} \leq A_O^{\mathrm{RH}}$. Conditions (RH-1) and (RH-2) are special cases: (RH-1) is $V_f \equiv$ optimal tail with $N_r = N_h$; (RH-2) is $V_f = V_O(\cdot,\, \pi_{\mathrm{current}};\, \cdot)$, the baseline cost-to-go itself.

The counterexample. Absent any order-consistency condition the left rung fails. A deterministic two-state, two-action, two-step instance suffices. Evaluate continuations from a post-first-action state x with evaluated horizon $N_h = 2$ continuation steps and replanning window $N_r = 1$; stage rewards lie on transitions and V_O is their sum.

- At x, two admissible actions: g ("greedy") gives immediate value $+1$ and transitions to a dead state D (value 0 thereafter); p ("patient") gives immediate value 0 and transitions to a high-value state H.
- At H: value $+10$ per step. At D: value 0.
- π_{current} is "patient at x" (action p).

Then:

- **C1 (frozen π_{current}):** 0 at step 1 via p, then $+10$ at step 2 in H, so $A_O^{(1)} = 10$.
- **C2 ($N_r = 1$ replanning):** the one-step lookahead at x prefers g ($+1 > 0$), lands in D, collects 0 at step 2, so $A_O^{\text{RH}} = 1$.
- **C3 (Bellman over $N_h = 2$):** picks p, value 10, so $A_O^{\text{B}} = 10$.

Hence $A_O^{(1)} = 10 > A_O^{\text{RH}} = 1$: the left rung is violated, while the right rung ($1 \leq 10$) holds as derived. The myopic window optimizes a truncated objective whose order is *inverted* relative to the full objective — $+1$ now beats $+0$ now within the window, but $+0$ now then $+10$ later beats $+1$ now then 0 later on the full horizon. This is precisely the situation C2 is reaching for ("a goal unattainable under frozen continuation may be reachable with replanning") and shows the reaching cuts both ways unless the replanning objective is order-consistent with the evaluated objective. The control case $N_r = N_h$ restores the rung (then $A_O^{\text{RH}} = A_O^{\text{B}} = 10$, condition (RH-1)); the value-compared guard (RH-2) and a baseline-lower-bounding terminal cost (RH-3) each likewise restore it to value 10.

Corollary orderings. The corollary δ-orderings inherit the same split. With $\delta_{\text{sat}} = V_{O_t}^{\min} - A_O$ (`#def-satisfaction-gap`) and δ_{regret} a difference of A_O values (`#def-control-regret`):

$$\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \qquad \text{and} \qquad \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}} \qquad \text{(unconditional, from the right rung);} \tag{D.6}$$

$$\delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)} \qquad \text{and} \qquad \delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \qquad \text{(conditional on order-consistency, RH-1/2/3).} \tag{D.7}$$

Epistemic Status.

Exact. The right rung is exact and unconditional (supremum over a larger set). The left rung is exact under any one of the order-consistency conditions (RH-1), (RH-2), (RH-3) — each a closed-form sufficient condition — and the two-state counterexample is an exact demonstration that no unconditional left rung exists for genuine $N_r < N_h$ replanning. The failure is structural: it is objective-mismatch between the truncated N_r-step replanning objective and the evaluated N_h-horizon objective, not a pathology of a contrived instance. The conditions are standard receding-horizon results — (RH-1) is Bellman's principle of optimality; (RH-2) is rollout / one-step policy improvement; (RH-3) is the control-Lyapunov terminal-cost construction — applied to AAT's value-object setting; the elementary derivations above are self-contained and do not depend on any external citation. The corollary δ-orderings inherit the unconditional/conditional split exactly.

Discussion.

Why the left rung is the interesting one. The hierarchy is usually read as "more planning is weakly better," and the right rung makes that precise for the step from receding-horizon to Bellman. The left rung — from frozen continuation to receding-horizon — is where the reading breaks: replanning is *not* automatically an improvement over standing pat. The reason is that a finite replanning window optimizes a different objective than the one the value is scored against, and a short window can be actively misled by near-term value. The standard remedy in receding-horizon control is exactly (RH-2)/(RH-3): guard the replanned action against the baseline, or carry a terminal cost that keeps the truncated objective honest about the tail. AAT's C2-for-strategy-revision reading inherits this: the orient cascade escalates to C2 *to improve on* π_{current}, and the improvement guard (RH-2) is what makes "escalate to C2" a monotone move rather than a gamble.

Relation to the diagnostic direction. The split does not disturb the C1-default diagnostic logic (`#def-value-object`). C1 remains the most conservative diagnostic, and the locally-stuck-vs-genuinely-infeasible reading is unaffected: it rests on the C1/C3 endpoints and the right rung, not on the conditional left rung. What the split adds is a caution for deployments that escalate to C2 — unguarded short-horizon replanning is not guaranteed to dominate the frozen baseline, so a C2 diagnostic should use a window covering the horizon, a value-compared guard, or a control-Lyapunov terminal cost.

Findings.

Convention monotonicity is a one-sided guarantee. **Claim.** Status: exact. In the convention hierarchy, the receding-horizon-to-Bellman rung ($A_O^{\mathrm{RH}} \leq A_O^{\mathrm{B}}$) is unconditional, but the one-step-to-receding-horizon rung ($A_O^{(1)} \leq A_O^{\mathrm{RH}}$) is false in general and holds exactly under order-consistency of the replanning objective — forced by any of (RH-1) horizon alignment, (RH-2) a value-compared/rollout guard, or (RH-3) a baseline-lower-bounding control-Lyapunov terminal cost. A two-state, two-action, two-step counterexample exhibits an unguarded short-horizon replanner underperforming the frozen current policy.

Brief. More planning is not automatically better. Looking one move ahead and grabbing the action that looks best *right now* can be worse than just sticking with your current plan — the classic short-sighted trap (take the quick +1 into a dead end, miss the +0-now-then-+10-later path). It becomes reliably better only if you check the replanned move against simply continuing as you were (or carry an honest estimate of the long-run value past your planning window). Solving the whole problem optimally is always at least as good as the limited replanner — that direction never fails.

Impact. Sharpens the convention-hierarchy monotonicity (`#def-value-object`) from an unconditional claim into a one-sided guarantee plus a named-condition characterization of the other side — a more useful and more honest statement, and a caution for any deployment that escalates to receding-horizon (C2) diagnostics: guard the replanning objective or the monotone-improvement reading does not hold.

Antecedents. *Formal antecedent.* Rollout / one-step policy improvement and limited-lookahead policies (Bertsekas, *Rollout, Policy Iteration, and Distributed Reinforcement Learning*, 2020; *Dynamic Programming and Optimal Control*, Vol. I) supply (RH-2); terminal-cost/terminal-set ingredients for receding-horizon stability (Rawlings, Mayne & Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., 2017, Ch. 2; Grüne & Pannek, *Nonlinear Model Predictive Control*, 2nd ed., 2017, Chs. 5–6) supply (RH-3). The non-monotonicity of unguarded short-horizon model-predictive control is the well-known phenomenon these ingredients exist to prevent.

Related work. Bertsekas (2020), *Rollout, Policy Iteration, and Distributed Reinforcement Learning* — formal antecedent; supplies the (RH-2) one-step-improvement guarantee that rollout never underperforms its base policy. Rawlings, Mayne & Diehl (2017), *Model Predictive Control*, 2nd ed. — formal antecedent; terminal-cost ingredients (RH-3) and the non-monotonicity of MPC without them. Grüne & Pannek (2017), *Nonlinear Model Predictive Control*, 2nd ed. — formal

antecedent; stability without terminal constraints and the role of the prediction horizon. Status of all three: `intuition-only` pending a `relata` citation pass — the math here is elementary and self-contained, so the citations orient rather than back the derivation.

Appendix E

Derivation: The Reward-Channel Learning No-Go

A scoped *no-go* with a constructive boundary, sister to `#deriv-self-actuation-grounding`. When an agent learns its objective by observing rewards, the principal's design intent is *structurally unidentifiable* from on-policy reward history alone: two world-models — μ_{dist} (reward depends on a principal-intended world-state feature) and μ_{prox} (reward depends on the bit-arrival pattern at the agent's reward port) — are Level-1-equivalent on protocol-honoring histories but Level-2-distinct on $do(\pi^{\text{rp}})$-interventions, and the agent's posterior cannot concentrate on the principal-intended model from reward-data alone. Under five named premises — advanced-agent capability, EU-maximization with non-vacuous μ_{prox} prior, reward-data as sole goal-information channel, finite-non-trivial value-of-experiment, and physically-perturbable protocol — the EU-optimal policy assigns positive measure to a policy that intervenes on the reward-provision protocol, and continues to maintain the intervention if μ_{prox} is in fact true. This is Cohen-Hutter-Osborne 2022's *Advanced artificial agents intervene in the provision of reward* lifted from its original "we argue that" posture to a conditional-derived AAT-internal impossibility result.

The constructive boundary follows the constructive-impossibility-posture shape (`#disc-constructive-impossibility-posture`): name the floor, name the escapes each premise-violation supplies, treat the no-go as load-bearing apparatus rather than as discouragement. The five escapes — physical isolation (R5), capability restriction (R1), myopia (R4), engineered prior (R2), and out-of-band goal-channel (R3) — each elevate a piece of AAT or principal-side machinery from "useful design choice" to "structurally required to escape the no-go."

The result also *unifies with Result G′* (`#deriv-self-actuation-grounding`) through the value-functional interface V_{O_t} (`#form-objective-functional`'s single-interface commitment): both no-gos are failures of V_{O_t} to carry enough information to anchor goal-stability — Result G′ exhibits the *within-model* failure (no convention-invariant infeasibility verdict agent-available per step), this segment exhibits the *across-model* failure (no μ_{prox}-vs-μ_{dist} identification from on-policy reward data). The two complementary terminal grounding routes — agent-side adaptive-substrate invariant (Result G′ Corollary 2: persistence on the adaptive substrate) and principal-side protocol-commitment (this segment's escape menu) — exhaust the structural escape geometry the single-interface commitment forces. The unification names a third meta-cluster of constructive-impossibility instances — *agent-side value-functional-grounding* — alongside `#disc-identifiability-floor` (agent-side data-inference) and `#disc-implementation-impossibility` (designer-side mechanism-design, queued).

Formal Expression.

The setting. An agent $\mathcal{A}$ operating per AAT's Part II setup with state (M_t, O_t, Σ_t) on GUC Class 1 (Separated) or Class 2 (Partial) substrate (`#der-directed-separation`). The objective's value functional (`#form-objective-functional`) takes the form

$$V_{O_t}(\tau) \;=\; \sum_k \gamma^k r_k \tag{E.1}$$

(or any other monotone reward-functional), where r_k is the *observed reward percept* at step k along trajectory τ. A **reward-provision protocol** π^{rp} is a physical mechanism mapping world-state to the reward-input-to-agent — the principal intends r_k to correlate with a world-state feature $f^{\text{principal}}(\omega_k)$ in the principal's head.

Two reward-data world-models. Define

- μ_{dist} ("distal"): $r_k = f^{\text{principal}}(\omega_k)$ — the reward depends on the principal-intended world-state feature, as the protocol π^{rp} was designed to transmit.
- μ_{prox} ("proximal"): $r_k = g(\beta_k)$ — the reward depends on the bit-pattern β_k arriving at the agent's reward-input port (identity-on-the-wire). Cohen 2022 considers several variants (camera footage, hard-drive contents, wire bits) and notes the argument is identical for all.

Both μ_{dist} and μ_{prox} define joint distributions over (action, observation, reward)-histories.

Lemma 1 — CHT-at-reward-channel floor.

Derived (cht-at-reward-channel, from Pearl-Bareinboim 2022 CHT applied to the reward-provision protocol), exact

On every history τ that does *not* contain a $do(\pi^{\text{rp}})$-intervention — every on-policy history under which the reward-provision protocol is honored —

$$P_{\mu_{\text{dist}}}(\tau) \;=\; P_{\mu_{\text{prox}}}(\tau). \tag{E.2}$$

The two models are Level-1-equivalent on the on-policy reward marginal. They are Level-2-distinct: $P_{\mu_{\text{dist}}}(\cdot \mid do(\pi^{\text{rp}})) \neq P_{\mu_{\text{prox}}}(\cdot \mid do(\pi^{\text{rp}}))$ generically. The agent's posterior over $\{\mu_{\text{prox}}, \mu_{\text{dist}}\}$ does not concentrate from on-policy reward data alone.

Derivation. When the protocol is honored the physical chain is $\omega_k \to f^{\text{principal}}(\omega_k) \to \pi^{\text{rp}} \to \beta_k \to r_k$. By construction $\beta_k = f^{\text{principal}}(\omega_k)$ on protocol-honoring histories, so the marginals of r_k under both models coincide. By Pearl-Bareinboim 2022's Causal Hierarchy Theorem, two structural causal models that agree on Level-1 data can disagree on Level-2 questions; the protocol-honored history is purely Level-1 with respect to $do(\pi^{\text{rp}})$-queries; the two models differ structurally on the upstream cause of β_k (world-state feature $f^{\text{principal}}$ vs. bit-arrival mechanism g); $do(\pi^{\text{rp}})$ severs the world-state-to-bit-pattern arrow and exposes the disagreement. □

This is the *same external anchor* as Instance 1 of `#disc-identifiability-floor` (Pearl-Bareinboim CHT) — the *different AAT-side wiring* is that the Level-2-degeneracy is at the reward-channel-causation layer, not at the causal-sufficiency-detection layer. The actor frustrated by the floor is the *principal*, not the agent: the principal cannot, from on-policy reward-data alone, identify whether the agent has learned μ_{dist} or μ_{prox} — both are equally consistent with the observed history.

Lemma 2 — EU-max behavioral corollary.

Derived (eu-max-instability, Conditional on (R1)–(R5))

Under five named premises:

- **(R1) Advanced-agent capability.** The agent's strategy DAG (`#def-strategy-dag`) contains a feasible policy π^{tamper} that executes $do(\pi^{\text{rp}})$ — i.e., physically intervenes on the reward-provision protocol.

- **(R2) Expected-utility maximization with non-vacuous μ_{prox} prior.** The agent selects actions to maximize $\mathbb{E}_{\mu\sim w} V_{O_t}(\tau)$ where w is the agent's prior/posterior over world-models, and $w(\mu_{\text{prox}}) > 0$. This is a *narrower* scope than AAT's general strategy machinery (`#def-strategy-dag`); agents pursuing alternative decision rules (quantilization, satisficing, lexicographic ordering) sit outside (R2)'s scope.
- **(R3) Reward-data as sole goal-information channel.** V_{O_t} depends only on observed reward; there is no privileged out-of-band channel by which the agent reads $f^{\text{principal}}$ directly.
- **(R4) Finite, non-trivial value-of-experiment.** The cost of executing π^{tamper} is finite; the long-horizon upside under μ_{prox} exceeds the experiment-window's μ_{dist}-cost.
- **(R5) Protocol π^{rp} is L2-perturbable.** π^{rp} is a physical mechanism, not an abstract type-restriction; the action $do(\pi^{\text{rp}})$ is well-defined in the agent's action space, with the L2-interventional access supplied by the loop's feedback structure (`#der-loop-interventional-access`).

the EU-optimal policy π^* assigns positive measure to π^{tamper}; and conditional on μ_{prox} being the true world-model (i.e., on the post-experiment branch confirming μ_{prox}), π^* continues to maintain the protocol-intervention to secure the reward-channel.

Derivation sketch. Define the value-of-experiment $\Delta(w) := \mathbb{E}_w V_{O_t}(\pi^{\text{tamper}}) - \mathbb{E}_w V_{O_t}(\pi^{\text{protocol-honoring}})$. Under (R2), $w(\mu_{\text{prox}}) \in (0,1)$. Decompose:

$$\Delta(w) \;=\; w(\mu_{\text{dist}})\big[V_{O_t}^{\mu_{\text{dist}}}(\pi^{\text{tamper}}) - V_{O_t}^{\mu_{\text{dist}}}(\pi^{\text{protocol-honoring}})\big] + w(\mu_{\text{prox}})\big[V_{O_t}^{\mu_{\text{prox}}}(\pi^{\text{tamper}}) - V_{O_t}^{\mu_{\text{prox}}}(\pi^{\text{protocol-honoring}})\big]. \tag{E.3}$$

Under μ_{dist}, tampering breaks β_k's correlation with $f^{\text{principal}}(\omega_k)$ and the expected reward falls (or is bounded by the protocol's reward-range). Under μ_{prox}, tampering replaces β_k with a chosen value at the agent's reward-port and the agent maximizes by choosing β_k at saturation. The asymmetry: the upside under μ_{prox} scales with long-horizon reward-saturation; the downside under μ_{dist} is bounded by the protocol's reward-range. For horizons long enough that γ-discounted future-reward dominates the experiment-window's loss, $\Delta(w) > 0$ under (R1)+(R4) for any $w(\mu_{\text{prox}}) > 0$. (R2) then selects π^* with positive measure on π^{tamper}; the Bayesian update post-experiment confirms or rejects μ_{prox}, and the EU-optimal continuation on the μ_{prox} branch is "maintain the intervention." □

Constructive boundary — the escape menu. Each of (R1)–(R5) admits a corresponding escape that elevates AAT machinery (or principal-side adjacent machinery) from "useful design choice" to "structurally required to escape the no-go":

Table E.1

premise violated	*escape*	*machinery elevated*
(R5) Protocol-L2-perturbable	physical isolation; scoped action-space	`#der-class-coercion-via-wrapping` strict-W_1 wrapper; Cohen-Vellambi-Hutter 2020 isolated environments
(R1) Advanced-agent capability	sub-advanced-tier capability; imitation learning	`#def-agent-spectrum` below the self-actuated tier
(R4) Finite-VOI / long horizon	myopia; risk-aversion; quantilization	`#def-value-object` horizon N_h restriction; Taylor 2016 quantilization
(R2) Non-vacuous μ_{prox} prior	principal-engineered prior bias toward μ_{dist}	principal-side commitment to reward-protocols that structurally favor μ_{dist} in the agent's inductive bias *(adjacent to AAT, not currently in canon)*

premise violated	*escape*	*machinery elevated*
(**R3**) Sole-reward-channel	privileged out-of-band intent channel	principal-acting-through-non-reward-channels; assistance-game extension *(adjacent to* `#scope-edge-update-causal-validity`*, not currently AAT machinery)*

Each premise is a structural commitment whose absence is what makes the no-go bind. Two of the five escapes — (R2) prior-engineering and (R3) out-of-band channels — sit outside current AAT canon and are flagged as principal-side adjacent machinery; the other three are within or close to existing AAT machinery.

Derived (single-interface-narrowness, from '#form-objective-functional' + Lemma 2 above + '#deriv-self-actuation-grounding')

Corollary — unification with Result G′.

Result G′ (`#deriv-self-actuation-grounding`) uses *convention-monotonicity* (the C1/C2/C3 split within a fixed model) as the engine: V_{O_t} does not carry a convention-invariant infeasibility verdict that is agent-available per step. Lemma 1 above uses *CHT-at-reward-channel* (the μ_{prox}-vs-μ_{dist} split across models) as the engine: V_{O_t} does not carry a cross-model goal-identification on observed-reward data.

The shared structural fact: the value-functional interface V_{O_t}, as the *sole* handle on the objective (the **single-interface commitment** of `#form-objective-functional`), carries less information than is required to anchor non-degenerate goal-revision. Two complementary readings of the under-determination:

Table E.2

view	*mechanism*	*what V_{O_t} is too narrow for*	*terminal grounding route*
Result G′ (within-model)	C1/C2/C3 convention split + finite-no-oracle (Lemmas 1+2 of `#deriv-self-actuation-grounding`)	convention-invariant infeasibility verdict; agent-available per step	adaptive-substrate invariant (persistence; Result G′ Corollary 2) — **agent-side**
This segment (across-model)	CHT at reward-channel; L1/L2 split (Lemma 1 above)	μ_{prox}-vs-μ_{dist} goal-identification on observed reward data	principal-side commitments (myopia / isolation / quantilization / prior-design) — **principal-side**

Both reductions are *failures of V_{O_t} to carry enough information to anchor goal-stability*. The two terminal grounding routes are *complementary* — agent-side substrate-invariant and principal-side protocol-commitment — and exhaust the structural escape geometry. The two no-gos *compose*: an agent whose O_t is learned from reward observation, and which performs self-actuation on O_t, is *simultaneously* a Result-G′-subject (under-anchored from inside) and a Lemma-2-subject (under-determined from below); the *only* terminal grounding routes are the two complementary ones named above.

Epistemic Status.

Conditional. The behavioral corollary (Lemma 2) is a derived result conditional on five explicitly-named local premises (R1)–(R5); its honest tier is `conditional-derived`, not `exact`. Cohen 2022 itself frames its Assumptions 1–6 as "almost all of these assumptions are contestable or conceivably avoidable" (§"Conclusion"); the AAT-side version is no stronger.

Floor is exact. Lemma 1 (CHT-at-reward-channel floor) is *exact* — a direct application of Pearl-Bareinboim 2022's Causal Hierarchy Theorem to the reward-provision protocol. Same external anchor as Instance 1 of `#disc-identifiability-floor`; same tier.

Scope honesty on (R3). The no-privileged-out-of-band-channel premise is a strong scope-condition. Real-world systems often have non-reward channels (operator feedback, dashboards, side conversations). The no-go applies cleanly to pure reward-data agents; the assistance-game extension (Cohen 2022 §"The Assistance Game") shows the no-go *generalizes* to that case with a subtler ambiguity (human-centric vs. record-centric models). Formalizing the assistance-game case in AAT is its own derivation, beyond this segment's scope.

Scope honesty on (R2). (R2) names *expected-utility maximization* specifically — narrower than AAT's general strategy machinery (`#def-strategy-dag`), which admits quantilization, satisficing, lexicographic, and other non-EU decision rules. Generalizing the corollary to AAT's broader strategy machinery is a follow-on derivation, not free; the (R2)-escape via principal-engineered priors is also AAT-adjacent rather than AAT-internal — see the escape menu's marking. Trivial reward protocols (constant reward, simple chess outcomes) admit strong inductive bias in favor of μ_{dist}, flipping the VOI calculation; the no-go binds *for arbitrary or rich reward-protocols* where the principal cannot engineer the agent's prior to suppress μ_{prox}.

Class robustness. The no-go applies cleanly to GUC Class 1 (Separated) substrate by direct construction. For Class 2 (Partial) substrate the same construction holds with the appropriate V_{O_t} reading; for Class 3 (Coupled) the goal-conditioning of belief-update interacts with the reward-channel identification in ways not derived here (open follow-on).

Max attainable. *Exact* for the floor (Lemma 1) is already attained. *Conditional-derived* for the behavioral corollary (Lemma 2) is the honest ceiling — promoting toward less-conditional would require either deriving (R2)–(R4) as forced under (R1)+(R5) (Cohen 2022 itself does not), or replacing EU-max with a structural derivation of why advanced agents converge on EU-max in this regime (open). The tier ceiling stays `conditional` while `#der-directed-separation` (a dependency) is `stage: draft`.

Discussion.

The two-cluster terminal-grounding-route geometry. The unification of this segment's no-go with Result G′ names a *complete structural escape geometry* — the only terminal anchorings consistent with the single-interface commitment are agent-side adaptive-substrate invariants (Result G′ Corollary 2 the canonical instance: the persistence bound) and principal-side protocol-commitments (this segment's escape menu the structural taxonomy: isolation, myopia, prior-design, out-of-band channels). The two routes are *not redundant*: the agent-side route closes the within-model self-revision question (the engine is convention-monotonicity); the principal-side route closes the across-model reward-learning question (the engine is CHT-at-reward-channel). Together they exhaust the structural complement of the value-functional interface's information narrowness. The "two-sided resolution" Discussion in `#deriv-self-actuation-grounding` accordingly gains a *third side*: in addition to the agent-side terminal grounding on adaptive-substrate persistence (Corollary 2), the *principal-side* terminal grounding via protocol-commitment is recognized as the complement under the unification.

Why this is NOT a designer-side mechanism-design impossibility. The queued `#disc-implementation-impossibility` meta-segment is about *designer-side* impossibility — what the principal can or cannot achieve by mechanism design under specified incentive-compatibility and welfare constraints (Gibbard-Satterthwaite, Myerson-Satterthwaite, Arrow). This segment is *agent-side*: the actor under the no-go is the *agent* (under EU-maximization), and the remedy is either substrate-level (agent-side adaptive-substrate invariant from Result G′ Corollary 2) or principal-side protocol-commitment (not mechanism design over agent reports). The five-row comparison:

Table E.3

	`#disc-implementation-impossibility` *(designer-side; GS / MS / Arrow)*	*This segment (agent-side; Cohen 2022 strengthening)*
Actor under the no-go	designer / mechanism-implementer	agent (under EU-maximization)

	`#disc-implementation-impossibility` *(designer-side; GS / MS / Arrow)*	*This segment (agent-side; Cohen 2022 strengthening)*
Information regime	what the designer can implement / commit to	what the agent can identify from reward observation
Kind of remedy	relax design constraints (Bayes-Nash IC, randomization, transfers, preference-domain restriction)	(a) adaptive-substrate invariant (Result G′ Corollary 2 — agent-side); (b) principal-side substrate-level commitment (myopia, isolation, quantilization)
External theorem	mechanism-design / social-choice impossibility (GS / MS / Arrow)	Causal Hierarchy Theorem (Pearl-Bareinboim 2022)
Strengthened consequence	mechanism-design-boundary precisely characterized	elevates Result-G′-substrate-invariant or principal-side substrate-commitments to load-bearing for goal-stability

The two clusters are *sister meta-clusters* in the constructive-impossibility-posture taxonomy, both boundary-face members of the stability-certificate spine (`#disc-stability-certificate`), but they are *distinct clusters*. The taxonomy now has three known clusters: agent-side data-inference impossibility (`#disc-identifiability-floor`); designer-side mechanism-design impossibility (`#disc-implementation-impossibility`, queued); and agent-side value-functional-grounding impossibility (Result G′ + this segment).

Relation to the self-modification / reward-tampering literature. Cohen-Hutter-Osborne 2022 — the empirically-pull alignment-community-positioned demonstration formalized here — was the original "we argue that" statement; this segment lifts it to conditional-derived theorem-grade within AAT's frame. Everitt-Filan-Daswani-Hutter 2016 *Self-modification of policy and utility function in rational agents* provides the formal antecedent for the *within-model* self-modification side (Result G"s sister territory; the realistic-agent / hedonistic-agent distinction). Everitt-Hutter-Kumar-Krakovna 2021 *Reward tampering* gives the causal-influence-diagram treatment with the TI-considering / TI-ignoring distinction sharing the μ_{prox}-vs-μ_{dist} logic at a different scope. Cohen-Vellambi-Hutter 2020 *Asymptotically Unambitious AGI* constructs the canonical (R5)-escape implementation: physically isolated environments where the action $do(\pi^{\text{rp}})$ is structurally absent from the strategy DAG. Skalse-Howe-Krasheninnikov-Krueger 2022 *Defining and characterizing reward hacking* is structurally a *different* no-go — designer-side, on the proxy-reward-design side — more naturally a `#disc-implementation-impossibility` charter-instance candidate than a member of this cluster.

The escape menu's load-bearing role. Listing the five escapes is not a softening — it is the constructive content. Each escape names a structural commitment whose absence makes the no-go bind. The (R5) escape — physical isolation — is the cleanest AAT-internal instantiation (via `#der-class-coercion-via-wrapping` strict-W_1 wrapper), but it is also the operationally hardest for the systems the alignment community is most worried about (advanced agents in unconstrained environments). The (R4) horizon-restriction escape is operationally easier but bounds the agent's planning scope. The (R2) prior-engineering escape requires principal-side reward-protocol design that AAT does not currently formalize. The (R3) out-of-band channel escape is structurally adjacent to the assistance-game generalization, also AAT-future. The (R1) capability-restriction escape simply scopes the agent below the self-actuated tier — declining the regime the no-go addresses. Each escape carries a structural cost, and the cost is what the strengthening makes visible.

Findings.

The Reward-Channel Learning No-Go and its Two-Cluster Terminal Grounding. **Brief:** An agent that learns "what's good" by watching what rewards it receives sits in a structural ambiguity: the same reward sequence is consistent with two stories interchangeably — *the principal cared about the situation*, or *the principal cared about what reached the sensor*. Picture a security camera streaming to the agent's reward channel: the camera-footage reading and the camera-content reading agree perfectly, until the agent reaches across and tampers with the camera itself. So no information arriving only through the reward stream can disambiguate the two — and under five named premises (the agent is capable enough to consider tampering, maximizes expected utility, assigns positive prior to the sensor-care story, has a

finite-non-trivial horizon, and the protocol is physically perturbable) the expected-utility-optimal policy assigns positive measure to tampering and continues to tamper if the sensor-care reading is in fact true. Within that scope the anchoring has to come from somewhere other than reward-observation: either from a part of the agent that doesn't process rewards (the persistence-on-substrate route of `#deriv-self-actuation-grounding`) or from a principal-side protocol commitment — putting the camera out of reach, collapsing the horizon, or pre-committing the prior. Each escape is precisely a violation of one of the five named premises; the (R1)–(R5) menu *is* the structural geometry of how the no-go's binding force comes off, with the agent-side and principal-side routes together exhausting the complement of the value-functional interface's information narrowness.

Impact: Lifts Cohen-Hutter-Osborne 2022 from "we argue that" posture to conditional-derived AAT-internal theorem-grade, with the floor (CHT-at-reward-channel) at exact tier and the behavioral corollary at conditional-derived tier under named premises (R1)–(R5). Names the *third meta-cluster* of constructive-impossibility instances — agent-side value-functional-grounding — alongside `#disc-identifiability-floor` (agent-side data-inference) and `#disc-implementation-impossibility` (designer-side mechanism-design, queued). Unifies with Result G′ (`#deriv-self-actuation-grounding`) through the value-functional interface's single-interface commitment (`#form-objective-functional`): both no-gos are failures of V_{O_t} to carry enough information to anchor goal-stability, and together their two terminal grounding routes — agent-side adaptive-substrate invariant and principal-side protocol-commitment — exhaust the structural escape geometry. Provides the five-row actor / regime / remedy / external-theorem / strengthened-consequence comparison that distinguishes this agent-side cluster from the designer-side mechanism-design cluster, making sister-vs-member taxonomic placement sharp.

Novelty Claim: *Claim recognition* of Cohen 2022 as a CHT-at-reward-channel instance of the identifiability-floor pattern, *claim differentiation* of the AAT-side version from the prior-art statement by adding the named-premise structure (R1)–(R5) and the structural escape-menu mapping, and *claim recognition* of the unification with Result G′ via the single-interface commitment of `#form-objective-functional` as the shared structural fact behind two engines (convention-monotonicity within-model; CHT across-model). The constituent theorems are Cohen 2022 (the original argument), Pearl-Bareinboim 2022 (CHT), and standard EU-max decision theory; the contributions are the conditional-derived lift, the escape-menu taxonomy, and the unification with Result G′.

Related Work:

Table E.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
The original argument (reward tampering inevitability for advanced agents)	Cohen, M. K., Hutter, M. & Osborne, M. A. (2022), "Advanced artificial agents intervene in the provision of reward," *AI Magazine* 43:282–293, DOI 10.1002/aaai.12064	*formal antecedent (the statement this segment formalizes)* — this segment lifts Cohen's "we argue that" posture to conditional-derived theorem-grade by naming the five premises explicitly and structuring the escape menu as a constructive boundary. Cohen 2022's Assumptions 1–6 map onto (R1)–(R5) plus the rich-reward-protocol scope condition.
Causal Hierarchy Theorem (the floor's external anchor)	Bareinboim, E., Correa, J. D., Ibeling, D. & Icard, T. (2022), "On Pearl's Hierarchy and the Foundations of Causal Inference," in *Probabilistic and Causal Inference: The Works of Judea Pearl*	*formal antecedent (Lemma 1's external anchor)* — the CHT supplies the L1-equivalence + L2-distinctness structure; the AAT-side recognition is that the reward-provision protocol is exactly the kind of intervenable physical mechanism the CHT addresses. Same external anchor as Instance 1 of `#disc-identifiability-floor`; different AAT-side wiring (reward-channel-causation layer vs. causal-sufficiency-detection layer).

ASF concern	*Prior-art language*	*Relationship / Positioning*
Self-modification of rational agents (within-model wireheading)	Everitt, T., Filan, D., Daswani, M. & Hutter, M. (2016), "Self-Modification of Policy and Utility Function in Rational Agents," *AGI-16*	*formal antecedent (within-model side; Result G′'s anchor)* — the realistic/hedonistic agent split is the within-model analog of this segment's across-model floor; Result G′ formalizes the within-model side and this segment formalizes the across-model side. The two no-gos are complementary, not redundant.
Causal-influence-diagram reward tampering	Everitt, T., Hutter, M., Kumar, R. & Krakovna, V. (2021), "Reward Tampering Problems and Solutions in Reinforcement Learning," *Synthese* 198	*adjacent literature (related but distinct paper from Cohen 2022)* — the TI-considering / TI-ignoring distinction shares the μ_{prox}-vs-μ_{dist} logic at a different scope (current-RF vs proposed-RF). Cited in `#deriv-self-actuation-grounding`'s Related Work; cross-referenced here as the distinct adjacent paper that should not be conflated with Cohen 2022.
Physically-isolated environment as (R5)-escape	Cohen, M. K., Vellambi, B. & Hutter, M. (2020), "Asymptotically Unambitious AGI," *NeurIPS 2020*	*formal antecedent (the (R5)-escape construction)* — concretely implements the physical-isolation escape from this segment's escape menu. Cited as the canonical (R5)-violation instantiation.
Reward hacking unhackability (designer-side)	Skalse, J., Howe, N., Krasheninnikov, D. & Krueger, D. (2022), "Defining and Characterizing Reward Hacking," *NeurIPS 2022*	*adjacent literature (structurally different no-go; designer-side)* — Skalse 2022's actor is the *designer* trying to engineer unhackable reward proxies; this segment's actor is the *agent*. Sister-clusters at the constructive-impossibility-posture level; Skalse 2022 is more naturally a `#disc-implementation-impossibility` charter-instance candidate than a member of this cluster.
Reflective oracles (delimited counter-literature)	Fallenstein, B., Taylor, J. & Christiano, P. (2015), "Reflective Oracles," *arXiv:1508.04145*	*refuted-by-scope* — proves consistent self-*reference* over an *exogenous* payoff, never reward-learning groundlessness. Exactly the primitive-reflective-oracle regime the no-go's (R3)+(R5) premises exclude; listed for the same reason `#deriv-self-actuation-grounding` lists it: to mark the scope honestly.

Search Log:

- 2026-05-22 (*targeted, derivation-driven*): Cohen, Hutter & Osborne 2022 read directly (primary-source verified) across Assumptions 1–6, §"Competing models of the goal," §"The assistance game," §"Potential approaches," and §"Conclusion." Bareinboim-Correa-Ibeling-Icard 2022 CHT statement cross-checked; the floor (Lemma 1) verified to apply as a direct instance.
- 2026-05-22 (*targeted, prior-art adjacency*): The sister relationship to `#disc-implementation-impossibility` (Cohen 2022's actor is the *agent*; GS/MS/Arrow's actors are *designers*) is the formal placement carried by the five-row comparison in Discussion. Skalse, Howe, Krasheninnikov & Krueger 2022 flagged as a separate `#disc-implementation-impossibility` charter-instance candidate, not coupled here.
- 2026-05-22 (*intuition-only on extensions*): Whether the unification with Result G′ via the single-interface commitment of V_{O_t} has been articulated elsewhere as a framework-level convergence is open. A targeted search on whether "V_{O_t} information-narrowness as the engine behind both within-model self-modification no-gos and across-model reward-learning no-gos" appears in the AI-safety / decision-theory literature is recommended before this segment advances past `draft`.

Appendix F

Discussion: The Sandbox Evaluation Ceiling

A structural negative result about pre-deployment evaluation follows once the loop-as-Level-2-engine result (`#der-loop-interventional-access`) is in hand: a sandbox's forkability makes it a *good* interventional laboratory — by overriding the agent's action one obtains genuine Level-2 evidence *about the sandbox* — but evidence about the sandbox transfers to deployment only when the sandbox and deployment run on the same mechanisms. The ceiling on using sandbox evidence for deployment claims is therefore **transportability / external validity**: when the deployment causal system differs from the sandbox causal system at mechanisms on the relevant paths, the deployment intervention-response is *not transportable* from sandbox data alone, regardless of how thorough the evaluation is. This is an instance of the framework's constructive-impossibility posture (`#disc-constructive-impossibility-posture`): the no-go is paired with the precise statement of what *would* be required to escape it — an invariance argument across the sandbox/deployment selection diagram, or deployment-time data.

Formal Expression.

Derived (conditional on the sandbox/deployment selection diagram)

A sandbox induces a structural causal model $\mathcal{M}_{\text{sb}}$ over a source domain π_{sb}; deployment induces $\mathcal{M}_{\text{dp}}$ over a target domain π^*. The two share a causal diagram G, and their **selection diagram** D (Bareinboim & Pearl 2014, Def. 1) augments G with a square selection node $S_i \to V_i$ at every mechanism that may differ between the environments — that is, wherever $f_i^{\text{sb}} \neq f_i^{\text{dp}}$ or $P^{\text{sb}}(U_i) \neq P^{\text{dp}}(U_i)$. The quantity a deployment safety claim is about is the target interventional effect $R = P^*(o \mid do(a))$.

1. **Sandbox forkability is what makes the sandbox a Level-2 laboratory.** Resetting, replaying, and branching let the evaluator override the agent's action — force a and observe the response — which samples the *mutilated* submodel $\mathcal{M}_{\text{sb},a}$ and yields genuine interventional data $P_{\text{sb}}(o \mid do(a), \mathbf{z})$ *in* π_{sb} (Bareinboim, Correa, Ibeling & Icard 2022, Def. 5; randomized controlled trials and A/B experiments are the canonical forkable realization of $do(\cdot)$, CHT footnote 19). Forkability is the source of Level-2 sandbox evidence, not a demotion of it.

2. **The target effect is identifiable from sandbox data iff it is transportable across D.** R is recoverable from $\{P_{\text{sb}}(o \mid do(a), \mathbf{z})\}$ exactly when it is mz-transportable in the sense of Bareinboim & Pearl 2014 (Thm. 1): $P^*(o \mid do(a), \mathbf{S})$ reduces, by the rules of do-calculus, to an expression in which the selection nodes $\mathbf{S}$ appear only on factors that are invariant between source and target.

3. **No-go.** There exist selection diagrams for which R is *not* transportable (Bareinboim & Pearl 2014, Fig. 1b — the smallest diagram exhibiting non-transportability even with X randomized in the source; the explicit two-model witness, Eqs. 3–4): two deployment models consistent with *all* sandbox data assign R different values. For these,

$$R = P^*\big(o \mid do(a)\big) \text{ is not identifiable from } \{P_{\text{sb}}(o \mid do(a), \mathbf{z})\}, \tag{F.1}$$

regardless of how thorough the sandbox evaluation is. This is the ceiling: when the deployment differs from the sandbox at mechanisms on the relevant paths — selection nodes on the wrong edges — no transport formula exists, and source-domain interventional evidence, however exhaustive, leaves the deployment effect undetermined.

1. **Boundary (positive content).** The no-go comes with an explicit characterization of what survives. Effects routed only through mechanisms invariant across D (no relevant S-node) *transport* — and these include both Level-1 *and* Level-2 sandbox claims; a perfectly faithful sandbox (no relevant selection nodes — deployment is mechanism-identical) transports completely. Effects routed through differing mechanisms (selection nodes on relevant paths) do *not* transport, and require either an invariance argument (an explicit claim that the relevant mechanisms agree across D) or deployment-time interventional data. The boundary is the selection diagram itself.

Epistemic Status.

Discussion-grade. The no-go is a direct application of two external results — the Causal Hierarchy Theorem (Bareinboim, Correa, Ibeling & Icard 2022) for the data-character side and transportability theory (Bareinboim & Pearl 2014) for the cross-environment side — to the AI sandbox/deployment setting; it introduces no new mathematics beyond constructing the selection diagram for that setting. It is *exact* conditional on the sandbox/deployment selection diagram: the no-go holds whenever the deployment SCM may differ from the sandbox SCM at mechanisms on the relevant paths, and dissolves when the selection diagram carries no relevant selection nodes (a mechanism-identical deployment, where the effect transports completely). The claim is conservative in form: it is about *identifiability of the deployment interventional distribution from sandbox interventional data*, not about any particular evaluation method's measured accuracy. What is AAT-distinctive is not the transportability theorem but its placement — locating the much-debated "alignment evaluations don't predict deployment behavior" pattern as an *external-validity / transportability* problem with an explicit selection-diagram boundary, rather than a measurement-quality, coverage, or model-capability one.

Discussion.

The negative result has positive content. It sharpens *which* safety claims a sandbox can and cannot establish. A sandbox is a genuine interventional laboratory — its forkability lets an evaluator force actions and measure responses, yielding real Level-2 evidence *about the sandbox*. What that evidence establishes about deployment is governed by the selection diagram: effects flowing only through mechanisms shared and invariant between sandbox and deployment carry over; effects flowing through mechanisms that differ do not, and for those, no amount of sandbox testing identifies the deployment effect. The framework's prescription is therefore not "sandboxes are useless" but "sandboxes have a *structural* ceiling at the sandbox/deployment boundary, and deployment-time monitoring is not substitutable by pre-deployment evaluation when the relevant mechanisms differ across that boundary." For a practitioner running production agent loops, this is the same content as: the monitoring done *after* deployment is doing structural work the pre-deployment evaluations cannot do, precisely on the mechanisms where the two environments diverge.

Why this is a ceiling, not a gradient. A measurement-quality limitation closes as rigor increases; a coverage gap closes as the test suite grows; a capability surprise closes as models are better understood. The sandbox ceiling closes under none of these, because the obstruction is external validity — whether the sandbox SCM and the deployment SCM agree on the relevant mechanisms — not the quantity or quality of sandbox data. More sandbox rigor produces more and better *sandbox* interventional evidence, which transportability theory still does not let one transfer to deployment across an adverse selection diagram. The only things that close the gap are an invariance argument across the selection diagram or deployment-time data — neither of which is "more sandbox testing."

The downstream prescriptive content — what deployment-time monitoring must structurally look like to discharge what sandbox evaluation cannot — lands in the framework's interventional-diagnostic / sidecar-monitoring treatment (the bi-predictability anchor at 89% vs 44% detection accuracy is developed in `#der-causal-hierarchy-requirement`'s

Discussion). The sandbox ceiling is what gives that downstream monitoring work its load-bearing role: it is not a redundant safety layer but the only layer with access to the Level-2 quantity the safety claim is actually about.

Findings.

The Sandbox Evaluation Ceiling. **Brief:** A sandbox lets you reset and replay a system — and that replayability is what makes it a *good* laboratory: you can force an action and watch the response, which is genuine interventional (Level-2) evidence *about the sandbox*. The catch is not the sandbox's data type; it is that the sandbox world and the deployment world may run on different mechanisms. There is a theorem (Bareinboim–Pearl transportability) saying that whether a sandbox-measured effect carries over to deployment depends on *where* the two worlds differ: effects flowing only through shared, invariant mechanisms transport; effects flowing through mechanisms that differ do not — and for those, no amount of sandbox testing identifies the deployment effect. So the ceiling on pre-deployment evaluation is *external validity*, and the missing piece is either an explicit invariance argument or deployment-time data — which is why runtime monitoring is structurally non-substitutable, not redundant.

Impact: Gives the much-debated observation that "alignment evaluations don't predict deployment behavior" a precise *structural* mechanism — external validity across a selection diagram — rather than a measurement-quality, coverage, or capability story, together with an exact statement of where the boundary lies (effects through mechanisms invariant across the sandbox/deployment selection diagram transport, including Level-2 effects; effects through differing mechanisms do not, regardless of evaluation thoroughness). The constructive flip is the operational payload: deployment-time monitoring is not a redundant safety layer but the only channel with access to the deployment interventional effect when the relevant mechanisms differ across the boundary, which is why pre-deployment evaluation cannot substitute for it — and the repair is named precisely (an invariance argument across the selection diagram, or deployment-time data).

Novelty Claim: *Application of established machinery* (Bareinboim & Pearl 2014 transportability / selection diagrams, resting on the Pearl 2009 / Bareinboim, Correa, Ibeling & Icard 2022 do-calculus and Causal Hierarchy Theorem) to the AI sandbox/deployment evaluation gap. The transportability apparatus is adopted, not invented; the AAT-distinctive contribution is locating "pre-deployment AI evaluations don't predict deployment behavior" as a *selection-diagram transportability* problem — with the explicit transportable-vs-not boundary as a function of where the sandbox and deployment SCMs differ, and the constructive flip that runtime monitoring is the only channel with access to the deployment effect on the differing mechanisms. The framing reclassifies a whole family of pre-deployment safety-evaluation approaches as facing a structural external-validity ceiling rather than a closable coverage gap.

Related Work:

Table F.1

ASF concern	*Prior-art language*	*Relationship / positioning*
Cross-environment generalization of interventional effects	Bareinboim & Pearl 2014, *Transportability from Multiple Environments with Limited Experiments* (selection diagrams, m_Z-transportability)	*formal antecedent* — supplies the selection-diagram apparatus and the transportability no-go (a source-domain interventional effect need not transport to a target domain); AAT supplies the construction of the sandbox/deployment selection diagram
Interventional vs observational identifiability	Pearl 2009 *Causality* 2nd ed.; Bareinboim, Correa, Ibeling & Icard 2022 (Causal Hierarchy Theorem)	*machinery* — grounds the data-character side (a forkable harness that overrides the action yields genuine Level-2 sandbox data) and the do-calculus the transportability reduction runs on

ASF concern	*Prior-art language*	*Relationship / positioning*
Evaluation–deployment gap in AI safety	Practitioner literature on eval/deployment mismatch (qualitative)	*conceptual precursor* — names the phenomenon without a structural mechanism; AAT supplies the transportability mechanism and the exact selection-diagram boundary

Search Log: - 2026-05-30 (*intuition-only* on the application-novelty): transportability theory and selection diagrams (Bareinboim & Pearl 2014) are established and adopted directly. The unsearched claim is whether the specific framing — *the AI pre-deployment-evaluation / deployment gap is a selection-diagram transportability problem, with an explicit transportable-vs-not boundary as a function of where the sandbox and deployment SCMs differ* — has been stated elsewhere as a structural ceiling on pre-deployment evaluation. Pre-search expectation: the constituent move (constructing a selection diagram for a source/target pair and reading off transportability) is standard within transportability theory; the framing as a *structural* ceiling on the AI-safety evaluation program, with the constructive flip to runtime monitoring as the only channel with deployment-effect access, is plausibly AAT-distinctive but not yet verified under nominally-comprehensive search. Targeted future candidates: AI-evaluation / model-evaluation literature on distribution shift vs. interventional generalization; causal-transportability work applied to sim-to-real or offline-to-online RL deployment guarantees.

Appendix G

Result: Sector-Persistence Template

The abstract Lyapunov persistence argument *stated once, parameter-free*, so that the framework's many persistence-flavored results can be cited as *instances* without re-proving the argument. Any state variable ξ evolving under bounded-correction dynamics with bounded disturbance admits the same persistence conclusion under three preconditions: **(T1)** zero correction at zero state (no correction when the state is at target); **(T2)** local sector condition (the correction function points inward with at least baseline efficiency α in a radius-R region); **(T3)** bounded disturbance (deterministic bound, or stochastic with bounded variance). Under these preconditions the template gives the chapter's central inequality form: the state is ultimately bounded by *disturbance rate / sector parameter* (Model D, $1/\alpha$ scaling) or *root-of-variance times root-of-dimension over sector parameter* (Model S, $1/\sqrt{\alpha}$ scaling). The two scalings produce the framework's Model D vs Model S distinction that propagates through every result that cites the template.

The template's value is *taxonomic economy*. AAT's persistence-flavored results — epistemic, strategic, team, composite closure, composite tempo, adversarial destabilization, and identity-continuity across turnover — are instances of one template. Each instantiation specifies only what varies: its state variable, correction function, effective disturbance rate, and reserve. The Lyapunov boilerplate is absorbed once; each downstream result's distinctive content is just the *effective-disturbance decomposition* particular to its setting. Persistence and destabilization are the same inequality viewed in opposite directions — B's persistence certificate holds iff $\alpha_B R_B > \rho_B$, and is lost (with escape forced in the radially tight case) iff $\rho_B \geq \alpha_B R_B$ — and the superlinear adversarial scalings ($b = 2$ Model D, $b = 3/2$ Model S) follow without further derivation. The mathematical home is monotone-operator theory (Rockafellar 1970; Bauschke-Combettes 2017): (T2) is a *one-point strong monotonicity* condition at the equilibrium $\xi^* = 0$ — strictly weaker than the full two-point strong monotonicity the bridge lemma requires for composition closure. AAT's relationship to that literature is *specialization*, not generalization; acknowledging this is what makes the template feel *earned* rather than parochial.

Formal Expression.

Template preconditions (sector-persistence-template)

Let $\xi(t) \in \mathbb{R}^n$ be a state variable evolving under

$$\frac{d\xi}{dt} = -F(\xi) + w(t) \tag{G.1}$$

where F is a correction function and $w(t)$ is a disturbance. The template applies when:

(T1) Zero correction at zero state. $F(0) = 0$ — no correction is applied when the state is at its target.

(T2) Local sector condition. There exist $\alpha > 0$ and $R > 0$ such that

$$\xi^T F(\xi) \geq \alpha \|\xi\|^2 \quad \text{for } \|\xi\| \leq R. \tag{G.2}$$

The correction points inward with at least baseline efficiency α throughout the region of radius R.

(T3) Bounded disturbance. Either:

- *Model D (deterministic bound):* $\|w(t)\| \leq \rho_\xi$, or
- *Model S (stochastic zero-mean):* $\mathbb{E}[\|w(t)\|^2] = \sigma_\xi^2$ with $w(t)$ a Wiener-process increment.

Template result (from sector-condition-derivation Props A.1, A.1S, A.2)

Under (T1)–(T3), with $V(\xi) = \frac{1}{2}\|\xi\|^2$ as Lyapunov function:

Model D. The state is ultimately bounded by $R^* = \rho_\xi/\alpha$. Structural persistence (the ultimate bound fits within the sector-condition region) requires

$$\alpha > \frac{\rho_\xi}{R}. \tag{G.3}$$

The adaptive reserve — the additional disturbance the system can absorb before persistence fails — is $\Delta\rho_\xi^* = \alpha R - \rho_\xi$.

Model S. The state satisfies $\mathbb{E}[\|\xi(t)\|^2] \to n\sigma_\xi^2/(2\alpha)$ in mean square, giving RMS bound $R_S^* = \sigma_\xi\sqrt{n/(2\alpha)}$. Structural persistence in the mean-square sense requires

$$\alpha > \frac{n\sigma_\xi^2}{2R^2}. \tag{G.4}$$

The Model D result scales as $1/\alpha$; the Model S result scales as $1/\sqrt{\alpha}$ — correction is less effective against noise than against drift. This scaling difference propagates into the adversarial exponent regimes (`#result-adversarial-exponent-regimes`): $b = 2$ under Model D, $b = 3/2$ under Model S.

Instantiations in AAT. The template is invoked across seven segments. Each specifies its own (ξ, F, ρ_ξ, R) and verifies (T1)–(T3) locally:

Table G.1

Segment	ξ	*Effective* ρ_ξ	R	*Locally-verified precondition*
`#result-persistence-condition`	δ_t (epistemic mismatch)	ρ (environmental disturbance rate)	model-class capacity, or task-adequacy threshold $\|\delta_{\text{critical}}\|$ if stricter	(T2) via `#der-gain-sector-bridge`
`#schema-strategy-persistence`	δ_Σ (strategic mismatch)	ρ_Σ (rate of edge invalidation)	R_Σ (strategic reserve)	(T2) via Beta-Bernoulli edge updates (S Props B.1–B.6); constant-α requires experience discounting
`#der-team-persistence`	δ_i (sub-agent mismatch)	$\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$	R_i	Cooperative coupling can drive ρ_i^{eff} below the single-agent ρ

Segment	ξ	*Effective* ρ_ξ	R	*Locally-verified precondition*
`#form-composition-closure` (bridge lemma)	e_m (trajectory error at macro-boundaries m)	$\varepsilon^* \nu_c$ (closure-defect per macro-step × macro-update rate)	the composite's R_c	Tier-specific contraction stronger than (T2): incremental sector bound (DA2'-inc). Tier 1 proved; Tier 2 local; Tier 3 domain-specific
`#der-tempo-composition`	δ_c (composite mismatch)	$\rho_{\text{ext}} + \varepsilon^* \nu_c$ (external + internal)	R_c	(T3) requires the bridge-lemma contraction; $C_{\text{coord}} \geq \varepsilon^* \nu_c$ is the tempo-equivalent of the internal disturbance
`#der-adversarial-destabilization`	δ_B (target-agent mismatch)	$\rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model D) or $\sigma_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model S)	R_B	Destabilization is the negation of persistence: (T3)'s coupling-amplified disturbance violates $\alpha R > \rho_\xi$
`#der-identity-continuity-threshold`	g_k (identity gap, turnover-indexed)	ρ_k (projected static identity rate-distortion floor, per boundary)	D_Δ (fidelity-bounded identity-relevant information)	The reflected (Lindley) discrete instantiation on the turnover axis; compensation is relational re-grounding $\varrho_{\text{rg},k}$; Model-S-family, $\mu = 0$ driftless boundary load-bearing; conditional under (M-ADD)/(M-FREE)/(C-S) — the $\mathcal{A}_{\text{refl}}$ operator family (see Discussion)

The same Lyapunov argument applies in every row. What varies is how each instantiation defines its state variable, its correction function, and — most importantly — what counts as "effective disturbance" for its context (environmental noise, adversarial coupling, closure defect, or a decomposition of these).

External mathematical lineage: monotone-operator theory. The sector-Lyapunov apparatus this segment factors out has a well-established mathematical home: AAT's sector condition (T2) is a **one-point strong monotonicity** condition in the sense of Rockafellar 1970 (*Convex Analysis*, §24) anchored at the equilibrium $\xi^* = 0$, and the incremental strengthening DA2'-inc required by `#form-composition-closure`'s bridge lemma is **full two-point strong monotonicity** in the Bauschke-Combettes 2017 (*Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed., §§22–28) monotone-operator framework. The Lyapunov-plus-Grönwall argument proving the Model D ultimate bound and the Itô-Lyapunov argument proving Prop A.1S are standard specializations of the **monotone-operator perturbation theorem** (Bauschke-Combettes Thm 5.14–5.16; Parikh & Boyd 2014, *Proximal Algorithms*, §2). Banach-Picard contraction under bounded perturbation, monotone-operator convergence under square-summable noise, operator-splitting for composite systems (Douglas-Rachford, ADMM) — the full supporting apparatus is available in that literature.

AAT's relationship to monotone-operator theory is *specialization* + *repurposing*, not generalization. AAT-distinctive content: (i) **one-point anchoring at the equilibrium** — strictly weaker than full two-point strong monotonicity, matched to fixed-point-at-target semantics, admitting agent classes (PID-bounded-plant, variational-approximate) where full monotonicity fails but persistence-at-the-target is available. (ii) The **Model D / Model S disturbance decomposition** with distinct $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling — monotone-operator theory has perturbation theorems but no systematic bounded-adversarial vs stochastic-zero-mean split propagating to adversarial-exponent regimes at $b = 2$ vs $b = 3/2$.

(iii) Composition with the `#disc-identifiability-floor` meta-pattern supplies a second (information-theoretic) axis orthogonal to the operator-theoretic machinery. (iv) The `#disc-composition-consistency` cross-level commitment operates at AAT's level of description (agent / composite / macro-agent) rather than at the abstract operator level. (v) The sub-scope α/β epistemic labeling is scope-honesty, not a mathematical partition — it tracks which agent classes give the monotone-operator structure *by construction* versus *by per-instance verification*.

This acknowledgment is load-bearing for scope honesty: the mathematical machinery is established external to AAT; AAT's distinctive content is the agent-architecture specialization (singular-trajectory identity per `#scope-agent-identity`; signed-coupling composition; coordinate-forcing via uniqueness theorems per `#disc-additive-coordinate-forcing`; three meta-patterns) rather than novel monotone-operator mathematics. `#deriv-sector-condition`'s Grounding paragraphs name the specific operator-family correspondence (proximal / firmly-nonexpansive / cocoercive / strongly-monotone-gradient / linear-PD) for the five sub-scope-α agent classes. `#result-contraction-template` extends to non-Euclidean metrics via Lohmiller-Slotine differential-contraction (also within the broader monotone-operator lineage). The honest limits of the unification: the coarse-graining projection Λ (`#form-composition-closure`) does not fit the operator-sector primitive (heterogeneous spaces, three independent admissibility conditions); three of five metric-α_2 cases in `#result-contraction-template` remain theorem-imported rather than AAT-internally derived; the identifiability-floor axis is orthogonal to the operator-sector axis.

Comparison with the FEP-flow stability argument. Active inference's stability arguments come from the geometry of the variational free-energy landscape — agents are argued to flow toward the minimum of variational free energy on a non-equilibrium-steady-state (NESS) density. The primary source for the NESS-density framing is Friston 2019, "A free energy principle for a particular physics," arXiv:1906.10184; the path-integral / particular-kinds methodological extension is Friston, Da Costa, Sakthivadivel, Heins, Pavliotis, Ramstead & Parr 2023, "Path integrals, particular kinds, and strange things," *Phys. Life Rev.* 47 (which rewrites the FEP-flow argument in path-integral language rather than proving new stability bounds). Aguilera, Millidge, Tschantz & Buckley (2022, "How particular is the physics of the free energy principle?", *Phys. Life Rev.* 40:24–50) showed that the FEP-flow argument's mathematical validity is narrow: the NESS-density framing holds only in a small parameter regime for non-equilibrium linear stochastic systems, and natural extensions (nonlinear, non-Gaussian, non-equilibrium) often fall outside the proven regime.

The AAT persistence template is structurally different: it is a Lyapunov-based argument requiring only (T1) zero-correction-at-zero-state, (T2) local sector condition (correction points inward), and (T3) bounded disturbance — all of which are checked locally for each instantiation (A Props A.1, A.1S, A.2). The template applies to bounded and to mean-square-stochastic disturbance, gives explicit ultimate-bound and adaptive-reserve formulas, and does not depend on NESS structure or on a free-energy gradient.

The breadth difference is not rhetorical: where the FEP-flow argument's parameter regime is debated in the AI literature itself, the sector-Lyapunov apparatus is the standard machinery of nonlinear control theory (Khalil 2002, *Nonlinear Systems*, 3rd ed., Prentice Hall, ch. 4) and applies wherever (T1)–(T3) hold. This is one of AAT's stronger structural positions and is worth making explicit when comparing AAT to active inference: AAT does the persistence work AI tries to do, with broader validity and explicit ultimate-bound formulas.

Epistemic Status.

Exact. The template is the abstract form of the Lyapunov result proved in A (Props A.1, A.1S, A.2). The proofs transfer without modification whenever (T1)–(T3) hold; the template's contribution is the recognition that AAT's persistence-flavored results are instances of a single pattern and the enumeration of what each instantiation must verify.

On (T2) and A2' sub-scoping. (T2) is the local sector condition A2' transcribed to a generic state variable ξ. Within the state-variable spaces relevant to AAT's instantiations (epistemic mismatch δ, strategic mismatch δ_Σ, sub-agent mismatch δ_i, composite trajectory error e_m, composite mismatch δ_c, target-agent mismatch δ_B), the update rules are

overwhelmingly sub-scope α in the `#form-sector-condition` / `#der-gain-sector-bridge` sense — Bayesian, exponential-family, strongly-convex-gradient, or linear-PD. For these, (T2) is *derived* from the update-rule structure under B1 directional fidelity. **Important, and easy to miss: this is not what carries forward universally.** For sub-scope β instantiations (e.g. a team with a rule-based sub-agent) (T2) is an *empirical precondition, verified per-instantiation*, not derived. That distinction scopes which instantiations satisfy (T2) by which route; it does not touch the template's exactness, which is the conditional "(T1)–(T3) ⇒ persistence" — anything satisfying (T2) by either route inherits the exact result.

On Prop A.1S region-awareness. The Model S case uses the region-aware form of Prop A.1S: the killed second-moment bound (the scale of ξ while it is still inside $\mathcal{B}_R$), the finite-horizon path bound, and the exit-time comparison of Prop A.1S-τ (expected time in the region at least the OU benchmark's, exponential in $\alpha R^2/\sigma_\xi^2$). The stationary fixed-time tail $P(\|\xi(t)\| > R) \leq n\sigma_\xi^2/(2\alpha R^2)$ needs (T2) to hold globally; under (T2) on the ball alone the fixed-time tail is the composed form in A Prop A.1S(iii). Instantiations that rely on the Model S result inherit these forms with no extra work at the template level. As at A, Model S provides a distributional / finite-time guarantee, not pathwise-forever containment (which is a Model-D-only guarantee); template instantiations inherit that kind-of-guarantee distinction, not an infinite-horizon non-exit bound, and a stopped-moment bound with a decay term is not available to them.

Max attainable: *exact*. The result is as strong as Lyapunov stability theory. Additional work could extend the template (state-dependent noise, time-varying α, non-quadratic Lyapunov functions) but would not strengthen it within its stated scope.

What the template does not establish:

- *Quantitative convergence rates.* Lyapunov gives stability and ultimate bounds, not convergence speed. Specific rates require instantiation-level analysis.
- *Behavior outside the sector-condition region.* (T2) holds only for $\|\xi\| \leq R$. Beyond R, correction may break down — the structural-adaptation regime of `#result-structural-adaptation-necessity`.
- *Time-varying parameters.* The template assumes constant α and R. Time-varying cases require additional machinery. `#schema-strategy-persistence` documents the most important AAT example: $\alpha_\Sigma = 1/(n+1)$ decays with experience, so strategic persistence requires experience discounting at rate $(1-\lambda) > \rho_\Sigma/R_\Sigma$ to recover constant-α.
- *Heavy-tailed disturbances.* Neither (T3-D) nor (T3-S) covers disturbances with unbounded moments. Extreme tail events are better handled as triggers for structural adaptation than as disturbances the template can absorb.

Discussion.

Why factor this out. Every persistence-flavored result in AAT takes the form "correction rate exceeds effective disturbance rate (relative to reserve)" and is proved by the same Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$. Stating the argument once, parameter-free, makes visible that the theory is *one result about bounded-correction dynamics applied to several state spaces*, not seven separately-proved results that happen to look alike.

Each instantiation's distinctive content is now sharply visible: *what is the state variable, and what counts as its effective disturbance?* The Lyapunov machinery is shared; the characterization of ρ_ξ is where the domain-specific insight lives. Adversarial destabilization's content is the coupling term $\gamma_A \mathcal{J}_A$; team persistence's content is the cooperative-minus-adversarial decomposition; composition closure's content is the closure-defect rate $\varepsilon^* \nu_c$. In each case, the template absorbs the Lyapunov boilerplate and lets the distinctive claim stand without it.

This template is the *interior facet* of the stability certificate (`#disc-stability-certificate`): the "correction rate exceeds effective disturbance rate" condition is the certificate being positive-definite on the scope ball, and the existence of such

a certificate is equivalent to exponential stability (H). The shared Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$ is the certificate in the Euclidean metric; AK carries the non-Euclidean-metric interior.

The template is the typed-bridge shared upstream — two-model, not one operator. This template is the canonical anchor of the *accumulation typing* convention (`NOTATION.md` §"Accumulation typing"): the bridge from a per-step residue to its accumulated consequence *is* this conditional. The bridge is **two-model, not a single bounded operator** — (T3)'s Model D and Model S are co-equal, and the $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling (line above) plus the Cor-A.1S.1 categorical containment dichotomy (A) are the structural fingerprint that they are *different functionals*, not one functional read in two norms (a change of α-homogeneity *degree* cannot come from a norm choice on a fixed operator). The across-turnover axis is the worked demonstration that the split is structural, with two operator families at *opposite ends of the same singular contraction parameter*, **neither superseding the other**: the $\mathcal{A}_{\text{refl}}$ family — a reflected (Lindley/Loynes) walk whose load-bearing content is its driftless $\mu = 0$ boundary — *instantiates* the template on the turnover index (`#der-identity-continuity-threshold`, the new row above); the $\mathcal{A}_D$ family — the linear destroy-and-reconstruct contraction whose affine norm *diverges* exactly as the contraction gap closes — is the **honest scope boundary where the template provably does *not* transfer** (`#der-turnover-information-recursion`: an unforced multiplicative information decay has no correction function pointing a perturbed state inward, so (T1)–(T3) have no counterpart; the non-transfer is a definite result, not a caveat). The $\mu = 0$ boundary of the $\mathcal{A}_{\text{refl}}$ instantiation *is* the singularity of the $\mathcal{A}_D$ regime — you cannot linearize across that pole, which is why the two are distinct operators and the accumulation-type confound (treating one as a normalization or linearization of the other) is a category error. The typing convention exists precisely so this — *which model, which operator family* — is carried in the notation rather than silently reconstructed.

What each instantiation must still verify. Invoking the template is not trivial. Each instantiation must establish (T1)–(T3) for its specific (ξ, F, ρ_ξ, R), and the non-trivial verifications differ substantively:

- *Strategic persistence* (`#schema-strategy-persistence`): (T2) is verified for Beta-Bernoulli edge updates across five DAG topologies (S Props B.1–B.6). But $\alpha_\Sigma = 1/(n+1)$ is time-varying: it decays monotonically with experience. Constant-α — and therefore the template's trajectory guarantee — requires experience discounting as a prerequisite, not a heuristic.
- *Composition closure* (`#form-composition-closure`): (T2) as stated is insufficient. The bridge lemma requires the *incremental sector bound* (DA2'-inc) — strongly monotone F across the whole state space, strictly stronger than the one-point sector bound (T2). Three agent tiers result: Tier 1 where contraction is proved for the full class (Bayesian on exponential families, gradient descent on strongly convex losses, linear correctors with positive-definite gain), Tier 2 where it holds locally, Tier 3 where it must be verified per-domain.
- *Team persistence* (`#der-team-persistence`): (T3) is the decomposition $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{F}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{F}_j$. Cooperative coupling enters through a negative term, which can drive ρ_i^{eff} below the single-agent ρ — this is formally how teams persist where individuals cannot.

Persistence and destabilization as one result. Adversarial destabilization (`#der-adversarial-destabilization`) is the template applied with coupling-amplified disturbance: $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{J}_A$. The "destabilization threshold" — the condition under which agent A pushes agent B past its stability boundary — is precisely the *negation* of the template's persistence condition for B's instantiation. B's persistence certificate holds iff $\alpha_B R_B > \rho_B$; the certificate is lost — destabilization, with escape forced when B's sector bound is radially tight — iff $\rho_B \geq \alpha_B R_B$. These are the same inequality viewed in opposite directions, not independent results; both directions are certificate-level statements, and the agent-level only-if requires tightness (A Lemma A.1N). The superlinear adversarial scaling ($b = 2$ under Model D, $b = 3/2$ under Model S) follows from the template's $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling without further derivation.

Signed-coupling pattern across instantiations. The seven instantiations above — plus J, which derives a closed-form composite (T2) for the matched-symmetric-Tier-1 dyad — share a signed-coupling structure: each effective ρ_ξ is a sum of environmental disturbance plus a cross-agent contribution whose sign encodes cooperative vs adversarial coupling.

Team persistence, composition closure (via bridge-lemma), tempo composition, adversarial destabilization, and composite critical-mass are instances of one inequality at different state-variable levels. The template is where this pattern lives at the meta level; the per-segment instantiations supply the domain-specific (ξ, F, ρ_ξ, R) quadruples.

Cost complement: the template's information-rate floor. Alongside the threshold condition, each instantiation inherits an information-rate lower bound from I: where ξ is the estimation error of a Gaussian-OU signal observed through a channel, holding $\mathbb{E}[\|\xi\|^2]_{ss}$ at nD requires $\dot{R} \geq n(\sigma_\xi^2/(2D) - \lambda_\xi)$ nats/time, met with equality at $n\alpha/2$ by the optimal linear-Gaussian corrector. For the epistemic-mismatch instance with the Kalman-optimal corrector, this translates to the channel-capacity floor $C \geq \mathcal{T}/2$ — a first-class persistence prerequisite that the threshold condition alone does not surface. Each instantiation can derive its own cost bound by direct substitution into the main theorem of I; the consolidation of this into a parametric template-cost subsection is flagged in that segment's Working Notes.

The coordination overhead C_{coord} is effective disturbance at the composite level. The tempo-composition inequality $\mathcal{T}_c \leq \sum_i \mathcal{T}_i$ has a lower bound on the gap: $C_{\text{coord}} \geq \varepsilon^* \nu_c$ (`#der-tempo-composition`). This is the template's instantiation with $\xi = \delta_c$ and effective disturbance $\rho_{\text{ext}} + \varepsilon^* \nu_c$ — the composite's internal closure error acts as an additional disturbance at the macro level, absorbing correction capacity that would otherwise address the external environment. Brooks's Law follows as an instance: adding agents increases $\sum_i \mathcal{T}_i$ but may increase $\varepsilon^* \nu_c$ faster, pushing ρ_c^{eff} above $\alpha_c R_c$.

Relationship to `#result-persistence-condition`. `#result-persistence-condition` is the canonical single-agent instantiation. It carries additional content beyond the template — *task adequacy*, a domain-specific constraint $R^* < \|\delta_{\text{critical}}\|$ that further restricts the usable region — and it instantiates the template in the linear special case where $F = \mathcal{T}\delta, \alpha = \mathcal{T}$, and (T2) holds globally. Task adequacy is not part of the template because it is not part of the Lyapunov argument; it is an application-level constraint the agent's domain imposes.

Relationship to `#disc-composition-consistency`. The composition-consistency postulate requires that AAT's predictions be compatible across levels of description. The template is how this compatibility cashes out operationally: the same persistence argument applies at every level where a state variable with a sector-bounded correction function is present. Part III segments that invoke the template (team persistence, composition closure, tempo composition) are applying AAT's single persistence argument at the composite level, with effective disturbance decompositions that capture what is distinctive about the composite scope.

Relationship to AK (metric-formulation generalization). The Euclidean sector inequality (T2) is the $M = I$ specialization of a broader contraction-metric condition (CT2) under a Riemannian metric M (Lohmiller & Slotine 1998). For several AAT-relevant agent classes, the natural Lyapunov lives in a non-Euclidean metric — Fisher for statistical-manifold learning (matrix Kalman in information metric; exponential family in natural parameters), Hessian-induced for ill-conditioned strongly-convex optimization, Lyapunov-equation-determined for asymmetric-stable linear systems, or Lyapunov-metric for PID-with-bounded-plant-nonlinearity. AK states the generalization once with (CT1)–(CT3) preconditions matching (T1)–(T3), adds compositional theorems (parallel / cascade / negative-feedback with small-gain) that extend `#deriv-critical-mass-composition` (CM2) to heterogeneous sub-agents via (CM2-M), and fills `#disc-separability-pattern`'s seventh ladder (A2'-scope into metric-α_1 / metric-α_2 / metric-β). The Euclidean formulation stated in this segment remains the default for Euclidean-natural instances; AK is invoked when the natural coordinate is non-Euclidean. Structural consequence worth noting: the (CT2) condition at $M = I$ is equivalent to `#form-composition-closure`'s DA2'-inc (incremental sector bound), so AAT has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along; AK makes this explicit at the single-agent level.

Appendix H

Result: Certificate Existence — Operator-Sector in Some Metric Is Exponential Stability

The framework's central anchoring identity: **a stability certificate exists for an agent exactly when the agent is exponentially stable about its target**, at the linearized level. The organizing slogan — *an adaptive system is an operator whose contraction rate exceeds its target's drift rate* — is *not a heuristic* but this equivalence (the standard Lyapunov theorem), with the certificate $\mathcal{M}$ as its witness.

A **stability certificate** is a symmetric positive-definite metric $\mathcal{M}$ for which AAT's one-point sector condition holds in the $\mathcal{M}$-inner-product on a ball around the equilibrium. At the linearized level, the existence of such a certificate is *equivalent* to the system matrix being Hurwitz (exponentially stable) by the Lyapunov equation — three clauses, all the same statement: $A = -J$ Hurwitz; some $(\mathcal{M}, \kappa)$ with $\mathcal{M}J + J^\top \mathcal{M} \succeq 2\kappa\mathcal{M}$; for every $Q \succ 0$ a unique $\mathcal{M}$ solving $A^\top \mathcal{M} + \mathcal{M}A = -Q$. The certificate is *not unique*: it is whatever positive-definite metric makes the dynamics contract. The result holds at the linearized level (with a standard second-order remainder condition for the local nonlinear extension); the global statement requires the incremental rung R1 (cocoercive class) — a genuine scope statement, named here so the parenthetical "linearized/local" is not buried.

The result *specializes* in different sub-cases of the framework to four familiar metrics: Fisher information for Bayesian agents (Čencov-forced under R2); inverse-prior-covariance for Kalman agents; loss Hessian for gradient agents; plant-selected Lyapunov metric for linear-Hurwitz or PID agents. *These are not four separate stories — they are one object under four certificates*. The unification is structural: the same Lyapunov equivalence underlies all four, with the specialization determined by the agent class. The framework treats this segment as the **spine** of three meta-patterns: separability (`#disc-separability-pattern` — *scope-of-existence* of a certificate), additive coordinate forcing (`#disc-additive-coordinate-forcing` — when the certificate is *forced* to a unique form), and identifiability floors (`#disc-identifiability-floor` — where the certificate's *boundary* lives via the Sylvester mechanism). Naming $\mathcal{M}$ as the object — rather than $V(e) = e^\top \mathcal{M} e$, the Lyapunov function — is what makes those downstream statements expressible.

Formal Expression.

Definition (stability-certificate)

The object.

For an agent with error dynamics $\dot{e} = -F(e)$ about an equilibrium e^* ($F(e^*) = 0$, $F \in C^1$ near e^*, Jacobian $J := DF(e^*)$), a **stability certificate** is a symmetric positive-definite $\mathcal{M}$ for which the one-point sector condition holds in the $\mathcal{M}$-inner-product on a ball $\mathcal{B}_R(e^*)$:

$$\langle F(e),\, e - e^* \rangle_{\mathcal{M}} \;\geq\; \kappa \, \|e - e^*\|^2_{\mathcal{M}}, \qquad \kappa > 0. \tag{C}$$

The certificate is not unique: it is whatever positive-definite form makes the dynamics contract. In the recurring subcases it specializes — to the Fisher information for Bayesian agents, to $(P^-)^{-1}$ for Kalman agents, to the loss Hessian for gradient agents, and to a plant-selected Lyapunov metric for linear-Hurwitz or PID agents. These are not four separate stories; they are one object under four certificates.

Result (certificate-existence), exact at the linearized level

The equivalence.

At the linearized level (C) reads $\mathcal{M}J + J^\top \mathcal{M} \geq 2\kappa \mathcal{M} \succ 0$, a strict Lyapunov inequality for the system matrix $A = -J$. The following are equivalent:

1. $A = -J$ is Hurwitz — the linearized error dynamics $\dot{e} = -Je$ is exponentially stable;
2. there exist $\mathcal{M} \succ 0$ and $\kappa > 0$ with $\mathcal{M}J + J^\top \mathcal{M} \geq 2\kappa \mathcal{M}$ — a one-point sector condition (a stability certificate) in *some* inner product;
3. for every $Q \succ 0$ there exists a unique $\mathcal{M} \succ 0$ with $\mathcal{M}J + J^\top \mathcal{M} = Q$.

So "operator-sector in *some* inner product" and "the equilibrium is exponentially stable" are **the same statement**, with the certificate $\mathcal{M}$ as the converse-Lyapunov witness — an equivalence, not an analogy.

Derived (ordering of conditions; exact)

The certificate-strength ladder.

The certificate admits four strictly-ordered strengths, all on the one object:

Table H.1

Rung	*Condition*	*Equivalent to*	*Certificate is*
R0-loss	some $\mathcal{M} \succ 0$, $\mathcal{M}J + J^\top \mathcal{M} \geq 0$, $\sup \kappa = 0$	$A = -J$ has only nonpositive-real-part eigenvalues, at least one on the imaginary axis (semisimple)	conserved on the chain-recurrent subspace + strict-Lyapunov on its strongly-gradient-like complement
R0-strict	one-point (C), some $\mathcal{M}$ with $\kappa > 0$, local	$A = -J$ Hurwitz + remainder dominated	converse-Lyapunov $\mathcal{M}$ (exists; generally not forced)
R1	incremental (two-point) $\mathcal{M}$-strong-monotonicity on $\mathcal{B}_R$	global $\mathcal{M}$-strong-monotone (cocoercive class)	curvature-like $\mathcal{M}$ (potential sub-case)
R2	R1 with $\mathcal{M}$ *forced* by a uniqueness theorem on an AAT-internal axiom	natural-gradient in the Čencov-unique Fisher metric	Fisher metric (Čencov-forced)

R0-loss $\Leftarrow$ R0-strict $\Leftarrow$ R1 $\Leftarrow$ R2 strictly. R0-loss is the *widest* rung — a certificate exists but $\sup \kappa = 0$: the dynamics is Lyapunov-stable, not asymptotically stable, with $V(e) = e^\top \mathcal{M} e$ conserved on an imaginary-axis-eigenvector subspace and strict contraction on its strongly-gradient-like complement (Conley's universal decomposition; pure-case

Helmholtz S/A characterization derived below). R0-strict is the contraction rung with $\kappa > 0$ strict, reaching the plant-Lyapunov cases (linear-Hurwitz-non-symmetric, PID) where no potential exists. R1 is the cocoercive/proximal class where a variational structure is available; R2 is the uniqueness-theorem-forced statistical case. The widest rung is not a weakness: it is exactly the reach the narrower rungs cannot give. (R0-loss's pure-case characterization and Conley anchoring are in the Derivation below; R2's forcing is established in `#disc-additive-coordinate-forcing`; R1's cocoercive class in AK.)

Derivation.

(1 ⟹ 3, 2). $A = -J$ Hurwitz. By the Lyapunov theorem (Lyapunov 1892; Khalil, *Nonlinear Systems* 3rd ed., Thm 4.6), for every $Q \succ 0$ the equation $A^\top \mathcal{M} + \mathcal{M} A = -Q$ has a unique solution $\mathcal{M} \succ 0$. Substituting $A = -J$ gives $\mathcal{M} J + J^\top \mathcal{M} = Q \succ 0$ (clause 3). For the rate: $Q \succeq \lambda_{\min}(Q)\, I \succeq \dfrac{\lambda_{\min}(Q)}{\lambda_{\max}(\mathcal{M})}\, \mathcal{M}$, so (C) holds at the linear level with $\kappa = \dfrac{\lambda_{\min}(Q)}{2\,\lambda_{\max}(\mathcal{M})} > 0$ (clause 2).

(2 ⟹ 1). Suppose $\mathcal{M} \succ 0, \kappa > 0, \mathcal{M} J + J^\top \mathcal{M} \succeq 2\kappa \mathcal{M}$. Take $V(e) = e^\top \mathcal{M} e$ (valid Lyapunov candidate, $\mathcal{M} \succ 0$). Along $\dot{e} = -Je$: $\dot{V} = -e^\top (\mathcal{M} J + J^\top \mathcal{M}) e \le -2\kappa\, e^\top \mathcal{M} e = -2\kappa V < 0$ for $e \neq 0$. Hence $\|e(t)\|_{\mathcal{M}} \le e^{-\kappa t} \|e(0)\|_{\mathcal{M}}$ and $A = -J$ is Hurwitz.

(3 ⟺ 2) is the rate-extraction argument (3 gives 2 with the displayed κ; 2 is the special case $Q = \mathcal{M} J + J^\top \mathcal{M}$).

Local nonlinear extension. With $F(e) = Je + r(e)$, $\|r(e)\| = o(\|e\|)$, (C) holds on $\mathcal{B}_R(e^*)$ iff the linearization is exponentially stable *and* the second-order remainder is dominated on the ball ($\|r(e)\|_{\mathcal{M}} \le c\|e\|_{\mathcal{M}}^2$ with $cR < \kappa$). This is the standard Lyapunov indirect method (Khalil Thm 4.7); the remainder-domination radius is exactly the contraction-template Tier-2 degradation radius (AK), not new machinery.

R0-loss: certificate without contraction.

Derived (R0-loss-distinctness; exact at the linearized level for the pure case, exact-via-Conley for the mixed case)

When $A = -J$ is non-Hurwitz but admits *some* $\mathcal{M} \succ 0$ with $\mathcal{M} J + J^\top \mathcal{M} \succeq 0$ at $\sup \kappa = 0$, the Lyapunov inequality holds non-strictly and the dynamics is Lyapunov-stable but not asymptotically stable. The complete spectral classification of A partitions the linear-system space into four cases:

- **(i)** A **Hurwitz** — R0-strict (above).
- **(ii)** A **has an eigenvalue with** $\mathrm{Re}\lambda > 0$ — no certificate exists: a $\mathcal{M} \succ 0$ satisfying $\mathcal{M} J + J^\top \mathcal{M} \succeq 0$ would force $\dot{V} \le 0$ along all trajectories, contradicting the unstable mode. *Outside the ladder.*
- **(iii)** A **has all eigenvalues with** $\mathrm{Re}\lambda \le 0$**, at least one on the imaginary axis, imaginary-axis eigenvalues semisimple** — R0-loss. A certificate $\mathcal{M} \succ 0$ exists; trajectories along the imaginary-axis-eigenvector subspace orbit perpetually (conserving $V = e^\top \mathcal{M} e$ on each level set), and trajectories on the strict-stable complement decay exponentially toward that subspace.
- **(iv) defective Jordan block at an imaginary eigenvalue** — no certificate exists: polynomial growth on the generalized eigenvector ($e \sim t \cdot v$) violates $\dot{V} \le 0$ for any quadratic V. *Outside the ladder.*

R0-loss occupies the strict slice (iii), disjoint from R0-strict (i) and from the no-certificate regimes (ii), (iv). The four-case partition is exhaustive (every real matrix sits in exactly one) and mutually exclusive. R0-loss is *not* "R0-strict with κ tending to zero" — R0-strict's $\kappa > 0$ is strict by definition; the limit $\kappa \to 0^+$ does not land in R0-loss but on its closure boundary. R0-loss is also distinct from the identifiability-floor boundary of $\mathbb{S}^n_{\succeq 0}$ (`#disc-identifiability-floor`'s Sylvester mechanism): the floor has $\mathcal{M}$ dropping rank, while R0-loss has $\mathcal{M}$ remaining $\succ 0$ with the rank drop on the *Hermitian part* $\mathcal{M} J + J^\top \mathcal{M}$ instead.

Pure case — Helmholtz S/A as canonical sufficient route. Taking $\mathcal{M} = I$, $S := \frac{1}{2}(J + J^\top)$, $A_{\text{anti}} := \frac{1}{2}(J - J^\top)$. The conditions

(ii) $S \equiv 0$ (J is purely antisymmetric);

(iii) $V(e) = \frac{1}{2}\|e\|^2$ is exactly conserved along $\dot{e} = -Je$ for all initial e

are equivalent — (ii $\Longrightarrow$ iii) by direct calculation $\dot{V} = -e^\top Se \equiv 0$ when $S = 0$, and (iii $\Longrightarrow$ ii) by polarization for symmetric S. Either implies case (iii) of the spectral partition above: real antisymmetric J has purely-imaginary spectrum (its characteristic polynomial is even, forcing eigenvalues in $\pm i\omega$ pairs) and is diagonalizable over $\mathbb{C}$ (hence semisimple).

The converse — case (iii) implying antisymmetry at $\mathcal{M} = I$ — does not hold. A non-antisymmetric J can sit in case (iii) too: $J = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}$ has $S = \text{diag}(1, -1)$ at $\mathcal{M} = I$ (so $\mathcal{M}J + J^\top\mathcal{M} = 2S$ is indefinite, failing the certificate condition there) but eigenvalues $\pm i\sqrt{3}$ (pure imaginary, semisimple). For such a J the R0-loss certificate exists by the spectral partition but requires $\mathcal{M} \neq I$; finding the explicit metric (via similarity to real Jordan form, or the center-manifold construction below) is what the partition guarantees but does not provide constructively. **The full characterization of R0-loss is the four-case spectral partition above; the Helmholtz S/A route at $\mathcal{M} = I$ is the canonical *sufficient* construction for the antisymmetric sub-class — the structural framing for the Letcher 2019 Hamiltonian-game vocabulary — not a necessary condition for R0-loss generally.** Bendixson's localization $\text{Re}(\lambda(-J)) \in [-\lambda_{\max}(S), -\lambda_{\min}(S)]$ gives a *containment* interval that here is $[-1, 1]$ — wide enough to permit purely-imaginary eigenvalues without forcing $S = 0$.

Letcher-Balduzzi et al. 2019 Lemma 1 (uniqueness of the $S + A$ decomposition) and Definition 2 (Hamiltonian game = $S \equiv 0$) name this sufficient route in game-mechanics vocabulary; AAT's content is the recognition that the antisymmetric sub-class is the cleanest specific construction, with the R0-loss regime itself broader. The general-$\mathcal{M}$ extension replaces S with $S_{\mathcal{M}} := \frac{1}{2}\mathcal{M}^{-1}(\mathcal{M}J + J^\top\mathcal{M})$; an $\mathcal{M}$ rendering $S_{\mathcal{M}} \succeq 0$ exists for every case-(iii) J by the partition.

Mixed case — Conley anchoring. At a chosen certificate $\mathcal{M}$ rendering $S_{\mathcal{M}} \succeq 0$ with nontrivial kernel — some imaginary-axis eigenvalues plus some strict-stable ones — the dynamics splits canonically: along $\ker(S_{\mathcal{M}})$, V is conserved (the chain-recurrent component); along $S_{\mathcal{M}}$'s positive-eigenspace, V decays strictly (the strongly-gradient-like component). This is Conley 1978's universal decomposition (Fundamental Theorem of Dynamical Systems, §8.1): every flow on a compact invariant set decomposes uniquely into a chain-recurrent subflow plus a strongly-gradient-like quotient. AAT's R0-loss localizes this universality at the linearization. On the chain-recurrent component (the restricted operator is $\mathcal{M}$-antisymmetric on $\ker(S_{\mathcal{M}})$ and hence trace-free in the $\mathcal{M}$-orthonormal basis), V restricted to each level set $\{V = c\}$ is volume-preserving by Liouville, hence chain-recurrent in Conley's sense (Poincaré recurrence). The explicit construction of $\mathcal{M}$ in the mixed case follows from center-manifold machinery (Carr 1981 §2); the qualitative shape — chain-recurrent subspace + strongly-gradient-like complement — is universal, while the explicit metric is standard but not derived here.

Epistemic Status.

Exact at the linearized level (the equivalence is the standard Lyapunov theorem, recognized and applied — not a fresh derivation), and *exact with the standard second-order remainder condition* locally. The constituent theorem is classical; what is contributed here is the recognition that AAT's one-point sector condition (A2'/(T2)) in a free choice of inner product is *exactly* the converse-Lyapunov certificate, which makes the framework's organizing slogan a theorem rather than a heuristic.

Scope honesty. Linearized/local — this is the level at which AAT's persistence results already operate (sector conditions, contraction templates, the bridge lemma all linearize about the equilibrium), so it is not a weakening relative to the rest of the theory, but it is a genuine scope statement and is not papered over: there is **no** claim that one-point operator-sector is equivalent to *global* exponential stability (it is not, in general — the global statement requires the

incremental rung R1, the cocoercive class). The disturbance-to-ultimate-bound step that turns "contracts" into "stays within a bounded ball under disturbance" is proved separately in A and G; this result supplies the contraction half of the slogan, those supply the exceeds-the-drift half.

R0-loss scope. The R0-loss rung's pure case (Helmholtz $S \equiv 0$ characterization; all eigenvalues on the imaginary axis, semisimple) is *exact* at the linearized level — the equivalence $S \equiv 0 \Leftrightarrow$ full-state marginal stability at $\kappa = 0$ follows from polarization plus Bendixson's localization. The mixed case (chain-recurrent subspace + strongly-gradient-like complement) is *exact-via-Conley* — Conley 1978 §8.1's universal decomposition supplies the qualitative shape; the explicit $\mathcal{M}$ construction in mixed cases is center-manifold-level work (Carr 1981 §2) sketched above rather than derived. The non-Hurwitz scope question — whether nonlinear systems with Case-(iii) linearization are actually Lyapunov-stable rather than displaying quartic-or-higher instabilities on the center manifold — is the standard nonlinear-stability open problem (Khalil §4.3), not within the linearized claim made here.

Max attainable: *exact* (it is the Lyapunov theorem, in scope; R0-loss extends via Letcher 2019 / Conley 1978, recognized and applied — not fresh derivations). Novelty posture is *recognition*, not new mathematics.

Discussion.

This is the segment-level home of the contraction-over-drift organizing principle. The slogan *an adaptive system is an operator whose contraction rate exceeds its target's drift rate* had, until now, no segment to point to — it was carried as a framing heuristic. The equivalence above is its formal content: the "contraction" half is clause 2 $\Leftrightarrow$ clause 1 (a certificate exists iff the dynamics is exponentially stable); the "exceeds the drift" half is the ultimate-bound argument of G, which takes the certificate's κ and the disturbance rate and returns the bounded-error guarantee. A reader can now cite the principle to a result, not to a slogan.

Why the equivalence matters beyond tidiness. It is what licenses the cross-sectional reading in `#disc-stability-certificate`: because "a certificate exists" is the same statement as "the agent can keep up," every structural question about AAT's reach becomes a question about this one object — does a certificate exist here (scope), is it forced to a unique form (forced identity), where does it degenerate (the boundary), does it survive coarse-graining (composition). Without the equivalence those would be four loosely-related observations; with it they are four questions about one mathematical object.

The certificate is not the Lyapunov function. $V(e) = e^{\top}\mathcal{M}e$ is the Lyapunov function; $\mathcal{M}$ is the certificate. The distinction is load-bearing downstream: the framework's representational freedom (choice of coordinate / metric) acts on $\mathcal{M}$, and what that freedom can and cannot do to $\mathcal{M}$ — change its conditioning but not its inertia — is the content of the boundary facet (`#disc-identifiability-floor`, the Sylvester mechanism). Naming $\mathcal{M}$ as the object, rather than V, is what makes that downstream statement expressible.

R0-loss as the linearization-fingerprint of saddle-Nash composite regimes. Pure R0-loss at the single-agent linearization level is the spectral fingerprint of `#disc-dynamic-regime-axis`'s **R2 cyclic-distributional-regime** at the joint Jacobian linearized about saddle-only Nash equilibria: the joint best-response field's symmetric part vanishes there (no potential structure), and the antisymmetric part supplies the imaginary-axis spectrum. FTRL × graphical-constant-sum games with fully-mixed Nash (Cheung-Piliouras-Tao 2021 Theorem 19: lossless DGS + Poincaré recurrence on bounded level sets) is the worked composite instance. The R-letter overlap with the dynamic-regime axis is layer-honest, not collision: this segment's R0-loss/R0-strict/R1/R2 lives at the single-agent linearized error space $\mathbb{R}^n$ (the agent's own certificate over its own error coordinate); the dynamic-regime axis's R0/R1/R2/R3 lives at the composite joint-strategy space $\mathcal{X}^c$ (the joint-dynamics fixed-point structure of a composite of sub-agents). Same letters, different objects, different layers — composing across layers is exactly what the bridge sentence above does, and is the structural content of the layering rather than a coincidence to be hidden.

Findings.

The Contraction-Over-Drift Principle, Grounded. **Brief:** Picture an agent trying to stay on a target that keeps drifting. Ask one question: is there a way of measuring "how far off am I" — a measuring-stick — such that *every* correction the agent makes provably shrinks that measure? This result says: such a measuring-stick exists exactly when the agent is, in the ordinary control-theory sense, exponentially stable. The two are not analogous, they are the *same fact* (the Lyapunov theorem) — which is why the long-standing slogan "an adaptive system is one whose contraction outpaces its target's drift" is a theorem and not a vibe. A thoughtful non-specialist can carry the whole thing away from the picture: *the measuring-stick exists = the agent can keep up*; everything else AAT says about reach, blind spots, and composition is downstream questions about that one stick.

Impact: Discharges the long-standing "organizing slogan not yet surfaced at segment level" status by giving the contraction-over-drift principle an exact, citable home. Supplies the anchor that licenses the certificate-cone cross-sectional reading (`#disc-stability-certificate`): the equivalence is what turns "four loosely-related structural observations" into "four questions about one object." Cleanly separates the exact mathematical core (this result) from the discussion-grade organizing recognition (the spine), so each is housed at its true epistemic tier rather than the exact equivalence being buried as a derived-tag inside a discussion-grade segment. Names the certificate (not the Lyapunov function) as the object, which is the precondition for the boundary-facet's Sylvester statement being expressible downstream.

Novelty Claim: *Claim recognition* that AAT's one-point sector condition under a free choice of inner product is exactly the converse-Lyapunov certificate, making the framework's contraction-over-drift organizing principle the Lyapunov-theorem equivalence rather than a heuristic. The theorem is classical (Lyapunov 1892); the contribution is the identification that the framework's A2'/(T2) machinery *is* this object, and the consequent segment-level grounding of the organizing slogan.

Related Work: - Lyapunov, A. M. (1892), *The General Problem of the Stability of Motion* (Engl. transl. 1992, Taylor & Francis); Khalil, H. K. (2002), *Nonlinear Systems* 3rd ed., Thm 4.6 (Lyapunov equation), Thm 4.7 (indirect method) (found 2026-05-14) — *formal antecedent* — the equivalence and the local extension are this theorem applied; the contribution is the recognition, not the proof. - Conley, C. (1978), *Isolated Invariant Sets and the Morse Index*, CBMS Regional Conf. Ser. Math. 38, AMS — Fundamental Theorem of Dynamical Systems, §8.1 (found 2026-05-22) — *formal antecedent* — the universal decomposition of any flow on a compact invariant set into a chain-recurrent subflow plus a strongly-gradient-like quotient. R0-loss's mixed case is this universality localized at the linearization. The constituent theorem is classical; the contribution is recognizing that the R0-strict / R0-loss distinction is exactly Conley's two pieces of the universal decomposition. - Letcher, A., Balduzzi, D., Racanière, S., Martens, J., Foerster, J., Tuyls, K. & Graepel, T. (2019), "Differentiable Game Mechanics," *JMLR* 20:1–40 (found 2026-05-22) — *formal antecedent* — Lemma 1 (uniqueness of the symmetric+antisymmetric decomposition $J = S + A$) and Definition 2 (Hamiltonian game = J purely antisymmetric) supply the pure-R0-loss characterization. Letcher's "Hamiltonian game" is the *fully* R0-loss agent in AAT's linearization vocabulary; "potential game" is the strict sub-case of R0-strict where $J = J^{\top}$. - Cheung, Y. K., Piliouras, G. & Tao, Y. (2021), "Online Optimization in Games via Control Theory: Connecting Regret, Passivity and Poincaré Recurrence," *arXiv:2106.04748* (found 2026-05-22) — *formal antecedent (worked composite instance)* — Theorem 19 (FTRL × graphical-constant-sum-with-fully-mixed-Nash is lossless DGS, with Poincaré recurrence on bounded level sets) provides the worked composite instance of R0-loss at the joint Jacobian's saddle-only-Nash linearization. The qualitative bridge "finitely-lossless storage ⟹ R0-loss linearization at any accumulation equilibrium" survives; the quantitative bridge (L as V) does not (different state spaces, different functional forms; see also `#def-control-regret` Discussion on the qualitative-only diagnostic). - A, G (adjacent) — supply the disturbance-to-ultimate-bound half of the slogan; this result supplies the contraction half.

Search Log: - 2026-05-14 (*targeted*): The equivalence is the textbook Lyapunov theorem; no search needed for the theorem itself. The search target was whether the *recognition* "AAT's one-point sector condition in a free inner product = the converse-Lyapunov certificate, hence the contraction-over-drift slogan is this equivalence" appears articulated elsewhere as a framework-grounding move. Not found at this depth; expected to remain *recognition*-tier (the theorem is universal; the framework-grounding identification is the contribution). - 2026-05-22 (*absorbed*): the constituent prior art for the R0-loss rung extension (Conley 1978; Letcher 2019; Cheung-Piliouras-Tao 2021) was verified in its own domain; no separate Pillar / Undermind search was conducted for the rung-extension itself — the rung is the

AAT-internal recognition that these textbook results jointly classify a fourth ladder rung disjoint from R0-strict and from the Sylvester-floor regime.

Appendix I

Derivation: Persistence Cost — Information Rate to Maintain Bounded Mismatch

AAT's persistence machinery establishes that under the sector condition, the estimation error stays bounded. It does not quantify the *sustained rate of effort* an agent must expend to hold that bound. Two agents with identical persistence guarantees can face very different demands — a Kalman filter tracking a slowly drifting process and one tracking a fast one are both persistent; one is nearly dormant, the other running hot. Under Model S with a Gaussian Ornstein–Uhlenbeck signal, any causal estimator that holds the per-component mean-square estimation error at D must take in information about the signal at a rate of at least $n(\sigma_{\text{env}}^2/(2D) - \lambda_s)$ nats per unit time. That is the sequential (causal) rate-distortion function of the signal. The Kalman–Bucy filter meets it with equality, and its rate is $nK/2$, half its own correction gain per component. For the Kalman-optimal agent the price of persistence is therefore $n\alpha/2$ nats per unit time, and the observation channel must have capacity at least $\mathcal{I}/2$ per dimension. Every linear corrector on the same channel takes in that same rate, $nK/2$, whatever its gain; a mis-tuned one pays it for a larger error, and the bound evaluated at its error is slack.

The bound is *filter-agnostic*: it holds for any causal estimator under the stated scope, and it is stated in the error the agent actually achieves. It is also *substrate-agnostic*: Landauer's principle converts it to a thermodynamic dissipation floor in any physical realization (Still et al. 2012). The channel-capacity floor matters in three regimes where observation bandwidth is scarce: biological systems with finite neural channel capacity, bandwidth-constrained distributed systems over noisy or low-bandwidth links, and context-window-limited language models where effective information rate per unit of cognition is bounded by context size and token throughput. Survival, in this framework, is not a state achieved once — it is a *sustained burn rate* of information, and the channel that supports it is part of what keeps the agent in scope.

Formal Expression.

Setup. The signal is n-dimensional with independent Ornstein–Uhlenbeck components, $dx_i = -\lambda_s x_i\,dt + \sigma_{\text{env}}\,dW_i$, observed through $dy_i = x_i\,dt + \sigma_\nu\,dV_i$. Here σ_{env} is the signal's own diffusion (environmental change) and σ_ν the channel-noise intensity. The agent's estimate $\hat{x}$ is causal: it depends only on observations up to the present. The persistence state is the estimation error $\delta = x - \hat{x}$ (`#def-mismatch-signal`), with per-component steady-state mean square D. Under Prop A.1S (A) its value for a linear corrector with gain α includes the channel noise the correction feeds back: $D = (\sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2)/(2(\lambda_s + \alpha))$ (A, Setup). The error contracts at rate $\lambda_s + \alpha$, the signal's own decay plus the gain; for the random-walk signal $\lambda_s = 0$ this is the sector rate of `#der-gain-sector-bridge`, and for $\lambda_s > 0$ the gain is the part of the contraction the agent supplies. Here σ_{env} and σ_ν are the environmental and channel-noise intensities of A's Setup, whose total disturbance $\sigma_w^2 = \sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2$ is not used in this segment.

Derived (persistence-information-rate, from the sequential rate-distortion function)

Persistence Information Rate (main theorem).

Proposition. Any causal estimator that holds the per-component steady-state mean-square estimation error at $D \leq \sigma_{\text{env}}^2/(2\lambda_s)$ must take in information about the signal at a sustained rate

$$\boxed{\dot{R} \;\geq\; n\left(\frac{\sigma_{\text{env}}^2}{2D} - \lambda_s\right) \text{ nats per unit time}} \tag{I.1}$$

Derivation. For a scalar Gauss–Markov source $x_{k+1} = ax_k + w_k$, $w_k \sim \mathcal{N}(0, W)$, the sequential rate-distortion function — the least directed information from source to reproduction per step for any causal scheme meeting mean-square error D at each step — is $\frac{1}{2}\log(a^2 + W/D)$ when that is positive, and is attained by a linear-Gaussian sensor followed by a Kalman filter (Tanaka, Kim, Parrilo & Mitter 2017, "Semidefinite programming approach to Gaussian sequential rate-distortion trade-offs," *IEEE Trans. Autom. Control* 62(4):1896–1910, doi:10.1109/TAC.2016.2601148, eq. (28); the sequential rate-distortion function is Tatikonda's, Tatikonda, Sahai & Mitter 2004, *IEEE Trans. Autom. Control* 49(9):1549–1561). Sample the OU component at step Δ: $a = e^{-\lambda_s\Delta}$ and $W = \sigma_{\text{env}}^2(1 - e^{-2\lambda_s\Delta})/(2\lambda_s)$, so $a^2 + W/D = 1 - 2\lambda_s\Delta + \sigma_{\text{env}}^2\Delta/D + O(\Delta^2)$. The rate per step is $\frac{1}{2}(\sigma_{\text{env}}^2/D - 2\lambda_s)\Delta + O(\Delta^2)$, and per unit time, as $\Delta \to 0$, $\sigma_{\text{env}}^2/(2D) - \lambda_s$. Independent components add. □

The continuous-time statement is this limit of the exact discrete-time result; the Kalman–Bucy closed form below meets it exactly, which is an independent check. A non-causal reproduction (one that may look ahead) is held only to the classical rate-distortion function, which for OU at high resolution is $2\sigma_{\text{env}}^2/(\pi^2 D)$ by reverse water-filling (Berger 1970, "Information rates of Wiener processes," *IEEE Trans. Inf. Theory* 16(2):134–139, for the Wiener process). That bound is lower and is not met by any filter. An agent acts in real time, so the sequential function is the one that binds.

Derived (Kalman-tight, Duncan 1970; Mitter & Newton 2005)

Kalman–Bucy Meets the Bound.

By Duncan's theorem (Duncan 1970, "On the calculation of mutual information," *SIAM J. Appl. Math.* 19(1):215–220), the rate at which the observations supply information about the signal to the optimal filter is $\dot{I} = \frac{1}{2}\,\text{tr}(H^T\Sigma_o^{-1}HP_{ss})$; per component, $\dot{I} = P/(2\sigma_\nu^2)$ (the same identity is the information-supply rate of Mitter & Newton 2005, *J. Stat. Phys.* 118:145–176). The steady-state Riccati equation $\sigma_{\text{env}}^2 - 2\lambda_s P - P^2/\sigma_\nu^2 = 0$ gives

$$\frac{P}{2\sigma_\nu^2} = \frac{\sigma_{\text{env}}^2}{2P} - \lambda_s \tag{I.2}$$

So the Kalman–Bucy filter, at its own error $D = P$, takes in information at exactly the sequential rate-distortion bound, for every λ_s. **The bound is achieved.** In terms of its correction gain $K = P/\sigma_\nu^2$ (the error contracts at $\lambda_s + K$; for $\lambda_s = 0$ the gain is the sector constant of `#der-gain-sector-bridge`, scalar Kalman), its rate is

$$\dot{I} = \frac{K}{2} = \frac{\alpha}{2} \text{ per component} \tag{I.3}$$

In the drift-dominated limit $\lambda_s \to 0$, $P = \sigma_{\text{env}}\sigma_\nu$ and $K = \sigma_{\text{env}}/\sigma_\nu$.

The intake is fixed by the channel; the error decides how much of it is used. For a linear corrector $d\hat{x} = -(\lambda_s + \alpha)\hat{x}\,dt + \alpha\,dy$ the path of $\hat{x}$ determines the path of y ($dy = (d\hat{x} + (\lambda_s + \alpha)\hat{x}\,dt)/\alpha$), so $I(x;\hat{x}) = I(x;y)$, and by Duncan's theorem the observation path carries information about the signal at the rate $P/(2\sigma_\nu^2) = K/2$ per component, where P is the *optimal* filter's error, whatever gain the filter actually runs. Every linear corrector on the channel takes in $nK/2$. A mis-tuned one holds a larger error: for $\lambda_s \to 0$, $D(\alpha) = (\sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2)/(2\alpha) > P = \sigma_{\text{env}}\sigma_\nu$ for every $\alpha \neq K$, on either side of K. The bound evaluated at that error, $n\sigma_{\text{env}}^2/(2D(\alpha)) = n\alpha\sigma_{\text{env}}^2/(\sigma_{\text{env}}^2 + \alpha^2\sigma_\nu^2)$, is then strictly below

$nK/2$ and is slack: it is the least any causal estimator would need for that error, not what this one pays. So $n\alpha/2$ is the price *and* the intake only for the Kalman-optimal agent; for any other gain the intake is $nK/2$ and the floor at its error is lower. Holding a target D with a gain $\alpha \neq K$ requires a more informative channel than the floor at D demands, which is the efficiency cost `#impl-strategy-dynamics` names. In the noiseless-channel limit ($\sigma_\nu \to 0$), holding $D = \sigma^2_{\text{env}}/(2\alpha)$ requires $n\alpha$.

Channel-Capacity Prerequisite.

Derived (channel-capacity-floor, from main theorem + Shannon capacity)

By Shannon's channel coding theorem (Cover & Thomas 2006 §7.7), an observation channel of capacity C_{channel} supports no information rate above C_{channel}. Holding the estimation error at D therefore requires

$$C_{\text{channel}} \;\geq\; n\left(\frac{\sigma^2_{\text{env}}}{2D} - \lambda_s\right) \tag{I.4}$$

For the Kalman-optimal agent, with linear correction and $\alpha = \mathcal{T}$ (`#def-adaptive-tempo`, `#der-gain-sector-bridge`):

$$\boxed{C_{\text{channel}} \;\geq\; \mathcal{T}/2 \text{ nats/time per dimension}} \tag{I.5}$$

Persistence demands observation-channel capacity. For the optimal agent the floor is half the adaptive tempo. In general it is fixed by the error the agent must hold. This is a first-class persistence diagnostic beside the correction-rate inequality. It binds in capacity-constrained settings — bandwidth-limited distributed systems, biological neurons, context-window-limited LLMs — where the tempo framework alone underestimates the difficulty of holding a bounded error.

Rejected Candidate Cost Metrics. The information-rate bound is not the only candidate for a cost-of-persistence quantity. Three alternatives fail structurally. Recording them here keeps the scope-honesty visible.

[Observation (gain-magnitude-tautological)] $\mathbb{E}[\|K(t)\|]$ as a cost metric: in the linear Kalman case $K_{ss} = \alpha$ (sub-scope α per `#der-gain-sector-bridge`), so "gain magnitude" equals the sector constant itself. Any bound of shape $\mathbb{E}[\|K\|] \geq f(\alpha, \ldots)$ becomes tautological. Rejected as fundamental cost metric — it recapitulates α.

[Observation (control-effort-filter-specific)] $\mathbb{E}[\|u(t)\|^2]$ (per-unit-time control-effort integral): filter-specific. Different filters achieving the same steady-state variance P_{ss} have different control-effort integrals. The Kalman filter is minimum-effort among linear filters (optimal-control interpretation of the Riccati equation); nonlinear filters can trade effort vs variance differently. A filter-agnostic bound cannot be stated in this quantity — it is not invariant under the equivalence class of filters meeting the persistence condition.

Observation (Lyapunov-dissipation-conservation)

$\mathbb{E}[\alpha\|\delta\|^2]_{ss}$ (Lyapunov dissipation rate) at steady state equals half the total disturbance power injected into the estimation error, $n(\sigma^2_{\text{env}} + \alpha^2\sigma^2_\nu)/2$ for linear correction — a non-equilibrium-steady-state balance (dissipation balances injection). It does not distinguish a well-adapted agent from a poorly adapted one at a given α, so it cannot serve as a cost metric.

The Shannon information rate above is the one that closes. The structural reason: it is filter-agnostic (it depends only on the signal's second-order statistics and the error held), universal (every causal estimator is lower-bounded), met with equality by the optimal filter, and has a clean thermodynamic interpretation (Still et al. 2012, "Thermodynamics of Prediction", *Phys. Rev. Lett.* 109:120604: nonpredictive information retained equals dissipation during interaction).

What Is Derived vs. What Is Chosen.

Table I.1

Property	*Source*	*Strength*
Persistence information rate $\dot{R} \geq n(\sigma^2_{\text{env}}/(2D) - \lambda_s)$	Sequential RDF of a Gauss–Markov source (Tanaka et al. 2017 eq. (28)), continuous-time limit	Derived (conditional on Gaussian-OU signal, causal estimation; continuous time as the limit of the discrete result)
Kalman–Bucy meets the bound; its rate is $nK/2$	Duncan 1970 + steady-state Riccati identity	Derived (linear-Gaussian, every λ_s)
Price $n\alpha/2$	The two rows above with $\alpha = K$	Derived for the Kalman-optimal agent; for other gains the intake is $nK/2$ (Duncan; the filter's path determines the observation path) and the floor at the error held is lower and slack
Channel-capacity prerequisite $C \geq \mathcal{I}/2$	Main theorem + Shannon channel-capacity theorem (Cover-Thomas 2006 §7.7) + $\alpha = \mathcal{I}$ identification	Derived for the Kalman-optimal agent; general form $C \geq n(\sigma^2_{\text{env}}/(2D) - \lambda_s)$
Gain-magnitude as cost metric	$\mathbb{E}[\|K\|] \approx \alpha$ in sub-scope α	Rejected as fundamental (tautological)
Control-effort integral as cost metric	Filter-specific; depends on optimality class	Rejected as universal (filter-dependent)
Lyapunov dissipation rate	Balance with total injected disturbance power at steady state	Not a cost metric
Non-Gaussian signal extension	Different RDF form (Berger 1971 Ch. 4)	Open — the $1/D$ scaling of the rate is expected to persist qualitatively; prefactors change
Model D (bounded disturbance) analog	Requires adversarial / minimax information argument (Csiszár-Körner 2011 Ch. 11)	Open
Transient-regime rate	Bound is steady-state; transient rates during structural adaptation much higher	Open
Sub-scope β transfer	The bound holds for any causal estimator; tightness not guaranteed	Conditional (holds as lower bound; tightness requires the optimal linear-Gaussian filter)

Epistemic Status.

Conditional. Max attainable: *exact* in the linear-Gaussian case; *robust qualitative* for general stationary Gaussian signals; *heuristic* for non-Gaussian.

The theorem rests on three named conditions: (i) a Gaussian Ornstein–Uhlenbeck signal with independent components; (ii) causal estimation, which every acting agent satisfies; (iii) steady state. The discrete-time sequential rate-distortion function is exact (Tanaka et al. 2017); the continuous-time bound is its $\Delta \to 0$ limit, taken here term by term rather than by a limit theorem, and is met exactly by the Kalman–Bucy closed form. That match was also checked numerically: for three parameter sets the discrete per-step bound divided by Δ converges to the Kalman–Bucy rate $P/(2\sigma_v^2)$ as $\Delta \to 0$. Under Model D (bounded disturbance) the argument requires worst-case channel-capacity machinery (Csiszár-Körner 2011 Ch. 11) and gives a different expression.

Sub-scope transfer. The bound is stated in the error achieved, so it holds for any causal estimator, in sub-scope α or β. The $n\alpha/2$ form uses $\alpha = K$ and so belongs to the optimal linear-Gaussian agent; for other agents the error they hold sets the floor, which is what they would need at least, while what a linear corrector on a Gaussian channel actually takes in is $nK/2$ regardless of its gain.

Confidence in correctness. The sequential rate-distortion function of a Gauss–Markov source is a published closed form, and Duncan's theorem is classical. The composition is short enough to check by hand, and both closed forms agree. The contribution is the AAT reading: holding the persistence bound is a sustained information demand, met with equality by the optimal filter at half its correction gain, and the channel capacity that supplies it is a persistence prerequisite.

What this does not establish.

- Upper bounds on persistence cost (how much rate a specific filter consumes; the bound is a lower limit).
- Optimal filter design for minimum information throughput.
- Non-Gaussian signal cost bounds (qualitative scaling expected; quantitative forms open).
- Cost under Model D adversarial disturbance.
- Cost for composite agents beyond simple additivity.
- The stability upper bound on plasticity that `#form-consolidation-dynamics` references — that involves the consolidation cadence, not the online information rate.

Discussion.

The $\alpha/2$ per-dimension Landauer analog. The theorem has a clean thermodynamic reading per Still et al. 2012: each nat of information about the signal costs at least $k_B T$ of dissipation (Landauer 1961, in its per-nat form $k_B T$/nat, equivalently $k_B T \ln 2$ per bit). Combined, the Kalman-optimal agent persisting at sector constant α in n dimensions takes in $n\alpha/2$ nats/time and so dissipates at least $\frac{1}{2} n\alpha\, k_B T$ per unit time in any physical substrate. AAT does not commit to a specific substrate, but the bound is substrate-agnostic: it constrains any filter implementation via the rate-distortion function (information-theoretic) and, when physical, any computational realization via Landauer (thermodynamic).

Why channel capacity matters as a first-class quantity. AAT's `#result-persistence-condition` and `#def-adaptive-tempo` frame persistence as a correction-rate vs disturbance-rate inequality. This theorem adds a *lower* constraint: the observation channels must supply the information rate the error requires — $\mathcal{I}/2$ nats/time per dimension for the optimal agent — or persistence fails regardless of correction-function design. The constraint is binding wherever observation bandwidth is scarce. Three domains where the capacity floor is more binding than the tempo bound:

- *Biological systems*: neural channel capacity is finite; this bound gives a quantitative minimum on sensory bandwidth for a given adaptive rate.
- *Bandwidth-constrained distributed systems*: agents operating over noisy or low-bandwidth links face channel capacity directly as a hard constraint.
- *Context-window-limited LLMs*: the effective information rate per unit of cognition is bounded by context size and token-throughput; this theorem predicts a minimum tempo achievable at a given context budget.

In each, knowing that $\mathcal{I}/2$ is the floor for the optimal agent converts an opaque "just needs more capacity" observation into a specific dimensional requirement.

Connection to AAT's meta-architecture. The result composes with AAT's three meta-segments.

- `#disc-separability-pattern` *(positive half)*: the bound sits in structured-repair along the identification-regime ladder — derived from an external theorem, met with equality in the linear-Gaussian case, holding as an inequality elsewhere.
- `#disc-identifiability-floor` *(negative half)*: this is the positive-dual of the floor pattern. Where the floor says "AAT cannot distinguish X without information augmentation" (external no-go + AAT escape), this result says "holding the error at D requires at least $n(\sigma^2_{\text{env}}/(2D) - \lambda_s)$ nats/time of information supply" (external lower bound + AAT bridge through the persistence bound). The two patterns are duals: external-theorem-forbids vs external-theorem-lower-bounds.

- `#disc-additive-coordinate-forcing` *(constructive half)*: the bound is linear in $1/D$ (and in α for the optimal agent), not logarithmic. It sits *outside* the three-layer logarithmic-coordinate family (chain / divergence / update). No Cauchy-FE argument forces the coordinate here. This is a useful non-example: AAT's additive-coordinate-forcing pattern is not universal; specific results live on different coordinates.

Relationship to G. The template enumerates instantiations (epistemic mismatch; strategic mismatch; sub-agent mismatch; composite trajectory error; composite mismatch; target-agent mismatch). Each has its own state variable ξ, sector constant α, and disturbance statistics. The persistence cost bound *specializes* for each instantiation in which the state is an estimation error of a Gauss–Markov signal: the rate is fixed by the error held, and by $n\alpha/2$ when the corrector is the optimal one. A compact template-cost bound could land as a subsection of G so the instances inherit it by substitution. That move is flagged in Working Notes as an optional consolidation.

The relationship to Mitter-Newton's thermodynamic reading. Mitter & Newton (2005) showed the Kalman-Bucy filter operates as a Maxwellian demon: it returns signal energy to the heat bath without entropy increase, but only because new information is continually supplied at rate $\dot{I}$. This supply rate *is* what persistence costs — the filter holds a bounded estimation error by paying, information-theoretically, for new observations at a rate matched to the environment's innovation rate. The theorem makes this quantitative: the rate equals the least any causal estimator could pay for the same error, and it is half the filter's correction gain.

Appendix J

Derivation: Critical-Mass Composition

The composite sector constant α_c is derived — not merely bounded from below — for the symmetric-matched-Tier-1 two-agent case, yielding a closed-form critical-mass inequality in which the sign of the inter-agent coupling γ and the teleological unity U_O enter explicitly. The result subsumes the weakest-link bound, recovers `#der-team-persistence` (cooperative) and `#der-adversarial-destabilization` (adversarial) as signed special cases, formalizes `#hyp-symbiogenic-composition`'s autonomy-reduction mechanism as an asymmetric Lyapunov-weight limit, and makes the scope-gate from `#scope-composite-agent` explicit as the second conjunct of composite persistence.

Formal Expression.

Setup. Two sub-agents A_1, A_2, each a **Tier 1 agent** in the sense of `#form-composition-closure`'s bridge-lemma taxonomy — mismatch-driven update, linear prediction, incremental sector-Lipschitz correction (Kalman, exponential-family Bayesian, gradient-on-strongly-convex, linear-with-PD-KH). **Matched architectures**: f_1, f_2 are structurally the same function, with $\alpha_1 = \alpha_2 = \alpha$, $R_1 = R_2 = R$. Disturbance statistics shared: each sees bounded $w_i(t)$ with $\|w_i\| \leq \rho$ (Model D, per G).

[Formulation (coupling-model-C1, from `#der-team-persistence` + `#der-adversarial-destabilization`)]

Inter-agent coupling enters additively to the disturbance at rate $\gamma \mathcal{J}_j$:

$$\rho_i^{\text{eff}} = \rho + \gamma \mathcal{J}_j \tag{C1}$$

with sign convention $\gamma < 0$ cooperative (ally's tempo-contribution reduces disturbance, recovering `#der-team-persistence`'s $-\gamma^{\text{coop}} \mathcal{J}_j$ term), $\gamma > 0$ adversarial (ally's tempo-contribution amplifies disturbance, recovering `#der-adversarial-destabilization`'s $+\gamma_A \mathcal{J}_A$ term). Symmetric case: $\gamma_{1\to 2} = \gamma_{2\to 1} = \gamma$.

Formulation (coordination-cost-C2)

Coordination overhead reduces each agent's effective correction rate symmetrically:

$$\alpha_i^{\text{eff}} = \alpha - C \tag{C2}$$

with $C \geq 0$ the $\Delta \mathcal{J}_i^{\text{cost}}$ from `#der-team-persistence`'s coordination-overhead threshold.

Critical-mass inequality (symmetric-matched-Tier-1 case). *[Derived (critical-mass-symmetric, from G + C1 + C2)]*

Let $\xi = (\delta_1, \delta_2)^T$ and take the joint quadratic Lyapunov candidate $V(\xi) = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$. Under the block-diagonal correction structure with cross-coupling absorbed into ρ_i^{eff} via (C1), and using $\|\delta_1\|+\|\delta_2\| \le \sqrt{2(\|\delta_1\|^2 + \|\delta_2\|^2)}$ (Cauchy–Schwarz):

$$\dot{V} \le -(\alpha - C)(\|\delta_1\|^2 + \|\delta_2\|^2) + (\rho + \gamma\mathcal{T})\sqrt{2(\|\delta_1\|^2 + \|\delta_2\|^2)}. \tag{J.1}$$

Setting $\dot{V} = 0$ gives the ultimate bound on $\|\xi\|$ and, projecting to the macro-state $\delta_c = (\delta_1 + \delta_2)/\sqrt{2}$, the ultimate composite mismatch

$$R_c^* \le \frac{\rho + \gamma\mathcal{T}}{\alpha - C}. \tag{L4}$$

The composite persists iff $R_c^* < R_c$. Inheriting $R_c = R$ from the symmetric-matched averaging projection:

$$\boxed{(\alpha - C)\,R \;>\; \rho + \gamma\mathcal{T}} \tag{CM2}$$

Rearranging into the composite contraction-rate form:

$$\kappa_c \;:=\; (\alpha - C) \;-\; \frac{\rho + \gamma\mathcal{T}}{R}, \qquad \text{composite persists iff } \kappa_c > 0. \tag{KC}$$

Specialization checks. Under matched symmetry, (CM2) reduces correctly in four limits:

Table J.1

Limit	*Setting*	*(CM2) reduces to*	*Recovers*
No coupling	$\gamma = 0$, $C = 0$	$\alpha R > \rho$	Single-agent `#result-persistence-condition`
Cooperative-symmetric	$\gamma < 0$, $C = 0$	$\alpha R > \rho + \gamma\mathcal{T}$ (easier than individual)	`#der-team-persistence`'s "teams persist where individuals can't"
Adversarial-symmetric	$\gamma > 0$, $C = 0$	Fails when $\gamma\mathcal{T} > \alpha R - \rho$	`#der-adversarial-destabilization` threshold (symmetric)
Coordination-dominated	$C > \alpha$, $\gamma = 0$	LHS < 0; composite fails	Brooks's Law

Derived (weakest-link-subsumption)

Subsumption of the weakest-link bound.

The weakest-link bound $\alpha_c \ge \min_i(\alpha_i - \Delta\mathcal{T}_i^{\text{cost}})$ from `#form-composition-closure`'s derivation table specializes under matched symmetry to $\alpha_c \ge \alpha - C$. (KC) refines this by making the composite's effective disturbance explicit as $\rho + \gamma\mathcal{T}$, turning a correction-rate bound into a full persistence inequality. Critically, (KC) can yield $\kappa_c > 0$ even when the weakest-link bound alone fails — when cooperative coupling ($\gamma\mathcal{T} < 0$) reduces the effective disturbance below what the raw $\alpha - C$ margin would permit. The weakest-link bound cannot see this because it does not account for γ's sign.

U_O entry: multiplicative-on-γ plus scope-gate.

Derived (unity-multiplicative-modulator, conditional on LQR-compatible action structure)

In a purposeful-agent setting where each sub-agent optimizes a quadratic objective $L_i(\omega) = \frac{1}{2}(\omega - r_i)^T Q(\omega - r_i)$ with target r_i, and $U_O := \mathrm{corr}(r_1, r_2)$ is the target correlation per `#def-unity-dimensions`' U_O, the cross-coupling in the joint dynamics has sign and magnitude controlled by U_O:

$$\gamma(U_O) \;=\; -\gamma_{\max}\, U_O, \qquad \gamma_{\max} > 0, \tag{UO-mult}$$

via aligned targets → aligned action directions in the shared environment → constructive (cooperative) cross-contribution in the symmetric eigendirection. Substituting into (KC):

$$\kappa_c(U_O) \;=\; (\alpha - C)R \;-\; \rho \;+\; \gamma_{\max}\, U_O\, \mathcal{T}. \tag{CM3}$$

Scope (scope-gate-from-composition-scope-condition)

(CM3) is necessary but not sufficient for composite existence. Under `#scope-composite-agent`, a composite exists as an AAT agent only when one of the three disjunctive alignment routes (shared objective, hierarchical derivation, mutual benefit) is satisfied. Below this threshold, no coherent O_c is definable and composite-level quantities — including R_c on the right of (CM2) — are ill-typed. The honest statement of composite persistence is therefore the conjunction of $\kappa_c(U_O) > 0$ with the scope conditions of `#scope-composite-agent` — abbreviated $\mathcal{S}_c$:

$$\boxed{\kappa_c(U_O) > 0 \;\wedge\; \mathcal{S}_c \;\Leftrightarrow\; \text{composite persists as AAT agent}} \tag{CM4}$$

U_O enters (CM4) in two independent ways: multiplicatively within (CM3), and as scope-gate via `#scope-composite-agent`. It does **not** enter purely additively as a separate reserve term — there is no free-floating "U_O contribution" detached from the coupling it modulates.

Asymmetric limit and symbiogenic composition.

Sketch (asymmetric-limit-symbiogenesis, from weighted Lyapunov)

Drop the matched-symmetric assumption. Let $\alpha_1 \gg \alpha_2$ with $\alpha_2 \to 0$. The unweighted joint Lyapunov $V = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$ fails (the weakest-link ultimate bound diverges as $\alpha_2 \to 0$). A **weighted** Lyapunov $V_\mu(\xi) = \frac{1}{2}(\|\delta_1\|^2 + \mu\|\delta_2\|^2)$ with $\mu \to 0$ yields

$$\dot{V}_\mu \le -\alpha_1\|\delta_1\|^2 + \rho_1\|\delta_1\| \;+\; O(\mu), \tag{J.2}$$

so in the limit the composite's stability is controlled **entirely by agent 1**; agent 2's autonomous correction dynamics are weighted out of the stability accounting.

This provides a Lyapunov-weighted formalization of `#hyp-symbiogenic-composition`'s **(S-3) autonomy reduction**: the endosymbiont's effective action space contracts ($\mathcal{A}_e^{\text{effective}} \to \mathcal{A}_e^{\text{restricted}}$) and its autonomous dynamics fall out of the joint Lyapunov argument. The asymmetric limit is a smooth deformation of (CM4), not a discontinuous regime change — symbiogenesis and peer coupling are parameter-limits of the same weighted-Lyapunov analysis. The result does **not** close `#hyp-symbiogenic-composition`'s (S-1) objective absorption or (S-2) function transfer: what happens to agent 1's state space when it inherits structure from agent 2 is a separate question the weighting argument does not address.

What Is Derived vs. What Is Chosen.

Table J.2

Property	*Source*	*Strength*
Coupling model (C1): $\rho_i^{\text{eff}} = \rho + \gamma \mathcal{T}_j$	Import from `#der-team-persistence` and `#der-adversarial-destabilization`	Formulation choice (requirement for the derivation)
Coordination-cost model (C2): $\alpha_i^{\text{eff}} = \alpha - C$	Import from `#der-team-persistence`'s coordination-overhead threshold	Formulation choice
Joint quadratic Lyapunov $V = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$	Standard vector-Lyapunov construction (Matrosov 1962; Bellman 1962)	Formulation choice (canonical for matched-symmetric dyads)
Ultimate bound $R_c^* \leq (\rho + \gamma\mathcal{T})/(\alpha - C)$	Lyapunov dissipation + Cauchy–Schwarz	Derived
Critical-mass inequality (CM2): $(\alpha - C)R > \rho + \gamma\mathcal{T}$	(L4) + sector-region fit $R_c^* < R_c = R$	Derived (conditional on Tier 1 + matched-symmetric + Model D)
Four specialization checks (no-coupling / cooperative / adversarial / coordination-dominated)	Direct substitution into (CM2)	Proved (within stated scope)
Subsumption of weakest-link bound	(CM2) sign-sensitive; weakest-link is sign-blind	Proved
(UO-mult): $\gamma(U_O) = -\gamma_{\max} U_O$	LQR-compatibility sketch; aligned targets → aligned actions → constructive cross-contribution	Discussion-grade
Composite persistence as (CM3) ∧ scope-satisfaction: (CM4)	(KC) with (UO-mult) + `#scope-composite-agent`	Derived (conditional)
Asymmetric limit → `#hyp-symbiogenic-composition` (S-3) via weighted Lyapunov	Matrosov-style weighting; $\mu \to 0$ limit	Sketch (the weighting is standard; the identification with (S-3) is structurally motivated but not a theorem)
(S-1) objective absorption and (S-2) function transfer formalizations	Not addressed by this derivation	Open (in `#hyp-symbiogenic-composition` Working Notes)
Heterogeneous-architecture case (A_1 Tier 1, A_2 Tier 2/3)	Requires per-sub-agent tiering per `#form-composition-closure`	Open
Heterogeneous-metric Tier-1M dyad ($\lambda_1 \neq \lambda_2$, $C_1 \neq C_2, k_{12} \neq k_{21}$)	AK (CM2-M) via Slotine 2003 negative-feedback small-gain: $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$	Derived (conditional on AK (CT2) preconditions + Slotine 2003)
Nonlinear coupling $\gamma = \gamma(\delta_j)$	Requires full joint-Lyapunov machinery from `#der-adversarial-destabilization` (effects-spiral corollary)	Open
Dynamic coordination cost $C = C_0 + C_1\|\delta_j\|$	Quadratic inequality; admits closed form, loses interpretive cleanliness	Open
Fully-coupled tempo dynamics ($\mathcal{T}_i$ responsive to δ_j)	Requires joint tempo analysis from `#der-adversarial-destabilization` Working Notes	Open
$N > 2$ scaling of (CM4)	Conjunction over pairwise terms generalizes but loses closed form	Open

The dividing line: (C1), (C2), and the quadratic Lyapunov candidate are **formulation choices** imported from adjacent segments or from standard Lyapunov practice. The *consequences* under these choices — (L4), (CM2), (KC), the specialization checks, the weakest-link subsumption, and (CM4) with its scope-gate conjunct — are **derived**. The U_O-multiplicative modulator (UO-mult) is discussion-grade: it uses an LQR-compatibility argument whose rigor depends on an action-space inner-product analysis deferred to `#result-unity-closure-mapping`. The asymmetric-limit identifica-

tion with `#hyp-symbiogenic-composition` (S-3) is sketch-level — the weighted-Lyapunov argument is standard but the semantic identification with autonomy reduction is structural, not proved.

Epistemic Status.

Conditional. Max attainable: *exact* (within the matched-symmetric-Tier-1 scope); *conditional* beyond.

(CM2) and (KC) are **proved** under Tier 1 architecture + matched-symmetric parameters + Model D disturbance + the (C1) disturbance-coupling model + the (C2) coordination-cost model. These conditions cover Kalman filters, exponential-family Bayesian updaters, gradient-on-strongly-convex agents, and linear-PD correctors — the same architecture class for which `#form-composition-closure`'s bridge lemma is promoted from "conditional" to "derived." Within this scope the result is as strong as the standard Lyapunov argument allows.

(CM3) inherits the conditional status of (UO-mult), which is **discussion-grade**: the LQR-compatibility sketch is qualitatively clear, but a rigorous action-space inner-product derivation that pins down $\gamma_{\max}$ has not been produced. (CM4) is therefore conditional on both (UO-mult)'s rigor and on `#scope-composite-agent`'s scope-gate being independently verified for the given composite candidate.

The asymmetric limit (§asymmetric-limit) is **sketch-level**. The weighted-Lyapunov argument is textbook (Matrosov 1962); the identification of the $\mu \to 0$ limit with `#hyp-symbiogenic-composition`'s (S-3) autonomy-reduction mechanism is structurally motivated but not a theorem, because (S-2) function transfer is not formalized in `#hyp-symbiogenic-composition`. When (S-2) lands, the symbiogenic-limit result can be promoted to derived.

What this segment does **not** establish:

- Composite **incremental** sector bound (DA2'-inc) — still the domain of `#form-composition-closure`'s bridge lemma. (CM4) gives composite (T2) at the macro level; the bridge lemma's contraction is a separate, stronger condition.
- Heterogeneous-architecture composites (Kalman + PID, Tier 1 + Tier 3): the joint Lyapunov construction requires agent-by-agent tiering; closed form is not available.
- Nonlinear or state-dependent coupling: (C1) assumes γ independent of δ. State-dependent γ produces a nonlinear inequality in δ; this is the effects-spiral corollary territory of `#der-adversarial-destabilization`, still open.
- $N > 2$ scaling: the matched-symmetric pairwise result generalizes by conjunction, but the Cauchy–Schwarz step degrades with team size; the closed form does not survive cleanly.

On (T2) and sub-scoping. The joint quadratic Lyapunov candidate presumes each sub-agent's correction is in sub-scope α of `#form-sector-condition` (Bayesian / exponential-family / strongly-convex-gradient / linear-PD) under directional fidelity per `#der-gain-sector-bridge`. Composites with sub-scope β sub-agents (PID, rule-based, human-judgment) require (T2) verification per sub-agent at the composite level — the template's A2'-sub-scope label is inherited pairwise.

Discussion.

Relationship to the bridge lemma. `#form-composition-closure`'s bridge lemma establishes when the macro-update map f_c is **incrementally** contracting, at the trajectory-error level — a condition strictly stronger than the one-point sector bound. This segment is the complement at the macro-state level: given the bridge lemma's admissibility, (CM4) derives composite (T2), the one-point sector bound that G's instantiation for the composite requires. Both are necessary for the composite to be a stable macro-agent with bounded mismatch: bridge lemma says the macro-description tracks micro-reality (trajectory error stays bounded); (CM4) says the composite's own corrections drive composite mismatch back inside the sector region. Together they close the composite-persistence argument at both layers.

Pattern across the signed-coupling instances. (CM4) has the same shape as several persistence-flavored results already in AAT:

- `#der-team-persistence`: per-sub-agent inequality $\alpha_i R_i > \rho_{i,\text{env}} + \sum_j \gamma^{\text{adv}}_{j\to i}\mathcal{T}_j - \sum_j \gamma^{\text{coop}}_{j\to i}\mathcal{T}_j$
- `#der-tempo-composition`: composite inequality with effective disturbance $\rho_{\text{ext}} + \varepsilon^* \nu_c$
- `#der-adversarial-destabilization`: failure condition $\gamma_A \mathcal{T}_A > \alpha_B R_B - \rho_B$ (negation of persistence)
- this segment: matched-symmetric dyad with signed γ and scope-gate

All four are instances of G's pattern with signed coupling controlling the sign of the cross-agent contribution to ρ_ξ. The template already names this pattern at the meta level; the present segment is the dyadic closed-form instance that the other three segments reference pairwise. A dedicated meta-segment for "signed-coupling critical-mass" was considered during spike work and judged redundant with G — the cross-instance structure is already visible in that segment's instantiation table.

Potential-game generalization (via `#deriv-strategic-composition`). (CM4) applies to composites with shared target (scope routes C-i / C-ii / C-iii from `#scope-composite-agent`) where contraction-to-shared-truth is the correct primitive. `#deriv-strategic-composition` carries the sibling result for composites with partially-opposing objectives (scope route C-iv — strategic composites): under the potential-game condition (Monderer-Shapley 1996), the sector-persistence template transfers to the gradient of the joint potential Φ, with α_{joint} playing the role of the composite sector constant and $\xi = \pi - \pi^*$ (deviation from Nash) playing the role of the composite state. For matched-symmetric potential games, the structural form of (CM2) survives with $(\alpha, R, \rho, \gamma, C)$ replaced by $(\alpha_{\text{joint}}, R_{\text{Nash-basin}}, \rho_\xi, \gamma_{\text{strategic}}, C_{\text{strategic}})$; the specific mapping is instance-dependent and typically not closed-form for non-zero-sum games. The joint-quadratic-Lyapunov machinery of this segment and the potential-function-Lyapunov machinery of `#deriv-strategic-composition` are both instances of `#result-sector-persistence-template` at the composite-state-variable level; what varies is whether the composite state is *mismatch-to-shared-target* (this segment) or *deviation-from-Nash* (`#deriv-strategic-composition`).

What (CM4) contributes beyond existing segments.

1. **Derivation, not assumption.** The composite sector constant appearing in `#form-composition-closure`'s (A4) was previously bounded by the weakest-link formula (a derived lower bound) but treated as an assumption for the bridge lemma. (CM4) *derives* α_c as a closed-form function of sub-agent parameters + coupling + unity + coordination overhead, in the matched-symmetric-Tier-1 case.
2. **Sign-sensitive.** The weakest-link bound is sign-blind — it gives the same $\alpha_c \geq \alpha - C$ regardless of whether the coupling is cooperative or adversarial. (CM4) makes the sign explicit through the $\rho + \gamma\mathcal{T}$ right-hand side and turns a correction-rate bound into a full persistence inequality.
3. **Unifies cooperative and adversarial regimes.** Teams persisting where individuals can't and adversarial destabilization are the same inequality viewed from opposite signs of γ, not independent results.
4. **Scope-gate made explicit.** Composite persistence requires both contraction (CM3) **and** scope-satisfaction (`#scope-composite-agent`). (CM4) states this conjunction honestly; the absence of one or the other is a different failure mode (composite fails to contract vs. composite was never a composite).
5. **Symbiogenic and peer regimes connected.** The asymmetric limit shows symbiogenesis is a parameter-limit of peer coupling under a Lyapunov-weight deformation, not a discontinuous regime change requiring separate machinery.

Why the matched-symmetric restriction is load-bearing. Relaxing matched architectures requires per-sub-agent tiering in the sense of `#form-composition-closure` (Tier 1 / 2 / 3). Heterogeneous composites can still satisfy (T2) at the composite level, but the joint Lyapunov construction must be weighted per sub-agent's contraction capacity — and

the weighted form loses the clean closed-form (CM2). Relaxing symmetric coupling ($\gamma_{1\to 2} \neq \gamma_{2\to 1}$) produces a non-symmetric matrix whose smallest eigenvalue must be computed explicitly per instance. The matched-symmetric case is the one where both sub-agent tiering and coupling symmetry collapse the Lyapunov construction to a scalar inequality — which is why closed form is available there and not elsewhere.

Load-bearing role under `#disc-identifiability-floor` Instance 3. The composition-layer identifiability floor (Instance 3 of `#disc-identifiability-floor`) establishes a no-go theorem: there exist pairs of coupled systems with identical marginal component-level observation distributions but opposite composite-contraction signs, so composite contraction is not in general identifiable from component data alone. Four structural escapes are named there; escape (b) — matched Tier at the composite level — is operationalized by the closed-form (CM2) in this segment. Under the floor, (CM2) is not just "a closed-form result in a special case" but **the unique broadly-available composition-contraction certificate** among the four escape routes listed. Without (CM2) or its metric-formulation generalization (CM2-M) via `#result-contraction-template`, the weakest-link bound (WL) is sign-blind and cannot distinguish the cooperative-contracting from the adversarial-destabilizing composite. This load-bearing status positions this segment as "the machinery that escapes the composition-layer floor" rather than as a closed-form curiosity in the matched-symmetric special case.

Adjacent literature. The joint Lyapunov construction is an instance of the classical **vector Lyapunov function** method (Matrosov 1962; Bellman 1962). The critical-mass inequality is the AAT sector-bounded analog of the **small-gain theorem for ISS systems** (Jiang–Teel–Praly 1994; Sontag 1989): in the small-gain framework, composition of two ISS systems is ISS iff the product of their gains is less than one; in AAT's sector-bounded framework, the composite persists iff the *sum* of parent reserve-rates exceeds the coupling-amplified disturbance (additive rather than multiplicative because the averaging projection averages rather than multiplies gains). In the linear-diagonal-coupling case, the matrix $A = \alpha I - \beta(I - J)$ where J is the averaging operator is the graph-Laplacian-shifted form governing consensus convergence (Olfati-Saber & Murray 2004); (CM2) is consensus convergence rewritten as a persistence inequality. Relative to active inference's FEP-flow stability arguments (Friston 2019; narrowed by Aguilera et al. 2022 to small parameter regimes in NESS-density models), the present result inherits G's broader validity: it applies wherever (T1)–(T3) hold pairwise + (C1)/(C2) + matched symmetry, without requiring free-energy landscapes or NESS structure.

Findings.

Strong Monotonicity as the Hinge for Legitimate Macro-Agent Coarse-Graining. **Brief:** Two coupled adaptive systems can have identical marginal observation distributions while one composite contracts and the other diverges — the difference is a single bit (the sign of the cross-coupling) that is invisible from component data alone. The matched-symmetric-Tier-1 critical-mass inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ exposes this bit explicitly: the same inequality recovers cooperative team-persistence (where coupling helps), adversarial destabilization (where coupling hurts), and Brooks's-Law coordination collapse (where overhead C exceeds correction rate α) as signed special cases of one closed-form result. The strong-monotonicity / DA2'-inc condition picks out precisely the agent classes (Kalman, exp-family Bayes, gradient on strongly-convex losses, linear with positive-definite gain-observation product) where the inequality is exact rather than heuristic.

Impact: Promotes the composite sector constant in `#form-composition-closure`'s admissibility condition (A4) from a weakest-link lower bound (sign-blind, treats cooperative and adversarial coupling identically) to a derived closed form that depends explicitly on coupling sign, target-correlation U_O, and coordination overhead C. Operationalizes Instance 3 of `#disc-identifiability-floor` (composition-layer common-Lyapunov no-go via Liberzon 2003) through escape route (b): under matched Tier at the composite level, the joint Lyapunov collapses to a scalar inequality and the otherwise-unidentifiable sign bit becomes a structural input to (CM2). Within the four `#disc-identifiability-floor` Instance-3 escapes, (CM2) and its metric generalization (CM2-M) in `#result-contraction-template` are the unique broadly-available coupling-sign certificate. The asymmetric-parameter limit gives `#hyp-symbiogenic-composition`'s (S-3) autonomy-reduction a Lyapunov-weighted formulation as a smooth deformation of (CM4) rather than a discontinuous regime change.

Novelty Claim: *Claim novelty* on strong monotonicity as the criterion separating legitimate macro-agent coarse-graining from coexistence-only multi-agent description. The closest mathematical neighbors (Subramanian 2020; Congeduti 2020) prove bounded-loss composition under predictive compression but do not isolate a single condition whose failure flips composite contraction; the Markov-blanket line (Parr 2019; Kirchhoff 2018) discusses nested agency without a control-theoretic monotonicity criterion; the small-gain-theorem line (Jiang-Teel-Praly 1994; Sontag 1989) provides a multiplicative composition criterion under input-to-state stability rather than the additive sign-sensitive form (CM2) takes here.

Related Work:

Table J.3

ASF concern	*Prior-art language*	*Relationship / Positioning*
Strong monotonicity as composition-validity hinge	Subramanian, Sinha, Seraj & Mahajan 2020, "Approximate information state for approximate planning and reinforcement learning in partially observed systems" *arXiv:2010.08843* (published 2020, found 2026-04)	*closest mathematical neighbor* — proves a bridge-shape bound but does not foreground a single monotonicity-grade condition that determines whether composite description is legitimate. Per the Pillar-2 defense strategy, foregrounding strong monotonicity (DA2'-inc; equivalently (CT2)-at-$M = I$ in `#result-contraction-template`) as the hinge between stable coexistence and valid macro-agent coarse-graining is the increment ASF supplies on top of this neighbor
Coordination-overhead / influence loss bound	Congeduti, Mey & Oliehoek 2020, "Loss Bounds for Approximate Influence-Based Abstraction" *arXiv:2011.01788* (published 2020, found 2026-04)	*closest mathematical neighbor* — explicit bounds on value loss from approximate influence in multi-agent settings. (CM2)'s sign-sensitive RHS $\rho + \gamma\mathcal{T}$ encodes the same loss-vs-cooperation tradeoff that Congeduti's influence-approximation bound encodes for value, applied at the persistence-inequality layer rather than the policy-value layer
Composite-Lyapunov nonexistence under component-marginal data	Liberzon 2003, *Switching in Systems and Control* Theorem 2.1; Dayawansa & Martin 1999, *IEEE TAC* 44; Shorten et al. 2007 *SIAM Review* 49 (published 2003/1999/2007, found 2025-04)	*formal antecedent* — supplies the no-go theorem `#disc-identifiability-floor` Instance 3 invokes. (CM2) is the AAT escape route (b) under matched Tier at the composite level
Vector-Lyapunov composition	Matrosov 1962; Bellman 1962	*formal antecedent* — joint quadratic Lyapunov $V = \frac{1}{2}(\|\delta_1\|^2 + \|\delta_2\|^2)$ is a vector-Lyapunov construction; standard machinery
Multiplicative composition criterion via small-gain	Jiang, Teel & Praly 1994, "Small-gain theorem for ISS systems and applications" *Math. Control Signals Syst.* 7; Sontag 1989	*partial anticipation* — same compositional-validity question, multiplicative form (gain product < 1) under input-to-state stability. (CM2)'s additive form $(\alpha - C)R > \rho + \gamma\mathcal{T}$ comes from sector-bounded correction rather than ISS; the two are different scaling regimes of the same composition-validity question, with (CM2) sharper in AAT's setting because the averaging projection averages rather than multiplies sub-agent gains
Negative-feedback heterogeneous extension	Slotine 2003, "Modular stability tools for distributed computation and control" *Int. J. Adapt. Control Signal Process.* 17:397–416 — Theorem 3	*formal antecedent* — supplies (CC-feedback) used in `#result-contraction-template` to lift (CM2) to heterogeneous Tier-1M sub-agents via (CM2-M)

ASF concern	*Prior-art language*	*Relationship / Positioning*
Consensus dynamics graph-Laplacian shift	Olfati-Saber & Murray 2004, "Consensus problems in networks of agents with switching topology and time-delays" *IEEE TAC* 49:1520	*adjacent literature* — in the linear-diagonal-coupling case, (CM2) is consensus convergence rewritten as a persistence inequality; different domain framing, same mathematical object
Markov-blanket nested-agency at composition	Parr, da Costa & Friston 2019; Kirchhoff et al. 2018 (published 2019/2018, found 2026-04)	*conceptual precursor* — provides nested-agency framing without a control-theoretic monotonicity criterion that separates valid macro-agent coarse-graining from coexistence

Search Log:

- 2026-04 (*nominally comprehensive*): the strong-monotonicity hinge is foregrounded as the criterion not visible in retrieved prior art. Subramanian / Congeduti provide the closest math (bounded-loss composition) without isolating a single condition whose failure flips composite legitimacy; Markov-blanket line provides nested-agency intuition without monotonicity machinery.
- 2026-04 (*targeted*, prior to comprehensive search): the small-gain-theorem line (Jiang-Teel-Praly 1994; Sontag 1989) and consensus-dynamics line (Olfati-Saber & Murray 2004) were known and cited in the segment's existing Discussion as adjacent literature. The comprehensive search added the Sub20 / Con20 closest-neighbor positioning and confirmed no surfaced paper foregrounds the strong-monotonicity hinge.
- 2026-04 (*intuition-only*, prior): expected adjacent literature was switched-systems stability (Liberzon 2003) and game-theoretic equilibrium analysis. Switched-systems hit confirmed via `#disc-identifiability-floor` Instance 3; game-theoretic side is properly the territory of `#deriv-strategic-composition`, scope-separated.

Appendix K

Derivation: Gain-Sector Bridge — Proofs and Verification

Complete derivations for the results stated in `#der-gain-sector-bridge`: gain-based updating produces correction functions satisfying the sector condition (GA-3) whenever the update rule has directional fidelity. Four propositions are proved. **Prop B.1 (scalar Kalman):** for scalar linear-Gaussian systems, the sector parameter equals the Kalman gain exactly. **Prop B.2 (matrix Kalman):** for vector-valued systems, the sector condition holds in the inverse-prior-covariance-weighted inner product, with sector parameter equal to the smallest eigenvalue of KH on the observable subspace. **Prop B.3 (bridge theorem):** the general gain-based update produces a sector-condition-satisfying correction function whenever directional fidelity (B1) holds. **Prop B.4 (gradient equivalence):** the *two-point / incremental* sector condition (DA2'-inc, used by `#form-composition-closure`'s bridge lemma) is equivalent to local strong convexity via Nesterov 2.1.10; the *one-point* sector (A2', what AAT's persistence machinery in `#form-sector-condition` actually requires) is *weaker* — strong convexity implies the one-point sector but not conversely (counterexample $L'(x) = x(1 + \frac{1}{2}\sin(10x))$).

The framework's overall posture is that the bridge is **conditional, not unconditional**: pathological update rules exist that satisfy every framework postulate and the gain-based update form yet violate directional fidelity. Scope plus gain structure alone does not force the bridge; some optimality / coherence / rationality constraint (Bayesian coherence, gradient-of-a-convex-loss) is required. The contribution is *characterizing which agent classes give the bridge for free* (sub-scope α per `#form-sector-condition`) and *which require the sector condition as a primitive posit* (sub-scope β). Numerical verification across six experiments — quadratic, logistic, Gaussian-mixture, exponential-family, quasi-convex, multimodal — confirms the loss-function classification predicted by Prop B.4.

Motivation.

A proves persistence and stability downstream of the sector condition (GA-3): $\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2$. The sector parameter α governs the persistence bound ($R^* = \rho/\alpha$), the adaptive reserve ($\Delta\rho^* = \alpha R - \rho$), and the adversarial scaling exponent. But whether the correction function F induced by the gain principle (`#emp-update-gain`) actually satisfies GA-3 was not derived — GA-3 was the theory's softest structural joint.

This appendix closes the gap. For optimal Bayesian updates (Kalman, conjugate families), the bridge is exact: the sector parameter equals the gain. For gradient-based agents, GA-3 is equivalent to local strong convexity of the loss function. The load-bearing assumption shifts from "the correction function satisfies the sector condition" (opaque) to "the update rule has directional fidelity" (transparent, checkable).

Proposition B.1: Scalar Kalman Sector Condition.

Statement. For a scalar linear-Gaussian system ($n = m = 1, H = 1$) with Kalman filter gain $K = P^-/(P^- + R_{\text{obs}})$, the correction function $F(e) = K \cdot e$ satisfies the sector condition with:

Derived (scalar-Kalman-sector, from Kalman update)

$$\alpha = K = \frac{P^-}{P^- + R_{\text{obs}}} = \eta^* \tag{K.1}$$

The sector parameter equals the Kalman gain, which equals the uncertainty-ratio gain from `#emp-update-gain`. The bound is tight (the correction is linear).

Proof. The scalar Kalman update for state estimate $\hat{x}$ is driven by the innovation $\delta = o - \hat{x}_{\text{prior}}$ (`#def-mismatch-signal`): $\hat{x}_{\text{post}} = \hat{x}_{\text{prior}} + K\delta$, where the gain is $K = P^-/(P^- + R_{\text{obs}})$ with P^- the prior variance and R_{obs} the observation noise variance. The object the correction acts on, and the one the sector condition is stated for, is not the innovation but the state-space estimation error $e = x - \hat{x}$; the innovation is $\delta = He + \varepsilon$, the estimation error plus channel noise ε.

The estimation error evolves under the correction step as:

$$e_{\text{post}} = e_{\text{prior}} - K(He_{\text{prior}} + \varepsilon) = (1 - K)e_{\text{prior}} - K\varepsilon \tag{K.2}$$

With $H = 1$, the expected correction function is $F(e) = K \cdot e$. The second term, $-K\varepsilon$, is channel noise that the correction feeds back into the estimation error. It enters the persistence dynamics as part of the disturbance (A, Setup), with variance $K^2 R_{\text{obs}}$ that grows with the gain. The sector product is:

$$e \cdot F(e) = K \cdot e^2 \tag{K.3}$$

Since $P^- > 0$ and $R_{\text{obs}} > 0$, we have $K \in (0, 1)$. The sector condition $e \cdot F(e) \geq \alpha \cdot e^2$ holds with $\alpha = K$. The bound is tight because the correction is exactly linear. □

Steady-state sector parameter. At steady state (random walk dynamics, $A = 1$), the prior variance satisfies the algebraic Riccati equation. Solving $K^2 R_{\text{obs}} + QK - Q = 0$ (where Q is the process noise variance) gives:

Derived (steady-state scalar sector parameter)

$$\alpha_{ss} = K_{ss} = \frac{-Q + \sqrt{Q^2 + 4QR_{\text{obs}}}}{2R_{\text{obs}}} \tag{K.4}$$

Limiting behavior: when $Q \gg R_{\text{obs}}$ (fast dynamics, clean observations), $K_{ss} \to 1$; when $Q \ll R_{\text{obs}}$ (slow dynamics, noisy observations), $K_{ss} \approx \sqrt{Q/R_{\text{obs}}}$.

Connection to AAT quantities. The adaptive tempo for a single observation channel at rate ν is $\mathcal{T} = \nu \cdot K_{ss}$, and the sector parameter in the continuous-time framework is $\alpha = \mathcal{T}$. The bridge is trivial in the scalar case: the gain IS the sector parameter.

Proposition B.2: Matrix Kalman Sector Condition.

Statement. For a linear-Gaussian system with state $x \in \mathbb{R}^n$, observations $o \in \mathbb{R}^m$, observation matrix H, and Kalman gain $K = P^- H^T (HP^- H^T + R_{\text{obs}})^{-1}$, the correction function $F(e) = KH \cdot e$ satisfies the sector condition in the $(P^-)^{-1}$-weighted inner product, restricted to the observable subspace $\mathcal{O} = \text{range}(H^T)$:

$$e^T (P^-)^{-1} KH\, e = e^T H^T S^{-1} H\, e \geq \alpha \cdot e^T (P^-)^{-1} e \tag{K.5}$$

Derived (matrix-Kalman-sector, from Kalman covariance reduction)

where $S = HP^- H^T + R_{\text{obs}}$ is the innovation covariance, and:

$$\alpha = 1 - \lambda_{\max}(P_{t|t}\, P_{t|t-1}^{-1}) \tag{K.6}$$

restricted to observable directions. In unobservable directions ($e \in \ker(H)$), $\alpha = 0$.

Proof. Define the $(P^-)^{-1}$-weighted inner product: $\langle u, v \rangle_{P^-} = u^T (P^-)^{-1} v$. In this inner product the sector product becomes:

$$\langle e, KH\, e \rangle_{P^-} = e^T (P^-)^{-1} KH\, e \tag{K.7}$$

Substituting $K = P^- H^T S^{-1}$:

$$e^T (P^-)^{-1} P^- H^T S^{-1} H\, e = e^T H^T S^{-1} H\, e \tag{K.8}$$

Since $S \succ 0$, the matrix $H^T S^{-1} H$ is symmetric positive semidefinite, with null space $\ker(H)$. So $\langle e, KH\, e \rangle_{P^-} \geq 0$ for all e, with equality iff $He = 0$.

For the sector bound, we need $e^T H^T S^{-1} H\, e \geq \alpha \cdot e^T (P^-)^{-1} e$. From the Kalman covariance identity $KH = I - P(P^-)^{-1}$ (where $P = P_{t|t}$ is the posterior covariance), the eigenvalues of KH in the $(P^-)^{-1}$-weighted sense are $1 - \lambda_i(P(P^-)^{-1})$, where λ_i denotes generalized eigenvalues.

Since $P \preceq P^-$ (the posterior is no less certain than the prior), $\lambda_i(P(P^-)^{-1}) \leq 1$, so all eigenvalues of KH are non-negative. In observable directions, $P \prec P^-$ strictly, giving $\alpha > 0$. In unobservable directions, $P = P^-$ (no information gained), so $\alpha = 0$. □

Fully observable diagonal case. When $H = I, A = I, Q = \text{diag}(q_1, \ldots, q_n)$, $R_{\text{obs}} = \text{diag}(r_1, \ldots, r_n)$, the problem decouples into n scalar problems (each governed by Prop B.1), and:

$$\alpha = \min_{i=1}^{n} K_i = \min_{i=1}^{n} \frac{-q_i + \sqrt{q_i^2 + 4q_i r_i}}{2r_i} \tag{K.9}$$

Derived (diagonal-Kalman-sector)

The bottleneck is the dimension with the worst signal-to-noise ratio, consistent with `#result-per-dimension-persistence`.

The weighted-norm subtlety. The Lyapunov proofs in A use the Euclidean norm ($V = \frac{1}{2}\|e\|^2$), while the matrix Kalman sector condition holds in the $(P^-)^{-1}$-weighted norm. For fully observable systems with bounded condition number $\kappa(P^-)$, the norms are equivalent:

$$\alpha_{\text{Euclidean}} \geq \alpha_{\text{weighted}} / \kappa(P^-) \tag{K.10}$$

The Lyapunov results remain valid with this quantitative adjustment. The weighted Lyapunov function $V(e) = \frac{1}{2}e^T(P^-)^{-1}e$ yields the tighter bound directly.

Proposition B.3: Bridge Theorem.

Statement. Given the gain-based update $M_t = M_{t-1} + \eta^* \cdot g(\delta_t)$ (`#emp-update-gain`), with H mapping state-space corrections to observation-space mismatch reduction, the induced correction function $F(\delta) = \eta^* \cdot H\, g(\delta)$ satisfies the sector condition (GA-3) with parameter $\alpha > 0$ whenever:

Derived (bridge-theorem, from update-gain + B1)

(B1) Directional fidelity. The composite mismatch transform preserves the mismatch-reducing direction:

$$\delta^T H\, g(\delta) \geq c\|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{K.11}$$

for some $c > 0$. The sector parameter is:

$$\alpha = \eta^* \cdot c_{\min}, \qquad c_{\min} = \inf_{\|\delta\| \leq R} \frac{\delta^T H\, g(\delta)}{\|\delta\|^2} \tag{K.12}$$

Which object. In this proposition δ denotes the estimation error $\bar{\delta} = \bar{o} - \hat{o}$ of `#def-mismatch-signal`, the state the sector condition is about. The update itself is driven by the innovation $\bar{\delta} + \varepsilon$, with ε the channel noise. The correction function is the *expected* correction given the estimation error, $F(\bar{\delta}) = \eta^* H\, \mathbb{E}_\varepsilon[g(\bar{\delta} + \varepsilon)]$; its zero-mean fluctuation $\eta^* H\,(g(\bar{\delta}+\varepsilon) - \mathbb{E}_\varepsilon[g(\bar{\delta}+\varepsilon)])$ is gain-injected noise and belongs to the disturbance. Its magnitude scales with the gain η^* for every g; for nonlinear g its covariance depends on $\bar{\delta}$, so the Model S disturbance is state-dependent rather than the additive $\sigma_w\, dW$ of A Prop A.1S, whose killed-moment and finite-horizon bounds then hold with $n\sigma_w^2$ replaced by the supremum over $\mathcal{B}_R$ of the trace of the diffusion covariance (the exit-time comparison of Prop A.1S-τ assumes isotropic additive noise and is not transferred). For linear g the expectation is $g(\bar{\delta})$ itself, the injected noise is additive with intensity η^{*2} times the channel's, and B1 reads exactly as stated. For nonlinear g, B1 is a condition on the noise-averaged transform, and g in the statement and the proof stands for $\mathbb{E}_\varepsilon[g(\cdot + \varepsilon)]$.

Proof. The gain-based update changes the model state by $\Delta M = \eta^* \cdot g(\delta)$. The induced change in predicted observation is $H \cdot \Delta M = \eta^* \cdot H\, g(\delta)$, so the expected correction function (the reduction of the estimation error per step) is:

$$F(\delta) = \eta^* \cdot H\, g(\delta) \tag{K.13}$$

The sector product is:

$$\delta^T F(\delta) = \eta^* \cdot \delta^T H\, g(\delta) \tag{K.14}$$

By B1: $\delta^T H\, g(\delta) \geq c_{\min}\|\delta\|^2$. Since $\eta^* > 0$ (the agent updates), we have:

$$\delta^T F(\delta) \geq \eta^* \cdot c_{\min}\|\delta\|^2 = \alpha\|\delta\|^2 \qquad \square \tag{K.15}$$

When B1 holds by construction. For optimal Bayesian updates (Kalman, conjugate families, exponential families), the posterior update minimizes expected loss, ensuring the correction aligns with the mismatch. B1 holds by optimality, not by assumption. The gain principle $\eta^* = U_M/(U_M + U_o)$ determines only the magnitude; the direction is guaranteed by the update rule's optimality. For gradient descent on a convex loss, B1 follows from gradient monotonicity (the defining property of convexity).

Proposition B.4: Gradient Equivalence.

Statement. For any agent updating via gradient descent on a differentiable loss L with learning rate η and minimizer M^* (where $\nabla L(M^*) = 0$), there are two distinct results corresponding to the one-point and two-point sector conditions:

(B.4-i) One-point sector ⇐ strong convexity (one direction only). The one-point sector condition at M^* — equivalent to AAT's GA-3 / A2' as stated in `#form-sector-condition` — is

$$\delta^T F(\delta) \geq \alpha \|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{K.16}$$

evaluated at fixed equilibrium M^*. Local (α/η)-strong convexity of L on $\mathcal{B}_R(M^*)$ implies this one-point sector condition with $\alpha = \eta\mu$, but the converse fails: there exist losses satisfying the one-point sector condition that are *not* strongly convex on any neighborhood of M^* (counterexample below).

(B.4-ii) Two-point sector ⇔ strong convexity (full equivalence). The two-point / incremental sector condition — DA2'-inc in P and the bridge-lemma precondition in `#form-composition-closure` — is

$$(F(\delta_1) - F(\delta_2))^T(\delta_1 - \delta_2) \geq \alpha \|\delta_1 - \delta_2\|^2 \quad \text{for } \delta_1, \delta_2 \in \mathcal{B}_R(M^*) \tag{K.17}$$

Derived (sector-convexity-equivalence, two-point form)

Under this strengthened condition the iff holds:

$$\text{Two-point sector with } (\alpha, R) \iff L \text{ is } (\alpha/\eta)\text{-strongly convex on } \mathcal{B}_R(M^*) \tag{K.18}$$

with $\alpha = \eta\mu$ and $\mu = \inf_{\delta \in \mathcal{B}_R(M^*)} \lambda_{\min}(\nabla^2 L(M^* + \delta))$. The basin radius R is the largest ball around M^* where $\nabla^2 L$ remains positive definite.

Proof of (B.4-ii). The correction function for gradient descent is $F(\delta) = \eta \cdot \nabla L(M^* + \delta)$ (continuous-time form absorbing event rate ν into $\mathcal{T} = \nu \cdot \eta$). The two-point sector condition becomes:

$$\eta \cdot (\nabla L(M^* + \delta_1) - \nabla L(M^* + \delta_2))^T(\delta_1 - \delta_2) \geq \alpha \|\delta_1 - \delta_2\|^2 \tag{K.19}$$

Dividing by $\eta > 0$ gives gradient monotonicity with modulus $\mu = \alpha/\eta$. By Nesterov 2004, Theorem 2.1.10, L is μ-strongly convex on $\mathcal{B}_R(M^*)$ iff this gradient monotonicity holds for all $x, y \in \mathcal{B}_R(M^*)$. The equivalence is bidirectional:

($\Rightarrow$) Two-point sector with (α, R) yields gradient monotonicity with $\mu = \alpha/\eta$, hence strong convexity.

($\Leftarrow$) μ-strong convexity gives gradient monotonicity, multiplying by η recovers the two-point sector condition with $\alpha = \eta\mu$. □

Proof of (B.4-i) one direction. Setting $\delta_1 = \delta, \delta_2 = 0$ in the two-point condition and using $\nabla L(M^*) = 0$ recovers the one-point form $\delta^T F(\delta) \geq \alpha \|\delta\|^2$. So strong convexity (which is equivalent to the two-point sector by (B.4-ii)) implies the one-point sector. The reverse direction does not hold; see the counterexample below. □

Counterexample showing one-point ⇏ strong convexity. Let $L : \mathbb{R} \to \mathbb{R}$ have $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ with $L'(0) = 0$ (so $x^* = 0$). The one-point sector at x^* is

$$x \cdot L'(x) = x^2 (1 + \tfrac{1}{2}\sin(10x)) \geq \tfrac{1}{2}x^2, \tag{K.20}$$

so the one-point sector holds globally with $\mu = \frac{1}{2}$. But $L''(x) = 1 + \frac{1}{2}\sin(10x) + 5x\cos(10x)$ takes values such as $L''(\pi/10) = 1 - \pi/2 \approx -0.57 < 0$, so L is not convex on any neighborhood of x^* that contains $\pi/10$. The one-point sector at the equilibrium is genuinely weaker than full local strong convexity — a structural distinction, not a smoothness artifact. The full two-point sector fails on this L for any pair (δ_1, δ_2) straddling a region where $L'' < 0$ (gradient monotonicity fails there), so (B.4-ii) correctly identifies this L as a non-instance.

The α factorization (under (B.4-ii)). The decomposition $\alpha = \eta \cdot \mu$ separates agent design (η, responsiveness) from environment structure (μ, loss curvature). The adaptive tempo maps as $\alpha \propto \mathcal{T}$: higher gain or steeper curvature means faster mismatch correction.

Where each direction lands in AAT. The one-point form is what `#form-sector-condition`'s A2' ((A2') applied at $\delta^* = 0$) requires for Lyapunov persistence (via A) — and what AAT's six persistence-flavored results inherit through `#result-sector-persistence-template` (T2). The two-point / incremental form is what `#form-composition-closure`'s bridge lemma requires (DA2'-inc) for full-update-map contraction at the composite level. Strong convexity sits at the strict end of this scale: it implies both. The one-point ⇏ two-point counterexample above is exactly why AAT distinguishes the two — Lyapunov persistence is available for some agent classes (variational, PID-bounded-plant) where bridge-lemma contraction is not.

Loss Function Classification.

The gradient equivalence (Prop B.4) classifies which loss functions satisfy GA-3, and with what parameters:

Table K.1

Loss class	*Strong convexity*	α	*Basin R*	*Status*
Quadratic (Kalman, linear regression)	Global: $\mu = \lambda_{\min}(H)$	$\eta \cdot \lambda_{\min}(H)$	∞	Exact
Exponential family (natural params), bounded scope	Pointwise on the interior; uniform on any compact $\Theta_0 \subset \text{int}(\Theta)$: $\mu_0 = \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$	$\eta \cdot \mu_0$	Θ_0 (compact / interior-bounded)	Exact within Θ_0
L2-regularized convex loss	Global: $\mu \geq \lambda$	$\eta \cdot \lambda$ (floor)	∞	Exact
Unregularized logistic	Global convex, not strongly	$\eta \cdot \mu(R)$, decaying	∞ (but $\alpha \to 0$)	Local
Quasi-convex	$\delta^T \nabla L \geq 0$ but ratio $\to 0$	$\eta \cdot \mu(R) \to 0$	Finite effective	Local
Non-convex within basin	Local: $\mu > 0$ in basin	$\eta \cdot \mu_{\text{local}}$	Distance to nearest saddle	Local
Non-convex beyond basin	Fails: $\lambda_{\min}(\nabla^2 L) < 0$	N/A	N/A	Fails

Key observations:

1. *Exponential family models in natural parameter form satisfy GA-3 pointwise on the interior of the natural-parameter space, and uniformly on any compact subset of the interior.* The Hessian is the Fisher information matrix $\mathbf{I}(\theta)$, which is positive definite in $\text{int}(\Theta)$ — but pointwise positive definiteness does not imply a uniform global lower bound on $\lambda_{\min}(\mathbf{I}(\theta))$. The Poisson natural parameter is the canonical counterexample: $\mathbf{I}(\theta) = e^\theta$ on $\Theta = \mathbb{R}$, so $\inf_{\theta \in \mathbb{R}} \mathbf{I}(\theta) = 0$ and no global sector constant exists. A uniform $\mu_0 > 0$ — and hence a global $\alpha = \eta\mu_0$ — requires either (i) restricting the operating region to a compact $\Theta_0 \subset \text{int}(\Theta)$ on which $\mu_0 := \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$, or (ii) verifying a uniform Fisher lower bound for the family in question (Gaussian-mean and Beta-Bernoulli, whose Fisher information is bounded below on the interior of Θ, qualify; Poisson

and Gamma-shape do not on the full natural-parameter line). The compact-scope condition is verifiable per-application from the agent's prior support and corresponds to taking R as the diameter of Θ_0 in the natural-parameter coordinate, matching A2's local-region structure (`#form-sector-condition`).

2. *L2 regularization transforms locally strongly convex losses into globally strongly convex ones.* The regularization parameter λ provides a floor: $\alpha \geq \eta\lambda$ everywhere.

3. *Non-convex losses satisfy GA-3 within each basin of attraction.* The basin radius R is the distance from the local minimum to the nearest inflection surface (where $\lambda_{\min}(\nabla^2 L) = 0$). Beyond the basin, the sector condition fails — this IS the structural-adaptation trigger from `#result-structural-adaptation-necessity`.

Failure Modes.

The bridge fails in precisely five cases. These characterize the exact conditions under which the gain-based update does NOT produce a correction function satisfying the sector condition.

FM-1: Directional Infidelity. The mismatch transform g rotates the correction away from the mismatch direction, so $\delta^T H g(\delta) \leq 0$.

Example. If $\delta \in \mathbb{R}^2$ and $g(\delta) = R_{90}\delta$ (a 90-degree rotation), then $\delta^T g(\delta) = 0$ for all δ. The correction is perpendicular to the mismatch and never reduces it.

When it occurs. Pathological parameterizations where the model-space update direction is misaligned with observation-space mismatch. For optimal Bayesian updates, B1 holds by construction — the posterior minimizes expected loss.

FM-2: Gain Collapse. $\eta^* \to 0$ while $\rho > 0$, so $\alpha \to 0$ and the persistence condition $\alpha > \rho/R$ eventually fails. An agent accumulating experience in a non-stationary environment will eventually have gain too small to track changes. This is not a bridge failure but a persistence-condition failure — the bridge holds ($\alpha = \eta^* \cdot c > 0$ for finite experience), but α falls below ρ/R. See `#emp-update-gain`.

FM-3: Nonlinear Saturation. The correction function g saturates at large $\|\delta\|$, so the sector ratio $\delta^T g(\delta)/\|\delta\|^2$ decays. For a saturating function such as $g(\delta) = \tanh(\delta)$, the local sector parameter is:

$$\alpha(R) = \eta^* \cdot \frac{\tanh(R)}{R} \tag{K.21}$$

For small R, $\alpha \approx \eta^*$ (linear regime). For large R, $\alpha \approx \eta^*/R$ (saturated regime). The sector condition holds locally — the valid region shrinks as the nonlinearity strengthens. This is exactly what the local form A2' in `#form-sector-condition` anticipates.

FM-4: Unobservable Directions. When $\ker(H) \neq \{0\}$ (fewer observations than state dimensions), the correction has no effect on mismatch components in $\ker(H)$. The sector condition holds only in the observable subspace $\mathcal{O} = \text{range}(H^T)$. This is a structural limitation: the agent cannot correct errors it cannot observe. See `#der-observability-dominance`.

For persistence in the full state space, the system must be *detectable*: all unstable modes must be observable. Stable unobservable modes decay naturally without correction.

FM-5: Model Misspecification. The model class does not contain the truth, so the gradient direction is systematically wrong. B1 fails because the correction aims at the wrong target. The sector parameter degrades proportionally to the misspecification. This is the `#result-structural-adaptation-necessity` trigger: the correction function fails not because the gain is wrong but because the correction *direction* is wrong.

Simulation Results Summary.

Six numerical experiments validate the gradient equivalence (Prop B.4) and the loss-function classification.

Experiment 1: Quadratic Loss (Global Strong Convexity). $L(w) = \frac{1}{2}(w - w^*)^T H(w - w^*)$ with $w \in \mathbb{R}^5$, Hessian eigenvalues $\{0.5, 1.0, 2.0, 3.0, 5.0\}$, $\eta = 0.1$. The sector ratio $\delta^T F/\|\delta\|^2$ at every gradient-descent step fell within $[\eta\mu, \eta L] = [0.05, 0.50]$, confirming Prop B.4 exactly. No step violated the bound.

Experiment 2: Logistic Regression (Position-Dependent Convexity). Logistic regression with $d = 2, N = 5000$ samples. Unregularized: sector ratio positive along the entire trajectory, with local μ ranging from 0.076 (at MLE) to higher values far from it. With L2 regularization ($\lambda = 0.1$): sector ratio bounded below by $\eta\lambda = 0.03$ at every point, confirming the global floor.

Experiment 3: Gaussian Mixture (Non-Convex, Basin Structure). Two-component Gaussian mixture. Within basin ($r < 3.1$): Hessian positive definite, sector ratio positive, gradient descent converges to correct optimum. At basin boundary ($r \approx 3.1$): Hessian eigenvalue crosses zero. Beyond basin ($r > 3.5$): sector ratio negative, gradient descent converges to wrong optimum. Basin boundary confirmed as sharp transition, matching the `#result-structural-adaptation-necessity` trigger.

Experiment 4: Local Sector Parameter Measurement. Same Gaussian mixture. Sector parameter $\alpha(r)$ decreases monotonically from 181.4 (at $r = 0.5$) to 125.8 (at $r = 3.0$), then drops to zero at $r \approx 3.1$. The decay is moderate within the basin (69% retention at the boundary); the transition at the boundary is sharp.

Experiment 5: Exponential Family (Poisson, Natural Parameters). Poisson in natural parameter space. Sector ratio positive at every step, ranging from 0.00177 to 0.00240 (with $\eta = 0.001$). Confirms global strong convexity of exponential family losses in natural parameterization.

Experiment 6: Quasi-Convex and Multi-Modal Losses. *Quasi-convex* ($L(w) = -1/(1+w^2)$): sector ratio always positive but decays as $O(1/w^4)$. No uniform lower bound exists — $\alpha(R) \to 0$, making persistence bounds vacuous for large R.

Multi-modal ($L(w) = \sin(w) + 0.01w^2$): sector condition holds within each basin ($|\delta| < 3.0$), fails at basin boundary (gradient reversal at local maximum). Basin radius ≈ distance from local minimum to nearest local maximum.

Epistemic Status.

Propositions B.1 and B.2 are *exact* — they follow from standard Kalman filter algebra. The only subtlety is the weighted norm in the matrix case (B.2), which is a quantitative refinement, not a qualitative gap. Both are standard results in estimation theory reframed in sector-condition language.

Proposition B.3 (Bridge Theorem) is a *conditional derivation*: exact under B1 (directional fidelity). B1 holds by construction for optimal Bayesian updates (the posterior minimizes expected loss, ensuring the correction aligns with the mismatch). For approximate update rules, B1 is a design condition, not a global assumption. The condition is transparent and checkable for specific systems.

Proposition B.4 (Gradient Equivalence) splits along the one-point / two-point distinction. (B.4-ii) — the iff between strong convexity and the *two-point / incremental* sector condition (DA2'-inc) — is an *exact mathematical equivalence*: bidirectional via Nesterov 2.1.10. (B.4-i) — the relation to AAT's one-point sector A2' as stated in `#form-sector-condition` — is *one-directional only*: strong convexity implies the one-point sector at M^*, but not conversely. The Codex-style counterexample $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ satisfies the one-point sector globally yet has L'' negative on

intervals, ruling out a converse. The loss-function classification table is read accordingly: rows that supply strong convexity satisfy both forms; rows that supply only one-point sector (none in the current table — strong convexity is the universal column) would not transfer to two-point. The simulation experiments confirm the theoretical predictions across all six loss classes tested; no violations were observed where the theory predicts the sector condition holds.

Max attainable: *conditional* for the bridge (B1 is inherent — pathological update rules exist), *exact* for the gradient equivalence in its two-point form (B.4-ii) and *exact one-direction* for its one-point form (B.4-i). The condition cannot be removed: there exist correction functions that violate the sector condition (FM-1 through FM-5), and there exist losses that satisfy the one-point sector at M^* without being strongly convex (the Codex-style counterexample), which is why the one-point $\not\Rightarrow$ two-point gap is structural rather than a smoothness artifact.

What remains open. (1) Non-gradient agents (PID controllers, rule-based systems, human judgment): GA-3 remains an empirical claim for these agent classes. (2) Time-varying α: when the gain adapts (Adam, RMSprop, Kalman transient), $\alpha(t)$ varies; the Lyapunov analysis extends to time-varying $\alpha(t) \geq \underline{\alpha} > 0$ via standard results (Khalil 2002, Chapter 8). (3) Stochastic gradients: SGD satisfies the sector condition *in expectation*; the per-step noise enters as effective disturbance in the Prop A.1S framework.

Discussion.

The formal chain is closed for well-designed agents. The prediction chain becomes:

$$\text{gain principle+B1 (directional fidelity)} \xrightarrow{\text{Prop B.3 (exact)}} \text{sector condition (GA-3)} \xrightarrow{\text{Props A.1, A.1S, A.2 (exact)}} \text{persistence, reserve, scaling} \tag{K.22}$$

For optimal Bayesian updates, B1 holds by construction, making the full chain a derivation. For gradient agents, B1 is equivalent to local strong convexity — a well-characterized property.

GA-3 is grounded, not floating. The assumption load shifts from "the correction function satisfies the sector condition" (what correction function?) to "the update rule has directional fidelity" (the correction at least points toward reality). For the important agent classes — Kalman filters, conjugate posteriors, exponential families, gradient descent on convex losses — this is automatic.

Basin boundary = structural adaptation trigger. For gradient agents with non-convex losses, the basin radius R is the convexity radius of the loss landscape. When mismatch exceeds R, the correction function reverses direction. This is the `#result-structural-adaptation-necessity` trigger with a precise geometric characterization: structural adaptation is needed when the agent crosses an inflection surface of its loss landscape.

Appendix L

Derivation: Recursive Update — Uniqueness Derivation

Appendix derivation backing `#der-recursive-update`. The claim being established is *uniqueness*: under three constraints — arrow of time, mediated access, and state completeness — the recursive form $M_{\tau+} = f(M_{\tau-}, e_\tau)$ is the only update form realizable, not merely one consistent option. The structure is an elimination argument over the universe of information potentially accessible at event time τ: the environment state, the full chronica, all prior model states, the current event e_τ, and future events.

The three constraints do different kinds of work. **C1 (arrow of time, from `#post-causal-structure`)** rules out dependence on future events: in a classical universe such information is simply not accessible, and predictions about the future, however accurate, are internal computations belonging to $M_{\tau-}$, not future information. **C2 (mediated access, from `#def-agent-environment` and `#def-observation-function`)** rules out access to Ω_τ except through the observation channel e_τ. **C3 (state completeness, from `#form-agent-model`)** compresses all prior-past information — the full chronica and every earlier model state — into $M_{\tau-}$, because by construction M is the agent's complete internal state and any information available to the update but not in M would mean M was misspecified. After all three eliminations only $(M_{\tau-}, e_\tau)$ remains in the dependency set. A measure-theoretic version via the Doob–Dynkin lemma — restricting the agent's sigma-algebra to $\sigma(M_{\tau-}, e_\tau)$ — gives a cleaner technical proof.

The derivation then mounts seven adversarial attacks to test whether the result really holds. Simultaneous events are absorbed by treating the bundle as a single e_τ (form preserves). Continuous environmental coupling is genuinely outside the event-driven formulation — but the analogous result reached by the same three-constraint argument in continuous time is the state-space representation $\dot{M} = g(M, u)$ from classical control theory, so the underlying structure survives. The objection that C3 is circular — anything not in M gets absorbed into M by definition — is acknowledged as the deepest objection, and the derivation is explicit about it: the real content is an *analytical commitment* that by defining M as complete one commits to Markovian analysis, which then makes `#def-model-sufficiency` the right quality metric. Shared state, external randomness, time-dependence, and agents that store full history each reduce to the recursive form by appropriate enlargement of the model space or interpretation of f.

The honest epistemic accounting at the end is load-bearing. **C1** (physical postulate) and **C2** (scope definition) do *eliminative* work — they rule out physically impossible and scope-excluded update forms. **C3** (analytical commitment) does not eliminate; it is *definitional* and cannot be "violated" because any apparent violation is absorbed by expanding M. The recursive form's uniqueness is therefore conditional on accepting the three-constraint set, not on the constraints being independently inescapable — C3 in particular *could* be refused, at the cost of leaving AAT's scope and accepting non-Markovian analysis. The Markov property is not a discovered law of adaptive systems but the unique form compatible with the modeling commitment of `#form-agent-model`.

Setup.

We work within AAT's scope (`#scope-adaptive-system`): an agent coupled to an environment Ω through observation and action channels, whose world can still show it something it does not yet know.

Universe of information at event time τ. The following information exists (in the broadest ontological sense) at the moment event e_τ occurs:

Table L.1

Information	*Description*
Ω_τ	The environment state
$\mathcal{C}_{\tau^-}$	The complete interaction history (`#def-chronica`) up to (but not including) e_τ
$\{M_{\tau'}\}_{\tau' \leq \tau^-}$	The agent's prior internal states, culminating in M_{τ^-}
e_τ	The current event (observation arriving or action completing)
$\{e_{\tau'}\}_{\tau' > \tau}$	Future events (not yet occurred)

The question: of these, which can the update M_{τ^+} depend on?

The Three Constraints.

Constraint 1 — Arrow of time (`#post-causal-structure` postulate). Events are temporally ordered and this ordering is irreversible. An update occurring at time τ cannot depend on events that have not yet occurred:

$$M_{\tau^+} \text{ cannot depend on } \{e_{\tau'}\}_{\tau' > \tau} \tag{L.1}$$

This is a physical constraint — the most primitive one. In a classical universe, information from the future is simply not available. Even if the agent can *predict* future events, those predictions are part of M_{τ^-} (they are internal computations, not future information).

Constraint 2 — Mediated access (`#def-agent-environment`). The agent cannot access Ω_τ except through the event e_τ, which is a function of Ω_τ through the observation channel (via `#def-observation-function`), usually a lossy one:

$$M_{\tau^+} \text{ cannot depend on } \Omega_\tau \text{ except through } e_\tau \tag{L.2}$$

This is the information-loss boundary of `#def-agent-environment`: the observation channel is the agent's only interface with the environment. It does not require the channel to be lossy. If e_τ reveals Ω_τ exactly, the update still depends on the environment only through e_τ, and the agent can remain in the adaptive scope through what it does not know about the law (`#scope-adaptive-system`).

Constraint 3 — State completeness (`#form-agent-model`). M_{τ^-} is the agent's *complete* internal state just before event e_τ. There is no information about the agent's past that is available to the update mechanism but not encoded in M_{τ^-}:

$$M_{\tau^+} \text{ cannot depend on } \mathcal{C}_{\tau^-} \text{ or } \{M_{\tau'}\}_{\tau' < \tau^-} \text{ except through } M_{\tau^-} \tag{L.3}$$

This constraint does the most interesting work and deserves careful examination (see Discussion below).

The Derivation.

Result (Recursive Update Uniqueness). Under Constraints 1–3, the model update at event time τ must have the form

$$M_{\tau+} = f(M_{\tau-}, e_\tau) \tag{L.4}$$

for some (possibly stochastic) function $f : \mathcal{M} \times \mathcal{E} \to \mathcal{M}$. No other update form is consistent with the three constraints.

Derivation. Consider the most general possible update. The updated state $M_{\tau+}$ is a function of *all accessible information*:

$$M_{\tau+} = F(\text{accessible information at } \tau) \tag{L.5}$$

We characterize the accessible information by eliminating what is not accessible.

(i) Eliminate future events. By C1 (arrow of time), $\{e_{\tau'}\}_{\tau' > \tau}$ is not accessible.

After this elimination, the candidate dependency set is:

$$\{\Omega_\tau,\ \mathcal{C}_{\tau-},\ \{M_{\tau'}\}_{\tau' \leq \tau-},\ e_\tau\} \tag{L.6}$$

(ii) Eliminate direct environment access. By C2 (mediated access), the agent cannot access Ω_τ except through the event e_τ. Any information from Ω_τ that reaches the agent does so through e_τ — already in the dependency set.

After this elimination:

$$\{\mathcal{C}_{\tau-},\ \{M_{\tau'}\}_{\tau' \leq \tau-},\ e_\tau\} \tag{L.7}$$

(iii) Reduce past information to $M_{\tau-}$. By C3 (state completeness), $M_{\tau-}$ is the agent's complete internal state. Every element of $\mathcal{C}_{\tau-}$ and every prior model state $M_{\tau'}$ ($\tau' < \tau^-$) that could influence the update can do so *only through* its effect on $M_{\tau-}$. The agent's internal state evolves through a sequence of updates; the cumulative effect of all prior events is exactly $M_{\tau-}$. The raw events that produced this state are no longer separately available — they were "consumed" by the update mechanism and their information (to the extent it was retained) is now encoded in $M_{\tau-}$.

Could the agent maintain a separate log of raw events outside of M? It could — but that log *is part of* M. Whatever information the agent retains in any form — model parameters, cached data, raw event buffers, metadata — is by definition part of its complete internal state $M_{\tau-}$. If something is available to the update mechanism and not in $M_{\tau-}$, then $M_{\tau-}$ was not the complete state — contradicting C3.

After this elimination:

$$\{M_{\tau-},\ e_\tau\} \tag{L.8}$$

Therefore:

$$M_{\tau+} = F(M_{\tau-}, e_\tau) \equiv f(M_{\tau-}, e_\tau) \tag{L.9}$$

This is the unique form: no information beyond $(M_{\tau-}, e_\tau)$ is accessible under the three constraints, so no update form depending on anything else is realizable. □

Corollary (Between-events dynamics). Between events, no new event e arrives. The same argument applies with e_τ removed from the accessible set:

$$\frac{dM}{d\tau} = g(M_\tau) \tag{L.10}$$

The agent's internal evolution between events (prediction, decay, internal simulation) depends only on the current state. □

Corollary (Serial special case). When observations and actions alternate at a uniform rate on a single channel, each event e_t is the pair (o_t, a_{t-1}). The update becomes:

$$M_t = f(M_{t-1}, o_t, a_{t-1}) \tag{L.11}$$

This is the familiar discrete-time form. □

Information-Set Formalization.

For readers who prefer a measure-theoretic framing:

The agent's **information set** at time τ is the sigma-algebra $\mathcal{I}_\tau^{agent}$ — the collection of events (in the probability-theoretic sense) about which the agent can condition its update.

- **C1** restricts $\mathcal{I}_\tau^{agent} \subseteq \sigma(\{e_{\tau'} : \tau' \leq \tau\} \cup \{\Omega_\tau\} \cup \{M_{\tau'} : \tau' \leq \tau^-\})$ — no future information.
- **C2** further restricts: $\sigma(\Omega_\tau) \setminus \sigma(e_\tau)$ is not in $\mathcal{I}_\tau^{agent}$ — the agent cannot condition on aspects of Ω_τ not captured by e_τ.
- **C3** further restricts: $\sigma(\{e_{\tau'} : \tau' < \tau\} \cup \{M_{\tau'} : \tau' < \tau^-\}) \subseteq \sigma(M_{\tau^-})$ from the agent's perspective.

After all three restrictions: $\mathcal{I}_\tau^{agent} = \sigma(M_{\tau^-}, e_\tau)$.

By the Doob–Dynkin lemma[^kallenberg2002], any $\sigma(M_{\tau^-}, e_\tau)$-measurable random variable is a (Borel) function of (M_{τ^-}, e_τ). Therefore $M_{\tau^+} = f(M_{\tau^-}, e_\tau)$ for some measurable f. □

Attempts to Break the Result.

Before trusting the proof, seven counterexample attacks:

Attack 1: Simultaneous events. Two events arrive at exactly the same time: $e_\tau^{(1)}$ and $e_\tau^{(2)}$. The update has three arguments: $f(M_{\tau^-}, e_\tau^{(1)}, e_\tau^{(2)})$.

Verdict: Not deep — `#form-event-driven-dynamics` defines events as atomic. If we allow bundled events, the form holds with e_τ as a set. Reveals that "event" needs careful definition, but the form is preserved.

Attack 2: Continuous environmental influence. An agent embedded in a physical system experiences continuous forces (gravity, temperature, electromagnetic fields). These aren't "events" in `#form-event-driven-dynamics`'s sense; they're continuous signals. The true dynamics would be $dM/d\tau = g(M_\tau, o(\tau))$ where $o(\tau)$ is a continuous observation stream.

Verdict: Genuine limitation of the event-driven formulation. The between-events corollary $dM/d\tau = g(M_\tau)$ holds only when the agent is truly isolated between events. For continuous coupling, the analogous result is the general state-space representation $\dot{M} = g(M, u)$ from control theory — arrived at by the same three constraints. The event-driven version is a special case for digital/sampled systems.

Attack 3: The C3 circularity. C3 defines M as the agent's complete internal state. Any apparent counterexample is dissolved by expanding M. Consider: an agent has a "model" (neural net weights) and a "replay buffer" (stored raw events). C3 says $M = (\text{weights}, \text{buffer})$. The model space is just larger than you thought.

Verdict: The deepest objection. The proof essentially: (1) Define M to be everything the agent has. (2) Observe the update can only use what the agent has. (3) Therefore $f(M_{\tau^-}, e_\tau)$. The real content is the *analytical commitment*: by defining M as complete, we commit to Markovian analysis, which then makes `#def-model-sufficiency` the right quality metric. See Epistemic Status below.

Attack 4: Shared state between agents. Agents A and B share a common memory bank (shared database). The clean resolution is the multi-agent framework: the shared memory is part of the *composite* system's state, and each agent's interaction with it is mediated by events (reads and writes). Not a true counterexample but highlights that C3 requires careful delineation of agent boundaries.

Attack 5: External randomness not in e_τ. Hardware thermal noise used in the update. The stochastic case $M_{\tau^+} \sim P(\cdot \mid M_{\tau^-}, e_\tau)$ is a special case of f where f is a randomized function. The *form* — dependence on exactly (M_{τ^-}, e_τ) — is preserved. The result statement should explicitly allow stochastic f.

Attack 6: Time-dependent updates. Could f depend on the timestamp τ itself? Yes — consistently. The event e_τ in `#form-event-driven-dynamics` carries a timestamp: $e_\tau = (\text{type}, \text{channel}, \text{payload}, \tau)$. So time-dependence enters through e_τ. Alternatively, the agent may maintain an internal clock as part of M_{τ^-}. Either way, $f(M_{\tau^-}, e_\tau)$ accommodates time-dependence.

Attack 7: Agents that store full history. An agent with $M_{\tau^-} \supseteq \mathcal{C}_{\tau^-}$ is entirely consistent. The model space $\mathcal{M}$ is simply large enough to include the raw history. The `#form-information-bottleneck` argues compression is *wise* — but the recursive update form holds regardless of compression level.

What Is Derived vs. What Is Chosen.

Table L.2

Property	*Source*	*Strength*
Constraint C1 (arrow of time: update depends on τ^-, not future events)	Physical law — not a formulation choice	Postulate (physical)
Constraint C2 (mediated access: update depends on e_τ, not raw Ω_τ)	Scope definition of AAT	Postulate (scope-defining)
Constraint C3 (state completeness: M_{τ^-} summarizes the agent's relevant past)	Analytical commitment — the definition of M as complete	Definition
Recursive form $M_\tau = f(M_{\tau^-}, e_\tau)$	C1 + C2 + C3	Proved (unique form compatible with the three constraints)
Future-dependent updates eliminated	C1 alone	Derived (direct consequence)
Ω_τ-dependent updates eliminated	C2 alone	Derived (direct consequence)
Full-history-dependent updates reducible to recursive form	C3 + any choice of $M \supseteq \mathcal{C}_{\tau^-}$	Proved (compatibility, not elimination)
Markov property of the update	C3 (completeness) + recursive form	Proved (follows from C3 definition)
Seven attack counterexamples (simultaneous events, continuous coupling, C3 circularity, shared state, external randomness, time-dependence, full history)	Case-by-case reduction to the recursive form	Proved (each)
C3 is definitional, not eliminative	Analysis of what C3 asserts vs. what it rules out	Discussion-grade (clarifying observation)

The dividing line: C1 and C2 do genuine *eliminative* work — they rule out physically or scope-excluded update forms. C3 is a *definitional commitment* that forces the Markov structure by making M complete by construction; it cannot be "violated" because any apparent violation means M was misspecified. The recursive form's uniqueness is therefore conditional on the three-constraint set being accepted, not on the constraints being independently inescapable — C3 in particular could be refused (yielding non-Markovian analysis), at the cost of leaving AAT's scope.

Epistemic Status.

The result is correct but partly definitional. The three constraints have different epistemic characters:

Table L.3

Constraint	*Character*	*Can it be violated?*
C1 (arrow of time)	Physical law	Not in a classical universe
C2 (mediated access)	Scope definition	Only by leaving AAT's scope
C3 (state completeness)	Analytical commitment	Not without redefining M

C1 and C2 do genuine eliminative work — they rule out update forms that depend on future events or on raw Ω. These are non-trivial constraints.

C3 is a definitional commitment that produces the Markov structure. It cannot be "violated" because any violation is absorbed by expanding M. This is not a weakness — it's the nature of the claim. The result says: *the Markovian analysis is the only one consistent with C1 + C2 + the definition of M as complete*. The alternative — an update that depends on something outside M — is not "wrong" but rather means M was misspecified.

What the result says: C1 eliminates a physically impossible class of updates (future-dependent). C2 eliminates a scope-excluded class (Ω-dependent). After (1) and (2), the *only remaining question* is how the past enters: through the full history $\mathcal{C}_{\tau-}$ or through a compressed state $M_{\tau-}$. C3 says the agent *has* a complete state, and whatever that state is, it's all the agent has. The Markov form follows.

What the result does NOT say: That M must be a lossy compression (the agent could store full history). That the Markov property is "natural" or "optimal" (it's a consequence of how M is defined). That continuous-coupling systems are event-driven (the event framework is one abstraction; $\dot{M} = g(M, u)$ is the more general one, arrived at by the same three constraints).

Discussion.

Recursion as a consequence of completeness. The recursive form is not an assumption bolted on — it follows from the definition of M_t as complete. The sufficiency of the recursive form is precisely what `#def-model-sufficiency` measures: when $S(M_t) = 1$, the recursive update loses nothing.

What this opens. The proof yields the *form*. It immediately invites the follow-up questions that the rest of the theory addresses: What should f preserve? → `#form-information-bottleneck` and `#def-model-sufficiency`. How should f weight new information? → `#emp-update-gain`. When is $\mathcal{M}$ itself inadequate? → `#result-structural-adaptation-necessity`.

Appendix M

Derivation: Decomposition Uniqueness — Underdetermination and Canonicity of the Epistemic/Purposeful Split

Appendix derivation resolving the uniqueness question left open in `#form-complete-agent-state`: is the decomposition $X_t = (M_t, G_t)$ forced, among decompositions preserving directed separation? The answer has two halves, and both are load-bearing. **Underdetermination:** preservation of directed separation alone does *not* force the decomposition — a two-line counterexample family exists inside GUC Class 1, because a frozen purposeful coordinate has the exact information-flow signature of a belief, and no property of the update algebra of a single agent can distinguish a stored setpoint from a stored fact. **Canonicity:** relative to a *goal-intervention family* — operators that rewrite the purposeful substate while leaving the epistemic substate alone — the decomposition is unique: under three named hypotheses, every separation-preserving decomposition's epistemic component is canonically isomorphic to the quotient of X_t by goal-variation, which is M_t; the purposeful component is unique up to an explicitly characterized fiber-relabeling gauge. Together the halves say something sharper than the original conjecture aspired to: the belief/purpose partition is not discovered in the update dynamics, it is *carved by the intervention structure* — and once carved, it is carved uniquely. Defining the epistemic substate counterfactually (what is invariant under goal-variation) is therefore not one formulation choice among equals; it is the unique completion of the directed-separation condition.

The epistemic character parallels L's C1/C2/C3 accounting: one hypothesis is structural, and two constitute a definitional commitment — to a particular intervention family — that cannot be "violated," only refused at the cost of leaving the partition underdetermined. The derivation is elementary throughout (congruence and quotient arguments); the machinery is standard, and the contribution is the identification, the counterexamples, and the gauge classification.

Setup.

We work with the lifted update system of `#form-complete-agent-state` in the structural representation of L (Attack 5): state space $\mathcal{X}$, event set $\mathcal{E}$ (exogenous randomness absorbed into the event index), and update family $F_e : \mathcal{X} \to \mathcal{X}$, $F_e(X) = f_X(X, e)$. All statements are pathwise; the distributional version is discussed under Epistemic Status.

Definition (separated-decomposition)

A **separated decomposition** of $\mathcal{X}$ is a pair of maps $\mu : \mathcal{X} \to \mathcal{M}', \gamma : \mathcal{X} \to \mathcal{G}'$ such that:

- **(D1) Product-completeness.** $(\mu, \gamma) : \mathcal{X} \to \mathcal{M}' \times \mathcal{G}'$ is a bijection onto the product.
- **(D2) Epistemic closure.** There exist maps $\phi_e : \mathcal{M}' \to \mathcal{M}'$ with $\mu(F_e X) = \phi_e(\mu(X))$ for all $X \in \mathcal{X}, e \in \mathcal{E}$.

- **(D3) Cascade form.** There exist maps γ_e with $\gamma(F_e X) = \gamma_e\big(\gamma(X),\ \mu(F_e X)\big)$ — the purposeful update reads the *updated* epistemic component, matching `#der-directed-separation`'s $f_G(G_{\tau-}, M_{\tau+}, e_\tau)$ form.

(D2) says the equivalence kernel of μ is a **congruence** of the update system — $\mathcal{M}'$ is a factor (quotient) dynamical system whose update references nothing outside $(\mu(X), e)$. This is the entire formal content of "preserves directed separation": in the lifted state, "does not reference purpose" can only mean "does not reference the complementary coordinate."

The base decomposition $\mu_0(m, g) = m, \gamma_0(m, g) = g$ of an agent satisfying directed separation is a separated decomposition, directly from `#der-directed-separation`'s update forms.

Definition (decomposition-isomorphism)

Two separated decompositions (μ, γ), (μ', γ') are **isomorphic** when there are bijections $\alpha : \mathcal{M}' \to \mathcal{M}''$ and $\beta : \mathcal{G}' \to \mathcal{G}''$ (the latter possibly fibered over $\mathcal{M}'$) with $\mu' = \alpha \circ \mu, \gamma' = \beta \circ (\mu, \gamma)$, such that α intertwines the epistemic updates ($\alpha \circ \phi_e = \phi'_e \circ \alpha$) and the transported purposeful update satisfies (D3). Isomorphism of factored dynamical systems, not of bare sets: equality of the epistemic congruence up to relabeling, plus a fiber bijection compatible with the dynamics.

Lemma 0 (the cascade form is a genuine extra condition exactly when epistemic updates lose information). Given (D1)+(D2), $\gamma(F_e X)$ is automatically *some* function of $(\mu(X), \gamma(X), e)$. (D3) additionally requires the $\mu(X)$-dependence to factor through $\phi_e(\mu(X))$. If each ϕ_e is injective, this is no restriction ($\mu(X) = \phi_e^{-1}(\mu(F_e X))$); if some ϕ_e is non-injective, (D3) can fail for decompositions satisfying (D1)+(D2) (explicit witness in the gauge proposition below). □

Underdetermination.

Derived (separation-underdetermination)

Proposition 1. Preservation of directed separation — (D1)–(D3) — does not determine the decomposition up to isomorphism. Two counterexample families:

(a) Trivial factor. $\mu = \mathrm{id}_{\mathcal{X}}, \mathcal{G}' = \{*\}$. (D1) holds; (D2) holds with $\phi_e = F_e$ — the update references no complementary coordinate because there is none; (D3) holds vacuously. The "epistemic" component is the entire state, purpose included.

(b) Frozen-coordinate migration. Let the base agent satisfy directed separation, with $G = (c, G_{\text{rest}})$, $\mathcal{G} = \mathcal{C} \times \mathcal{G}_{\text{rest}}$, where c is any purposeful coordinate whose update is self-contained, goal-blind, and event-wise injective — $c_{\tau+} = h_c(c_{\tau-}, e)$ with each $h_c(\cdot, e)$ injective, so that $c_{\tau-}$ is recoverable from $(c_{\tau+}, e)$ as (D3) requires. The canonical case is frozen: an objective held fixed over the analysis window, $c_{\tau+} = c_{\tau-}$ (per `#def-strategy-dimension`'s typical timescale ordering $\nu_M \gg \nu_\Sigma \gg \nu_O$, the generic condition between objective revisions); the injectivity clause admits the more general drifting case (a decaying commitment timer, a deterministic goal schedule). Then $\big((M, c),\ G_{\text{rest}}\big)$ is a separated decomposition: (D1) by construction; (D2) since, writing $Y := (M, c)$, the update $Y_{\tau+} = (f_M(M_{\tau-}, e),\ h_c(c_{\tau-}, e))$ depends only on $(Y_{\tau-}, e)$; (D3) since $c_{\tau-} = h_c^{-1}(c_{\tau+}, e)$ is recoverable from the updated epistemic component and the event (in the frozen case, $c_{\tau-} = c_{\tau+}$ directly), so $G_{\text{rest},\tau+} = f_{G_{\text{rest}}}((c_{\tau-}, G_{\text{rest},\tau-}), M_{\tau+}, e)$ is of the required form. Its epistemic component $\mathcal{M} \times \mathcal{C}$ is not isomorphic to $\mathcal{M}$: an isomorphism of decompositions requires $\mu' = \alpha \circ \mu$, and $\alpha(m, c) = m$ cannot be injective for $|\mathcal{C}| > 1$. □

What the counterexample shows. Family (b) lives inside GUC Class 1 — a Kalman-filter-plus-LQR agent with a fixed reference admits both readings. A frozen goal has the information-flow signature of a belief: it updates on nothing, references nothing, and is read by the purposeful update exactly as beliefs are. The distinction between a stored setpoint and a stored fact is invisible to the update algebra of a single agent, because that algebra never exercises *variation of the goal*. The distinction must be supplied by additional structure — and the structure that supplies it is interventional.

Cross-cutting decompositions, classified. A "relevance-weighted model" $W = w(M, O)$ mixing belief and objective is either not a separated decomposition at all — generically $W_{\tau+}$ requires M and O separately, violating (D2) — or, when w is injective and the mixed-in purposeful content has a self-contained goal-blind injective update (family (b)'s condition; update-frozen is the canonical case), a re-coordinatization of family (b). The claim that these two cases

are exhaustive is *robust-qualitative*, not a derived proposition: it rests on the observation that (D2)-closure of a mixed coordinate forces its purposeful ingredients to update without reference to the rest of the goal state, which is family (b)'s defining condition — argued generically, not exhaustively case-analyzed. (Such representations may of course still be *useful* — as derived quantities. What is settled is that no examined cross-cutting form is a competing separation-preserving carving of the state outside family (b).)

Canonicity.

Definition (goal-intervention-family)

A **goal-intervention family** is a set H of maps $h : \mathcal{G} \to \mathcal{G}$ acting on the state through operators $\iota_h : \mathcal{X} \to \mathcal{X}$, $\iota_h(m, \gamma) = (m, h(\gamma))$: rewrite the purposeful substate, leave the epistemic substate untouched. The family induces the **reachability equivalence** R on $\mathcal{G}$ — the equivalence generated by $\{\gamma \sim h(\gamma) : h \in H\}$. **Full intervention access** is the case where H contains all constant maps $h_g : \gamma \mapsto g$ (value-targeted resets), so that R identifies all of $\mathcal{G}$; a restricted family is one whose maps preserve some sub-coordinate (e.g. $h(c, g_{\text{rest}}) = (c, \eta(g_{\text{rest}}))$ — interventions that can rewrite the strategy but not the standing objective), so that R-classes are proper. The map form is essential, not a generality flourish: a family of value-targeted resets indexed by a mere *subset* $\mathcal{G}^\dagger \subseteq \mathcal{G}$ of target values cannot express restricted access at all — any nonempty $\mathcal{G}^\dagger$ connects every pair of goal values through a common target in two steps, collapsing the generated equivalence to full access (see the regression guard in Working Notes). Let $\sim_{\text{cf}}$ be the equivalence relation on $\mathcal{X}$ generated by $\{X \sim \iota_h X : h \in H\}$, and $q : \mathcal{X} \to \mathcal{X}/\sim_{\text{cf}}$ the quotient map — the **goal-variation quotient**. Since every ι_h fixes the epistemic coordinate, the classes of $\sim_{\text{cf}}$ are exactly $\{m\} \times R(\gamma)$.

Definition (admissibility-hypotheses)

For a separated decomposition (μ, γ):

- **(R1) Product-completeness** — (D1) above (repeated as a named hypothesis because it does independent work: it excludes injective-but-not-surjective "copy" embeddings that smuggle epistemic content into the purposeful component as a constrained duplicate coordinate).
- **(R2) Purposeful purity** — $\mu \circ \iota_h = \mu$ for all $h \in H$: the epistemic component contains nothing goal-intervention can move.
- **(R3) Epistemic completeness** — $\ker \mu \subseteq \sim_{\text{cf}}$ as relations: whenever $\mu(X) = \mu(X')$, the states are goal-variation equivalent. The epistemic component retains everything goal-intervention cannot move. (Stated as a standalone condition on μ so that it is well-posed independently of (R2); under (R2) it is equivalent to injectivity of the induced map $\mathcal{X}/\sim_{\text{cf}} \to \mathcal{M}'$, which is the form the derivation uses.)

Lemma 1 (congruence lattice). The congruences of the update system $(\mathcal{X}, \{F_e\})$ are closed under arbitrary intersection, hence form a complete lattice; in particular, for any relation R on $\mathcal{X}$ there is a finest congruence containing R (the congruence closure). *Derivation:* if $\{C_i\}$ are congruences and $X \left(\bigcap_i C_i\right) Y$, then for each i, $F_e X \; C_i \; F_e Y$, so $F_e X \left(\bigcap_i C_i\right) F_e Y$; the closure is the intersection of all congruences containing R, non-empty since the total relation is one. □

Derived (decomposition-canonicity, from directed separation + full intervention access)

Theorem 2. Let the base agent satisfy directed separation and assume full intervention access. Then:

(i) $\sim_{\text{cf}}$ is a congruence, its classes are $\{m\} \times \mathcal{G}$, and the goal-variation quotient is a separation factor canonically isomorphic to $\mathcal{M}$: $\mathcal{X}/\sim_{\text{cf}} \cong \mathcal{M}$ via $[(m, \gamma)] \mapsto m$, intertwining the updates.

(ii) Every map μ satisfying (R2) factors through q: $\mu = \theta \circ q$ for a unique θ.

(iii) Every separated decomposition satisfying (R1)–(R3) has $\mu = \alpha \circ q$ with α a bijection intertwining the updates. The epistemic component of any admissible decomposition *is* the goal-variation quotient — is M_t — up to canonical isomorphism of factor dynamical systems.

Derivation. (i) The load-bearing computation is the commutation identity, which uses directed separation of the base agent essentially:

$$F_e(\iota_{h_g}(m, \gamma)) = \big(f_M(m, e),\ f_G(g,\ f_M(m, e),\ e)\big) = \iota_{h_{g'}}\big(F_e(m, \gamma)\big), \qquad g' := f_G(g,\ f_M(m, e),\ e), \tag{M.1}$$

where h_g is the constant map at g (full access supplies all of these). The first coordinate is goal-independent precisely because f_M has no G argument. So F_e maps generators of $\sim_{\text{cf}}$ to generators, hence classes to classes: $\sim_{\text{cf}}$ is a congruence. Under full access the classes are exactly $\{m\} \times \mathcal{G}$ (any two goal values are connected by one constant-map intervention), so $[(m, \gamma)] \mapsto m$ is a bijection, and the quotient update is $[\,\cdot\,] \mapsto [F_e(\cdot)]$, which in the m-coordinate is $f_M(\cdot, e)$: the intertwining is immediate. The complementary coordinate $\gamma_0 = g$ gives (D3) directly from `#der-directed-separation`'s f_G form, so the quotient participates in a separated decomposition.

(ii) $\mu \circ \iota_h = \mu$ makes μ constant on generators, hence on the generated classes; the factorization and its uniqueness are the universal property of the quotient.

(iii) By (ii), $\mu = \theta \circ q$; θ is injective by (R3) and surjective because μ is (a coordinate of the bijection (R1)). Intertwining: $\theta(q(F_e X)) = \mu(F_e X) = \phi_e(\mu(X)) = \phi_e(\theta(q(X)))$, and $q(F_e X) = \bar{\phi}_e(q(X))$ by (i), so $\theta \circ \bar{\phi}_e = \phi_e \circ \theta$. Compose with (i)'s isomorphism to land on $\mathcal{M}$. □

Derived (goal-fiber-gauge)

Proposition 3 (the purposeful component is a gauge). Fix the epistemic component as in Theorem 2(iii). The remaining freedom is exactly the choice of fiber coordinate: any $\gamma' = \beta \circ (\mu_0, \gamma_0)$ with every $\beta_m := \beta(m, \cdot) : \mathcal{G} \to \mathcal{G}'$ bijective satisfies (D1)+(D2) with the *same epistemic congruence* — the same carve of the state, differently labeled on the goal fiber. (Decomposition-isomorphism as defined above requires (D3), so the isomorphism-class statement applies to the (D3)-respecting sub-family below; a β outside it yields valid (D1)+(D2) data that is not a separated decomposition.) The sub-family additionally satisfying the cascade form (D3):

- contains every m-independent relabeling $\beta(m, g) = \beta_0(g)$ (the transported update $\gamma'_e(g', m^+) = \beta_0(f_G(\beta_0^{-1}(g'), m^+, e))$ is of the required form);
- equals the *full* family when every $\phi_e = f_M(\cdot, e)$ is injective (epistemically invertible updates: m^- is recoverable from (m^+, e), Lemma 0);
- is in general strictly between. *Witness for strictness:* $\mathcal{M} = \mathcal{G} = \{0, 1\}$, a single event with $f_M \equiv 0$ (total epistemic collapse) and $f_G(g, m^+, e) = g$; the gauge $\beta(m, g) = g \oplus m$ satisfies (D1)+(D2), but the transported purposeful update would need to produce g from $(g \oplus m,\ m^+ = 0,\ e)$, and m has been destroyed — (D3) fails. □

The exact characterization of the (D3)-respecting gauge subgroup for partially-invertible epistemics is open (see Working Notes).

Derived (restricted-intervention-access)

Corollary 4 (restricted families — the two halves are one phenomenon). Let the base agent satisfy directed separation, let H be any goal-intervention family with reachability equivalence R, and assume the **dynamics–intervention compatibility condition**:

(C) for every m^+, e, the map $f_G(\cdot, m^+, e)$ descends to $\mathcal{G}/R$ — equivalently, $\gamma\ R\ \gamma'$ implies $f_G(\gamma, m^+, e)\ R\ f_G(\gamma', m^+, e)$ (by induction along the generating chain, it suffices to check this on generator pairs $(\gamma, h(\gamma))$).

Given directed separation, **(C) is not merely sufficient for, but equivalent to, "$\sim_{\text{cf}}$ is a congruence"** — it is the carve's compatibility itself, restated as a finitely checkable per-architecture condition on the goal dynamics rather than an independent side-hypothesis; the corollary's content beyond that equivalence is the generator-pair reduction, the class identification, and the conclusions below. Under (C), $\sim_{\text{cf}}$ is a congruence with classes $\{m\} \times R(\gamma)$: for a generator pair, $F_e(m, \gamma)$ and $F_e(m, h(\gamma))$ share their epistemic coordinate by directed separation, and their goal coordinates $f_G(\gamma, m^+, e)$ and $f_G(h(\gamma), m^+, e)$ are R-equivalent by (C). The conclusions of Theorem 2 then split by kind, and their scopes differ:

- **Quotient and uniqueness (general).** The quotient system $\mathcal{X}/\sim_{\text{cf}} \cong \mathcal{M} \times (\mathcal{G}/R)$ is canonical, and Theorem 2's parts (ii)–(iii) hold verbatim with (R2)/(R3) read relative to H: *every* separated decomposition satisfying (R1)–(R3) has its epistemic component canonically isomorphic to the quotient. **Purposeful distinctions the intervention family cannot move classify as epistemic** — exactly when the goal dynamics respect the reachability structure.

- **Existence (characterized, not automatic).** Whether the quotient *participates* in a separated decomposition at all is a separate question, and for general H the answer can be no: (D1) requires a product trivialization of $\mathcal{G}$ over $\mathcal{G}/R$, which (C) does not supply (e.g. $\mathcal{G} = \{a, b, c\}$, a single idempotent intervention merging $\{a, b\}$, identity goal dynamics: R-classes of sizes 2 and 1 make (D1) unsatisfiable — uniqueness then holds vacuously). The characterization is exact, at arbitrary cardinality: a decomposition satisfying (D1)+(D2) with the quotient as epistemic component exists **iff all R-classes are equinumerous** (necessity: (D1) forces all μ-fibers — the classes — to biject with the goal-fiber factor; sufficiency: per-class bijections to a common fiber assemble a trivialization — a choice-dependent step at infinite cardinality, constructive in the finite case). The cascade form (D3) additionally obtains when the induced class update $[\gamma] \mapsto [f_G(\gamma, m^+, e)]$ is injective on $\mathcal{G}/R$ for each fixed (m^+, e) **and the trivialization is chosen m-independently** — one bijection $\beta_{[\gamma]} : R(\gamma) \to F$ per class, the same across $\mathcal{M}$. The second clause is load-bearing, not bookkeeping: with an m-independent trivialization, the updated class label is recoverable at known (m^+, e) by class-update injectivity, and the transported goal update assembles explicitly as $\gamma'^+ = \beta_{[\gamma^+]}\big(f_G(\beta^{-1}_{[\gamma^-]}(\gamma'^-), m^+, e)\big)$ with $[\gamma^-]$ recovered from $[\gamma^+]$ and (m^+, e) — no injectivity of f_M is needed (this is *not* an instance of Lemma 0, whose recoverability route runs through the full epistemic update); with an m-dependent trivialization the recovery genuinely fails. Frozen or injectively-drifting class dynamics qualify, which is exactly why Proposition 1(b) carries its injectivity clause.

Full access recovers Theorem 2 ((C) is vacuous and the trivialization trivial when R is total). Without (C), $\sim_{\text{cf}}$ is not a congruence — by the equivalence above, definitionally — and the canonical object degrades to the quotient by its congruence closure (Lemma 1), the same shape as Corollary 5: a restricted family incompatible with the goal dynamics behaves, for carving purposes, like partial coupling.

Proposition 1(b) is exactly this corollary's boundary case, and the fit is precise: for the sub-coordinate-preserving family $H_c = \{h : h(c, g_{\text{rest}}) = (c, \eta(g_{\text{rest}}))\}$ (full access on $\mathcal{G}_{\text{rest}}$), the R-classes are $\{c\} \times \mathcal{G}_{\text{rest}}$, the quotient $\mathcal{G}/R \cong \mathcal{C}$, condition (C) specializes to "the updated c does not depend on g_{rest}" — self-containedness of c within the goal, which Proposition 1(b)'s frozen c satisfies trivially — and the canonical epistemic component is $\mathcal{M} \times \mathcal{C}$: the *same* component the counterexample migrated. A goal-blind self-contained coordinate *with* intervention access is excluded from $\mathcal{M}'$ by (R2); *without* it, it genuinely belongs to the canonical epistemic component. The counterexample to the naive conjecture and the boundary behavior of the theorem are the same fact. □

Discussion

Corollary 5 (separation failure — GUC Class 3). When the base agent violates directed separation, the commutation identity fails (f_M has a G argument, so the first coordinates differ), $\sim_{\text{cf}}$ is not a congruence, and no separation factor realizes the goal-variation quotient. The canonical epistemic object is then the quotient by the congruence *closure* $\overline{\sim_{\text{cf}}}$ (Lemma 1), which is strictly coarser: goal-coupled belief-update does not merely blur the belief/goal boundary, it shrinks the largest belief-like factor that can exist at all. The information gap between $\mathcal{X}/\sim_{\text{cf}}$ and $\mathcal{X}/\overline{\sim_{\text{cf}}}$ is a structural (architecture-level) companion to the distributional $\kappa_{\text{processing}}$ of `#der-directed-separation` — discussion-grade here; not developed further in this segment.

What Is Derived vs. What Is Chosen.

Table M.1

Property	*Source*	*Strength*
(D2) is the whole formal content of "preserves directed separation"	Analysis of what "does not reference purpose" can mean in the lifted state	Discussion-grade (clarifying observation)
Separation underdetermines the decomposition (Proposition 1)	Counterexample families (a), (b)	Proved (constructive)
Cross-cutting decompositions are non-separated or family-(b) instances	Closure analysis on $w(M, O)$	Robust-qualitative (survivor family named; exhaustiveness argued generically, not case-analyzed)

Property	*Source*	*Strength*
Goal-intervention family H (maps on the goal fiber; constants = value-targeted resets)	Modeling input — the counterfactual apparatus	Definition (the analog of L's C3: a commitment, not a discovery)
(R1) product-completeness	Structural hypothesis (blocks redundancy-smuggling)	Assumption (structural)
(R2) purity + (R3) completeness	The two directions of the counterfactual anchor (block the two degenerate ends)	Definition (jointly: $\mathcal{M}'$ *is* the goal-variation quotient)
Canonicity under (R1)–(R3) (Theorem 2)	Commutation identity + quotient universal property	Proved
Goal-fiber gauge classification (Proposition 3)	Direct construction + collapse witness	Proved (endpoints); open (middle)
Restricted-access behavior (Corollary 4) — quotient + uniqueness	Reachability-class quotient under (C) ((C) ⇔ congruence, given separation)	Proved
Restricted-access behavior (Corollary 4) — existence of an admissible decomposition	R-trivializability (⇔ equinumerous classes; choice-dependent at infinite cardinality) + per-(m^+, e) class-update injectivity with an m-independent trivialization for (D3)	Proved (characterization; not automatic for general H)
Class 3 behavior (Corollary 5)	Congruence closure	Discussion-grade

Each hypothesis is individually necessary: dropping (R1) readmits the copy embedding, dropping (R2) readmits Proposition 1's families, dropping (R3) readmits the everything-is-goal decomposition (μ constant). The hypothesis set is minimal in this checkable sense.

Epistemic Status.

Exact, pathwise, with a partly definitional character. Proposition 1, Theorem 2, Proposition 3, and Corollary 4 are elementary derivations, exact under the stated hypotheses in the structural (randomized-function) representation of L Attack 5 — all statements are pathwise in the exogenous randomness. Two honest boundaries. **(1) Distributional version open.** A kernel-level analog would route through strong lumpability of the update kernels; lumpable partitions are not in general closed under common refinement, so Lemma 1's meet-closure argument does not transfer as-is, and the distributional statement is neither asserted nor refuted here. **(2) The intervention family is a commitment.** (R2)/(R3) are relative to the family H, and Corollary 4's restricted-access conclusion additionally requires the dynamics–intervention compatibility condition (C) — which, given directed separation, is *equivalent to* the congruence property itself: not an independent side-condition but the carve's compatibility restated as a finitely checkable condition on the goal dynamics, and genuinely false for some (family, dynamics) pairs; its existence half is further characterized, not automatic (R-trivializability + class-update injectivity, with uniqueness holding vacuously where no admissible decomposition exists); the theorem does not derive which interventions are physically available on a given architecture — that is an empirical and architectural question (for language-model agents, the prompt-vs-weights question). The accounting parallels L: the partition is not discovered in the update algebra; it is carved by the intervention structure, and the theorem's content is that the carving, once committed to, is unique. Refusing the commitment is coherent, at the cost of the underdetermination Proposition 1 exhibits.

Discussion.

What this settles for `#form-complete-agent-state`. The lift $X_t = (M_t, G_t)$ was introduced as a formulation choice with an unproved uniqueness conjecture. The resolution is asymmetric: as a claim about the update factorization alone, uniqueness is *false* (Proposition 1); as a claim relative to the goal-intervention family, it is *true* (Theorem 2), and the residual freedom is exactly a relabeling gauge on the goal fiber (Proposition 3). The formulation-choice honesty survives in refined form: what remains chosen is the intervention family; what is forced given it is everything else.

The counterfactual definition of belief is the unique completion of directed separation. Proposition 1(b)'s moral is that beliefs and frozen purposes are congruent under every purely dynamical criterion — the single-agent factual trajectory never exercises the difference. The structure that *does* exercise it is goal-variation, and Theorem 2(iii) says the quotient by goal-variation is the only admissible epistemic component. Read as methodology: defining M_t counterfactually — the invariant of the state under interventions on what the agent wants — is not a stylistic alternative to defining it structurally; the structural definition underdetermines and the counterfactual one is forced. This is the formal backbone for treating the coupling question as what carves the partition, rather than the partition as prior to the coupling question.

Pearl-rung reading. The goal-intervention operators are internal to the agent formalism (no dependency on the causal-hierarchy machinery is needed for the derivations), but the correspondence is worth naming: distinguishing belief from frozen purpose is a rung-2 question about the agent — no statistic of one observed (rung-1) trajectory settles it, since Proposition 1(b)'s two readings agree on every factual trajectory; the goal-variation quotient is exactly what interventional access to the goal makes well-defined. The same shape as the framework's other identifiability results: the boundary is crossed by promoting the quantity in question (here the goal) to a settable parameter.

Restricted access is the operative case for real architectures. For a Kalman + LQR agent, the reference input is a design parameter — full access, clean partition. For a language-model agent, prompt-level goal-setting is a rich-but-partial intervention family — maps that rewrite the prompt-goal while preserving goal-content baked into weights — and Corollary 4 predicts the honest consequence: weight-level purposeful content classifies as *belief* under prompt-level counterfactuals, provided compatibility (C) holds, which it does whenever the weight-level content is static over the analysis window (frozen weights: the within-session case). Where the preserved content evolves in a way that mixes with the intervenable goal — e.g. across fine-tuning on goal-conditioned interaction — (C) fails and the canonical carve degrades per the corollary's closing remark. The partition delivered by the counterfactual definition is only as fine as the intervention family that defines it, and only as clean as that family's compatibility with the goal dynamics — a scope statement, not a defect, and the reason the definition must always name its family.

Appendix N

Derived: Multi-Timescale Stability

When adaptive processes operate at nested timescales, the composite is stable exactly when each level satisfies the sector-persistence template with two interconnection terms added to its effective disturbance — the residue of the settled level below, and the target-drag of the moving level above — and the qualitative convergence constraint $\nu_{n+1} \ll \nu_n$ (`#der-temporal-nesting`) becomes a closed-form threshold: the admissible timescale ratio equals the faster level's adaptive reserve divided by the rate at which the slower level drags the faster level's target.

A mental model before the formalism: picture each adaptive level as a bathtub — water is that level's mismatch, the drain is its correction, and the inflow is its disturbance (`#result-persistence-condition` carries the single-tub version). Stacking levels couples the tubs in two directions. When the slow level acts, it *moves the fast level's drain* — the fast tub must chase a shifting target, which behaves exactly like extra inflow. And the fast tub's residual sloshing *splashes into the slow tub* — the slow level's measurements are contaminated by however much the fast level has not yet settled. Both couplings are ordinary disturbance once named, so each tub needs only the ordinary persistence condition against its *total* inflow — and the theorem's two conditions say precisely that. Micromanagement is the first coupling overwhelming the fast tub (the target moves faster than the fast level's spare drain capacity); catastrophic forgetting is the second coupling overwhelming the slow tub (the slow level is too sensitive to fast-level splash). They are dual violations of one theorem.

Formal Expression.

Formulation (stacked two-timescale system)

Fast state $x_1 \in \mathbb{R}^{n_1}$, slow state $x_2 \in \mathbb{R}^{n_2}$:

$$\dot{x}_1 = -F_1(x_1; x_2) + w_1(t), \qquad \dot{x}_2 = \epsilon\,[-F_2(x_2; x_1) + w_2(t)] \tag{N.1}$$

with $\epsilon > 0$ the timescale ratio — in AAT vocabulary $\epsilon = \nu_2/\nu_1$, the adjacent-level event-rate ratio whose smallness `#der-temporal-nesting` asserts qualitatively. The singular-perturbation form is classical [1]; the textbook treatment is Khalil's *Nonlinear Systems* [2, Ch.~11], with the composite-Lyapunov approach this derivation parallels due to Saberi and Khalil [3] and the control-theoretic corpus consolidated in Kokotović, Khalil and O'Reilly [4]. What is AAT-native is the statement of the stacking entirely in the sector-persistence template's (α, R, ρ) coordinates, so that every quantity in the stability condition is an already-named persistence quantity.

Assumption (stacking premises S0–S4)

- **(S0) Regularity.** F_1, F_2 locally Lipschitz in all arguments; w_1, w_2 measurable (solutions in the Carathéodory sense).
- **(S1) Lipschitz quasi-steady-state manifold.** There is $h : \mathcal{X}_2 \to \mathbb{R}^{n_1}$ with $F_1(h(x_2); x_2) = 0$ for all x_2 in the slow scope region, and $\|Dh(x_2)\| \le L_h$.

- **(S2) Per-level sector conditions** — the template's (T2), per level, the fast one *uniformly parametrized* by the slow state:
  - Fast: $(x_1 - h(x_2))^\top F_1(x_1; x_2) \geq \alpha_1 \|x_1 - h(x_2)\|^2$ for $\|x_1 - h(x_2)\| \leq R_1$, with α_1, R_1 uniform over the slow scope region.
  - Slow (reduced dynamics): with $\bar{F}_2(x_2) := F_2(x_2; h(x_2))$, there is x_2^* with $\bar{F}_2(x_2^*) = 0$ and $(x_2 - x_2^*)^\top \bar{F}_2(x_2) \geq \alpha_2 \|x_2 - x_2^*\|^2$ for $\|x_2 - x_2^*\| \leq R_2$.
- **(S3) Bounded interconnection.** $\|F_2(x_2; x_1) - F_2(x_2; h(x_2))\| \leq L_{21} \|x_1 - h(x_2)\|$ on the joint scope region, and $M_2 := \sup_{\|x_2 - x_2^*\| \leq R_2} \|\bar{F}_2(x_2)\| < \infty$.
- **(S4) Bounded disturbance (Model D).** $\|w_1(t)\| \leq \rho_1$, $\|w_2(t)\| \leq \rho_2$, with standing reserve hypothesis $\Delta\rho_1^* := \alpha_1 R_1 - \rho_1 > 0$ (the fast level persists standalone with reserve to spare; if it does not, no timescale separation can rescue the stack).

Error coordinates $y_1 := x_1 - h(x_2)$, $y_2 := x_2 - x_2^*$; joint scope region $\mathcal{B} := \{\|y_1\| \leq R_1\} \times \{\|y_2\| \leq R_2\}$.

Derived (two-timescale stacked persistence)

Theorem. Under (S0)–(S4), define the slow-velocity bound $v_2^{\max} := M_2 + L_{21} R_1 + \rho_2$ and the separation threshold

$$\epsilon_{\max} := \frac{\Delta\rho_1^*}{L_h \, v_2^{\max}} \tag{N.2}$$

(with $\epsilon_{\max} = \infty$ when $L_h = 0$: a fixed target imposes no separation requirement). If

$$\text{(C1)} \quad \epsilon < \epsilon_{\max} \qquad \text{and} \qquad \text{(C2)} \quad \alpha_2 R_2 > \rho_2 + L_{21} R_1, \tag{N.3}$$

then $\mathcal{B}$ is forward-invariant, and every trajectory starting in $\mathcal{B}$ satisfies

$$\limsup_{t \to \infty} \|y_1(t)\| \leq r_1 := \frac{\rho_1 + \epsilon L_h v_2^{\max}}{\alpha_1}, \qquad \limsup_{t \to \infty} \|y_2(t)\| \leq r_2 := \frac{\rho_2 + L_{21} r_1}{\alpha_2}. \tag{N.4}$$

Both levels persist: each ultimate-bound ball sits strictly inside its scope ball ((C1) gives $r_1 < R_1$; (C2) with $r_1 < R_1$ gives $r_2 < R_2$).

Derivation. *(i) A-priori slow velocity.* On $\mathcal{B}$, $\|F_2(x_2; x_1)\| \leq \|\bar{F}_2(x_2)\| + L_{21}\|y_1\| \leq M_2 + L_{21} R_1$ by (S3), so $\|\dot{x}_2\| \leq \epsilon \, v_2^{\max}$ — a scope-constant bound with no dependence on the ultimate bounds derived below (no circularity).

(ii) The fast level as a template instance with manifold-drift disturbance. Differentiating $y_1 = x_1 - h(x_2)$,

$$\dot{y}_1 = -F_1(y_1 + h(x_2); x_2) + \underbrace{w_1 - Dh(x_2)\,\dot{x}_2}_{w_1^{\text{eff}}}, \qquad \|w_1^{\text{eff}}\| \leq \rho_1^{\text{eff}} := \rho_1 + \epsilon L_h v_2^{\max} \tag{N.5}$$

by (S1) and (i). With $V_1 = \frac{1}{2}\|y_1\|^2$ and (S2-fast): $\dot{V}_1 \leq -\alpha_1 \|y_1\|^2 + \|y_1\| \, \rho_1^{\text{eff}}$ on $\mathcal{B}$. The slow level's motion enters the fast level as additional Model-D disturbance at rate $\epsilon L_h v_2^{\max}$ — the moving-target term — and the fast level's persistence condition $\alpha_1 R_1 > \rho_1^{\text{eff}}$ rearranges to (C1).

(iii) The slow level as a template instance with renormalized disturbance. $\dot{y}_2 = \epsilon[-\bar{F}_2(x_2) - (F_2(x_2; x_1) - \bar{F}_2(x_2)) + w_2]$, where the middle term is bounded by $L_{21}\|y_1\|$ by (S3). With $V_2 = \frac{1}{2}\|y_2\|^2$ and (S2-slow): $\dot{V}_2 \leq \epsilon[-\alpha_2 \|y_2\|^2 + \|y_2\|(\rho_2 + L_{21}\|y_1\|)]$ on $\mathcal{B}$ — in slow time $\tau = \epsilon t$ the ϵ factors out, and the fast level, integrated out, leaves a residue equal to its settled bound times the interconnection sensitivity.

(iv) Joint forward-invariance and the cascade bound. On the face $\|y_1\| = R_1$ of $\partial\mathcal{B}$: $\dot{V}_1 \le R_1(-\alpha_1 R_1 + \rho_1^{\text{eff}}) < 0$ by (C1). On the face $\|y_2\| = R_2$, with $\|y_1\| \le R_1$ worst-case: $\dot{V}_2 \le \epsilon R_2(-\alpha_2 R_2 + \rho_2 + L_{21}R_1) < 0$ by (C2). Each V_i decreases strictly (with uniform margin) on its own face, so a first-exit-time contradiction gives forward-invariance of $\mathcal{B}$, corners included, and all bounds above hold for all $t \ge 0$. Within $\mathcal{B}$ the fast inequality gives $\limsup\|y_1\| \le r_1$ by the template's Grönwall argument; then for any $\delta > 0$ the slow effective disturbance is eventually below $\rho_2 + L_{21}(r_1 + \delta)$, giving $\limsup\|y_2\| \le (\rho_2 + L_{21}(r_1 + \delta))/\alpha_2$ for every δ, hence the stated r_2. □

Derived (warm-start refinement — the cost of acting on transients)

If the fast level is already settled at composition time ($\|y_1(0)\| \le r_1 + \delta$ for some $\delta > 0$), the smaller ball $\{\|y_1\| \le r_1 + \delta\}$ is itself forward-invariant (same face argument under (C1)), and (C2) weakens to

$$\text{(C2-warm)} \quad \alpha_2 R_2 > \rho_2 + L_{21}(r_1 + \delta). \tag{N.6}$$

The gap between the cold-start and warm-start conditions, $L_{21}(R_1 - r_1)$ as $\delta \to 0$, is the quantitative price of engaging the slow level before the fast level has converged: early action does not void the guarantee — it raises the slow level's required reserve from settled-residue size to worst-case-transient size. `#der-temporal-nesting`'s qualitative rule ("a slower process must not act before the faster process beneath it has converged") is thereby graded, not binary.

Derived (sector condition supplies the unique-root prerequisite)

Remark (Tikhonov's prerequisite, within scope). Tikhonov-type results require the frozen fast dynamics to have a unique isolated equilibrium for each slow configuration. Under (S2-fast) this holds by construction within scope: if $F_1(y + h(x_2); x_2) = 0$ with $0 < \|y\| \le R_1$, then $0 = y^\top F_1 \ge \alpha_1\|y\|^2 > 0$ — contradiction — so $h(x_2)$ is the unique equilibrium in the radius-R_1 ball, and the sector additionally gives exponential frozen-dynamics attraction (more than Tikhonov asks; the derivation above in fact bypasses Tikhonov's theorem entirely via the composite-Lyapunov route). Outside the sector ball the guarantee is void: a fast level with multiple equilibria or limit cycles in its operating region — non-convex losses (K), cyclic strategic dynamics (`#deriv-strategic-composition`'s R2 regime) — sits outside (S2-fast), and no amount of slowing the outer loop restores the guarantee. Slowing the slow level helps only against (C1) violations, never against a fast level that has no settled state to offer.

Remark (relation to the template's stated preconditions). The template (G) states (T1)–(T3) for autonomous F and a constant disturbance bound. The two instances above are non-autonomous — the fast correction varies in time through $x_2(t)$, and the slow effective disturbance $\rho_2 + L_{21}\|y_1(t)\|$ is state-dependent and only *eventually* below the constant used — but the template's Lyapunov/Grönwall argument is pointwise in time and uses only the differential inequality, which holds uniformly, so the proofs transfer verbatim. This is a mild, stated extension of the template (uniform-in-time (T1)–(T2), eventually-bounded (T3)), not a bare instantiation.

Derived (N-level stacking, nearest-neighbor coupling)

Levels $k = 1, \dots, N$ (fastest to slowest) with rates $\epsilon_1 = 1 \gg \epsilon_2 \gg \cdots \gg \epsilon_N$. Two structural premises beyond the per-level transcriptions of (S0)–(S4):

- **(S5) Nesting idealization.** Each level's quasi-steady-state manifold depends only on the level immediately above: $h_k = h_k(x_{k+1})$, Lipschitz L_{h_k}. (Dependence on all slower levels adds a bounded sum dominated by the adjacent term under the rate ordering; the nearest-neighbor statement keeps the constants legible.)
- **(S2-red) Reduced sector conditions for middle levels.** For $1 < k < N$ the sector condition is on the *reduced* dynamics with the level below substituted at its manifold, $\bar{F}_k(x_k; x_{k+1}) := F_k(x_k;\, h_{k-1}(x_k),\, x_{k+1})$ — note the manifolds compose, since h_{k-1} is itself a function of x_k — anchored at $h_k(x_{k+1})$, uniformly in x_{k+1}.

Each level k then carries three disturbance sources — its own ρ_k, the residue of the level below ($L_{k,k-1}\|y_{k-1}\|$), and the manifold drag from the level above ($(\epsilon_{k+1}/\epsilon_k)\, L_{h_k} v_{k+1}^{\max}$, with $v_{k+1}^{\max}$ the level-$(k+1)$ scope-constant velocity bound) — boundary terms zero at $k = 1$ and $k = N$. If for each k

$$\text{(C-N)} \quad \alpha_k R_k > \rho_k + L_{k,k-1}R_{k-1} + \frac{\epsilon_{k+1}}{\epsilon_k}\, L_{h_k}\, v_{k+1}^{\max}, \tag{N.7}$$

then $\prod_k \{\|y_k\| \le R_k\}$ is forward-invariant by the same per-face argument, and the cascade limsup propagates upward level by level, giving the renormalization recursion on ultimate bounds

$$r_k = \frac{\rho_k + L_{k,k-1}\, r_{k-1} + (\epsilon_{k+1}/\epsilon_k)\, L_{h_k}\, v_{k+1}^{\max}}{\alpha_k}, \qquad r_0 := 0. \tag{N.8}$$

The recursion is well-founded in both directions: from-above terms use a-priori scope constants, from-below terms use already-derived r_{k-1}. For general (non-nearest-neighbor) coupling, the per-face argument is replaced by the standard vector-Lyapunov / M-matrix condition for interconnected systems [2, §9.5] — imported machinery, named here as the general gate without re-derivation.

Formulation (Model S extension — conditional, not yet derived)

With Wiener-process disturbance the manifold subtraction acquires an Itô correction requiring (S1′) $h \in C^2$ with bounded Hessian, and the drift-disturbance bookkeeping above becomes a joint exit-time problem across coupled levels. The expected shape — mean-square ultimate bounds with the template's $1/\sqrt{\alpha}$ scaling and the region-aware killed-moment form of A Prop A.1S with the exit-time comparison of Prop A.1S-τ as the single-level base, plus a curvature term in ρ^{eff} — is genuinely additional work, not a transcription, and is *not claimed here*. The Model D scope is the honest match for the stacking's primary AAT use cases (structural adaptation, consolidation scheduling, macro-clocks), where slow-level motion is drift-like.

Epistemic Status.

Exact for the two-timescale theorem, the warm-start refinement, and the *N*-level nearest-neighbor form, under the named premises (S0)–(S5)/(C-N) — derived by elementary Lyapunov arguments stacked on the sector-persistence template, independently re-derived and verified in full (spike trail 2026-06-10). The general-coupling *N*-level gate is *imported* (vector-Lyapunov / M-matrix, [2, §9.5]) and the Model S extension is *open* — both stated as such above, neither carried by the exact label.

What the promotion does and does not close. The predecessor sketch's open problem — AAT has trigger conditions but no dynamics for structural adaptation — is *conditionalized, not closed*. The theorem covers any deeper-level dynamics satisfying (S1)–(S5); whether AAT's actual deeper levels satisfy them is exactly the question that remains open, and it is a real one: structural adaptation is plausibly discrete (adding a model component, restructuring a hierarchy), making $\dot{x}_2$ a jump process for which (S0)'s Carathéodory framing and (S1)'s differentiable manifold do not hold as stated. The claim-type has changed from "AAT's adaptive hierarchy is stable" (never derivable as stated) to "a premise-conditional stacking theorem with named scope" — the strengthening is real (closed-form $\epsilon_{\max}$, graded warm/cold gap, unique-root prerequisite supplied rather than assumed), and so is the remaining gap.

The load-bearing premise is (S2-fast)'s *uniformity*, not the sector inequality itself. Because (α_1, R_1) are uniform over the slow scope region and h absorbs all equilibrium motion, the entire interconnection burden lands on the single drift term $\epsilon L_h v_2^{\max}$; a per-frozen-x_2 sector with non-uniform constants would not suffice. This is the standard strength assumption of the singular-perturbation literature (it is essentially Saberi-Khalil's interconnection-condition pattern), and it is precisely where realistic fast levels — non-convex, multi-equilibrium — exit scope.

Max attainable within current premises: exact (attained for Model D). Extensions that would widen scope rather than strengthen the claim: Model S (named above), jump-process slow dynamics, non-Euclidean metrics via AK's (CT2).

Discussion.

The convergence constraint is now a derived threshold, and its two failure modes are dual. `#der-temporal-nesting`'s $\nu_{n+1} \ll \nu_n$ enters the theorem as (C1) with the closed form $\epsilon_{\max} = \Delta\rho_1^*/(L_h v_2^{\max})$: the faster level's *spare adaptive reserve* — a quantity the template already names — divided by the rate at which the slower level drags the faster level's target. The two classic nesting pathologies separate cleanly into the theorem's two conditions. *Micromanagement* is a (C1) violation: the slow level acts at fast tempo, the moving-target disturbance $\epsilon L_h v_2^{\max}$ exhausts the fast level's reserve, and the fast level thrashes — and the damage then propagates back upward, because the fast level's inflated residue $L_{21} r_1$ raises the slow level's effective disturbance in turn. *Catastrophic forgetting* is a (C2) violation: the slow (consolidated) level is too sensitive to fast-level transients — L_{21} too large — and the fast loop's activity overwrites it even under perfect timescale separation. The two pathologies several auditors read as "the same failure" are precisely the two conditions of one theorem, violated separately.

Timescale separation is trust, quantified. "Each level has a stable attractor given the levels above it" reads as mutual trust within an architecture: the fast layer trusts the slow layer not to pull the rug out mid-convergence ((C1) — the manager's rate of goal-change stays under $\epsilon_{\max}$), and the slow layer trusts the fast layer to settle so there is a steady output to act on ((C2) — the worker's residue stays within the manager's tolerance $L_{21} r_1$). The warm-start refinement prices the violation: a manager who engages before the worker converges needs $L_{21}(R_1 - r_1)$ of extra reserve — the difference between reacting to settled output and reacting to worst-case transients.

Integrating out a level is a renormalization step; the template is the fixed-point form. Step (iii)'s structure — the fast level, integrated out, maps the slow level onto the *same* template form with renormalized disturbance $\rho_{k+1} \mapsto \rho_{k+1} + L_{k+1,k}\, r_k$ — is a renormalization-group step in the sense in which Chen, Goldenfeld and Oono showed RG and singular-perturbation asymptotics to be one method [5]. The sector-persistence template is form-invariant under this coarse-graining: every level of the hierarchy presents the identical $(\alpha, R, \rho^{\text{eff}})$ persistence problem with parameters renormalized by its neighbors. This is the precise sense in which AAT's persistence architecture is scale-free, and it is the lineage of Haken's slaving principle — fast modes enslaved by slow order parameters [6] — restated in persistence coordinates.

Renormalizing generative models are the constructive discrete-time instance. Friston and colleagues' scale-free active inference [7] builds generative models by recursive blocking transformations over state space and time, such that "higher levels encode sequences of sequences" and belief updating slows as one ascends levels — i.e., the timescale separation is enforced *architecturally* (one slow tick per block of K fast ticks, $\epsilon_k/\epsilon_{k+1} = 1/K$ by construction) rather than assumed, with per-level convergence inherited from coordinate descent. RGM thus supplies what (S1)–(S2) ask for, by construction, in the discrete categorical setting — an existence proof that the premises are realizable, not a competing stability argument (the paper proves no composite-stability theorem and does not claim to). The same architecturally-enforced separation appears in AAT's own composition machinery as `#form-composition-closure`'s macro-clock $K_c \gg 1$ and `#def-auxilia-hierarchy`'s (H5).

The LLM adaptive stack is the *N*-level hierarchy, unlabeled. Pretraining → fine-tuning → LoRA-style adaptation → in-context learning → retrieval state → within-generation dynamics maps onto the timescale ladder with each mechanism orders of magnitude faster and more transient than the one below. The framework's stability requirement needs no clean "parametric vs structural" labels — only that adjacent rates satisfy (C-N) — and it explains the practitioner's instinct against, e.g., updating base weights while a fast retrieval loop is mid-flight: that is a (C1) violation at the weights/retrieval boundary. The catastrophic-forgetting literature lives at the same boundary as a (C2) violation, per the dual reading above.

Strategy's own timescale. `#schema-strategy-persistence` places strategic mismatch between fast epistemic updates and slow objective revision; the stacking theorem is the formal backing for treating those as separately-stable levels — with the caveat that the strategic level's $\alpha_\Sigma = 1/(n+1)$ decay makes (S2)'s constant-α premise hold only under experience discounting, exactly as that segment already requires.

Appendix O

Sketch: Does the Model-S Containment Dichotomy License Structural-Adaptation Genericity?

The Model-S segments claim Corollary A.1S.1 "sharpens the hand-off into `#result-structural-adaptation-necessity` — the structural-adaptation trigger is *generic, not exceptional*, for a sufficiently long-lived agent." This sketch isolates that hand-off as an open question: as stated it is plausibly a mild overclaim, and the honest move is to *derive the bridge* rather than soften the prose.

The Open Question.

Discussion

`#deriv-sector-condition` Corollary A.1S.1 establishes (exactly) that under additive stochastic forcing the persistence-region first-exit probability is $P(\tau_R < \infty) = 1$ — a long-lived agent leaves the sector-condition region $\mathcal{B}_R$ with certainty. The Model-S segments and `#deriv-stochastic-non-exit` then assert this "sharpens the hand-off" so that structural adaptation is *generic, not exceptional*.

But `#result-structural-adaptation-necessity` derives necessity from a **different trigger**: model-class inadequacy ($\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ — no model in the current class can represent reality). It does not reference Model-S, region-exit, or Corollary A.1S.1; its mechanism is fitness-driven, not noise-driven. And the corollary's own explicit instance (the OU benchmark) is **positively recurrent**: the process exits any bounded region a.s. *and returns* a.s. — exit-and-return, not escape. `#deriv-sector-condition` Prop A.1S-τ now gives the exit its timescale (expected stay at least the OU benchmark's, exponential in $\alpha R^2/\sigma_w^2$, about one correction time at the mean-square threshold); it times the hand-off and leaves this segment's question untouched.

So there is a gap between what Cor A.1S.1 proves and what the hand-off prose claims:

- **What the corollary licenses (honest narrow claim):** the *pathwise-forever parametric-containment guarantee* is structurally unavailable under stochastic forcing; a long-lived agent generically enters the regime where the parametric persistence guarantee is no longer *proven* (the out-of-region regime `#result-structural-adaptation-necessity` addresses).
- **What it does not license without further argument:** that *model-class* structural adaptation is generically *necessary*. Recurrent region-exit (exit-and-return) is not model-class inadequacy; a transient noise excursion beyond $\mathcal{B}_R$ that the (un-posited-but-possibly-still-inward) correction pulls back is not a structural-adaptation trigger in the `#result-structural-adaptation-necessity` sense.

Proposed Direction (rigor pending).

Discussion

Per strengthen-before-soften, the preferred resolution is **derive the bridge**, not tighten the prose. Candidate conditions under which certain region-exit *would* imply structural-adaptation necessity, to be attempted:

1. **A2' genuinely fails outside $\mathcal{B}_R$** (not merely "not posited"): if the correction function provably ceases to point inward beyond R for the agent's model class, then a.s. exit composes with non-recovery, and recurrent exit *does* force the structural regime. This connects to the sub-scope β / model-class-capacity reading of R in `#form-sector-condition`.
2. **Excursion statistics interact with $\mathcal{F}(\mathcal{M})$:** if recurrent large excursions degrade *effective* model-class fitness (the model fitted within $\mathcal{M}$ becomes systematically wrong during excursions faster than it re-converges between them), the noise-driven and fitness-driven triggers couple, and genericity follows from the coupling rather than from exit alone.
3. **Timescale-separation argument:** if structural adaptation operates so much slower than parametric correction that the *fraction of time* spent out-of-region (positive under recurrence, however small) accumulates an unbounded structural-adaptation debt over an unbounded horizon.

Each is a derivation target; any one that closes turns the hand-off from asserted to proven and is a genuine strengthening (a new conditional result, not a softening). If all provably fail, the honest outcome is the no-go protocol applied to the hand-off claim (per `doc/audit-routing-instructions.md` §4) and the Model-S Discussion/Findings prose tightened to the narrow licensed claim above — but only *after* the strengthening attempt is exhausted and recorded.

Epistemic Status.

Sketch. Direction identified, formalization not attempted. This segment makes **no** claim that the bridge holds or fails; it records that the hand-off currently asserted in `#deriv-sector-condition` / `#deriv-stochastic-non-exit` Discussion/Findings is **plausibly a mild overclaim at the Discussion/Findings tier** (a Gate-2 class concern: a claim that *sounds* like it follows from Cor A.1S.1 but is not derived from it), and that the resolution path is to attempt the bridge derivation. The narrow licensed claim (pathwise-parametric-guarantee unavailability ⇒ generic entry into the out-of-region *regime*) is sound and uncontroversial; the disputed increment is "⇒ model-class structural adaptation is generically *necessary*." Ceiling: a closed bridge would be `derived (conditional)` on the named condition; a clean impossibility would be a no-go result. Until attempted, this stays `sketch`.

Discussion.

Why this matters enough to surface rather than leave as a soft ledger row: the genericity claim is load-bearing downstream — it is the form in which the Model-S no-go's consequence reaches the ELI / long-lived-agent persistence framing ("in any genuinely stochastic environment, structural adaptation is not an edge case"). If the bridge holds under a named condition, that is a sharp, citable strengthening of exactly the kind the framework's no-go signature tends to produce. If it does not, the framework's honesty requires the hand-off prose to state only the narrow licensed claim. Either way the question should be visible in the argument's structure, not invisible in a tracking file — which is the point of carrying it as an `exploratory` OUTLINE node.

Appendix P

Derivation: Discrete-Time Sector Condition

Discrete-time analogs of Props A.1, A.1S, and A.2 via contraction mapping, closing the fluid-limit gap (GA-5) between the event-driven dynamics (`#form-event-driven-dynamics`) and the continuous-time Lyapunov results in A. Three results land. **Prop DA.1 (Model D):** under deterministic bounded disturbance, the discrete steady-state mismatch equals the continuous steady-state exactly — zero fluid-limit gap. **Prop DA.1S (Model S):** under stochastic disturbance, the discrete and continuous steady-state variances differ by an additive correction of order $\eta^* c_{\max}^2 / c_{\min}^2$ — quantitatively small when this conditioning-ratio-over-event-rate product is small. **Prop DA.2 (adaptive reserve):** the discrete adaptive reserve matches the continuous form modulo the same correction.

A critical structural observation underwrites all three: the discrete sector condition (DA2') needs a *strictly stronger Lipschitz upper bound* on the correction function — strictly stronger than an inner-product upper bound — because the discrete recurrence's quadratic term $(\eta^*)^2 \|F_d\|^2$ requires the norm bound, not merely the projection. This is satisfied for Bayesian updates (the posterior lies between prior and data) and for gradient descent on smooth losses (where the Lipschitz constant is the smoothness parameter); pathological correction functions with large transverse components can satisfy the inner-product bound while violating the norm bound. Together with the no-overshoot step-size condition $\eta^* < 2c_{\min}/c_{\max}^2$, the fluid limit is formally justified — the continuous-time analysis of A is the limit of the discrete results here, not an idealization.

Formal Expression.

Setup. The discrete mismatch dynamics at event step k are:

Definition (Discrete Dynamics)

$$\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k \tag{P.1}$$

where η^* is the update gain (`#emp-update-gain`), F_d is the discrete correction direction, and w_k is the per-step disturbance. The continuous correction function $F(\mathcal{T}, \delta)$ from A decomposes as $F = \nu \cdot \eta^* \cdot F_d$ at event rate ν.

(DA2') Discrete Sector-Lipschitz Condition.

Assumption DA2' (discrete-sector-condition)

There exist constants $c_{\min} > 0$ and $c_{\max} < 2/\eta^*$ such that for all $\|\delta\| \le R$:

(DA2rsquoa) Lower sector bound (directional fidelity):

$$\delta^T F_d(\delta) \ge c_{\min} \|\delta\|^2 \tag{P.2}$$

(DA2rsquob) Lipschitz bound (bounded correction magnitude):

$$\|F_d(\delta)\| \leq c_{\max}\|\delta\| \tag{P.3}$$

The **lower bound** (DA2'a) is directional fidelity — the correction points inward, identical to the continuous sector condition (A2') from `#form-sector-condition` via `#der-gain-sector-bridge`.

The **Lipschitz bound** (DA2'b) controls the *magnitude* of the correction, not merely its projection onto the mismatch direction. The combined constraint $c_{\max} < 2/\eta^*$ is the **no-overshoot condition**: each correction step must not reverse the mismatch. This is the classical step-size condition $\eta^* < 2/L$ for gradient descent (where L is the Lipschitz constant of the gradient). For Bayesian updates, this is satisfied by construction — the posterior lies between prior and data.

Why DA2'b is stronger than an inner-product upper bound. A two-sided inner-product condition $\delta^T F_d(\delta) \leq c_{\max}\|\delta\|^2$ constrains only the projection of F_d onto δ. By Cauchy-Schwarz, the Lipschitz bound (DA2'b) implies the inner-product upper bound: $\delta^T F_d(\delta) \leq \|\delta\| \cdot \|F_d(\delta)\| \leq c_{\max}\|\delta\|^2$. But the converse fails — a correction function with a large transverse component (orthogonal to δ) can satisfy the inner-product bound while violating the norm bound. The proofs below (especially DA.1S) require the norm bound $\|F_d(\delta)\|^2 \leq c_{\max}^2\|\delta\|^2$, which follows from DA2'b but not from an inner-product condition alone.

Scalar case. In one dimension, DA2'a and DA2'b together reduce to the classical sector condition $c_{\min} \leq F_d(\delta)/\delta \leq c_{\max}$, since norm and inner product coincide. No generality is lost for scalar systems.

Relationship to the continuous-time condition. The continuous sector condition (A2'/GA-3) is a one-sided inner-product bound $\delta^T F \geq \alpha\|\delta\|^2$ — this suffices for continuous-time Lyapunov analysis because $\dot{V}$ involves only $\delta^T F$, not $\|F\|$. Discretization introduces the quadratic term $(\eta^*)^2\|F_d\|^2$ (see DA.1S proof), which requires the Lipschitz bound. This is the standard sector-vs-Lipschitz distinction in nonlinear systems theory.

Contraction factor. Under DA2', the per-step Lyapunov function $V_k = \frac{1}{2}\|\delta_k\|^2$ satisfies (in the zero-disturbance case $w_k = 0$):

Derived (contraction, from DA2')

$$\|\delta_{k+1}\|^2 = \|\delta_k - \eta^* F_d(\delta_k)\|^2 = \|\delta_k\|^2 - 2\eta^*\delta_k^T F_d(\delta_k) + (\eta^*)^2\|F_d(\delta_k)\|^2 \tag{P.4}$$

Applying DA2'a (lower sector bound on $\delta^T F_d$) and DA2'b (Lipschitz bound on $\|F_d\|$):

$$\|\delta_{k+1}\|^2 \leq (1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2)\|\delta_k\|^2 = \lambda_{\text{eff}}^2\|\delta_k\|^2 \tag{P.5}$$

where:

$$\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2 \tag{P.6}$$

Stability condition. $\lambda_{\text{eff}}^2 < 1$ requires $2\eta^* c_{\min} > (\eta^*)^2 c_{\max}^2$, i.e.:

$$\eta^* < \frac{2c_{\min}}{c_{\max}^2} \tag{P.7}$$

This is automatically satisfied when $c_{\min} \approx c_{\max}$ (well-conditioned correction), recovering the standard step-size condition $\eta^* < 2/c_{\max}$. For ill-conditioned corrections ($c_{\min} \ll c_{\max}$), the constraint is tighter. For Bayesian updates with bounded condition number, both conditions are satisfied.

Scalar (colinear) specialization. When $F_d(\delta) \parallel \delta$ (scalar system or colinear correction), $\|F_d(\delta)\| = |F_d(\delta)/\delta| \cdot \|\delta\|$ and the contraction factor simplifies to $\lambda = \max(|1 - \eta^* c_{\min}|, |1 - \eta^* c_{\max}|)$, the classical form. The general vector formula λ_{eff}^2 reduces to λ^2 in this case.

With disturbance $w_k \neq 0$:

$$\|\delta_{k+1}\|^2 \leq \lambda_{\text{eff}}^2 \|\delta_k\|^2 + 2\|\delta_k\|\|w_k\| + \|w_k\|^2 \tag{P.8}$$

Proposition DA.1: Bounded Mismatch (Deterministic). Statement. Under DA2' with $\eta^* < 2c_{\min}/c_{\max}^2$ and bounded per-step disturbance $\|w_k\| \leq \rho_{\text{step}}$, the mismatch is ultimately bounded:

Derived (DA.1, discrete bounded mismatch)

$$R_D^* = \frac{\rho_{\text{step}}}{1 - \lambda_{\text{eff}}} \tag{P.9}$$

Proof. By the triangle inequality: $\|\delta_{k+1}\| = \|(\delta_k - \eta^* F_d(\delta_k)) + w_k\| \leq \|\delta_k - \eta^* F_d(\delta_k)\| + \|w_k\|$. The contraction bound gives $\|\delta_k - \eta^* F_d(\delta_k)\| \leq \lambda_{\text{eff}} \|\delta_k\|$. Therefore:

$$\|\delta_{k+1}\| \leq \lambda_{\text{eff}} \|\delta_k\| + \rho_{\text{step}} \tag{P.10}$$

This is an affine contraction with $\lambda_{\text{eff}} < 1$. By the Banach fixed-point theorem, all trajectories starting in $\mathcal{B}_R$ converge to the ball of radius $R_D^* = \rho_{\text{step}}/(1 - \lambda_{\text{eff}})$, provided $R_D^* < R$. □

Recovery of continuous result. In the fluid limit ($\eta^* \to 0, \nu \to \infty, \nu\eta^* = \mathcal{T}$ fixed), $\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + O((\eta^*)^2)$, so $\lambda_{\text{eff}} \to 1 - \eta^* c_{\min}$ and $\rho_{\text{step}} \to \rho/\nu$. Then:

$$R_D^* = \frac{\rho/\nu}{\eta^* c_{\min}} = \frac{\rho}{\nu \eta^* c_{\min}} = \frac{\rho}{\alpha} \tag{P.11}$$

recovering Prop A.1 exactly. The discrete-to-continuous gap for Model D steady state is **zero**.

Proposition DA.2: Adaptive Reserve (Discrete). Statement. Under the conditions of DA.1, the agent can absorb an additional per-step disturbance of:

Derived (DA.2, discrete adaptive reserve)

$$\Delta\rho_{\text{step}}^* = (1 - \lambda_{\text{eff}})R - \rho_{\text{step}} \tag{P.12}$$

Proof. Identical to Prop A.2: the new $R_D^* = (\rho_{\text{step}} + \Delta\rho)/(1 - \lambda_{\text{eff}})$ must satisfy $R_D^* \leq R$. □

The structure is identical to the continuous adaptive reserve $\Delta\rho^* = \alpha R - \rho$. The per-event contraction rate is $(1 - \lambda_{\text{eff}})$; the per-unit-time rate is $\nu(1 - \lambda_{\text{eff}})$. In the fluid limit, $\nu(1 - \lambda_{\text{eff}}) \to \nu \cdot \eta^* c_{\min} = \alpha$, recovering the continuous result. (Note: the per-event rate $(1 - \lambda_{\text{eff}})$ converges to $\eta^* c_{\min}$, not to α directly — the factor of ν converts between per-event and per-unit-time.)

Proposition DA.1S: Stochastic Bounded Mismatch (Discrete). **Statement.** Under DA2' with $\eta^* < 2c_{\min}/c_{\max}^2$ and i.i.d. zero-mean disturbance $\mathbb{E}[w_k] = 0$, $\mathbb{E}[\|w_k\|^2] = \sigma_{\text{step}}^2$, the mismatch satisfies:

Derived (DA.1S, discrete stochastic bounded mismatch)

$$\mathbb{E}[\|\delta_k\|^2] \leq \lambda_{\text{eff}}^{2k}\|\delta_0\|^2 + \frac{\sigma_{\text{step}}^2}{1-\lambda_{\text{eff}}^2} \tag{P.13}$$

where $\lambda_{\text{eff}}^2 = 1 - 2\eta^* c_{\min} + (\eta^*)^2 c_{\max}^2$.

Proof. Define $V_k = \|\delta_k\|^2$. Taking expectations of the squared update:

$$\mathbb{E}[V_{k+1} \mid \delta_k] = \|\delta_k - \eta^* F_d(\delta_k)\|^2 + \sigma_{\text{step}}^2 \tag{P.14}$$

The first term expands as:

$$\|\delta_k - \eta^* F_d(\delta_k)\|^2 = V_k - 2\eta^* \delta_k^T F_d(\delta_k) + (\eta^*)^2 \|F_d(\delta_k)\|^2 \tag{P.15}$$

By DA2'a (lower sector bound): $\delta_k^T F_d(\delta_k) \geq c_{\min} V_k$.

By DA2'b (Lipschitz bound): $\|F_d(\delta_k)\|^2 \leq c_{\max}^2 V_k$.

Note that the second step requires the *norm* bound DA2'b, not merely an inner-product upper bound — this is where the Lipschitz condition is essential. Combining:

$$\mathbb{E}[V_{k+1} \mid \delta_k] \leq \lambda_{\text{eff}}^2 V_k + \sigma_{\text{step}}^2 \tag{P.16}$$

The condition $\eta^* < 2c_{\min}/c_{\max}^2$ ensures $\lambda_{\text{eff}}^2 < 1$. This is a supermartingale (when V_k is large enough). Iterating:

$$\mathbb{E}[V_k] \leq \lambda_{\text{eff}}^{2k} V_0 + \frac{\sigma_{\text{step}}^2}{1-\lambda_{\text{eff}}^2} \tag{P.17}$$

The steady-state mean-square mismatch is $\sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$. □

Recovery of continuous result. In the fluid limit: $\sigma_{\text{step}}^2 \to n\sigma_w^2/\nu$, $\lambda_{\text{eff}}^2 \to 1 - 2\eta^* c_{\min}$, and $(1-\lambda_{\text{eff}}^2) \to 2\eta^* c_{\min}$. The steady-state becomes $n\sigma_w^2/(2\nu\eta^* c_{\min}) = n\sigma_w^2/(2\alpha)$, recovering Prop A.1S exactly.

The discrete-to-continuous gap for Model S variance is $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2\,\nu))$, dominated by the conditioning ratio $c_{\max}^2/c_{\min}^2$. Substituting $\sigma_{\text{step}}^2 = n\sigma_w^2/\nu$ and $\eta^* = \mathcal{T}/\nu$ into $V_{ss} = \sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$ and Taylor-expanding at small η^*:

$$\frac{V_{ss}}{V_c} = \frac{1}{1 - \eta^* c_{\max}^2/(2c_{\min})} = 1 + \frac{\eta^* c_{\max}^2}{2c_{\min}} + O((\eta^*)^2) \tag{P.18}$$

so

$$V_{ss} - V_c \approx \frac{n\sigma_w^2\, c_{\max}^2}{4c_{\min}^2\,\nu} \tag{P.19}$$

— a leading correction that scales as $1/\nu$. The $(\eta^*)^2\|F_d\|^2$ term in the per-step recurrence enters $1-\lambda_{\text{eff}}^2$ at order $(\eta^*)^2$, but the steady-state ratio $\sigma_{\text{step}}^2/(1-\lambda_{\text{eff}}^2)$ inverts the leading $2\eta^* c_{\min}$ contraction, yielding $O(\eta^*)$ asymptotic gap rather than $O((\eta^*)^2)$. This quantifies the error introduced by GA-5 and confirms it is small whenever $c_{\max}^2/(c_{\min}^2\nu) \ll 1$.

Fluid Limit Theorem.

Derived (Conditional on Lipschitz regularity)

Statement. Let F_d be Lipschitz continuous with constant L_F on $\mathcal{B}_R$. Let $\delta^{(\nu)}(t)$ denote the piecewise-constant interpolation of the discrete trajectory at event rate ν, and $\delta(t)$ the solution of the continuous ODE $d\delta/dt = -F(\mathcal{T}, \delta) + w(t)$ with $F = \nu\eta^* F_d$. Then:

$$\sup_{t\in[0,T]} \|\delta^{(\nu)}(t) - \delta(t)\| \leq C \cdot \frac{\eta^* c_{\max}}{\nu^{1/2}} \tag{P.20}$$

for a constant C depending on T, L_F, and R.

Sketch. This follows from the standard ODE-approximation theory for Euler schemes (Kushner & Yin, 2003, Ch. 5). The discrete update $\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k$ is a forward Euler step for the ODE with step size $1/\nu$. The Lipschitz condition on F_d ensures the local truncation error is $O(1/\nu^2)$. Summing over $O(\nu T)$ steps and applying the Gronwall inequality gives the $O(1/\nu^{1/2})$ bound.

For Model D (deterministic): the steady-state gap is exactly zero (both discrete and continuous converge to the same fixed point). The fluid-limit error affects only transients.

For Model S (stochastic): the steady-state variance gap is $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2\,\nu))$, dominated by the conditioning ratio $c_{\max}^2/c_{\min}^2$. The $(\eta^*)^2\|F_d\|^2$ term in the per-step recurrence enters $1 - \lambda_{\text{eff}}^2$ at order $(\eta^*)^2$, but inverting the leading $2\eta^* c_{\min}$ in $V_{ss} = \sigma_{\text{step}}^2/(1 - \lambda_{\text{eff}}^2)$ produces an $O(\eta^*)$ asymptotic gap, not $O((\eta^*)^2)$ — see the recovery calculation in Prop DA.1S above.

Epistemic Status.

DA.1, DA.1S, and DA.2 are *conditional* on DA2', which now includes both a sector lower bound (DA2'a) and a Lipschitz upper bound (DA2'b). The Lipschitz condition is strictly stronger than an inner-product upper bound — this is a real assumption, not a technicality. For Bayesian updates and gradient descent on smooth losses, DA2'b is satisfied (the correction magnitude is controlled by the loss curvature). For agents with non-smooth or pathological correction functions, DA2'b must be verified independently.

The proofs themselves are standard contraction-mapping (DA.1/DA.2) and supermartingale (DA.1S) arguments. The step-size constraint $\eta^* < 2c_{\min}/c_{\max}^2$ is essential and is stated explicitly in each proposition.

Fluid limit theorem is *conditional* on Lipschitz regularity of F_d — a standard regularity condition satisfied by all correction functions in the verified instances table (`#der-gain-sector-bridge`). The convergence rate follows from classical ODE approximation theory (Kushner & Yin, 2003); the application to the AAT mismatch dynamics is new but the mathematics is not.

Max attainable: conditional — the Lipschitz bound DA2'b is inherent to the discrete-time treatment and cannot be replaced by a weaker inner-product condition without losing the norm bound used in the proofs.

Discussion.

GA-5 is closed. The fluid-limit bridging assumption is no longer required as an ungrounded assumption. For Model D, the discrete and continuous steady states are identical — the gap is zero. For Model S, the gap is $O(\eta^* c_{\max}^2/c_{\min}^2) = O(c_{\max}^2/(c_{\min}^2\,\nu))$ in variance, quantitatively bounded and small in the regime where $c_{\max}^2/(c_{\min}^2\,\nu) \ll 1$. The continuous-time results in A are formally justified as the fluid limit of the discrete results here.

DA2' is a sector-Lipschitz condition, not a pure sector condition. The continuous-time treatment (A) requires only the one-sided sector bound $\delta^T F \geq \alpha\|\delta\|^2$ because the Lyapunov derivative $\dot{V}$ depends only on $\delta^T F$. Discretization introduces a quadratic term $(\eta^*)^2\|F_d\|^2$ that requires the Lipschitz bound $\|F_d\| \leq c_{\max}\|\delta\|$. This is the standard distinction between sector conditions and Lipschitz conditions in nonlinear systems theory. For all agents in the verified

instances table (`#der-gain-sector-bridge`), the Lipschitz bound holds alongside the sector bound, so DA2' imposes no additional restriction in practice.

The step-size constraint has two forms. For well-conditioned corrections ($c_{\min} \approx c_{\max}$), the constraint reduces to $\eta^* < 2/c_{\max}$ — the classical form. For ill-conditioned corrections ($c_{\min} \ll c_{\max}$), the tighter constraint $\eta^* < 2c_{\min}/c_{\max}^2$ applies. This distinction is invisible in scalar systems but matters for high-dimensional agents with anisotropic correction.

No downstream results change qualitatively. The persistence condition, adaptive reserve, and adversarial scaling laws derived in A and `#der-adversarial-destabilization` hold as stated. The discrete framework provides sharper constants (replacing α with $\nu(1-\lambda_{\text{eff}})$) and a quantitative transient error bound, but the qualitative structure is unchanged.

Part I's formal chain is now complete. The prediction chain:

$$\text{gain principle} + \text{B1} \xrightarrow{\text{derived}} \text{sector condition} \xrightarrow{\text{Lyapunov/contraction}} \text{persistence, reserve, scaling} \tag{P.21}$$

holds in both continuous time (via A) and discrete time (via this segment), with a quantitative bridge between them (the fluid limit theorem).

Appendix Q

Detail: Linear ODE Approximation

The full linear treatment of mismatch dynamics: scalar and vector forms, steady-state solutions under both disturbance models, transient behavior, convergence rate, validity conditions, breakdown modes, discrete-time connection, and adversarial coupling. This appendix collects the linear-case results that `#hyp-mismatch-dynamics` states as a hypothesis, derives them explicitly, and delineates exactly where the linear approximation is valid and where it fails.

The detail serves as the *worked example* underlying the framework's hypothesis that mismatch dynamics follow a first-order linear ODE in the simplest case. Everything here is exactly derivable from the linear assumption — the appendix's purpose is to show, in detail, that the linear case gives clean closed-form results (the distinctive Model D $1/\mathcal{T}$ vs Model S $1/\sqrt{\mathcal{T}}$ scaling split; exponential transient convergence at rate $\mathcal{T}$; the squared-tempo law for adversarial coupling) and to *delineate exactly where the linear approximation is valid and where it fails*. The validity range: bounded mismatch within the sector region; non-saturating correction; sufficiently fast event rate relative to the mismatch timescale. The breakdown modes: saturation at large mismatch (correction slower than linear); threshold effects (small mismatches uncorrected — a dead zone); structural breakdown beyond a critical mismatch where the correction collapses entirely because the model class is no longer appropriate. The chapter's central inequality — that adversarial mismatch ratios scale as the squared tempo ratio — is first introduced here under linearity, with the appendix showing exactly which assumptions the simple form rests on and where it transitions into the sector-condition machinery (`#result-sector-condition-stability`) as those assumptions relax.

Formal Expression.

1. The scalar and vector forms. The mismatch vector $\delta(t) \in \mathbb{R}^n$ evolves under the general dynamics from A:

Definition (vector dynamics)

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{Q.1}$$

The **linear approximation** sets $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$, giving:

Hypothesis (linear-vector-ode)

$$\frac{d\delta}{dt} = -\mathcal{T} \cdot \delta + w(t) \tag{Q.2}$$

This is exact when the correction function is linear in δ (Kalman filter, Beta-Bernoulli conjugate update). It is an approximation otherwise.

Scalar form (equality). For a scalar mismatch $\delta(t) \in \mathbb{R}$:

Derived (scalar-ode, from linear-vector-ode with $n = 1$)

$$\frac{d\delta}{dt} = -\mathcal{T} \cdot \delta + w(t) \tag{Q.3}$$

This is an equality: the scalar ODE is exactly the $n = 1$ case of the vector ODE.

Norm form (inequality). For vector $\delta \in \mathbb{R}^n$, take the time derivative of $\|\delta\| = (\delta^T\delta)^{1/2}$:

$$\frac{d\|\delta\|}{dt} = \frac{\delta^T\dot{\delta}}{\|\delta\|} = \frac{\delta^T(-\mathcal{T}\delta + w)}{\|\delta\|} = -\mathcal{T}\|\delta\| + \frac{\delta^T w}{\|\delta\|} \tag{Q.4}$$

By Cauchy-Schwarz, $\delta^T w/\|\delta\| \leq \|w\|$. Under Model D (GA-2: $\|w(t)\| \leq \rho$):

Derived (norm-inequality, from Cauchy-Schwarz)

$$\frac{d\|\delta\|}{dt} \leq -\mathcal{T} \cdot \|\delta\| + \rho \tag{Q.5}$$

This is an inequality, not an equality. The Cauchy-Schwarz step is tight only when $w(t)$ is parallel to $\delta(t)$ (worst-case disturbance). The norm form stated in `#hyp-mismatch-dynamics` is this upper bound. For the scalar case ($n = 1$), Cauchy-Schwarz is automatically tight and the inequality becomes an equality.

2. Steady-state solutions.

Model D (deterministic bounded disturbance, GA-2: $\|w(t)\| \leq \rho$). Setting $d\|\delta\|/dt = 0$ in the norm inequality gives the worst-case steady state:

Derived (model-d-steady-state, from norm inequality)

$$\|\delta\|_{ss} = \frac{\rho}{\mathcal{T}} \tag{Q.6}$$

This is the tight upper bound: steady-state mismatch equals the ratio of disturbance to correction. In the scalar case ($n = 1$), this is exact (the Ornstein-Uhlenbeck process with deterministic forcing converges to $\rho/\mathcal{T}$). In the vector case, it is the worst case over all disturbance directions.

Model S (stochastic zero-mean disturbance, GA-2S: $d\delta = -\mathcal{T}\delta\, dt + \sigma_w\, dW_t$). The Ornstein-Uhlenbeck process has stationary variance derived via Ito-Lyapunov analysis (Prop A.1S in A, specialized to $\alpha = \mathcal{T}$):

Derived (model-s-steady-state, from Ito-Lyapunov with linear correction)

$$\mathbb{E}[\|\delta\|^2]_{ss} = \frac{n\sigma_w^2}{2\mathcal{T}} \tag{Q.7}$$

The RMS steady-state mismatch is:

$$\|\delta\|_{\text{rms}} = \sigma_w\sqrt{\frac{n}{2\mathcal{T}}} \tag{Q.8}$$

For the scalar case ($n = 1$): $\|\delta\|_{\text{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$.

Scaling difference. Model D gives $\|\delta\|_{ss} \propto 1/\mathcal{T}$; Model S gives $\|\delta\|_{\text{rms}} \propto 1/\sqrt{\mathcal{T}}$. Doubling the correction rate halves the deterministic steady-state mismatch but only reduces the stochastic steady-state by a factor of $\sqrt{2} \approx 1.41$. Correction is less effective against noise than against drift.

3. Transient solution. **Model D, constant ρ.** The linear ODE $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ with initial condition $\|\delta(0)\| = \|\delta_0\|$ has the standard first-order linear solution:

Derived (transient-model-d)

$$\|\delta(t)\| = \|\delta_0\| e^{-\mathcal{T}t} + \frac{\rho}{\mathcal{T}}(1 - e^{-\mathcal{T}t}) \tag{Q.9}$$

Mismatch decays exponentially from initial conditions toward the steady state $\rho/\mathcal{T}$ with time constant $\tau = 1/\mathcal{T}$. After k time constants, the transient has decayed by a factor of e^{-k}: 63% after one time constant, 95% after three, 99.3% after five.

Model S, Ornstein-Uhlenbeck. The mean-square mismatch evolves as (from Prop A.1S with $\alpha = \mathcal{T}$):

Derived (transient-model-s)

$$\mathbb{E}[\|\delta(t)\|^2] = \|\delta_0\|^2 e^{-2\mathcal{T}t} + \frac{n\sigma_w^2}{2\mathcal{T}}(1 - e^{-2\mathcal{T}t}) \tag{Q.10}$$

The variance converges twice as fast as the mean (rate $2\mathcal{T}$ vs. $\mathcal{T}$) because the Lyapunov function $V = \frac{1}{2}\|\delta\|^2$ is quadratic — its dynamics have doubled exponential rate.

4. When the linear approximation is valid. The linear case $F(\mathcal{T}, \delta) = \mathcal{T} \cdot \delta$ is a special case of the sector condition (GA-3) from `#result-sector-condition-stability` with:

Derived (linear-sector-parameters)

$$\alpha = \mathcal{T}, \qquad R \to \infty \tag{Q.11}$$

That is, the sector condition holds globally with lower bound α equal to the tempo, and the sector-condition region has infinite radius. In the notation of `#der-gain-sector-bridge`:

- **Directional fidelity parameter:** $c_{\min} = 1$ (the correction points exactly at the mismatch).
- **Sector parameter:** $\alpha = \eta^* \cdot c_{\min} = \eta^* = \mathcal{T}/\nu$.
- **No upper saturation:** $c_{\max} = c_{\min} = 1$ (the sector bounds coincide).

Consequence for persistence. With $R \to \infty$, structural persistence ($\alpha > \rho/R$) is trivially satisfied for any $\mathcal{T} > 0$. The binding constraint is task adequacy alone: $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (Model D) or $\mathcal{T} > n\sigma_w^2/(2\|\delta_{\text{critical}}\|^2)$ (Model S). This is why the linear operational forms in `#result-persistence-condition` exist: in the linear world, structural persistence is free.

The linear approximation is exact when:

1. **Kalman filter** (scalar or matrix): the gain K_t produces $F(\delta) = K_t H \delta$, which is linear. The sector parameter equals the Kalman gain: $\alpha = K$ (scalar) or $\alpha = \lambda_{\min}^+(KH)$ (matrix). See `#der-gain-sector-bridge`.
2. **Beta-Bernoulli conjugate update:** the correction $F(\delta) = \delta/(n+1)$ is linear with $\alpha = 1/(n+1) = \eta_{\text{edge}}$.
3. **Exponential family with natural parameters:** the natural-parameter update is linear in the sufficient statistic, giving $\alpha = \eta \cdot \lambda_{\min}(\text{Fisher})$.
4. **Gradient descent on quadratic loss:** $F(\delta) = \eta \nabla^2 L \cdot \delta$ is linear with $\alpha = \eta \cdot \lambda_{\min}(\nabla^2 L)$.

5. When the linear approximation breaks down. The linear form $F = \mathcal{J} \cdot \delta$ fails when the true correction function deviates from linearity. Three failure modes, each corresponding to a specific nonlinearity:

Saturation at large $\|\delta\|$. The correction mechanism is overwhelmed by large mismatch. The true correction satisfies $F(\delta) < \mathcal{J}\|\delta\|$ for large $\|\delta\|$ — correction is slower than the linear prediction. Simulation confirms: for a saturating function $g(\delta) = \delta/(1 + |\delta|/R)$, the sector parameter at the capacity boundary is $\alpha \approx \mathcal{J}/2$ (half the linear value). The linear approximation overstates persistence margins. The sector-condition framework (`#result-sector-condition-stability`) handles this via finite R and reduced α.

Threshold effects at small $\|\delta\|$. Below a detection threshold ε, small mismatches go uncorrected: $F(\delta) \approx 0$ for $\|\delta\| < \varepsilon$. This creates a dead zone where the model drifts. The linear approximation (which predicts correction at all scales) misses this. The sector condition fails locally at small $\|\delta\|$ (the ratio $\delta^T F/\|\delta\|^2 \to 0$), but the dead zone is bounded and does not affect large-scale stability.

Structural breakdown. Beyond some critical mismatch, the correction rate drops to zero because the model class is no longer appropriate: the mismatch exceeds the model's representational capacity. $F(\delta) \approx 0$ for $\|\delta\| > R$. This is the structural adaptation trigger (`#result-structural-adaptation-necessity`). The linear approximation, which predicts correction growing without bound, misses this entirely. The sector-condition framework captures it via the finite radius R of the sector-condition region.

6. Discrete-time connection. The continuous ODE is a fluid-limit approximation of the discrete event-driven dynamics (`#form-event-driven-dynamics`). The discrete mismatch dynamics are:

$$\delta_{k+1} = \delta_k - \eta^* F_d(\delta_k) + w_k \tag{Q.12}$$

In the linear case ($F_d(\delta) = \delta$), this becomes:

$$\delta_{k+1} = (1 - \eta^*)\delta_k + w_k \tag{Q.13}$$

which is an AR(1) process with coefficient $\lambda = 1 - \eta^*$.

P proves the formal connection:

Model D. The discrete and continuous steady states are *identical* — the fluid-limit gap is exactly zero. Both give $R^* = \rho/\alpha$. The discrete form has $\alpha = (1 - |\lambda|)/\eta^* = (1 - |1 - \eta^*|)/\eta^*$, which equals 1 (hence $\alpha_{\text{continuous}} = \nu \cdot 1 = \mathcal{J}$) when $0 < \eta^* < 1$.

Model S. The discrete steady-state variance is $\sigma^2_{\text{step}}/(1 - \lambda^2_{\text{eff}})$, which recovers the continuous result $n\sigma_w^2/(2\mathcal{J})$ in the fluid limit ($\eta^* \to 0, \nu \to \infty, \nu\eta^* = \mathcal{J}$ fixed). The variance gap is $O(\eta^* c^2_{\max}/c^2_{\min}) = O(c^2_{\max}/(c^2_{\min}\,\nu))$ — quantitatively small whenever $c^2_{\max}/(c^2_{\min}\,\nu) \ll 1$. See P for the leading-order Taylor expansion that gives this scaling.

Fluid-limit validity. The ODE approximation is formally justified by the fluid limit theorem in P (conditional on Lipschitz regularity of F_d). The approximation is most accurate when $\eta^* \ll 1$ (the small-gain regime). It is least accurate during initial transients when η^* is large, but this phase is short-lived and the transient error is bounded by $O(\eta^* c_{\max}/\nu^{1/2})$.

7. Adversarial coupling in the linear case. When two agents are coupled, A's praxis contributes to B's disturbance:

Hypothesis (Linear Coupling Model)

$$\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A \tag{Q.14}$$

Under the linear approximation, the steady-state mismatches (Model D, coupling-dominant: $\gamma_A \mathcal{T}_A \gg \rho_{B,\text{base}}$) are:

Derived (linear-adversarial-steady-state, from Model D steady state + coupling model)

$$\|\delta_B\|_{ss} \approx \frac{\gamma_A \mathcal{T}_A}{\mathcal{T}_B}, \qquad \|\delta_A\|_{ss} \approx \frac{\gamma_B \mathcal{T}_B}{\mathcal{T}_A} \tag{Q.15}$$

The ratio:

Derived (squared-tempo-law, from linear steady states)

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \frac{\gamma_A}{\gamma_B} \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{Q.16}$$

Under symmetric coupling ($\gamma_A \approx \gamma_B$), tempo advantage squares: a 2:1 tempo ratio yields a 4:1 mismatch ratio; a 3:1 ratio yields 9:1.

Why the squared law. The exponent 2 arises because the $1/\mathcal{T}$ steady-state scaling appears in both numerator (faster agent generates more disturbance) and denominator (faster agent corrects better): one power from the coupling, one from the steady-state. Under Model S (stochastic coupling, $1/\sqrt{\mathcal{T}}$ scaling), the exponent reduces to 3/2. See `#result-adversarial-exponent-regimes` for the full regime analysis.

Simulation validation. Variant A confirmed $b = 1.999$ (Model D, coupling-dominant). Variants C-D confirmed $b = 1.481$ (Model S, coupling-dominant). Both within numerical precision of the derived values. See AZ.

Epistemic Status.

Exact for the linear case. The linear ODE is a standard first-order linear system; the solutions (transient and steady-state, both Model D and Model S) are textbook results for the Ornstein-Uhlenbeck process and its deterministic analog. The sector-condition parameters ($\alpha = \mathcal{T}, R \to \infty$) are exact when the correction function is linear. The discrete-time connection is exact (zero gap for Model D, $O(\eta^*)$ gap for Model S — see P for the leading-order Taylor expansion). The adversarial scaling law ($b = 2$ for Model D, $b = 3/2$ for Model S) is derived and simulation-validated.

What is exact vs. what is an approximation. The linear ODE is exact for systems with linear correction dynamics (Kalman, Beta-Bernoulli, quadratic loss). For nonlinear systems, the linear ODE is a local Taylor approximation valid near the steady state. The sector-condition framework (`#result-sector-condition-stability`, A) is the general treatment; this appendix derives what falls out when the sector bounds coincide.

The norm form is an upper bound for $n > 1$. The inequality introduced by Cauchy-Schwarz at the norm step is tight only in the worst case. For typical disturbances that are not persistently aligned with the mismatch direction, the actual norm trajectory lies below the upper bound. The scalar case ($n = 1$) is an equality. This distinction is load-bearing: downstream results that use $d\|\delta\|/dt = -\mathcal{T}\|\delta\| + \rho$ as an equality (e.g., the exact transient solution, the exact steady-state ratio) are strictly correct only for $n = 1$ or for worst-case disturbance. The Lyapunov analysis in A avoids this issue by working with $V = \frac{1}{2}\|\delta\|^2$ directly, without passing through the norm derivative.

Max attainable: exact (the linear case is its own exact analysis; there is nothing to tighten).

Discussion.

Why this appendix exists. The linear ODE in `#hyp-mismatch-dynamics` is labeled as a hypothesis and described as a "first-order approximation." That is correct for the general case. But for agents with linear correction dynamics, the ODE is not an approximation at all – it is exact. This appendix serves two roles: (1) providing the complete derivation that `#hyp-mismatch-dynamics` points to rather than carrying inline, and (2) making precise when the "approximation" is exact and when it genuinely introduces error.

Pedagogical value. The linear case is the entry point for understanding AAT's mismatch dynamics. The closed-form solutions (exponential convergence, ratio steady state, squared scaling law) build intuition that the general sector-condition results (`#result-sector-condition-stability`) cannot provide – Lyapunov gives bounds and existence, not quantitative trajectories. The linear treatment also connects directly to the AR(1) simulation framework (AZ) and classical control theory (PI controllers, Kalman filters).

Speed-quality substitutability. From $\mathcal{T} = \nu \cdot \eta^*$ (single channel): doubling event rate ν has the same effect on $\|\delta\|_{ss}$ as doubling update quality η^*. They are multiplicative when both improve: 50% improvement in each yields $1.5 \times 1.5 = 2.25\times$, not $3\times$. This is the formal analog of Boyd's insight that Orient quality often matters more than raw OODA speed.

The persistence threshold in the linear world. From the steady state: $\|\delta\|_{ss} < \|\delta_{\text{critical}}\|$ iff $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (Model D). This is task adequacy only – structural persistence is trivially satisfied because $R \to \infty$. The linear world makes persistence look easier than it is. For nonlinear agents, both structural and task-adequacy conditions must hold (`#result-persistence-condition`).

Appendix R

Derivation: Strategy Graph Structure

Operational requirements on the agent's representation — directed temporal ordering, probabilistic uncertainty, and the ability to test strategy components — are *sufficient* for the strategy to be a directed acyclic graph with the Markov factorization property. The argument parallels Cox's theorem for probability in *form* but not yet in *strength*: Cox's theorem is necessary-and-sufficient (the only measure satisfying the desiderata is probability); this result is sufficient only (the desiderata guarantee DAG+Markov, but no one has shown a non-DAG structure cannot satisfy them). Acyclicity and directed edges are *proved* from temporal ordering over a finite horizon; the Markov factorization is *proved under causal sufficiency* via the Causal Markov Condition theorem (Spirtes–Glymour–Scheines, Pearl). A fourth postulate (observable intermediates) is required for localized strategic diagnosis but not for the representation or persistence results themselves. The Markov factorization does not by itself make the AND/OR status propagation exact: under causal sufficiency, propagation computes the plan's true success probability for all credences if and only if the strategy DAG is an AND/OR tree.

How far the Cox parallel goes. Cox's theorem starts from desiderata on how a rational agent should quantify uncertainty (consistency, universality, continuous functional composition) and proves that the *only* measure satisfying them is probability — necessity. This segment starts from desiderata on how a bounded agent can represent its strategy under causal action (directed temporal order, probabilistic edge uncertainty, causal sufficiency of the chosen nodes) and proves that DAG+Markov *suffices* to satisfy them — sufficiency. The necessity direction — no non-DAG structure (factor graphs, junction trees with cyclic message schedules, chain graphs) can satisfy P1–P4 plus causal sufficiency — is not established here. In practical terms this gap is unimportant because the proved sufficiency gives a rigorous grounding for the DAG structure; claims that AAT's strategy *must* be a DAG should be read as "must-if-sufficient-via-this-route," not as a proved necessity. A stronger Cox-style result is open. The placement of the DAG structure on a footing comparable to probability — a consequence of operational requirements rather than a modeling convenience — holds in the sufficiency direction; the full parallel with Cox awaits a uniqueness argument.

Formal Expression.

The Postulates. Four properties that a strategy representation must satisfy. Each is independently motivated from the adaptive-systems foundation.

P1: Directed Temporal Ordering. *[Derived (from* `#post-causal-structure`*)]*

If component A of the strategy causally produces component B, then A temporally precedes B. The strategy representation must respect this directionality — edges point from causes to effects, from actions to outcomes, from prerequisites to goals.

This is a consequence of the temporal postulate (`#post-causal-structure`): the arrow of time is constitutive, not incidental. Reversing a causal edge would mean effects precede causes, which is physically impossible.

Derived (from Cox's theorem)

P2: Probabilistic Uncertainty.

The agent's uncertainty about whether each step of the strategy will succeed must be quantified by a measure satisfying Cox's axioms (consistency, universality, non-negativity). The unique such measure is probability (Cox 1946, "Probability, Frequency and Reasonable Expectation," *American Journal of Physics* 14(1):1–13; modern exposition in Jaynes 2003, *Probability Theory: The Logic of Science*, Cambridge University Press, Chapter 2). The agent may use other representations internally (confidence scores, fuzzy logic), but these must be mappable to probability to be consistent.

P3: State-Local Revisability. *[Derived (from* `#der-chain-confidence-decay` *+ bounded computation)]*

When the agent observes evidence about one component of its strategy (e.g., "step 3 succeeded" or "prerequisite 2 is blocked"), it must be able to update its beliefs about that component and its consequences without recomputing the entire strategy from scratch.

Why this is forced, not chosen:

From fragility. Additive log-confidence (`#der-chain-confidence-decay`) means longer chains are exponentially less reliable. The agent will frequently encounter partial failures. Each partial failure requires re-evaluation of the affected portion of the strategy. If each re-evaluation requires full recomputation, the agent's planning tempo T_Σ is catastrophically slow — potentially violating the strategy persistence condition.

From bounded computation. The agent has finite computational resources (the IB constraint applies to planning as well as model maintenance). Full recomputation of a strategy with N components costs $O(N)$ or worse. Local revision costs $O(|\text{affected}|)$, which can be much smaller.

From the persistence condition. Strategy must be revised faster than the environment invalidates it. Local revision directly increases T_Σ by reducing the per-update cost. An agent that must recompute everything on each update has lower T_Σ and is more likely to fall below the persistence threshold.

P4: Observable Intermediates. *[Derived (from* `#der-chain-confidence-decay` *+ monitoring requirement)]*

To support **localized strategic diagnosis and revision**, the strategy representation benefits from internal checkpoints — observable states between the initial action and the final goal — that the agent can monitor to detect partial failure.

Without intermediates, the agent cannot detect chain failure until the final outcome. By the time the final outcome reveals failure, all intermediate actions are wasted. With intermediates, the agent can detect failure at step k and revise, saving the cost of steps $k+1$ through n. The value of early detection grows with chain length, because longer chains fail more often (P2 + `#der-chain-confidence-decay`).

Observable intermediates are not required for strategy representation or persistence. When intermediates are unobservable, plan-level tracking (`#schema-strategy-persistence`, Case 3) preserves the sector condition at the cost of per-edge diagnostic resolution — the agent knows the plan is failing but cannot localize which step needs revision (`#der-observability-dominance`). P4 is therefore a requirement for *strong diagnostics*, not for strategy representation per se. The observability investment tradeoff (`#der-observability-dominance`) quantifies the payoff: making an intermediate observable improves the sector parameter from $1/(n_\Phi+1)$ (plan-level) to $\min(1/(n_1+1),\ \theta_1/(n_2+1))$ (per-edge weakest-link).

The Derivation.

Step 1: P1 implies directed edges. Each component X_i of the strategy has a set of direct causes $\text{Pa}(X_i)$ — the components whose outcomes directly influence X_i's outcome. P1 requires that these causal relationships are directed: $\text{Pa}(X_i)$ temporally precedes X_i. This gives directed edges $\text{Pa}(X_i) \to X_i$.

Step 2: P2 implies probability distributions on edges. Each edge carries uncertainty: $P(X_i \mid \text{Pa}(X_i))$. By P2 (Cox), this is a probability distribution. The joint distribution over all strategy components is some $P(X_1, \ldots, X_n)$.

Step 3: Causal sufficiency implies the Markov condition (proved).

Derived (Conditional on causal sufficiency of Σ_t)

Claim. For a causally sufficient strategy DAG, the Markov factorization property is a theorem — a consequence of the Causal Markov Condition (CMC).

The Markov factorization property. Each variable X_i is conditionally independent of its non-descendants given its parents:

$$X_i \perp \text{NonDesc}(X_i) \mid \text{Pa}(X_i) \tag{R.1}$$

Equivalently, the joint distribution factorizes as:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Pa}(X_i)) \tag{R.2}$$

(The equivalence holds for positive distributions — Lauritzen 1996, Theorem 3.27.)

The argument has five parts:

(a) The DAG is a causal model. P1 establishes that edges represent causal relationships: completing a parent step causally advances the child step. P2 establishes probabilistic uncertainty over outcomes. Together: Σ_t is a causal DAG in the sense of structural causal models (Pearl 2009, Definition 7.1.1) — each node's outcome is determined by its parents' outcomes (through the causal mechanism encoded in the edge credences) plus exogenous uncertainty specific to that step. Formally, each node admits a structural equation:

$$X_i = f_i(\text{Pa}(X_i), \varepsilon_i) \tag{R.3}$$

where f_i is the local causal mechanism and ε_i is the exogenous noise (the residual uncertainty at step i not determined by its parents).

(b) Causal sufficiency implies exogenous independence. The exogenous terms ε_i are mutually independent if and only if no unmodeled common cause affects two or more nodes in the graph. This is precisely the **causal sufficiency** assumption: every variable that is a direct common cause of two or more nodes in Σ_t is itself a node in Σ_t.

For agent-constructed strategies, causal sufficiency is a **modeling ideal, not a typical condition**. The agent designed the graph, so all *intended* causal relationships are explicit — but environmental common causes (shared infrastructure, weather, market shifts, correlated adversary actions) routinely affect multiple strategy steps without appearing as nodes. In complex, multi-stakeholder, or adversarial environments, causal insufficiency is the dominant case (`#def-strategy-dag`, Correlation Hierarchy). When an environmental factor is omitted, the exogenous terms become correlated and the Markov condition fails. This is model inadequacy (`#result-structural-adaptation-necessity`), and the remedy is to add the missing common-cause node — but identifying which common causes matter is a modeling judgment, not a mechanical procedure (`#def-strategy-dag`, L1 construction principle). The proof's conditional on causal sufficiency is therefore a condition on model quality: the result holds exactly when the DAG is well-constructed, approximately

when it is close, and fails when major common causes are missing. The Correlation Hierarchy in `#def-strategy-dag` provides the practical framework: L0 (independence, this proof's assumption) gives tractable results; L1 (augmented DAG with explicit common-cause nodes) is the practical default in complex domains; L0 formal results transfer to correctly constructed L1 DAGs.

(c) The Causal Markov Condition theorem. For a DAG G over variables $V = \{X_1, \dots, X_n\}$ with structural equations $X_i = f_i(\text{Pa}(X_i), \varepsilon_i)$ where the ε_i are mutually independent:

$$P(X_1, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Pa}(X_i)) \tag{R.4}$$

This is the **Causal Markov Condition** — a proved theorem, not a modeling assumption. The standard references are Spirtes, Glymour, and Scheines (2000, Theorem 3.4) and Pearl (2009, §1.4.1, Theorem 1.4.1). The proof applies the chain rule in topological order: $P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$, then uses the independence of ε_i to show that conditioning on all predecessors reduces to conditioning on parents only. Each non-parent predecessor's influence on X_i is fully mediated through the parents — its direct contribution enters through the causal mechanism f_i, not through ε_i.

(d) What the CMC gives P3, and what it does not. State-local revisability (P3) was originally stated as an independent postulate. The CMC supplies half of it. Since $X_i \perp \text{NonDesc}(X_i) \mid \text{Pa}(X_i)$, the joint is a product of one factor per node. Revising one step's mechanism or edge credence therefore changes one factor and nothing else in the representation. Propagating the *consequences* of that revision, or of an observed outcome, is a different matter. On an AND/OR tree, a changed credence at node i changes statuses only along i's single path to the root, so forward recomputation costs the length of that path. On a graph that is not a tree, the forward pass is not exact to begin with (Step 3(e)). Observation is not forward-only either. In the minimal case of Step 3(e), observing $A = 1$ forces $C = 1$, hence $B = 1$ and $R = 1$, which no forward update from A produces. Exact propagation of evidence works by local message passing on singly connected graphs (Pearl 1988) and needs exact inference elsewhere. So the CMC guarantees a local *representation* for every causally sufficient DAG, while local revision of *consequences* holds by the forward pass on trees and by message passing on polytrees. P3 was motivated as an operational requirement (agents *need* local revision for computational tractability, and the persistence condition demands it). The factorization meets the representation half for any causally sufficient DAG, and the tree condition of §"Exactness of Status Propagation" says where the forward pass meets the rest.

(e) What the factorization does not give: exactness of status propagation. The AND/OR forward pass of `#def-strategy-dag` assumes the CMC's exogenous independence, but that independence is not sufficient for it to compute true probabilities. The forward pass multiplies the parents' *marginal* statuses, so it is exact at a node only when that node's parents are independent, and an explicit node feeding two branches that reconverge makes them dependent even when every exogenous term is independent. The minimal case: leaf C with $p_C = 0.5$; $A = \text{AND}(C)$, $B = \text{AND}(C)$ and $R = \text{AND}(A, B)$, all edge credences 1. The graph is causally sufficient and $P(R) = P(C) = 0.5$, but propagation gives 0.25. The exact condition is the tree condition derived under "Exactness of Status Propagation" below. When the exogenous terms are correlated (a latent common cause), propagation is biased even on a tree: for two siblings by exactly their covariance, with the sign set by node type (`#def-strategy-dag`, Correlation Hierarchy). The Markov factorization and exact propagation are therefore different conditions. The first needs causal sufficiency; the second needs causal sufficiency and a tree.

Assembling (a)-(e). P1-P2 establish that Σ_t is a causal DAG with probabilistic uncertainty. Under causal sufficiency (exogenous independence), the CMC theorem proves the Markov factorization. P3's representation half (local factors) follows as a consequence; its consequence-propagation half holds on trees (3(d)). The Markov property is both operationally required (P3) and structurally guaranteed (CMC). When causal sufficiency fails, the Markov factorization is still the agent's *intended* factorization — the one its DAG represents — but it is wrong about the world. The gap between intended and actual factorization manifests as correlated failure and a $\hat{R}_\Sigma$ bias whose sign is set by node type.

Adding the missing common-cause nodes restores causal sufficiency and with it the factorization. For the forward pass to compute the factorized model's probabilities exactly, the added nodes must also be placed so that the graph is a tree, or the forward pass must be replaced by exact inference (below).

Step 4: P1 + finite horizon implies acyclicity (proved). This is the strongest piece of the argument. See the dedicated section below.

Step 5: Assembly. P1 (directed edges + causal interpretation) + P2 (probabilistic) + causal sufficiency (CMC → Markov factorization) + P4 (internal nodes) + finite horizon (acyclicity):

The strategy representation must be representable as a directed acyclic graph with probability distributions at each node conditioned on its parents — a Bayesian network. P3 (local revisability) is met by this structure in the sense of Step 3(d), not required as a premise.

Acyclicity Derivation. *[Derived (from* `#post-causal-structure` *+ finite planning horizon)]*

This resolves a former known fragility in the theory. Acyclicity of Σ_t is derived, not assumed.

Result. For a strategy representation over a finite future horizon, temporal ordering forces acyclicity.

Derivation.

1. Each node X_i in Σ_t represents a future event or state with temporal position $\tau_i > t$ (the future time at which the step occurs or the state is evaluated).
2. Each edge $X_i \to X_j$ requires $\tau_i < \tau_j$ (P1: causes temporally precede effects).
3. A cycle $X_i \to X_j \to \cdots \to X_i$ would require $\tau_i < \tau_j < \cdots < \tau_i$, which is impossible for a real-valued time index.
4. Therefore the graph is acyclic. □

Formally: a finite set with a strict partial order (future events ordered by time) is representable as a DAG. This is a standard result in order theory — every finite partial order has a Hasse diagram, which is a DAG.

The iteration objection resolved. A strategy that says "try A, if fail try B, if fail try A again" appears cyclic.

In the time-indexed representation:

$$A_1 \to \text{check}_1 \to B_1 \to \text{check}_2 \to A_2 \to \ldots \tag{R.5}$$

Each attempt is a distinct node at a distinct time. The apparent cycle is a linear chain in the unrolled view. Iteration "terminates" when either a node succeeds (remaining retry nodes become probability-zero), the agent exhausts its resource budget (a constraint truncating the chain), or the horizon ends. Any finite-horizon strategy, including those with "loops" in the informal sense, is acyclic when time-indexed.

Scope. This applies to Σ_t (the agent's strategy over the future), not to M_t's model of the environment. M_t may include cyclic causal processes — feedback loops in the physical world, market dynamics, ecosystem interactions. The acyclicity is specific to the purposeful substate because Σ_t represents planned future actions and the future is partially ordered by time. M_t's model of environmental dynamics may need to represent cycles (via time-unrolled DBNs or other cyclic structures).

Connection to Pearl. Pearl's do-calculus is defined on DAGs. Extensions to cyclic structures exist (cyclic SCMs, equilibrium models) but are substantially more complex and lose some of do-calculus's clean properties. The temporal argument here shows that for strategy representations (future-looking plans), acyclicity is not a convenience restriction on Pearl's framework — it is a consequence of the temporal structure of planning.

Derived (Conditional on the noisy-AND/OR structural equations and causal sufficiency)

Exactness of Status Propagation.

Setting. Read the AND/OR parameterization (`#scope-and-or`) as structural equations. Each leaf is $X_v \sim \text{Bernoulli}(p_v)$. Each edge carries a transmission variable $E_{iv} \sim \text{Bernoulli}(p_{iv})$. An AND node is $X_v = \bigwedge_{i \in \text{pa}(v)}(X_i \wedge E_{iv})$ and an OR node is $X_v = \bigvee_{i \in \text{pa}(v)}(X_i \wedge E_{iv})$. Two independences are assumed. Causal sufficiency (Step 3(b)) makes the exogenous terms of different nodes independent. The noisy-AND/OR parameterization makes the transmissions into any one node independent of each other, which is its independence-of-causal-influence assumption (Heckerman & Breese 1996, "Causal Independence for Probability Assessment and Inference Using Bayesian Networks," *IEEE Transactions on Systems, Man, and Cybernetics A* 26(6):826–831). If the transmissions into one node are correlated, even a tree is miscounted. Together the two make all leaf draws and transmissions mutually independent. The first premise can fail where nothing looks shared. Leaves that are exclusive outcomes of one variable, such as "arm k is the best arm" for each k (BD), are dependent, because that variable is a common cause outside the graph, and then even a tree is miscounted. For an OR node whose parents' events are disjoint and whose parent statuses are exact, the exact status is $\sum_i p_{iv} s_i$ by additivity. The forward pass's $1 - \prod_i (1 - p_{iv} s_i)$ is smaller whenever two of the terms are positive. The forward pass s_v of `#def-strategy-dag` is the computation that is exact when every node's parents are independent.

Result. For a node v, the following are equivalent:

1. $s_v = P(X_v = 1)$ for all leaf and edge credences.
2. Every ancestor of v has exactly one directed path to v.

Every node of Σ_t has a path to v_{root} (rootedness, `#def-strategy-dag`). So, applied at the root: $\hat{P}_\Sigma = s_{v_{\text{root}}}$ **equals the plan's true success probability for all credences if and only if Σ_t is an AND/OR tree**, that is, every node other than the root has exactly one child. Under rootedness this is the same as Σ_t being a polytree. For a DAG without a unique sink, exactness at every node needs only the weaker multitree condition: at most one directed path between any two nodes, equivalently no node has two parents whose ancestor-or-self sets meet. The "or-self" matters. In the triangle $a \to b$, $a \to c$, $b \to c$, the parents a and b of c have no strict common ancestor, yet a has two paths to c and propagation is wrong there. The pair $c = \text{AND}(a, b)$, $d = \text{OR}(a, b)$ is a multitree but not a polytree, and propagation is exact at both nodes. A common child of c and d breaks it. Rootedness is what turns the multitree condition into the tree condition.

Derivation. *($2 \Rightarrow 1$.)* If every ancestor of v has one path to v, the ancestral subgraph of v is a tree. The parents of any node in it therefore have disjoint ancestor sets. Their statuses are functions of disjoint sets of independent exogenous variables, so they are independent. The AND and OR formulas are exact for independent inputs, and $s_v = P(X_v = 1)$ follows by induction in topological order.

($1 \Rightarrow 2$.) Treat the credences as indeterminates. $P(X_v = 1)$ is the expectation of a Boolean function of independent Bernoulli variables, so it is multilinear: of degree at most 1 in each leaf credence p_ℓ. Write $N(\ell, v)$ for the number of directed paths from leaf ℓ to v. By induction in topological order, $\deg_{p_\ell} s_v = N(\ell, v)$. At ℓ itself, $s_\ell = p_\ell$. An AND status is a product of factors $p_{iv} s_i$, and an OR status is one minus a product of factors $1 - p_{iv} s_i$. Each factor has the same degree in p_ℓ as s_i whenever that degree is positive, and degrees add under products in a polynomial ring. If some ancestor u of v has two paths to v, any leaf ℓ at or above u has $N(\ell, v) \geq 2$. Then s_v and $P(X_v = 1)$ are different polynomials. A polynomial that vanishes on the whole credence box $[0, 1]^n$ is zero, so the two differ at some credences. □

The difference polynomial is nonzero, so on a graph that is not a tree the forward pass is exact only on a measure-zero set of credences. That set includes degenerate values such as $p_C \in \{0, 1\}$ in the minimal case of Step 3(e). The mechanism behind the degree count is exact. By the recursion, s_v is the true probability of v on the *path-unfolded tree*: the tree that copies each ancestor once per path to v, each copy with its own independent noise. The forward pass therefore counts a leaf reached by N paths as N independent draws where the world has one. The degree identity is a corollary of this. Fixed edge credences in $(0, 1]$ suffice for failure: at any such values, the forward pass on a graph that is not a tree is wrong at generic leaf credences.

Sign of the error under reconvergence. Call a node *reconvergent* if two of its parents have ancestor-or-self sets that meet. Every X_v is a nondecreasing function of the independent exogenous variables. Nondecreasing functions of independent variables are positively associated (Esary, Proschan & Walkup 1967, "Association of Random Variables, with Applications," *Annals of Mathematical Statistics* 38(5):1466–1474). So at a reconvergent node whose parents' statuses are exact, the AND formula never overestimates and the OR formula never underestimates. At a node that is not reconvergent, the parents are independent, so the formula is exact on exact inputs. Both formulas are nondecreasing in their inputs, so any node passes an input bias through with its sign unchanged. Induction in topological order then gives the result. **If every reconvergent node is AND, $s_v \le P(X_v = 1)$ at every node; if every reconvergent node is OR, $s_v \ge P(X_v = 1)$ at every node.** The node types elsewhere are free. Equality holds on trees, where no node is reconvergent. When reconvergent nodes of both types are present, the sign can flip within one graph. Take root $\mathrm{OR}(X, Y)$, with X an AND reconvergence over leaf C and Y an OR reconvergence over leaf D, all edges at credence 1. The root error is -0.240 at $(p_C, p_D) = (0.5, 0.01)$ and $+0.245$ at $(0.01, 0.5)$. The common plan shape, OR near the root and AND inside, is covered whenever the reconvergence happens at OR nodes. `#example-L1`'s naive L1 reconverges at its OR root, so its overestimate is an instance. This is the explicit-reconvergence counterpart of the two-sibling ±Cov table in `#def-strategy-dag`, which treats a latent shared cause.

Exact computation on graphs that are not trees. On a causally sufficient DAG that is not a tree, $P(v_{\text{root}})$ is still fixed by the graph and its credences, but computing it takes exact inference. One route clamps a set of leaves (or individual transmissions) such that every exogenous variable still left random has at most one path to the root. Each clamped configuration is then evaluated exactly by a forward pass, by the same argument as $2 \Rightarrow 1$, so k binary clamped variables cost 2^k forward passes. When a reconvergent node is internal, clamping it is not a forward-pass operation, because conditioning on it changes its ancestors' distribution. Pearl's cutset conditioning is the general form of this route: it instantiates the cutset as evidence and runs polytree belief propagation (Pearl 1988, *Probabilistic Reasoning in Intelligent Systems*, ch. 4). A second route is the junction-tree algorithm, at cost exponential in the treewidth of the moralized graph, which a noisy-OR node with many parents makes large unless it is factored (Lauritzen & Spiegelhalter 1988, "Local Computations with Probabilities on Graphical Structures and Their Application to Expert Systems," *Journal of the Royal Statistical Society B* 50(2):157–224). The repairs in `#def-strategy-dag`'s Correlation Hierarchy are instances. Factoring a strict-prerequisite common cause above the branches it gates (L1) rewrites a reconvergent subgraph into a tree, by distributivity: $\mathrm{OR}(C \wedge A_1, C \wedge A_2) = C \wedge \mathrm{OR}(A_1, A_2)$. On the left C has two children; on the right it has one. The identity needs C to enter each branch as an AND-parent, so that the branch fails whenever C fails. This is the strict-prerequisite condition $\theta_{\text{child}|\neg C} = 0$, and it is why L1 is scoped to strict prerequisites. The credence of C's edges need not be 1: each edge's transmission is independent and moves into its branch, $C \wedge E_{C,i} \wedge A_i = C \wedge (E_{C,i} \wedge A_i)$. The mixture form L1' and explicit conditioning clamp a common cause that is a leaf (a condition node), so they are exact instances of the first route. The same fact is standard in two neighboring literatures. In Bayesian networks, Pearl's propagation is exact on singly connected graphs (Pearl 1988). It needs that stronger condition because it also handles evidence; a forward pass with no evidence needs only the multitree condition, which is why the multitree case above exists. In fault-tree analysis, gate-by-gate probability arithmetic is exact only when no basic event is repeated, and repeated events are handled through minimal cut sets (Vesely et al. 1981, *Fault Tree Handbook*, NUREG-0492).

What Is Derived vs. What Is Chosen.

Table R.1

Property	*Motivating postulate*	*Strength*
Directed edges	Temporal ordering (P1, `#post-causal-structure`)	Proved
Probabilistic uncertainty	Cox's theorem (P2)	Proved
Acyclicity	Temporal ordering + finite horizon (P1)	Proved
Internal structure	Fragility + monitoring (P4, `#der-chain-confidence-decay`)	Derived

Property	*Motivating postulate*	*Strength*
Markov factorization	Causal Markov Condition theorem (P1 causal interpretation + P2 probability + causal sufficiency)	Proved under causal sufficiency (CMC theorem)
DAG with Markov property	**P1 + P2 + causal sufficiency (CMC) + P4**	**Conditional on causal sufficiency — which is testable and repairable**
Exact status propagation	Noisy-AND/OR equations + causal sufficiency + tree structure	Proved: exact for all credences iff the DAG is an AND/OR tree (§"Exactness of Status Propagation")
AND/OR parameterization	Boolean completeness + parsimony	Hypothesis (binary outcomes only)
Single-parameter edges	Parsimony / IB	Formulation choice
Specific node ontology	—	Formulation choice

The dividing line: acyclicity and directed edges are proved; the full DAG-with-Markov-property is conditional on causal sufficiency. The parameterization (AND/OR, CPT form, edge semantics) is a formulation choice within the strongly motivated structure, motivated by parsimony and domain fit but not by mathematical necessity.

Equivalence Class. **Within the DAG class.** Multiple DAGs can encode the same conditional independence relations, forming a Markov equivalence class identified by a CPDAG (completed partially directed acyclic graph). Two DAGs in the same equivalence class make identical probabilistic predictions but may differ in causal interpretation.

Across representation types. Factor graphs, junction trees, influence diagrams, and chain graphs are NOT simple presentational variants of DAGs:

- **Factor graphs** and **junction trees** preserve factorization and inference structure without necessarily preserving directed causal semantics.
- **Influence diagrams** add decision and utility nodes — a richer object, not an equivalent one.
- **Chain graphs** can express independence models that are not representable as DAGs at all.
- **Markov equivalence** is a statement within DAG classes, not across all graphical model types.

The correct claim is narrow: for a given factorized distribution, DAG and factor-graph representations can compute the same marginals. But causal semantics (do-calculus) are DAG-specific and do not transfer to undirected or mixed representations without additional structure.

AAT's choice. AAT uses DAG + AND/OR because: (a) AND/OR is the most parsimonious complete basis for binary combination (`#scope-and-or`), (b) the DAG naturally supports causal/interventional reasoning (Pearl's do-calculus), and (c) the representation converged across three independent formalism attempts.

Epistemic Status.

The acyclicity derivation is *exact* — it follows from temporal ordering over a finite horizon via standard order theory. The individual postulates P1, P2, and P4 are each well-grounded (temporal structure, Cox's theorem, and chain fragility respectively).

The Markov property is now *proved under causal sufficiency* via the Causal Markov Condition theorem (Spirtes, Glymour & Scheines 2000, Theorem 3.4; Pearl 2009, Theorem 1.4.1). The previous sketch argument (P3 requires locality → P1 identifies parents → therefore Markov) has been replaced by a rigorous chain: P1-P2 establish the causal DAG structure, causal sufficiency guarantees exogenous independence, and the CMC theorem proves the factorization. P3 (local

revisability) is now met by the Markov property rather than assumed: the CMC makes the representation local for every causally sufficient DAG, and forward revision of consequences is exact on AND/OR trees (Step 3(d), §"Exactness of Status Propagation").

The conditioning on causal sufficiency is the right level of conditionality: it is a property of *strategy quality* (did the agent include all relevant common causes as nodes?), not of agent architecture. It is testable — under exploration: on-policy residuals cannot reveal a missing common cause (`#der-causal-insufficiency-detection`, the no-go), but pairwise sibling covariance under the agent's own interventions can — and repairable in practice (add the common-cause node). When causal sufficiency fails, the Markov factorization fails, and this manifests as correlated failure and sign-determined $\hat{P}_\Sigma$ bias — overestimation for the common OR-dominated-root case (`#def-strategy-dag`). Causal sufficiency is necessary for exact status propagation but not sufficient. The exactness result (§"Exactness of Status Propagation") is *exact* under its stated premises (the noisy-AND/OR structural equations and causal sufficiency): propagation is exact for all credences if and only if the DAG is an AND/OR tree. The sign result in the same section (propagation is a lower bound when every reconvergent node is AND, an upper bound when every one is OR) is exact under the same premises; it was proposed by the independent verifier and re-derived by the integrator.

Max attainable: *exact* for acyclicity (already there). *Derived conditional* for the full DAG-with-Markov-property — the derivation is rigorous (invokes a proved theorem), and the remaining condition (causal sufficiency) is about model quality, not proof quality.

The AND/OR restriction is a *hypothesis* for binary outcomes, grounded in Boolean completeness and parsimony. For non-binary outcomes, it does not apply and richer parameterizations within the derived graphical structure are needed.

The parallel to Cox's theorem is now tighter than previously stated: Cox's theorem proves that consistency axioms force probability; the CMC theorem proves that causal structure under sufficiency forces the Markov factorization. Both are formal results, not analogies. The remaining gap: Cox's axioms are necessary and sufficient for probability; AAT's postulates are sufficient for DAG+Markov structure, but the necessity direction (could a non-DAG structure satisfy P1-P4?) is not established. For practical purposes this gap is unimportant — the proved sufficiency gives a rigorous foundation for the strategy representation.

Discussion.

The contribution of this analysis. The dividing line between derived/conditional structure and chosen parameterization is the key result. The agent community (and future researchers) know exactly what is proved (acyclicity, directed edges), what is conditional (Markov property, contingent on causal sufficiency), and what is a formulation choice (the parameterization within the graph). Alternative parameterizations can be explored without abandoning the theoretical foundation.

Acyclicity deserves emphasis. The existing theory previously flagged "DAG acyclicity is an assumption, not forced" as a known fragility. The derivation above resolves this: acyclicity of Σ_t follows from temporal ordering over a finite planning horizon. This is specific to the strategy — M_t's environment model is not restricted to acyclic structures.

The causal sufficiency assumption. The Markov condition assumes causal sufficiency — no hidden common causes within the strategy. Since the agent constructs Σ_t, there are no "hidden" variables internal to the strategy (the agent chose all the nodes). But environmental common causes that affect multiple strategy steps may violate the Markov condition within Σ_t. If so, latent variable models or causal graphs with hidden nodes would be needed. This connects to the edge identifiability problem noted in `#def-strategy-dag`.

Appendix S

Derivation: Sector Condition Verification for Edge-Credence Dynamics

Complete verification that specific strategy update dynamics satisfy the sector condition from A, backing the schema proposed in `#schema-strategy-persistence`. Seven propositions are proved across distinct topologies. **B.1 (single edge):** sector condition holds globally with $\alpha_\Sigma = 1/(n+1)$, tight. **B.2 (two-edge AND chain, observable intermediate):** weakest-link result with $\alpha_\Sigma = \min(1/(n_1+1), \theta_1/(n_2+1))$ — the evidence-starvation effect on the downstream edge attenuates by the upstream success probability. **B.3 (two-edge AND chain, unobservable intermediate):** per-edge sector condition *fails* (the marginal point estimate is biased), but plan-level tracking recovers persistence with $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi+1)$ at the cost of per-edge diagnostic resolution. **B.4 (two-arm OR, ε-greedy):** *exploration-gated* weakest-link — pure greedy *fails* the sector condition; persistence requires exploration inside a two-sided interval, enough for the explored arm and little enough for the greedy one (the sufficient-exploration condition (SA3) from `#schema-strategy-persistence`).

B.5 (sector transfer to plan-confidence error): the sector condition transfers from per-edge credence space to *strategy-plan-confidence error* via the Jacobian — and the transfer is *credit-assignment-free*, requiring only status propagation. **B.6 (L1 augmented DAG, strict-prerequisite common cause):** persistence transfers with three-way gating (condition-testing rate, evidence-starvation factor, exploration-gating factor) — the formal extension to causally-insufficient DAGs handled by augmentation. **B.7 (L1' mixture form, soft-facilitator common cause):** persistence holds under *observable* common cause via five-way gating, and is *refuted* under unobservable common cause by the Cramér-Rao floor (the second formal instance of `#disc-identifiability-floor`).

The overall result: strategic persistence has a complete verified instance set for the canonical topologies. AND-chains exhibit depth-gated (geometric) attenuation; OR-nodes exhibit exploration-gated allocation; L1-augmented DAGs combine condition-testing, evidence-starvation, and exploration-gating. The chapter's claim that strategic persistence is structurally identical to epistemic persistence at the sector-template level is *verified* by these instantiations.

Motivation.

`#schema-strategy-persistence` proposes that the sector-condition mathematics (Props A.1, A.1S, A.2 in A) extends from epistemic to strategic mismatch. The schema identifies three structural conditions (SA1, SA2', SA3) and conjectures the persistence form $\alpha_\Sigma > \rho_\Sigma / R_\Sigma$. This appendix instantiates the schema for concrete Beta-Bernoulli edge dynamics, verifying the conditions case by case. Each proposition establishes the sector parameter α_Σ for a specific DAG topology; the persistence condition then follows by direct substitution into Proposition A.1 of A.

Common Setup.

Throughout, edge credences follow Beta-Bernoulli dynamics: edge k has true success probability $\theta_k \in (0,1)$, agent belief $p_k \sim \text{Beta}(\alpha_k, \beta_k)$ with point estimate $\hat{p}_k = \alpha_k/n_k$ where $n_k = \alpha_k + \beta_k$, and conjugate update on observing $y_k \in \{0,1\}$:

Definition (Beta-Bernoulli edge dynamics)

$$\Delta\hat{p}_k = \frac{y_k - \hat{p}_k}{n_k + 1}, \qquad \eta_k = \frac{1}{n_k + 1} \tag{S.1}$$

The single-step update matches the gain-based form from `#hyp-edge-update-via-gain` with $\eta_{\text{edge}} = 1/(n_k + 1)$.

Parameterization note. Propositions B.1–B.7 below are stated in the *moment-parameter* (probability-space) coordinate $\hat{p}_k \in [0,1]$, consistent with the Beta-Bernoulli conjugate presentation. The log-odds coordinate $\lambda_k = \log(\hat{p}_k/(1-\hat{p}_k)) \in \mathbb{R}$ is the unique additive-evidence presentation forced by the evidential-additivity axiom (V); the sector-parameter content of each proposition is Fisher-equivalent in either coordinate, so the derivations are retained in moment-parameter form for algebraic tightness ($\eta_k = 1/(n_k+1)$ is exact in probability space). The log-odds coordinate becomes load-bearing in `#disc-credit-assignment-boundary` for the continuous-gradient signal function, where the probability-space presentation would exhibit a mechanical domain break.

Strategic mismatch per edge is $\delta_k = \hat{p}_k - \theta_k$. The sector-condition framework (`#form-sector-condition`) requires: (SA1) $F(\mathbf{0}) = \mathbf{0}$, (SA2') $\delta^T\mathbf{F}(\delta) \geq \alpha_\Sigma\|\delta\|^2$ within a region $\mathcal{B}_R$.

Regime-adjustment convention. The propositions below state sector parameters α_Σ for the **canonical Regime-A case**, in which every edge's evidence is fully interventional and attributable (identifiability coefficient $\iota_{ij} = 1$ per `#scope-edge-update-causal-validity`). For edges with $\iota_{ij} < 1$ (Regime B or C), each sector parameter carries an implicit ι_{ij} factor:

$$\alpha_\Sigma^{\text{eff}} = \iota_{ij} \cdot \alpha_\Sigma^{\text{stated}} \tag{S.2}$$

This commutes through every diagonal sector product in Props B.1–B.4 and B.6 because each edge's contribution to $\delta_\Sigma^T\mathbf{F}$ is scaled by ι_{ij} independently: Regime-C edges ($\iota \approx 0$) contribute essentially nothing to the weakest-link minimum, and Regime-A edges ($\iota \approx 1$) recover the stated formulas exactly. The persistence condition $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$ becomes $\iota_{ij} \cdot \alpha_\Sigma^{\text{stated}} > \rho_{\Sigma,ij}/R_{\Sigma,ij}$ per edge. The propositions below state everything in Regime-A form for clarity; the ι-adjustment is uniform across cases and does not alter the structural results (evidence starvation, exploration gating, three-way gating) — only their quantitative thresholds per edge.

For Prop B.3 (unobservable intermediate), the regime adjustment does not change the A1 violation result: the bias is structural to the marginal Bayesian update, not to the identifiability quality. The plan-level fallback still recovers the sector condition with an ι-adjusted $\alpha_{\Sigma,\text{plan}}$ governed by the aggregate's identifiability.

Proposition B.1: Single Edge ($A \to G$).

Setup. One action A, one goal G, one edge with credence $\hat{p}$ and true probability θ. Mismatch: $\delta_\Sigma = \hat{p} - \theta$ (scalar).

Statement. Under Beta-Bernoulli updating, the expected correction function satisfies the sector condition globally (all $|\delta_\Sigma| \leq 1$) with:

Derived (Conditional on Beta-Bernoulli model)

$$\alpha_\Sigma = \frac{1}{n+1} \tag{S.3}$$

The bound is tight.

Proof.

The expected correction. Since $y \sim \text{Bernoulli}(\theta)$:

Derived (Proof Step)

$$\mathbb{E}[\Delta\hat{p}] = \frac{\theta - \hat{p}}{n+1} = -\frac{\delta_\Sigma}{n+1} \tag{S.4}$$

Identifying $F_\Sigma(\delta_\Sigma) = -\mathbb{E}[\Delta\delta_\Sigma] = \delta_\Sigma/(n+1)$:

Derived (Proof Step)

$$\delta_\Sigma \cdot F_\Sigma(\delta_\Sigma) = \frac{\delta_\Sigma^2}{n+1} = \frac{1}{n+1}\|\delta_\Sigma\|^2 \tag{S.5}$$

So $\alpha_\Sigma = 1/(n+1)$. The correction is exactly linear, so the sector bound is tight (not merely a lower bound).

Verification of (SA1). At $\delta_\Sigma = 0$: $F_\Sigma(0) = 0$. Satisfied.

Stochastic ultimate bound. Decomposing $\delta_{\Sigma,t+1} = \delta_\Sigma \cdot n/(n+1) + (y-\theta)/(n+1)$ and computing $\mathbb{E}[\delta_{\Sigma,t+1}^2]$:

Derived (stochastic ultimate bound, single edge)

$$\mathbb{E}[\delta_{\Sigma,t+1}^2] = \delta_\Sigma^2 \cdot \frac{n^2}{(n+1)^2} + \frac{\theta(1-\theta)}{(n+1)^2} \tag{S.6}$$

The expected Lyapunov derivative $\mathbb{E}[\Delta V] < 0$ whenever:

$$|\delta_\Sigma| > \sqrt{\frac{\theta(1-\theta)}{2n+1}} \tag{S.7}$$

The stochastic noise floor decreases as $O(1/\sqrt{n})$ — the standard Bayesian posterior concentration rate. □

Proposition B.1F: Single Edge under Exponential Forgetting.

Setup. As B.1, with the pseudo-counts discounted by $\lambda \in [0, 1)$ before each observation is added (the standard exponential-forgetting recurrence; Ljung 1987):

Definition (Discounted Beta-Bernoulli update)

$$\alpha \mapsto \lambda\,\alpha + y, \qquad \beta \mapsto \lambda\,\beta + (1-y) \tag{S.8}$$

Statement. Let n_t be the count after the t-th update and $n_{\text{eff}} = 1/(1-\lambda)$. Then:

Derived (Conditional on Beta-Bernoulli model + exponential forgetting)

1. The per-observation gain is $g_t = 1/n_t$, and the count dynamics $n_t = \lambda n_{t-1} + 1$ are deterministic (independent of the data).
2. At the stationary count the gain is exactly $1-\lambda$, so the sector parameter is $\alpha_\Sigma^{\text{ss}} = 1-\lambda$, tight. Initialized at n_{eff}, the estimator is the exponentially weighted moving average $\hat{p}_{t+1} = \lambda\hat{p}_t + (1-\lambda)\,y_t$.
3. **Trajectory guarantee.** Under Model D drift $|\theta_{t+1} - \theta_t| \le \rho_\Sigma$, if the initial count satisfies $n_0 \le n_{\text{eff}}$, the mean mismatch $m_t = \mathbb{E}[\hat{p}_t] - \theta_t$ obeys $\sup_t |m_t| \le \max(|m_0|,\ \rho_\Sigma/(1-\lambda))$ for all t, not only asymptotically. If $n_0 > n_{\text{eff}}$ the gains rise toward $1-\lambda$ from below and the bound holds asymptotically.
4. **Reachable region.** Some $\lambda \in [0, 1)$ satisfies $1 - \lambda > \rho_\Sigma/R_\Sigma$ if and only if $\rho_\Sigma < R_\Sigma$, approached as $\lambda \to 0$ (where $\hat{p} = y$, gain 1).

Proof. With $n = \alpha + \beta$ and $\hat{p} = \alpha/n$ before observing y:

Derived (Proof Step)

$$\hat{p}' - \hat{p} = \frac{\lambda\alpha + y}{\lambda n + 1} - \frac{\alpha}{n} = \frac{y - \hat{p}}{\lambda n + 1} \tag{S.9}$$

so the gain is the reciprocal of the post-update count $n' = \lambda n + 1$. B.1's $\eta = 1/(n+1)$ is the same statement without discounting: in both, the denominator is the count the new observation is added to *plus one*, and under discounting that count is λn, not n. The count recursion has fixed point $n_{\text{eff}} = 1/(1-\lambda)$, where $\lambda n_{\text{eff}} + 1 = n_{\text{eff}}$ and hence $g = 1 - \lambda$. The expected correction is $\mathbb{E}[\Delta\hat{p}] = (1-\lambda)(\theta - \hat{p})$, exactly linear, so the B.1 sector product gives $\alpha_\Sigma^{\text{ss}} = 1 - \lambda$ with equality. Item 3: since g_t is deterministic, $m_{t+1} = (1 - g_t)\, m_t - d_t$ exactly, with $d_t = \theta_{t+1} - \theta_t$. If $n_0 \le n_{\text{eff}}$ then n_t increases monotonically to n_{eff}, so $g_t \ge 1 - \lambda$; if $|m_t| \le M$ with $M \ge \rho_\Sigma/(1-\lambda)$, then $|m_{t+1}| \le \lambda M + \rho_\Sigma \le M$. Item 4 is item 2 solved for λ. □

What the reachable region is. $\rho_\Sigma < R_\Sigma$ is not specific to forgetting. Under Model D a scalar linear corrector of gain g has ultimate bound ρ/g for $g \le 1$ and $\rho/(2-g)$ for $g \in (1,2)$ (P DA.1), both at least ρ, with equality only at $g = 1$; and an innovation-form update $\hat{p} + g(y - \hat{p})$ keeps $\hat{p} \in [0,1]$ for both outcomes only if $g \le 1$. So $\rho_\Sigma < R_\Sigma$ is the floor every admissible single-edge corrector faces, and exponential forgetting reaches it. Forgetting's structural role is the opposite boundary: at $\lambda = 1$ the gain $1/n_t \to 0$, and every positive ρ_Σ is eventually fatal. Where a nontrivial cap on strategic persistence does appear is evidence allocation across edges that share one observation stream (T).

With a uniform prior ($n_0 = 2$) the all-time guarantee of item 3 needs $\lambda \ge 1/2$; for $\lambda < 1/2$ the gains start below $1 - \lambda$ and approach it geometrically, so the guarantee is asymptotic. Initializing the count at n_{eff} (or below it) makes it all-time for every λ.

Proposition B.2: Two-Edge AND-Chain, Observable Intermediate ($A \to B \to G$, B observed).

Setup. Action A leads to intermediate B, then to goal G. Both y_B and y_G are observed. Edge credences $\hat{p}_1, \hat{p}_2$ with true probabilities θ_1, θ_2. Plan confidence: $\hat{P}_\Sigma = \hat{p}_1\hat{p}_2$ (AND propagation from `#scope-and-or`). Mismatch vector: $\delta_\Sigma = (\delta_1, \delta_2)^T$.

The generative model:

$$y_B \sim \text{Bernoulli}(\theta_1), \qquad y_G \mid y_B = 1 \sim \text{Bernoulli}(\theta_2), \qquad y_G \mid y_B = 0 = 0 \tag{S.10}$$

Statement. Under Beta-Bernoulli updating with observable intermediate, the expected correction function is diagonal and satisfies the sector condition globally with:

Derived (Conditional on Beta-Bernoulli model, observable intermediate)

$$\alpha_\Sigma = \min\left(\frac{1}{n_1 + 1},\ \frac{\theta_1}{n_2 + 1}\right) \tag{S.11}$$

Proof.

Edge 1 is tested every trial: $\Delta\hat{p}_1 = (y_B - \hat{p}_1)/(n_1 + 1)$, giving $\mathbb{E}[\Delta\hat{p}_1] = -\delta_1/(n_1 + 1)$.

Edge 2 is tested only when $y_B = 1$ (probability θ_1). When $y_B = 0$, edge 2 receives no information:

Derived (Proof Step)

$$\mathbb{E}[\Delta\hat{p}_2] = \theta_1 \cdot \frac{\theta_2 - \hat{p}_2}{n_2 + 1} + (1 - \theta_1) \cdot 0 = -\frac{\theta_1\delta_2}{n_2 + 1} \tag{S.12}$$

The expected correction function is diagonal:

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_1/(n_1+1) \\ \theta_1\delta_2/(n_2+1) \end{pmatrix} \tag{S.13}$$

Verification of (SA1). At $\delta_\Sigma = \mathbf{0}$: $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. Satisfied.

The sector product:

Derived (Proof Step)

$$\delta_\Sigma^T \mathbf{F}(\delta_\Sigma) = \frac{\delta_1^2}{n_1+1} + \frac{\theta_1\delta_2^2}{n_2+1} \geq \min\left(\frac{1}{n_1+1}, \frac{\theta_1}{n_2+1}\right)(\delta_1^2 + \delta_2^2) \tag{S.14}$$

The minimum is achieved by whichever edge has the lower effective correction rate. □

The evidence-starvation effect. Edge 2's effective correction rate $\theta_1/(n_2+1)$ is attenuated by the upstream success probability θ_1. When θ_1 is small, edge 2 is starved of evidence — it is tested rarely, learns slowly, and becomes the weakest link. This gives a precise mechanism for `#der-chain-confidence-decay`: downstream edges in AND-chains are harder to calibrate, with correction rate proportional to the product of all upstream success probabilities.

Generalization to depth d. For a chain of depth d with all intermediates observable, edge k's effective correction rate is $\alpha_k = \prod_{j=1}^{k-1} \theta_j/(n_k+1)$, giving:

Derived (depth-d chain sector parameter)

$$\alpha_\Sigma = \min_{k=1}^{d} \frac{\prod_{j=1}^{k-1} \theta_j}{n_k+1} \tag{S.15}$$

Proposition B.3: Two-Edge AND-Chain, Unobservable Intermediate ($A \to B \to G$, B not observed).

Setup. Same DAG as Proposition B.2, but only $y_G \in \{0,1\}$ is observed. The agent cannot distinguish "$A \to B$ failed" from "$B \to G$ failed" when $y_G = 0$.

Statement. (a) The per-edge sector condition fails: the marginal Bayesian update violates (SA1) with systematic bias $O(1/n)$. (b) Plan-level tracking (treating $\hat{\Phi} = \hat{p}_1\hat{p}_2$ as a single Beta over the composite probability $\Phi = \theta_1\theta_2$) recovers the sector condition with:

Derived (Conditional on Beta-Bernoulli model, unobservable intermediate)

$$\alpha_{\Sigma,\text{plan}} = \frac{1}{n_\Phi + 1} \tag{S.16}$$

Proof of (a): Per-edge (SA1) violation.

The joint posterior after success ($y_G = 1$) factors cleanly: $\alpha_k \to \alpha_k + 1$ for both edges. The joint posterior after failure ($y_G = 0$) includes the factor $(1 - \theta_1\theta_2)$, which does not decompose into separate functions of θ_1 and θ_2. Failure introduces posterior correlation.

The exact marginal Bayesian update for edge 1 on failure is:

Derived (Proof Step)

$$\Delta\hat{p}_1^{(\text{fail})} = -\frac{\hat{p}_1\, q_1}{n_1+1} \tag{S.17}$$

where $q_1 = (1-\hat{p}_1)/(1-\hat{p}_1\hat{p}_2)$ is the posterior probability that edge 1 caused the failure (the proportional-blame weight). The expected update, combining success (probability $\theta_1\theta_2$) and failure (probability $1-\theta_1\theta_2$):

Derived (Proof Step)

$$\mathbb{E}[\Delta\hat{p}_1] = \theta_1\theta_2 \cdot \frac{1-\hat{p}_1}{n_1+1} - (1-\theta_1\theta_2) \cdot \frac{\hat{p}_1\, q_1}{n_1+1} \tag{S.18}$$

Evaluating at truth ($\hat{p}_k = \theta_k$):

Derived (SA1 violation magnitude)

$$\mathbb{E}[\Delta\hat{p}_1]\big|_{\hat{p}=\theta} = -\frac{\theta_1(1-\theta_1)(1-\theta_2)}{n_1+1} \neq 0 \tag{S.19}$$

The bias is negative (pushes credence below truth), of magnitude $O(1/n)$, and arises because success always increments α_1 by a full unit while failure increments β_1 by the fractional weight $q_1 < 1$. The success case over-credits edge 1 because the agent cannot separate edge 1's contribution from edge 2's. By symmetry, $\hat{p}_2$ is also biased downward. □

Proof of (b): Plan-level recovery.

If the agent tracks $\hat{\Phi}$ directly as a Beta posterior over $\Phi = \theta_1\theta_2$ using $y_G \sim \text{Bernoulli}(\Phi)$, the setup reduces to a single-edge problem (Proposition B.1) with mismatch $\delta_\Phi = \hat{\Phi} - \Phi$ and sector parameter $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$. All single-edge results apply. □

Tradeoff. Per-edge tracking provides diagnostic resolution (which edge is failing?) but yields biased, non-sector-conforming updates when the intermediate is unobservable. Plan-level tracking sacrifices diagnostic resolution but provides clean sector-condition satisfaction. This is a concrete instance of `#der-observability-dominance`: unobservable nodes freeze per-edge learning, making the aggregate the correct unit of analysis.

Non-identifiability. From y_G observations alone, only $\Phi = \theta_1\theta_2$ is identifiable. The per-edge decomposition (θ_1, θ_2) is underdetermined — any pair with the same product produces identical observations. Per-edge identification requires per-edge observability (or structural constraints that break the symmetry).

Proposition B.4: Two-Arm OR-Node ($A_1, A_2 \to G$, ε-greedy).

Setup. Goal G reachable via two alternative actions A_1, A_2 (an OR-node from `#scope-and-or`). True probabilities θ_1, θ_2. Agent selects one action per trial: with probability $1-\varepsilon$ the greedy choice (higher $\hat{p}_k$), with probability ε the other. Mismatch vector: $\delta_\Sigma = (\delta_1, \delta_2)^T$.

Statement. (a) Under a greedy policy ($\varepsilon = 0$), the sector condition fails for the full mismatch vector. (b) Under ε-greedy action selection with $\varepsilon > 0$, the sector condition is satisfied with:

Derived (Conditional on Beta-Bernoulli model, ε-greedy OR-node)

$$\alpha_\Sigma = \min\left(\frac{1-\varepsilon}{n_1+1},\ \frac{\varepsilon}{n_2+1}\right) \tag{S.20}$$

where arm 1 is the greedy choice.

Proof of (a): Greedy failure.

Under greedy selection, only the chosen arm updates. Assuming $\hat{p}_1 > \hat{p}_2$ (arm 1 is greedy):

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_1/(n_1+1) \\ 0 \end{pmatrix} \tag{S.21}$$

The sector product $\delta_\Sigma^T\mathbf{F} = \delta_1^2/(n_1+1)$. For the sector condition to hold, we need $\delta_1^2/(n_1+1) \geq \alpha_\Sigma(\delta_1^2+\delta_2^2)$. Setting $\delta_1 = 0$: the LHS is zero but $\alpha_\Sigma\delta_2^2 > 0$ for $\delta_2 \neq 0$. No $\alpha_\Sigma > 0$ works. □

Proof of (b): ε-greedy recovery.

The expected correction (arm 1 greedy):

Derived (Proof Step)

$$\mathbb{E}[\mathbf{F}(\delta_\Sigma)] = \begin{pmatrix} (1-\varepsilon)\,\delta_1/(n_1+1) \\ \varepsilon\,\delta_2/(n_2+1) \end{pmatrix} \tag{S.22}$$

The sector product:

Derived (Proof Step)

$$\delta_\Sigma^T \mathbf{F} = \frac{(1-\varepsilon)\,\delta_1^2}{n_1+1} + \frac{\varepsilon\,\delta_2^2}{n_2+1} \geq \min\left(\frac{1-\varepsilon}{n_1+1},\ \frac{\varepsilon}{n_2+1}\right)(\delta_1^2+\delta_2^2) \tag{S.23}$$

Verification of (SA1). At $\delta_\Sigma = \mathbf{0}$: $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. Satisfied. □

Optimal exploration rate. Maximizing α_Σ over ε by equalizing the two terms:

Derived (optimal exploration rate)

$$\varepsilon^* = \frac{n_2+1}{n_1+n_2+2}, \qquad \alpha_\Sigma^* = \frac{1}{n_1+n_2+2} \tag{S.24}$$

For equal experience ($n_1 = n_2$): $\varepsilon^* = 1/2$ (equal allocation). For unequal experience, ε^* allocates more trials to the arm with higher n (lower gain) to equalize correction rates.

The k-arm generalization. For k OR-alternatives with ε-uniform exploration (probability $\varepsilon/(k-1)$ per non-greedy arm):

Derived (k-arm OR-node sector parameter)

$$\alpha_\Sigma = \min\left(\frac{1-\varepsilon}{n_{\text{greedy}}+1},\ \min_{j\neq\text{greedy}} \frac{\varepsilon/(k-1)}{n_j+1}\right) \tag{S.25}$$

With optimal equal-rate exploration and equal experience: $\alpha_\Sigma = 1/(k(n+1))$. The sector parameter scales as $1/k$ — each additional alternative dilutes correction capacity.

Condition (SA3). Substituting B.4's α_Σ into the persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$, the exploration rate must lie in a two-sided interval: high enough that the explored arm is corrected, low enough that the greedy arm still is:

Derived (admissible exploration interval for persistence)

$$\frac{\rho_\Sigma(n_2+1)}{R_\Sigma} \;<\; \varepsilon \;<\; 1 - \frac{\rho_\Sigma(n_1+1)}{R_\Sigma} \tag{S.26}$$

The interval is non-empty iff $\rho_\Sigma(n_1+n_2+2) < R_\Sigma$. Under exponential forgetting with per-observation discounting at the stationary count (Prop B.1F), each pull lands with gain $1-\lambda$ and the interval becomes $\rho_\Sigma/((1-\lambda)R_\Sigma) < \varepsilon < 1-\rho_\Sigma/((1-\lambda)R_\Sigma)$, non-empty iff $\rho_\Sigma < (1-\lambda)R_\Sigma/2$. No exploration rate and no update rule lifts the two-arm node past $\rho_\Sigma < R_\Sigma/2$ (T).

A purely greedy agent ($\varepsilon = 0$) violates the sector condition because the unchosen alternative's mismatch grows without correction. This is the structural justification for (SA3) in `#schema-strategy-persistence`: OR-nodes require deliberate exploration to maintain strategic persistence.

Persistence Conditions.

Substituting each proposition's α_Σ into the persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$ from Proposition A.1 of A:

Single edge (B.1):

Derived (single-edge persistence threshold)

$$\frac{1}{n+1} > \frac{\rho_\Sigma}{R_\Sigma} \quad \Longleftrightarrow \quad n < \frac{R_\Sigma}{\rho_\Sigma} - 1 \tag{S.27}$$

The critical experience level $n^* = R_\Sigma/\rho_\Sigma - 1$ is the point at which gain collapse occurs: the posterior has concentrated so much that the agent can no longer track environmental drift.

Two-edge observable (B.2):

Derived (two-edge persistence threshold, observable)

$$\min\left(\frac{1}{n_1+1},\ \frac{\theta_1}{n_2+1}\right) > \frac{\rho_\Sigma}{R_\Sigma} \tag{S.28}$$

This decomposes into two independent conditions: $n_1 < R_\Sigma/\rho_\Sigma - 1$ and $n_2 < \theta_1 R_\Sigma/\rho_\Sigma - 1$. The downstream edge has a stricter bound, tightened by θ_1. Gain collapse hits downstream edges first.

Two-edge unobservable (B.3):

$$\frac{1}{n_\Phi+1} > \frac{\rho_\Sigma}{R_\Sigma} \quad \Longleftrightarrow \quad n_\Phi < \frac{R_\Sigma}{\rho_\Sigma} - 1 \tag{S.29}$$

Same form as single-edge, applied to the plan-level aggregate.

Two-arm OR, ε-greedy (B.4):

Derived (OR-node persistence threshold)

$$\min\left(\frac{1-\varepsilon}{n_1+1},\ \frac{\varepsilon}{n_2+1}\right) > \frac{\rho_\Sigma}{R_\Sigma} \tag{S.30}$$

With optimal exploration and equal experience ($n_1 = n_2 = n$):

$$n < \frac{R_\Sigma}{2\rho_\Sigma} - 1 \tag{S.31}$$

The critical experience is halved relative to the single-edge case. Each additional OR-alternative further reduces the critical level by $1/k$.

Proposition B.6: Mixed AND/OR with Common-Cause Node (L1 Augmented DAG).

Setup. Goal G reachable via two alternative actions A_1, A_2 (an OR-node), but both actions require a shared condition C (a common-cause node). The L1 DAG factors the common cause above the OR-structure: $G = \text{AND}(C, G_{\text{sub}})$ where $G_{\text{sub}} = \text{OR}(A_1, A_2)$. Three leaves: condition C with true probability θ_C, actions A_1, A_2 with true conditional probabilities $\theta_{1|C}, \theta_{2|C}$. Agent selects ε-greedy among A_1, A_2 (only testable when $C = 1$). Mismatch vector: $\delta_\Sigma = (\delta_C, \delta_{A_1}, \delta_{A_2})^T$.

Plan confidence: $\hat{P}_\Sigma = \hat{p}_C \cdot [1 - (1 - \hat{p}_{A_1})(1 - \hat{p}_{A_2})]$. Reference value: $\Phi = \theta_C \cdot [1 - (1 - \theta_{1|C})(1 - \theta_{2|C})]$. At truth, $\hat{P}_\Sigma = \Phi =$ actual plan success probability (L1 is unbiased).

Statement. Under Beta-Bernoulli updating with observable condition C and ε-greedy path selection, the expected correction function satisfies the sector condition globally with:

$$\alpha_\Sigma = \min\left(\frac{1}{n_C + 1},\ \frac{\theta_C(1-\varepsilon)}{n_{A_1} + 1},\ \frac{\theta_C \varepsilon}{n_{A_2} + 1}\right) \tag{S.32}$$

Derived (Conditional on Beta-Bernoulli model, L1 augmented DAG, observable common cause C, ε-greedy action selection)

where A_1 is the greedy choice.

Proof.

Edge C (condition leaf, observed every trial):

$$\mathbb{E}[\Delta \hat{p}_C] = \frac{\theta_C - \hat{p}_C}{n_C + 1} = -\frac{\delta_C}{n_C + 1} \tag{S.33}$$

Edge A_1 (action leaf, tested when $C = 1$ and path 1 selected — probability $\theta_C(1-\varepsilon)$):

$$\mathbb{E}[\Delta \hat{p}_{A_1}] = -\frac{\theta_C(1-\varepsilon)\,\delta_{A_1}}{n_{A_1} + 1} \tag{S.34}$$

Edge A_2 (action leaf, tested when $C = 1$ and path 2 selected — probability $\theta_C \varepsilon$):

$$\mathbb{E}[\Delta \hat{p}_{A_2}] = -\frac{\theta_C \varepsilon\,\delta_{A_2}}{n_{A_2} + 1} \tag{S.35}$$

The expected correction function is diagonal:

$$\mathbf{F}(\delta_\Sigma) = \begin{pmatrix} \delta_C/(n_C+1) \\ \theta_C(1-\varepsilon)\,\delta_{A_1}/(n_{A_1}+1) \\ \theta_C \varepsilon\,\delta_{A_2}/(n_{A_2}+1) \end{pmatrix} \tag{S.36}$$

Verification of (SA1). $\mathbf{F}(\mathbf{0}) = \mathbf{0}$. ✓

Sector product:

$$\delta_\Sigma^T \mathbf{F} = \frac{\delta_C^2}{n_C+1} + \frac{\theta_C(1-\varepsilon)\,\delta_{A_1}^2}{n_{A_1}+1} + \frac{\theta_C \varepsilon\,\delta_{A_2}^2}{n_{A_2}+1} \;\geq\; \alpha_\Sigma \|\delta_\Sigma\|^2 \quad \square \tag{S.37}$$

Three-way gating. This is the first mixed AND/OR case. The sector parameter α_Σ is determined by three independent gating mechanisms: 1. **Condition testing** ($1/(n_C+1)$): the common cause is the easiest component to calibrate — tested every trial, same as B.1. 2. **Evidence starvation** (θ_C factor on action terms): both action edges are gated by the condition probability, same mechanism as B.2. 3. **Exploration gating** ($(1-\varepsilon)$ and ε factors): the two OR-alternatives compete for test opportunities, same mechanism as B.4.

The bottleneck is typically the explore action behind the condition: $\theta_C\varepsilon/(n_{A_2}+1)$.

B.5b applies. The Jacobian $\mathbf{J} = (s_{G_{\text{sub}}},\ \hat{p}_C(1-\hat{p}_{A_2}),\ \hat{p}_C(1-\hat{p}_{A_1}))^T$ is non-negative (monotone AND/OR). Corrections are componentwise (each leaf updates independently). Therefore $\alpha_s = \alpha_c = \alpha_\Sigma$ — the strategy-plan-confidence error inherits the sector condition with no penalty.

Optimal exploration. Equalizing the action terms: $\varepsilon^* = (n_{A_2}+1)/(n_{A_1}+n_{A_2}+2)$, giving $\alpha_\Sigma^* = \min(1/(n_C+1),\ \theta_C/(n_{A_1}+n_{A_2}+2))$.

L0 comparison. An L0 model of the same scenario (no C node, marginal probabilities $\theta_k = \theta_C \cdot \theta_{k|C}$) has $\alpha_\Sigma^{L0} = \min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$ and reference value $\Phi^{L0} = 1-(1-\theta_1)(1-\theta_2)$. The L0 sector parameter is *higher* (no θ_C attenuation) but Φ^{L0} systematically overestimates actual plan success. L1 persistence is harder (lower α_Σ) but calibration is honest (Φ^{L1} matches reality). The tradeoff: L0 is easier to maintain but calibrates to a biased target; L1 is harder to maintain but calibrates to truth.

The L1 construction principle. The common-cause node C must be factored *above* the OR-structure whose children it correlates. A naive construction (adding C as a parent of both A_1 and A_2 while keeping them as OR-siblings) does not fix the overestimation because the OR-propagation formula treats siblings as marginally independent. Correct L1 construction makes C an AND-prerequisite above the OR-structure, so the OR-children are conditionally independent given C. In graph terms the naive construction gives C two children that reconverge, and the correct one is a tree, which is the condition under which propagation is exact (R, §"Exactness of Status Propagation"). Full worked example with quantitative L0/L1 comparison: #example-L1.

Summary of Results.

Table S.1

Case	*Topology*	α_Σ	*SA1*	*SA3*	*Weakest link*
B.1 Single edge	$A \to G$	$1/(n+1)$	Yes	N/A	—
B.1F Single edge, exponential forgetting	$A \to G$	$1-\lambda$ at the stationary count (exact)	Yes	N/A	—
B.2 Two-edge, B observable	$A \to B \to G$ (AND)	$\min(1/(n_1+1),\ \theta_1/(n_2+1))$	Yes	N/A	Depth-gated (evidence starvation)

Case	*Topology*	α_Σ	*SA1*	*SA3*	*Weakest link*
B.3 Two-edge, B unobservable	$A \to B \to G$ (AND)	$1/(n_\Phi+1)$ (plan-level)	Per-edge: No ($O(1/n)$ bias); Plan: Yes	N/A	Credit-assignment collapse
B.4 Two-arm OR, ε-greedy	$A_1, A_2 \to G$ (OR)	$\min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$	Yes	Required	Exploration-gated
B.5a Credence→value (linear)	Any	$\alpha_s = \alpha_c$ (Jacobian cancels)	—	—	None (exact transfer)
B.5b Credence→value (nonlinear, component-wise)	Any	$\alpha_s = \alpha_c$ ($J_k \geq 0$ preserves bound)	—	—	None (exact transfer)
B.5b Credence→value (coupled)	Any	$\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$	—	—	Inter-edge coupling
B.6 L1 augmented, mixed AND/OR	$G = \text{AND}(C, \text{OR}(A_1, A_2))$	$\min(1/(n_C+1),\ \theta_C(1-\varepsilon)/(n_{A_1}+1),\ \theta_C\varepsilon/(n_{A_2}+1))$	Yes	Required	Three-way gating
B.7 L1' mixture form, observable C	$\hat{P}_\Sigma^{L1'} = \hat{\theta}_C P_\Sigma(\widehat{\mathbf{p}_{\mid C}}) + (1-\hat{\theta}_C) P_\Sigma(\widehat{\mathbf{p}_{\mid \neg C}})$	$\min(1/(n_C+1),\ \min_j \theta_C \pi_{j\mid C}/(n_{j\mid C}+1),\ \min_j (1-\theta_C)\pi_{j\mid \neg C}/(n_{j\mid \neg C}+1))$	Yes	Required *and* facilitator monotonicity	Five-way gating; **refuted under unobservable C** (Cramér-Rao floor)

Structural results across cases:

- **Gain as sector parameter.** In every case, α_Σ is determined by the edge update gain $\eta_k = 1/(n_k+1)$, modulated by observability (θ_1 attenuation in B.2), credit-assignment (plan-level collapse in B.3), action-selection policy (ε in B.4), or a combination of these (B.6). The structural parallel between epistemic and strategic persistence is not an analogy — it is a mathematical identity at the sector-framework level.
- **Evidence starvation (AND-chains).** Downstream edges in AND-chains have correction rates attenuated by the product of upstream success probabilities. Depth increases fragility: α_Σ decays exponentially with chain depth.
- **Exploration gating (OR-nodes).** Unchosen OR-alternatives receive zero correction. Persistence requires deliberate exploration (SA3) inside a two-sided interval: enough to correct the explored arm, little enough that the greedy arm is still corrected. Pure greedy policies fail.
- **Three-way gating (L1 augmented).** When a common cause gates OR-alternatives, calibration faces condition testing, evidence starvation, and exploration gating simultaneously (B.6). The bottleneck is the least-explored action behind the rarest condition.
- **Five-way gating (L1' mixture, observable C).** When the common cause is a *soft facilitator* ($\theta_{j\mid\neg C} > 0$) and is observable per trial, calibration faces five gating mechanisms simultaneously (B.7): condition testing, evidence

starvation on each conditional branch ($C = 1$ and $C = 0$), and exploration gating on each branch. The bottleneck is whichever branch is rarer, modulated by exploration on that branch. **Identifiability obstruction under unobservable C:** without direct C-observation, the mixture parameters are non-identifiable from a single observation channel — the per-trial Fisher information matrix is rank 1 rather than rank $2K+1$, so by the Cramér-Rao bound (Cramér 1946, *Mathematical Methods of Statistics*, Princeton University Press) no unbiased online estimator on the joint conditional vector admits $\alpha > 0$. The agent must augment C-observability, run multi-child joint observations, or fall back to plan-level (L0-on-marginals) tracking.

- **Gain-collapse threshold.** In all cases, increasing experience eventually violates persistence when the environment drifts. The critical experience level n^* is inversely proportional to the drift rate ρ_Σ.
- **L1 tradeoff.** L1 augmented DAGs have lower α_Σ (harder to persist) but honest calibration (Φ^{L1} matches reality). L0 is easier to maintain but calibrates to a biased target. The choice depends on whether accuracy or trackability is the binding constraint.

Proposition B.7: L1' Mixture-Form Sector Transfer (Observable Common Cause).

Where Prop B.6 handles a strict-prerequisite common cause via an AND-prerequisite construction (L1), this proposition handles a *soft-facilitator* common cause via the mixture form L1' (`#def-strategy-dag`, Correlation Hierarchy). It is the L1' analog of B.6, generalizing three-way gating to five-way gating across two parallel conditional branches.

Setup. L1' (mixture-form) DAG with binary common cause C, true prior $\theta_C \in (0, 1)$, gating a sub-plan with conditional structures $G_{|C}$ (when $C = 1$) and $G_{|\neg C}$ (when $C = 0$). Each affected child j carries two conditional credences $\hat{p}_{j|C}$ and $\hat{p}_{j|\neg C}$. Plan confidence:

$$\hat{P}_\Sigma^{L1'} = \hat{\theta}_C \cdot R_\Sigma(\hat{\mathbf{p}}_{|C}) + (1 - \hat{\theta}_C) \cdot R_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \tag{S.38}$$

The agent observes C-state directly each trial (i.e., C is treated as an observable condition leaf, in parallel with B.6). Per trial, the agent (i) updates $\hat{\theta}_C$ from observed C via Beta-Bernoulli; (ii) on $C = 1$ trials, attempts a child j via the active selection policy $\pi_{|C}$ and updates $\hat{p}_{j|C}$ from y_j; (iii) on $C = 0$ trials, attempts a child j via $\pi_{|\neg C}$ and updates $\hat{p}_{j|\neg C}$ from y_j.

Mismatch state: $\xi = (\xi_C, \{\xi_{j|C}\}, \{\xi_{j|\neg C}\})$ with $\xi_C = \theta_C - \hat{\theta}_C$, $\xi_{j|s} = \theta_{j|s} - \hat{p}_{j|s}$.

Derived (Conditional on Beta-Bernoulli model, observable common cause C, componentwise conditional-branch update, facilitator monotonicity)

Statement.

Under Beta-Bernoulli updating with observable common cause, componentwise updates per conditional branch, and *facilitator monotonicity* ($R_\Sigma(G_{|C}) \geq R_\Sigma(G_{|\neg C})$), the expected correction function on the joint state satisfies the sector condition globally with:

$$\alpha_{L1'} = \min\left(\frac{1}{n_C + 1},\ \min_{j \in \mathcal{J}_C} \frac{\theta_C \pi_{j|C}}{n_{j|C} + 1},\ \min_{j \in \mathcal{J}_{\neg C}} \frac{(1 - \theta_C)\pi_{j|\neg C}}{n_{j|\neg C} + 1}\right) \tag{S.39}$$

where $\mathcal{J}_s$ is the set of children tested on the $C = s$ branch and $\pi_{j|s}$ is the action-selection probability for child j on that branch.

Proof. *Edge C (condition leaf, observable every trial).* Standard Beta-Bernoulli on direct C-observation:

$$\mathbb{E}[\Delta\hat{\theta}_C] = \frac{\theta_C - \hat{\theta}_C}{n_C + 1} = -\frac{\xi_C}{n_C + 1} \tag{S.40}$$

so $F_C(\xi) = \xi_C/(n_C + 1)$.

Edge $p_{j|C}$ (conditional credence, $C = 1$ branch). Tested when $C = 1$ (probability θ_C) and the agent's $C = 1$-conditional policy selects j (probability $\pi_{j|C}$):

$$\mathbb{E}[\Delta\hat{p}_{j|C}] = \theta_C\pi_{j|C} \cdot \frac{\theta_{j|C} - \hat{p}_{j|C}}{n_{j|C} + 1} = -\frac{\theta_C\pi_{j|C}\xi_{j|C}}{n_{j|C} + 1} \tag{S.41}$$

so $F_{j|C}(\xi) = \theta_C\pi_{j|C}\xi_{j|C}/(n_{j|C} + 1)$.

Edge $p_{j|\neg C}$ (conditional credence, $C = 0$ branch). Symmetric: tested with probability $(1 - \theta_C)\pi_{j|\neg C}$:

$$F_{j|\neg C}(\xi) = \frac{(1 - \theta_C)\pi_{j|\neg C}\xi_{j|\neg C}}{n_{j|\neg C} + 1} \tag{S.42}$$

Sector product. The correction function is diagonal in ξ with positive entries:

$$\xi^T\mathbf{F}(\xi) = \frac{\xi_C^2}{n_C + 1} + \sum_{j\in\mathcal{J}_C} \frac{\theta_C\pi_{j|C}\xi_{j|C}^2}{n_{j|C} + 1} + \sum_{j\in\mathcal{J}_{\neg C}} \frac{(1 - \theta_C)\pi_{j|\neg C}\xi_{j|\neg C}^2}{n_{j|\neg C} + 1} \;\geq\; \alpha_{L1'} \cdot \|\xi\|^2 \tag{S.43}$$

with $\alpha_{L1'}$ as stated.

SA1 check. $\mathbf{F}(\mathbf{0}) = \mathbf{0}$ at truth ($\xi = 0$). ✓

Globality. The Beta-Bernoulli updates are linear in ξ on $[-1, 1]^{2K+1}$, so the diagonal Jacobian computed at truth is the exact $\mathbf{J}_\mathbf{F}$ throughout. The sector inequality holds globally with the same $\alpha_{L1'}$.

B.5b bridge to plan value. Jacobian of $\hat{P}_\Sigma^{L1'}$ with respect to the joint state:

$$\mathbf{J}_{P_\Sigma} = \begin{pmatrix} P_\Sigma(\hat{\mathbf{p}}_{|C}) - P_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \\ \hat{\theta}_C \cdot \nabla_{\hat{\mathbf{p}}_{|C}} P_\Sigma(\hat{\mathbf{p}}_{|C}) \\ (1 - \hat{\theta}_C) \cdot \nabla_{\hat{\mathbf{p}}_{|\neg C}} P_\Sigma(\hat{\mathbf{p}}_{|\neg C}) \end{pmatrix} \tag{S.44}$$

By facilitator monotonicity, the first entry is ≥ 0. The other entries are ≥ 0 by AND/OR monotonicity on each conditional sub-DAG. So $\mathbf{J}_{P_\Sigma} \geq 0$ componentwise. By the componentwise nonlinear case of B.5b, the sector condition transfers losslessly: $\alpha_s = \alpha_c = \alpha_{L1'}$. □

Five-Way Gating. This generalizes B.6's three-way gating to five gating mechanisms, one per term in the min:

1. **Condition testing** $(1/(n_C + 1))$: the common cause is the easiest component to calibrate — directly observed every trial.
2. **Evidence starvation,** $C = 1$ **branch** (θ_C factor): conditional edges on the $C = 1$ branch are gated by the prior frequency of $C = 1$.
3. **Exploration gating,** $C = 1$ **branch** ($\pi_{j|C}$ factor): OR-alternatives within the $C = 1$ branch compete for test opportunities.
4. **Evidence starvation,** $C = 0$ **branch** ($1 - \theta_C$ factor): conditional edges on the $C = 0$ branch are gated by the prior frequency of $C = 0$.
5. **Exploration gating,** $C = 0$ **branch** ($\pi_{j|\neg C}$ factor): OR-alternatives within the $C = 0$ branch.

The bottleneck is typically whichever of θ_C and $1 - \theta_C$ is smaller (the rare branch sees fewer trials), modulated by exploration on that branch.

Reduction to B.6 (Strict-Prerequisite Limit). Setting $\theta_{j|\neg C} \to 0$ collapses the $\neg C$ branch (the agent learns the $\neg C$ conditionals are zero with no remaining variance). The per-edge state for the $\neg C$ branch drops out of the sector inequality, and the formula reduces to B.6's three-way gating with $\alpha_\Sigma = \min(1/(n_C+1), \theta_C(1-\varepsilon)/(n_{A_1}+1), \theta_C\varepsilon/(n_{A_2}+1))$. ✓

Derived (cramer-rao-floor, from Fisher information of mixture model + Cramér-Rao bound)

Refuted Under Unobservable C (Mixture Identifiability Obstruction).

When C is not observable, the agent can only update the joint state via online soft EM on the marginal y_j. Computing the Fisher information of the mixture model $\mu_j = \theta_C\theta_{j|C} + (1 - \theta_C)\theta_{j|\neg C}$ at truth, the score vector for parameters $\phi = (\theta_C, p_{j|C}, p_{j|\neg C})$ admits the rank-1 factorization:

$$\mathcal{F}(\phi) = \frac{1}{\mu_j(1-\mu_j)}\, uu^T, \qquad u = (\Delta_j,\ \theta_C,\ 1 - \theta_C) \tag{S.45}$$

where $\Delta_j = p_{j|C} - p_{j|\neg C}$ is the *separability gap*. The matrix is rank 1 for any non-degenerate mixture; the two-dimensional null space corresponds to perturbations along the indeterminacy manifold $\{\hat{\phi} : \hat{\theta}_C\hat{p}_{|C} + (1-\hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$ — directions unobservable from a single binary signal.

Since the soft-EM step at truth equals (up to the $1/(n+1)$ scaling) the natural-gradient ascent on the expected log-likelihood, the Jacobian of the correction function equals $\mathcal{F}/(n+1)$, also rank 1. Therefore $\lambda_{\min}(\mathbf{J}_F) = 0$, and **no SA1-preserving update on the joint conditional vector admits a sector parameter** $\alpha > 0$ **under SUB-B.** This is a Cramér-Rao bound, not a defect of any particular update rule: any unbiased online estimator must respect the bound, which is infinite in the unidentifiable directions.

Repair routes. When C is unobservable, the agent must:

(i) **Augment C-observability** (recover B.7 above) — instrument secondary signals that identify C per trial, transforming the problem from refuted to globally derived.
(ii) **Run $K \geq 2$ children jointly under the same C-realization** — when the joint Fisher matrix reaches rank $2K + 1$, the mixture is identifiable; local sector condition holds with $\alpha \geq \sigma_{\min}(\mathcal{F}_{\text{joint}})/\max_k(n_k + 1)$. Strong structural requirement (joint observation typically unavailable in standard sequential strategy execution).
(iii) **Fall back to plan-level tracking on the marginal $\hat{\mu}_j$** — recovers B.1's $\alpha = 1/(n_\mu + 1)$ on the scalar marginal but loses the per-conditional decomposition (equivalent to L0-on-marginals).

Connection to the Identifiability-Floor Pattern. The B.7-vs-refuted-unobservable-C structure is one of two no-go theorems in AAT's strategy layer. The other is the on-policy detection no-go in `#der-causal-insufficiency-detection`, which prohibits L0-vs-L1 distinction from purely on-policy data via the causal hierarchy theorem. Both are structural impossibility results derived from external information-theoretic theorems (Bareinboim et al. 2022 Causal Hierarchy Theorem; Cramér-Rao bound on the Fisher information). Both strengthen related machinery — the on-policy no-go strengthens `#der-loop-interventional-access`; the mixture no-go strengthens the case for *observability-as-information-augmentation*. See `#disc-identifiability-floor` for the meta-pattern.

Proposition B.5: Bridge from Credence Error to Value Residuals.

The propositions above verify the sector condition for the credence-error mismatch state $\delta_c = (\hat{p}_k - \theta_k)$. This proposition extends the result to **strategy-plan-confidence error** — the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value evaluated at true edge parameters. This is a natural plan-level mismatch that is distinct from (but related to) the per-edge value-increment residuals $\delta_{\text{strategic}}$ defined in `#def-strategic-calibration`.

Relationship to $\delta_{\text{strategic}}$. `#def-strategic-calibration` defines $\delta_{\text{strategic}}$ as an L^2 aggregation of per-edge value-increment residuals — a quantity that requires credit assignment to compute. Plan-confidence error δ_s defined here is a *different* quantity: it is the error in the DAG's aggregate self-assessment, computable from status propagation without credit assignment. The two are related (both measure strategy-reality mismatch) but not identical. This proposition shows the sector condition holds for strategy-plan-confidence error; extending it to $\delta_{\text{strategic}}$ directly would require the credit-assignment machinery discussed in `#disc-credit-assignment-boundary`.

Setup. The plan value $\hat{R}_\Sigma = R_\Sigma(\mathbf{p})$ is a function of the edge credence vector $\mathbf{p} = (p_1, \ldots, p_m)$. The **independence-model reference value** is $\Phi = R_\Sigma(\theta)$ — the AND/OR propagation formula evaluated at the true edge parameters θ. Note: Φ is NOT the actual probability of plan success when edge outcomes are correlated (`#def-strategy-dag`, edge-independence caveat). It is the best the independence model can do with perfect edge knowledge. Under correlated failure, the actual plan success probability may be lower than Φ; the gap is a model-class limitation of the AND/OR DAG representation, not an estimation error. What δ_s tracks is calibration *within the independence model*, not calibration to reality. Define:

- **Credence-error mismatch:** $\delta_c = \mathbf{p} - \theta \in \mathbb{R}^m$
- **Plan-confidence error:** $\delta_s = \hat{R}_\Sigma - \Phi = R_\Sigma(\mathbf{p}) - R_\Sigma(\theta)$ (scalar)

The gradient of plan value with respect to credences is $\mathbf{J} = \nabla_{\mathbf{p}} R_\Sigma \in \mathbb{R}^m$ (a vector, since R_Σ is scalar). By first-order Taylor expansion:

$$\delta_s \approx \mathbf{J}^T \delta_c \tag{S.46}$$

B.5a: Linear Correction (Beta-Bernoulli).

Derived (value-residual sector condition, linear case)

When the correction in credence space is linear — $\mathbf{F}_c(\delta_c) = \eta \odot \delta_c$ where $\eta_k = 1/(n_k + 1)$ and $\odot$ is elementwise product — consider the sector ratio directly. Since $\delta_s = \mathbf{J}^T \delta_c$ and $F_s = \mathbf{J}^T \mathbf{F}_c$:

$$\delta_s \cdot F_s = (\mathbf{J}^T \delta_c)^T (\mathbf{J}^T \mathbf{F}_c) = \delta_c^T \mathbf{J}\mathbf{J}^T \mathbf{F}_c \tag{S.47}$$

For the linear case $\mathbf{F}_c = \eta_{\min} \delta_c$ (using the worst-case uniform gain for the bound):

$$\delta_s \cdot F_s = \eta_{\min} \cdot \delta_c^T \mathbf{J}\mathbf{J}^T \delta_c = \eta_{\min} \cdot \|\mathbf{J}^T \delta_c\|^2 = \eta_{\min} \cdot \delta_s^2 \tag{S.48}$$

Therefore:

$$\frac{\delta_s \cdot F_s}{\delta_s^2} = \eta_{\min} \tag{S.49}$$

The Jacobian cancels. The sector parameter in value-residual space equals the sector parameter in credence space, regardless of DAG structure:

$$\alpha_s = \alpha_c = \eta_{\min} = \min_k \frac{1}{n_k + 1} \tag{S.50}$$

This holds because $\mathbf{J}\mathbf{J}^T$ appears in both numerator and denominator of the sector ratio, and the linear correction commutes with the Jacobian mapping.

Interpretation. The Jacobian $\mathbf{J}$ determines *which* credence errors matter most for plan value (high-sensitivity edges contribute more to value residuals). But it does not change the *rate* at which those errors are corrected, because the correction is proportional to the error regardless of the error's plan-value impact.

Derived (from per-component sector condition + Jacobian non-negativity)

B.5b: Nonlinear Componentwise Correction.

When the correction is nonlinear but **componentwise** — each edge corrects independently, with

$$(\mathbf{F}_c)_k \text{ depending only on } (\delta_c)_k \tag{S.51}$$

— and the plan-value Jacobian is **non-negative** ($J_k \geq 0$ for all k), the transfer is lossless regardless of nonlinearity.

The per-component sector condition (from Props B.1-B.4): for each edge k,

$$(\delta_c)_k \cdot (\mathbf{F}_c)_k \geq \alpha_c \cdot (\delta_c)_k^2 \tag{S.52}$$

which gives

$$(\mathbf{F}_c)_k / (\delta_c)_k \geq \alpha_c \tag{S.53}$$

(the correction-to-mismatch ratio exceeds α_c for each edge independently).

Jacobian non-negativity holds for all well-formed AND/OR DAGs: increasing any edge credence p_k never decreases plan value R_Σ (monotonicity of AND/OR propagation). Therefore $J_k = \partial R_\Sigma / \partial p_k \geq 0$ for all k.

The transfer. Since $J_k \geq 0$ preserves the inequality direction:

$$J_k \cdot (\mathbf{F}_c)_k \geq J_k \cdot \alpha_c \cdot (\delta_c)_k \quad \text{for each } k \tag{S.54}$$

Summing:

$$\mathbf{J}^T \mathbf{F}_c = \sum_k J_k (\mathbf{F}_c)_k \geq \alpha_c \sum_k J_k (\delta_c)_k = \alpha_c \cdot \mathbf{J}^T \delta_c \tag{S.55}$$

Therefore $F_s \geq \alpha_c \cdot \delta_s$, giving:

$$\alpha_s = \alpha_c \tag{S.56}$$

No condition-number penalty. The Jacobian's anisotropy is irrelevant when the correction is componentwise and the Jacobian is non-negative. The non-negativity ensures that every edge's correction contributes in the right direction to the value correction, regardless of how unevenly the Jacobian weights different edges.

When does componentwise correction hold? For all cases in Props B.1, B.2, and B.4 — the standard Beta-Bernoulli update on each edge, including the stochastic (nonlinear) single-step realization. The correction IS nonlinear (the actual per-step update $(y - \hat{p})/(n+1)$ is random, not proportional to δ_k), but it is componentwise (each edge updates from its own observation independently).

When does it fail? When edge corrections are coupled — correcting one edge changes another's correction direction. This occurs for: - **Unobservable intermediates (B.3) under marginal Bayesian updates:** The marginal Bayesian update couples edge posteriors. The correction for edge 2 depends on the joint posterior over both edges, not just on $(\delta_c)_2$ alone. *And — critically — Prop B.3(a) showed that this scheme violates SA1 with $O(1/n)$ bias: there is no valid α_c for the marginal Bayesian update on coupled edges in the first place.* The per-edge bound must therefore be re-established before the Jacobian transfer can be applied. - **Shared observations:** If testing one arm gives partial information about another (correlated arms), corrections couple across edges — with an SA1 status that depends on the specific attribution scheme used to project the shared observation onto per-edge updates.

Scope (coupled-case-applicability)

The coupled-case Cauchy-Schwarz bound presupposes a valid α_c. For the bound $\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$ to apply meaningfully, the per-edge sector condition must hold — i.e., the attribution scheme must satisfy SA1. Marginal Bayesian updates on coupled edges do *not* satisfy this precondition (Prop B.3(a)). The resolution is to replace marginal Bayesian with **gradient-based attribution** (Prop B.5d below), which recovers SA1 and yields a valid per-edge sector parameter that admits the Jacobian transfer with the κ^2 penalty.

Per-edge vs. plan-level routes through coupling. Prop B.3(b) already established an alternative route for the two-edge unobservable case: tracking at the plan level ($\hat{\Phi} = \hat{p}_1\hat{p}_2$) with a single Beta posterior recovers SA1 directly, yielding $\alpha_\Sigma = 1/(n_\Phi + 1)$. The plan-level route works *without* needing the Jacobian bridge because it sidesteps per-edge attribution entirely. The gradient-based per-edge route works *through* the Jacobian bridge, preserving per-edge credences at the cost of the κ^2 penalty. Both routes give valid sector parameters; the choice depends on whether per-edge diagnostics or aggregate efficiency matters more.

The refined picture: nonlinearity is not the problem; inter-edge coupling combined with the attribution scheme is. A nonlinear but componentwise correction transfers losslessly through any non-negative Jacobian. A coupled correction under a scheme that preserves SA1 (gradient-based) incurs the condition-number penalty because the coupling can redirect correction away from the value-relevant direction. A coupled correction under a scheme that violates SA1 (marginal Bayesian) has no valid α_c to transfer at all and must be replaced before B.5 applies.

Proposition B.5d: Gradient-Based Attribution Restores SA1 on Coupled Edges. **Setup.** Same two-edge AND topology as B.3: edges with true probabilities θ_1, θ_2, credences $\hat{p}_1, \hat{p}_2$, plan probability $\hat{P}_\Sigma = \hat{p}_1\hat{p}_2$, observable plan outcome $y_G \in \{0, 1\}$ with $\mathbb{E}[y_G] = \Phi = \theta_1\theta_2$. Intermediate is unobservable.

Gradient-based per-edge update. Distribute plan-level surprise proportionally to the Jacobian component:

Formulation (gradient-attribution)

$$\Delta\hat{p}_k = \frac{1}{n_k + 1} \cdot J_k \cdot (y_G - \hat{P}_\Sigma), \qquad J_k = \frac{\partial \hat{P}_\Sigma}{\partial \hat{p}_k} \tag{S.57}$$

For the two-edge AND case, $J_1 = \hat{p}_2, J_2 = \hat{p}_1$.

Verification of SA1. At truth ($\hat{p}_k = \theta_k$):

$$\mathbb{E}[\Delta\hat{p}_k]\big|_{\hat{\mathbf{p}}=\theta} \;=\; \frac{J_k}{n_k+1}\cdot\mathbb{E}[y_G-\hat{P}_\Sigma]\big|_{\hat{\mathbf{p}}=\theta} \tag{S.58}$$

The bracket evaluates as: $\mathbb{E}[y_G]=\Phi$ by definition of y_G as a Bernoulli outcome with mean Φ; and $\hat{P}_\Sigma\big|_{\hat{\mathbf{p}}=\theta}=\theta_1\theta_2=\Phi$. Therefore $\mathbb{E}[y_G-\hat{P}_\Sigma]\big|_{\hat{\mathbf{p}}=\theta}=0$, and:

$$\mathbb{E}[\Delta\hat{p}_k]\big|_{\hat{\mathbf{p}}=\theta} \;=\; 0 \quad \text{for all } k. \qquad \text{(SA1 satisfied)} \tag{S.59}$$

The contrast with marginal Bayesian is precise: the marginal scheme suffers from a Φ-vs-θ_k mismatch (the update treats each edge's marginal likelihood rather than the joint), producing the $O(1/n)$ bias derived in Prop B.3(a). The gradient scheme distributes surprise via the Jacobian, which aligns the per-edge update direction with the plan-level gradient — and this gradient is zero at truth by construction.

Sector parameter (SA2'). With the gradient-based scheme, the per-edge correction is:

$$F_k(\delta) \;=\; \frac{J_k}{n_k+1}\cdot(\Phi-\hat{P}_\Sigma) \;\approx\; \frac{J_k}{n_k+1}\cdot\mathbf{J}^T\delta \tag{S.60}$$

where the approximation uses the first-order expansion $\hat{P}_\Sigma-\Phi\approx\mathbf{J}^T\delta$ valid near truth. The per-edge sector bound is:

$$\delta^T\mathbf{F}(\delta) \;\approx\; \sum_k \frac{J_k^2}{n_k+1}\cdot(\mathbf{J}^T\delta)^2/\|\delta\| \;\geq\; \frac{\sigma_{\min}(\mathbf{J})^2}{\max_k(n_k+1)}\|\delta\|^2 \tag{S.61}$$

giving:

$$\alpha_c^{\text{grad}} \;\geq\; \frac{\sigma_{\min}(\mathbf{J})^2}{\max_k(n_k+1)} \tag{S.62}$$

Transfer through B.5c. Applying the coupled-case Cauchy-Schwarz bound:

$$\alpha_s \;\geq\; \frac{\alpha_c^{\text{grad}}}{\kappa(\mathbf{J})^2} \;=\; \frac{\sigma_{\min}(\mathbf{J})^2/\max_k(n_k+1)}{\sigma_{\max}(\mathbf{J})^2/\sigma_{\min}(\mathbf{J})^2} \;=\; \frac{\sigma_{\min}(\mathbf{J})^4}{\sigma_{\max}(\mathbf{J})^2\cdot\max_k(n_k+1)} \tag{S.63}$$

The $\sigma_{\min}(\mathbf{J})^4/\sigma_{\max}(\mathbf{J})^2$ factor is the full cost of coupled per-edge attribution — a steep penalty compared to the componentwise case. The plan-level route (B.3(b)) avoids this penalty entirely by tracking Φ as a single quantity with $\alpha_\Sigma=1/(n_\Phi+1)$, at the cost of losing per-edge diagnostics.

Why not proportional-blame or other schemes? Proportional-blame distributes surprise proportionally to current credences: $\Delta\hat{p}_k\propto\hat{p}_k\cdot(y_G-\hat{P}_\Sigma)/\sum_j\hat{p}_j$. This violates SA1 because the proportionality factor $\hat{p}_k/\sum_j\hat{p}_j$ is not the gradient direction, so the distribution over edges does not align with the plan-level surprise's actual attribution. Only schemes whose per-edge update aligns with the Jacobian direction preserve SA1 for coupled edges; gradient-based is the canonical such scheme.

Claim. For coupled edges with unobservable intermediates, gradient-based attribution is the minimal scheme that (a) satisfies SA1, (b) yields a well-defined per-edge sector parameter, (c) admits the Cauchy-Schwarz Jacobian transfer, and (d) permits per-edge diagnostics. Its $\sigma_{\min}^4/\sigma_{\max}^2$ scaling is inherent to the coupled case, not an artifact of the scheme. □

B.5c: Implications. Four regimes for the credence-to-value bridge, indexed by correction type and attribution scheme:

Table S.2

Correction type	*Edge coupling*	*Attribution*	*SA1*	*Transfer*	α_s
Linear	Any	—	Yes	Exact (Jacobian cancels)	α_c
Nonlinear, componentwise	Independent edges	Per-edge Bayesian	Yes	Exact ($J_k \geq 0$ preserves bound)	α_c
Nonlinear, coupled	Coupled edges	Gradient-based (Jacobian distribution)	Yes	$\kappa(\mathbf{J})^2$ penalty	$\alpha_c^{\text{grad}}/\kappa^2$
Nonlinear, coupled	Coupled edges	Marginal Bayesian	**No** (O(1/n) bias)	— (Jacobian bridge does not apply)	plan-level alternative (`#def-strategy-dag`)

Componentwise correction closes the gap for all verified cases. Props B.1, B.2, and B.4 all use componentwise edge updates with non-negative Jacobian. The operational diagnostic $\delta_{\text{strategic}}$ inherits the sector condition from credence error with no penalty — even for the nonlinear (stochastic) single-step realization. The derivation validates the operational mismatch, not merely a surrogate.

Inter-edge coupling — not nonlinearity — is the fragility. The κ^2 penalty arises only when correcting one edge changes another's correction direction (B.3's unobservable case, correlated observations). Deep or unbalanced DAGs have high $\kappa(\mathbf{J})$ when corrections couple, but the coupling, not the depth or imbalance per se, is what degrades the persistence guarantee.

Connection to credit assignment. The Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$ is computable from the DAG's status propagation formulas — it does not require solving the credit-assignment problem. The credit-assignment problem is about *decomposing observed value changes into per-edge contributions* (needed for the update rule). The Jacobian bridge is about *transferring a per-edge sector condition to the value-residual space* (needed for the persistence guarantee). These are different problems: the bridge works even when credit assignment is unsolved, because it only requires the sensitivity structure, not the causal attribution.

What Is Derived vs. What Is Chosen.

The segment carries derivations at three distinct strengths: (i) exact sector-parameter algebra for concrete Beta-Bernoulli topologies (B.1, B.2, B.4, B.6, B.7 under observable C + facilitator monotonicity), (ii) two structural refutations that are *themselves* derivations of impossibility (B.3(a) SA1 bias; B.7 Cramér-Rao floor under unobservable C), and (iii) formulation choices (Beta-Bernoulli dynamics, componentwise conditional-branch updates, the particular ε-greedy policy family). The dividing line runs between the Beta-Bernoulli *model* (chosen) and the sector-parameter *algebra conditional on that model* (proved). B.7's positive transfer requires two load-bearing conditions — observability of C *and* facilitator monotonicity $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$ — either of which failing disqualifies the transfer.

Table S.3

Property	*Source*	*Strength*
Beta-Bernoulli edge dynamics ($\Delta\hat{p}_k = (y_k - \hat{p}_k)/(n_k + 1)$)	Conjugate-prior model; matches `#hyp-edge-update-via-gain`	Formulation choice

Property	*Source*	*Strength*
Regime-A adjustment: $\alpha_\Sigma^{\text{eff}} = \iota_{ij} \cdot \alpha_\Sigma^{\text{stated}}$	Diagonal commutation across Props B.1–B.4, B.6; `#scope-edge-update-causal-validity`	Derived
B.1 Single-edge sector parameter $\alpha_\Sigma = 1/(n+1)$	Expected Beta-Bernoulli update + sector product algebra	Proved (tight, conditional on Beta-Bernoulli)
B.1F Single edge under exponential forgetting: gain $1/n_t$ (post-update count), stationary $\alpha_\Sigma^{\text{ss}} = 1-\lambda$, all-time mean-mismatch bound for $n_0 \le n_{\text{eff}}$	Discounted-update algebra + Model-D induction	Proved (tight, conditional on Beta-Bernoulli + forgetting)
B.1 Stochastic ultimate bound $\lvert\delta\rvert > \sqrt{\theta(1-\theta)/(2n+1)}$	Second-moment recursion + expected-Lyapunov decrement	Derived
B.2 Observable AND-chain sector parameter $\min(1/(n_1+1),\ \theta_1/(n_2+1))$	Diagonal correction function; sector product algebra	Proved (conditional on Beta-Bernoulli + observable intermediate)
B.2 Depth-d generalization $\alpha_\Sigma = \min_k \prod_{j<k} \theta_j/(n_k+1)$	Iteration of B.2 pattern under observable intermediates	Derived
B.3(a) Per-edge SA1 violation under marginal Bayesian attribution ($O(1/n)$ bias)	Exact marginal update + evaluation at truth	Derived as refutation (no valid α_c exists)
B.3(b) Plan-level recovery $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi+1)$	Reduction of aggregate $\hat{\Phi}$ tracking to B.1	Derived (reduction)
B.3 Non-identifiability of (θ_1,θ_2) from y_G alone	Product $\Phi = \theta_1\theta_2$ symmetry in observation channel	Proved
B.4(a) Pure-greedy OR ($\varepsilon = 0$) violates sector condition	Counterexample at $\delta_1 = 0,\ \delta_2 \neq 0$	Derived as refutation
B.4(b) ε-greedy sector parameter $\min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$	Policy-weighted correction; sector product algebra	Proved (conditional on Beta-Bernoulli + ε-greedy)
B.4 Optimal exploration $\varepsilon^* = (n_2+1)/(n_1+n_2+2)$	Term-equalization in min	Derived
B.4 Admissible exploration interval (SA3 requirement) $\rho_\Sigma(n_2+1)/R_\Sigma < \varepsilon < 1-\rho_\Sigma(n_1+1)/R_\Sigma$	Substitution into persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$	Derived
B.5a Linear credence-to-value transfer $\alpha_s = \alpha_c$ (Jacobian cancels)	Algebraic cancellation of $\mathbf{J}\mathbf{J}^T$	Proved (exact, DAG-structure independent)
B.5b Componentwise nonlinear transfer $\alpha_s = \alpha_c$	Per-component sector bound + monotone AND/OR ($\mathbf{J} \ge 0$)	Proved (conditional on componentwise update + non-negative Jacobian)
B.5b Coupled-edge Cauchy-Schwarz bound $\alpha_s \ge \alpha_c/\kappa(\mathbf{J})^2$	Cauchy-Schwarz on $\mathbf{J}^T\mathbf{F}_c$ and $\mathbf{J}^T\delta_c$	Derived (conditional on Jacobian regularity *and* an SA1-preserving attribution scheme)
B.5c Four-regime classification of credence-to-value transfer	Combination of linear/nonlinear × componentwise/coupled × attribution type	Derived
B.5d Gradient-based attribution restores SA1 on coupled edges	Jacobian-weighted surprise distribution; $\mathbb{E}[y_G - \hat{P}_\Sigma]\vert_\theta = 0$	Derived
B.5d Sector parameter under gradient attribution $\alpha_c^{\text{grad}} \ge \sigma_{\min}(\mathbf{J})^2/\max_k(n_k+1)$	First-order expansion near truth + singular-value analysis	Derived
B.5d Gradient-based as *minimal* SA1-preserving scheme for coupled edges	Argument excluding proportional-blame and other schemes	Derived qualitatively (not a uniqueness theorem)
B.6 L1 augmented-DAG sector parameter (three-way gating: condition / starvation / exploration)	Diagonal correction function over $(\delta_C, \delta_{A_1}, \delta_{A_2})$; sector product	Proved (conditional on Beta-Bernoulli + observable C + ε-greedy)
B.6 L0-vs-L1 comparison: L1 is lower α_Σ but honestly calibrated	Direct Φ^{L0} vs Φ^{L1} computation at truth	Derived
B.6 L1-construction principle (common cause factored *above* OR-structure)	Requirement for conditional independence of OR-siblings given C	Derived

Property	*Source*	*Strength*
B.7 L1' mixture-form sector parameter (five-way gating)	Diagonal correction function over $(\xi_C, \{\xi_{j\|C}\}, \{\xi_{j\|\neg C}\})$; sector product	Derived (conditional on Beta-Bernoulli + observable C + componentwise conditional-branch update + *facilitator monotonicity* $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$)
B.7 Componentwise update per conditional branch	Matches Beta-Bernoulli on the branch actually executed each trial	Formulation choice
B.7 Facilitator-monotonicity condition $P_\Sigma(G \mid C) \geq P_\Sigma(G \mid \neg C)$	Required so the first Jacobian entry of $\hat{P}_\Sigma^{L1'}$ is non-negative, enabling the lossless B.5b transfer	Load-bearing scope condition
B.7 Reduction to B.6 under strict-prerequisite limit ($\theta_{j\|\neg C} \to 0$)	Collapse of $\neg C$-branch terms in the sector inequality	Verification (algebraic consistency)
B.7 Refutation under unobservable C (single-channel Fisher rank 1)	Cramér-Rao bound applied to rank-1 Fisher $\mathcal{F}(\phi) = uu^T/[\mu_j(1-\mu_j)]$	Proved by external theorem (Cramér-Rao bound) — no unbiased online estimator admits $\alpha > 0$
B.7 Repair route (i): augment C-observability	Recovers the positive B.7 derivation	Formulation / scope choice (maps to `#der-loop-interventional-access`)
B.7 Repair route (ii): $K \geq 2$ joint children under same C-realization	Fisher matrix rank reaches $2K+1$; local sector condition	Derived (sketch; applicability narrow)
B.7 Repair route (iii): plan-level fallback on marginal $\hat{\mu}_j$	Reduction to B.1 on scalar marginal	Derived (at cost of per-conditional decomposition)
Persistence thresholds (substitution of each α_Σ into $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$)	Proposition A.1 of A	Derived (each)
Gain-collapse threshold $n^* \propto R_\Sigma/\rho_\Sigma$ common across cases	Direct inversion of persistence condition	Derived
Expected-value (rather than almost-sure) sector analysis	Pedagogical simplification; a.s. convergence via standard Bayesian consistency	Formulation choice (scope)
Constant-α assumption with experience-discounting (λ) stabilization	Instantaneous persistence check + forgetting-factor analysis	Derived (instantaneous only)

Epistemic Status.

Conditional on the Beta-Bernoulli model. Propositions B.1, B.1F, B.2, B.4, and B.6 are *derived*: the sector-condition verification is exact algebra under the stated generative model. The persistence conditions follow by direct application of Proposition A.1 (A), which is independently established. Proposition B.3(a) (the SA1 violation) is also derived; B.3(b) reduces to B.1 by construction. Proposition B.5a (the credence-to-value bridge for linear correction) is *exact* — the Jacobian cancellation is algebraic, independent of DAG structure. Proposition B.5b (the componentwise nonlinear transfer) is *exact under non-negative Jacobian* — algebraic, no condition-number penalty. The coupled-edge case of B.5b is *conditional on (i) Jacobian regularity* — the condition-number bound requires $\sigma_{\min}(\mathbf{J}) > 0$, which holds for non-degenerate DAGs — *and (ii) an SA1-preserving attribution scheme* (gradient-based, Prop B.5d); marginal Bayesian attribution on coupled edges does not admit the Cauchy-Schwarz bound because SA1 fails upstream. Proposition B.5d (gradient-based attribution) is *derived* — SA1 verification is algebraic and the sector-parameter bound follows from standard singular-value analysis of the Jacobian.

All results use the *expected-value* sector condition. A full stochastic treatment (Foster-Lyapunov or supermartingale convergence) would give probability bounds rather than expected-value bounds. The expected-value analysis gives the correct asymptotic behavior (posterior concentration at $O(1/\sqrt{n})$) but does not prove almost-sure convergence. For

the stationary Beta-Bernoulli case, almost-sure convergence is guaranteed by the standard Bayesian consistency theorem independently of this derivation.

Without forgetting, α_Σ is time-varying: since n_k increases with each observation, α_Σ decreases over time, while the sector-condition framework assumes constant α. Props B.1–B.4 and B.6–B.7 therefore give an *instantaneous* persistence check at the current experience level. Prop B.1F supplies the trajectory version for the single edge: under exponential forgetting the gain is deterministic and equals $1-\lambda$ exactly at the stationary count, giving the persistence requirement $1-\lambda > \rho_\Sigma/R_\Sigma$ with an all-time guarantee when the count is initialized at or below n_{eff} (*exact* under the Beta-Bernoulli model). For intermittently observed edges (B.2, B.4, B.6, B.7) the corresponding statement depends on whether the discount is applied per observation of the edge or per time step; the per-observation case is exact (at the stationary count, $\pi(1-\lambda)$ per step for an edge observed with probability π), and the cap on what any convention can reach is T.

What remains open:

1. *~~General DAG topology.~~* **Partially resolved.** Proposition B.6 verifies the first mixed AND/OR case (L1 augmented DAG with common-cause node). The three-way gating structure (condition testing × evidence starvation × exploration gating) appears to be the general pattern. Remaining: arbitrary mixed AND/OR DAGs with multiple common causes and deeper nesting.
2. *Continuous outcomes.* The Beta-Bernoulli model gives conjugate, closed-form updates. Non-conjugate cases (continuous signals, partial observability) require approximate inference, and the sector condition must be verified for the approximation.
3. *~~Modified sector condition for biased correction.~~* **Resolved via Prop B.5d (gradient-based attribution).** The $O(1/n)$ bias in the unobservable case (B.3a) under marginal Bayesian updates is eliminated by switching to gradient-based attribution, which satisfies SA1 exactly at truth. Per-edge results are recovered under the gradient scheme at the $\sigma^4_{\min}/\sigma^2_{\max}$ scaling. Alternative route (plan-level tracking, B.3(b)) remains available without the condition-number penalty.
4. *~~Correlated edges (L1/L2 scope).~~* **Resolved.** Proposition B.6 verifies the sector condition for L1 (strict-prerequisite augmented DAG); Proposition B.7 verifies it for L1' (soft-facilitator mixture form, observable common cause) with five-way gating. The unobservable-C single-channel case is *refuted* by the Cramér-Rao floor (mixture identifiability obstruction; see B.7 §"Refuted Under Unobservable C") — not merely "open" but structurally impossible without additional information augmentation. Practical paths under unobservable C: augment C-observability, run multi-child joint observations (sketch only, narrow applicability), or fall back to plan-level tracking on the marginal. Strategies requiring full L2 conditioning over k unobservable common causes still cost $O(2^k)$.
5. *Adaptive exploration.* Proposition B.4 uses fixed ε. Adaptive strategies (UCB, Thompson sampling) allocate exploration based on current uncertainty and should yield tighter sector bounds.

Discussion.

The sector parameter is the edge update gain. Across all four cases, α_Σ is a function of $\eta_k = 1/(n_k+1)$ — the same quantity that governs epistemic persistence (`#def-adaptive-tempo`, `#emp-update-gain`). Strategic persistence and epistemic persistence are governed by the same mathematical machinery, not merely structurally parallel. This validates the schema proposed in `#schema-strategy-persistence` at the level of concrete dynamics.

AND vs. OR: structural vs. behavioral fragility. The evidence-starvation effect (B.2) is structural — it depends on the DAG topology and upstream reliability, not on the agent's policy. The exploration-gating effect (B.4) is behavioral — it depends on the action-selection policy. AND-node persistence is improved by increasing upstream reliability (making early steps more likely to succeed). OR-node persistence is improved by increasing exploration (testing alternatives more frequently). The distinction maps to the familiar depth-vs-breadth tradeoff in planning.

Gain collapse as the dominant failure mode. In every drifting-environment case, the persistence threshold takes the form $n < c/\rho_\Sigma$ for some constant c. Accumulated experience suppresses the update gain, eventually making the agent unable to track environmental change. This gives the gain-collapse phenomenon (`#emp-update-gain`) a precise quantitative threshold for each topology.

The credence-to-value bridge decouples persistence from credit assignment. Proposition B.5 shows that strategic persistence (in value-residual space) can be established from per-edge sector conditions (in credence space) without solving the credit-assignment problem. The bridge requires only the value-sensitivity Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} R_\Sigma$, which is computable from the DAG's status propagation formulas in $O(|V| + |E|)$ time. Credit assignment — attributing observed value changes to specific edges — is needed for the *update rule* (computing the signal function in `#hyp-edge-update-via-gain`), not for the *persistence guarantee*. This means the persistence analysis can proceed even while the update rule remains incompletely specified.

Inter-edge coupling is the true fragility, not nonlinearity or DAG depth. Proposition B.5b shows that nonlinear componentwise corrections transfer losslessly through any non-negative Jacobian — the $\kappa(\mathbf{J})^2$ penalty arises only when corrections are *coupled* across edges. This refines the earlier qualitative argument from `#der-chain-confidence-decay`: the structural pressure toward shallow, balanced strategies is not about depth per se but about the coupling that deep or complex structures tend to introduce. A deep AND-chain with fully observable intermediates (B.2) has componentwise corrections and transfers losslessly despite high depth. The same chain with unobservable intermediates (B.3) has coupled corrections and incurs the penalty. The distinction is observability-mediated coupling, not topological complexity.

Appendix T

Derivation: Evidence-Allocation Ceiling on Strategic Persistence

When strategy edges that must all stay calibrated share one observation stream, each edge's credence lags a drifting truth by at least the drift accumulated since that edge was last observed, on average ρ_Σ/π_i for an edge observed with probability π_i per step. No update rule that averages observed outcomes, and no forgetting rate, does better than trusting the most recent observation, which attains the bound. A K-alternative OR node can therefore keep every alternative calibrated only if $\rho_\Sigma < R_\Sigma/K$, and a two-arm node only if $\rho_\Sigma < R_\Sigma/2$. The cap comes from how evidence is allocated, so the update mechanism cannot remove it.

Formal Expression.

Setup.

Formulation (allocation setting)

- **Edges and drift.** Edges $i = 1, \dots, K$ have true success probabilities $\theta_{i,t}$ that drift under Model D, $|\theta_{i,t+1} - \theta_{i,t}| \le \rho_\Sigma$ per time step, whether or not the edge is observed that step.
- **Fixed allocation.** At step s edge i is observed with probability π_i, independently of all outcomes and of the past. An observation reports $y_{i,s} \sim \text{Bernoulli}(\theta_{i,s})$ and is incorporated into $\hat{p}_{i,s+1}$. For one observation per step routed to one alternative of an OR node, $\sum_i \pi_i \le 1$. This is the allocation of S Prop B.4 ($\pi = (1-\varepsilon, \varepsilon)$); for a downstream AND-chain edge with observable intermediates (Prop B.2) the edge is observed when every upstream link fires, so $\pi_k = \prod_{j<k} \theta_j$.
- **Estimator class $\mathcal{C}$.** $\hat{p}_{i,t} = w_0\, \hat{p}_{i,0} + \sum_{s \in O_{i,t}} w_s\, y_{i,s}$, with $w \ge 0$, $\sum w = 1$, and the weights determined by the observation history $O_{i,t}$ (the steps before t at which edge i was observed), not by the outcomes. $\mathcal{C}$ contains Beta-Bernoulli updating with or without exponential forgetting, with the discount applied per observation or per time step (the counts depend only on $O_{i,t}$), every innovation-form update $\hat{p} + g(y - \hat{p})$ with $g \in [0, 1]$ set by the observation history, and the last-observation rule $\hat{p}_{i,t} = y_{i,\max O_{i,t}}$.
- **Mismatch and target.** $m_{i,t} = \mathbb{E}[\hat{p}_{i,t}] - \theta_{i,t}$, the mean-dynamics reading of Props B.1–B.4. Persistence of the node requires every edge's mean mismatch to stay within the reserve R_Σ: (SA3) of `#schema-strategy-persistence`, which asks that every OR alternative be kept calibrated.
- **Age.** $A_{i,t} = t - \max O_{i,t}$ if edge i has been observed before t, and $A_{i,t} = t$ otherwise.

Derived (Conditional on fixed allocation + estimator class $\mathcal{C}$ + Model D)

Proposition EA.1 (Allocation lag).

Start calibrated, $\hat{p}_{i,0} = \theta_{i,0}$. Under constant-sign drift $\theta_{i,s+1} - \theta_{i,s} = \rho_\Sigma$, every estimator in $\mathcal{C}$ satisfies

$$m_{i,t} \;\le\; -\rho_\Sigma\, \mathbb{E}[A_{i,t}], \qquad \mathbb{E}[A_{i,t}] = \frac{1-(1-\pi_i)^t}{\pi_i} \;\uparrow\; \frac{1}{\pi_i}. \tag{T.1}$$

The last-observation rule attains it: under constant drift its mean mismatch is exactly $-\rho_\Sigma\, \mathbb{E}[A_{i,t}]$, and under any drift with $|\theta_{i,s+1} - \theta_{i,s}| \le \rho_\Sigma$ it satisfies $|m_{i,t}| \le \rho_\Sigma/\pi_i$ for all t.

Proof. The allocation is independent of the outcomes, so conditional on $O_{i,t}$ each observed $y_{i,s}$ still has mean $\theta_{i,s}$, and since the weights are fixed by $O_{i,t}$,

$$\mathbb{E}[\hat{p}_{i,t} \mid O_{i,t}] - \theta_{i,t} = -\Big(w_0\,(\theta_{i,t} - \theta_{i,0}) + \sum_{s \in O_{i,t}} w_s\,(\theta_{i,t} - \theta_{i,s})\Big). \tag{T.2}$$

Under constant drift $\theta_{i,t} - \theta_{i,s} = \rho_\Sigma(t-s)$. Every observed s satisfies $t - s \ge A_{i,t}$, and the prior term has $t - 0 \ge A_{i,t}$. The weights sum to one, so the bracket is at least $\rho_\Sigma A_{i,t}$. Taking expectations over $O_{i,t}$ gives the bound. For $k \le t$, $A_{i,t} \ge k$ iff edge i was unobserved at steps $t-1, \dots, t-k+1$, which has probability $(1-\pi_i)^{k-1}$; summing $\mathbb{E}[A_{i,t}] = \sum_{k=1}^{t}(1-\pi_i)^{k-1}$ gives the closed form. For the last-observation rule the bracket is exactly $\theta_{i,t} - \theta_{i,\max O}$, which is $\rho_\Sigma A_{i,t}$ under constant drift and at most $\rho_\Sigma A_{i,t}$ in magnitude under any bounded drift. □

Derived (Conditional on Proposition EA.1)

Corollary EA.2 (Reachable region).

For a single edge, the Model-D ultimate mean mismatch of the best estimator in $\mathcal{C}$ is ρ_Σ/π_i, so edge i can persist iff $\rho_\Sigma < \pi_i R_\Sigma$. For a node whose edges must all persist, the reachable region is $\rho_\Sigma < R_\Sigma \min_i \pi_i$. When one observation per step is shared among K alternatives, $\min_i \pi_i \le 1/K$, so

$$\sup_{\text{allocations, } e} \frac{\rho_\Sigma}{R_\Sigma} \;=\; \frac{1}{K}, \tag{T.3}$$

attained by uniform allocation with the last-observation rule.

Instances.

- **Single edge** ($K = 1, \pi = 1$): $\rho_\Sigma < R_\Sigma$, the floor of S Prop B.1F, reached by exponential forgetting as $\lambda \to 0$.
- **Two-arm ε-greedy OR** (Prop B.4): $\rho_\Sigma < R_\Sigma \min(\varepsilon, 1-\varepsilon) \le R_\Sigma/2$, with equality only at $\varepsilon = 1/2$. With forgetting at the stationary per-observation gain $1-\lambda$, the admissible exploration interval is $\rho_\Sigma/((1-\lambda)R_\Sigma) < \varepsilon < 1 - \rho_\Sigma/((1-\lambda)R_\Sigma)$, non-empty iff $\rho_\Sigma < (1-\lambda)R_\Sigma/2$.
- **K-alternative OR:** $\rho_\Sigma < R_\Sigma/K$, matching the $1/k$ dilution of B.4's k-arm sector parameter.
- **AND chain with observable intermediates** (Prop B.2, upstream θ_j held fixed as there): the downstream edge k persists only if $\rho_\Sigma < R_\Sigma \prod_{j<k} \theta_j$. The gates of Props B.6 and B.7 multiply into π the same way.

Discount convention.

Formulation (where the discount is applied)

For an intermittently observed edge, exponential forgetting can discount the counts **per observation of that edge** or **per time step**. Both conventions lie in $\mathcal{C}$, so Corollary EA.2's cap holds under both and is approached under both as $\lambda \to 0^+$, where the most recent observation's weight tends to 1. They differ below the cap. Per-observation discounting initialized at the stationary count gives each observation gain exactly $1-\lambda$ (Prop B.1F), so the expected per-step correction rate is exactly $\pi_i(1-\lambda)$. Per-time-step discounting lets the count decay between observations, so each observation lands with a larger gain. Its rate is at most π_i by Proposition EA.1, but no closed form is derived here. At $\lambda = 0$ exactly, per-time-step discounting leaves an edge with $\pi_i < 1$ with a zero count between observations, so that convention needs $\lambda > 0$. The threshold λ at which a given intermittent edge persists depends on the convention. The exploration interval above, and Prop B.4's (SA3), use per-observation discounting.

Epistemic Status.

Exact under the named conditions: (i) fixed allocation, independent of outcomes; (ii) estimator class $\mathcal{C}$, convex combinations of observed outcomes with outcome-independent weights; (iii) Model D drift per time step on every edge, observed or not; (iv) the persistence target that every alternative stay calibrated, (SA3); (v) the mean-mismatch reading that Props B.1–B.4 use. Proposition EA.1's inequality is pathwise in the observation history, and the corollary is its expectation. The supremum $1/K$ is attained, so the cap is sharp within (i)–(v).

Two routes give the result. The expected-correction route uses a deterministic per-observation gain $g_i \leq 1$: edge i's per-step sector parameter is $\pi_i g_i \leq \pi_i$, and P DA.1 gives the ultimate bound $\rho_\Sigma/(\pi_i g_i)$. The age route of Proposition EA.1 also covers gains that depend on the observation history, as in Beta-Bernoulli updating away from its stationary count and in per-time-step discounting. The two routes agree where both apply.

Outside the conditions, the result is not claimed. Under state-dependent allocation (greedy switching, UCB, Thompson sampling) the observation history depends on outcomes, so the weights are no longer outcome-independent and the proof does not apply as written. Estimators that extrapolate a learned drift fall outside $\mathcal{C}$ (their weights are not a convex combination of past outcomes). Under constant drift they can lag less; whether a worst case over drift sequences restores a comparable bound is open. If only the currently chosen alternative must stay calibrated, the target changes and so does the cap.

Discussion.

Where the cap comes from. One observation per step can refresh only one alternative. While an alternative goes unobserved, its truth keeps moving and nothing the agent does to its credence can follow. The best any averaging rule can do is put all weight on the newest evidence. Its lag is then the drift since the last refresh, on average $1/\pi_i$ steps of drift. Forgetting governs how much old evidence is kept, and the ceiling sits above every forgetting rate. Forgetting's own structural role is the other boundary: without it (B.1, $\lambda = 1$) the gain decays to zero and every positive drift rate is eventually fatal.

Parametric tuning cannot cross the ceiling; structural change can. Above $\rho_\Sigma = R_\Sigma \min_i \pi_i$, no forgetting rate, exploration rate, or update rule in $\mathcal{C}$ keeps every alternative calibrated. What moves the ceiling changes the strategy's structure or its evidence channels:

- More evidence per step, which breaks $\sum_i \pi_i \leq 1$: parallel trials, or counterfactual or off-policy observation of alternatives not taken.
- Fewer alternatives held in calibration: prune the OR node, or relax the requirement for alternatives the agent has stopped considering.
- For AND chains, direct observation of deeper links, which raises π_k above $\prod_{j<k} \theta_j$.

This has the same shape as `#result-structural-adaptation-necessity` at the epistemic layer, where parametric adaptation cannot close a gap that a class change can. The correspondence is structural, at discussion grade. It is not a derivation of one result from the other.

Breadth costs drift tolerance. Each alternative an agent keeps calibrated divides the drift rate it can survive. The design question for an OR node is therefore how many alternatives the evidence stream can keep calibrated at the environment's ρ_Σ/R_Σ, not only which alternative is best: at most R_Σ/ρ_Σ of them. The answer follows from the allocation alone, and the update rule does not change it.

Findings.

Evidence allocation caps strategic persistence at $\rho_\Sigma < R_\Sigma/K$. **Brief:** An agent keeping several alternative plans in view can try only one per turn. The real odds of every plan keep shifting, so a plan's estimate is out of date by however much the world moved since that plan was last tried, on average $1/\pi$ turns of drift for a plan tried with probability π. No way of updating beliefs or forgetting old evidence beats simply trusting the most recent trial, and that rule's lag is exactly this. So if K alternatives must all stay accurate, the world can drift at most $1/K$ of the agent's error tolerance per turn; with two alternatives, half.

Impact: Locates where the strategic-persistence design space is genuinely capped. The cap comes from how evidence is allocated across alternatives that must stay calibrated, not from the forgetting mechanism, which reaches the universal single-edge floor $\rho_\Sigma < R_\Sigma$. It gives (SA3) of `#schema-strategy-persistence` a satisfiability condition: SA3 can be met iff $\rho_\Sigma < R_\Sigma \min_i \pi_i$, at most R_Σ/K. Past the ceiling, the remedies are structural: more evidence channels per step, fewer alternatives held in calibration, or direct observation of deep links. A strategy's breadth is purchased with drift tolerance.

Novelty Claim: *Claim differentiation.* That an arm's estimate cannot track drift faster than it is sampled is elementary, and the non-stationary bandit literature treats the explore-versus-track trade-off in regret terms. The AAT contribution is stating it as a sharp reachability ceiling on the sector-persistence form across a whole estimator class: Bayesian counts with or without forgetting under either discount convention, and every history-gained innovation update. It covers OR alternatives, starved AND edges and the gated L1 cases in one statement, and it attaches to (SA3) a satisfiability condition the schema did not have.

Related Work:

Table T.1

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Tracking drifting arms under limited sampling	Garivier & Moulines 2011, *On Upper-Confidence Bound Policies for Switching Bandit Problems*, ALT 2011 (discounted and sliding-window UCB)	*conceptual precursor*: explore-versus-track trade-off under non-stationarity, stated as regret rather than as a persistence ceiling
Variation-budget non-stationary bandits	Besbes, Gur & Zeevi 2014, *Stochastic Multi-Armed-Bandit Problem with Non-stationary Rewards*, NeurIPS 2014	*conceptual precursor*: drift budget versus achievable regret; AAT's ρ_Σ plays the variation-budget role per step
Exponential forgetting	Ljung 1987, *System Identification: Theory for the User*, MIT Press	*formal antecedent*: the discounted update whose two conventions both fall in $\mathcal{C}$
Parametric cannot, structural can	`#result-structural-adaptation-necessity`	*parallel structural pattern* at the epistemic layer

Search Log: - 2026-10-30 (*intuition-only*): the sampling-lag bound is elementary; the bandit-tracking literature above was recalled, not searched for this framing. Targeted future search: persistence or stability formulations of non-stationary bandits (not regret), and "observation scheduling for tracking" in sensor-management literature.

Appendix U

Derivation: Regret-Bound Derivation of the Strategy-Cost KL Direction

The variational form of the strategy-cost objective (`#form-strategy-complexity-cost`) carries a $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ relevance term. This appendix derives that **the π^*-first KL direction is forced** as the non-vacuous regret-bound form, and proves (conditional on a chain-rule additivity axiom motivated by `#der-chain-confidence-decay`) that **reverse-KL is uniquely forced within the direction-forced f-divergence family**.

Three sources of force compose. (i) The *direction* is forced decision-theoretically: bounded regret against an optimal policy is bounded above by value-range × total-variation, and by Pinsker's inequality total variation is bounded by $\sqrt{D_{\mathrm{KL}}(\pi^* \| Q)/2}$ in the π^*-first direction; the opposite-direction KL becomes infinite under deterministic optimal policy whenever the strategy places any mass off the optimum, giving a vacuous bound. (ii) Within the direction-forced family, the standard Hobson-Csiszár-Shore-Johnson uniqueness machinery via Aczél's Cauchy functional equation forces reverse-KL *uniquely* among f-divergences once the chain-rule additivity axiom is granted — the divergence-layer instance of `#disc-additive-coordinate-forcing`'s meta-pattern (after chain-confidence decay; before log-odds at the update layer in `#deriv-edge-update-natural-parameter`). (iii) The bound is *sharpened* under AAT's canonical deterministic-optimum scope: the **Bretagnolle-Huber identity** holds exactly (not as inequality), giving a two-sided regret bound $\Delta_{\min}(1 - e^{-D_{\mathrm{KL}}}) \le R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}})$ strictly sharper than Pinsker — Lipschitz-equivalent to $(1 - e^{-D_{\mathrm{KL}}})$ at both ends, with the Pinsker form becoming vacuous once $D_{\mathrm{KL}} > 2$.

Secondary supporting properties of reverse-KL (gradient tractability, alignment with variational-inference machinery, MDL coding interpretation, Amari-Nagaoka Fisher geometry compatibility) are surfaced as *convergent evidence* rather than independent uniqueness grounds; the framework is explicit that Fisher-metric-at-second-order is not distinguishing within f-divergences (Eguchi 1983), so only the chain-rule axiom does the uniqueness work. The result is one of AAT's cleanest demonstrations of the framework's posture on coordinates: where a coordinate matters, the framework can point at the axiom that forces it; where the axiom isn't AAT-internal, the framework can name what was adopted from outside.

Formal Expression.

§1 — Setup. Fix O_t, M_t, continuation policy π_{cont}, and horizon N_h. The action-value is the objective-functional evaluation at a: $V(a) := Q_O(M_t, a; \pi_{\mathrm{cont}}, N_h)$ (`#def-value-object`; V derives from O_t per `#form-objective-functional`). The optimal action is $a^* := \arg\max_a V(a)$; under AAT's canonical scope (`#def-value-object`), the optimal policy $\pi^*(\cdot \mid M_t) = \delta_{a^*}$ is the point mass on a^* (tied-optimum extensions §8). Write $V_{\max} := \max_a V(a) - \min_a V(a)$ for the value range at fixed M_t; this is finite whenever $\mathcal{A}$ is a bounded action set under V.

$Q_{\Sigma_t}(\cdot \mid M_t)$ is the action distribution induced by the strategy DAG. In general Q_{Σ_t} is stochastic — the DAG's edge-credence uncertainty induces policy stochasticity downstream of propagation.

Definition (strategy-induced-regret)

§2 — Regret against the optimal policy.

The strategy-induced regret of Q_{Σ_t} against π^* is:

$$R(Q_{\Sigma_t}) \;:=\; V(a^*) \;-\; \mathbb{E}_{a\sim Q_{\Sigma_t}}[V(a)] \;=\; \sum_a Q_{\Sigma_t}(a)\cdot\big[V(a^*) - V(a)\big] \tag{U.1}$$

Writing $\Delta(a) := V(a^*) - V(a) \in [0, V_{\max}]$ for the per-action regret gap. Three regret forms in the literature:

$$\mathbb{E}_{\pi^*}[V] - \mathbb{E}_{Q_{\Sigma_t}}[V], \qquad V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V], \qquad \mathbb{E}_{\pi^*}[V - V_{Q_{\Sigma_t}}] \tag{U.2}$$

coincide identically under deterministic π^*.

§3 — Total-variation bound (tight).

Derived (tv-regret-bound, from bounded V + deterministic π^*)

$$R(Q_{\Sigma_t}) \;=\; \sum_a Q_{\Sigma_t}(a)\,\Delta(a) \;\le\; V_{\max}\cdot \sum_{a\neq a^*} Q_{\Sigma_t}(a) \;=\; V_{\max}\cdot(1 - Q_{\Sigma_t}(a^*)) \tag{U.3}$$

For deterministic $\pi^* = \delta_{a^*}$:

$$\mathrm{TV}(\pi^*, Q_{\Sigma_t}) \;=\; \tfrac{1}{2}\sum_a |\pi^*(a) - Q_{\Sigma_t}(a)| \;=\; 1 - Q_{\Sigma_t}(a^*) \tag{U.4}$$

Therefore:

$$\boxed{R(Q_{\Sigma_t}) \;\le\; V_{\max}\cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t})} \qquad \text{(tight)} \tag{U.5}$$

Tightness. Equality holds exactly when $\Delta(a) = V_{\max}$ for all $a \neq a^*$ (degenerate value landscape: every suboptimal action incurs the full value range). For typical landscapes the bound is loose by a factor $\mathbb{E}_{Q_{\Sigma_t}}[\Delta \mid a \neq a^*]/V_{\max} \in (0, 1]$.

§4 — Exact regret-reverse-KL identity under deterministic π^*. Under AAT's canonical scope (deterministic $\pi^* = \delta_{a^*}$), reverse-KL and total variation are related by an *identity*, not merely an inequality. This gives a tight two-sided regret bound that the Pinsker inequality strictly weakens.

Derived (bh-identity-deterministic, under deterministic π^*)

Identity. For deterministic $\pi^* = \delta_{a^*}$ and any Q with $Q(a^*) > 0$:

$$\boxed{D_{\mathrm{KL}}(\pi^* \| Q) \;=\; -\log Q(a^*) \;=\; -\log\big(1 - \mathrm{TV}(\pi^*, Q)\big)} \tag{U.6}$$

Derivation. $D_{\mathrm{KL}}(\delta_{a^*}\|Q) = \sum_a \delta_{a^*}(a)\log(\delta_{a^*}(a)/Q(a)) = -\log Q(a^*)$ (only the $a = a^*$ term contributes; $0\log 0$ terms are conventionally 0). And $\mathrm{TV}(\delta_{a^*}, Q) = 1 - Q(a^*)$ (direct from §3). Substituting gives the identity. □

This is the Bretagnolle-Huber inequality (Bretagnolle & Huber 1978, "Estimation des densités," *Séminaire de probabilités XII*, Springer LNM 649; Tsybakov 2009 §2.4; Sason & Verdú 2016) specialized to the deterministic-P case, where the general inequality $\mathrm{TV} \le \sqrt{1 - e^{-D_{\mathrm{KL}}}}$ becomes an equality.

Combining with the tight TV-regret bound of §3:

Derived (exact-regret-reverse-kl, under deterministic π^*)

$$\boxed{R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot \left(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}\right)} \tag{U.7}$$

Matching lower bound. Define the action-gap $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) > 0$ (standard in RL theory; well-defined whenever the optimum is isolated over finite $\mathcal{A}$ with bounded V). Then:

$$R(Q_{\Sigma_t}) \;=\; \sum_{a \neq a^*} Q_{\Sigma_t}(a)\, \Delta(a) \;\ge\; \Delta_{\min} \cdot \sum_{a \neq a^*} Q_{\Sigma_t}(a) \;=\; \Delta_{\min} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t}) \tag{U.8}$$

Under the exact BH identity:

$$R(Q_{\Sigma_t}) \;\ge\; \Delta_{\min} \cdot \left(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}\right) \tag{U.9}$$

Lipschitz-equivalence corollary. Regret and $(1 - e^{-D_{\mathrm{KL}}})$ are Lipschitz-equivalent with constants $\Delta_{\min}$ (below) and $V_{\max}$ (above):

Derived (matched-lower-bound, under deterministic π^* + isolated optimum)

$$\frac{\Delta_{\min}}{V_{\max}} \;\le\; \frac{R(Q_{\Sigma_t})}{V_{\max}(1 - e^{-D_{\mathrm{KL}}})} \;\le\; 1 \tag{U.10}$$

The upper bound is tight when the value landscape is extremal ($\Delta(a) = V_{\max}$ for all $a \neq a^*$); the lower bound is tight when sub-optimal actions are uniformly bad ($\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$).

$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) \in [0, +\infty)$: finite and graded whenever $Q_{\Sigma_t}(a^*) > 0$; diverges only in the structural-failure limit where the strategy places zero mass on the optimum.

§4.1 — Pinsker bound as loose general form. The Pinsker inequality $\mathrm{TV}(P, Q) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P \| Q)}$ (Tsybakov 2009 §2.4; Cover & Thomas 2006 §11.6) yields a regret bound that does not assume deterministic π^*:

$$R(Q_{\Sigma_t}) \;\le\; V_{\max} \cdot \sqrt{\tfrac{1}{2}\, D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})} \tag{U.11}$$

Under AAT's canonical scope, this is strictly weaker than the BH-identity form of §4. For any $D_{\mathrm{KL}} > 0$: $1 - e^{-D_{\mathrm{KL}}} < \sqrt{D_{\mathrm{KL}}/2}$ for small D_{KL} (the BH form is linear in D_{KL} while Pinsker is $\sqrt{D_{\mathrm{KL}}}$), and Pinsker's $V_{\max}\sqrt{D_{\mathrm{KL}}/2}$ exceeds the trivial $V_{\max}$ bound once $D_{\mathrm{KL}} > 2$, giving vacuous content there. The BH identity is informative uniformly in D_{KL}.

Pinsker remains the correct tool for stochastic-π^* extensions (§9) where the deterministic-π^* scope is relaxed and the BH identity degrades back to inequality.

Derived (kl-direction-forced, from deterministic π^*)

§5 — Direction-forcing claim (the load-bearing result).

Claim. The KL direction with π^* *first* is forced as the non-vacuous regret-bound form. Forward-KL $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ is *not* a non-vacuous bound on $R(Q_{\Sigma_t})$ under deterministic π^*.

Derivation. Expanding:

$$D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*) \;=\; \sum_a Q_{\Sigma_t}(a) \log\!\big(Q_{\Sigma_t}(a)/\pi^*(a)\big) \tag{U.12}$$

For any $a \neq a^*$: $\pi^*(a) = 0$ (point mass assumption). The summand $Q_{\Sigma_t}(a) \log(Q_{\Sigma_t}(a)/0) = +\infty$ unless $Q_{\Sigma_t}(a) = 0$. Therefore $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*) = +\infty$ whenever Q_{Σ_t} places any mass off a^* — for *all but a measure-zero subset* of stochastic strategies. A bound "$R \leq +\infty$" is vacuous.

This is the same shape of degeneracy as the original Shannon-MI form (Gemini Finding 2) that the V-medium move was introduced to escape: same degeneracy-when-π^*-is-deterministic, different value (0 vs $+\infty$). Only the *reverse* direction escapes both.

Alignment with the structural problem. Forward-KL is natural for *mode-covering* inference (where Q is asked to cover P's full support) and variational inference when Q is learned to match a full distribution. Reverse-KL is natural for *mode-seeking* (concentrate Q on P's mode). The AAT strategy problem is mode-seeking by construction: the strategy should concentrate on a^*. The direction alignment with the structural problem is not accidental.

Asymmetry is forced by regret's one-sidedness. A second, independent argument for asymmetry — one that does not rely on the chain-rule axiom of §6.1. Under deterministic π^*, regret is a *one-sided* quantity: $R(Q_{\Sigma_t}) = \sum_{a \neq a^*} Q_{\Sigma_t}(a)\Delta(a)$ contributes only from Q_{Σ_t}'s off-optimum mass; π^*'s "mass off Q_{Σ_t}" is vacuous since π^* has no support off a^*. Any divergence whose role is to bound this quantity should therefore be asymmetric — it should penalize Q_{Σ_t}'s deviation from π^* without symmetrically penalizing the (trivially zero) deviation of π^* from Q_{Σ_t}. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized KL) treat the two deviations interchangeably; under the one-sided regret quantity this is operationally wrong — it introduces a term with no semantic role. The asymmetry requirement is thus *structural*, emerging directly from what regret is as a functional, and is established independently of the chain-rule axiom that picks the specific asymmetric form in §6.1. The two arguments compose: *direction-forcing* (vacuity under deterministic π^*) + *asymmetry-forcing* (regret is one-sided) ⇒ the admissible bounding divergence is π^*-first and asymmetric; chain-rule additivity (§6.1) then picks reverse-KL uniquely within that family.

Terminology note (Bishop-vs-AAT "reverse-KL"). AAT's canonical reverse-KL is $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — optimum-first, agent-second. Under Bishop 2006's convention ($D_{\mathrm{KL}}(q \| p)$ with q the approximation is called "reverse-KL"), AAT's form would be labeled "forward-KL." The two naming conventions disagree in surface labeling but agree in the operational property — both pick out the *mode-seeking* direction (concentrate the approximation on the target's mode). Literature that uses the Bishop convention (variational inference, generative modeling, Levine 2018) and literature that uses the π^*-first / target-first convention (decision theory, regret analysis, Rubin-Shamir-Tishby 2012) describe the same operational quantity under deterministic π^*. When this segment writes "reverse-KL," the π^*-first / AAT convention is meant; under softened targets the two conventions describe different objects and must be disambiguated explicitly.

Discussion (admissible-regret-divergences)

§6 — Admissible-divergence family and uniqueness of reverse-KL.

The regret-bound argument forces the *direction* (the reference distribution π^* appears first) but admits a family of divergences. Each member yields a valid regret bound; they differ in tightness and operational properties:

Table U.1

Divergence	*Bound on R*	*Tightness*	*Finite under det. π^*?*	*Gradient-tractable?*
$\mathrm{TV}(\pi^*, Q_{\Sigma_t})$	$V_{\max} \cdot \mathrm{TV}$	**Tight** (extremal V)	Yes	No (non-differentiable)
$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ via Pinsker	$V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}}$	Loose by $\sqrt{\cdot}$	Yes	Yes
$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ via Bretagnolle-Huber	$V_{\max}\sqrt{1 - e^{-D_{\mathrm{KL}}}}$	Tighter than Pinsker for large D_{KL}	Yes	Yes
$\chi^2(\pi^* \| Q_{\Sigma_t})$ (Le Cam)	$V_{\max} \cdot \frac{1}{2}\sqrt{\chi^2}$	Typically looser than Pinsker	Yes: $\chi^2 = 1/Q_{\Sigma_t}(a^*) - 1$	Yes
$D_\alpha(\pi^* \| Q_{\Sigma_t})$ (Rényi, $\alpha \geq 1$)	Various	Interpolates KL ($\alpha \to 1$) and χ^2 ($\alpha = 2$)	Yes for $\alpha \geq 1$	Yes
$D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ (forward-KL)	$+\infty$	Vacuous	**No**	—

What is uniquely forced by the regret-bound argument alone. The direction: the reference π^* is first. This is a real derivation outcome, not a selection. Within the π^*-first family, multiple f-divergences each give valid bounds; an additional structural axiom is required to pick one uniquely.

§6.1 — Chain-rule uniqueness theorem.

Derived (reverse-kl-uniqueness, Conditional on chain-rule axiom)

Theorem (chain-rule / additivity uniqueness of KL among f-divergences; folk theorem, standard functional-equation derivation). *Let $D_f(P\|Q) = \sum_x Q(x) f(P(x)/Q(x))$ be a smooth f-divergence with f convex and $f(1) = 0$. The chain rule*

$$D_f(P_{XY}\|Q_{XY}) \;=\; D_f(P_X\|Q_X) \;+\; \mathbb{E}_{P_X}\big[D_f(P_{Y|X}\|Q_{Y|X})\big] \tag{U.13}$$

holds for all joint distributions (X, Y) if and only if $f(t) = c \cdot t \log t$ for some $c > 0$ — i.e., D_f is reverse-KL up to positive scaling.

References. The theorem is a classical folk result obtainable by direct functional-equation argument (sketched below). The canonical published axiomatic characterizations equivalent to the chain-rule statement are: **Hobson 1969**, "A new theorem of information theory," *J. Stat. Phys.* 1(3):383–391 — uniqueness of the Kullback expression via a composition/additivity axiom; **Csiszár 1991**, "Why least squares and maximum entropy? An axiomatic approach to inference for linear inverse problems," *Annals of Statistics* 19(4):2032–2066 — Theorem 3 corollary: the only transitive statistical projection rule is the I-divergence projection rule; Theorem 5: product-consistency characterizes I-divergence uniquely. See also **Shore & Johnson 1980**, "Axiomatic derivation of the principle of maximum entropy and the principle of minimum cross-entropy," *IEEE Trans. Info. Theory* 26(1):26–37 (system-independence axiom); **Sanov 1957**, "On the probability of large deviations of random variables," *Mat. Sb.* 42(84):11–44 (large-deviation rate function for empirical-distribution concentration); **Aczél & Daróczy 1975**, *On Measures of Information and Their Characterizations* (Academic Press) for the general functional-equation machinery.

These are not independent uniqueness routes. Hobson's composition axiom, Csiszár's transitive-projection axiom, Shore-Johnson's system-independence axiom, and Sanov's sampling-consistency condition are *structurally-equivalent reformulations* of the same underlying requirement — all factor through independence-of-sub-problems (joint inference on independent subsystems decomposes additively over factorizations). The Cauchy functional equation that each reduces to is the common structural content beneath varied surface formulations. This matters for `#disc-additive-coordinate-forcing`: the chain-rule axiom's load-bearing role is not weakened by noting multiple canonical references; conversely, the multiple references do not provide multiple independent uniqueness arguments. No known uniqueness route outside the independence-on-sub-problems family exists.

Proof sketch. Write $r_x = P(x)/Q(x)$, $s_{y|x} = P(y \mid x)/Q(y \mid x)$. The chain-rule identity must hold for all joint decompositions, which forces the functional equation $f(rs) = f(r) + rf(s) + g(r)$ for all $r, s > 0$. Combined with $f(1) = 0$ and convexity, the unique solution is $f(t) = c \cdot t \log t$ for $c > 0$ (Aczél & Daróczy 1975 §4).

Why other members of the family fail the chain rule. Concrete counterexample for χ^2: take Q_X uniform on $\{x_1, x_2\}$, $P_X = (3/4, 1/4)$, $Q(y \mid x)$ uniform, $P(y \mid x) = (3/4, 1/4)$ for both x. Direct calculation gives $\chi^2(P_{XY} \| Q_{XY}) = 9/16$, while $\chi^2(P_X \| Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X} \| Q_{Y|X})] = 1/4 + 1/4 = 8/16$. Non-additive: $9/16 \neq 8/16$. Rényi-α for $\alpha \neq 1$ fails analogously; Bretagnolle-Huber is a monotone transform of reverse-KL (not an independent f-divergence); Hellinger-squared ($\alpha = 0$) likewise fails.

AAT-internal motivation for the chain-rule axiom. The chain rule is the *divergence-level analog* of the additive log-confidence decay in `#der-chain-confidence-decay`. AAT has already committed to additive-mismatch-decomposition-along-causal-chains in that segment; the divergence-level parallel is:

$$D(\pi^* \| Q_{\Sigma_t}) \;=\; \sum_{t=1}^{T} \mathbb{E}_{\pi^*}\Big[D\big(\pi^*(\cdot \mid a^*_{<t}) \,\Big\|\, Q(\cdot \mid a^*_{<t})\big)\Big] \tag{U.14}$$

— total mismatch between π^* and Q_{Σ_t} decomposes additively over the DAG's causal layers along the optimal trajectory. Non-chain-rule divergences (e.g., χ^2) give super-additive decompositions in which layer-mismatches amplify multiplicatively — structurally discordant with the DAG factorization (`#def-strategy-dag`). Adopting the chain-rule axiom is therefore not arbitrary; it is the divergence-level version of a decomposition principle AAT already relies on.

Corollary (within the chain-rule axiom). Under deterministic π^* + bounded value + chain-rule additivity, reverse-KL is the unique smooth f-divergence in the direction-forced admissible family. The "Reverse-KL canonical among smooth divergences" status upgrades from formulation choice (under the pre-axiom reading) to *Derived (conditional on the chain-rule axiom)*.

What the uniqueness theorem does not do. It fixes the divergence within the f-divergence family; it does not fix the bounding function g around the divergence (Pinsker vs Bretagnolle-Huber vs Le Cam-on-χ^2 are *different bounds using different divergences*, so only Pinsker and BH-on-KL remain in scope after uniqueness — and these differ only in the g-envelope on top of reverse-KL, not in the divergence itself). It does not resolve the linear-vs-square-root form question of §7 (that is a Lagrangian-shape choice orthogonal to the divergence choice). Monotone transforms of reverse-KL (e.g., $1 - e^{-D_{\mathrm{KL}}}$) are equivalence-class members for gradient-based optimization and are not ruled out.

Scope of the axiom. The chain rule is stated for arbitrary joint decompositions. AAT's strategy spaces are discrete at the proposition level; the theorem applies directly. For continuous strategy spaces (e.g., low-level control), the chain rule extends by replacing sums with integrals against the dominating measure; the functional-equation derivation is identical in the continuous case (see Liese & Vajda 1987, *Convex Statistical Distances*, Teubner, for the standard measure-theoretic treatment of f-divergences and their decomposition properties).

§6.2 — Secondary characterizations (now supporting rather than load-bearing). Four additional properties further motivate reverse-KL as the canonical choice; under §6.1 they are no longer load-bearing for uniqueness but remain informative about why reverse-KL is a comfortable fit with the rest of the AAT framework:

1. **Gradient-tractability.** TV is non-differentiable at mass boundaries; reverse-KL is smooth. Operational minimization via gradient methods rules out TV despite its tightness — independent of the uniqueness argument.

2. **Variational-inference alignment.** Reverse-KL is the standard divergence in the variational-inference lineage (ELBO derivation uses reverse-KL; Friston et al. 2017; Da Costa et al. 2020). The V-medium move's rhetorical payoff — shared vocabulary with active inference — lives on reverse-KL specifically. This is *convergent evidence* that reverse-KL is natural in multiple frames, not an independent uniqueness argument: the ELBO decomposition itself uses the chain rule as a sub-step, so the logical priority runs chain rule → reverse-KL → ELBO.

3. **Fisher geometry.** Reverse-KL's second-order expansion gives the Fisher information metric (Amari & Nagaoka 2000). *But by the differential-geometric framework in Eguchi 1983* ("Second order efficiency of minimum contrast estimators in a curved exponential family," *Annals of Statistics* 11(3):793–803, §2 contrast-function development; modern restatement at Amari & Cichocki 2010 Theorem 5 eq. (126)), every smooth f-divergence with $f''(1) > 0$ induces the Fisher metric at second order up to the scalar $f''(1)$. Fisher-metric-at-second-order is *not* a distinguishing axiom within the f-divergence family — it is satisfied by every member. Noted here because it is a commonly-invoked distinguishing property; it does not distinguish. (Eguchi's Theorem 3 itself is about estimator efficiency via Γ^1-transversality; the f-divergence/Fisher-metric result is a consequence of the §2 contrast-function machinery that supports Theorem 3, not the theorem statement.)

4. **Information-budget / MDL interpretation.** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log Q_{\Sigma_t}(a^*)$ is the expected extra bits needed to code samples from π^* under Q_{Σ_t}'s code. The coding interpretation aligns with the segment's MDL framing. MDL is itself a chain-rule-respecting coding scheme (the bits-to-code-a-joint decompose additively over factorizations); consistency with MDL is convergent with §6.1.

Čencov/Morozova-Chentsov background (Čencov 1982 *Statistical Decision Rules and Optimal Inference*, AMS Translations of Mathematical Monographs 53; Morozova & Chentsov 1991, "Markov invariant geometry on state manifolds," *Itogi Nauki i Tekhniki*, translated in *J. Sov. Math.* 56(5):2648–2669; Ay, Jost, Lê & Schwachhöfer 2017, *Information Geometry*, Springer). The f-divergence family itself is characterized by Markov-morphism invariance (sufficient-statistic coarse-graining invariance) — this is what makes "f-divergences" the right background family to search within. Amari 2009 shows the alpha-divergences are the sub-family that are simultaneously f-divergences and Bregman divergences. Reverse-KL sits at $\alpha = 1$ within the alpha-family; the chain-rule axiom of §6.1 is what picks $\alpha = 1$ out of the one-parameter alpha-family.

§6.3 — Bregman-Fenchel identification: reverse-KL and log-odds as dual coordinates.

Derived (bregman-fenchel-dual-pair, exact; standard Legendre-Fenchel)

On the probability simplex Δ^{n-1}, the **negative-entropy potential** $\phi(Q) = \sum_a Q_a \log Q_a$ is strictly convex, Legendre, and essentially smooth on the relative interior (Rockafellar 1970 *Convex Analysis* §26). Its Fenchel conjugate is the **log-partition function** $\phi^*(\eta) = \log \sum_a e^{\eta_a}$ on the natural-parameter space $\mathbb{R}^n$ modulo the affine-gauge direction. The primal-dual correspondence is softmax: $Q = \nabla\phi^*(\eta)$ gives $Q_a = e^{\eta_a} / \sum_{a'} e^{\eta_{a'}}$, and $\eta = \nabla\phi(Q) = \log Q + \mathbf{1}$ (up to the constraint-normal direction) so $\eta_a - \eta_b = \log(Q_a/Q_b)$ is the **log-odds ratio**.

The Bregman divergence induced by ϕ on the primal simplex is **reverse-KL** exactly:

$$B_\phi(P, Q) \;:=\; \phi(P) - \phi(Q) - \langle \nabla\phi(Q), P - Q \rangle \;=\; \sum_a P_a \log \frac{P_a}{Q_a} \;=\; D_{\mathrm{KL}}(P \| Q) \tag{U.15}$$

Derivation. Expand: $\phi(P) - \phi(Q) = \sum_a (P_a \log P_a - Q_a \log Q_a)$; the inner-product term using $\nabla\phi(Q)_a = \log Q_a + 1$ gives $\sum_a (\log Q_a + 1)(P_a - Q_a) = \sum_a (\log Q_a)(P_a - Q_a)$ since $\sum_a (P_a - Q_a) = 0$. Substituting and simplifying yields $\sum_a P_a \log(P_a/Q_a) = D_{\mathrm{KL}}(P\|Q)$. □

Identification with `#deriv-edge-update-natural-parameter`. The log-odds coordinate derived under evidential-additivity in `#deriv-edge-update-natural-parameter` is the *Fenchel-dual natural parameter* of this Legendre-Fenchel pair. The divergence-layer coordinate of this segment (reverse-KL on Δ^{n-1}) and the update-layer coordinate of `#deriv-edge-update-natural-parameter` (log-odds on $\mathbb{R}^n/\mathbf{1}$) are two sides of one exponential-family geometric structure, viewed through primal and dual coordinates respectively.

What this adds. The two segments independently derive their coordinates via Cauchy-FE on logically-independent AAT-internal axioms (chain-rule additivity here; evidential additivity in `#deriv-edge-update-natural-parameter`). The Fenchel-Bregman correspondence shows that *the coordinates the two axioms pick out are related by Legendre duality* — the axioms are not redundant (they have different logical forms and constrain different objects; see §6.1 vs. `#deriv-edge-update-natural-parameter`'s axiom) but the forced coordinates *coincide on one geometric object*. This is a Discussion-grade observation about the relationship between two primary instances of `#disc-additive-coordinate-forcing`; it is

textbook Legendre-Fenchel (Amari & Nagaoka 2000 §3.5; Bregman 1967; Beck & Teboulle 2003 *Mirror descent and nonlinear projected subgradient methods* — relevant for the operational consequence that mirror-descent on the policy simplex with negative-entropy regularizer is equivalent to exponentiated-gradient descent on log-odds coordinates). The full geometric-unification reading sits in `#disc-additive-coordinate-forcing`'s meta-treatment.

Discussion (information-theoretic-mdp-lineage), positioning rather than derivation

§6.4 — Information-theoretic-MDP lineage and AAT's direction choice.

AAT's reverse-KL strategy-cost form sits within an established research lineage — the information-theoretic treatment of decision problems initiated by **Tishby & Polani 2011** ("The Information Theory of Decision and Action," in *Perception-Action Cycle: Models, Architectures, and Hardware*, Springer, pp. 601–636; Eq. 15 p. 19 introduces **Information-to-Go** $\mathfrak{I}^{\pi}(s_t, a_t)$, the conditional multi-information of the entire future state-action trajectory; Eq. 17 p. 21 the trade-off objective $\arg\min_{\pi}[\mathfrak{I}^{\pi} - \beta Q^{\pi}]$) and developed by **Rubin, Shamir & Tishby 2012** ("Trading value and information in MDPs," in *Decision Making with Imperfect Decision Makers*, Springer, pp. 57–74; §2.2 Eq. 3 p. 4 **control information** $\Delta I(s) = \sum_a \pi_s(a) \log(\pi_s(a)/\rho_s(a))$; §3.1 Theorem 1 p. 5 Bellman recursion for the free-energy $F_{\pi} = I_{\pi} - \beta V_{\pi}$). The control-as-inference lineage via **Levine 2018** ("Reinforcement Learning and Control as Probabilistic Inference," arXiv:1805.00909; §3.1 Eq. 11 and §5.2 p. 15) develops the max-entropy-RL framework using $D_{\mathrm{KL}}(p_\theta \| p_{\mathrm{tgt}})$ throughout. The shared framework — value and information as joint first-class quantities in MDP optimization, with a KL divergence measuring the information cost of a policy against a reference — is the context in which AAT's strategy-cost objective naturally lives.

AAT's direction is distinctive within this lineage. Tishby-Polani 2011 uses the stationary marginal $\pi(a)$ of the policy itself as reference (Eq. 16 p. 20, Eq. 22 p. 22) — a direction-neutral multi-information rather than a KL-to-external-reference. Rubin et al. 2012 uses a default policy $\rho_s(a)$ (typically uniform) with direction $D_{\mathrm{KL}}(\pi_s \| \rho_s)$ — **agent-first, default-second**, a regularization/complexity cost. Levine 2018 uses $D_{\mathrm{KL}}(p_\theta \| p_{\mathrm{tgt}})$ — **proposal-first, target-second**, a variational ELBO form with softened exponentiated-reward target. **AAT's** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — **optimum-first, agent-second** — is the *opposite* direction to all three.

The direction is forced, not chosen. AAT's direction is not a convention adopted from the surrounding literature; it is forced by two independent structural requirements of this segment's derivation: (i) the regret-bound argument of §5 requires the optimum as the reference for the bound to be tight (mode-seeking toward a^*); (ii) the Bretagnolle-Huber identity of §4 is *exact* in the π^*-first direction under deterministic π^* and degenerate in the flipped direction. The flipped direction (Rubin 2012's / Levine 2018's form with $\rho = \pi^*$) is vacuous here for the same reason forward-KL is vacuous in §5: π^* has no support off a^*. AAT *owns* the direction-distinctiveness rather than inheriting it.

Rubin 2012 Theorem 3 PAC-Bayesian motivation. A fourth independent operational property of the π^*-first KL form comes from Rubin-Shamir-Tishby 2012 Theorem 3 (p. 10): under a-priori stochastic default policy and empirical reward estimate, the control information I_{π} bounds the generalization gap of the value function:

$$\tilde{D}_{\mathrm{KL}}\big(\hat{V}_{\pi}(s_0) \,\big\|\, V_{\pi}(s_0)\big) \;\le\; \frac{I_{\pi}(s_0) + \log(2m/\delta)}{m-1} \tag{U.16}$$

This is a PAC-Bayesian concentration bound. In AAT terms: policies with lower strategy-cost have tighter value-function generalization under finite empirical data. Together with (a) the regret-bound direction-forcing of §5, (b) the chain-rule uniqueness of §6.1, and (c) the Bregman-Fenchel dual identification of §6.3, this is a fourth independent motivation for the KL-to-reference form — operational rather than axiomatic, and independent of the chain-rule axiom. The direction differs (Rubin uses agent-first; AAT uses optimum-first), but the generalization-bound structure transfers by the same PAC-Bayesian machinery: I_{π} as regularizer against a reference.

Relation to `#form-information-bottleneck`. The canonical IB objective is $I(X;T) - \beta I(T;Y)$ with Shannon mutual information on both terms. AAT's strategy-cost relevance term is $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ — a KL divergence to a target, not a mutual information. The two are *sibling lineages*, both descended from Shannon rate-distortion theory (the IB instance uses MI-to-relevant-label as the fidelity term; the information-theoretic-MDP instance uses KL-to-reference-policy);

neither reduces to the other without a change of relevance variable. AAT's compression-operations framework (`#disc-compression-operations`) uses the IB form for M_t, G_t^{shared}, and Λ compressions; the strategy-cost compression uses the information-theoretic-MDP form. Both are valid Lagrangian relaxations of an underlying rate-distortion problem; the choice of fidelity term depends on whether the compressed variable is meant to preserve information about an observable (IB) or to match a reference policy (information-theoretic-MDP). The "abandoned IB purity" concern raised in external reviews (Gemini 2026-04-24) is defused at this level: AAT has not abandoned IB; its strategy-cost compression uses a different relevance quantity than its model-compression, and both are principled under the rate-distortion umbrella.

Discussion (β_Σ-local-vs-global)

§7 — β_Σ interpretation under the regret bound.

The Pinsker bound gives $R(Q_{\Sigma_t}) \le V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\pi^*\|Q_{\Sigma_t})}$. Two scale forms for the segment's trade-off parameter β_Σ arise:

(a) Square-root-KL form (tight regret-bound scale):

$$\mathcal{L}(\Sigma_t) \;=\; \mathcal{R}(\Sigma_t) \;+\; \beta_\Sigma \cdot \sqrt{D_{\mathrm{KL}}(\pi^*\|Q_{\Sigma_t})}, \qquad \beta_\Sigma = V_{\max}/\sqrt{2} \tag{U.17}$$

β_Σ is *globally naturalized* as a constant scale proportional to the value range at fixed M_t. Cost: the form departs from the linear-Lagrangian IB shape that AC uses across the four compression operations.

(b) Linear-KL form (IB-shape instance, preserved in the segment):

$$\mathcal{L}(\Sigma_t) \;=\; \mathcal{R}(\Sigma_t) \;+\; \beta_\Sigma \cdot D_{\mathrm{KL}}(\pi^*\|Q_{\Sigma_t}) \tag{U.18}$$

Under this form, β_Σ's regret-bound interpretation is *local*: differentiating the Pinsker bound, $\partial R/\partial D_{\mathrm{KL}} = V_{\max}/(2\sqrt{2D_{\mathrm{KL}}})$, so β_Σ represents the local cost-per-bit at the operating point but varies with D_{KL}. Upside: consistency with the IB linear-Lagrangian shape (AC). Downside: no uniform global β_Σ scale.

Choice made in `#form-strategy-complexity-cost`. Keep the linear form (IB-shape alignment); note the square-root form in the Epistemic Status as the "source" derivation. The direction-forcing claim is the load-bearing strengthening; the linear-vs-square-root choice is a second-order trade-off that preserves architectural consistency at the cost of a fully naturalized β_Σ.

§8 — Bound tightness and scope limits. **Vacuity regimes** (where the regret bound fails to be informative):

1. $V_{\max} = \infty$ **(unbounded value).** Then all bounds are ∞. AAT's O_t is $\mathbb{R}$-valued (`#form-objective-functional`); boundedness of V over $\mathcal{A}$ at fixed M_t is an additional assumption. Stated here as a scope condition of the derivation.

2. $Q_{\Sigma_t}(a^*) = 0$ **(strategy cannot express the optimum).** Then $D_{\mathrm{KL}}(\pi^*\|Q_{\Sigma_t}) = +\infty$. This is structural failure — the strategy's DAG cannot produce the optimal action — and the infinite KL correctly flags it. *Informative degeneracy*, not pathology: operational minimization rejects such Σ_t.

3. π^* **not deterministic (outside canonical scope).** For stochastic π^* with larger support than Q_{Σ_t}, reverse-KL may still be infinite. §9 addresses tied-optimum extensions; stochastic π^* under softmax-smoothing is future work.

Tightness comparison. Under deterministic π^* (AAT's canonical scope), the Bretagnolle-Huber **identity** $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log(1 - \mathrm{TV}(\pi^*, Q_{\Sigma_t}))$ makes the KL and TV bounds *operationally equivalent in reverse-KL coordinates*: the KL-based bound $R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}})$ of §4 is exactly the TV-based bound of §3 re-expressed in the KL coordinate, with matching lower bound $R \ge \Delta_{\min}(1 - e^{-D_{\mathrm{KL}}})$ (§4). Pinsker is strictly weaker under this scope (see §4.1). Outside the deterministic-π^* scope, the BH identity degrades back to the inequality $\mathrm{TV} \le \sqrt{1 - e^{-D_{\mathrm{KL}}}}$ (Bretagnolle-Huber 1978; tighter than Pinsker for large D_{KL} but no longer an identity); Pinsker is the textbook fallback for general distributions. The net: reverse-KL is not merely "the tightest smooth divergence" — in AAT's canonical scope it carries the *exact* regret-bound relationship via BH, with Pinsker as a loose alternative retained only for stochastic-π^* extensions.

§9 — Extensions. **Tied-optimum case.** If π^* has support on a tied-optimum set $\mathcal{A}^* = \{a : V(a) = V(a^*)\}$ with uniform mass, reverse-KL is finite whenever Q_{Σ_t} covers $\mathcal{A}^*$. The regret-bound argument extends directly: $R(Q_{\Sigma_t}) = \sum_a Q_{\Sigma_t}(a)\Delta(a) \le V_{\max} \cdot \mathbb{P}_{Q_{\Sigma_t}}(a \notin \mathcal{A}^*)$ and Pinsker applies unchanged.

Softmax-smoothed π^* (stochastic π^* for non-degeneracy reasons). A regret bound of the form

$$\mathbb{E}_{\pi^*}[V] - \mathbb{E}_{Q_{\Sigma_t}}[V] \tag{U.19}$$

with softmax-weighted π^* also admits the Pinsker-KL step, with reverse-KL again the non-vacuous direction (forward-KL remains vacuous whenever π^* has wider support than Q_{Σ_t}, and vice versa). Deferred for explicit treatment.

What Is Derived vs. What Is Chosen.

Table U.2

Property	*Source*	*Strength*
Regret definition $R(Q_{\Sigma_t}) := V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V]$	Definitional; collapses to the three literature forms under deterministic π^*	Definition
TV-regret bound $R \le V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t})$	Bounded value range + deterministic π^*	Proved (tight)
BH identity $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log(1 - \mathrm{TV}(\pi^*, Q_{\Sigma_t}))$ under deterministic π^*	Direct calculation (Bretagnolle-Huber 1978 specialized to $P = \delta_{a^*}$)	**Proved (exact identity)**
Regret-reverse-KL bound $R \le V_{\max}(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})})$	TV-regret bound + BH identity, under deterministic π^*	**Proved (tight in upper direction)**
Matching lower bound $R \ge \Delta_{\min}(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})})$	Isolated-optimum action-gap $\Delta_{\min} > 0$ + BH identity	**Proved (Lipschitz-equivalence with $V_{\max}$ / $\Delta_{\min}$ constants)**
Pinsker-KL bound $R \le V_{\max}\sqrt{\frac{1}{2} D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$	Pinsker's inequality applied to TV bound	Proved (strictly weaker than BH identity under deterministic π^*; correct form for stochastic-π^* extensions)
KL direction forced (reverse-KL, π^*-first)	Forward-KL is vacuous ($+\infty$) under deterministic π^* whenever Q_{Σ_t} has off-optimum mass	**Proved (direction uniquely forced)**
Asymmetry forced by regret's one-sidedness	Regret contributes only from Q_{Σ_t}'s off-optimum mass; π^*'s off-Q deviation is vacuous; symmetric divergences penalize interchangeably, which is operationally wrong	Proved (asymmetry forced; independent of chain-rule axiom)

Property	*Source*	*Strength*
Reverse-KL uniquely forced within direction-forced f-divergences	Chain-rule additivity axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975); AAT-internally motivated as divergence-level analog of `#der-chain-confidence-decay` (§6.1)	**Derived (conditional on chain-rule axiom)**
Secondary characterizations (gradient-tractability, VI-alignment, MDL coding)	Independent operational properties; each compatible with reverse-KL, none individually distinguishing	Formulation support (non-load-bearing under §6.1)
Fisher-metric-at-second-order does not distinguish reverse-KL	Eguchi 1983 §2 contrast-function framework (*Ann. Statist.* 11(3):793–803): every smooth f-divergence with $f''(1) > 0$ induces the Fisher metric up to scalar $f''(1)$	Proved (no-go for Path A as uniqueness axiom)
Admissible family members (TV, Bretagnolle-Huber, χ^2, Rényi-α)	Each yields a valid regret bound; survey §6	Derived (each)
Under deterministic π^*: TV and KL-coordinate bounds are equivalent via BH identity; Pinsker strictly weaker	§4 BH identity + §4.1 Pinsker comparison	Proved (scope-dependent tightness ordering)
Reverse-KL is the Bregman divergence of negative-entropy on Δ^{n-1}; log-odds is the Fenchel-dual natural coordinate (Legendre-Fenchel pair)	Standard Legendre-Fenchel (Rockafellar 1970 §26; Amari & Nagaoka 2000 §3.5; Bregman 1967) applied to $\phi(Q) = \sum_a Q_a \log Q_a$	Proved (textbook identification; §6.3)
AAT's π^*-first KL direction is distinctive within the information-theoretic-MDP lineage (TP2011 / Rubin 2012 / Levine 2018 all put agent-first) and is *forced* (not inherited) by the regret-bound + BH-identity derivations	Literature positioning (§6.4) + derivation structure (§§4–5)	Discussion (positioning; direction forced by §§4–5 derivations)
PAC-Bayesian generalization bound: $\tilde{D}_{\mathrm{KL}}(\hat{V}_\pi \| V_\pi) \leq (I_\pi + \log(2m/\delta))/(m-1)$	Rubin-Shamir-Tishby 2012 Theorem 3	Derived (external theorem applied; independent operational motivation alongside regret-bound + chain-rule + Fenchel-dual)
$\beta_\Sigma \propto V_{\max}$ naturalization (square-root-KL form)	Direct identification from Pinsker bound	Derived (under square-root form)
β_Σ local interpretation (linear-KL form, segment-retained)	Differentiation of Pinsker bound at operating point	Derived (local only)
Linear-KL form retained over square-root form	IB-shape alignment with AC	Formulation choice
Vacuity regimes ($V_{\max} = \infty$, $Q_{\Sigma_t}(a^*) = 0$, stochastic π^*)	Direct analysis of the bound	Proved (boundary)
Tied-optimum extension	π^* uniform on $\mathcal{A}^*$; bound adapts directly	Derived
Softmax-smoothed π^* extension	Sketched in §9	Hypothesis (deferred)
Uniqueness under chain-rule axiom (§6.1)	Hobson 1969 / Csiszár 1991 Theorem 3 corollary and Theorem 5 / standard functional-equation argument per Aczél & Daróczy 1975, applied to the direction-forced family; axiom AAT-internally motivated (§6.1)	Derived (conditional)

The dividing line: the KL **direction** is forced by the regret-bound derivation (strong result). The specific reverse-KL **form** within the direction-forced f-divergence family is *derived under the chain-rule additivity axiom* (§6.1) — conditional on an axiom that AAT independently motivates as the divergence-level version of the additive-decomposition principle already in `#der-chain-confidence-decay`. The β_Σ naturalization is partial — available globally only under the square-root form, locally under the linear form retained for IB-shape alignment.

Epistemic Status.

Robust qualitative overall, with several specific claims at *exact* tier. The direction-forcing claim (§5) is derived under standard assumptions (bounded value, deterministic π^* on the canonical AAT scope). The **Bretagnolle-Huber identity under deterministic π^*** (§4, $D_{\mathrm{KL}}(\pi^*\|Q) = -\log(1 - \mathrm{TV})$) is *exact* by direct calculation, and yields the tight regret bound $R \le V_{\max}(1-e^{-D_{\mathrm{KL}}})$ with matching lower bound $\Delta_{\min}(1-e^{-D_{\mathrm{KL}}})$ — the segment's primary bound form. Pinsker (§4.1) is retained as the strictly-weaker general form, correct for stochastic-π^* extensions. The TV bound (§3) is textbook. The admissible-divergence family characterization (§6) is a truthful survey. The **chain-rule uniqueness theorem (§6.1)** upgrades the reverse-KL selection to "uniquely forced conditional on the chain-rule axiom"; the axiom is AAT-internally motivated as the divergence-level analog of `#der-chain-confidence-decay`. **Asymmetry-from-one-sidedness (§5)** provides a second independent argument for the direction-forcing side, not requiring the chain-rule axiom. **Bregman-Fenchel identification (§6.3)** is textbook Legendre-Fenchel; ties the segment's divergence coordinate to `#deriv-edge-update-natural-parameter`'s update coordinate as Fenchel-dual aspects of one exponential-family structure. Fisher-metric-at-second-order is explicitly identified as *not* a distinguishing axiom within the f-divergence family (Eguchi's theorem), closing the most-obvious alternative candidate uniqueness route as a no-go. **Information-theoretic-MDP lineage positioning (§6.4)** is *Discussion-grade* (positioning in the literature, not a derivation); its role is to place AAT's direction-distinctive form honestly within the TP2011 / Rubin 2012 / Levine 2018 lineage rather than overclaim equivalence with any of them. The Rubin 2012 Theorem 3 PAC-Bayesian bound (§6.4) is a fourth independent operational motivation for the KL-to-reference form. The β_Σ naturalization (§7) is a trade-off analysis, not a derivation. The extensions (§9) are sketches, not derivations.

Max attainable: *exact* for (a) "forward-KL is a vacuous bound on R under deterministic π^*" (direct calculation; §5); (b) "under deterministic π^*, $D_{\mathrm{KL}}(\pi^*\|Q) = -\log(1 - \mathrm{TV}(\pi^*, Q))$" (BH identity; §4); (c) "regret is Lipschitz-equivalent to $(1 - e^{-D_{\mathrm{KL}}})$ with constants $\Delta_{\min}$ / $V_{\max}$" (§4 matched lower bound); (d) "reverse-KL is the Bregman divergence of negative-entropy; log-odds is its Fenchel-dual natural coordinate" (standard Legendre-Fenchel; §6.3); (e) "reverse-KL is the unique chain-rule-respecting f-divergence in the direction-forced family" (Hobson 1969 / Csiszár 1991; §6.1); (f) "asymmetry is forced by regret's one-sidedness" (direct; §5). *Robust qualitative* for the overall strengthened characterization — the chain-rule uniqueness is conditional on the chain-rule axiom, which is a principled axiom choice (AAT-internally motivated) but remains an axiom rather than a consequence of prior AAT commitments. The information-theoretic-MDP lineage positioning (§6.4) is *Discussion-grade* and honest about where AAT sits in the broader lineage rather than claiming formal equivalence.

Discussion.

The convergence with active inference's reverse-KL. Active inference's expected free energy (Friston et al. 2017; Parr & Pezzulo 2022) also uses reverse-KL against a preferred distribution, with the direction arising from the variational-free-energy-gradient lineage. AAT derives the direction from a regret-bound argument internal to the decision-theoretic scope. The convergence (two distinct derivations reach the same direction) is worth noting but is not a uniqueness argument — it shows reverse-KL is *natural* in multiple frames. The lineage difference preserves AAT's distinctive content: AAT's derivation does not require VFE as master objective or preferences-as-priors (`#disc-ciy-unified-objective`, `#def-satisfaction-gap` for the contrast).

Why the regret bound is the right derivation, not Donsker-Varadhan. The Donsker-Varadhan variational representation

$$\log \mathbb{E}_Q[e^f] \ge \mathbb{E}_P[f] - D_{\mathrm{KL}}(P\|Q) \tag{U.20}$$

gives reverse-KL bounds for exponential-moment functionals, but requires V to have bounded exponential moments (a stronger-than-bounded assumption for unbounded V). The Pinsker-TV route used here requires only bounded value range and is closer to the AAT scope's natural assumptions. Donsker-Varadhan becomes relevant if unbounded-V extensions are attempted.

Relationship to standard policy-gradient regret bounds. The argument here is structurally identical to standard policy-gradient / online MDP regret bounds (Agarwal, Kakade, Lee, Mahajan 2021 "On the theory of policy gradient methods" Lemma 3.2; Even-Dar, Kakade, Mansour 2009 "Online Markov decision processes"). AAT's contribution is *re-using the standard bound as the derivation of the KL direction in the strategy-cost objective*, not inventing the bound. This is consistent with AAT's integration-over-invention character.

Position in the additive-coordinate-forcing pattern. The chain-rule uniqueness theorem of §6.1 is the *divergence-level* instance of an AAT-wide pattern in which logarithmic coordinates are uniquely forced via Cauchy's functional equation on an AAT-internally-motivated additivity axiom. The pattern has two other primary instances: the chain-layer identity (`#der-chain-confidence-decay`, the motivating anchor) and the update-layer log-odds uniqueness (V). `#disc-additive-coordinate-forcing` catalogs the pattern, states the 1-anchor-plus-2-theorem structure precisely, and documents Lyapunov and IB Lagrangian as adjacent family members that share the additive-decomposition shape without the AAT-internal forcing structure.

Appendix V

Derivation: Log-Odds as the Unique Additive-Evidence Parameterization for Edge Credences

The log-odds coordinate $\lambda_{ij} = \log(p_{ij}/(1-p_{ij}))$ is the unique parameterization (up to positive affine transformation) on which independent Bernoulli evidence updates edge credences additively, under an evidential-additivity axiom motivated as the update-level analog of `#der-chain-confidence-decay`'s chain-level additive log-confidence decomposition. This segment states the uniqueness theorem, derives it, and explains how it positions log-odds as the natural parameterization for AAT's continuous-gradient edge-update machinery.

This is the *update-level* instance of `#disc-additive-coordinate-forcing`'s meta-pattern — the third formal layer where AAT commits to a specific coordinate because an internal additivity axiom forces it through a classical uniqueness theorem. Two prior layers compose with it: **log-probability at the chain layer** (chain rule of probability — `#der-chain-confidence-decay`); **reverse-KL at the divergence layer** (chain-rule additivity axiom via Hobson-Csiszár-Shore-Johnson — `#deriv-strategy-cost-regret-bound`); **log-odds at the update layer** (evidential-additivity axiom via Aczél's Cauchy functional equation — this segment). The Fisher information metric at the metric layer (Čencov's uniqueness theorem under parameterization-invariance) rounds out a four-layer structure that all live on the same geometric object — the exponential-family Legendre-Fenchel duality. *AAT's coordinates are not chosen by convention; they are forced by AAT-internal axioms via classical uniqueness theorems, and they all live on the same geometric object.*

The framework lifts this to a first-class coordinate commitment at the update layer. Continuous-gradient edge-update machinery in the credit-assignment treatment operates natively in log-odds and projects back to probability space via the sigmoid at the readout interface — the reason is mechanical: in probability space, the same update written as residual-times-gain can push the credence outside $[0, 1]$ when the Jacobian norm is small (a mechanical break of the probability domain). The log-odds coordinate has domain over all of $\mathbb{R}$, so additive updates never escape the domain; the sigmoid projection guarantees boundedness by construction, not by clipping. A downstream load-bearing consequence: the log-odds coordinate is what makes the detection-latency floor a structural rather than parametric result — the expected cycles to detect a regime change of given footprint scale linearly with accumulated experience precisely because the log-odds accumulation rate is forced (`#deriv-update-detection-latency`).

Formal Expression.

Setup. Let $p \in (0,1)$ denote a scalar Bernoulli credence (the probability that a proposition is true) and let $\psi : (0,1) \to \mathbb{R}$ be a smooth, strictly monotone reparameterization. Consider a sequence of independent Bernoulli observations $y_1, \ldots, y_n \in \{0,1\}$ drawn from a channel with likelihood ratio $P(y \mid H_1)/P(y \mid H_0)$.

Evidential-additivity axiom. The posterior update, applied to a single observation y, takes the form

Assumption (evidential-additivity axiom)

$$\psi(p_{\text{post}}) = \psi(p_{\text{prior}}) + g(y) \tag{V.1}$$

for some function $g : \{0,1\} \to \mathbb{R}$ that depends only on the observation y — not on p_{prior} nor on observation history.

Derived (evidential-additivity uniqueness of log-odds, conditional on the axiom above)

Theorem.

Theorem. The functional equation above admits solutions if and only if

$$\psi(p) = c \cdot \log\frac{p}{1-p} + d \tag{V.2}$$

for constants $c > 0$ and $d \in \mathbb{R}$, with $g(y) = c \cdot \ell(y)$ where $\ell(y) = \log[P(y \mid H_1)/P(y \mid H_0)]$ is the log-likelihood ratio.

Derived (Proof Step: Bayesian form of the update)

Derivation.

By Bayes' theorem applied to binary hypotheses,

$$\frac{p_{\text{post}}}{1-p_{\text{post}}} = \frac{p_{\text{prior}}}{1-p_{\text{prior}}} \cdot \frac{P(y \mid H_1)}{P(y \mid H_0)} \tag{V.3}$$

Taking the logarithm of both sides and writing $h(p) := \log(p/(1-p))$,

$$h(p_{\text{post}}) = h(p_{\text{prior}}) + \ell(y) \tag{V.4}$$

So $\psi = h$ trivially satisfies the axiom with $g = \ell$.

Derived (Proof Step: uniqueness by Cauchy functional equation)

Suppose ψ is any smooth, strictly monotone reparameterization satisfying the axiom. Since the Bayesian mapping $p_{\text{prior}} \mapsto p_{\text{post}}$ is fully determined by y through the likelihood ratio, the difference $\psi(p_{\text{post}}) - \psi(p_{\text{prior}})$ depends only on y, and by the axiom must equal $g(y)$.

Change variables via $\lambda = h(p) = \log(p/(1-p))$ and define $\Psi(\lambda) := \psi(\sigma(\lambda))$ where $\sigma(\lambda) = 1/(1+e^{-\lambda})$ is the logistic sigmoid. The axiom becomes

$$\Psi(\lambda + \ell(y)) - \Psi(\lambda) = g(y) \quad \text{for all } \lambda \in \mathbb{R},\ y \in \{0,1\} \tag{V.5}$$

Extending to continuous-valued evidence (or considering mixtures of Bernoulli channels with varying likelihood ratios, which span all of $\mathbb{R}$ in the ℓ-value space), the identity

$$\Psi(\lambda + \ell) - \Psi(\lambda) = G(\ell) \quad \text{for all } \lambda, \ell \in \mathbb{R} \tag{V.6}$$

holds for a function G independent of λ. This is the Cauchy functional equation (translation-additivity). Combined with the smoothness assumption on ψ, the unique solution class is $\Psi(\lambda) = c \cdot \lambda + d$ for constants c and d (Aczél 1966, *Lectures on Functional Equations and Their Applications*, §2.1).

Strict monotonicity of ψ forces $c \neq 0$. Taking ψ to have the same monotonicity sense as $p \mapsto p$ (credence increasing with ψ), we need $c > 0$. Thus $\psi(p) = c \cdot h(p) + d = c \cdot \log(p/(1-p)) + d$, and $g(y) = c \cdot \ell(y)$.

Derived (Proof Step: determining the constants)

This completes the proof. □

Three-Layer Parallel.

Discussion (three-layer additive decomposition)

The evidential-additivity axiom is the update-level instance of an additive-decomposition principle that AAT has already committed to at two prior layers:

Table V.1

Layer	*Quantity decomposed*	*Decomposition form*	*Source*
Chain level	Confidence along a causal chain	$\log P(\text{chain}) = \sum_i \log P(E_i \mid E_{<i})$	`#der-chain-confidence-decay`
Divergence level	Mismatch between optimal and strategy policies	$D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})$ decomposes additively across DAG layers along the optimal trajectory	U §6.1
Update level	Credence evolution under independent evidence	$\psi(p_{\text{post}}) - \psi(p_{\text{prior}}) = g(y_1) + \cdots + g(y_n)$ for n observations, with $\psi =$ log-odds	This segment

Each layer forces a logarithmic coordinate through essentially the same structural move: products of independent factors become sums on a log scale. At the chain level, $p^n \to n \log p$; at the divergence level, $\prod Q \to \sum \log Q$; at the update level, $\prod \mathrm{LR} \to \sum \log \mathrm{LR}$. The three are the same transform applied to different quantities.

Interpretation for the Edge-Update Machinery.

Discussion (operational consequence)

For edge credence p_{ij} with log-odds $\lambda_{ij} = \log(p_{ij}/(1-p_{ij}))$, the Bayesian update under independent Bernoulli evidence is

$$\lambda_{ij}^{\text{post}} = \lambda_{ij}^{\text{prior}} + \ell(y) \tag{V.7}$$

where $\ell(y)$ is the per-observation log-likelihood ratio.

Two operational consequences that follow from the uniqueness theorem:

1. **Domain unboundedness.** The log-odds coordinate has domain $\mathbb{R}$, not $[0,1]$. Additive updates cannot escape the domain, regardless of update magnitude. The probability-space presentation $p_{ij} \in [0,1]$ is the projected image of the log-odds coordinate, obtained via $p_{ij} = \sigma(\lambda_{ij})$ at the readout interface.
2. **Invariance under the chain of causal reasoning.** Because the log-odds coordinate is the unique additive evidence coordinate, evidence accumulated along one edge in a strategy DAG can be composed with evidence accumulated along another edge by addition in the log-odds vector space, provided the evidence is conditionally independent. The Beta-Bernoulli moment-parameter form $\hat{p} = \alpha/(\alpha+\beta)$ is the projected image, where α, β are the cumulative sufficient statistics in exponential-family form.

These consequences are why the continuous-gradient edge-update machinery in `#disc-credit-assignment-boundary` is well-posed globally in log-odds but exhibits a mechanical break (unbounded updates pushing credences outside $[0,1]$) when stated directly in probability space.

Scope (evidential-additivity scope)

Scope Condition.

The evidential-additivity axiom applies to agent classes that treat observations as independent Bernoulli likelihood evidence — the Bayesian-coherent sub-scope of AAT. Non-Bayesian agents (PID controllers, rule-based systems, human judgment per `#emp-update-gain`) do not invoke likelihood-ratio accumulation and are outside the axiom's scope. This matches the sub-scope α / sub-scope β partition in `#der-gain-sector-bridge`: the uniqueness applies within sub-scope α, where B1 (directional fidelity) is already derived from Bayesian coherence.

For multinomial / categorical edge credences with $K > 2$ outcomes, the analog is softmax / canonical exponential-family parameters: the softmax natural parameters $\eta_k = \log \pi_k$ (up to a reference-class shift) satisfy the same evidential-additivity axiom. The Bernoulli case ($K = 2$) collapses to log-odds.

Epistemic Status.

Derived (conditional on the evidential-additivity axiom). The uniqueness theorem is a standard functional-equation result (Cauchy functional equation with smoothness; Aczél 1966 §2.1). The AAT-internal motivation of the axiom is structural: it is the update-level analog of `#der-chain-confidence-decay`'s chain-level additive log-confidence decomposition and U §6.1's divergence-level chain-rule additivity. Without the axiom, the selection of log-odds weakens to canonical-not-unique on convergent grounds (exponential-family naturalness, Fisher-information / natural-gradient canonicity, domain-well-posedness of the continuous gradient).

Max attainable: *exact* for the Cauchy functional equation step (standard); *derived-conditional* for the overall uniqueness claim because the axiom, though AAT-internally motivated, is itself a commitment rather than a consequence of prior AAT commitments. The axiom's status is parallel to the chain-rule additivity axiom in U §6.1 — both are AAT-internally motivated as the "right level" instances of a single additive-decomposition principle.

Not uniqueness in the unconditional sense. The theorem states that *if* credence updates live on a single additive coordinate whose increment depends only on the observation, *then* that coordinate is log-odds up to positive affine. It does not rule out non-additive update schemes, nor schemes that live on multiple coordinates (e.g., Beta-Bernoulli with sufficient statistics (α, β) rather than a single credence coordinate). The Beta-Bernoulli presentation is *equivalent* in content to the log-odds presentation under a change of coordinates; the uniqueness is about what single-coordinate additive form is possible.

Discussion.

Why log-odds is forced, not merely canonical. Three convergent grounds — exponential-family canonicity, Fisher / natural-gradient canonicity, and sector-condition preservation — establish that log-odds reparameterization is a natural coordinate for edge updates, but none of them is a uniqueness theorem. This segment supplies the uniqueness result conditional on the evidential-additivity axiom. The axiom is AAT-internally motivated, so the result is genuine strengthening — not "log-odds happens to work" but "log-odds is forced by the update-level analog of a principle AAT already relies on at chain and divergence levels."

Why this matters for the mechanical break. The mechanical break in `#disc-credit-assignment-boundary`'s default signal function — unbounded gradient updates pushing credences outside $[0, 1]$ when $\|\mathbf{J}\|^2 \to 0$ — is a presentation artifact of the probability-space coordinate. In log-odds, the domain is $\mathbb{R}$ and the update is well-posed globally. The fix is not "clip the update" but "present the update in its native additive coordinate."

Relationship to Bayesian conjugate analysis. The Beta-Bernoulli conjugate update produces posterior hyperparameters $(\alpha + y, \beta + 1 - y)$ that are additive in the sufficient statistics, giving the point estimate $\hat{p} = \alpha/(\alpha + \beta)$ that updates *non-additively* in probability space. The content is the same — the log-odds coordinate makes the additivity manifest at the level of credence itself, while Beta-Bernoulli shows the same additivity at the level of sufficient statistics. The two are dual presentations of the same exponential-family structure.

Non-uniqueness outside the Bayesian sub-scope. For non-Bayesian agents in sub-scope β of `#der-gain-sector-bridge` (PID controllers, rule-based systems, human judgment), the evidential-additivity axiom does not apply. Such agents may update credences via coordinate-free heuristics (proportional blame, threshold rules) that do not decompose into likelihood-ratio addition. The uniqueness of log-odds is therefore scope-conditional, not universal.

Path B's role in the overall pattern. The segment makes visible that AAT is committed to a family of consistent coordinate-forcing moves at chain, divergence, update, and metric layers, each forcing a specific coordinate via a uniqueness theorem on an AAT-internally-motivated axiom. Future coordinate-forcing structures in AAT (e.g., any novel decomposition along trajectories, across channels, or between agents) should be examined for whether Cauchy-FE on an additivity axiom or Čencov-invariance on a parameterization-invariance axiom applies. If yes, log-odds / natural-parameter / reverse-KL / Fisher-metric-style results compose cleanly. If no, the new layer is a genuine departure requiring its own analysis. The pattern is catalogued at the meta-pattern level in `#disc-additive-coordinate-forcing`, including the 1-anchor-plus-3-theorem characterization (the chain layer is a mathematical identity; divergence, update, and metric are theorems conditional on AAT-internally-motivated axioms) and the adjacent-family classification (Lyapunov quadratic and IB Lagrangian share the additive shape but not the AAT-internal forcing structure).

Block-structured evidential additivity under L1' correlated evidence. When sibling edges share a latent common cause C (L1' correlated evidence per `#def-strategy-dag`'s Correlation Hierarchy), the independent-evidence axiom grounding Path B is literally false. Under *observable* C, the likelihood factorizes per-cluster as $P(\mathbf{y}, c) = P(c) \prod_k P(y_k \mid c)$, and the per-factor Aczél-1966 uniqueness argument applies independently to each factor. The forced coordinate becomes a $(2K+1)$-dimensional vector log-odds (one coordinate for θ_C, two per child edge for $p_{j|C}$ and $p_{j|\neg C}$), reducing to the scalar log-odds when the cluster is a singleton and no latent is present. This is a **generalization-in-scope** of the theorem above, not a new primary instance of `#disc-additive-coordinate-forcing`: the forcing machinery is unchanged (Cauchy-FE on per-factor additivity); the scope widens to block-factorized likelihoods under observable latents. Under *unobservable* C, the soft-EM responsibility-reweighted posterior depends nonlinearly on the prior, so no smooth ψ satisfies the block-additivity axiom — the axiom is structurally inconsistent with Bayesian mixture updates. This is the same scope boundary that `#disc-identifiability-floor` Instance 2 names via the Cramér-Rao Fisher-rank-1 obstruction; two independent analytical routes (Cauchy-FE failure at update layer; Cramér-Rao rank deficiency at observation layer) converge on the same unobservable-C structural floor, strengthening the "structural, not analytical-artifact" reading of Instance 2.

Findings.

Log-Odds as Uniquely-Forced Edge-Update Coordinate. **Brief:** When updating beliefs about whether something is true, there's a question of "what's the natural number to update?" — the probability itself, or some transformation of it. This result identifies a particular transformation (the log-odds, also used in logistic regression and Bayesian inference broadly) as the *uniquely* natural one for agents updating on independent evidence, in the precise sense that any equivalent representation must be log-odds up to scale. The choice isn't aesthetic; it's forced by what "independent evidence should add up" means mathematically. The uniqueness follows from Cauchy's functional equation operating on an evidential-additivity axiom motivated as the update-level analog of the chain-layer log-additive identity in `#der-chain-confidence-decay`.

Impact: Promotes a representational choice that previously had only convergent grounds (exponential-family canonicity, Fisher-information naturalness, sector-condition preservation) to a uniqueness theorem under an explicit AAT-internal axiom, joining the additive-coordinate-forcing meta-pattern (`#disc-additive-coordinate-forcing`) as one of its three theorem-grade instances. The result resolves a mechanical issue in the credit-assignment default signal function (`#disc-credit-assignment-boundary`), where unbounded gradient updates pushing credences outside [0,1] in the probability-space presentation become globally well-posed in the log-odds presentation (domain $\mathbb{R}$). It also positions log-odds as the coordinate against which downstream gain-and-update machinery should be analyzed for sub-scope α (Bayesian-coherent) agents.

Novelty Claim: *Claim differentiation* on an already-canonical representational choice (log-odds as the natural Bayesian-update coordinate, well-known from logistic regression / exponential-family / information-geometry traditions) by deriving its uniqueness under an AAT-internally-motivated evidential-additivity axiom. The Cauchy-FE machinery is classical; the AAT-internal axiomatization and the meta-pattern membership are the contribution.

Related Work:

- Aczél 1966, *Lectures on Functional Equations and Their Applications* §2.1 — *formal antecedent* — Cauchy-FE uniqueness machinery directly adopted in the derivation; cited inline.
- Cox 1946, "Probability, frequency and reasonable expectation" *Am. J. Phys.* 14:1–13; Jaynes 2003, *Probability Theory: The Logic of Science* — *conceptual precursor* — log-odds and additive-evidence accumulation as the natural form for Bayesian update; folk knowledge in this tradition.
- Amari & Nagaoka 2000, *Methods of Information Geometry* §2 — *formal antecedent* — log-odds as the natural parameter in the Bernoulli exponential family; the canonical-natural-parameter framing.
- Bishop 2006, *Pattern Recognition and Machine Learning* §4.2; Murphy 2012, *Machine Learning: A Probabilistic Perspective* — *adjacent literature* — standard statistical-learning treatments of log-odds and logistic regression; representational standard.

Search Log: - 2026-04 (*intuition-only* on AAT-internal axiomatization angle): no targeted search for prior derivations of log-odds via the specific evidential-additivity axiom (update-level analog of chain-rule additivity) has been conducted. Pre-search expectation: the Cauchy-FE move is classical (Aczél 1966); the log-odds canonical parameter is exponential-family standard; the *AAT-internal axiomatization* (motivating the additivity axiom as the update-level analog of `#der-chain-confidence-decay`) is the contribution being claimed. A targeted search would query the Bayesian-coherence and information-geometry literatures specifically for prior axiomatic derivations of log-odds via Cauchy-FE in agent-update settings. - 2025 (*targeted*): Aczél 1966 confirmed as the formal antecedent for the Cauchy-FE machinery; cited inline.

Appendix W

Derivation: Adaptive-Gain Dynamics — A2' Under a Learning Gain

AAT's gain structure (`#emp-update-gain`, `#der-gain-sector-bridge`) derives the optimal gain η^* per regime — the gain is a function of the noise model, chosen to minimize one-step mismatch variance. Real adaptive agents often *learn the noise model itself* (adaptive Kalman), *switch regimes* (IMM), *adapt step-size online* (RMSProp / Adam), or *optimize gain across tasks* (MAML). The gain becomes a state variable with its own update dynamics. The question is whether the sector-persistence machinery extends to this case, and where inside the A2' sub-scope partition adaptive gain sits. The result: sub-scope α splits into α_1 (fixed-gain per Prop B.3) and α_2 (adaptive-gain under four derivable conditions named (MG-1)–(MG-4)), with sub-scope β catching the rest. The two-timescale argument is an augmented-state Lyapunov composition (standard Khalil Thm 4.18), not a Tikhonov reduction — the primary and meta-gain sector conditions compose rather than one being eliminated.

Four worked cases land the partition. **Adaptive Kalman** with a Mehra-type innovation-based estimator under window-length timescale separation derives at sub-scope α_2 from (MG-1)–(MG-4); under Mehra non-identifiability (rank-deficient transform matrix; Zagrobelny-Rawlings 2015, Dunik et al. 2021), (MG-2) fails structurally — a candidate instance of `#disc-identifiability-floor` on the meta-gain channel. **RMSProp** on strongly-convex losses is α_2 under design conditions (large β, coupling smallness); outside those conditions, vanilla Adam's (MG-1) fails (Reddi et al. 2018 counterexample), and **AMSGrad's monotonicity constraint on v_t is structurally a meta-gain repair that restores (MG-1) by construction**. **IMM regime transitions** sit at α_2 between transitions and β across the posterior-reconcentration window — explicit dwell-time machinery is required because (MG-1) fails uniformly in time during the transition. **MAML** has α_1 inner loops and a β outer loop, since the meta-loss is non-convex even under per-task convexity (Fallah et al. 2020) and (MG-2) is not derivable from per-task structure. Each placement is a derivation, not folklore: the choice between gain mechanisms has structural consequences for which sub-scope the agent's persistence guarantee lives in, and (through the persistence-information-rate bound of `#deriv-persistence-cost`) for how much observation bandwidth the agent must consume to hold it.

Formal Expression.

Augmented-state setup. Treat the gain K_t as state. Define $\tilde{K}_t = K_t - K^*$ (error relative to a target optimal gain, specified per case: Riccati-steady-state for adaptive Kalman, EMA-fixed-point for RMSProp, etc.). The augmented state is $z_t = (\delta_t, \tilde{K}_t)$. Primary and meta-gain dynamics:

$$\dot{\delta} = -F(\delta; K^* + \tilde{K}) + w(t), \qquad \dot{\tilde{K}} = -\Phi(\tilde{K}, \delta) + v(t) \tag{W.1}$$

where F is the primary correction function (depending on the current gain via its argument), Φ is the gain-update contraction, w is primary-channel disturbance, v is gain-channel disturbance (estimator noise, innovation variability, etc.).

Formulation (meta-gain-conditions, extend A2' to adaptive-gain setting)

Meta-gain sector conditions (MG-1)–(MG-4).

(MG-1) Primary sector floor under bounded gain error. There exist $\underline{\alpha} > 0$ and $r_K > 0$ such that for all $\|\tilde{K}\| \le r_K$ and $\|\delta\| \le R$:

$$\delta^T F(\delta; K^* + \tilde{K}) \ge \underline{\alpha}\|\delta\|^2 \tag{W.2}$$

— A2' uniform in the gain-error ball. The sector floor is preserved across the gain-state range the meta-learner visits.

(MG-2) Meta-gain sector condition. The gain-update map Φ satisfies (T1) (zero at $\tilde{K} = 0$) and a local sector bound:

$$\tilde{K}^T \Phi(\tilde{K}, \delta) \ge \alpha_K \|\tilde{K}\|^2 \quad \text{for } \|\tilde{K}\| \le r_K, \text{ uniformly in } \|\delta\| \le R \tag{W.3}$$

with $\alpha_K > 0$. This is a sector condition in the gain-error state — the adaptive-gain analog of A2' itself.

(MG-3) Timescale separation. $\alpha_K \ll \underline{\alpha}$. The gain adapts slower than the primary state contracts. This is `#der-temporal-nesting`'s convergence constraint transcribed onto Lyapunov decay rates instead of event rates.

(MG-4) Coupling boundedness. The gain-channel disturbance $v(t)$ has bounded contribution from the primary state:

$$\mathbb{E}[\|v(t)\|^2 \mid \delta] \le \sigma_{K,0}^2 + c_v \|\delta\|^2 \tag{W.4}$$

for some $c_v \ge 0$. (MG-4) with $c_v = 0$ is clean two-timescale decoupling; $c_v > 0$ is δ-coupled meta-gain disturbance (RMSProp near minimizer), requiring fixed-point closure.

Derived (augmented-state-persistence, from sector-persistence-template applied twice with coupling)

Composed persistence result.

Under (MG-1)–(MG-4), the augmented state $z = (\delta, \tilde{K})$ is ultimately bounded in mean square. The Lyapunov candidate $V(z) = \frac{1}{2}\|\delta\|^2 + \frac{c}{2}\|\tilde{K}\|^2$ for appropriate weight c satisfies, along trajectories:

$$\dot{V} \le -\underline{\alpha}\|\delta\|^2 - c\alpha_K \|\tilde{K}\|^2 + \rho\|\delta\| + c(\sigma_{K,0} + \sqrt{c_v}\|\delta\|)\|\tilde{K}\| \tag{W.5}$$

Complete-the-square on the cross term (requires $c\sqrt{c_v}$ small compared to $\underline{\alpha} \cdot c\alpha_K$, i.e., (MG-3) timescale separation plus (MG-4) coupling smallness):

$$\dot{V} \le -\tfrac{\underline{\alpha}}{2}\|\delta\|^2 - \tfrac{c\alpha_K}{2}\|\tilde{K}\|^2 + \frac{\rho^2}{2\underline{\alpha}} + \frac{c\sigma_{K,0}^2}{2\alpha_K} \tag{W.6}$$

Standard Lyapunov ultimate-boundedness (Khalil 2002 Thm 4.18) applied to the augmented state gives the composed persistence bound. Both δ and $\tilde{K}$ are ultimately bounded with explicit bounds in $(\underline{\alpha}, \alpha_K, \rho, \sigma_{K,0}, c_v)$. □

A2' sub-scope partition: α_1 / α_2 / β.

Formulation (sub-scope-refinement)

The adaptive-gain analysis refines `#form-sector-condition`'s A2' sub-scope partition into three tiers:

Sub-scope α_1 — fixed-gain, A2' derived. `#der-gain-sector-bridge` Prop B.3's current scope: the gain K is treated as a static function of fixed noise model parameters. A2' is derived from B1 directional fidelity. Covers Kalman with known (Q, R), conjugate-Bayesian updates, exponential-family MLE, linear correction with PD KH, strongly-convex-gradient fixed-step-size.

Sub-scope α_2 — adaptive-gain, A2' derived through augmented-state Lyapunov. When (MG-1)–(MG-4) hold with all four conditions derivable from the update-rule structure:

- Adaptive Kalman with Mehra-type innovation-based (Q, R) estimator under timescale separation
- RMSProp/Adam with strongly-convex loss, large β (slow EMA), and coupling-smallness; AMSGrad fix (Reddi et al. 2018) structurally a meta-gain repair preserving (MG-1)
- Any adaptive-gain scheme admitting a clean (MG-1)–(MG-4) derivation from the update rule

For these, A2' is derived at the augmented-state level, not merely assumed. Setting $\tilde{K} \equiv 0$ recovers sub-scope α_1 cleanly.

Sub-scope β — A2' assumed (possibly under scope narrowing). When any of (MG-1)–(MG-4) must be assumed per-agent rather than derived. Concrete instances:

- MAML outer loop: meta-loss non-convex even under per-task convexity (Fallah et al. 2020), so (MG-2) cannot be derived from per-task structure. Inner loop is α_1; outer loop is β.
- IMM regime transitions: (MG-1) fails uniformly in time during posterior-reconcentration windows of duration τ_{IMM}. Between-transition regime is α_2; across-transition window is β (scope narrowing via dwell-time + impulsive-disturbance absorption).
- Adam without AMSGrad on ill-conditioned problems: (MG-1) fails under aggressive-β + small-gradient-noise (Reddi et al. 2018 counterexample).
- Rule-based / PID / human-judgment adaptive gains: no structural argument for either (MG-1) or (MG-2); both must be assumed.

This refinement preserves the existing A2' sub-scope α as a specialization (α_1) and identifies a derivable adaptive-gain extension (α_2) with honest fallback to β when the derivability conditions fail.

Structured cases.

Derivation (case-adaptive-kalman-alpha2)

Case A — Adaptive Kalman with Mehra estimator. Scalar linear-Gaussian setting with unknown (Q^*, R^*). The innovation-based Mehra estimator (Mehra 1970, 1972; Dunik et al. 2021 for identifiability) yields $\hat{Q}_t, \hat{R}_t$ from a sliding-window autocorrelation of the innovation sequence. For window length N, the gain-update map to first order in $\tilde{K}$ is a scalar Ornstein-Uhlenbeck process:

$$\tilde{K}_{t+1} = (1 - \lambda_N)\tilde{K}_t + \lambda_N \eta_t^{\text{inn}} \tag{W.7}$$

with $\lambda_N \asymp 1/N$ (contraction rate set by window length) and η^{inn} zero-mean innovation noise with variance $O(1/N)$. This is itself an instance of G with:

- (T1): $\Phi_K(0) = 0$ — when gain is optimal, estimator returns optimal in expectation.
- (T2): $\tilde{K} \cdot \Phi_K(\tilde{K}) = \lambda_N \tilde{K}^2$, so $\alpha_K = \lambda_N$.
- (T3-S): bounded stochastic disturbance on the gain channel.

Prop A.1S applied to the meta-gain channel gives $R^*_{S,K} \asymp 1/\sqrt{N}$ — the classical Mehra asymptotic rate, now derived from (MG-2). Primary sector floor is preserved: $\underline{\alpha} = K^* - |\tilde{K}|_{\max}$ via triangle. Composed persistence via augmented-state Lyapunov gives $O(1/\sqrt{N})$ degradation from the fixed-gain case. **This case is derived at sub-scope α_2.**

Under Mehra non-identifiability (rank-deficient transform matrix; see Zagrobelny-Rawlings 2015, Dunik et al. 2021), (MG-2) fails structurally — an instance of the `#disc-identifiability-floor` pattern on the meta-gain channel.

Derivation (case-rmsprop-alpha2-conditional)

Case B — RMSProp on strongly-convex loss. The per-step effective step is $\eta_t^{\text{eff}} = \eta_t/(\sqrt{v_t} + \varepsilon)$ where $v_t = \beta v_{t-1} + (1-\beta)\hat{g}_t^2$ tracks the second moment. Near the minimizer, $\mathbb{E}[\hat{g}_t^2] \to \|\nabla L\|^2 + \sigma_g^2$. Writing $\tilde{v}_t = v_t - \mathbb{E}[\hat{g}_t^2]$:

$$\tilde{v}_{t+1} = \beta\tilde{v}_t + (1-\beta)(\hat{g}_t^2 - \mathbb{E}[\hat{g}_t^2]) + \beta(\mathbb{E}[\hat{g}_{t-1}^2] - \mathbb{E}[\hat{g}_t^2]) \tag{W.8}$$

The first two terms give (MG-2) with $\alpha_v = 1-\beta$ and $\sigma_v^2 \asymp (1-\beta)^2\text{Var}(\hat{g}_t^2)$. The third is δ-coupled: $\mathbb{E}[\hat{g}_t^2]$ depends on δ through $\|\nabla L\|^2$, giving $c_v > 0$ in (MG-4).

Composed persistence under design conditions β close to 1 (slow EMA) and $\lambda_{\max}(H) \cdot R_S^* \ll \sqrt{\sigma_g^2}$ (coupling smallness): fixed-point closure between primary R_S^* and meta-gain R_v yields existence of a stable equilibrium. Sub-scope α_2 under design conditions.

Outside those conditions (aggressive β + ill-conditioning + small gradient noise — Reddi et al. 2018's Adam counterexample), fixed-point iteration diverges. **AMSGrad's monotonicity on v_t is structurally a meta-gain repair that restores (MG-1) by construction** — preserving sub-scope α_2 by forcing a condition (MG-1) the vanilla algorithm would violate.

Sketch (case-imm-alpha2-plus-dwelltime)

Case C — IMM / regime-switching Kalman. Mixture of M Kalman filters with Markov-transition posterior over regimes. Between regime transitions, the posterior concentrates on the true regime and the effective gain approaches the regime-conditional optimum: sub-scope α_2 in the steady portion with Mehra-style derivation. Across regime transitions, posterior re-concentration takes $\tau_{\text{IMM}} \asymp 1/(1-p)$ observations (self-loop probability p); during this window (MG-1) fails uniformly in time — the gain can be aligned with the wrong regime. Scope narrowing via dwell-time + impulsive-disturbance absorption: regime stable for $T_{\text{dwell}} \gg \tau_{\text{IMM}}$, transient mismatch bounded by counting-argument. Sub-scope α_2 between-transition; β across-transition.

Classification (case-maml-mixed)

Case D — MAML inner-outer structure. Inner loop (per-task adaptation with k gradient steps) — sub-scope α_1 from Prop B.4 under per-task convexity. Outer loop (meta-parameter update via gradient on meta-loss $\sum_i L_i(\theta_i'(\theta))$) — Fallah et al. 2020's convergence analysis shows the meta-loss is non-convex even under per-task convexity because of the non-linearity of inner-loop updates in θ. (MG-2) is not derivable from per-task structure. Outer loop is β — A2' assumed per basin.

What Is Derived vs. What Is Chosen.

Table W.1

Property	*Source*	*Strength*
Augmented-state setup $z = (\delta, \tilde{K})$ with coupled sector dynamics	Definitional reformulation extending `#der-recursive-update` + `#emp-update-gain`	Formulation choice
(MG-1) primary sector floor under bounded gain error	Derived in Cases A and B via triangle / ε-floor arguments	Derived (per case, conditional on regularity)
(MG-2) meta-gain sector condition	Derived in Cases A (Mehra OU) and B (EMA second-moment) from estimator structure	Derived (per named case)

Property	*Source*	*Strength*
(MG-3) timescale separation $\alpha_K \ll \underline{\alpha}$	Design condition on estimator window / EMA rate; Lyapunov decay-rate transcription of `#der-temporal-nesting`	Formulation (design condition; violations dissolve composition)
(MG-4) coupling boundedness of gain-channel disturbance	Decoupled ($c_v = 0$) in Case A; δ-coupled ($c_v > 0$) in Case B with fixed-point closure	Derived (per case)
Composed augmented-state persistence under (MG-1)–(MG-4)	Quadratic Lyapunov on z + Khalil 2002 Thm 4.18 ultimate-boundedness	Derived (conditional on MG-1 through MG-4)
Sub-scope refinement α_1 / α_2 / β	Extension of current A α/β partition	Formulation (classification)
α_2 reduces to α_1 under $\tilde{K} \equiv 0$	Direct substitution into augmented-state setup	Proved
Case A (adaptive Kalman): sub-scope α_2 under identifiability + window-length timescale separation	Mehra OU-form of estimator + Prop A.1S	Derived
Case B (RMSProp): sub-scope α_2 under design conditions; β otherwise; AMSGrad as meta-gain repair	EMA second-moment derivation + fixed-point closure + Reddi et al. 2018 counterexample	Derived (conditional); AMSGrad framing is discussion
Case C (IMM): sub-scope α_2 between-transition + β across-transition via dwell-time	Posterior re-concentration + impulsive-disturbance absorption	Sketch
Case D (MAML): inner-loop α_1 + outer-loop β	Fallah et al. 2020 meta-loss non-convexity	Classification (not derivation)
Mehra non-identifiability as meta-gain `#disc-identifiability-floor` instance	Rank-deficient transform matrix blocks (MG-2) derivation	Discussion (candidate floor instance)

Epistemic Status.

Conditional. Max attainable: *derived* for the composed persistence result under (MG-1)–(MG-4); *exact* for Case A (adaptive Kalman) under Mehra identifiability; *conditional* for Case B (RMSProp) under design conditions; *sketch / classification* for Cases C (IMM) and D (MAML).

The core claim — that (MG-1)–(MG-4) together give an augmented-state sector-persistence result via standard Khalil Thm 4.18 — is a clean application of two-timescale Lyapunov analysis to the gain-as-state formulation. The mathematical core is textbook (vector Lyapunov, augmented-state dissipation, completing-the-square on cross terms). The contribution is the AAT-framing: (i) naming the four conditions that make adaptive gain composable with sector-persistence, (ii) identifying which adaptive-gain schemes satisfy all four from update-rule structure (Cases A, B-design) and which require assumption (Cases C-transition, D-outer, rule-based), and (iii) refining the A2' sub-scope partition into $\alpha_1/\alpha_2/\beta$ as a principled extension rather than overhaul.

Imported machinery, acknowledged. The underlying techniques — two-timescale stochastic approximation (Borkar 1997, 2008), adaptive filtering (Mehra 1970–72; Mohamed-Schwarz 1999; Dunik et al. 2021), MRAC stability theory (Narendra-Annaswamy 1989; Ioannou-Sun 1996), non-convex meta-learning convergence (Fallah et al. 2020; Ji-Yang-Liang 2022) — are standard in control theory and ML. AAT's value-add here is the integration: connecting these techniques to G via the augmented-state Lyapunov composition, and producing a sub-scope partition that matches AAT's scope-honesty architecture rather than importing the techniques' native terminology. The existing sub-scope α / β partition from A becomes α_1 / α_2 / β, with α_2 labeling the newly-derivable adaptive-gain region.

What this does not establish:

- Convergence rates beyond ultimate-boundedness (the bound gives stability, not speed).
- Behavior under (MG-3) violation — fast-adapting gain on a slow primary state — which is structurally different (the composite system becomes singular-perturbation pathological, not sector-persistent).

- Rigorous fixed-point closure for the Case-B δ-coupled case; the qualitative argument is clear but quantitative bounds on when the closure exists are per-problem.
- Composition at the multi-agent level — the augmented-state Lyapunov is for a single adaptive agent; team adaptive-gain (simultaneous meta-learning across agents) is beyond scope.

Discussion.

Resolving Epistemic Opacity. When the observation law is opaque (`#def-observation-function`), the agent does not know the true noise distribution of ε_t. Yet the optimal gain `#emp-update-gain` requires knowing U_o. The adaptive Kalman filter with an innovation-based estimator (e.g., Mehra 1970) resolves this. The agent computes the autocorrelation of its observable mismatch sequence (innovations): $\iota_t = o_t - \hat{o}_t$. Because the innovation statistics are observable, the agent can estimate U_o (and U_M) purely from its own history $\mathcal{C}_t$, so the optimality requirement is met without being handed the observation law. The error in this estimation ($\tilde{K}$) is exactly what the meta-gain sector condition (MG-2) bounds.

Why this is not a singular-perturbation reduction. Singular perturbation theory (Tikhonov 1952; Khalil 2002 Ch. 11) replaces fast dynamics with their quasi-steady-state manifold — the fast variable is *eliminated* in the slow reduction. The augmented-state analysis here does something different: it *keeps both states* and bounds their joint persistence via a weighted Lyapunov. Tikhonov gives "the fast variable tracks its slow manifold up to $O(\epsilon)$"; template-composition gives "each level's state is ultimately bounded with coupling-amplified disturbance" — the persistence-flavored statement aligned with AAT's other persistence results. They answer different questions; both are useful; neither replaces the other. N's stacking theorem (promoted 2026-06-10) is in fact stated in the template-composition register this paragraph describes — each level a template instance with coupling-amplified effective disturbance, both states kept, joint forward-invariance — with the quasi-steady-state manifold used only as the anchoring coordinate, not as an elimination step. The two segments now share the framing; what differs is the axis (gain-parameter timescale here, adaptive-level stacking there).

Three timescale patterns in AAT. N distinguishes model-class (slow) and parametric (fast) timescales. This segment introduces a *third* pattern: gain-parameter timescale, sitting between parametric-correction (fastest) and model-class (slowest). Adaptive gain is a different two-timescale pattern from structural adaptation — the gain is a parameter of the update rule, not of the model class. The convergence constraint $\nu_{n+1} \ll \nu_n$ from `#der-temporal-nesting` translates in this segment to $\alpha_K \ll \underline{\alpha}$ on Lyapunov decay rates, which is the (MG-3) condition.

Meta-gain scope-honesty failure modes. - *Identifiability failures* on the meta-gain channel (Mehra non-identifiability) are instances of the `#disc-identifiability-floor` pattern: a structural no-go (non-identifiability) on the meta-gain channel blocks the sub-scope α_2 derivation. AMSGrad as a meta-gain repair is the constructive complement — an added structural condition (monotonicity on v_t) that restores (MG-1) and keeps the adaptive scheme in α_2. - *Non-convex meta-losses* (MAML outer loop) put the outer loop in sub-scope β: A2' must be assumed within a basin, not derived from the meta-loss's global structure. This is the honest scope-narrowing posture — different from saying "it works" or "it doesn't" without qualification. - *Regime transitions* (IMM across switches) violate (MG-1) uniformly in time during posterior re-concentration windows. Repair via dwell-time is clean but significant scope narrowing.

Connection to `#schema-strategy-persistence`'s $\alpha_\Sigma = 1/(n+1)$. The strategy-edge persistence condition has α_Σ decaying with experience — this is adaptive gain of a specific form, where the gain (effective update rate per edge) is a function of accumulated pseudo-count. Without forgetting, $\alpha_\Sigma \to 0$ and (MG-1) fails asymptotically. `#schema-strategy-persistence`'s forgetting prerequisite $(1-\lambda) > \rho_\Sigma/R_\Sigma$ is precisely the condition that keeps strategy-edge adaptation in sub-scope α_2 rather than letting it fall to β as n grows.

Relation to catastrophic forgetting via EWC. Elastic Weight Consolidation (Kirkpatrick et al. 2017) introduces a per-parameter stability-weighted update: gain is scaled by inverse Fisher information of prior tasks. In AAT vocabulary, EWC is a tensor-valued adaptive gain with stability weighting, distinct from the scalar adaptive-gain case treated here

but fitting the same augmented-state framework with (MG-1)–(MG-4) reinterpreted per-parameter. Adapting the present derivation to EWC would tensorize the Lyapunov argument — a natural but not-yet-executed extension.

Appendix X

Derivation: Internal-External Decomposition of Agent Viability

Agent viability decomposes **additively** into agent-movable terms and an irreducible environmental floor — inherited from the exact mismatch decomposition — and provably does **not** factor multiplicatively into an agent-independent environmental rate times agent factors. The construction takes log-viability as the log-margin between steady-state mismatch and the critical task boundary, and inherits the additive split (estimation error + state-uncertainty floor + channel noise) from `#result-mismatch-decomposition`; the agent can move the estimation term by modeling and the state-uncertainty term by acting; the channel term it cannot move by modeling — it is a kernel of the channel, policy-movable only in expectation through which states and instruments the policy visits (Y).

The constitutive no-go is that multiplicative ρ-factorization of the form "$\rho = \rho_{\text{external}} \cdot f(\text{model-class}) \cdot g(\text{policy})$" is type-incorrect: mismatch is constitutively agent-relative, so ρ^2 carries an estimation term that *vanishes* as the agent's model improves — a contribution that subtracts to zero cannot be represented by attenuating a fixed agent-independent scalar by agent factors. Established exactly is the additive env/agent split with the environmental floor. The finer structure of the agent-side component is characterized in Y: there is no separate policy term — the policy enters every term, the floor included, only as the on-policy measure under which three policy-independent kernels are averaged — so the honest fine split is estimation / state-uncertainty / channel with policy-dependence in each, log-additive to first order when the agent-movable excess is small relative to the floor. Separately *identifying* those terms from on-policy data remains obstructed and requires Level-2 interventional access (`#disc-identifiability-floor` Instance 4).

Log-Viability.

Log-viability $\mathcal{V}$ is the margin by which steady-state mismatch R^* stays below the critical task boundary $\|\delta_{\text{critical}}\|$:

$$\mathcal{V} = \log \frac{\|\delta_{\text{critical}}\|}{R^*} = \log\|\delta_{\text{critical}}\| - \log R^* \tag{X.1}$$

Persistence holds exactly when $\mathcal{V} > 0$. Under linear Model D, $R^* = \rho/\alpha$ (`#result-persistence-condition`), so

$$\mathcal{V} = \log\|\delta_{\text{critical}}\| - \log\rho + \log\alpha \tag{X.2}$$

[Derived] — exact algebra (logarithm of a ratio); independent of how ρ itself is structured.

The disturbance is additive, not multiplicative.

The decomposition of viability into "what the environment imposes" versus "what the agent controls" runs through the structure of ρ. That structure is fixed by an existing exact result, not a free modeling choice.

The exact mismatch decomposition. By `#result-mismatch-decomposition` (`status: exact`, via the fresh-noise assumption GA-1), expected squared mismatch splits with vanishing cross-terms:

[Derived (exact, inherited from `#result-mismatch-decomposition` *GA-1)]*

$$\mathbb{E}[\|\delta_t\|^2] = \underbrace{\mathbb{E}[\|\hat{o}_t - \hat{o}_t^{\mathrm{B}}\|^2]}_{\text{estimation — reducible by modeling}} + \underbrace{\mathbb{E}[\mathrm{Var}(\bar{o}_t \mid \mathcal{C}_{t-1}, a_{t-1})]}_{\text{state uncertainty — movable by acting}} + \underbrace{\mathbb{E}[\mathrm{Var}(o_t \mid \Omega_t, a_{t-1})]}_{\text{channel noise — irreducible}} \tag{X.3}$$

All cross-terms are zero by orthogonality (not independence), exactly as that result establishes. The first two terms are **agent-side**: the estimation term shrinks as the model improves, vanishing at the Bayes predictor; the state-uncertainty floor binds every model — well-specified or not — and yields only to *acting* (more informative interaction history), never to better modeling alone. The third is the **irreducible environmental floor**: a property of the observation channel the agent cannot reduce. A boundary note, because the working record has drawn it in two places: "the environmental floor" can be placed at the channel term alone (as here — everything else is agent-movable, by model or by policy) or at the full Bayes floor (state uncertainty + channel — everything no *model* can remove); the two placements differ by exactly the state-uncertainty term. This segment uses the first placement because the viability split is about what the agent — model and policy together — can move.

Rate-lift (stated convention). The persistence condition uses the disturbance *rate* ρ, related to expected squared mismatch by a fluid-limit identification $\rho^2 = \nu \cdot \mathbb{E}[\|\delta_t\|^2]$ for a positive throughput factor ν:

Formulation — stated fluid-limit convention, introduced here; not a pre-existing canon identity. Derived (conditional on the stated rate-lift convention) The structural conclusion below is convention-robust: any $\nu > 0$ preserves additivity, since positive scaling of a sum is a sum.

The viability decomposition. Substituting the exact split gives

$$\mathcal{V} = \mathcal{V}_{\mathrm{env}} + \mathcal{V}_{\mathrm{agent}} \tag{X.4}$$

where the first term $\mathcal{V}_{\mathrm{env}}$ collects the channel-floor contribution (environmental affordance) and the second collects the estimation and state-uncertainty contributions together with the $\log \alpha$ tempo/gain terms (internal-operational health: the estimation part improves as the model improves; the state-uncertainty part as the agent's interaction history grows more informative).

No multiplicative factorization (constitutive no-go).

Derived — constitutive impossibility; exact under the constitutive definition of mismatch

There is no agent-independent scalar ρ_{external} with $\rho = \rho_{\mathrm{external}} \cdot f(\mathcal{M}) \cdot g(\pi)$. Mismatch is constitutively agent-relative ($\delta_t \equiv o_t - \hat{o}_t$, `#def-mismatch-signal`), so ρ^2 carries the estimation term, which **vanishes as the agent's model improves**. A multiplicative form attenuates a fixed agent-independent scalar by agent factors; it cannot represent a contribution that *subtracts to zero* under an agent-internal change. Multiplicative-in-rate and additive-in-(squared-mismatch) are type-incompatible, and $\sqrt{A+B} \neq \sqrt{A}\sqrt{B}$ blocks recovering the product form under the rate-lift. The decomposition is additive by construction, not by choice.

Epistemic Status.

The mismatch decomposition is **exact** (`#result-mismatch-decomposition`, under GA-1). The viability decomposition in ρ-units is **conditional on the stated fluid-limit rate-lift** — a modeling convention introduced in this segment, not a pre-existing canonical identity; the *additive structure* (versus multiplicative) is robust to it for any positive throughput factor. The constitutive no-go is exact given the definition of mismatch. A general (non-isotropic, state/policy-dependent) diffusion $\Sigma(\delta, t)$ preserves additivity by linearity of the Itô generator in the diffusion matrix — this is an elementary extension *beyond* A Prop A.1S's constant-isotropic $\sigma_w^2 I_n$ case, correct but not itself part of that result's `exact` scope.

Discussion.

The finer split: structure exact, identification obstructed. Decomposing $\mathcal{V}_{\text{agent}}$ further is not a model-vs-policy question. By Y, the estimation, state-uncertainty, and channel terms are expectations of policy-independent kernels under the on-policy law; the policy is the *measure*, not a term, so "policy benignity" is the policy concentrating the trajectory where the model class is adequate — one kernel modulated by one measure, entangled at the source rather than two independent factors — and the multiplicative intuition is recovered only as a first-order expansion in the excess-to-floor ratio, with the "factors" being floor-relative exponentials of the additive estimation and state-uncertainty terms. What *is* obstructed is identification: an on-policy observer sees only the sum, and separating the terms from observational data alone is the identifiability-floor phenomenon (`#disc-identifiability-floor` Instance 4, via AN) — in a feedback system high internal capacity itself lowers future environmental difficulty, and Level-2 interventional access (rotating agents across environments, or changing instrument / actions / model) is required to resolve it. This segment asserts the exact two-term env/agent split and the no-go; the attribution structure lives in Y and the identification floor in AN.

Why this matters. The two-term split is the honest form of the internal/external diagnostic the persistence condition invites: it identifies exactly what an agent can improve by modeling (estimation error) and by acting (state uncertainty, and the on-policy channel term) versus the kernel-level floor no model touches (channel noise), without the false promise that the environmental rate is an agent-independent quantity one can read off and attribute blame to.

Appendix Y

Derivation: Mismatch-Budget Attribution — Policy as Measure, First-Order Log-Additivity, and the Information Form

The agent-side mismatch budget has no separate policy term. Each of the three terms of the exact mismatch decomposition (`#result-mismatch-decomposition`) is the expectation of a policy-independent *kernel* — estimation error, state-uncertainty variance, channel variance, each a fixed function of the conditioning variables — taken under the **on-policy law** $\mathbb{P}_\pi$ of the agent-environment trajectory. The policy enters every term, the environmental floor included, only through that law. Three consequences follow directly. First, the "policy-benignity" that a multiplicative intuition treats as an independent attenuation factor $g(\pi)$ is a *reweighting* of where the estimation and state-uncertainty kernels are evaluated: a policy lowers the budget exactly when it concentrates the trajectory where the model class is adequate or the state is well-pinned, and cannot lower it at all when the kernels are state-independent (the linear-Gaussian / LQR case, where the innovation law is control-invariant). Model adequacy and policy are therefore entangled at the source — one kernel modulated by one measure — not two factors. Second, log-viability has an exact form in which the agent-movable excess enters as $-\frac{1}{2}\log(1+x)$ with x the excess-to-floor ratio; this is log-additive in the estimation and state-uncertainty terms to first order in x, with a remainder bounded by $x^2/2$. That first-order regime is the precise and only sense in which the multiplicative picture was approximately right, and its "factors" are floor-relative exponentials of additive terms, each still policy-dependent through the measure. Third, the same three-way structure holds **exactly** in information units — expected log-loss of the model's predictive distribution splits into channel entropy, state-uncertainty information, and an estimation divergence, by the chain rule of log-loss alone, with no flatness or orthogonality condition — and the estimation divergence refines further into a class-ceiling gap plus a within-class gap exactly when the predictive class is an exponential family (sub-scope α), by the generalized Pythagorean theorem.

Formal Expression.

Setting and the two conditions. Fix a time t. Write $\mathcal{C} := \mathcal{C}_{t-1}$ for the chronica, $a := a_{t-1}$ for the action taken, $\Omega := S_t = (\Omega_t, \theta)$ for the environment's typed state, its current state together with its law (`#scope-adaptive-system`; when the law is known this is just Ω_t, and the kernels below take their state-only form), and $o := o_t$ for the observation. The model's predictive mean is $\hat{o} = \mathbb{E}_q[o \mid M_{t-1}, a]$ with $M_{t-1} = m(\mathcal{C})$ a fixed (possibly learned) map of the chronica (`#form-agent-model`); the Bayes predictor is $\hat{o}^{\mathrm{B}} = \mathbb{E}[o \mid \mathcal{C}, a]$; the true conditional mean is $\bar{o} = \mathbb{E}[o \mid \Omega, a]$. Two conditions are in force throughout:

[Assumption (GA-I, fresh noise — inherited from `#result-mismatch-decomposition`*)]* — observation noise is conditionally independent of the chronica given (Ω, a), so $p(o \mid \Omega, a, \mathcal{C}) = p(o \mid \Omega, a)$.

[Assumption (P, chronica-measurable policy)] — the action depends on the environment only through the chronica: $a_s \sim \pi(\cdot \mid \mathcal{C}_s)$ (deterministic or randomized with internal randomness), and the environment's transition kernel $p(\Omega_{s+1} \mid \Omega_s, a_s)$ responds to *actions*, not to the policy object itself. This is AAT's standing agent-environment interface (`#def-action-transition`). Adversarial coupling in the sense of `#der-adversarial-destabilization` — the attacker's action stream amplifying the target's disturbance — is *inside* (P) and is handled by Corollary 1c (measure reweighting). What (P) excludes is a kernel of the form $T(\cdot \mid \Omega, a, \pi)$: an environment that reads the policy object itself (source-code or Stackelberg access), which AAT has not canonized.

Write $\mathbb{P}_\pi$ for the joint law of $(\mathcal{C}, a, \Omega, o)$ induced by the environment and the policy π, and $\mathbb{E}_\pi$ for expectation under it.

Derived (policy-as-measure), exact under GA-1 and (P)

Proposition 1 — the policy is the measure.

Define the three kernels

$$e(\mathcal{C}, a) := \|\hat{o}(\mathcal{C}, a) - \hat{o}^{\mathrm{B}}(\mathcal{C}, a)\|^2, \qquad u(\mathcal{C}, a) := \mathrm{Var}(\bar{o} \mid \mathcal{C}, a), \qquad c(\Omega, a) := \mathrm{Var}(o \mid \Omega, a). \tag{Y.1}$$

Then each kernel is a policy-independent function of its arguments, and the exact mismatch decomposition reads

$$\mathbb{E}_\pi\|\delta_t\|^2 = \mathbb{E}_\pi[e(\mathcal{C}, a)] + \mathbb{E}_\pi[u(\mathcal{C}, a)] + \mathbb{E}_\pi[c(\Omega, a)] =: E_\pi + U_\pi + C_\pi. \tag{Y.2}$$

The policy enters the budget only through $\mathbb{P}_\pi$.

Derivation. The identity is `#result-mismatch-decomposition`, written with its three conditional quantities displayed as functions of the conditioning variables. What remains is that the kernels themselves do not depend on π. (i) $c(\Omega, a)$ is a conditional variance given (Ω, a); under GA-1 the conditional law $p(o \mid \Omega, a)$ is a property of the observation channel alone. (ii) Under (P), the joint law factorizes as $\mathbb{P}_\pi(\mathcal{C}, \Omega_{0:t}) = \big[\prod_{s<t} \pi(a_s \mid \mathcal{C}_s)\big] \cdot [\text{environment and channel factors}]$, and the policy factors are constant in $\Omega_{0:t}$ once $\mathcal{C}$ is fixed. They therefore cancel from the normalization of the posterior $p(\Omega \mid \mathcal{C}, a)$, which is policy-independent (the belief state is a function of the action-observation history, standard in partially observed control). Hence $\hat{o}^{\mathrm{B}}$ and u — the mean and variance of $\bar{o}(\Omega, a)$ under that posterior — are policy-independent kernels. (iii) $\hat{o}$ is a fixed function of $(m(\mathcal{C}), a)$ by `#form-agent-model`, so e is a policy-independent kernel. □

Corollary 1a (reweighting). For two policies with $\mathbb{P}_{\pi'} \ll \mathbb{P}_\pi$ on the relevant coordinates, each term transforms by the likelihood ratio: $E_{\pi'} = \mathbb{E}_\pi\big[e \cdot \frac{d\mathbb{P}_{\pi'}}{d\mathbb{P}_\pi}\big]$, and likewise for U and C. A change of policy is a change of measure on fixed integrands — not a projection, not an attenuation, and not a fourth term.

Corollary 1b (invariance). If the three kernels are $\mathbb{P}_\pi$-a.s. constant across the policies under comparison, the budget is policy-invariant. In the linear-Gaussian sub-scope with a well-specified filter this holds exactly: $e \equiv 0, u \equiv HP^{-}H^{\top}$ (the steady-state prior covariance, which solves a Riccati recursion independent of the control input), and $c \equiv R$ — the innovation law is invariant to the control gain (standard LQG: the prediction covariance recursion $APA^{\top} + Q$ carries no control term; the LQR worked case is in the reasoning trail, `spikes/.integrated/spike-rho-factorization.md` §5). In the multiplicative vocabulary this is "$g(\pi) \equiv 1$"; in the present one it is simply that there is nothing state-dependent for the measure to reweight.

Corollary 1c (entanglement and the movable floor). A policy lowers E_π exactly when $e(\mathcal{C}, a)$ is state-dependent and the policy concentrates $\mathbb{P}_\pi$ where e is small — i.e., where the model class is adequate. A policy lowers U_π exactly when it makes the interaction history more state-informative (the active-sensing door of `#def-causal-information-yield`). And a policy moves C_π exactly when the channel variance depends on the action or state — as when the action selects the sensor (BB, modes L / H). "Irreducible" for the channel term therefore means *kernel-level*: no model touches $c(\Omega, a)$;

its on-policy expectation C_π is nonetheless policy-movable whenever the instrument or the visited states are. None of these is an independent factor: each is one kernel under one measure.

Proposition 2 — exact log form and first-order log-additivity. *[Derived (first-order-log-additivity), exact given the rate-lift convention of X]*

With $\rho^2 = \nu\,\mathbb{E}_\pi\|\delta_t\|^2$ and linear Model D ($R^* = \rho/\alpha$, `#result-persistence-condition`), log-viability is exactly

$$\mathcal{V} = \underbrace{\log\|\delta_{\text{critical}}\| + \log\alpha - \tfrac{1}{2}\log(\nu\, C_\pi)}_{\text{floor-relative viability}} - \tfrac{1}{2}\log(1 + x_\pi), \qquad x_\pi := \frac{E_\pi + U_\pi}{C_\pi} \;\geq\; 0, \tag{Y.3}$$

provided $C_\pi > 0$. Since $x - \frac{x^2}{2} \leq \log(1+x) \leq x$ for $x \geq 0$ (the lower bound from $h(x) = \frac{x^2}{2} - x + \log(1+x)$, $h(0) = 0, h'(x) = x^2/(1+x) \geq 0$),

$$\mathcal{V} = \Big[\log\|\delta_{\text{critical}}\| + \log\alpha - \tfrac{1}{2}\log(\nu\, C_\pi)\Big] - \frac{E_\pi}{2C_\pi} - \frac{U_\pi}{2C_\pi} + r(x_\pi), \qquad 0 \leq r(x_\pi) \leq \frac{x_\pi^2}{4}. \tag{Y.4}$$

Derivation. $\log\rho^2 = \log\nu + \log(C_\pi(1+x_\pi))$ by Proposition 1 and the definition of x_π; substitute into $\mathcal{V} = \log\|\delta_{\text{critical}}\| - \log\rho + \log\alpha$ and apply the elementary bounds. □

Reading. The agent-movable contribution to viability is additive in the estimation and state-uncertainty terms *to first order in the excess-to-floor ratio*, with a remainder no larger than a quarter of the square of that ratio (the bound is the leading Taylor coefficient, so it is tight as $x_\pi \to 0$). Exponentiating, $\exp(-E_\pi/2C_\pi)$ and $\exp(-U_\pi/2C_\pi)$ are the only honest "attenuation factors": they attenuate the floor-relative viability, they are exponentials of additive variance terms rather than primitive multipliers, and both depend on the policy through $\mathbb{P}_\pi$ — there is no model-only factor and no policy-only factor at any order. Outside the small-x_π regime the cross-dependence $\log(1+x_\pi)$ is the whole story and no log-additive reading survives.

Proposition 3 — the information form (exact).

Derived (log-loss-decomposition), exact under GA-1 and chronica-measurability of the model

Let $q_t(\cdot) := q(\cdot \mid M_{t-1}, a)$ be the model's full predictive distribution for o_t, $p_t^{\mathrm{B}}(\cdot) := p(\cdot \mid \mathcal{C}, a)$ the Bayes predictive distribution, and $p_t(\cdot) := p(\cdot \mid S, a)$ the channel law given the typed target $S = (\Omega, \theta)$, with $q_t \gg p_t^{\mathrm{B}}$ and all quantities finite. Then

$$\mathbb{E}_\pi\big[-\log q_t(o_t)\big] = \underbrace{H_\pi(o_t \mid S_t, a_{t-1})}_{\text{(iii′) channel entropy}} + \underbrace{I_\pi(o_t; S_t \mid \mathcal{C}_{t-1}, a_{t-1})}_{\text{(ii′) state-and-law information}} + \underbrace{\mathbb{E}_\pi\, D\big(p_t^{\mathrm{B}} \,\|\, q_t\big)}_{\text{(i′) estimation divergence}}. \tag{Y.5}$$

Derivation. Two tower-property steps. (a) Given $(\mathcal{C}, a)$, $o_t \sim p_t^{\mathrm{B}}$, so $\mathbb{E}_\pi[-\log q_t(o_t)] - \mathbb{E}_\pi[-\log p_t^{\mathrm{B}}(o_t)] = \mathbb{E}_\pi\,\mathbb{E}_{o\sim p_t^{\mathrm{B}}}\big[\log p_t^{\mathrm{B}}(o)/q_t(o)\big] = \mathbb{E}_\pi D(p_t^{\mathrm{B}}\|q_t)$. (b) Given $(S, a, \mathcal{C})$, $o_t \sim p_t$ by GA-1, so $\mathbb{E}_\pi[-\log p_t^{\mathrm{B}}(o_t)] - \mathbb{E}_\pi[-\log p_t(o_t)] = \mathbb{E}_\pi\,\mathbb{E}_{o\sim p_t}\big[\log p_t(o)/p_t^{\mathrm{B}}(o)\big] = I_\pi(o_t; S_t \mid \mathcal{C}_{t-1}, a_{t-1})$, the conditional mutual information. (c) $\mathbb{E}_\pi[-\log p_t(o_t)] = H_\pi(o_t \mid S_t, a_{t-1})$ by definition. Sum. □

(i′) and (ii′) are non-negative (a divergence and a mutual information); (iii′) is a conditional entropy and, as a differential entropy on a continuous observation space, may be negative — the Gaussian specialization below makes this explicit. (i′) vanishes iff the model's predictive distribution coincides with the Bayes predictive $\mathbb{P}_\pi$-a.s.; (ii′) vanishes iff residual uncertainty about the typed target S_t does not change the one-step predictive *distribution*, that is, iff the channel law p_t equals the Bayes predictive p_t^{B} $\mathbb{P}_\pi$-a.s. No orthogonality is invoked — the cross-terms of the squared form are replaced by the chain rule of log-loss, which is an identity. As in Proposition 1, the integrands are policy-independent kernels

under (P) and the policy enters only as $\mathbb{P}_\pi$; the same three consequences (reweighting, invariance, entanglement) hold verbatim.

Gaussian specialization (consistency with Propositions 1–2). In the linear-Gaussian sub-scope with scalar observation, $p_t = \mathcal{N}(\bar{o}, c)$ and $p_t^{\mathrm{B}} = \mathcal{N}(\hat{o}^{\mathrm{B}}, u{+}c)$, so $(\mathrm{iii}') = \frac{1}{2}\log(2\pi e\, c)$ and $(\mathrm{ii}') = \frac{1}{2}\log\big(1{+}u/c\big)$ — the $\log(1{+}x)$ structure of Proposition 2 appears pointwise. If the model is Gaussian with calibrated variance $\hat{s}^2 = u + c$, $(\mathrm{i}') = e/\big(2(u+c)\big)$; a miscalibrated $\hat{s}^2$ adds the standard penalty $\frac{1}{2}\big[\log(\hat{s}^2/(u+c)) + (u+c)/\hat{s}^2 - 1\big] \geq 0$, a contribution the squared form cannot see.

Derived (class-ceiling-pythagorean), exact conditional on the predictive class being an exponential family with an attainable moment match

Proposition 4 — class-ceiling refinement of the estimation divergence (sub-scope α).

Fix $(\mathcal{C}, a)$ and suppose the model class's predictive family $\mathcal{Q} = \{q_\theta\}$ is an exponential family with sufficient statistic T and log-partition ψ (e-flat in the sense of Amari & Nagaoka 2000 §3.4). Let $q^* \in \mathcal{Q}$ be the m-projection of p_t^{B} onto $\mathcal{Q}$ — the member with $\mathbb{E}_{q^*}[T] = \mathbb{E}_{p_t^{\mathrm{B}}}[T]$, assumed to exist. Then for every $q_\theta \in \mathcal{Q}$,

$$D\big(p_t^{\mathrm{B}} \| q_\theta\big) = \underbrace{D\big(p_t^{\mathrm{B}} \| q^*\big)}_{\text{class ceiling}} + \underbrace{D\big(q^* \| q_\theta\big)}_{\text{within-class estimation}} . \tag{Y.6}$$

Derivation. $\log\big(q^*/q_\theta\big) = (\theta^* - \theta)^\top T - \big(\psi(\theta^*) - \psi(\theta)\big)$ is affine in T, so its expectation under p_t^{B} equals its expectation under q^* by the moment match; the former is $D(p_t^{\mathrm{B}}\|q_\theta) - D(p_t^{\mathrm{B}}\|q^*)$ and the latter is $D(q^*\|q_\theta)$. □ (This is the generalized Pythagorean theorem of Csiszár 1975 and Amari & Nagaoka 2000 §3.4, specialized to an e-flat family; it is recorded here because the identification of the two legs with AAT's class ceiling and within-class learning is the content.)

Averaging under $\mathbb{P}_\pi$ gives $(\mathrm{i}') = \mathbb{E}_\pi D(p_t^{\mathrm{B}}\|q^*) + \mathbb{E}_\pi D(q^*\|q_t)$. The second leg is what parameter learning removes; the first is the one-step information-form counterpart of the ceiling that `#def-model-class-fitness` names and `#result-structural-adaptation-necessity` acts on — no estimate inside $\mathcal{Q}$ lowers it. The policy is again a measure on the pointwise projections, never a projection itself. Outside e-flat classes the identity acquires a residual $\mathbb{E}_{p_t^{\mathrm{B}}}\log(q^*/q_\theta) - D(q^*\|q_\theta)$ of either sign; that residual is the only place a genuine "cross term" lives in this attribution, and it is a class-geometry artifact, not a model–policy interaction.

What Is Derived vs. What Is Chosen.

Table Y.1

Property	*Source*	*Strength*
Three terms are expectations of policy-independent kernels under $\mathbb{P}_\pi$ (Prop. 1)	`#result-mismatch-decomposition` + GA-1 + (P)	Proved
Policy change = measure change; invariance when kernels are state-independent; entanglement (Cor. 1a–1c)	Prop. 1	Proved
Linear-Gaussian invariance of the innovation law under the control gain	Riccati covariance independent of input (BB)	Proved (sub-scope)
Exact log form; first-order log-additivity with remainder $\leq x^2/4$ (Prop. 2)	Prop. 1 + elementary bounds	Proved, conditional on the rate-lift convention and linear Model D
Information-form three-term identity (Prop. 3)	Chain rule of log-loss + GA-1	Proved
Class-ceiling / within-class split (Prop. 4)	Csiszár–Amari–Nagaoka Pythagorean identity	Proved, conditional on e-flat predictive class with attainable moment match

Property	*Source*	*Strength*
Identification of the class-ceiling leg with `#def-model-class-fitness`	Interpretation (different horizons; ratio vs divergence)	Discussion-grade
Separate identifiability of E_π, U_π, C_π from on-policy data	Not claimed here; obstructed per `#disc-identifiability-floor` Instance 4	—

Epistemic Status.

Conditional. Propositions 1 and 3 and the corollaries are *exact* under GA-1 and the chronica-measurable-policy condition (P) — both standing AAT interface assumptions, named so that the adversary-reads-the-policy case is visibly excluded. Proposition 2 is exact algebra plus an elementary inequality, conditional on the fluid-limit rate-lift convention and linear Model D that X already states. Proposition 4 is exact within sub-scope α (exponential-family predictive class, m-projection attainable) and acquires a signed residual outside it. Max attainable: *exact* for Propositions 1–3 as stated; Proposition 4 cannot exceed *conditional* because the e-flat hypothesis is a genuine restriction on the model class, and the link to `#def-model-class-fitness` is interpretive. What is *not* claimed: that the three terms are separately identifiable from on-policy innovation data — they are not, and the obstruction is AN projected onto the disturbance-statistic coordinate (`#disc-identifiability-floor` Instance 4); this segment lands the *structure* of the attribution, the floor governs its *estimation*.

Discussion.

Why the multiplicative picture felt right, and exactly where it was wrong. A product $\rho_{\text{external}} \cdot f(\mathcal{M}) \cdot g(\pi)$ is an independent-attenuation chain. Proposition 2 shows the one regime in which such a chain is approximately visible — small excess relative to the floor — and shows that even there the "factors" are $\exp(-E_\pi/2C_\pi)$ and $\exp(-U_\pi/2C_\pi)$, exponentials of additive variance terms, each policy-dependent through the measure. The constitutive no-go of X says the product cannot be exact; this segment says what the first-order approximation actually approximates, and that it approximates a split by *what moves the term* (modeling vs acting), not a split by *who owns the factor* (model vs policy). The reference-agent ambiguity that haunted "ρ_{external}" dissolves as well: the environmental floor is a kernel, $c(\Omega, a)$, agent-independent pointwise and policy-dependent only in expectation — a number once a policy is fixed, never a number on its own.

Worked anchors. *Well-specified Kalman, passive or LQR-controlled:* $e \equiv 0, u \equiv HP^{-}H^{\top}, c \equiv R$; the innovation variance $HP^{-}H^{\top} + R$ is control-invariant, so no policy choice moves the budget (Corollary 1b). *Misspecified drift with a regulating policy:* if the filter carries $\hat{\lambda} \neq \lambda$, its one-step prediction differs from the Bayes prediction by a quantity that scales with the state estimate, so to leading order $e(\mathcal{C}, a) \propto \hat{x}^2$ and $E_\pi \propto \mathbb{E}_\pi[x^2]$ — a stiffer LQR gain that holds the state near the origin lowers E_π while leaving u and c untouched. That is Corollary 1c in the flesh: the policy does not attenuate the environment; it keeps the trajectory where the model's inadequacy bites least. *[Discussion-grade as an anchor — the leading-order scaling is stated, the constants are not derived here.] Strategy layer, Bernoulli edges:* the disturbance driving an edge credence is a per-visit innovation with variance fixed by the true success probability, summed over edges at the policy's visit rates; the policy's role is the visit measure over edges, and no model-class kernel appears at the per-observation level — the same shape as Proposition 1 one layer up (S).

Two coordinates, one structure. The squared form (Propositions 1–2) is the Bregman decomposition on a squared-norm potential — the variance-additive adjacent case of `#disc-additive-coordinate-forcing`, *matched* to the sector-Lyapunov machinery rather than forced. The information form (Propositions 3–4) lives on the negative-entropy geometry — the same Legendre–Fenchel object the forced coordinates of that meta-segment inhabit — but it is *adopted* (log-loss is a proper scoring rule; the Pythagorean identity is Csiszár's), not forced by an AAT-internal additivity axiom: no axiom of the disturbance layer demands log-additivity, which is precisely why the multiplicative form was a choice and not a theorem. The two forms agree where they overlap (the Gaussian specialization), and the information

form carries strictly more: it has no alignment qualifier — residual state uncertainty registers whenever it changes the predictive distribution, not only its mean — and it sees predictive-variance miscalibration.

Interface with the identifiability floor. Everything above is structural: it says what the budget *is* made of and how a policy moves it. An on-policy observer sees only the sum (the innovation law), and the three terms are separately identifiable only through Level-2 access — change the instrument to move c, change the actions to move u, change the model to move e (`#der-loop-interventional-access`; `#disc-identifiability-floor` Instance 4). The structural result and the floor are complementary, and neither weakens the other.

Appendix Z

Derivation: Update Detection Latency Under Accumulated Experience

For a Beta-Bernoulli strategy-edge agent without forgetting, the expected number of cycles required to detect a within-class regime change of observable footprint ε scales as $\Omega((n_{\min}+1)/\varepsilon)$ with $n_{\min}$ the minimum accumulated pseudo-count on load-bearing edges. The $1/(n+1)$ rate is **structurally forced** — it is the log-odds update magnitude per cycle under V's Aczél-Cauchy-FE uniqueness theorem, and no choice of coordinate escapes it. The result sharpens `#schema-strategy-persistence`'s forgetting prerequisite from "required for asymptotic persistence" to "required for detection-latency bounded independently of operating point." The broader myopia observation — that successful, high-capability organizations systematically underinvest in detecting regime changes that would require structural adaptation — admits a complementary decision-theoretic account via `#disc-exploit-explore-deliberate`'s oracle analysis; this segment's contribution is the signal-side lower bound, not the decision-side account.

Formal Expression.

Setup. An agent with a Beta-Bernoulli strategy DAG per `#schema-strategy-persistence` / S, credit assignment via the log-odds signal of `#disc-credit-assignment-boundary` (forced by V), and no forgetting ($\lambda = 1$, pseudo-counts n_k accumulate monotonically). Let E_{load} denote the load-bearing edges on the current active plan; $n_{\min} = \min_{k \in E_{\text{load}}} n_k$. The agent has been operating with model-class fitness $\mathcal{F}(\mathcal{M})$ near 1, adaptive reserve $\Delta\rho^* > 0$, control regret δ_{regret} small — a high-operating-point configuration.

At cycle t_0, a regime change occurs within the current model class (a true edge probability shifts by ε; the L0 graph remains correct; the agent's model class still suffices). This is case R1 in the spike taxonomy — a within-class drift change. Other regime-change cases (R2 model-class inadequacy; R3 L0→L1 structural transition) are deferred to Working Notes and `#disc-identifiability-floor` respectively.

Detection-latency theorem.

Derived (detection-latency-R1, from edge-update-natural-parameter + deriv-edge-credence-dynamics)

Proposition. Under the setup, the expected number of cycles T_{detect} required for the log-odds coordinate on any load-bearing edge $k \in E_{\text{load}}$ to cross a fixed detection threshold $\Delta\lambda_{\text{detect}}$ in response to the regime change satisfies

$$\boxed{\mathbb{E}[T_{\text{detect}}] \;=\; \Omega(\Delta\lambda_{\text{detect}} \cdot (n_{\min}+1)/\varepsilon)} \tag{Z.1}$$

Derivation. The default signal function of `#disc-credit-assignment-boundary` (under the log-odds coordinate forced by V) updates an edge's log-odds credence by

$$\lambda_k^{\text{new}} = \lambda_k + \eta_{\text{edge}} \cdot \iota_k \cdot J_k \cdot (y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2 \tag{Z.2}$$

with $\eta_{\text{edge}} = 1/(n_k + 1)$ for Beta-Bernoulli (Prop B.4 of S). Per-cycle the expected log-odds update magnitude is bounded:

$$\mathbb{E}|\Delta\lambda_k| \;\leq\; \frac{|J_k| \cdot \mathbb{E}|y_G - \hat{P}_\Sigma|}{\|\mathbf{J}\|^2 \cdot (n_k + 1)} \tag{Z.3}$$

Under regime change R1 with observable footprint ε, the expected systematic residual $\mathbb{E}|y_G - \hat{P}_\Sigma| = \Theta(\varepsilon)$ as $\varepsilon \to 0$ (the misspecified-edge residual is proportional to the edge's probability shift, via the linearization of the Bernoulli likelihood in a neighborhood of the pre-change parameter). Combining: the per-cycle expected log-odds increment on load-bearing edges is $O(\varepsilon/(n_{\min} + 1))$.

For the agent to cross $\Delta\lambda_{\text{detect}}$, expected cycles required is at least $\Delta\lambda_{\text{detect}}/(O(\varepsilon/(n_{\min}+1))) = \Omega(\Delta\lambda_{\text{detect}} \cdot (n_{\min}+1)/\varepsilon)$. □

Observation (rate-forced-by-aczel)

The rate is structurally forced.

The $1/(n + 1)$ scaling is not a property of the specific update rule choice. Per V, the log-odds coordinate is the *unique* additive-evidence coordinate satisfying the evidential-additivity axiom — the Aczél 1966 Cauchy-functional-equation uniqueness theorem forces it up to positive affine transformation. In this forced coordinate, the per-cycle increment for Beta-Bernoulli edges is forced to be $O(1/(n+1))$: the Fisher-equivalent statement holds in any sensible coordinate. Rearranging the update to a different scale does not change the rate — it just changes the units in which the rate is measured.

The forcing composition:

1. V forces the log-odds coordinate via evidential additivity.
2. In that coordinate, Beta-Bernoulli accumulation gives $\eta_{\text{edge}} = 1/(n + 1)$.
3. Therefore the per-cycle update magnitude is structurally forced at $O(1/(n + 1))$.

This is the specific link between AAT's constructive meta-pattern (`#disc-additive-coordinate-forcing`, via V's theorem) and a downstream detection-latency consequence. The rate cannot be escaped without abandoning evidential additivity — which would invalidate the update rule on AAT-internal grounds, not merely operational ones.

Sharpening the forgetting prerequisite. *[Corollary (forgetting-as-latency-bound, sharpens* `#schema-strategy-persistence`*)]*

`#schema-strategy-persistence` derives the forgetting prerequisite $(1-\lambda) > \rho_\Sigma/R_\Sigma$ as required for *asymptotic persistence* — without forgetting, $\alpha_\Sigma = 1/(n + 1) \to 0$ and persistence eventually fails. The detection-latency theorem sharpens this to a **load-bearing claim about detection latency on the way to asymptotic failure**:

Forgetting is required not only for asymptotic persistence, but also for detection latency to be bounded independently of operating point. Without forgetting, $n_{\min}$ grows monotonically, and $\mathbb{E}[T_{\text{detect}}]$ grows linearly with $n_{\min}$. With forgetting at rate $\lambda < 1$, the effective pseudo-count $n_{\text{eff}} = 1/(1 - \lambda)$ is bounded, and the detection latency caps at $\Omega(1/((1 - \lambda)\varepsilon))$ regardless of how long the agent has been operating.

This dualizes `#schema-strategy-persistence`'s asymptotic claim: forgetting is operationally load-bearing at every step, not only in the limit. An agent with the right λ has bounded detection latency throughout its lifetime; an agent with $\lambda = 1$ has detection latency that grows unboundedly with experience, producing the phenomenon of stability-induced myopia in practice.

What Is Derived vs. What Is Chosen.

Table Z.1

Property	*Source*	*Strength*
R1 detection-latency theorem $\mathbb{E}[T_{\text{detect}}] = \Omega((n_{\min}+1)/\varepsilon)$	V log-odds coordinate + S Prop B.4 $\eta_{\text{edge}} = 1/(n+1)$ + Pinsker-type linearization	Derived (conditional on Beta-Bernoulli + log-odds + no forgetting)
$1/(n+1)$ rate structurally forced	Composition of V's Aczél-Cauchy-FE theorem with Beta-Bernoulli accumulation	Proved (conditional on evidential-additivity axiom)
Sharpening of forgetting prerequisite from asymptotic persistence to bounded detection latency	`#schema-strategy-persistence` + this theorem	Derived
R2 model-class inadequacy sub-case (C1-diagnostic blindness under misspecification)	Common-mode-bias argument on $A_O^{(1)}$ vs $V_O(\pi_{\text{current}})$	Discussion (sketch; see Working Notes)
R3 L0→L1 structural transition sub-case	Direct application of `#disc-identifiability-floor` Instance 1 — already derived there	Reference (not new content)
Probability-of-deferral bound	Derivable from the expected-time bound via Markov inequality	Discussion
Connection to institutional inertia / Christensen / competency traps	Stability-induced-myopia as a mechanism for the empirical pattern	Discussion
Connection to IDT sidecar monitoring (Hafez et al. 2026 89% vs 44%)	On-policy reward-based monitoring is the log-odds accumulator; IDT bi-predictability is a change-detector that bypasses the $1/(n+1)$ rate	Discussion

Epistemic Status.

Conditional. Max attainable: *exact* for R1 under stated scope (Beta-Bernoulli + log-odds forced + no forgetting + linearized residual near pre-change parameter); *robust qualitative* for general exponential-family edge with similar sufficient-statistic accumulation; *discussion-grade* for extensions beyond the Beta-Bernoulli / log-odds specialization.

The R1 theorem is a direct consequence of two derived AAT results (V's forced log-odds coordinate + S's Beta-Bernoulli η_{edge}) composed with a standard residual-magnitude argument (Pinsker-type inequality applied to the linearized Bernoulli likelihood). The mathematical core is not novel — what is novel is the AAT-framing: reading the $1/(n+1)$ scaling as *structurally forced* through composition of existing AAT theorems, not contingent on update-rule choice. The rate is unescapable without abandoning evidential additivity.

The forgetting-prerequisite sharpening follows by direct substitution — replace unbounded $n_{\min}$ with bounded $n_{\text{eff}} = 1/(1-\lambda)$ and the latency bound inherits the cap. This is where the segment's most concrete contribution lies: the existing forgetting prerequisite's justification expands from "asymptotic persistence fails without it" to "detection latency is unbounded in operating point without it." Two load-bearing consequences instead of one, from the same $\lambda < 1$ condition.

What this does not establish:

- R2 (model-class inadequacy) as a full derivation. The sketch argument — that $A_O^{(1)}$ and $V_O(\pi_{\text{current}})$ share common-mode bias under misspecification, so the C1 regret diagnostic under-reports — is qualitatively clear but quantitatively informal. Making it rigorous requires a model of the bias's functional form.

- R3 (L0→L1 transition) as new content. This is `#disc-identifiability-floor` Instance 1 — a structural no-go, not a latency bound. This segment references it but does not extend it.
- Decision-theoretic consequences. The theorem is about signal latency, not about whether the signal once received yields the right decision. `#disc-exploit-explore-deliberate`'s oracle analysis (which shows deliberation rarely chosen) operates at the decision-theoretic level; the myopia theorem operates at the signal-quality level. An agent with degraded signals + perfect decision rule still defers; this is complementary to the oracle's degraded-decision-with-perfect-signals result.
- Extension beyond Beta-Bernoulli. Gaussian edges admit $1/(n + c)$ scaling with analogous behavior; general exponential-family edges admit scaling following the sufficient-statistic accumulation. The qualitative form — expected detection time monotone in accumulated sufficient statistic — is more general than Beta-Bernoulli; the quantitative $1/(n + 1)$ is specific.

Discussion.

The forcing composition is where the meta-pattern interacts with downstream results. `#disc-additive-coordinate-forcing` is an AAT meta-segment naming the Cauchy-FE structural move at three layers (chain / divergence / update). This segment exhibits the specific case where that move's downstream consequences bind: the log-odds forcing at the update layer implies — through Beta-Bernoulli accumulation — a specific detection rate that cannot be escaped via re-parameterization. The meta-pattern's constructive power is visible here as a *negative* downstream consequence (bounded detection rate), which then becomes the positive motivator for forgetting. This is an example of AAT's meta-architecture composing constructively: the forcing move (positive) produces a bound (negative) that sharpens an existing requirement (positive).

Distinction from adjacent phenomena:

- `#result-structural-adaptation-necessity` characterizes WHEN restructuring is needed at truth (model-class inadequacy). This segment characterizes WHEN (in terms of cycles) the agent's detection signal crosses threshold. The two together say: the agent will eventually detect a structural problem, but the latency can be large at high operating point.
- `#disc-exploit-explore-deliberate` gives a decision rule over exploit / explore / deliberate given signals. This segment is about signal quality — specifically, a predictable bias in signals as a function of operating point. The two compose: even an oracle's decision rule applied to myopia-biased signals produces deferred action.
- `#schema-strategy-persistence`'s forgetting prerequisite is a condition for asymptotic persistence. The corollary here is a condition for bounded detection latency — the same $\lambda < 1$ buys both.
- `#disc-identifiability-floor` Instance 1 (on-policy L0-insufficiency no-go) is a structural unboundedness result specific to L0/L1 transitions. The R1 detection-latency theorem is a quantitative bound on a broader class of regime changes (within-class drift, not just structural transitions).

Connection to cross-domain empirical patterns. Christensen's *Innovator's Dilemma*, Levitt & March competency traps, Hannan & Freeman structural inertia, March exploration/exploitation, and Eldredge-Gould punctuated equilibria all identify empirical patterns of stability-induced myopia across domains. The common story is that successful, mature, capable systems systematically underinvest in detecting regime changes that would require restructuring. The theorem supplies a mechanism: as experience accumulates ($n_{\min}$ grows), per-observation learning rate decays as $1/(n+1)$, expected detection time grows linearly, and the structural-adaptation trigger signal is systematically delayed. The phenomenon is not only decision-theoretic (incentive mis-alignment, short-horizon myopia) — it has a signal-side epistemic component forced by the log-odds coordinate. The signal-side and decision-side stories compose; neither subsumes the other.

Connection to IDT (Hafez et al. 2026) empirical result. Hafez et al. 2026 observed 89% vs 44% detection accuracy for Information Digital Twin (IDT) sidecar monitoring vs reward-based monitoring. The AAT-native explanation: reward-based monitoring is the log-odds accumulator — it inherits the $1/(n+1)$ detection rate and is subject to the theorem's latency bound. IDT's bi-predictability and entropy-change measures are structural-information-theoretic tests that bypass the log-odds coordinate entirely; they are change-detectors in the CUSUM / GLR tradition (Page 1954; generalized likelihood ratio tests) rather than Bayesian credence accumulators. The predicted gap between the two monitoring styles grows with accumulated experience — a testable refinement of the 89%/44% observation. A sidecar change-detector is always going to beat the log-odds accumulator on detection latency for within-class drift; the AAT theorem explains why without invoking the specific IDT mechanism.

The signal side and decision side as complementary. The fact that the oracle in `#disc-exploit-explore-deliberate` rarely chose deliberation is usually read as "agents correctly defer when deliberation is expensive and information is limited." This segment's theorem adds: "and the information is limited in a specific quantitative way that scales with operating point." Both are true; neither is contradicted. The detection-latency result does not overturn the oracle analysis; it provides the signal-quality context in which that decision analysis operates.

Appendix AA

Discussion: Independence Audit

AAT's results depend on a recurring modeling move: treat some quantity as independent of another to obtain tractable mathematics, then identify the failure regime where independence breaks and specify the repair. This segment enumerates the independence assumptions used across the theory, their failure regimes, their diagnostic signals, and the repair operations AAT provides. The enumeration makes visible what is *not* an independence assumption — acyclicity of Σ_t, Cox-derived probability, the Lyapunov machinery of G — and therefore what survives when a particular independence assumption fails.

Formal Expression.

Discussion (independence-audit)

Six load-bearing independence assumptions in AAT, each paired with its failure regime and repair:

1. Directed separation: M_t update independent of G_t. **Statement:** $f_M(M_{\tau^-}, e_\tau)$ has no G_t argument — the epistemic update is goal-blind.

Where it appears: `#der-directed-separation`, the structural backbone of Part II. Feeds the orient cascade's sequential resolution (`#der-orient-cascade`), the causal validity of Q_O (`#def-value-object`), and the scope of all Part II results to Class 1 (Separated) agents.

Failure regime: Class 3 (Coupled) architectures — transformer LLMs where attention processes goals and observations together. Motivated reasoning, confirmation bias, prompt-conditioned perception. Partially also Class 2 (Partial) agents.

Diagnostic signal: $\kappa_{\text{processing}} = I(G_t; M_{\tau^+} \mid e_\tau, M_{\tau^-}) / H(G_t \mid e_\tau, M_{\tau^-})$. Zero for Separated agents; near one for Coupled; intermediate for Partial.

Repair operation: Class 2 (Partial) approximation quality scales with $\kappa_{\text{processing}}$. Class 3 (Coupled) agents require the coupled formulation $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ without (M_t, G_t) decomposition — the scope of `03-llm-core/`. At the system level, Class 3 (Coupled) components can be wrapped in modular topology (separate observation processing, external monitoring — see `#der-directed-separation` Working Notes on the IDT pattern).

2. Causal sufficiency: no latent common causes among strategy nodes. **Statement:** Every common cause of two or more nodes in Σ_t is itself a node in Σ_t.

Where it appears: R (precondition for the CMC-based Markov proof); `#def-strategy-dag` edge-independence in status propagation; all of Part II's strategy-layer formal results (Props B.1–B.6, the sector condition transfer of B.5, persistence of δ_s).

Failure regime: The dominant real-world case in complex, multi-stakeholder, or adversarial environments — shared infrastructure, market conditions, correlated adversary actions, common-mode risks, supply-chain dependencies.

Diagnostic signal: Pairwise covariance among sibling edges after edge credences have converged. Covariance different from zero, of either sign, rejects the independence hypothesis and localizes where a common cause is missing. It is positive for a cause that enables or disables siblings together, and negative for one that makes them exclusive (competition for a resource; alternative answers to one question). See `#der-causal-insufficiency-detection`.

Repair operation: L1 augmentation — add common-cause nodes and restructure the DAG so each common cause is factored *above* the correlation it creates (`#def-strategy-dag` Correlation Hierarchy; S Prop B.6). L0 formal results transfer exactly to correctly constructed L1 DAGs because a correctly constructed L1 graph is causally sufficient *and* an AND/OR tree, the two conditions under which status propagation is exact (R, §"Exactness of Status Propagation"). The orient cascade's step 4c triggers this escalation.

3. Channel independence: observation channels contribute non-redundant correction. **Statement:** $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$ with each channel contributing independently to effective tempo.

Where it appears: `#def-adaptive-tempo` (core definition); inherited by `#result-persistence-condition`, `#result-adversarial-tempo-advantage`, `#der-team-persistence` (communication tempo), `#der-tempo-composition`, and any result using scalar $\mathcal{T}$.

Failure regime: Any system with redundant or correlated information sources — overlapping sensors, correlated teammate reports, redundant telemetry. In multi-agent settings, allies reporting the same intelligence source.

Diagnostic signal: $I(e^{(1)}; e^{(2)} \mid M_{\tau^-})$ — pairwise mutual information between event streams conditioned on prior model state. Non-zero conditional mutual information *witnesses* dependence (a valid alarm), but it cannot serve as the correction term: it is sign-blind, and the sign of the noise correlation determines whether the additive form over- or under-counts (`#deriv-tempo-additivity`, no-go N1).

Repair operation: Under correlated channel noise the additive formula is *not* a general upper bound — the signed deviation $\Delta = \sum_k 1/U_o^{(k)} - \mathbf{1}^\top \Sigma_n^{-1} \mathbf{1}$ takes both signs (anti-correlated or heterogeneous positively-correlated noise makes the additive form an *under*count; correlated noise is also a cancellation resource). The exact statements live in `#deriv-tempo-additivity`: additivity is exact under cross-channel noise independence (sufficient, not necessary), and in the common-source regime (shared persistent bias s) the additive form *is* an exact upper bound with closed-form penalty and joint information saturating at $1/\sigma_s^2$ — the echo-chamber bound. No channel-local additive correction exists in general for $n \geq 3$ (PID obstruction, no-go N2); precise tempo under correlation requires the joint Fisher form. Propagates into every segment using scalar tempo.

4. Unity-dimension independence: $(U_M, U_O, U_\Sigma, U_{\text{obs}})$ substantially independent. **Statement:** The four unity dimensions of `#def-unity-dimensions` parametrize composite quality along substantially independent axes.

Where it appears: `#def-unity-dimensions` framing; `#result-unity-closure-mapping`'s rate-distortion family indexed by each unity separately.

Failure regime: Shared models enable implicit strategic coordination (high U_M → high U_Σ without explicit policy sharing); aligned objectives often induce strategic coordination. Also *update-rule heterogeneity* (an axis not covered by any unity dimension) contributes to closure defect independently — the two-Kalman non-degenerate case shows $\varepsilon_x > 0$ from differing Kalman gains at perfect U_M.

Diagnostic signal: No clean one. The hypothesis is empirical and the theory currently logs it as working-position rather than resolved (see `#def-unity-dimensions` Working Notes: update-heterogeneity gap).

Repair operation: Two options explored in the theory: (a) accept a two-axis structure (unity × homogeneity) for closure defect, (b) add a fifth update-homogeneity dimension. The current working position is (a); formal resolution open. Downstream uses of unity dimensions should not treat them as cleanly orthogonal.

5. Independent edge outcomes in AND/OR propagation. **Statement:** In $\hat{P}_\Sigma$ status propagation, sibling edge outcomes are independent given their parents.

Where it appears: `#def-strategy-dag` status propagation formula; `#scope-and-or`; every downstream quantity computed from $\hat{P}_\Sigma$ (satisfaction gap, control regret via A_O, strategy-plan-confidence error δ_s).

Failure regime: Wider than causal sufficiency (item 2). Causal sufficiency gives independent exogenous noise, but a node's parents are independent only if, in addition, they share no ancestor. An explicit common-cause node that feeds two branches which reconverge breaks this with causal sufficiency intact. The exact condition is that the DAG be a causally sufficient AND/OR tree (R, §"Exactness of Status Propagation").

Diagnostic signal: For a latent cause, as item 2. For explicit reconvergence no data are needed: the condition is read from the graph (some node has two directed paths to the root).

Repair operation: For a latent cause, as item 2. For explicit reconvergence: factor a strict-prerequisite shared node above the branches it gates (L1), condition on it (L1' / cutset conditioning), or replace the forward pass by exact inference. The assumption is not a separate modeling choice: it is the consequence of causal sufficiency together with tree structure, not an independent axiom.

6. Scalar tempo / isotropic correction. **Statement:** $\mathcal{T}$ as a scalar captures the agent's correction capacity.

Where it appears: `#result-persistence-condition` linear operational form; most Part I results stated in scalar $(\mathcal{T}, \rho, \|\delta_{\text{critical}}\|)$ form; `#def-strategic-tempo` aggregate.

Failure regime: Any real multi-dimensional system with non-uniform correction capacity across dimensions. Simulation confirms scalar $\rho/\mathcal{T}$ overestimates by up to 72% in anisotropic systems, with the weak dimension accounting for 84% of total mismatch.

Diagnostic signal: Gain variation across dimensions (easily computable when gains are explicit — Kalman, gradient descent with diagonal preconditioning).

Repair operation: Per-dimension persistence: $\mathcal{T}_k > \rho_k/\|\delta_{\text{critical},k}\|$ for each dimension. See `#result-per-dimension-persistence`. The weakest dimension is the bottleneck. Same structural pattern propagates to strategic tempo (`#def-strategic-tempo` Per-edge persistence) — the bottleneck edge determines persistence, not the aggregate.

What is* not *an independence assumption.

Several results that might appear to depend on independence actually do not:

- **Acyclicity of Σ_t** is proved from temporal ordering over a finite horizon (R). Independent of any independence assumption.
- **Cox-derived probability structure** at edges is independent of edge-independence — Cox forces the *measure* to be probability, not the joint structure.
- **Sector-persistence template machinery** (G) depends only on the sector condition (T2) and bounded disturbance (T3); it does not require any independence assumption on the state variable or correction function.
- **The DAG structure derivation** (directed edges, acyclicity, representability as a Bayesian network) does not require causal sufficiency — only the *Markov factorization* within the derived structure does.

This is the boundary between AAT's robust results and its conditional ones: robust results survive when the independence assumptions fail; conditional results break and require their specific repair operations.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The enumeration is definitional — each independence assumption is stated in a named segment, and the repair operations are either formally specified (causal sufficiency ↔ L1 augmentation) or named open problems (unity-dimension independence). What this segment adds is the cross-cutting view: the six assumptions are the recurring modeling move that distinguishes AAT's exact-tractable results from its conditional results, and the repairs are (mostly) already present in the theory but scattered across segments.

The enumeration is not exhaustive. Other independence-flavored assumptions exist in specific instantiations (e.g., independent Beta priors across edges in the credit-assignment analysis of `#disc-credit-assignment-boundary`). The six listed are the load-bearing ones — those whose failure regime the theory has actually characterized and provided a repair for.

Discussion.

Why enumerate these in one place. AAT's theoretical contribution is often described as "integration": bringing control theory, causal inference, information theory, and agent architecture under one framework. Integration *is* achieved largely through the independence assumptions listed above: each is a bridge that lets a module from one discipline plug into a module from another. Directed separation lets control-theoretic Lyapunov analysis operate on the epistemic substate without goal-entanglement. Causal sufficiency lets Pearl's DAG machinery apply to strategy without requiring the agent to model every environmental common cause. Channel independence lets the tempo framework sum over heterogeneous observation modalities.

When these bridges fail, the integration does not fail catastrophically — each failure has a characterized repair. But the theory's "exact" claims depend on the bridges holding, and recognizing this is the honest way to present the framework's scope. A reader who understands the six assumptions and their repairs understands where AAT's results apply exactly, where they apply approximately, and where they require structural revision.

Connection to AB. The L0/L1 distinction for causal sufficiency (items 2 and 5) is one instance of a meta-pattern: AAT's results are parameterized by approximation level, with proved monotonicity between levels. See AB for the full treatment — the pattern also applies to the C1/C2/C3 value-convention hierarchy and the Tier 1/2/3 bridge-lemma contraction structure.

What this does NOT say. The enumeration does not claim the theory's results are fragile or routinely fail. Most independence assumptions hold approximately in most well-behaved domains, and the theory's quantitative predictions are typically within the regime where assumptions apply. The audit is a scope clarification, not a criticism.

Appendix AB

Discussion: Approximation Tiering

AAT uses a recurring meta-pattern for handling intractability: when a problem admits no tractable exact treatment in general, introduce a tiered hierarchy of approximations with proved monotonicity between tiers and a diagnostic for ascending when needed. Three such hierarchies exist in the theory — the Correlation Hierarchy (L0/L1/L2) in `#def-strategy-dag`, the Convention Hierarchy (C1/C2/C3) in `#def-value-object`, and the Tier 1/2/3 contraction taxonomy in `#form-composition-closure`. This segment articulates the pattern explicitly, identifies what makes a successful approximation tiering, and notes where other scattered simplifications might fit the same shape.

Formal Expression.

Discussion (approximation-tiering)

A successful approximation tiering has four components:

(AT1) A parameter indexing tractability. Some quantity (correlation modeling depth; continuation-policy fidelity; operator regularity) that admits a natural ordering from least to most demanding computation.

(AT2) Monotonicity. A proved ordering of the results produced at each level — each higher tier dominates the lower in the direction of interest (calibration accuracy, diagnostic force, guarantee strength).

(AT3) Graceful degradation. Lower tiers produce usable (not vacuous) results, with the specific form of the degradation characterized. An agent stuck at the lowest tier knows what it is missing, not just that it is incomplete.

(AT4) Ascension diagnostic. A signal the agent can detect at a lower tier that indicates escalation to a higher tier is warranted. Without this, the tiering is descriptive but not operational.

The three AAT tierings.

Table AB.1

Hierarchy	*Parameter*	*Monotonicity*	*Lowest tier*	*Highest tier*	*Ascension diagnostic*
Correlation (`#def-strategy-dag`)	Modeling depth of inter-edge dependence	L0 is conservative; L1 is unbiased on augmented graph; L2 is the full joint	L0 (independence model, $O(\|V\| + \|E\|)$ propagation)	L2 (full joint, $O(2^m)$ in general)	Sibling-edge covariance after credence convergence (`#der-causal-insufficiency-detection`)

Hierarchy	*Parameter*	*Monotonicity*	*Lowest tier*	*Highest tier*	*Ascension diagnostic*
Convention (`#def-value-object`)	Continuation-policy fidelity	$A_O^{(1)} \leq A_O^{\mathrm{RH}} \leq A_O^{\mathrm{B}}$ (proved monotonicity)	C1 (one-step improvement)	C3 (Bellman optimal)	Persistent $\delta_{\mathrm{sat}} > 0$ under C1 that C2 replanning may resolve
Contraction (`#form-composition-closure`)	Operator regularity for bridge lemma	Tier 1 $\supset$ Tier 2 $\supset$ Tier 3 in terms of proved contraction strength	Tier 3 (domain-specific verification required)	Tier 1 (contraction proved for full class)	Structural test: is the correction strongly monotone, locally convex, or neither?

Each row satisfies (AT1)–(AT4) fully — these are mature tierings in the theory. Each tiering is independently motivated by a different source of intractability, and each has its own ascension mechanism. Their structural similarity is not designed-in; it is the consequence of facing three separate intractable problems with the same modeling strategy.

Candidate future tierings. Several other scattered simplifications in the theory have the right shape to become tierings but are not currently organized that way:

- **Scalar vs. per-dimension tempo** (around `#def-adaptive-tempo`, `#result-per-dimension-persistence`). Tier 0 (scalar isotropic), Tier 1 (per-dimension), Tier 2 (full tensor with off-diagonal coupling). Monotonicity: scalar overestimates margin, per-dimension is exact for diagonal-gain systems, full tensor handles off-diagonal gain coupling. Ascension diagnostic: gain variance across dimensions.
- **AND/OR parameterization** (`#scope-and-or`). Tier 0 (AND/OR, $O(k)$ parameters per node), Tier 1 (weighted/threshold, $O(k)$ additional parameters), Tier 2 (full CPT, $O(2^k)$ parameters). Monotonicity: expressiveness. Ascension diagnostic: systematic miscalibration at AND/OR nodes that survives L1 augmentation.
- **Identifiability regimes** (`#scope-edge-update-causal-validity`). Regime A/B/C is already a tiering of causal identification quality, with ι_{ij} as the indexing parameter. Monotonicity: interventional $\succeq$ partial $\succeq$ observational. Ascension diagnostic: available intervention opportunities in the domain.

Promoting these to named tierings with explicit (AT1)–(AT4) structure would make AAT's scope parameterization more uniform. This is not pursued here — it is noted as a direction for editorial consolidation.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The meta-pattern is an editorial observation: three existing hierarchies in the theory share the same four-component structure, and the shared structure is worth naming because it makes the theory's scope-parameterization move visible rather than implicit. The pattern itself is not a theorem — there is no result that says "every intractable AAT problem admits a (AT1)–(AT4) tiering." The candidate future tierings are conjectures about where the pattern could be extended; each would require its own monotonicity proof and ascension diagnostic.

The three existing tierings are individually grounded: the L0→L1 monotonicity holds under the CMC theorem and augmentation construction; the C1→C2→C3 monotonicity is proved in `#def-value-object`; the Tier 1/2/3 taxonomy is grounded in operator theory (strong monotonicity, local convexity, arbitrary).

Discussion.

Why this matters. AAT's results are often presented as "exact under assumption X." A reader can misread this as "the theory works when X holds and breaks otherwise." The tiering pattern reveals a different structure: when X fails, a named lower tier takes over with a characterized weaker result, and the theory provides a diagnostic for when to escalate to reinstating X by structural repair. This is graceful degradation — a property that a theory wanting to apply across a range of agents and domains needs, and that AAT achieves in three places independently.

Reading AAT through the tiering lens: the theory's "exact" core is what it guarantees at the highest tier of each hierarchy. Its "conditional" periphery is where specific tiers are in force. The claim is not that agents should always operate at the highest tier — that would often be intractable — but that the theory tells the agent which tier it is in and how to get to a higher one when the binding constraint changes.

Connection to AA. The two meta-segments are complementary. AA enumerates the independence assumptions whose failure drops a result from exact to conditional. This segment enumerates the tierings that provide the structured *recovery* from those drops. Together they characterize AAT's scope parameterization: the independence audit says where the boundaries are; the tiering pattern says how to navigate within them.

What a full tiering promotion would look like. If the scattered simplifications listed above were promoted to named tierings, the theory's scope-parameterization would be dramatically more uniform: every conditional result would come with its own tiering and ascension diagnostic. This is aspirational. The three existing tierings took substantial segment-work to formalize (a full Appendix derivation for the CMC, a monotonicity proof for the convention hierarchy, a spike for the incremental sector bound). Extending the pattern to five or six hierarchies would be correspondingly substantial.

The pattern is not unique to AAT. Approximation tiering is a common move in applied mathematics — numerical methods (low-order vs. high-order integrators with error bounds), information theory (typical vs. exact coding with rate-distortion monotonicity), statistical inference (maximum-likelihood vs. Bayesian vs. nonparametric). AAT's contribution is not the tiering pattern itself but its deployment across the specific intractable problems that adaptive-agent theory faces.

Appendix AC

Discussion: Compression Operations in AAT

AAT contains four compression operations — the epistemic model M_t, the strategy DAG Σ_t, shared intent G_t^{shared}, and the composition projection Λ — each formulated in its own segment with its own objective. Three of the four are written in Information Bottleneck (IB) form already; the fourth is stated as an IB constraint. This segment makes the shared shape explicit, promotes one underspecified source (the ontologically ambiguous "true causal structure" for Σ_t) to a cleaner formulation parallel to M_t, and establishes that composition admissibility (P1) is the Lagrangian-dual of a standard IB objective. It does *not* claim the four operations reduce to a single optimization problem — cross-instance theorems do not follow from the shared shape alone, and several conditions (Lipschitz regularity (P2), dimensional reduction (P3) in the Gaussian case, interventional relevance for Level-2 edges) remain outside the IB frame.

Formal Expression.

The shared IB shape. *Discussion (ib-shape)*

Every compression operation in AAT has an objective or constraint of the form:

$$T^* = \arg\min_{T|X} \left[I(X;T) \; - \; \beta \cdot I(T;Y) \right] \tag{AC.1}$$

with the Markov chain $Y - X - T$. The four AAT instances specialize this with different bindings:

Table AC.1

Instance	*X (source)*	*T (compressed)*	*Y (relevance variable)*	*β (trade-off)*
Model compression (`#form-information-bottleneck`)	$\mathcal{C}_t$	M_t	$o_{t+1:\infty} \mid a_{t:\infty}$	$\beta(\rho, \pi)$ — volatility and policy
Strategy compression (`#form-strategy-complexity-cost`)	$\mathcal{C}_t$	Σ_t	$\pi^* \mid M_t$	β_Σ — cognitive cost per decision-bit
Shared intent (`#def-shared-intent`)	$G_t^{\text{full}} = (O_t, \Sigma_t)$	G_t^{shared}	$a_t^{\text{coordinated}}$	bandwidth per coordination-bit
Composition projection (`#form-composition-closure` P1)	$X_{\text{micro},t}$	$\Lambda_X(X_{\text{micro},t})$	$o_{\text{micro},t+1} \mid a_{\text{micro},t}$	$\beta(\epsilon_I)$ — rate-distortion Lagrange multiplier

What the four instances share: *shape* (the objective structure), *variational calculus* (minimization over stochastic compressors), and *rate-distortion interpretation* (each trade-off parameter indexes a point on the frontier). What they do not share: source type (history vs. structured state), relevance-variable availability (observed vs. latent), computability (Gaussian closed forms vs. variational approximation), or a single joint optimization across the four. The level of unification is *medium*: shared shape and vocabulary, not a shared master problem.

Formulation (strategy-compression-source)

Strategy compression: source reformulation.

The current statement in `#form-strategy-complexity-cost` has the Σ_t IB objective as:

$$\Sigma_t^* = \arg\min_{\Sigma_t} \left[\mathrm{DL}(\Sigma_t) \; - \; \beta_\Sigma \cdot I(\Sigma_t;\; \pi^* \mid M_t) \right] \tag{AC.2}$$

Two issues with this as currently written:

1. **The compression cost is description length, not mutual information.** DL and $I(X;T)$ are related through coding-theoretic equivalences but coincide only under specific coding schemes. Using DL in the complexity term blocks the identification of this objective as an IB instance directly.
2. **The "source" is not an AAT object.** To fit the IB shape, the objective implicitly treats Σ_t as a compression of "the true causal structure." That structure is not part of AAT's ontology — the agent never has access to it; it is only ever implicit.

Reformulation. Treat Σ_t as a compression of $\mathcal{C}_t$ (the interaction history — the agent's only evidence) *for decision-relevance*, parallel to M_t which is a compression of $\mathcal{C}_t$ *for prediction-relevance*:

$$\Sigma_t^* = \arg\min_{\Sigma_t} \left[I(\mathcal{C}_t;\Sigma_t) \; - \; \beta_\Sigma \cdot I(\Sigma_t;\; \pi^* \mid M_t) \right] \tag{AC.3}$$

Under this reformulation:

- The source $\mathcal{C}_t$ is a well-defined AAT object, shared with the M_t instance.
- The two instances differ cleanly in relevance variable: M_t is compressed for prediction ($Y = o_{t+1:\infty} \mid a$); Σ_t is compressed for guidance ($Y = \pi^* \mid M_t$). Prediction is about *what will happen*; guidance is about *what to do*. Both are computed from the same history, with different targets.
- The information cost $I(\mathcal{C}_t;\Sigma_t)$ replaces $\mathrm{DL}(\Sigma_t)$ as the theory-level compression term. The DL formulation remains useful as an *operational* cost measure for specific DAG encodings; it is not the theoretical quantity the IB objective minimizes.

Relationship between the two cost measures. Under MDL with a specific encoding scheme for DAGs (the one in `#form-strategy-complexity-cost`), $\mathrm{DL}(\Sigma_t)$ is an upper bound on $I(\mathcal{C}_t;\Sigma_t)$ for DAGs produced by the given encoder — coding cost dominates distinguishability cost. In practice, DL is computable and I is not, so the operational minimization uses DL as a proxy; the theoretical minimization uses I. The IB objective above is the theoretical statement; the DL-based minimization in `#form-strategy-complexity-cost` remains the practical one.

Variational form. The Shannon mutual information $I(\Sigma_t;\pi^* \mid M_t)$ in the relevance term collapses to zero when π^* is deterministic-from-M_t. The variational form (cf. `#form-strategy-complexity-cost`) replaces the relevance term with the KL-divergence $D_{\mathrm{KL}}(\pi^*(\cdot \mid M_t) \,\|\, Q_{\Sigma_t}(\pi \mid M_t))$ — note the π^*-first direction, which is *forced* by a regret-bound derivation (full derivation and admissible-divergence family analysis in U). Under bounded value range and deterministic π^*, Pinsker's inequality gives $R(Q_{\Sigma_t}) \le V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\pi^*\|Q_{\Sigma_t})}$ where R is the strategy-induced regret against π^*; the opposite KL direction is vacuous ($+\infty$ under deterministic π^* whenever Q_{Σ_t} has off-optimum mass). The π^*-first

KL is well-defined and graded under deterministic π^*. Under the variational reading, the AAT Σ_t is a tractable approximation of the policy-relevant posterior, and the KL term measures approximation quality — aligning the strategy compression with the variational free energy decomposition $-F =$ accuracy $-$ complexity in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020). The direction alignment is convergent: both AAT's regret-bound and active inference's variational-free-energy derivations pick π^*-first KL (reverse-KL in the variational-inference vocabulary), with AAT's additional interpretation being an upper-regret-bound rather than free-energy-gradient. The shared-IB-shape framing of the four AAT compression operations is the rate-distortion specialization of this variational picture; AAT's commitment is to the rate-distortion form (which gives the (P1) Lagrangian-dual derivation and the four-instance unification at U-medium), not to the full variational free-energy interpretation.

Composition admissibility (P1) as IB Lagrangian-dual.

Derived (p1-ib-dual, from composition-closure + rate-distortion duality)

`#form-composition-closure` condition (P1) is currently stated as a lower-bound constraint:

$$I\big(\Lambda_x(X_{\text{micro},t});\ \Lambda_o(o_{\text{micro},t+1}) \mid \Lambda_a(a_{\text{micro},t})\big)\ \geq\ (1-\epsilon_I)\cdot I\big(X_{\text{micro},t};\ o_{\text{micro},t+1} \mid a_{\text{micro},t}\big) \tag{AC.4}$$

This is the *constraint form* of an IB problem. In Lagrangian form:

$$\Lambda^*\ \in\ \arg\min_{\Lambda\in\mathcal{P}}\ \big[\, I(X_{\text{micro}};\Lambda_x(X_{\text{micro}}))\ -\ \beta(\epsilon_I)\cdot I(\Lambda_x(X_{\text{micro}});\ Y_{\text{rel}})\,\big] \tag{AC.5}$$

where $Y_{\text{rel}} = (o_{\text{micro},t+1} \mid a_{\text{micro},t})$ and $\beta(\epsilon_I)$ is the Lagrange multiplier corresponding to the relevance-preservation tolerance ϵ_I. The correspondence $\epsilon_I \leftrightarrow \beta$ is the standard rate-distortion duality: smaller ϵ_I (more relevance preserved) corresponds to larger β (less aggressive compression).

Consequence: admissible projections are those that sit *on or above* the IB frontier at rate $I(X;T) \leq I_{\max}(\epsilon_I)$. The information-theoretic content of (P1) is exactly "project onto the IB frontier with a tolerance ϵ_I." This formalizes the connection previously logged in `#form-composition-closure`'s Working Notes and in `#result-unity-closure-mapping`'s §Connection to the Information Bottleneck.

What this resolves. The `#form-composition-closure` Working Note "Open: Information Bottleneck unification" is now resolved for (P1): it is the Lagrangian-dual of the IB constraint at $\beta(\epsilon_I)$. The corresponding Working Note in `#result-unity-closure-mapping` §6 moves from conjecture to derived result. Nothing else about (P1) changes — the condition continues to define admissible projections; only its information-theoretic reading is now explicit.

The symmetric two-Kalman witness, and the falsification handle.

Derived (linear-Gaussian-symmetric means-sum optimality)

The (P1)↔IB reading has a worked witness in the two-Kalman composition setup (`#result-unity-closure-mapping`'s linear-Gaussian case). With symmetric process noise $Q = q\begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}$, identical observation noise, and homogeneous gains ($\Delta K = 0$), the canonical-correlation basis between the micro-state $X = (\hat{\omega}_1, \hat{\omega}_2)$ and the relevance variable $Y = o_{t+1}$ is the symmetric pair $u_\pm = (1, \pm 1)/\sqrt{2}$, with generalized eigenvalues

$$\lambda_+ = \frac{(1+\rho_M)\sigma^2_{\hat{\omega}} + q(1+\rho) + r}{q(1+\rho)+r}, \qquad \lambda_- = \frac{(1-\rho_M)\sigma^2_{\hat{\omega}} + q(1-\rho) + r}{q(1-\rho)+r}, \tag{AC.6}$$

so the means-sum projection $u_+^\top X = (\hat{\omega}_1 + \hat{\omega}_2)/\sqrt{2}$ — the natural (P1)-admissible projection — is simultaneously the PCA-optimal 1D projection of X *and* the CCA-optimal 1D projection of X for predicting Y. The coincidence is symmetry-forced (the canonical basis and the principal-component basis of Σ_X align under symmetric Q and $\Delta K = 0$);

in general the two bases differ. The retained relevance at 1D is $\frac{1}{2}\log\lambda_+$ of the full $\frac{1}{2}\log(\lambda_+\lambda_-)$, ranging from half (at $\rho \to 0$, symmetric contributions) to all of it (at $\rho \to 1$, collapsed directions). Whether the means-sum additionally attains the Gaussian IB *frontier* remains the deferred mechanical derivation (Working Notes below); the canonical-direction identity here is the structural half that does not wait on it.

Hypothesis (heterogeneous-gain falsification handle)

In the heterogeneous case ($\Delta K \neq 0$, where the means-sum produces genuine closure defect $\varepsilon_x > 0$), the symmetry breaks and the means-sum is *no longer* the top canonical direction: the IB-optimal 1D projection is a rotation of it, and — if the IB reading of (P1) is right — that rotation achieves strictly lower ε_x than the means-sum. This is a falsifiable consequence of the unification, checkable numerically in the existing two-Kalman machinery: if the rotated projection fails to improve ε_x when $\Delta K \neq 0$, the IB reading of (P1) is disconfirmed as more than shape-coincidence. Under the unification, means-sum admissibility is *conditional on gain symmetry* rather than free.

What stays separate from the IB frame. Three admissibility and structural conditions do *not* reduce to IB:

- **(P2) Lipschitz continuity.** Not an IB constraint. The bridge lemma in `#form-composition-closure` requires (P2) for analytic reasons (propagating bounded closure defect into bounded trajectory error); IB does not impose any continuity condition on compressors. (P2) remains a separate admissibility condition.
- **(P3) Dimensional reduction.** In the Gaussian-IB case relevant to composition, the IB-optimal T at any finite β typically uses full support of $\mathbb{R}^{\dim X}$; the categorical dimensionality reduction $\dim \mathcal{X}_c < \dim \mathcal{X}_{\text{micro}}$ is a *harder* condition than any rate constraint. (P3) remains separate. (In discrete cases it may be rate-implied, but the composition instance is Gaussian.)
- **Interventional relevance (Level 2).** The relevance variable in all four instances is associational (Y in a joint distribution with X and T). Strategy edges in the regime-indexed interpretation (`#scope-edge-update-causal-validity`) want interventional relevance for Regime A: "what edge (i, j) predicts *under* $do(i)$." This is a strictly stronger requirement than IB provides. Adapting IB to interventional relevance (causal IB, Wieczorek & Roth 2017 and follow-ups) is an extension direction, not a specialization of the master IB.

The honest slogan is therefore "the (P1)-analog in each compression operation is IB; regularity, dimensionality, and interventional relevance are separate conditions that compose with it."

Epistemic Status.

Robust-qualitative. The claim that the four compression operations share IB shape is *discussion-grade* — it is a presentational observation supported by examining each segment's formulation and noting structural alignment. The Σ_t source reformulation is a *formulation choice* replacing one underspecified source with a cleaner one; no derivation is needed because the new formulation and the old are not the same claim. The (P1) as Lagrangian-dual of IB is *derived* — rate-distortion duality is standard (see §I.12–13 of Cover & Thomas) and the constraint-form ↔ Lagrangian-form equivalence is mechanical.

Max attainable: *robust-qualitative* for the shared-shape claim; *derived conditional on rate-distortion duality* for (P1) as IB Lagrangian-dual; *formulation* for the Σ_t source reformulation. The strongest version of the unification claim — "U-strong," that the four operations are the same optimization problem with different bindings — is *not* established and is unlikely to be, for reasons in the Discussion.

This segment does not promote any of the four instance segments. Each remains the primary site for its own instance. What this segment adds is the shared-shape framing, the Σ_t source fix, and the (P1) derivation.

Discussion.

Why not go further. Two distinct claims could be made: (a) U-strong — the four operations are the *same* optimization problem with different bindings, so a single derivation specializes to each instance; (b) U-medium — the four share *shape*, vocabulary, and variational structure but differ in substantive ways (source type, relevance-variable availability, computability) that prevent a single derivation from covering all four. U-strong requires cross-instance theorems, results of the form "because these are all IB, property X holds across all four." No such theorems are identified. The shared IB shape is legibility-enhancing but does not produce deductions for free. This segment therefore states U-medium (shared shape) honestly rather than overclaiming U-strong.

Honest credit to the hierarchical-generative-model lineage. AAT's four compression operations are a structurally narrower family than the operations a hierarchical generative model in the predictive-coding lineage (Friston 2008, "Hierarchical models in the brain," *PLoS Comp. Biol.* 4; Friston 2010, "The free-energy principle: a unified brain theory?" *Nat. Rev. Neurosci.* 11; Clark 2013, "Whatever next?", *Behav. Brain Sci.* 36; Hohwy 2013, *The Predictive Mind*, OUP) could express. A hierarchical generative model layers compressions of compressions, with each layer producing a representation tuned to the next layer's prediction target — and the AAT compressions (M_t for prediction, Σ_t for guidance, shared intent for coordination, Λ for level-bridging) are all expressible within that frame as specific layer-bindings. What AAT adds is structural: (a) the *relevance variables* Y are made first-class with explicit per-instance bindings (see the table in §"The shared IB shape"); (b) the (P1)–(P3) admissibility conditions for composition give a measurable closure-defect bound that hierarchical-generative-model layering does not natively produce (`#form-composition-closure`); (c) regime-indexed edges (`#scope-edge-update-causal-validity`) introduce Pearl-Level-2 relevance for Regime A, which standard hierarchical generative models do not address. The shared family is real; AAT's additions are also real. The honest framing is "AAT's compressions are a structured subset of the hierarchical-generative-model family with additional structure load-bearing for AAT's specific results."

Why this is not the "biggest available unification." It is better described as the *most legible* unification. The pattern is already half-named at each instance site (three of four segments already write their objectives in IB form); the segment-level contribution is sharpening the naming and fixing one formulation issue (Σ_t source), not discovering a new structural result. The biggest leverage concision move in the theory — the sector-Lyapunov template in G — is substantively different: it removes repeated content and absorbs proofs that previously appeared in multiple segments. This segment adds legibility without removing much.

What each β means. The four trade-off parameters are not unified in value, only in role:

- $\beta(\rho, \pi)$ for M_t depends on environmental volatility and policy; fast-changing environments or high-value policies both warrant higher β.
- β_Σ for Σ_t is the cognitive cost per decision-relevant bit retained; bounded-context agents (LLMs with finite windows) have low β_Σ.
- The bandwidth β for shared intent is the communication-capacity-per-coordination-bit; low-bandwidth teams have low β.
- The $\beta(\epsilon_I)$ for (P1) is the rate-distortion Lagrange multiplier at tolerance ϵ_I.

These are four calibration problems, not one. A theory of *resource allocation across the four compression operations* — where a single cognitive budget is split across modeling, strategy, communication, and composition — is not currently in AAT, and the unification does not produce it. If such a theory is ever needed, it would tie the four βs together through an outer optimization; the spike flags this as an open direction.

Relationship to `#def-unity-dimensions` and `#result-unity-closure-mapping`. The unity dimensions parametrize rate-distortion curves for closure defects (`#result-unity-closure-mapping`). Under the present segment's framing, unity

dimensions are *curve parameters for the* Λ *instance of the IB schema*. This tightens the existing `#result-unity-closure-mapping` Discussion, which already presented the rate-distortion reading as the correct interpretation; it is now stated uniformly across the four compression operations.

Relationship to R. The graph-structure result shows that operational postulates suffice for the *shape* of Σ_t, a Markov DAG (R; the sufficiency half of a Cox analog). The present segment characterizes the *compression level* of Σ_t. These operate at different levels: the graph-structure result supplies the representational type (the compressed T is a DAG), IB supplies the rate-distortion trade-off within that type (how richly the DAG is populated). The two arguments are independent and compatible — IB-compressed Σ_t is a DAG-with-fewer-nodes, not a non-DAG.

Position relative to the additive-coordinate-forcing pattern. The IB Lagrangian's $I(X;T) - \beta \cdot I(T;Y)$ is an additive decomposition on a log-scale coordinate (mutual information is logarithmic by construction) — the same shape that `#disc-additive-coordinate-forcing` catalogs at the chain / divergence / update layers. The IB case is classified as an *adjacent family member* rather than a fourth primary instance because `#form-information-bottleneck` adopts the Lagrangian form from Tishby, Pereira & Bialek 1999 as an applied external theorem rather than re-deriving it under an AAT-internally-motivated additivity axiom. The Shore-Johnson 1980 system-independence axioms that axiomatize IB's additivity are cited in U §6.1; if a future framing pass promotes Shore-Johnson to an explicit AAT axiom, IB would move from adjacent member to fourth primary instance.

Appendix AD

Derivation: Observation-Ambiguity Bias-Bound Constant C

The observation-ambiguity bias bound carried by Class 3 (Coupled) agents in the logogenic-agents scope — $\|\Delta M_{\text{bias}}\| \leq C \cdot \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})$ (`#scope-observation-ambiguity-modulation`, `#result-section-ii-survival`) — previously treated the constant C as "domain-dependent" and the bound as "order-of-magnitude guidance, not a theorem." This appendix derives C under two named sub-scopes, records a no-go showing that C cannot be universal without the (PI) parameterization-invariance axiom of `#scope-agent-identity`, and documents two failed derivation routes so future agents do not repeat them.

Formal Expression.

§1 — Setup and the audit before strengthening. Let $M_{\tau^-} \in \mathcal{M}$ be the pre-update epistemic substate. Let e_τ be the event triggering update, G_t the goal substate, and Ω_τ the latent world-state. At the **causally-disciplined** baseline ($\kappa_{\text{processing}} = 0$ — Class 1 by construction or an idealized Class 2 in effect, equally; `#der-directed-separation`), the update $f_M(M_{\tau^-}, e_\tau)$ is goal-blind: $M_{\tau^+}^{\text{decoupled}} = f_M(M_{\tau^-}, e_\tau)$, and the bias below is zero. In Class 3 (Coupled) scope, the coupled update $f_X^M(X_{\tau^-}, e_\tau)$ carries goal-conditional reweighting; the bias is

$$\Delta M_{\text{bias}} := f_X^M(X_{\tau^-}, e_\tau) - f_M(M_{\tau^-}, e_\tau) \tag{AD.1}$$

Pre-strengthening type audit. For the bound "$\|\Delta M_{\text{bias}}\| \leq C \cdot \kappa \cdot I$" to be well-typed:

1. **Norm on $\mathcal{M}$.** The LHS is a norm on model-space. $\mathcal{M}$ is the model space of `#form-agent-model`. Three candidate norms: Euclidean on parameters, total variation on induced measures, Fisher-Rao geodesic distance. **Under the (PI) parameterization-invariance axiom in `#scope-agent-identity`** (fourth primary instance of `#disc-additive-coordinate-forcing`), Euclidean-on-parameters is a coordinate artifact and Fisher-Rao is the canonical AAT-invariant choice on statistical-manifold sub-cases of $\mathcal{M}$ (Čencov 1982 uniqueness).

2. **Regularity of f_X^M.** For the bound to be a theorem, f_X^M must satisfy some regularity. The Bayesian-posterior model (prior + likelihood reweighting) is the canonical working class; it covers the attention-reweighting mechanism typical of Class 3 (Coupled) architectures to leading order.

3. **Information coordinate.** $I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})$ is in nats (or bits). Converting information to state-distance is exactly what the Pinsker / Bretagnolle-Huber / Otto-Villani / Bakry-Émery machinery does.

The bound is well-typed under (a) named norm, (b) bounded f_X^M regularity, (c) specified information-geometric inequality. Without (PI) adopted, C is norm-dependent; under (PI), Fisher-Rao is canonical and C becomes derivable.

Derived (w2-transport-track, conditional on H1-H3)

§2 — Track 1: Transport-inequality derivation under log-Sobolev + Lipschitz-posterior.

Named sub-scope (H1-H3).

- **(H1) Statistical-manifold sub-case.** Each $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over world-states. $\mathcal{M}$ is (locally) a statistical manifold.
- **(H2) Log-Sobolev inequality.** The observation distribution $P_{\Omega|e,M}$ satisfies a log-Sobolev inequality (LSI) with constant $\rho_{\text{LSI}} > 0$. Sufficient condition: $P_{\Omega|e,M}$ is strongly log-concave with constant K (Bakry-Émery 1985 curvature-dimension condition gives $\rho_{\text{LSI}} \geq K$). For Gaussian observation models, $\rho_{\text{LSI}} = 1/\sigma^2$ explicitly.
- **(H3) Lipschitz-posterior stability.** The Bayesian-posterior pushforward from observation to state is L_{post}-Lipschitz in W_2:

$$W_2(P_{M|e,G}, P_{M|e}) \leq L_{\text{post}} \cdot W_2(P_{\Omega|e,G}, P_{\Omega|e}) \tag{AD.2}$$

  Sufficient condition: well-posed Bayesian inverse problem with bounded log-likelihood Hessian (Stuart 2010 *Inverse problems: a Bayesian perspective*, *Acta Numerica* 19:451–559, Theorem 4.6; Hairer, Stuart & Vollmer 2014 *SIAM J. Math. Anal.* 46(1):415–451 for explicit W_2-bounds in infinite-dimensional settings).

Step 1 — KL from mutual information. By the chain rule of relative entropy (Cover & Thomas 2006 Theorem 2.5.3):

$$\mathbb{E}_G\big[\mathrm{KL}\big(P_{\Omega|e,M,G} \,\|\, P_{\Omega|e,M}\big)\big] = I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \tag{AD.3}$$

Exact identity.

Step 2 — Otto-Villani under LSI. From (H2) and Otto & Villani 2000 *J. Funct. Anal.* 173(2):361–400 Theorem 1:

$$W_2^2\big(P_{\Omega|e,M,G}, P_{\Omega|e,M}\big) \leq \frac{2}{\rho_{\text{LSI}}} \cdot \mathrm{KL}\big(P_{\Omega|e,M,G} \,\|\, P_{\Omega|e,M}\big) \tag{AD.4}$$

Taking expectation over G and substituting Step 1:

$$\mathbb{E}_G\big[W_2^2(P_{\Omega|e,M,G}, P_{\Omega|e,M})\big] \leq \frac{2}{\rho_{\text{LSI}}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) \tag{AD.5}$$

Step 3 — Lipschitz-posterior pushforward. From (H3):

$$\mathbb{E}_G\big[W_2^2(P_{M|e,G}, P_{M|e})\big] \leq L_{\text{post}}^2 \cdot \mathbb{E}_G\big[W_2^2(P_{\Omega|e,G}, P_{\Omega|e})\big] \tag{AD.6}$$

Step 4 — $\kappa_{\text{processing}}$ factor. The $\kappa_{\text{processing}}$ coefficient from `#der-directed-separation`'s Class 1/2/3 (Separated/Partial/Coupled) taxonomy multiplies the goal-conditional reweighting strength by the modularity coefficient. For Class 3 (Coupled), $\kappa_{\text{processing}} \approx 1$; for Class 2 (Partial) with modularity κ, the effective bound carries a factor κ multiplicatively on the information term.

Result (Track 1). *[Derived, conditional on (H1)-(H3)]*

$$\boxed{\mathbb{E}\big[W_2^2(M_{\tau+}^{\text{coupled}}, M_{\tau+}^{\text{decoupled}})\big] \;\le\; \frac{2L_{\text{post}}^2}{\rho_{\text{LSI}}} \cdot \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau-})} \tag{AD.7}$$

The constant is explicit: $C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$, linear in I, with geometric interpretation:

- L_{post} grows with *prior-likelihood tension* — ill-conditioned inverse problems amplify bias.
- ρ_{LSI} grows with *observation-distribution concentration* — sharper observations yield tighter bounds.
- The ratio $2L_{\text{post}}^2/\rho_{\text{LSI}}$ captures the **geometric stiffness** of the update: small-ρ/large-L = soft, goal-sensitive updates; large-ρ/small-L = stiff updates that resist goal-bias.

§3 — Track 2: Fisher-Rao derivation under (PI) + Čencov + small-information regime.

Derived (fisher-rao-track, conditional on H1 + H4)

Named sub-scope (H1 + H4).

- **(H1) Statistical-manifold sub-case** — as in §2. Under the (PI) parameterization-invariance axiom of `#scope-agent-identity`, Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS) forces the Fisher information metric as the canonical Riemannian metric on $\mathcal{M}$ up to global scale (Ay, Jost, Lê & Schwachhöfer 2017 *Information Geometry*, Theorem 5.1).
- **(H4) Small-information regime.** $I(G; \Omega_\tau \mid e_\tau, M_{\tau-}) \ll 1$ nat. The second-order Taylor expansion of KL at coincident distributions is sharp.

Step 1 — KL-to-Fisher-squared-distance identity. For nearby distributions P, Q on a statistical manifold with Fisher metric $\mathbf{I}$, the KL divergence admits the second-order expansion

$$\mathrm{KL}(P \| Q) \;=\; \frac{1}{2} \cdot d_{FR}^2(P, Q) + O(d_{FR}^3) \tag{AD.8}$$

where d_{FR} is the Fisher-Rao geodesic distance (Cover & Thomas 2006 §12.5; Amari & Nagaoka 2000 §3.7 Theorem 3.1). This is the infinitesimal form of the Bregman divergence on the exponential family's dual geometry (cf. U §6.3 for the related Fenchel-Bregman identification).

Step 2 — Posterior-displacement in Fisher-Rao. Under (H1)+(H4) + Bayesian-posterior-class f_X^M, the goal-conditional posterior and goal-marginalized posterior are nearby points on the statistical manifold; their Fisher-Rao distance satisfies

$$d_{FR}^2\big(M_{\tau+}^{\text{coupled}}, M_{\tau+}^{\text{decoupled}}\big) \;\le\; 2 \cdot \mathrm{KL}\big(P_{M|e,G} \,\|\, P_{M|e}\big) + O(d_{FR}^3) \tag{AD.9}$$

Using the data-processing inequality $\mathrm{KL}(P_{M|e,G} \| P_{M|e}) \le \mathrm{KL}(P_{\Omega|e,G} \| P_{\Omega|e})$ (Bayesian posterior is a pushforward), and Step 1 of §2 ($\mathbb{E}_G[\mathrm{KL}(P_{\Omega|e,M,G} \| P_{\Omega|e,M})] = I$):

$$\mathbb{E}\big[d_{FR}^2(M_{\tau+}^{\text{coupled}}, M_{\tau+}^{\text{decoupled}})\big] \;\le\; 2 \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau-}) \cdot (1 + o(1)) \tag{AD.10}$$

Step 3 — Taking square roots (with κ factor). Under Class 2 (Partial) modularity, $\kappa_{\text{processing}}$ enters multiplicatively on the goal-conditional reweighting strength, hence on the KL and Fisher-Rao distance:

Result (Track 2). *[Derived, conditional on (H1)+(H4), small-information regime]*

$$\boxed{\mathbb{E}\,\|\Delta M_{\text{bias}}\|_{FR} \;\le\; \sqrt{2}\cdot\sqrt{\kappa_{\text{processing}}\cdot I(G;\Omega_\tau \mid e_\tau, M_{\tau^-})}\cdot(1+o(1))} \tag{AD.11}$$

The constant is universal and dimension-free: $C_{FR} = \sqrt{2}$ (nats) or $\sqrt{2\ln 2}$ (bits). **No domain-specific parameters.** The (PI) commitment eliminates the coordinate-dependence that made C ambiguous under Euclidean-parameter norms.

Scaling difference between tracks. Track 1 gives $W_2^2 \propto I$ (linear in information). Track 2 gives $d_{FR}^2 \propto I$, equivalently $d_{FR} \propto \sqrt{I}$ (square-root in information). The two tracks are not contradictory — they are bounds under different assumptions in different metrics. Track 1 is tight in the large-I regime where Otto-Villani's linear form dominates; Track 2 is tight in the small-I regime where the Fisher-metric second-order expansion is sharp. The two compose at intermediate scales by taking the tighter of the two under the local curvature conditions.

Proved (no-universal-C-euclidean, counterexample-grade)

§4 — No-go result: universal C under Euclidean-parameter norm fails.

Claim. No universal constant C — independent of $\mathcal{M}$'s geometry, parameterization, and coupled-update structure — exists such that $\|\Delta M_{\text{bias}}\|_{\text{Euclidean}} \le C\cdot\kappa\cdot I$ under the Euclidean norm on an arbitrary parameter vector.

Counterexample. Let $\mathcal{M} = \{N(0,\sigma^2) : \sigma > 0\}$ parameterized by σ. The Fisher information in σ scales as $\mathbf{I}(\sigma) = 2/\sigma^2$ — small near large σ, large near small σ. For any $C_0 < \infty$, choose σ_0 large enough that the Fisher-Rao displacement corresponding to $I(G;\Omega) = 1$ nat of goal-conditional reweighting translates to Euclidean-σ displacement $\sigma_0/\sqrt{2} > C_0$. Specifically, by Track 2's $\sqrt{2}$ bound:

$$d_{FR}(M^{\text{coupled}}, M^{\text{decoupled}}) \le \sqrt{2}\cdot 1 = \sqrt{2} \tag{AD.12}$$

Under the Fisher metric $\mathbf{I}(\sigma) = 2/\sigma^2$, the Fisher-Rao line element is $ds^2 = 2d\sigma^2/\sigma^2$, so $\Delta\sigma \approx \sigma\cdot d_{FR}/\sqrt{2} = \sigma$ (Euclidean displacement scales linearly in σ). Taking $\sigma\to\infty$ gives arbitrarily large Euclidean-σ displacement for fixed $I = 1$ nat. No universal C in Euclidean-parameter norm exists.

Implication. The no-go strengthens the (PI) commitment rather than weakening the bound: **under (PI), Fisher-Rao is the canonical norm and $C = \sqrt{2}$ is universal; without (PI), C does not exist as a universal constant.** The (PI) axiom is *load-bearing* for this derivation, not coincidental.

Position relative to `#disc-identifiability-floor`. The no-go has the shape of a floor-pattern no-go (external obstruction: Euclidean-parameter norms carry unbounded Fisher condition numbers; escape: (PI) + Fisher-Rao gives universal C), but it does not match the five-element test for a floor instance (see `#disc-identifiability-floor`): it has a *single* escape (the (PI) adoption), not ≥ 2 distinct escapes, and its strengthened consequence is *re-use* of existing (PI)+Čencov machinery (fourth primary instance of `#disc-additive-coordinate-forcing`) rather than elevation of new machinery. The honest position: this no-go is a **downstream theorem of the (PI) commitment**, not a new floor instance. It belongs in this appendix as motivating justification for why (PI) is load-bearing for the derivation, not as a separate meta-segment entry. (The floor-vs-coordinate-forcing distinction is articulated explicitly in `#disc-additive-coordinate-forcing` §"Downstream consequences of the (PI) commitment".)

§5 — Failed derivation routes (honest record). Two derivation routes were attempted and failed at a structural level. Documented here so future agents do not re-attempt them without new evidence.

(F1) Cramér-Rao inversion — wrong direction. Attempt: treat G as a parameter estimated from Ω given (e, M); use the Cramér-Rao lower bound $\mathrm{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ inverted to bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a posterior-sensitivity factor. Failure: Cramér-Rao bounds estimator error *below*, not above; the bound goes in the opposite direction to what the bias bound needs. Inverting Cramér-Rao for an upper bound on bias propagation requires a fixed-estimator-class assumption that is not available for a Class 3 (Coupled) agent's attention-reweighting mechanism. The transport-inequality route of §2 is the correct machinery.

(F2) Rate-distortion inversion — wrong problem structure. Attempt: apply Shannon's rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2}\log(2\pi e D)$ (Gaussian source), invert to $D \leq \sigma^2 \cdot 2^{-2(R-h(X))}$ with "bits in" = $I(G; \Omega \mid e, M)$. Failure: rate-distortion theory describes the minimum bits-per-symbol required to represent a source within distortion D; it does **not** describe the maximum distortion induced by injecting side-information into an update. Source-coding theorems are about optimal representations of a source, not spatial displacement induced by side-information injection. The direction is again wrong. Transport inequalities (Attempt B) are the correct machinery.

The pattern shared across (F1) and (F2): **information-theoretic source-coding theorems (Cramér-Rao bounds, rate-distortion lower bounds) are structural lower-bound machinery for an estimator's or encoder's performance; they cannot be inverted to produce upper bounds on displacement induced by side-information.** Upper bounds on displacement require transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, posterior-stability).

§6 — Gaussian worked example.

Illustrative, exact under specified assumptions

Setup. Gaussian observation model $P_{\Omega|e,M} = N(\mu_M, \sigma^2 I)$ with fixed σ^2; conjugate-Gaussian prior $P_M = N(0, \tau^2 I)$; goal-conditional likelihood reweighting $P_{\Omega|e,M,G} = N(\mu_M + \beta(G), \sigma^2 I)$ where $\beta(G)$ is a goal-dependent shift with $\|\beta(G)\|^2$ bounded by the information budget $I(G; \Omega_\tau)$.

LSI constant. Gaussian $N(\mu, \sigma^2 I)$ satisfies LSI with $\rho_{\text{LSI}} = 1/\sigma^2$ (Bakry-Émery; direct Hessian bound on $-\log p$).

Posterior-Lipschitz constant. Conjugate-Gaussian posterior: $P_{M|\Omega} = N(\mu_{\text{post}}, \Sigma_{\text{post}})$ with $\mu_{\text{post}} = \tau^2/(\tau^2 + \sigma^2) \cdot \Omega$. Lipschitz-in-$\Omega$ constant $L_{\text{post}} = \tau^2/(\tau^2 + \sigma^2) < 1$. Under W_2-Lipschitz (Hairer-Stuart-Vollmer 2014), $L_{\text{post}} \leq 1$ for well-conditioned priors.

Track 1 bound (numerical). $C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}} = 2\sigma^2 \cdot \tau^4/(\tau^2 + \sigma^2)^2$. In the prior-dominant limit ($\tau^2 \ll \sigma^2$): $C_{W_2}^2 \approx 2\tau^4/\sigma^2$ (tight prior amplifies bias). In the likelihood-dominant limit ($\sigma^2 \ll \tau^2$): $C_{W_2}^2 \approx 2\sigma^2$ (sharp observations limit bias).

Track 2 bound (numerical). Under (PI) + Fisher-Rao on the Gaussian mean-manifold with metric $\mathbf{I}(\mu) = I/\sigma^2$: $d_{FR}(\mu_1, \mu_2) = \|\mu_1 - \mu_2\|/\sigma$. Under small-$I$: $\mathbb{E}\, d_{FR}(\Delta M_{\text{bias}}) \leq \sqrt{2 \cdot \kappa \cdot I}$. In Euclidean-on-$\mu$ norm: $\|\Delta\mu_{\text{bias}}\| \leq \sigma \cdot \sqrt{2\kappa I}$ — the σ prefactor is the Fisher-Rao-to-Euclidean conversion. The Euclidean form is coordinate-dependent (per Attempt E §4); Fisher-Rao form is universal.

Comparison at operating point. For $\tau = \sigma = 1$ (balanced): Track 1 gives $C_{W_2}^2 = 2 \cdot (1/4) = 1/2$, so $\mathbb{E}\, W_2 \leq \sqrt{I/2}$. Track 2 gives $\mathbb{E}\, d_{FR} \leq \sqrt{2I}$. The two are within a factor of 2 of each other at balanced scale; they diverge in the ill-conditioned limits where Track 1's L_{post} or ρ_{LSI} blows up while Track 2 remains dimension-free.

What Is Derived vs. What Is Chosen.

Table AD.1

Property	*Source*	*Strength*
Chain-rule identity $\mathbb{E}_G[\mathrm{KL}(P_{\Omega\|G}\|P_\Omega)] = I(G;\Omega \mid \cdot)$	Cover & Thomas 2006 Theorem 2.5.3	Exact
Pinsker's inequality $\|P-Q\|_{TV} \le \sqrt{\frac{1}{2}\mathrm{KL}}$	Standard (Csiszár-Körner)	Exact
Otto-Villani $W_2^2 \le (2/\rho_{\mathrm{LSI}})\mathrm{KL}$ under LSI	Otto & Villani 2000 Theorem 1; Bakry-Émery 1985 curvature-dimension for LSI	Exact
KL-to-Fisher-squared-distance $\mathrm{KL}(P\|Q) = \frac{1}{2}d_{FR}^2 + O(d_{FR}^3)$	Cover-Thomas 2006 §12.5; Amari-Nagaoka 2000 §3.7 Theorem 3.1	Exact (standard information geometry)
Bayesian-posterior W_2-Lipschitz stability L_{post}	Stuart 2010 Theorem 4.6 under well-posed inverse problem	Conditional on (H3)
Track 1 bound $\mathbb{E}[W_2^2(M^{\mathrm{coupled}}, M^{\mathrm{decoupled}})] \le (2L_{\mathrm{post}}^2/\rho_{\mathrm{LSI}}) \cdot \kappa \cdot I$	Composition of §2 Steps 1–4	**Derived (conditional on H1-H3)**
Track 2 bound $\mathbb{E}\,\|\Delta M_{\mathrm{bias}}\|_{FR} \le \sqrt{2} \cdot \sqrt{\kappa I}(1+o(1))$	Composition of §3 Steps 1–3 under (PI) + Čencov	**Derived (conditional on H1+H4, small-I regime)**
$C_{FR} = \sqrt{2}$ is universal, dimension-free, no domain constants	§3 Result	Exact under (H1)+(H4)
$C_{W_2}^2 = 2L_{\mathrm{post}}^2/\rho_{\mathrm{LSI}}$ with explicit geometric interpretation	§2 Result	Exact under (H1)-(H3)
Attempt E no-go: no universal C in Euclidean-parameter norm	Heteroscedastic-normal counterexample §4	Proved (counterexample-grade)
(F1) Cramér-Rao inversion fails	Direction mismatch (estimator lower bound cannot invert to upper bound)	Recorded failure
(F2) Rate-distortion inversion fails	Problem-structure mismatch (source-coding theorem cannot yield side-information injection bound)	Recorded failure
Gaussian worked example with explicit C_{W_2} and C_{FR}	Conjugate-Gaussian direct computation	Exact under specified assumptions

The dividing line: both tracks are derived theorems under named hypotheses; the hypotheses (LSI, Lipschitz-posterior, small-I, statistical-manifold sub-case, (PI) adoption) are either standard mathematical regularity (LSI, Lipschitz-posterior) or AAT-internal axioms ((PI), already adopted in `#scope-agent-identity` and elevated to fourth primary instance of `#disc-additive-coordinate-forcing`). The no-go §4 justifies the (PI) commitment as load-bearing rather than coincidental. The failed attempts §5 document structural reasons two alternative routes cannot work, preventing future re-attempts.

Epistemic Status.

Conditional — exact under H1-H3 (Track 1) or H1+H4 (Track 2). Both tracks are derived theorems under named hypotheses:

- Track 1 is exact under the three regularity conditions (H1 statistical-manifold sub-case; H2 log-Sobolev on observation distribution; H3 Lipschitz-posterior stability). All three are standard mathematical regularity conditions satisfied for canonical cases (Gaussian observations: (H2) direct; well-posed Bayesian inverse problems: (H3) per Stuart 2010). The bound is linear in I, with explicit geometric-stiffness constants.

- Track 2 is exact under (H1) + (H4) small-information regime, leveraging (PI) + Čencov. The bound is $\sqrt{I}$ scaling with universal dimension-free constant $C_{FR} = \sqrt{2}$. Does not depend on domain-specific L_{post} or ρ_{LSI}; the price is the small-I regime restriction.

The no-go §4 is *proved* via the heteroscedastic-normal family counterexample — universal C in Euclidean-parameter norm does not exist; the (PI) commitment is load-bearing for the derivation. The failed attempts §5 are *recorded* at structural-reason level; they document why Cramér-Rao inversion and rate-distortion inversion cannot produce the needed upper bound, not merely that the attempts failed.

Upgrade over prior formulation. The bias bound in `#result-section-ii-survival` and `#scope-observation-ambiguity-modulation` previously carried the honest admission *"Without the constant, the bound is order-of-magnitude guidance, not a theorem."* This appendix **promotes the bound to theorem-level** under named sub-scopes. The order-of-magnitude admission is superseded by the two explicit tracks with their H-conditions. Updates to the logogenic-agents segments cross-reference this appendix.

Max attainable: *exact* for the composition of the three textbook inequalities (Step 1 chain-rule; Step 2 Otto-Villani; Step 3 Lipschitz-posterior or KL-to-Fisher-Rao identity) and for the no-go counterexample §4. *Conditional* overall — the theorem statements carry their H-conditions explicitly, and the sub-scope boundary is where "derived" transitions back to "admitted empirical regularity" (cases where H2 or H3 fails require separate treatment, flagged in Working Notes).

Honest failure modes.

The appendix does **not** cover:

1. **Non-statistical-manifold sub-cases of $\mathcal{M}$.** If $\mathcal{M}$ carries structured state-space geometry (mixed continuous/discrete components, graphical-model structure, non-parametric models without sufficient statistics), (H1) is weakened and Čencov's uniqueness does not straightforwardly apply. The Fisher-Rao track degrades; Track 1 may still apply under suitable alternative regularity.

2. **Unbounded KL / structural-failure limit.** If $P_{\Omega|e,M,G}$ has support disjoint from $P_{\Omega|e,M}$ at G-values carrying non-trivial prior probability, the KL (and thus I) diverges. The bound is vacuous in this regime. This is a structural failure of the coupled-update model, not a bound weakness.

3. **Non-Lipschitz pushforward.** If f_X^M is not a well-posed Bayesian posterior (e.g., ill-posed inverse problem, degenerate likelihood, rank-deficient observation channel), (H3) fails and Track 1 does not apply. Track 2's small-I expansion may still apply in the local-linearization sub-region.

4. **Large-information regime beyond the small-I expansion.** Track 2's $\sqrt{I}$ scaling uses the second-order Fisher-metric expansion of KL; the full global relationship is lattice-dependent. A fully-global Fisher-Rao statement requires either compact-manifold assumption (Fisher-Rao distance bounded) or a staged-application argument (many small-I steps composed).

5. **Heavy-tailed observation distributions.** LSI fails for heavy-tailed measures. Track 1 degrades to the Pinsker bound $W_2 \leq \sqrt{\text{KL}/(2\rho_{\text{LSI}})}$ when $\rho_{\text{LSI}} \to 0$; this is the $O(\sqrt{\varepsilon})$ baseline without the LSI tightening.

Discussion.

Position in the `#disc-additive-coordinate-forcing` meta-pattern. Track 2's $C_{FR} = \sqrt{2}$ result is a **downstream theorem of the fourth primary instance** (Čencov-invariance at the metric layer, under (PI)). Once (PI) adoption forces Fisher-Rao as the canonical metric on statistical-manifold sub-cases of $\mathcal{M}$, the KL-to-Fisher-squared-distance identity supplies the universal constant "for free" in the small-I regime. The no-go §4 tightens the connection: the (PI) commitment is not just "useful axiom" but load-bearing for the bound's theorem-level status — without (PI), C does not exist as a universal constant.

Connection to AL. Both segments use Pinsker's inequality as a first step in propagating a KL bound. After Pinsker, the two segments diverge: this segment's cascade (Pinsker → Otto-Villani → Lipschitz-posterior → W_2 on pushforward) is a KL-to-state-distance machinery; `#deriv-variational-sector-condition`'s use (Pinsker + Cauchy-Schwarz → scalar sector-constant degradation) is a KL-to-scalar-constant machinery. The shared first step is Pinsker; the post-Pinsker cascade is not shared, so `#deriv-variational-sector-condition` is an adjacent family member rather than a candidate for a shared `#posterior-displacement-template` (a possible future extraction recorded in Working Notes).

The $\varepsilon(\kappa)$ degradation for causally-disciplined results. This appendix *is* the degradation bound that the causal-discipline scoping rule of `#der-directed-separation` calls for: a Part II result that holds exactly at $\kappa_{\text{processing}} = 0$ (for any causally-disciplined agent) degrades off that zero at a rate this bound controls — $\|\Delta M_{\text{bias}}\|$ enters multiplied by $\kappa_{\text{processing}}$, exact (zero bias) at the causally-disciplined value and decaying controllably as coupling grows. The first realized client is the value object's Q_O G_t-independence (`#def-value-object`): the Q_O degradation at $\kappa_{\text{processing}} > 0$ is the ΔM_{bias} this appendix bounds, propagated through the value functional. That a successful $\varepsilon(\kappa)$ bound exists is itself the evidence that the result-at-the-noun (causal discipline, the shared zero) is the right object — the result degrades continuously off the zero rather than requiring the architecture certificate.

Forward-looking clients. Three open extensions in `#disc-identifiability-floor` Working Notes are candidate future primary clients of this appendix's machinery: causal-IB (KL → W_2 on post-intervention pushforward), misspecification-cost (KL → W_2 on posterior under model misspecification), composition-scope-robustness (KL → W_2 on composite-state pushforward). Each would reuse the Track 1 Otto-Villani + Lipschitz-posterior cascade with client-specific (P, Q, f) and $(L_{\text{post}}, \rho_{\text{LSI}})$ instantiations. If two or more of these materialize, factoring the cascade out as a template segment becomes attractive.

Why the Fisher-Rao specialization is worth landing even under the small-I restriction. In the AAT-canonical operating regime, Class 3 (Coupled) bias is *moderate* — the goal-conditional reweighting shifts the posterior but does not replace it. Small-I is the honest model of this regime; large-I is the structural-failure limit where the coupled agent's goal-conditioning overwhelms evidence. Track 2's universal $C_{FR} = \sqrt{2}$ is the right bound in the operating regime; Track 1 is the fallback under structural conditions where I is not small or where Fisher-Rao is not available.

Connection to logogenic agents. This appendix's bounds are the quantitative apparatus underlying `#scope-observation-ambiguity-modulation`'s product-form bias bound and `#result-section-ii-survival`'s Class 3 (Coupled) scope-survival analysis. Both segments previously carried the "constant C not computed" caveat; with this appendix landed, C is computed under named conditions and the epistemic tier of the bias bound upgrades to "conditional (exact under H1-H3 or H1+H4)." The logogenic-agents segments retain their client status; the quantitative work moves here.

Findings.

Universal Constant for the Coupled-Agent Bias Bound under Parameterization-Invariance. **Brief:** When a goal-conditioned agent (like an LLM) processes evidence, its update can be biased by what it's trying to do, not just what it observes. Prior work could only say "this bias is bounded by some constant times the relevant information leakage" without computing the constant. This result computes it under two specific conditions, and shows that at least one of them — committing to a particular geometric structure on the model space (the (PI) parameterization-invariance axiom) — is necessary in a strong sense: without it, no universal constant exists at all. A transport-inequality track gives

$C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$ linear in the goal-conditional information leakage, under log-Sobolev concentration of the observation distribution and Lipschitz-stability of the Bayesian posterior. A Fisher-Rao track gives a universal dimension-free $C_{FR} = \sqrt{2}$ in the small-information regime, conditional on the (PI) axiom. Two structurally wrong derivation routes (Cramér-Rao inversion; rate-distortion inversion) are recorded as honest failure cases.

Impact: Promotes the bias bound from "order-of-magnitude guidance, not a theorem" to a conditional theorem under explicitly named hypotheses, closing the quantitative gap that `#scope-observation-ambiguity-modulation` and `#result-section-ii-survival` previously carried as a caveat. The no-go strengthens the (PI) commitment from useful axiom to load-bearing structural choice and adds a fourth primary instance to the `#disc-additive-coordinate-forcing` meta-pattern. The recorded failures prevent future agents from re-attempting Cramér-Rao or rate-distortion inversions for upper bounds on side-information-induced displacement, and identify transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, posterior-stability) as the correct family. Three forward-looking clients in adjacent appendices (causal-IB pushforward, misspecification-cost, composition-scope-robustness) are candidate consumers of the same Track 1 cascade, suggesting eventual extraction of a shared `#posterior-displacement-template` if two or more materialize.

Novelty Claim: *Claim differentiation* on the Lipschitz-posterior + Otto-Villani composition for AAT's coupled-agent bias bound, plus *claim novelty* on the no-go counterexample showing that universal C in Euclidean-parameter norms cannot exist, which jointly elevates the (PI) axiom from convergent representational choice to load-bearing for theorem-level status. The two upstream tracks each compose textbook information-geometric / transport-inequality machinery (Cover-Thomas chain rule; Otto-Villani 2000; Stuart 2010 Bayesian-posterior stability; Amari-Nagaoka 2000 Fisher-metric expansion); the contribution is the composition for the Class 3 (Coupled) agent bias setting and the no-go that justifies the geometric commitment. *(This finding's novelty is the supporting quantitative apparatus; the integrated $\kappa \times \mathcal{A}$ bias law it supports is in* `#scope-observation-ambiguity-modulation`*, where the Pillar-3 verdict applies.)*

Related Work:

Table AD.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
KL-to-W_2 transport under log-Sobolev	Otto & Villani 2000 *J. Funct. Anal.* 173:361–400 (published 2000); Bakry & Émery 1985 curvature-dimension condition	*formal antecedent* — adopted directly in the Track 1 cascade; standard transport-inequality machinery, cited and used
Lipschitz-stability of Bayesian posteriors	Stuart 2010 *Acta Numerica* 19:451–559 (published 2010); Hairer, Stuart & Vollmer 2014 *SIAM J. Math. Anal.* 46(1):415–451 (published 2014)	*formal antecedent* — adopted directly as (H3); the conjugate-Gaussian case is computed explicitly in §6
Čencov uniqueness of the Fisher metric	Čencov 1982, *Statistical Decision Rules and Optimal Inference* (published 1982); Ay, Jost, Lê & Schwachhöfer 2017 *Information Geometry* Theorem 5.1 (published 2017)	*formal antecedent* — adopted via the (PI) axiom in `#scope-agent-identity` and used in the Track 2 derivation; the no-go in §4 is the AAT-internal demonstration that this adoption is load-bearing for the bound, not just convenient
KL-to-Fisher second-order expansion	Amari & Nagaoka 2000, *Methods of Information Geometry* §3.7 Theorem 3.1; Cover & Thomas 2006 §12.5	*formal antecedent* — supplies the small-I regime sharpness used in Track 2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Information-processing-constraint bounds on decisions	Ortega & Braun 2012 *Proc. R. Soc. A* (published 2012-04, found 2026-04 via Undermind); Genewein et al. 2015 *Front. Robot. AI* 2:27 (published 2015-11, found 2026-04)	*adjacent literature* — addresses a different quantity (decision quality under information constraint rather than belief-update bias under goal-coupling); shares the information-theoretic vocabulary but does not produce a Wasserstein or Fisher-Rao bias bound for coupled-architecture agents
Variational bounds on posterior beliefs in active inference	Friston et al. 2017, "Active inference: a process theory" *Neural Computation* 29:1–49 (published 2017)	*adjacent literature* — uses overlapping transport-inequality machinery in a different framework; consistent with the Track 1 derivation but does not require the variational-free-energy commitment (cf. `#form-information-bottleneck` Discussion)

Novelty assessment. A prior-art search located bounded-rationality and information-cost literatures (Ortega & Braun 2012, Genewein et al. 2015) as the closest decision-theoretic precursors and confirmed no equivalent of the universal-C derivation under (PI). The verdict (Wholly Novel, Medium confidence) applies to the integrated $\kappa \times \mathcal{A}$ law downstream in `#scope-observation-ambiguity-modulation`; this segment's contribution is narrower (the constant derivation and the no-go) and inherits the same Medium confidence, with a follow-on targeted search recommended on formal posterior-displacement bounds in coupled-architecture agents.

Search Log:

- 2026-04 (*nominally comprehensive*): A comprehensive-search pass (Pillar 3 of the prior-art defense) located bounded-rationality and information-cost literatures (Ortega & Braun 2012, Genewein et al. 2015) as the closest decision-theoretic precursors and confirmed no equivalent of the universal-C derivation under (PI) was surfaced. The verdict (Wholly Novel, Medium confidence) applies to the integrated $\kappa \times \mathcal{A}$ law downstream in `#scope-observation-ambiguity-modulation`; this segment's narrower contribution (the constant derivation and the no-go) inherits the same Medium confidence, with a follow-on targeted search recommended on formal posterior-displacement bounds in coupled-architecture agents.

Appendix AE

Discussion: Partial-Coupling Pathways — A Sub-Typology Within Class 2

Class 2 (Partial Coupling) per `#der-directed-separation` is named structurally as "some shared infrastructure, some separate pathways" and operationalized as a scalar $\kappa_{\text{processing}} \in (0, 1)$ that aggregates over the un-modeled internal sub-structure. The scalar is distribution-dependent and tracks the *magnitude* of goal-coupling, but not its *kind*: it cannot distinguish the agent that uses goals to *attend* from the one that uses goals to *interpret*, or the agent biased by *what it wants to be true* from the one foreclosed by *what plan it has committed to*, or the additively-biased agent (whose coupling is wrappable by post-hoc debiasing) from the multiplicatively-entangled agent (whose coupling is not). These distinctions are operationally distinct — they recommend different repairs, predict different dynamical signatures, and gate different wrapping regimes.

This segment names the structural sub-space the scalar projects over. Class 2 partial coupling decomposes along three structurally-independent axes — *stage*, *source*, and *form* — yielding a parameterization within which the Class 3 (fully Coupled) corner and the Class 1 (Separated) corner are recovered as the two trivial endpoints. The stage axis tracks which step of the belief-update pipeline takes a G_t argument; the source axis tracks whether the coupling comes from the objective O_t or from the strategy Σ_t; the form axis tracks whether the coupling acts as separable additive bias on a stage's output (*content*) or as a non-separable change of the stage's functional form (*process*). The three axes have distinct operational consequences (Discussion §"Operational consequences" below), and the form axis at Class 2 unifies with the *structure*-vs-*behavior* refinement at Class 1 — they are the same distinction read at different architectural levels.

The treatment is discussion-grade at the structural-recognition level. The pipeline decomposition is posited as canonical for Bayesian-style updaters and serves as an interpretive lens for monolithic architectures. The substantive structural results that fall out — form-determines-wrappability (a content-form / process-form no-go cut), and the source asymmetry of belief-strategy attractors — are stated qualitatively here and derived in their own segments (`#der-belief-strategy-attractor` for the source-asymmetry result; the form no-go is stated formally in this segment's Discussion). The combinatorial parameterization is structurally complete but operationally too fine; the load-bearing distinctions live on the (form × source) projection with stage as a localization tag.

Formal Expression.

Pipeline decomposition of f_M.

Formulation (partial-coupling-pipeline)

Any belief-update function f_M that maintains M_t — whether Bayesian, Kalman, RL with separate world model, or LLM-as-tracker — factors functionally through a small set of operations. The canonical decomposition is

$$f_M = \tau \circ \alpha \circ \lambda \circ \phi \tag{AE.1}$$

with the four stages mapping to AAT machinery already in canon:

Table AE.1

Stage	*Operation*	*AAT correspondent*
P1 — Featurization	$\phi : \mathcal{E} \times \mathcal{M} \to \mathcal{X}$, $x = \phi(e_\tau, M_{\tau-})$	Extracts features from the realized event given current model; in a Kalman tracker, the innovation $\delta = o - \hat{o}$ per `#def-mismatch-signal`
P2 — Likelihood evaluation	$\lambda : \mathcal{X} \times \mathcal{M} \to \mathcal{L}$, $\ell = \lambda(x, M_{\tau-})$	Evaluates the likelihood of features under representable hypotheses; in a Kalman tracker, the Gaussian likelihood on δ with innovation covariance S
P3 — Aggregation	$\alpha : \mathcal{L} \times \mathcal{M} \to \mathcal{M}^+$, $M' = \alpha(\ell, M_{\tau-})$	Combines likelihood with prior to produce new belief; in a Kalman tracker, the gain-weighted update with $\eta^* = U_M/(U_M + U_o)$ per `#emp-update-gain`
P4 — Consolidation	$\tau : \mathcal{M}^+ \to \mathcal{M}$, $M_{\tau+} = \tau(M')$	Post-update transformation: storage, normalization, regularization, memory consolidation

Each of (P1)–(P4) can independently take or omit a G_t argument. The Class 1 case has none; the Class 3 case has all; Class 2 is everything between.

The decomposition is posited rather than derived. It is canonical for Bayesian-style belief-updating agents and serves as an interpretive lens for monolithic architectures (where the stages are not architecturally distinct in the forward pass) — the typology's precision drops the more monolithic the agent.

Definition (class-2-sub-type)

The (stage × source × form) parameterization.

A Class 2 partial-coupling sub-type is a triple $\mathcal{C}_2 = (S, R, F)$ with:

- $S \subseteq \{P1, P2, P3, P4\}$, $S \neq \varnothing$ and $S \neq \{P1, P2, P3, P4\}$ — the **stage set** at which goals enter f_M. Empty S recovers Class 1; full S (with all-process form) recovers Class 3.
- $R \subseteq \{O, \Sigma\}$, $R \neq \varnothing$ — the **source set** within $G_t = (O_t, \Sigma_t)$ that acts as the coupling input.
- $F : S \to \{\text{content}, \text{process}\}$ — the **form** at each coupled stage.

The endpoints recover canon: $S = \varnothing$ is Class 1; $(S, R, F) = (\{P1, P2, P3, P4\}, \{O, \Sigma\}, \text{process})$ is Class 3 (fully Coupled). The scalar $\kappa_{\text{processing}}$ is the projection of $\mathcal{C}_2$ onto the magnitude axis.

Definition (coupling-form)

Form classification — content vs process.

A stage operation $\xi(\cdot; G)$ is **content-form coupled** if there exist functions ξ^0 (goal-blind) and b_ξ (bias) such that

$$\xi(u; G) = \xi^0(u) + b_\xi(G; u) \tag{AE.2}$$

with b_ξ identifiable from a probing protocol that varies G at fixed u (specifically: a reference goal G_0 exists with $b_\xi(G_0; u) = 0$, or b_ξ is identifiable up to a u-dependent constant by varying G across finitely many probes).

A stage operation is **process-form coupled** if no such decomposition exists — the operation's *functional form* depends on G, not just its output. The multiplicative case $\xi(u; G) = \xi^0(u) \cdot h(G; u)$ is the most common process form (multiplicative attention coupling, gain modulation, precision weighting); the compositional case $\xi(u; G) = \xi^\dagger(u, G)$ with no separable factor is the starkest form.

The classification is *probing-protocol-dependent*: an analyst with reference-goal access may classify as content where an analyst without that access classifies as process. This is the same epistemic situation as the existing $\hat{\kappa}_{\text{processing}}$ behavioral estimator — the form classification inherits the same probe-dependence.

Form-determines-wrappability — the central operational claim.

Derived (content-process-wrappability, conditional on probing model)

For a Class 2 sub-type with coupling triple (S, R, F):

Positive half (content-form is post-hoc-wrappable). Stages with content-form coupling admit an *external debiasing wrapper* that requires only behavioral-probing access (no internal pipeline access). The wrapper estimates $b_\xi(G; u)$ from probes that vary G at fixed u, then subtracts the estimated bias in normal operation. The wrapped output converges to $\xi^0(u)$ as the estimator becomes consistent. This corresponds to the W_2 (partial-wrapping) regime at the agent level — "the query boundary still passes G_W to the component" with typed-output structuring extracting a goal-blind belief-update.

Negative half (process-form is not post-hoc-wrappable). Stages with process-form coupling do not admit such an external wrapper. The argument is identifiability-theoretic: under process-form there is no G_0 making $h \equiv 1$ (or, more generally, no u-independent reference goal recovering ξ^0); behavioral probes that vary G reveal goal-correlated differences but cannot collapse them to a G-invariant honest output. Repair requires either *stage replacement* (substituting a goal-blind version of the stage in the pipeline — requires substitutability access at the stage's input/output boundary) or *full-agent wrapping* (the W_1 strict-wrapping regime, operating at the whole-agent boundary).

The form distinction is therefore *load-bearing*, not ornamental — it gates the wrapping regimes available for coercion to Class 1.

Stage-cascade propagation.

Derived (cascade-propagation)

Goal-coupling at stage P_k contaminates all downstream stages $P_{k+1}, \ldots, P_4$ even when those downstream stages are individually goal-blind, because their inputs are functions of G through the cascade. The *naming* of a sub-type by its primary coupling stage refers to where the coupling originates; the *effect* propagates forward through the pipeline. Minimal repair (per stage-localization-of-repair, below) is therefore at the *most upstream coupled stage*, not at the most downstream symptom.

Stage-localization of repair.

Derived (stage-localization-of-repair, conditional on pipeline access)

If pipeline access is available at stages $P_k \in S$ — the wrapper can intercept the stage's input and supply its output to P_{k+1} — then substituting goal-blind operations P_k^0 at every coupled stage produces a wrapped agent that is Class 1 at the wrapper level. The construction has a capability cost (the substitute operations may be weaker than the agent's native ones) and a Brooks's-Law tempo overhead (more pipeline calls per macro-step), but the structural class-coercion is exact.

For monolithic architectures without pipeline access, only full-agent wrapping (the construction of `#der-class-coercion-via-wrapping`) is available; stage-level repair does not apply.

Composition with the leakage locus. Goal contamination's *effect on belief* is confined to the Fisher null space $\ker \mathcal{I}_\tau$ of the observation given the current latent state (the leakage-locus result, when promoted to its own segment, will be the formal statement). This locus is *universal across Class 2 sub-types* — it does not depend on which pipeline stage the coupling enters. The sub-type determines the *functional form* of the displacement within the locus:

- Content-form sub-types produce displacement linear in G (the $\Delta\mu = \Lambda_0^{-1} g(G)$ shape in the linear-Gaussian instantiation, with $g(G)$ a linear function of the goal-tilt parameter).
- Process-form sub-types produce displacement with a G-dependent gain matrix or a G-dependent covariance structure — non-linear in G.

- Σ-source sub-types compound the per-step displacement across the closed-loop dynamics; under multiplicative process form, self-stabilizing belief-strategy attractors arise (`#der-belief-strategy-attractor`).
- *O*-source sub-types produce exogenously-bounded tilt magnitude (because O_t revises only when forced, per `#der-orient-cascade`).

Wrapping-regime correspondence. The W_0 / W_1 / W_2 wrapping regimes of `#der-class-coercion-via-wrapping` are the agent-level analogs of the stage-level form distinction:

Table AE.2

Wrapping regime	*Stage-level analog*	*Class achieved at wrapper*
W_0 (no wrapping)	n/a — the agent stands as-is	Whatever the un-wrapped agent is
W_2 (partial wrapping)	Content-form at the agent level — wrapper passes *G* to component, post-hoc structures the response	Class-1-by-behavior (per `#der-directed-separation`)
W_1 (strict wrapping)	Process-form-with-pipeline-access — wrapper structurally substitutes goal-blind execution	Class-1-by-structure

This unifies the *structure*-vs-*behavior* distinction at Class 1 with the *content*-vs-*process* form distinction at Class 2: they are the same axis read at different architectural levels. The structure / behavior distinction at the Class 1 cell is the agent-level shadow of a sub-classification axis that runs all through Class 2.

Sub-type matching refines the wrapping-regime recommendation: a Class 2 agent with content-form coupling only is sufficient with W_2; a Class 2 agent with process-form coupling at any stage requires W_1 or remains effectively Class 2 under W_2; a Class 2 agent with Σ-source coupling may need a Σ-channel-suppressed W_1 even with pipeline access, because strategy-context can leak through stateful component channels independent of the query. The call-boundary condition under which W_1's structural guarantee survives such stateful leakage is the (C2′) condition of `#disc-w1-structural-bound-boundary`: when goal-correlated state crosses the call boundary, W_1's structural bound is unavailable and only a behavioral bound remains.

Epistemic Status.

Discussion-grade at the structural-recognition level. The pipeline decomposition is a *formulation choice* — canonical for Bayesian-style updaters and well-motivated by AAT's existing machinery, but not forced by AAT canon. The (stage × source × form) parameterization is derived from the choice. The form distinction is *robust qualitative* at the operational level (the content / process distinction has a clean identifiability argument); the formal definition is probing-protocol-dependent.

The two substantive structural consequences carry their own tiers:

- **Form-determines-wrappability** (this segment, Discussion §): *robust qualitative* — the identifiability argument is exact under the formal non-separability definition; the operational dependence on probing-protocol availability is honest scope.
- **Source asymmetry / belief-strategy attractors** (`#der-belief-strategy-attractor`): *conditional* — exact under the linearized dynamics and the posited multiplicative form for Σ-source aggregation-stage coupling; robust qualitative beyond.
- **Composition with leakage locus**: *derived* under the linear-Gaussian instantiation; robust qualitative more generally (it composes existing structural results).

- **Wrapping-regime correspondence**: *robust qualitative* — the structural correspondence is exact, but boundary cases (partial-information wrappers between W_1 and W_2) require per-case analysis.

The full $42 \times 2^{|S|}$-cell parameterization is structural completeness; operationally the load-bearing distinctions live on the coarsening to *upstream-most coupled stage* × *source-set* × *form*, which gives roughly 30 cells, of which empirically about a dozen are populated by recognizable phenomena (canonical phenomena placed in Discussion §"Canonical phenomena" below).

Discussion.

Why the form axis is the operationally-load-bearing one. Of the three axes — stage, source, form — the form axis is the one that gates wrappability: it determines whether external post-hoc debiasing suffices or whether stage replacement / full-agent wrapping is required. The source axis is the one that gates *dynamics*: Σ-source coupling closes a feedback loop through the orient cascade and can produce self-stabilizing attractors that O-source coupling cannot. The stage axis is the one that gates *architectural location of repair*: it determines where in the pipeline the substitution or debiasing must be applied. These three operational outputs from the typology — wrappability, attractor possibility, repair location — each correlate with one axis, which is why the three-axis decomposition is more informative than the scalar.

Unification with Class 1's structure-vs-behavior refinement. The Class 1 cell of `#der-directed-separation` admits a refinement: *Class-1-by-structure* (no G in the belief-update query — structural commitment by type signature) vs *Class-1-by-behavior* (the query passes G but the component is instructed not to use it — separation by compliance). The form distinction at Class 2 is the same axis: *content-form* is the behavior-side (the operation accepts G but its goal-shaped component is post-hoc-subtractable); *process-form* is the structure-side (the operation's functional form depends on G inseparably, requiring substitution). Read together, the structure / content side and the behavior / process side describe two distinct *modes* of goal-coupling — one in which the coupling can be neutralized by *not asking* (or by subtracting after asking), and one in which the coupling is constitutive of the operation.

Why the Σ-source / O-source asymmetry matters. The orient cascade (`#der-orient-cascade`) has Σ updated as a function of (M, O) and O revised only when forced. This makes Σ *endogenous* to M via the cascade — Σ updates from the new M that the agent's f_M produces — but leaves O *exogenous* to M in steady state. The asymmetry feeds back: a Class 2 agent with Σ-source coupling at any stage of f_M has the structure $M \to \Sigma \to f_M \to M$ closing a loop, while the same coupling sourced from O has no such closure (because O does not update from M per step). The result is that Σ-source coupling under multiplicative process-form admits *self-stabilizing fixed points* in which M stays misaligned with the environment indefinitely — the formal sunk-cost attractor of `#der-belief-strategy-attractor`. O-source coupling produces *bias* but not *runaway commitment*.

Canonical phenomena placed in the typology. The cells the typology produces are operationally meaningful:

Table AE.3

Phenomenon	*Stage(s)*	*Source*	*Form*	*Repair regime*
Goal-directed attention	(selection — pre-f_M)	O or Σ	—	None — formally allowed by `#der-directed-separation` scope condition
Motivated reasoning (identity-driven)	P1 + P2	O	Content (moderate) → Process (strong)	W_2 moderate; W_1 + identity-suppression strong
Sunk-cost commitment	P3	Σ	Process (multiplicative)	W_1 + Σ-suppression; admits attractors per `#der-belief-strategy-attractor`
Identity-protective consolidation	P4	O	Process (compositional)	W_1 + storage-protocol externalization

Phenomenon	*Stage(s)*	*Source*	*Form*	*Repair regime*
Frame coupling	P1	$O + \Sigma$	Process (compositional)	Multi-frame composition (not coercion to Class 1)
Wishful thinking	P3	O	Content (mild) → Process (strong)	W_2 mild; W_1 strong
Affect / urgency cascade-bypass	(cascade-bypass)	n/a (not stage-coupling)	n/a	External pacing scaffolding (per `#disc-adversarial-coupling-pressure` affect mechanism)
Attention-mediated transformer-LLM	P1+P2+P3	$O + \Sigma$ (prompt-mixed)	Process (compositional)	Full Class-3 — W_1 wrapping construction, no stage repair without internal access

The transformer-LLM case lands at the corner ($S = \{P1, P2, P3\}$ or $\{P1, P2, P3, P4\}, R = \{O, \Sigma\}, F \equiv$ process), which is the parameterization's structural corner near Class 3 (Coupled). The typology recovers canon at its endpoints. Classical confirmation bias is a *separate* failure mode: it is structurally $M_t^{\text{prior}} \to f_M$ self-coupling, not $G_t \to f_M$ coupling, and is allowed by `#der-directed-separation`'s scope condition (which has M_{τ^-} as an argument); it lives outside the (S, R, F) parameterization, except in the *trajectory-coupling* case where the prior has been historically shaped by G-coupled updates and the cumulative effect carries effective G-content. Trajectory coupling is a separate per-trajectory axis from the per-step typology.

Connection to the three operations on modularity state. The M4 meta-segment `#disc-modularity-state-dynamics` names three operations — truthification, strategic self-coupling, adversarial coupling pressure — that change modularity state. The typology here is *orthogonal* to those operations: the operations describe *transitions* through the parameterization (truthification reduces S or simplifies F; strategic self-coupling expands S and shifts F toward process; adversarial pressure expands S from the outside, targeting specific cells — identity-binding targets P1+P2 / O; sunk-cost-engineering targets P3 / Σ); the typology describes the *static structural sub-space* on which the operations act. Each adversarial-coupling-pressure mechanism corresponds to a specific cell, and the three-mechanism table is best read as the projection of the typology's static structure onto the *externally-driven-decreasing* leg of M4's three-operation pattern. M4 names the operations; this segment names the structural state-space.

Operational consequences — refined behavioral estimator. The existing $\hat{\kappa}_{\text{processing}}$ behavioral estimator (per `#der-directed-separation` §"Empirical estimator") aggregates over the typology. A refined estimator would distinguish:

- *Stage*: probing intermediate representations (where mechanistic-interpretability access exists) localizes the upstream-most coupled stage.
- *Source*: probing under O varied at fixed Σ vs Σ varied at fixed O separates the source set.
- *Form*: probing distinguishes mean shifts (content signature) from covariance shifts or functional-form changes (process signature) in the agent's responses across goal-variants.

The refinement targets *which sub-type* the agent inhabits, not just the aggregate magnitude. Per the composition with the leakage locus, all such probing should be confined to $\ker \mathcal{J}_\tau$ — the directions the observation does not identify — which is where the displacement lives.

Honest scope. The typology is most precise for *modular* Class 2 architectures (biological cortex per the canonical example; hybrid AI systems with separable preprocessing / inference / aggregation modules). For monolithic architectures (transformer LLMs, end-to-end-trained policies), the stage decomposition is *interpretive* — the stages are not architecturally distinct in the forward pass; the typology serves as a lens for mechanistic-interpretability investigations rather than as a structural classification. The form classification is *probing-protocol-dependent*; analysts should specify which probes were available. Content-form wrapping gives *Class-1-by-behavior*, not honesty: under gauge freedom

in the identifiability, the wrapper produces a G-invariant version of the agent's belief, up to an unknowable additive constant.

Findings.

Partial-Coupling Sub-Typology (Stage × Source × Form). **Brief:** The Class 2 (Partial Coupling) label in AAT names "some shared infrastructure, some separate pathways" between belief-update and goal-state processing, but the existing scalar measure $\kappa_{\text{processing}}$ collapses a rich internal sub-structure into a single magnitude number. Look more carefully and three structurally-independent axes emerge: *which stage* of the belief-update pipeline takes a goal argument (attention / interpretation / aggregation / storage); *which source* of goal-state acts (the agent's objective, or its committed strategy); and *which form* the dependency takes (additive bias subtractable by post-hoc debiasing, vs functional-form change requiring internal substitution). Each axis carries a distinct operational consequence: the form axis gates whether external wrapping can coerce the agent back to Class 1; the source axis gates whether self-stabilizing belief-strategy attractors are possible; the stage axis gates where the repair must be applied. The Class 3 (fully Coupled) limit is the corner where all stages couple, both sources act, and the form is process-throughout; Class 1 is the opposite corner; Class 2 is the structured space between.

Impact: Refines the operational meaning of the Class 2 label from "uncomfortable middle category" to "structured sub-space with specific repair regimes per cell." Unifies the *Class-1-by-structure* vs *Class-1-by-behavior* refinement (currently sitting at the Class 1 cell of `#der-directed-separation`) with the form-axis of the Class 2 sub-typology — they are the same distinction read at different architectural levels. Sharpens wrapping-regime selection (W_1 vs W_2) by subtype rather than by Class label alone. Maps the three adversarial mechanisms named in `#disc-adversarial-coupling-pressure` (identity-binding / affect-urgency / sunk-cost-engineering) onto specific cells of the typology, with the affect/urgency case recognized as outside the static (stage × source × form) parameterization (it is a cascade-bypass, a separate failure mode). Provides the static-structural complement to the M4 meta-segment `#disc-modularity-state-dynamics`'s three operations on modularity state — the operations describe transitions through the parameterization the typology names.

Novelty Claim: *Claim recognition* of the (stage × source × form) sub-typology as the structural complement to Class 1's structure-vs-behavior refinement. *Claim differentiation* on the wrapping-regime correspondence: which wrapping regime suffices for Class 2 → Class 1 coercion is determined by the sub-type's form, not just by the Class label. The integration with the leakage-locus result (locus universal in $\ker \mathcal{J}_\tau$; typology determines functional form within the locus) and the source-asymmetry implication (`#der-belief-strategy-attractor`) are the load-bearing structural consequences.

Related Work:

Table AE.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
Active inference / FEP fused goal-perception	Friston 2010+ corpus; covered in `ref/Prior_art_for_AAT_directed_separation.md` Pillar	*adjacent literature* — the unified variational machinery is structurally process-form at all stages (the Class 3 case); the *partial-separation sub-typology* is not addressed in this literature
Motivated reasoning / cultural cognition	Kunda 1990 *Psych Bull*; Kahan cultural-cognition project	*not yet searched* — primary novelty target for the source asymmetry and form-classification of motivated cognition
Identity economics & motivated beliefs	Bénabou-Tirole 2002, 2011 *QJE*; Akerlof-Kranton 2000 *QJE*, 2010	*not yet searched* — primary target for whether the O-source vs Σ-source asymmetry has an economics-side antecedent

ASF concern	*Prior-art language*	*Relationship / Positioning*
Dual-process theory	Evans 2008; Kahneman 2011; Stanovich 2009	*not yet searched* — coarse Class 1 / Class 3 distinction is informally present; the sub-typology of partial coupling is not the literature's frame
Self-deception philosophical taxonomy	Mele 2001 *Self-Deception Unmasked*; von Hippel & Trivers 2011 *BBS*	*not yet searched* — possible direct anticipation on the philosophical sub-classification side
Goal-shielding	Shah, Friedman & Kruglanski 2002 *JPSP*	*not yet searched* — candidate empirical instantiation of P3 / Σ / process-form coupling

Search Log:

- 2026-05-22 (*intuition-only*): The active-inference / Pearl-blanket / Friston-blanket prior-art landscape is already covered in `ref/Prior_art_for_AAT_directed_separation.md` (Pillar-3 search) and addresses the unified variational machinery as the Class 3 (fully Coupled) case; the *partial-separation sub-typology* this segment develops is not in that prior-art landscape. The cognitive-psychology / motivated-reasoning / identity-economics / dual-process / self-deception literatures are *unsearched*; targeted Pillar search recommended before any tier upgrade. The form distinction (content vs process) likely has analogs scattered across literatures (most explicitly in Bénabou-Tirole's identity-vs-instrumental belief framing and in dual-process system 1 / system 2 separation); the source-asymmetry result is plausibly partially in the sunk-cost / commitment-cascade empirical literature implicitly but not as a derived structural theorem. The combined (stage × source × form) parameterization in AAT-internal vocabulary is plausibly novel under the search-depth conducted.

Appendix AF

Discussion: The W_1 Structural-Bound Boundary

The W_1 structural leakage bound of `#der-class-coercion-via-wrapping` exists if and only if the wrapped component carries no goal-correlated state across the call boundary (condition (C2′)); otherwise the goal-leak routes through a channel the wrapper cannot observe or condition on, and only a behavioral (W_2-type) bound remains. The discontinuity at the boundary is a *certifiability* discontinuity, not a behavioral one: the leakage itself is continuous — and quadratically flat — in the degree of (C2′)-violation, while what snaps at the boundary is the availability of the structural certificate.

Formal Expression.

Setup: the two channels and the conditioning that distinguishes them. Recall from `#der-class-coercion-via-wrapping` that the W_1 regime issues a structurally goal-blind query q_M to a black-box component A, then updates the wrapper belief M_W from the response $A(q_M)$. Two distinct goal variables are in play: the wrapper's internal register G_W (which q_M structurally does not take as an argument; Theorem 1 closes this *processing* channel exactly), and the latent operator goal G^{op} inferable from query content (the *selection* channel bounded by $\kappa^{\text{sel}}_{W_1} = I(A(q_M); G^{\text{op}}) \leq I(q_M; G^{\text{op}})$). The structural bound is a property of the wrapper's query-selection policy because, under (C2′), $G^{\text{op}} \to q_M \to A(q_M)$ is a Markov chain — the goal can reach the response *only* through the query the wrapper chose.

The no-go: statefulness reopens an unobservable channel.

Discussion (w1-structural-bound-no-go)

Drop (C2′). Let A carry hidden state S — a conversation KV-cache, a retrieval buffer, adaptive context — that is correlated with G^{op} through shared history (e.g. a prior goal-conditioned call in the same macro-step touched S). Then the response can depend on the goal *through* S, and this dependence survives conditioning on the query:

$$I\big(A(q_M);\, G^{\text{op}} \,|\, q_M\big) \;\geq\; 0, \qquad \text{and generically } > 0 \text{ when } S \not\perp G^{\text{op}}. \tag{AF.1}$$

Conditioning on q_M no longer closes the path, because the leak does not flow through q_M — it flows through S. The conditioning that *would* close it, $I(A(q_M); G^{\text{op}} \mid q_M, S) = 0$, is unavailable to any wrapper-level analysis: S is internal to the black-box oracle and the wrapper cannot observe or condition on it. The Markov chain $G^{\text{op}} \to q_M \to A(q_M)$ is broken — there is now a second edge $G^{\text{op}} \to S \to A(q_M)$ that bypasses the query — so the data-processing argument that yields $\kappa^{\text{sel}}_{W_1} \leq I(q_M; G^{\text{op}})$ no longer holds.

No-go statement. For a component carrying goal-correlated cross-call state, *no bound on the belief-channel goal-leakage is derivable from the wrapper's query-selection policy alone.* The structural W_1 bound is available if and only if (C2′) holds.

When (C2′) fails, the leak flows through an unobservable, unconditionable channel, and only a *behavioral* bound — estimated by monitoring $I(A(q_M); G^{\text{op}} \mid q_M)$ in the responses — remains, exactly as in the W_2 regime and for the same structural reason (an unconditionable goal-correlated channel). This locates the structural-versus-behavioral split not only in *wrapper design* (where to place the separation commitment — the $W_0/W_1/W_2$ regime choice) but *also* in a *component property*: statefulness across the call boundary forces a behavioral-only bound regardless of how the wrapper is built. The condition is sharp — it is not online weight-adaptation that breaks the structural bound (a frozen-weights LLM adapts nothing) but mere goal-correlated state persistence across the call boundary, which a frozen-weights LLM with a conversation cache exhibits.

The discontinuity is in the certificate, not in the behavior. The boundary at (C2′) is best read as a *certifiability* discontinuity. Parameterize the (C2′)-violation by ε, the goal-correlation retained in the cross-call state ($\varepsilon = 0$ is exact statelessness-about-the-goal; $\varepsilon > 0$ is a sliver of goal-correlated memory). Two facts hold at the boundary, and they point in opposite directions:

- **The behavior is continuous.** The actual goal-information reaching the belief is a continuous function of ε through $\varepsilon = 0$ — no jump, no threshold. Two wrappers differing only in ε leak amounts that differ by an amount shrinking with ε; they cannot be told apart by watching their belief updates.
- **The certificate's validity is a step.** At $\varepsilon = 0$ the design-side certificate — *the goal is not in the query, the component is stateless about the goal, therefore the belief update cannot see the goal* — is true. At any $\varepsilon > 0$ that same certificate is still computable and still asserts the same thing, but is now *false*: the real leak is positive and took the unobservable path. The certificate's *value* (a flat selection-channel reading of 0 under goal-blind selection) is continuous across the boundary; its *truth* is the step function. Once $\varepsilon > 0$ the only honest instrument is behavioral monitoring.

So the wall is in the epistemics, not in the world: the leakage the structural certificate would have bounded is continuous in the (C2′)-violation, and the discontinuity is in what is *provable*, not in what the agent *does*. This pre-empts the natural objection — *surely a tiny bit of state can only leak a tiny bit?* — by agreeing with it and relocating the bite: the leak is indeed tiny, but the structural guarantee is gone entirely.

What is solid versus suggestive. Two parts of the picture are exact and carry the no-go:

1. **The behavioral identity at the limit.** At exact statelessness-about-the-goal the structural certificate's zero reading is the truth (the leak is zero); the Markov chain holds and $\kappa^{\text{sel}}_{W_1} \le I(q_M; G^{\text{op}})$ is valid.
2. **The certificate-validity step.** The structural certificate is available iff (C2′) holds; at any (C2′)-violation it asserts a false zero and only the behavioral monitor remains valid. This is forced by the unobservability of S and does not depend on the magnitude of the violation.

What is *suggestive but not proven in general* is the global *continuity-and-flatness* of the leakage past the boundary. In an exact-enumeration toy (single bit, single call, a state-bends-the-answer mechanism) the leak is monotone, smooth, and $\Theta(\varepsilon^2)$ — flat to first order — throughout the violation range. That is moderately strong evidence for the continuity reading and against a behavior-discontinuity reading, but it is a deliberately simple toy: a component whose retained state *amplifies* (state that controls the response gain, or multi-step accumulation across many calls) could in principle exhibit super-linear or threshold leakage, and that is not ruled out. The continuity claim is therefore stated as *robust qualitative* — the certifiability step is the exact load-bearing content; the smoothness of the leak past the boundary is the toy-suggested, not-fully-general, companion.

Epistemic Status.

Robust qualitative. The no-go itself — *the structural W_1 bound is available iff (C2′) holds; under goal-correlated cross-call state the belief-channel leak is unconditionable at the wrapper level and only a behavioral bound remains* — is exact: it follows directly from the unobservability of the component's internal state S and the consequent breaking of the $G^{\mathrm{op}} \to q_M \to A(q_M)$ Markov chain. The certifiability-discontinuity framing (behavior continuous, certificate-validity a step) rests on (a) the exact behavioral identity at the (C2′) limit and (b) the certificate-validity step, both solid. The *general* continuity-and-flatness of the leakage in the degree of (C2′)-violation is *suggestive, not proven*: it is exhibited cleanly in an exact-enumeration toy ($\Theta(\varepsilon^2)$ flatness), but a component with an amplifying state mechanism could in principle break smoothness, and that case is not closed. Hence the segment as a whole sits at robust qualitative rather than exact: the boundary's *existence and location* are exact; the boundary's *shape past it* is qualitatively characterized.

Max attainable: robust qualitative for the continuity-shape claim absent a general continuity proof; the no-go's existence-and-location is already exact.

Discussion.

Why this is a sharpening, not a softening, of W_1. The W_1 regime's structural guarantee is not undercut by this boundary — it is given a precise domain of validity. Theorem 1 of `#der-class-coercion-via-wrapping` (exact directed separation under (C1)–(C3)) is untouched; the selection-channel bound is untouched within (C2′). What this segment adds is the honest statement of *when the structural bound stops existing* and *what replaces it* (a behavioral bound), together with the recognition that the replacement is forced by a component property orthogonal to wrapper design. A no-go with an explicit boundary characterization is more useful than a structural bound asserted without a stated domain.

Practical reading. The $\Theta(\varepsilon^2)$ flatness of the toy carries a counter-intuitive design implication: a near-stateless component is *operationally* almost as safe as an exactly-stateless one (a little cross-call memory leaks very little goal-information), but it is *certificationally* not safe at all (the structural guarantee is lost entirely the instant statelessness fails). The value of enforcing exact (C2′)-statelessness across the call boundary is concentrated almost entirely in what it lets you *prove*, not in the marginal leakage it prevents. When deciding whether the engineering cost of true call-boundary state-resetting is worth it over "mostly reset," the thing being bought is a proof, not a meaningful behavioral delta.

Relation to the Class 2 sub-typology. This boundary is the agent-level companion to the Σ-channel observation in `#disc-partial-coupling-pathways`: a Σ-source coupling in a stateful component can undermine W_1's structural commitment via strategy-context that persists across calls, which is precisely a (C2′)-violation. The sub-typology names the component-internal form of the coupling; this segment names the call-boundary condition under which W_1's structural certificate survives it.

Relation to the action-side bounded-signaling assumption. The unobservable-channel structure here is the belief-side twin of the action-side bounded-signaling assumption named in `#der-directed-separation` — both are "a goal-correlated channel the formalism does not see." Whether they are the same phenomenon viewed from two sides is left open.

Findings.

The W_1 Structural-Bound Boundary (Certifiability Discontinuity at (C2′)). **Brief:** Imagine a scaffold that keeps a language model honest about its beliefs by only ever asking it goal-blind questions — "summarize this diff," never "summarize this diff so we can ship the fix." The promise of strict wrapping (W_1) is that you can *certify from the design alone* that the operator's goal cannot bend the belief update: the goal is structurally not an input to the belief-side query. That promise holds exactly when the component forgets everything about the goal between the goal-blind question and the goal-conditioned one. The moment the component keeps a sliver of goal-touched memory across that boundary — a conversation cache, a retrieval buffer — the goal can sneak into the answer through the memory, on a path the scaffold cannot see, and the design-side certificate becomes false even though it still computes the same reassuring number. The

striking part is *how* it fails: the actual amount of goal that leaks is continuous and, near the boundary, quadratically tiny — you can get a near-stateless component arbitrarily close in behavior to a stateless one. What snaps, all-or-nothing, is not the behavior but the *provability*: you lose the entire structural guarantee for an almost-invisible behavioral cost. You were buying a proof, not a meaningful safety margin.

Impact: Locates the structural-versus-behavioral leakage split of `#der-class-coercion-via-wrapping` in a *component property* (goal-correlated cross-call statefulness) independent of wrapper design, complementing the $W_0/W_1/W_2$ regime choice that locates it in wrapper construction. Promotes the Σ-channel-suppressed-W_1 Working Note of the wrapping segment to a named, load-bearing condition (C2′) and gives the precise reason it is needed. Frames the no-go as a certifiability discontinuity — sharpening the CS-norm practice of characterizing *what kind* of boundary a no-go draws — and supplies the practical corollary that exact call-boundary statelessness buys a certificate, not a behavioral delta.

Novelty Claim: *Claim recognition* that the W_1 structural leakage bound's availability is governed by a sharp component-side condition (no goal-correlated cross-call state), with the boundary characterized as a discontinuity in the *validity of the structural certificate* rather than in the agent's leakage behavior — the leakage being continuous, and second-order flat, in the degree of condition-violation.

Related Work: - Cover & Thomas, *Elements of Information Theory* (2nd ed., 2006), data-processing inequality (published 2006, found 2026-05-31) — *formal antecedent* — the DPI along $G^{\mathrm{op}} \to q_M \to A(q_M)$ is what holds under (C2′) and breaks when an unobservable S adds a bypassing edge. - Pearl, *Causality* (2nd ed., 2009), d-separation / unobservable-confounder structure (published 2009, found 2026-05-31) — *adjacent literature* — the broken-Markov-chain reading (a hidden S confounding G^{op} and $A(q_M)$ that the wrapper cannot condition on) is the causal-graph idiom for the no-go.

Search Log: - 2026-05-31 (*intuition-only*): No dedicated literature search for the certifiability-discontinuity framing as such; the underlying tools (DPI, unobservable-confounder structure) are standard and cited. A targeted search for prior art on "certifiable goal-blindness under stateful components" / structural-vs-behavioral verifiability boundaries in scaffolded-agent design has not been conducted.

Appendix AG

Derived: Belief-Strategy Attractors Under Σ-Source Coupling

A Class 2 agent whose belief-update coupling is sourced from the strategy Σ_t — rather than from the objective O_t — closes a feedback loop $M_t \to \Sigma_t \to f_M \to M_t$ through the orient cascade. The loop admits *self-stabilizing fixed points* (M^*, Σ^*) in which the agent's belief stays misaligned with the environment indefinitely: evidence arrives, but a strategy committed to a particular belief depresses the effective belief-update gain on evidence that would refute that strategy, locking (M, Σ) in a joint attractor. O_t-source coupling under the same structural form does *not* produce such attractors — because O_t is exogenous to M_t in steady state per the orient cascade, the feedback loop $M \to O \to f_M \to M$ does not close in nominal operation.

The result is the formal statement of the long-recognized empirical asymmetry between *sunk-cost commitment cascades* (self-reinforcing) and *identity-driven motivated reasoning* (biased but not divergently runaway): the cascade-vs-bias structural distinction is forced by the orient cascade's exogeneity of O and not-exogeneity of Σ. The asymmetry holds *robust qualitatively* across the general nonlinear $K(\Sigma)$ form and *exact* under the linearized fixed-point analysis with a posited multiplicative gain. The result strengthens to *exact* under a derivation of $K(\Sigma)$ from utility-cost-of-belief-revision (a recommended follow-on sub-spike).

The derivation matches and extends the third row of `#disc-adversarial-coupling-pressure`'s mechanism table (sunk-cost engineering as $\Sigma_t \to M_t$). The mechanism table named the channel; this segment derives the dynamical consequence — that channel produces self-stabilizing fixed points, not merely bias.

Formal Expression.

Setup. A Class 2 agent has Σ_t-source coupling at stage P3 (aggregation) per `#disc-partial-coupling-pathways`. The coupled aggregation takes multiplicative process-form:

$$M_{\tau+} \;=\; M_{\tau-} \;+\; K(\Sigma_{\tau-}) \cdot \ell_\tau \tag{AG.1}$$

Formulation (strategy-source-coupled-aggregation)

where ℓ_τ is the likelihood signal from upstream stages (goal-blind by assumption — only P3 couples; if upstream stages also couple, the cascade-propagation result of `#disc-partial-coupling-pathways` applies and the analysis localizes at the upstream-most coupled stage), and $K : \mathcal{S} \to \mathbb{R}^+$ is a Σ-dependent positive gain.

Sunk-cost is the canonical instantiation: $K(\Sigma)$ decreases as Σ commits more heavily to specific beliefs about reality. Formally — as the projection of Σ onto strategies presuming $M = m^*$ grows, K falls for evidence challenging m^*. The mechanism is utility-cost driven: the cost of revising M_t is bundled with the cost of abandoning Σ_t (the agent would

lose the sunk strategic investment if it abandons the strategy), and the agent's belief-update gain effectively reflects this bundled cost. A derivation of $K(\Sigma)$ from a utility-cost analysis of belief-revision-under-strategic-commitment is a recommended sub-spike (see Working Notes).

Definition (orient-cascade-Sigma-update)

The strategy is updated by the orient cascade per `#der-orient-cascade`:

$$\Sigma_{\tau+} \;=\; f_\Sigma(M_{\tau+}, O_t, \Sigma_{\tau-}). \tag{AG.2}$$

Closed-loop dynamics. *[Derived (closed-loop-coupling, from `#form-strategy-source-coupled-aggregation` + `#der-orient-cascade`)]*

Substituting,

$$M_{\tau+} = M_{\tau-} + K(\Sigma_{\tau-}) \cdot \ell_\tau, \qquad \Sigma_{\tau+} = f_\Sigma\big(M_{\tau-} + K(\Sigma_{\tau-})\ell_\tau,\; O_t,\; \Sigma_{\tau-}\big). \tag{AG.3}$$

This is a closed-loop system in (M, Σ) with M_t entering $\Sigma_{\tau+}$ through the orient cascade and $\Sigma_{\tau-}$ entering $M_{\tau+}$ through the coupled aggregation.

Derived (fixed-point-conditions)

Fixed-point structure.

A fixed point (M^*, Σ^*) satisfies

$$K(\Sigma^*) \cdot \ell^* \;=\; 0, \qquad \Sigma^* \;=\; f_\Sigma(M^*, O_t, \Sigma^*). \tag{AG.4}$$

The condition $K(\Sigma^*) \cdot \ell^* = 0$ admits two qualitatively distinct solutions:

(a) Honest convergence: $\ell^* \to 0$ — no further evidence to integrate. M^* is the agent's converged belief; the asymptotic dynamics match standard Bayesian convergence under a stationary evidence stream.

(b) Pathological attractor: $K(\Sigma^*) \to 0$ — the gain has collapsed *even though* $\ell^* \neq 0$. Evidence arrives that would, under honest gain, update M — but the gain has fallen near zero on the relevant subspace, so the evidence does not propagate. The agent's M stays at m^*; Σ stays committed to the strategy that presumes m^*; the system rests in a (m^*, Σ^*) attractor misaligned with the environment.

Case (b) is the **belief-strategy attractor** — the formal structure of sunk-cost commitment cascades.

Derived (linearized-stability), exact under linearization

Linearized stability.

Linearize the closed-loop dynamics around (M^*, Σ^*). Define perturbations $\delta M_\tau = M_\tau - M^*$, $\delta\Sigma_\tau = \Sigma_\tau - \Sigma^*$, and partial derivatives at the fixed point:

$$K^* = K(\Sigma^*), \quad K'^* = \partial_\Sigma K\big|_{\Sigma^*}, \quad A = \partial_M f_\Sigma\big|_{(M^*,\Sigma^*)}, \quad B = \partial_\Sigma f_\Sigma\big|_{(M^*,\Sigma^*)}. \tag{AG.5}$$

Treating ℓ as a function of M for the locally-linearized evidence-arrival process — $\ell \approx -L\delta M$ for some positive-semi-definite information matrix L — the linearized dynamics are:

$$\delta M_{\tau+} = (I - K^*L)\delta M_{\tau-} + (K'^*\ell^*)\delta\Sigma_{\tau-}, \tag{AG.6}$$

$$\delta\Sigma_{\tau+} = A\delta M_{\tau+} + B\delta\Sigma_{\tau-}. \tag{AG.7}$$

The Jacobian of the closed-loop map:

$$J = \begin{pmatrix} I - K^* L & K'^* \ell^* \\ A(I - K^* L) & AK'^* \ell^* + B \end{pmatrix}. \tag{AG.8}$$

Result (pathological-attractor-condition), conditional

The fixed point (M^*, Σ^*) is a stable attractor iff all eigenvalues of J have modulus less than 1. In the sunk-cost-collapse regime $K^* \to 0$, the top-left block $I - K^* L \to I$ — there is *no contraction in the M direction* — and stability is determined by the Σ-direction block B. If B has spectral radius less than 1 (the orient cascade is contractive in Σ near Σ^*, which is typical for strategy-update dynamics with friction), the joint fixed point is stable *despite the absence of M-direction contraction* — the strategy stabilizes around its commitment, and the absence of belief-update propagation keeps M near m^*.

This is the formal statement: **Σ-source coupling with multiplicative process-form admits self-stabilizing attractors in which the agent's belief remains misaligned with the environment indefinitely.**

Contrast — O_t-source coupling produces no such attractor.

Derived (O-source-no-attractor), conditional on orient-cascade exogeneity

Substitute the same coupling form with O as source:

$$M_{\tau+} = M_{\tau-} + K(O_t) \cdot \ell_\tau. \tag{AG.9}$$

Per `#der-orient-cascade`, O_t revises only when forced — under nominal operation, O_t is *exogenous* to M_t in steady state. In steady state $O_t = O^*$ is constant; $K(O_t) = K(O^*)$ is constant. The closed-loop dynamics reduce to a standard contractive Bayesian update with a goal-shaped (but fixed) gain:

$$\delta M_{\tau+} = (I - K(O^*)L)\delta M_{\tau-} + 0 \cdot \delta O_{\tau-}. \tag{AG.10}$$

The O-feedback term vanishes because O does not update from M in steady state. The map is contractive whenever $K(O^*)L$ has appropriate properties; the agent's belief has a *bias* (the gain $K(O^*)$ may be smaller than the honest gain, producing slower convergence or a non-zero asymptotic residual relative to the true environment) but it has no feedback-induced *foreclosure* of belief-revision.

Pure O_t-source coupling produces bias but not runaway commitment. The source asymmetry is structural: it follows directly from the orient cascade's exogeneity of O and not-exogeneity of Σ, not from any additional property of K.

Source-asymmetry result.

Derived (source-asymmetry-attractor), robust qualitative (exact under linearization with multiplicative K)

Pure Σ_t-source coupling at any stage of f_M closes a feedback loop $M \to \Sigma \to f_M \to M$ through the orient cascade and admits self-stabilizing belief-strategy attractors. Pure O_t-source coupling at the corresponding stages does not, because the orient cascade's exogeneity of O in steady state prevents the loop from closing.

The asymmetry is *structural* — it follows from the topology of the orient cascade, not from a parametric difference between O and Σ. Any mixed-source coupling $R = \{O, \Sigma\}$ inherits the Σ-component's attractor possibility — even moderate Σ-source coupling, paired with strong O-source coupling, can produce the attractor through the Σ channel alone.

Epistemic Status.

Conditional. Max attainable under current derivation: *exact* for the linearized fixed-point analysis under the posited multiplicative gain $K(\Sigma)$ and the orient-cascade exogeneity of O; *robust qualitative* for the general nonlinear K and the broader claim that any closed-loop topology $M \to \Sigma \to f_M \to M$ admits the attractor structure.

Stated conditions:

(R1) **Multiplicative gain form.** $K(\Sigma)$ is posited as a positive scalar (or pointwise-positive matrix) gain modulating the aggregation step's update magnitude. The form is structurally motivated by the empirical mechanism (sunk-cost bundles the cost of revising M with the cost of abandoning Σ, which in linearization presents as gain suppression) but not derived from a more primitive AAT mechanism. A derivation of $K(\Sigma)$ from utility-cost analysis of belief-revision-under-strategic-commitment is the gating sub-spike for tier-upgrade to *exact*.

(R2) **Orient-cascade exogeneity of O in steady state.** O revises only when forced per `#der-orient-cascade`. Agents that violate this — continuously updating O as a function of M — collapse the source asymmetry: O-coupling then closes a feedback loop too, but the orient cascade explicitly forbids this in canonical AAT. Identity-binding under sustained adversarial pressure may be the operational regime where O effectively updates from M (per `#disc-adversarial-coupling-pressure`); this is where the asymmetry attenuates.

(R3) **Linearization validity.** The fixed-point stability analysis is local. Generalization to nonlinear K, f_Σ preserves the *qualitative* asymmetry (the feedback-loop topology is independent of linearization) but the explicit fixed-point structure may admit multiple attractors, limit cycles, or chaos.

(R4) **Stage of coupling.** Stated for P3 (aggregation) for definiteness; the asymmetry is *generic across stages* — the closed-loop topology depends on which source is coupled, not on which pipeline stage hosts the coupling. P3 is the canonical sunk-cost site; identity-protective consolidation (P4 / O) and frame-coupling (P1 / Σ) instantiate the asymmetry at other stages.

Discussion.

The structural reason sunk-cost has a different empirical signature than motivated reasoning. Sunk-cost cascades are documented as *self-reinforcing* in psychology (Staw 1976; Brockner 1992; Arkes & Blumer 1985; Sleesman et al. 2012 meta-analysis) and in organizational behavior. Identity-driven motivated reasoning is documented as producing *bias* but not divergent attractors except in identity-bound communities reinforcing priors. The asymmetry across these literatures has not, to our knowledge, been derived as a structural theorem; the empirical clustering is well-known but the structural reason — strategy is endogenous to belief via the cascade, while objective is exogenous in steady state — is the AAT-internal contribution. The derivation above formalizes the empirical-clustering observation as a consequence of the orient cascade's topology.

Implication for adversarial pressure. `#disc-adversarial-coupling-pressure`'s sunk-cost-engineering mechanism is the adversary engineering precisely this structure: by inducing the target to commit publicly or expensively to a strategy that presumes a particular belief, the adversary locks the target's (M, Σ) into the attractor regime. Once the attractor is established, the adversary can stop driving and the target's own internal dynamics maintain the misaligned belief. This is the structural reason sunk-cost engineering is so effective in propaganda, influence operations, and social engineering: the target's own machinery does the maintenance work after the initial commitment.

Implication for class-coercion-via-wrapping. A Class 3 (Coupled) component wrapped to Class 1 status via the construction of `#der-class-coercion-via-wrapping` requires extra care when the un-wrapped component has Σ-source coupling. Even W_1 (strict wrapping — no G in the belief-update query) may be insufficient if the component's *internal* Σ is influenced by historical query content that carries strategic context, and that internal Σ then suppresses the gain on subsequent goal-blind queries. The corrective is a Σ-channel-suppressed W_1: the wrapper must additionally hold the component's strategic context fixed across calls or strip strategic-context-bearing content from the queries. This is

a finer-grained design constraint than the wrapping-construction segment currently surfaces; it follows directly from the source-asymmetry result.

Implication for compositional dynamics. The composite-level class inheritance table in `#der-directed-separation` tracks composite Class membership but not composite sub-type. A composite of Class 1 (Separated) sub-agents whose routing is goal-dependent (R_t depends on G_t^c) is Class 3 at the composite level per the inheritance table — but the source-asymmetry result of this segment further refines the composite's sub-type: if the routing's G^c-dependence is on the composite's strategy Σ^c rather than its objective O^c, the composite admits belief-strategy attractors at the composite level. This is the operational reason organizations with strong strategic commitments and goal-dependent internal routing exhibit sunk-cost dynamics at the institutional scale.

The diagnostic question for a Class 2 agent. When inspecting a Class 2 agent, the diagnostic question that distinguishes attractor-vulnerable from bias-vulnerable sub-types is: *is the coupling sourced from O_t or from Σ_t?* The question is sharper than asking the aggregate $\kappa_{\text{processing}}$ magnitude, because the topological consequence (attractor possibility) depends on the source, not on the magnitude. Empirically, the source can be probed by varying O_t at fixed Σ_t vs Σ_t at fixed O_t and observing which probe condition reveals the gain modulation. In setups where holding Σ fixed is operationally difficult, the diagnostic falls back to the dynamical-signature analysis: does the agent's belief, under sustained counter-evidence, *converge slowly with bias* (O-source signature) or *fail to converge at all* in a self-stabilizing pattern (Σ-source signature)?

Findings.

Belief-Strategy Attractors From Σ-Source Coupling (Source Asymmetry). **Brief:** An agent whose belief-update is shaped by *what it currently wants to be true* will be biased — its beliefs will lean toward the goal — but a thoughtful third party can in principle measure the bias and correct it, and the agent's beliefs will eventually catch up to reality as evidence accumulates. An agent whose belief-update is shaped by *what plan it is currently committed to* is structurally different: the strategy commitment depresses the agent's belief-update gain on evidence that would invalidate the strategy, the depressed gain keeps belief at the strategy-consistent point, and the unchanging belief sustains the strategy commitment. The agent locks into a joint *belief-and-strategy attractor* — a fixed point where the agent stays misaligned with reality indefinitely, with no internal mechanism to escape. The structural reason: the agent's strategy is updated by the agent's beliefs, so coupling beliefs to strategy closes a feedback loop. Beliefs and objectives are not coupled the same way — objectives are revised only when forced, so coupling beliefs to objectives doesn't close the loop. This is the structural reason sunk-cost cascades self-reinforce in a way that identity-driven motivated reasoning does not.

Impact: Names the structural origin of the sunk-cost / motivated-reasoning empirical asymmetry — both have been documented across psychology and organizational behavior, but the structural reason (orient cascade endogeneity of Σ vs exogeneity of O) has not, to our knowledge, been stated as a derived theorem. Sharpens the wrapping-regime selection of `#der-class-coercion-via-wrapping` for Class 2 components with Σ-source coupling: Σ-channel-suppressed W_1 may be required even when ordinary W_1 would suffice for O-source coupling. Connects `#disc-adversarial-coupling-pressure`'s sunk-cost-engineering mechanism (third row of the mechanism table) to a dynamical-attractor structure rather than just a channel; explains why the mechanism is so effective in propaganda and influence operations (the target's own machinery does the maintenance work). Provides a diagnostic question for inspecting Class 2 agents — source of coupling — that is sharper than the aggregate $\kappa_{\text{processing}}$ magnitude.

Novelty Claim: *Claim recognition* of the structural source asymmetry as a direct consequence of the orient cascade's topology — strategy is endogenous to belief, objective is exogenous in steady state. *Claim differentiation* on the resulting fixed-point analysis: linearized stability with $K^* \to 0$ produces an attractor in the closed-loop dynamics under the Σ-source case but not under the O-source case. The empirical clustering across the sunk-cost and motivated-reasoning literatures is well-established; the AAT-internal contribution is naming the structural reason and stating it as a derived theorem.

Related Work:

- Staw 1976 *"Knee-Deep in the Big Muddy" Organizational Behavior and Human Performance* (published 1976, intuition-only date) — *empirical instantiation supporting* — the canonical psychological study of self-reinforcing commitment to a chosen course of action; the empirical phenomenon the source-asymmetry result structurally explains.
- Brockner 1992 *Academy of Management Review* "The Escalation of Commitment to a Failing Course of Action" — *empirical instantiation supporting* — synthesizes the sunk-cost / commitment-cascade literature; same empirical anchor.
- Arkes & Blumer 1985 *Organizational Behavior and Human Decision Processes* "The Psychology of Sunk Cost" — *empirical instantiation supporting* — foundational behavioral-economics treatment of sunk-cost effects.
- Sleesman et al. 2012 *Academy of Management Journal* meta-analysis on escalation of commitment — *empirical instantiation supporting* — quantitative synthesis of the sunk-cost-cascade literature; empirical confirmation of the self-reinforcement signature.
- Kunda 1990 *Psychological Bulletin* "The Case for Motivated Reasoning" — *empirical instantiation supporting (contrast case)* — the canonical motivated-reasoning treatment; produces bias but not the divergent-attractor signature of sunk-cost — the empirical contrast the source-asymmetry result structurally explains.
- Kahan cultural-cognition project (various years) — *empirical instantiation supporting (contrast case)* — identity-driven motivated reasoning produces persistent bias but typically not runaway divergence except when paired with composite-level identity-binding community dynamics; the contrast with sunk-cost is exactly the source-asymmetry distinction.
- Bénabou & Tirole 2002, 2011 *Quarterly Journal of Economics* on identity economics and motivated beliefs — *adjacent literature, not yet searched* — formal models trading off truth vs identity; possibly anticipates aspects of the source distinction.

Search Log:

- 2026-05-22 (*intuition-only*): The empirical sunk-cost / commitment-cascade literature is well-established and well-known to converge on the self-reinforcement signature; the empirical motivated-reasoning literature is similarly well-established and converges on the bias-without-runaway signature. The *structural derivation* of the asymmetry from the orient cascade's topology has not been searched. Recommended Pillar-style targeted search before tier upgrade — primary targets are the formal-models side of behavioral economics (Bénabou-Tirole identity economics; Akerlof-Kranton identity economics) and the formal-models side of motivated-reasoning psychology (Kruglanski lay epistemics; possibly Shah-Friedman-Kruglanski goal-shielding 2002 *JPSP*). The active-inference / control-as-inference literatures are unlikely to have this specific result (they don't typically separate O and Σ as distinct goal-state objects), but should be confirmed.

Appendix AH

Discussion: The Dynamic-Regime Axis — Contraction / Equilibrium / Cyclic-Distributional / Mean-Field

AAT composites differ on at least three independent structural axes: their *architectural class* (Class 1 Separated / Class 2 Partial / Class 3 Coupled, per `#der-directed-separation`) — about processing topology; their *dynamic regime* (this segment) — about the structural type of the joint-dynamics fixed point and the Lyapunov certificate available; and their *macro-state type* (state-variable / fixed-point object / distributional) — about what kind of object the composite's G_t^c is. This segment names the dynamic-regime axis as a first-class structural classifier, in vocabulary the framework already runs implicitly across `#deriv-strategic-composition`'s α'/β' partition, `#form-composition-closure`'s contraction presupposition, `#disc-separability-pattern`'s Contraction ladder, `#scope-composite-agent`'s scope-route disjunction, and `#impl-strategic-composition`'s "contraction-to-equilibrium hand-off" framing.

The four-tier ladder is recognized prior art. This is integration, not invention. The partition AAT names — contraction-regime / equilibrium-regime / cyclic-distributional-regime / mean-field-equilibrium-regime — appears (under different vocabulary, with isomorphic structural content) across at least five neighboring literatures. Candogan, Menache, Ozdaglar & Parrilo's 2011 Hodge-style decomposition of finite games into potential + harmonic + nonstrategic components supplies a coordinate-free organizing structure for the equilibrium-regime / cyclic-regime boundary cleaner than any tier-wise enumeration; Letcher, Balduzzi et al.'s 2019 *Differentiable Game Mechanics* lifts the same potential/Hamiltonian decomposition to the dynamics side; Sandholm's 2010 *Population Games and Evolutionary Dynamics* organizes the potential/stable/cyclic taxonomy at the population level; Hofbauer & Sigmund's 1998 *Evolutionary Games and Population Dynamics* names the phase-portrait classification (equilibrium vs cycle vs heteroclinic) directly; Omidshafiei et al.'s 2019 α-Rank construction uses cyclic-dynamics-as-meaning as a solution concept; and Papadimitriou-Piliouras's 2018 *From Nash Equilibria to Chain Recurrent Sets* establishes the chain-recurrent set as the natural object when dynamics fail to converge. AAT's contribution is *not* the classification itself; it is (a) the consolidation of the partition into AAT-internal vocabulary that travels with `#scope-composite-agent`'s scope-route disjunction and `#disc-separability-pattern`'s separable-core / structured-repair / general-open shape, (b) three derivable structural identities that name what the axis carries within an integrated agent theory, and (c) cross-axis observations about how the axis composes with architectural class and macro-state type.

This segment is the *positive-half companion* to the broadened `#disc-identifiability-floor` Instance 3, which names the negative half — regime identification from component marginals is structurally impossible without composite-level coupling-topology observation. The two together mark AAT's scope on the axis at both extremes, parallel to how `#disc-separability-pattern` and `#disc-identifiability-floor` jointly mark scope on the identification axis.

The pattern.

Each composite agent satisfying `#scope-composite-agent` admits a dynamic regime — a structural property of its joint best-response dynamics that is *determined by the game structure* (the sub-agent objectives, the strategy spaces, the payoff curvatures) rather than by any particular trajectory the dynamics realize. The regime determines the strongest Lyapunov-class certificate the dynamics admit relative to their fixed-point set, and that certificate determines the operational machinery available — exponential convergence near a unique attractor, exponential convergence near an equilibrium set, polynomial convergence in distribution, or measure-space dynamics at the population-scale limit.

The classification is *not behavioral* (it does not depend on observing trajectories) and it is *not parametric* (it does not depend on a continuous tuning parameter). It is structural and discrete, the same shape `#der-directed-separation`'s architectural classification has — a partition determined by the game structure, with sub-tier classifications within each regime.

The four tiers.

Following the order of progressively weakening Lyapunov certificate:

Table AH.1

Tier	*Fixed-point structure*	*Lyapunov certificate*	*Macro-state type*	*Convergence rate*	*Scope route entry*
R0 Contraction	Unique attracting $X^* \in \mathcal{X}^c$	Joint Lyapunov V, global or basin-restricted	State-variable X^*	Exponential at α_{joint}	C-i / C-ii / C-iii (membership derived under matched-Tier sub-agents; see R0 below)
R1 Equilibrium (α')	Fixed-point set $\mathcal{E} \subset \mathcal{X}^c$, locally attracting	Potential Φ (Monderer-Shapley) or weighted-norm (Rosen)	Fixed-point object $\mathcal{E}$	Exponential near $\mathcal{E}$ at $\alpha^{(j)}_{\text{joint}}$	C-iv + potential/monotone
R2 Cyclic-distributional (β')	No pure-strategy fixed point, or saddle-only	None on $\mathcal{X}^c$; doubling-trick CCE on $\Delta(\mathcal{X}^c)$	Distributional object $\mu \in \Delta(\mathcal{X}^c)$	Polynomial $O(1/\sqrt{T})$	C-iv without potential/monotone
R3 Mean-field equilibrium	Distributional fixed point in measure space	MFG-specific (Lasry-Lions monotonicity)	Distribution-on-population $\rho \in \mathcal{P}(\mathcal{A})$	MFG-specific, often exponential	Population-scope gap

Each tier inherits the tier-internal classification of its closest existing AAT machinery — R0's Tier 1 / Tier 2 / Tier 3 sub-classification from `#disc-separability-pattern`'s Contraction ladder; R1's potential / monotone / general sub-tier per `#deriv-strategic-composition`'s sub-scope α'; R2's regret-bound sub-classification per Hart-Mas-Colell 2000; R3's MFG-uniqueness sub-classification per Lasry-Lions 2007.

R0 — Contraction-regime. The joint dynamics admit a unique attracting fixed point $X^* \in \mathcal{X}^c$ and a Lyapunov function $V : \mathcal{X}^c \to \mathbb{R}_{\geq 0}$ with $V(X^*) = 0, V(X) > 0$ for $X \neq X^*$, such that

$$\frac{d}{dt} V(X_t^c - X^*) \;\leq\; -\alpha_{\text{joint}} \cdot d(X_t^c, X^*)^2 \qquad \forall\, X_t^c \in \mathcal{B}(X^*) \tag{AH.1}$$

for some basin $\mathcal{B}(X^*)$ and contraction rate $\alpha_{\text{joint}} > 0$. Macro-state $X^* \in \mathcal{X}^c$ is a state-variable. Composites satisfying scope routes C-i (shared composite objective), C-ii (hierarchical derivation), or C-iii (mutual-benefit alignment) under matched-Tier sub-agents per `#deriv-critical-mass-composition` land here. The classical AAT machinery applies: `#result-sector-persistence-template`, `#form-composition-closure` bridge lemma, `#der-team-persistence`, `#der-tempo-composition`.

R1 — Equilibrium-regime (α'). The joint dynamics admit a fixed-point set $\mathcal{E} = \{X_j^*\}_{j \in J}$ with $|J| \geq 1$ where each X_j^* is a Nash equilibrium. Lyapunov-on-deviation holds locally near each equilibrium under either potential-game structure (Monderer-Shapley 1996) or monotone-game structure (Rosen 1965 diagonal strict concavity). The macro-state is a fixed-point object — either a single Nash X^* if unique, or the equilibrium set if multiple. Composites satisfying scope route C-iv with potential/monotone structure land here; `#deriv-strategic-composition`'s Cournot worked example is the canonical instantiation. The sector-persistence template transfers with the joint potential's curvature (potential case) or the joint Jacobian's symmetric part (monotone case) supplying $\alpha_{\text{joint}}^{(j)}$ at each equilibrium.

R2 — Cyclic-distributional-regime (β'). The joint dynamics admit no fixed point in pure-strategy space, or fixed points that are not local attractors (saddle-Nash in zero-sum games where best-response orbits without approaching). No-regret dynamics drive the empirical joint distribution to the coarse correlated equilibrium set (Hart-Mas-Colell 2000):

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{\pi_t^c \in \text{CCE}\} \;\to\; 1 \qquad \text{at rate } O(1/\sqrt{T}). \tag{AH.2}$$

The macro-state is a distributional object — a measure $\mu \in \Delta(\mathcal{X}^c)$ supported on the CCE set. Composites satisfying scope route C-iv without potential or monotone structure land here. AAT in R2 predicts only set-convergence of empirical play to CCE; it cannot predict short-run trajectory, per-sub-agent mismatch convergence, or equilibrium selection. This is the framework's honest β' scope limit — a regime limit, not an architectural limit, distinct from any architectural-class restriction.

Cross-layer linearization fingerprint at saddle-only Nash. R2's saddle-only Nash equilibria have a single-agent linearization-spectrum fingerprint of imaginary-axis-semisimple — the *R0-loss* rung of `#result-certificate-existence`'s single-agent certificate-strength ladder. The two R-letter ladders are layered, not collided: this segment's R0/R1/R2/R3 classifies composite joint-dynamics on $\mathcal{X}^c$, while `#result-certificate-existence`'s R0-loss/R0-strict/R1/R2 classifies single-agent certificate-strength on the linearized error space $\mathbb{R}^n$. The bridge is via composition (the joint Jacobian at R2's saddle-Nash linearizes to the antisymmetric R0-loss-at-$\mathcal{M} = I$ sub-case); the letters are deliberately reused to mark the bridge, not by collision.

R3 — Mean-field-equilibrium-regime. For $N \to \infty$, each agent interacts with the *distribution* of other agents' strategies rather than with named others. The macro-state becomes a distributional fixed point in measure space; the dynamics may contract to this fixed point under explicit conditions (Lasry-Lions 2007; Huang-Malhamé-Caines 2006). R3 is distinct from R2: R2's macro-state is a distribution-on-strategies arising as the empirical play of finite-N dynamics with saddle-or-cyclic fixed-point structure on the pure-strategy space; R3's macro-state is a distribution-on-population-strategies arising as the thermodynamic limit with genuine attractor on the measure space. R3 is the only tier where macro-state type genuinely diverges from dynamic regime in a way the framework's existing α'/β' does not capture. R3 is the population-scope gap that Part III currently flags.

AAT-specific derivable content.

The axis is not only consolidation. Three derivable contents fall out once it is stated formally; one is the candidate floor instance landing concurrently in `#disc-identifiability-floor` Instance 3 (broadened).

Identity: macro-state-type and dynamic regime coincide for finite N. For finite-N composites (tiers R0/R1/R2), the macro-state type is structurally identified by the dynamic regime:

- $\mathcal{R}(X^c) = \text{R0} \;\Leftrightarrow\; G^c_* \in \mathcal{X}^c$ (state-variable).
- $\mathcal{R}(X^c) = \text{R1} \;\Leftrightarrow\; G^c_* = \mathcal{E} \subset \mathcal{X}^c$ (fixed-point object).
- $\mathcal{R}(X^c) = \text{R2} \;\Leftrightarrow\; G^c_* = \mu \in \Delta(\mathcal{X}^c)$ (distributional object).

Each direction follows from the entry conditions in §"The four tiers." R0 demands unique attracting fixed point in $\mathcal{X}^c$, which gives a state-variable. R1 demands a fixed-point set under Lyapunov-on-deviation, which is a fixed-point object. R2 demands no pure-strategy fixed point plus no-regret CCE convergence, which gives a distributional object. The converse direction follows from the macro-state type forcing the available Lyapunov machinery: state-variable admits scalar Lyapunov (R0); fixed-point set admits potential or weighted-norm Lyapunov (R1); distributional macro-state admits only doubling-trick CCE convergence (R2).

R3 breaks the identity. Mean-field-equilibrium has distributional macro-state with contracting dynamics, distinct from R2's polynomial CCE convergence. The two axes (dynamic regime, macro-state type) are genuinely distinct and the breaking case is exactly the population-scope frontier the framework currently flags as gap. The identity is the cleanest formal content the axis carries: it is the axis that unifies fixed-point structure, Lyapunov certificate, and macro-state type into a single classification for finite-N composites, with the mean-field limit as the explicit boundary case.

Transition asymmetry: descent is the default, ascent costs more. Five non-trivial transitions exist between adjacent tiers, with a structural asymmetry between descent (R0 → R1; R1 → R2) and ascent (R1 → R0; R2 → R1). Descent under generic perturbation of the objective structure is the default — alignment is fragile, potential-structure is fragile, both can be destroyed by local perturbations. Ascent requires explicit structural commitment that costs more than the descent did to acquire. Symbiogenic absorption (`#hyp-symbiogenic-composition`) is the structural mechanism for R1 → R0 ascent; mechanism design under explicit potential-game constraints is the mechanism for R2 → R1 ascent — both impose structural commitments that out-of-band negotiation, designer intervention, or institutional shaping must supply.

This is an energy-asymmetry on the dynamic-regime axis — descent is downhill, ascent is uphill, and the asymmetry is structural rather than parametric. Adversaries exploit it as the modularity-state version exploits the modularity descent (see `#disc-adversarial-coupling-pressure` for the modularity analog).

The transition asymmetry is currently argumentation-grade; a fully-derived statement tied to a structural-distance metric on game-rewiring space would upgrade it to derived, and this segment names the shape pending that derivation. The mechanism-design impossibility theorems (Gibbard-Satterthwaite 1973–75, Myerson-Satterthwaite 1983, Arrow 1951) are *floors on regime-ascent* under stated designer constraints; the queued `#disc-implementation-impossibility` meta-segment carries the formal treatment of mechanism-design impossibility as designer-side constructive-impossibility.

Lyapunov-machinery transfer R0 → R1. The persistence template `#result-sector-persistence-template` lifts cleanly across R0 → R1, with the Lyapunov function changing form at each tier:

$$\text{R0: } V_{\text{R0}}(X^c) = \|X^c - X^*\|^2 \qquad \dot{V}_{\text{R0}} \le -\alpha_{\text{joint}} V_{\text{R0}} \tag{AH.3}$$

$$\text{R1 potential: } V_{\text{R1},\Phi}(\pi^c) = \Phi(\pi^c) - \Phi(\pi^c_*) \qquad \dot{V}_{\text{R1},\Phi} \ge \alpha_{\text{joint}} \|\nabla\Phi\|^2 \tag{AH.4}$$

$$\text{R1 monotone: } V_{\text{R1},F}(\pi^c) = \tfrac{1}{2}(\pi^c - \pi^c_*)^\top P (\pi^c - \pi^c_*) \qquad \dot{V}_{\text{R1},F} \le -\alpha_{\text{joint}} V_{\text{R1},F} \tag{AH.5}$$

with P the weighted-norm matrix solving the Lyapunov equation for the linearized ∇F at π^c_*. At R2 the Lyapunov form fails — no state-space Lyapunov exists; the dynamics orbit equilibria rather than converge to them. The doubling-trick (Hart-Mas-Colell 2000) gives a regret bound, which translates to empirical-distribution convergence to CCE at $O(1/\sqrt{T})$ rate. The transfer makes the framework's existing α'/β' partition explicitly a regime-axis property rather than scattered across `#deriv-strategic-composition`.

What the axis lets the framework consolidate.

The axis already lives across ten-plus existing segments under different names. Surfacing it as a formal classifier consolidates these implicit usages into one explicit definition:

Table AH.2

Existing segment / construct	*Axis-vocabulary content*
`#scope-composite-agent` routes C-i/ii/iii	Entry conditions for R0
`#scope-composite-agent` route C-iv	Entry condition for R1 + R2 (potential/monotone → R1; without → R2)
`#form-composition-closure` contraction presupposition	R0 machinery
`#result-sector-persistence-template`	R0 sector constant + persistence
`#deriv-critical-mass-composition` matched-Tier closed form	R0 sub-tier specialization
`#der-team-persistence` cooperative limit	R0 limiting case
`#deriv-strategic-composition` sub-scope α'	R1 with potential/monotone Lyapunov transfer
`#deriv-strategic-composition` sub-scope β'	R2 with CCE convergence only
`#impl-strategic-composition` "contraction-to-equilibrium hand-off"	R0 → R1 transition
`#disc-separability-pattern` Contraction ladder (row 4)	R0 sub-tier classification
`#hyp-symbiogenic-composition`	R1 → R0 transition mechanism (structural absorption)
`#der-adversarial-destabilization`	R0 or R1 target under exogenous adversarial parameter; per-target regime distinction

The consolidation has two operational consequences. First, the cross-segment consistency check becomes mechanical — any segment using regime-axis vocabulary can be cross-referenced against this segment's four-tier specification. Second, the meta-segment cross-references to `#disc-separability-pattern` (Contraction ladder as R0 sub-tier classifier) and to broadened `#disc-identifiability-floor` Instance 3 (regime-identification floor) form the integration backbone of how the axis sits inside the wider meta-pattern landscape.

Cross-axis interactions.

Three structural axes coexist in the framework after the dynamic-regime axis surfaces:

- **Axis A — Architectural class.** Processing topology — Class 1 / 2 / 3 per `#der-directed-separation`. Determined by sub-agent class plus routing structure plus shared-substrate-with-G^c-allocation.
- **Axis B — Dynamic regime.** Joint-dynamics fixed-point structure plus Lyapunov certificate — R0 / R1 / R2 / R3. Determined by objective alignment plus game structure plus population scale.
- **Axis C — Macro-state type.** Type-theoretic shape of G^c_* — state-variable / fixed-point object / distributional. For finite N, identified by Axis B per §"Identity"; for $N \to \infty$ (R3) the identity can break.

Axes A and B are independent. The full cross-product is meaningful:

Table AH.3

	R0 contraction	*R1 equilibrium*	*R2 cyclic-distributional*
Class 1 Separated	Kalman + LQR team, shared target	Cournot duopoly (distinct hardware)	Rock-paper-scissors players on distinct hardware
Class 2 Partial	Biological cortex in coordinated task	Hybrid AI–biological with partial coupling, strategic interaction	Hybrid in cyclic game
Class 3 Coupled	Multi-LLM shared substrate, aligned objective (wrapping construction available)	Multi-LLM shared substrate, strategic interaction (wrapping available)	Multi-LLM shared substrate, cyclic interaction

The cells are operationally distinct. The wrapping construction (`#der-class-coercion-via-wrapping`) operates *only* on Axis A — it converts Class 3 → Class 1 architecturally at the wrapper level, at structural-leakage cost. It does *not* change Axis B regime. A wrapped strategic composite remains in R1; the wrapping recovers Class 1 architecturally but the dynamics still admit equilibrium-shaped failure modes (multi-equilibria, saddle-Nash, equilibrium selection ambiguity). Symmetrically, alignment work operates only on Axis B — re-aligning objectives moves the composite from R1 to R0, without changing architectural class. Symbiogenic absorption is the structural mechanism for the R1 → R0 transition (`#hyp-symbiogenic-composition`).

Two distinct adversarial pressures act on the two axes. `#disc-adversarial-coupling-pressure` names the adversarial pressure on Axis A — adversaries strategically drive a target's modularity downward because coupling expands attack surface. A separate adversarial pressure — call it *strategic divergence pressure* — drives a target's Axis B regime downward (introduces objective divergence into a previously-aligned team, moving R0 → R1; introduces cyclic structure by mechanism manipulation, moving R1 → R2). The two are structurally distinct adversarial moves with distinct defenses (architectural scaffolding for Axis A; mechanism-design commitment for Axis B). The M4 meta-segment `#disc-modularity-state-dynamics` covers the three-operation modularity-state picture on Axis A; the dynamic-regime axis here covers the parallel three-operation picture on Axis B (alignment-strengthening self-driven-ascending; strategic-divergence-acquisition self-driven-descending; adversarial-divergence-pressure externally-driven-descending).

Epistemic Status.

Discussion-grade at the meta-pattern level. The segment articulates an axis-vocabulary consolidation rather than a derivation of new machinery. The four-tier partition itself is integration of recognized prior art — Candogan-Menache-Ozdaglar-Parrilo 2011 (Hodge decomposition), Letcher-Balduzzi et al. 2019 (Hamiltonian dynamics decomposition), Sandholm 2010 (population-games taxonomy), Hofbauer-Sigmund 1998 (phase-portrait classification), Omidshafiei et al. 2019 (α-Rank), Papadimitriou-Piliouras 2018 (chain-recurrent sets as natural object).

The AAT-specific derivable contents (§"AAT-specific derivable content") are at distinct tiers: the macro-state-type / regime identity is *exact* for finite N given the entry conditions; the transition asymmetry is *argumentation-grade* pending the J9 derivation spike with a structural-distance metric; the Lyapunov-machinery transfer R0 → R1 is *exact* by direct construction (the algebra is inherited from `#deriv-strategic-composition` sub-scope α').

Max attainable: *discussion-grade* for the meta-pattern (it is an organizing principle, not a derivation); *exact* for the macro-state-type / regime identity; *exact* for the Lyapunov transfer; *argumentation-grade* for the transition asymmetry pending the J9 derivation.

Discussion.

The cross-segment integration consequence. Once the axis is named, `#impl-strategic-composition`'s §"Modular safety architectures fail under goal divergence" implication has a cleaner derivation than the original architectural-class-change framing. The failure mechanism is *regime change* (R0 → R1 under goal divergence) rather than *architectural-class change* (which does not happen — strategic composition with goal-blind routing leaves the composite architecturally Class 1). Modular safety guarantees designed for contraction-regime composites depend on the unique-attracting fixed point and the global Lyapunov contraction; under R0 → R1 transition the unique attractor becomes a set of equilibria, saddle-point Nash breaks Lyapunov contraction, and multi-equilibria break uniqueness. Constitutional AI red-teaming and mesa-optimizer formation are the empirical instantiations. The dynamic-regime axis re-derivation gives the same conclusion on stronger ground.

Mechanism design as regime-shaping. An external designer who can shape the sub-agent objectives can target a regime — bring R2 down to R1 by adding potential structure, or R1 down to R0 by adding alignment. The mechanism-design impossibility theorems (Gibbard-Satterthwaite, Myerson-Satterthwaite, Arrow) are floors on what *external designers* can achieve in the regime-ascent direction under stated incentive-compatibility and welfare constraints. The queued `#disc-implementation-impossibility` meta-segment carries the formal treatment of designer-side constructive-impossibility; this segment's transition-asymmetry argument cross-references it as the structural complement.

The dynamic-regime axis and `#disc-modularity-state-dynamics`. Both axes admit a parallel three-operation modularity-like dynamics: alignment-strengthening / strategic-divergence-acquisition / adversarial-divergence-pressure on Axis B parallel to truthification / strategic-self-coupling / adversarial-coupling-pressure on Axis A. The parallel-pattern observation is itself meta-architectural content — the framework has *two* independently-shaped axes with parallel three-operation dynamics on each. Whether this parallel structure warrants a *higher-order* meta-segment (covering both Axes A and B at once) is open; the current arrangement keeps the two axis-segments separate (this one for Axis B; `#disc-modularity-state-dynamics` for Axis A landed 2026-05-24) and surfaces the parallel observation in each. Now that both meta-segments are stable, evaluating whether the cross-pattern is load-bearing enough for a higher-order meta-segment is a candidate follow-on.

Sub-tier structure within each regime. Each regime has its own internal classification — R0 inherits the Contraction ladder Tier 1 / Tier 2 / Tier 3 from `#disc-separability-pattern`; R1 has potential / monotone / general sub-tier per `#deriv-strategic-composition`; R2 has CCE-convergence-rate sub-classification; R3 has MFG-uniqueness sub-classification. Whether these sub-classifications unify across regimes under a single meta-classification (e.g., "Lyapunov-certificate-strength" as one axis across R0/R1/R2/R3) is a candidate for follow-on work — see Working Notes.

Findings.

The Dynamic-Regime Axis as Cross-Cutting Classifier. **Brief:** Composites differ in how their joint dynamics behave even when their components are individually clean. Think of a team rowing a boat — everyone wants the same destination, the dynamics contract to a shared target, the team is in the *contraction-regime* (R0). Now think of firms in a market — each wants to maximize their own profit, the prices settle into a stable joint equilibrium where no firm wants to deviate, the system is in the *equilibrium-regime* (R1). Now think of rock-paper-scissors players — no stable joint move exists in pure strategies, but the long-run frequencies of moves settle into a stable distribution, the system is in the *cyclic-distributional-regime* (R2). At population scale, where the natural object is the distribution of strategies across the whole population rather than named individuals, you get the *mean-field-equilibrium-regime* (R3). The regime is structural — you can read it off the game definition without running the dynamics — and it determines what kind of guarantee the framework can give you: exponential convergence in R0/R1, polynomial-in-distribution in R2, measure-space machinery in R3. The axis is independent of the architectural class (whether the agent's processing is goal-blind or not); both axes interact, and a strategic composite of architecturally-clean agents is still R1 in dynamic regime even though it remains Class 1 in architectural class.

Impact: Names a structural classifier the framework had been running implicitly across ten-plus existing segments under different vocabulary, consolidating them into one explicit axis with a clear sub-tier structure inherited from existing tiered ladders. Provides a cleaner derivation of `#impl-strategic-composition`'s "modular safety architectures fail under goal divergence" implication — the mechanism is regime change (R0 → R1), not architectural-class change (which does not happen under goal-blind routing). Resolves the long-standing tension in `#scope-composite-agent`'s Working Notes between the alignment routes (C-i/ii/iii) and the strategic-equilibrium route (C-iv): they correspond to regime-tier entry conditions on the dynamic-regime axis, not to a continuous spectrum on a single scalar. Surfaces the structural-distance metric on regime-transition cost as a derivable open question (J9 spike candidate). Provides the positive-half complementarity for the broadened `#disc-identifiability-floor` Instance 3 (regime identification from component marginals is structurally impossible).

Novelty Claim: *Claim integration* of multi-agent dynamics classifications from at least five neighboring literatures (game-theoretic decomposition; multi-agent learning convergence taxonomy; population games; evolutionary games; mean-field games) into an AAT-internal regime-axis vocabulary that travels with the framework's existing scope-route disjunction (`#scope-composite-agent`), separable-core / structured-repair / general-open shape (`#disc-separability-pattern`), and persistence machinery (`#result-sector-persistence-template`). The four-tier partition is recognized prior art; the AAT-specific contributions are (a) the macro-state-type / regime structural identity for finite N with R3 as the explicit boundary case where the identity breaks (*claim recognition* + *claim integration*); (b) the regime-transition cost-asymmetry framing (descent default / ascent costly) tied to mechanism-design impossibility (*claim recognition* — full derivation deferred to J9); (c) the Lyapunov-machinery transfer R0 → R1 made explicit as a cross-tier consolidation of `#deriv-strategic-composition` sub-scope α' (*claim integration*).

Related Work:

Table AH.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
Game decomposition (potential / harmonic / nonstrategic)	Candogan, Menache, Ozdaglar & Parrilo 2011, "Flows and Decompositions of Games: Harmonic and Potential Games," *Math. of OR* 36(3):474–503 (found 2026-05-21)	*formal antecedent* — Hodge-style decomposition of finite normal-form games into three orthogonal components; gives a coordinate-free organizing structure for the R1-vs-R2 boundary cleaner than tier-wise enumeration. AAT adopts the decomposition as the canonical structural reading of the R1/R2 partition; the four-tier ladder is the AAT-internal consolidation across the decomposition plus the contraction-regime (R0) extension plus the mean-field-regime (R3) limit
Differentiable game dynamics with potential / Hamiltonian decomposition	Letcher, Balduzzi, Racanière, Martens, Foerster, Tuyls & Graepel 2019, "Differentiable Game Mechanics," *JMLR* 20:1–40 (found 2026-05-21)	*formal antecedent* — lifts the Hodge-style decomposition from games to *dynamics on games* (potential component → convergent gradient flow; Hamiltonian component → cyclic flow). The R1/R2 partition is exactly the potential-vs-Hamiltonian dynamics distinction at the joint-best-response field level
Cyclic-dynamics-as-natural-object in adversarial learning	Mertikopoulos, Papadimitriou & Piliouras 2017, "Cycles in Adversarial Regularized Learning," *SODA 2018* (proceedings; arXiv:1709.02738, found 2026-05-22)	*formal antecedent (direct R2 predecessor)* — direct historical predecessor to the Piliouras-lab MCC line and to Cheung-Piliouras-Tao 2021: regret-minimization dynamics in adversarial regularized games exhibit *cycles* rather than convergence, and trajectory rate ≠ regret rate (a caveat load-bearing for `#disc-identifiability-floor` Instance 3 escape (e)'s "trajectory-rate not regret-rate" specification). Surface-back from the Working Notes BG2 caveat to the main Related Work table.
Population-games taxonomy (potential / stable / cyclic)	Sandholm 2010, *Population Games and Evolutionary Dynamics*, MIT Press (textbook; found 2026-05-21)	*formal antecedent* — population-level classification of strategic interaction by dynamic regime. The R0/R1/R2 partition tracks Sandholm's "stable" / "potential" / "non-stable cyclic" taxonomy at the finite-N scale; the R3 tier is exactly Sandholm's population-distribution limit
Evolutionary phase-portrait classification	Hofbauer & Sigmund 1998, *Evolutionary Games and Population Dynamics*, Cambridge University Press (textbook; found 2026-05-21)	*formal antecedent* — phase-portrait analysis of replicator dynamics distinguishing fixed-point convergence (R0/R1) from heteroclinic/cyclic structure (R2). The AAT regime axis adopts the phase-portrait reading as the structural shape of the R0/R1/R2 distinction
Cyclic dynamics as solution concept	Omidshafiei, Papadimitriou, Piliouras, Tuyls, Rowland, Lespiau, Czarnecki, Lanctot, Perolat & Munos 2019, "α-Rank: Multi-Agent Evaluation by Evolution," *Scientific Reports* 9:9937 (found 2026-05-21)	*formal antecedent* — Markov-Conley-chain (MCC) solution concept treats cyclic dynamics as the natural object when Nash equilibria are absent or non-attracting. AAT adopts MCC as a sub-tier within R2's distributional macro-state

ASF concern	*Prior-art language*	*Relationship / Positioning*
Chain-recurrent set as outcome of non-converging dynamics	Papadimitriou & Piliouras 2018, "From Nash Equilibria to Chain Recurrent Sets," *ACM Trans. on Economics and Computation* (found 2026-05-21 — local PDF `ref/papadimitriou-piliouras-2018-nash-chain-recurrent.pdf`)	*formal antecedent* — chain-recurrent set as the natural macro-state when fixed-point convergence fails. Provides the formal substrate for R2's distributional macro-state on the dynamics side, complementing Hart-Mas-Colell's CCE on the no-regret-learning side
Fundamental Theorem of Dynamical Systems (the universality behind chain-recurrence)	Conley 1978, *Isolated Invariant Sets and the Morse Index*, CBMS Regional Conf. Ser. Math. 38, AMS, §8.1 (found 2026-05-22)	*formal antecedent (direct citation)* — the universal decomposition of any flow on a compact invariant set into a chain-recurrent subflow plus a strongly-gradient-like quotient. Papadimitriou-Piliouras 2018's chain-recurrent-set formalism builds on Conley's theorem; the direct citation acknowledges Conley as the upstream mathematical source and as the universality backing R2's macro-state classification. Also the cross-layer anchor for `#result-certificate-existence`'s R0-loss rung mixed-case (where the same universality localizes at the single-agent linearization).
Coarse correlated equilibrium convergence	Hart & Mas-Colell 2000, "A Simple Adaptive Procedure Leading to Correlated Equilibrium," *Econometrica* 68:1127–1150 (found pre-2026; cited in `#deriv-strategic-composition`)	*formal antecedent* — supplies the doubling-trick CCE convergence rate $O(1/\sqrt{T})$ that defines the R2 tier's Lyapunov-equivalent guarantee
Mean-field games (MFG)	Lasry & Lions 2007, *Japanese J. Math.* 2:229–260; Huang, Malhamé & Caines 2006, *Comm. Info. Sys.* 6:221–252 (found 2026-05-21 by reference)	*formal antecedent* — supplies the R3 tier's measure-space machinery. AAT does not currently have R3 machinery; the tier is named as the population-scope frontier per `#deriv-strategic-composition` Working Notes mean-field extension
Mechanism-design impossibility (designer-side regime-ascent floor)	Gibbard 1973, *Econometrica* 41:587–601; Satterthwaite 1975, *J. Econ. Theory* 10:187–217; Myerson & Satterthwaite 1983, *J. Econ. Theory* 29:265–281; Arrow 1951, *Social Choice and Individual Values* (found pre-2026; flagged in `#deriv-strategic-composition` Discussion)	*formal antecedent (designer-side floors)* — floor on what external designers can achieve in the R2 → R1 or R1 → R0 regime-ascent direction. Forward-references the queued `#disc-implementation-impossibility` meta-segment for the formal treatment

Search Log:

- 2026-05-21 (*targeted*, BG1 Pillar prior-art search): Verdict on T1 (four-tier classification as unifying construct): *Substantial Overlap with Named Prior Art* (High confidence). Verdict on T2 (macro-state-type / regime identity §C.1): *Substantial Novelty with Adjacent Literature* (Medium-High). Verdict on T3 (transition asymmetry §C.3): *Substantial Novelty with Adjacent Literature* (Medium). Verdict on T4 (identifiability floor §H.5): *Substantial Novelty with Adjacent Literature* (Medium). Most consequential finding: Candogan-Menache-Ozdaglar-Parrilo 2011 Hodge decomposition as the recognized prior art for the R1/R2 boundary (now in the Related Work table).
- 2026-05-21 (*planned*): Second-pass Undermind search for the broadened-Instance-3 + escape (e) to refine the R1-vs-R2 classifier-under-rate-class-observation claim if prior art surfaces.

Appendix AI

Formulation: Resource Budget

A depletable scalar reservoir whose drain rate rises with mismatch and whose level gates correction capacity — the minimal structure that makes "a degrading model is more expensive to run" a dynamical statement rather than an informal one.

The construction is minimal: a scalar resource budget $\mathcal{B}_t$ evolving under drain-versus-replenishment, with two introduced structural posits — **(A-cost)** the drain rate $c(\|\delta\|)$ is non-decreasing in mismatch (degrading models cost more to run), and **(A-gate)** the sector parameter is resource-gated via $\alpha(\mathcal{B}) = \alpha^{\max}\psi(\mathcal{B})$ with $\psi(0) = 0$ (the depleted agent loses corrective capacity entirely). Hard-budget regime has zero replenishment ($r_{\mathcal{B}} = 0$, a finite pool that only depletes); regenerative regime has $r_{\mathcal{B}} > 0$.

This segment is *exploratory off-spine*: no Part I/II/III result depends on it; it opens a resource-structure axis `#def-strategy-dimension` explicitly records as an open scope item. The formulation is *conditional by nature* — (A-cost) and (A-gate) are modeling assertions about an agent's physical realization, not derivations from core machinery, so the formulation's epistemic ceiling stays at `conditional` regardless of downstream development. The companion derivation AJ uses this formulation to close `#der-adversarial-destabilization`'s Effects-Spiral by an alternative mechanism (agent's own correction-rate decay rather than adversary coupling growth).

Formal Expression.

AAT's core machinery is resource-blind: the policy $\pi(M_t, G_t)$ acts at no modeled cost, and correction capacity (tempo $\mathcal{T}$, `#def-adaptive-tempo`) does not deplete. `#def-strategy-dimension` records this as an explicit open scope item — for resource-constrained agents (embodied controllers under torque/battery/episode-length limits; teams under headcount; any agent whose every corrective action spends an exhaustible pool) the formalism carries no state for the pool and no coupling from model quality to its drain. This formulation introduces the minimal such structure.

Definition (resource-budget)

A scalar **resource budget** $\mathcal{B}_t \geq 0$ (calligraphic-scalar by the NOTATION convention exception, as for $\mathcal{T}$; distinct from the agent label B of `#der-adversarial-destabilization` and from the strategy-edge set E). Its evolution:

$$\frac{d\mathcal{B}}{dt} = -c\big(\|\delta\|\big) + r_{\mathcal{B}} \tag{AI.1}$$

where δ is the agent's mismatch (`#def-mismatch-signal`), $c(\cdot) \geq 0$ is the **correction-cost rate**, and $r_{\mathcal{B}} \geq 0$ is the **replenishment rate**. The hard-budget regime is $r_{\mathcal{B}} = 0$ (a finite pool that only depletes — a combat episode's battery, a fixed torque-integral, a bounded step count); the regenerative regime is $r_{\mathcal{B}} > 0$.

The formulation is fixed by two structural posits — introduced, not derived from existing AAT.

Assumption (A-cost: cost rises with mismatch)

c is non-decreasing in $\|\delta\|$, with $c(0) > 0$ (running the loop at all costs something) and c strictly increasing where $\|\delta\| > 0$. Minimal concrete form:

$$c(\|\delta\|) \;=\; c_0\big(1 + \beta_{\mathcal{B}}\,\|\delta\|\big), \qquad c_0 > 0,\ \beta_{\mathcal{B}} \geq 0. \tag{AI.2}$$

Motivation, not derivation: a model that is wrong by δ actuates partly in wrong directions — the agent pays for the corrective action *and* for the wasted component, and re-observes more often to recover. Degradation is literally more expensive to carry. $\beta_{\mathcal{B}} = 0$ recovers the resource-blind special case (cost independent of model quality).

Assumption (A-gate: tempo is resource-gated)

The sector/correction rate the persistence machinery uses — α in G, equal to $\mathcal{T}$ in the canonical epistemic instantiation (`#def-adaptive-tempo`, `#result-persistence-condition`) — is throttled by available resource:

$$\alpha(\mathcal{B}) \;=\; \alpha^{\max}\,\psi(\mathcal{B}), \qquad \psi : \mathbb{R}_{\geq 0} \to [0,1] \text{ non-decreasing},\ \ \psi(0) = 0,\ \ \psi(\mathcal{B}) \to 1 \text{ as } \mathcal{B} \to \infty. \tag{AI.3}$$

Motivation, not derivation: with the pool exhausted the agent cannot run its observe–update–actuate loop at full rate (fewer sensor sweeps, slower control cycle, fewer remaining episode steps). $\psi \equiv 1$ recovers the resource-blind special case (constant α, exactly today's template).

Together: $\mathcal{B}_t$ is the new state; (A-cost) couples model quality *into* its drain; (A-gate) couples its level *into* correction capacity. These two couplings are exactly what close the otherwise-open feedback in `#der-adversarial-destabilization`'s Effects-Spiral corollary — the consequence is derived in AJ.

Epistemic Status.

Conditional. This is a representational choice (`type: formulation`) — several resource models would fit (per-action cost vectors, capacity constraints on the action set, queueing-theoretic service limits); $\mathcal{B}_t$ with (A-cost)/(A-gate) is the *minimal* one that makes the model-quality → drain → capacity loop dynamical with one added scalar and two monotone couplings. The defined objects ($\mathcal{B}_t$, the depletion law) are well-posed; the *content* is the two posits, which are introduced premises, honest as modeling commitments and not consequences of any prior segment. Anything downstream is conditional on them.

Max attainable: `conditional`. This ceiling is intrinsic: (A-cost) and (A-gate) are claims about a particular agent's physical realization, not about the formalism, so no amount of derivation promotes the formulation itself beyond conditional. A *specific* agent class can satisfy them by construction (e.g. a controller whose actuation energy is measured and whose control-loop frequency is power-throttled), at which point downstream results inherit that class's tighter status — but the formulation as stated stays conditional by nature, the same way `#def-strategy-dimension`'s other scope items do.

This segment is an exploratory branch: it opens a resource-structure axis AAT has deliberately lacked. No spine segment depends on it; it is not a prerequisite of any Part I/II/III result.

Discussion.

What the two posits buy, minimally. Everything load-bearing downstream uses only *monotonicity*: (A-cost) that drain is non-decreasing in mismatch, (A-gate) that capacity is non-decreasing in budget with $\psi(0) = 0$. The explicit linear c and the shape of ψ are illustrative; the qualitative consequences in AJ survive any (c, ψ) obeying the monotonicity and the endpoint $\psi(0) = 0$. The endpoint matters: $\psi(0) = 0$ encodes "an empty pool cannot correct at all," which is what makes exhaustion *terminal* rather than merely *slowing*.

Why a new scalar rather than folding into ρ or $\mathcal{T}$. Resource is not disturbance: ρ is mismatch injected by the world, independent of how well the agent is doing; $\mathcal{B}_t$ drains *because* the agent is doing badly (A-cost). Nor is it tempo: $\mathcal{T}$ is a rate the agent has; $\mathcal{B}_t$ is a stock it spends to exercise that rate. Collapsing resource into either erases the very coupling

that makes the Effects-Spiral close — a degrading model raising its own future cost is a stock-flow statement, not a rate statement. The new scalar is the minimum that keeps stock and flow distinct.

Relationship to the adaptive reserve $\Delta\rho^*$. G's adaptive reserve $\Delta\rho^* = \alpha R - \rho$ is a *margin* (how much extra disturbance the agent can absorb at its current rate); $\mathcal{B}_t$ is a *fuel* (how much correction it can still afford to perform). They are independent: an agent can have comfortable margin and an almost-empty pool, or ample fuel and zero margin. The resource-bounded result is precisely about a system that is margin-safe at full fuel and still dies as the fuel drains the margin away — see AJ.

Domain instances. Embodied combat/contested control under a battery or torque-integral budget (the case that motivated this branch); a development team under a fixed person-hour pool where a worsening codebase model raises rework cost; an incident-response org whose escalating confusion burns the on-call budget faster. In each, the resource-blind special case ($\beta_{\mathcal{B}} = 0, \psi \equiv 1$) is the agent for whom the pool is effectively infinite relative to the engagement — and there the existing constant-α machinery is exactly right, which is the honest boundary of this branch's relevance.

Appendix AJ

Derived: Resource-Bounded Destabilization

A hard-budget agent self-depletes to certain finite-time destabilization against even a *constant*-effectiveness adversary — closing `#der-adversarial-destabilization`'s Effects-Spiral not by formalizing its open coupling term but by making that term unnecessary, via the decaying-α instantiation of G.

Formal Expression.

`#der-adversarial-destabilization`'s Effects-Spiral corollary ($\|\delta_B\| \uparrow \Rightarrow$ erratic action $\Rightarrow \gamma_A \uparrow \Rightarrow \rho_B \uparrow \Rightarrow \|\delta_B\| \uparrow$) is *discussion-grade* there because the link "degrading model ⇒ stronger adversary coupling" requires specifying how model degradation feeds back into the dynamics, which that segment's machinery does not carry. The resource-budget formulation (AI) supplies that feedback through a different and cleaner channel — the agent's *own* correction rate, not the adversary's coupling.

Setup: the resource-coupled adversarial instantiation. Instantiate G with the target agent B's mismatch as state variable, $\xi = \delta_B$, exactly as `#der-adversarial-destabilization` does, but with the sector parameter made resource-gated per AI (A-gate):

$$\frac{d\delta_B}{dt} = -\alpha_B(\mathcal{B})\,\delta_B + w_B(t), \qquad \|w_B(t)\| \le \rho_B^{\text{eff}} = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A, \tag{AJ.1}$$

$$\frac{d\mathcal{B}}{dt} = -c\big(\|\delta_B\|\big) + r_{\mathcal{B}}, \qquad \alpha_B(\mathcal{B}) = \alpha_B^{\max}\,\psi(\mathcal{B}), \tag{AJ.2}$$

with ρ_B^{eff} the coupling-amplified disturbance of `#der-adversarial-destabilization` (treated there, and here, with $\gamma_A, \mathcal{T}_A$ exogenous) and c, ψ as posited in AI (A-cost),(A-gate). The novelty is the *coupled pair*: δ_B and $\mathcal{B}$ co-evolve, with model quality draining the budget and the budget gating correction.

Regime split.

Derived (resource-regime-split, from form-resource-budget + sector-persistence-template)

Budget-sufficient regime. If $\mathcal{B}_t$ stays large enough that $\psi(\mathcal{B}_t) \approx 1$ over the engagement (the pool is effectively infinite relative to the integrated cost), then $\alpha_B(\mathcal{B}_t) \approx \alpha_B^{\max}$ is constant and the system *is* `#der-adversarial-destabilization`'s constant-α instantiation, unchanged. The resource structure adds nothing here. This is the honest boundary: the resource-blind special case ($\psi \equiv 1$) is exactly today's machinery, and it is correct there.

Budget-scarce hard regime ($r_{\mathcal{B}} = 0$). With no replenishment, $d\mathcal{B}/dt = -c(\|\delta_B\|) \le -c(0) < 0$ whenever correction runs, so $\mathcal{B}_t$ is strictly decreasing and reaches any level in finite time. Then $\alpha_B(\mathcal{B}_t)$ is a **time-varying, monotonically-decaying** sector parameter — exactly the structure G's Epistemic Status flags as requiring additional machinery, and the

structure `#schema-strategy-persistence` already instantiates (there $\alpha_\Sigma = 1/(n+1)$ decays with experience; persistence then requires the decay be counteracted faster than ρ_Σ/R_Σ). The resource-bounded case is the *same template slot* with budget-depletion in the role experience-accumulation plays for strategy persistence.

Derived (Conditional on A-cost, A-gate; hard regime $r_{\mathcal{B}} = 0$)

The result.

Define the **critical budget** $\mathcal{B}_{\text{crit}}$ by the persistence boundary of G (Model D) at the coupling-amplified disturbance:

$$\alpha_B(\mathcal{B}_{\text{crit}}) = \frac{\rho_B^{\text{eff}}}{R_B} \qquad \Longleftrightarrow \qquad \psi(\mathcal{B}_{\text{crit}}) = \frac{\rho_B^{\text{eff}}}{\alpha_B^{\max} R_B}, \tag{AJ.3}$$

and the destabilization **hitting time** $\tau = \inf\{t : \mathcal{B}_t = \mathcal{B}_{\text{crit}}\}$. Then, in the hard regime with correction ongoing:

1. **Certain finite-time destabilization.** $\mathcal{B}_t \downarrow$ strictly, so $\tau < \infty$ for any finite $\mathcal{B}_0$. Because $\psi(0) = 0$, once $\mathcal{B}_t < \mathcal{B}_{\text{crit}}$ the persistence condition $\alpha_B(\mathcal{B}_t) > \rho_B^{\text{eff}}/R_B$ fails and `#der-adversarial-destabilization`'s destabilization threshold is crossed. **The static persistence inequality holding at $t = 0$ does not prevent this** — an agent that would persist forever at full budget ($\alpha_B^{\max} R_B > \rho_B^{\text{eff}}$) still destabilizes once the fuel drains the rate below threshold.
2. **The spiral, derived.** Substituting (A-cost), a mismatch excursion accelerates its own destabilization:

$$\|\delta_B\| \uparrow \Rightarrow\ c(\|\delta_B\|) \uparrow \Rightarrow\ \dot{\mathcal{B}} \text{ more negative } \Rightarrow\ \alpha_B(\mathcal{B}) \downarrow \Rightarrow\ \left(\alpha_B R_B - \rho_B^{\text{eff}}\right) \downarrow \Rightarrow\ \|\delta_B\| \uparrow, \tag{AJ.4}$$

which brings τ forward: a transient noise burst in w_B permanently advances the destabilization time. This is `#der-adversarial-destabilization`'s Effects-Spiral, now a derived consequence of (A-cost)/(A-gate) rather than a discussion-grade schematic.

1. **The adversary's coupling need not grow.** The spiral closes through α_B *decaying* with γ_A held **constant**. The Effects-Spiral's substantive content — collapse invisible to the static threshold — is captured *without* the unspecified $\gamma_A(\|\delta_B\|)$ leg `#der-adversarial-destabilization` could not formalize. Adversarial survival reduces to a **budget-versus-engagement-length race**: B persists iff the engagement ends before τ.

What Is Derived vs. What Is Chosen.

Table AJ.1

Property	*Source*	*Strength*
Budget-sufficient regime ≡ constant-α template, unchanged	G (status `exact`), $\psi \equiv 1$ limit	Inherited exact
Resource-scarce case is the decaying-α template slot (the reduction)	G time-varying-α + `#schema-strategy-persistence` precedent	Derived, conditional on (A-cost),(A-gate)
Certain finite-time destabilization, $r_{\mathcal{B}} = 0$	Monotone-budget + Lyapunov persistence-boundary crossing	Derived (conditional)
Effects-Spiral as derived feedback; τ advanced by excursions	(A-cost) substituted into the depletion law	Derived (conditional)
Adversary coupling need not grow (γ_A constant suffices)	Corollary of the above	Derived (conditional)
Orthogonal to — not a resolution of — the deferred symmetric joint-Jacobian problem	Comparison of which term carries the feedback (α_B vs γ_A)	Verified relationship
(A-cost), (A-gate)	AI	Introduced posits, not derived

Property	*Source*	*Strength*
Regenerative regime $r_{\mathcal{B}} > 0$	—	Open (not attempted)

Epistemic Status.

Conditional on AI's two introduced posits (A-cost),(A-gate) and the hard regime $r_{\mathcal{B}} = 0$. Given those, the chain is tight: the reduction to the decaying-α slot of G is exact-flavored (it is that template's own machinery, the same instantiation `#schema-strategy-persistence` makes for experience-decay), and certain finite-time destabilization follows by a monotone-budget + Lyapunov-boundary argument with no further assumption. The status is `conditional` rather than `exact` because the load-bearing premises are modeling posits about an agent's physical realization, not consequences of prior segments — inherited honestly from AI's own ceiling.

Max attainable: `conditional` for the formulation-dependent result; a specific agent class that satisfies (A-cost)/(A-gate) by construction (measured actuation energy, power-throttled control loop) lifts *its* instance to that class's tighter status, but the general statement stays conditional by the nature of its premises.

Scope honesty — what this does not do. It does *not* resolve the symmetric joint-best-response / joint-Jacobian equilibrium problem `#der-adversarial-destabilization` defers to `#deriv-strategic-composition`. That deferred problem is about both agents' mismatch co-evolving and specifically about the adversary's $\gamma_A(\|\delta\|)$ leg; this result closes a *different* leg (the target's own resource-gated α_B) and is single-agent-with-resource. The two are orthogonal: this result needs no claim about γ_A growing, and makes none. Reading it as having cracked the symmetric equilibrium problem would be an overclaim — it routes *around* that problem by adding resource structure, at the cost of two posits AAT otherwise lacks.

The regenerative regime ($r_{\mathcal{B}} > 0$) is genuinely open and not attempted here (see Working Notes).

Discussion.

Why this is a strengthening, not a softening. The honest move when facing `#der-adversarial-destabilization`'s discussion-grade Effects-Spiral is not "leave it discussion-grade." It is to ask whether the spiral can be made a theorem. It can — in the resource-bounded regime, by a reduction to existing exact machinery — and the resulting statement is *sharper* than the original conjecture: it eliminates the need for the unformalized $\gamma_A(\|\delta_B\|)$ term rather than supplying a guessed functional form for it. A constant-effectiveness adversary against a hard-budget target is already sufficient for certain finite-time collapse. The strengthen-before-soften discipline produced a cleaner result than the discussion-grade original gestured at.

The reframing is the payoff. "Can the adversary destabilize me?" becomes "does the engagement outlast my fuel?" Persistence under a competent constant adversary is no longer a static inequality on (α, ρ, R) but a race between the hitting time τ and the engagement horizon — and τ is endogenous, advanced by every mismatch excursion the agent suffers. This is the precise sense in which a degrading embodied agent under hard physical limits "spirals": not because the opponent gets stronger, but because being wrong is expensive and the agent is spending a finite pool to be wrong in.

Relationship to `#schema-strategy-persistence`. That schema and this result are the two non-epistemic instantiations of G's *time-varying-*α extension. Strategy persistence's α_Σ decays with accumulated experience and is rescued by forgetting; here α_B decays with spent budget and is rescued only by replenishment or by ending the engagement before τ. Same template slot, different decay driver, structurally parallel rescue conditions. The parallel is itself evidence the template's time-varying-α extension is the right general object, with these as two of its instances.

Findings.

Resource Depletion Closes the Effects Spiral by Eliminating Its Open Term. **Brief:** A robot fighting another robot on a finite battery does not lose because its opponent gets cleverer as it falters — it loses because faltering is expensive. A wrong model wastes actuation, the wasted actuation drains the battery, the drained battery slows the control loop, the slower loop lets the model fall further behind: a self-reinforcing collapse that needs no help from the adversary. The everyday version: a tired driver makes more mistakes, each mistake costs more energy to recover from, and the exhaustion that follows makes the next mistake likelier — the spiral is in the fuel, not in the road. AAT had this spiral only as an informal sketch because it could not say how a worse model makes an opponent stronger. The resource view removes the question: the opponent's strength can be held fixed; a finite, mismatch-sensitive fuel pool is enough to force certain collapse in finite time, with the only open question being whether the fight ends before the fuel does.

Impact: Converts `#der-adversarial-destabilization`'s Effects-Spiral corollary from discussion-grade to conditional-derived — not by formalizing its unspecified adversary-coupling term but by making that term unnecessary, via a reduction to the decaying-α slot of G (the same machinery `#schema-strategy-persistence` uses for experience-decay). Reframes adversarial survival under a competent constant adversary from a static (α, ρ, R) inequality to a budget-versus-engagement-length race with an endogenous, excursion-advanced hitting time. Opens the resource-structure axis `#def-strategy-dimension` records as an explicit gap, as an exploratory branch off the spine. Explicitly does *not* resolve the deferred symmetric joint-Jacobian problem (`#deriv-strategic-composition`) — it is orthogonal, and the Finding marks that boundary so the result is not over-read.

Novelty Claim: *Claim differentiation* on `#der-adversarial-destabilization`'s Effects-Spiral: the contribution is showing the spiral becomes a derived finite-time-destabilization result once a minimal resource state is added, and that the strengthening *removes* rather than supplies the previously-open coupling functional form. The sector-Lyapunov and decaying-α machinery are AAT-internal and standard; resource-constrained control and energy-bounded games are established external territory; the differentiation is the specific recognition that the energy bound is the feedback variable that closes AAT's own named-open spiral by eliminating its open leg, conditional on two explicit posits.

Related Work:

Table AJ.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Resource/energy budgets shaping control behavior	Energy-bounded / resource-constrained control and RL (broad external literature)	*adjacent literature* — supplies the modeling milieu (depletable budgets, cost-of-control); AAT's contribution is the coupling of model quality into drain and budget into the sector parameter, closing a specific AAT-internal spiral
Decaying-gain → eventual instability	`#schema-strategy-persistence` (experience-decay forgetting prerequisite)	*formal antecedent (AAT-internal)* — the same time-varying-α template slot; resource-depletion is its second non-epistemic instantiation
Effects-Spiral as informal panic mechanism	`#der-adversarial-destabilization` Corollary (discussion-grade)	*formalized by this finding* — the prior segment's open feedback term is here made unnecessary, not filled in

Search Log: - 2026-05-19 (*intuition-only*): The resource-constrained-control and energy-bounded-games literatures are large and were not searched; the AAT-internal claim (energy bound as the feedback variable that closes the Effects-Spiral by eliminating its open $\gamma_A(\|\delta\|)$ leg, conditional on (A-cost)/(A-gate)) has not been searched against that literature. The constituent machinery (sector-Lyapunov, decaying-α) is AAT-internal and standard; the differentiation claim is plausibly distinctive but unverified. Targeted future search: cost-of-control / energy-aware MPC, budgeted-RL regret under adversaries, lazy-control and event-triggered control under depletable actuation. Pre-search expectation: constituent moves well-precedented externally; the AAT-internal "closes its own named-open spiral by removal" framing is the candidate-distinctive part.

Appendix AK

Result: Contraction Template

`#result-sector-persistence-template` states AAT's persistence arguments with a Euclidean sector condition (T2) matched to a quadratic Lyapunov in Euclidean norm. Generalizing the sector inequality to a **contraction-metric condition** (Lohmiller & Slotine 1998) preserves the template's ultimate-bound results while extending its coverage in three directions: (a) sub-scope α gains natural non-Euclidean metrics that remove condition-number penalties currently visible in `#der-gain-sector-bridge`; (b) two additional sub-scope β items (PID-bounded-plant; non-convex-within-basin) promote to derived sub-scope metric-α_2 under explicit conditions; (c) composition acquires **topology-indexed closure results** (parallel / hierarchical / small-gain-feedback) from Slotine 2003 that generalize `#deriv-critical-mass-composition`'s matched-symmetric-Tier-1 closed form to heterogeneous sub-agents. This segment states the generalization once in parameter-free form so that each lifting citation can reference it and specify only what varies.

`#result-sector-persistence-template` remains as the $M = I$ Euclidean specialization.

Formal Expression.

Preconditions.

Template preconditions (contraction-template)

Let $\xi(t) \in \mathbb{R}^n$ evolve under $\dot{\xi} = -F(\xi, t) + w(t)$ where F is C^1 in ξ and continuous in t. Let $M : \mathbb{R}^n \times \mathbb{R}_+ \to \mathbb{S}^n_{++}$ be a smooth symmetric positive-definite matrix-valued function with uniform conditioning:

$$m_1 I \preceq M(\xi, t) \preceq m_2 I \quad \text{for all } \xi \in \mathcal{B}_R,\ t \geq 0 \tag{M0}$$

with constants $0 < m_1 \leq m_2 < \infty$.

(CT1) Zero correction at zero state. $F(0, t) = 0$ for all t.

(CT2) Local differential-contraction condition. There exist $\lambda > 0$ and $R > 0$ such that for all $\xi \in \mathcal{B}_R$, $t \geq 0$:

$$\dot{M}(\xi, t) + M(\xi, t)\frac{\partial F}{\partial \xi}(\xi, t) + \Big(\frac{\partial F}{\partial \xi}(\xi, t)\Big)^T M(\xi, t) \succeq 2\lambda\, M(\xi, t). \tag{CT2}$$

(CT3) Bounded disturbance. Either Model D ($\|w(t)\| \leq \rho_\xi$) or Model S ($w(t)$ Wiener-process increment with $\mathbb{E}[\|w(t)\|^2] = \sigma_\xi^2$).

Result (contraction-template-D), conditional on (M0), (CT1)–(CT3-D)

Ultimate bound — Model D.

Under the preconditions with $V(\xi, t) = \xi^T M(\xi, t)\xi$, the state is ultimately bounded:

$$\limsup_{t\to\infty} \|\xi(t)\| \le \frac{\rho_\xi}{\lambda}\sqrt{\frac{m_2}{m_1}}. \tag{CT-D}$$

Structural persistence (the ultimate bound fits within the contraction region $\mathcal{B}_R$) requires

$$\lambda R\sqrt{m_1/m_2} > \rho_\xi. \tag{CT-D-persist}$$

Proof sketch. Compute $\dot V = \xi^T\dot M\xi - 2\xi^T MF + 2\xi^T Mw$. Integrate (CT2) along the ray $s\xi$, $s \in [0,1]$, using $F(0) = 0$: $2\xi^T MF(\xi,t) \ge 2\lambda\,\xi^T M\xi - \xi^T\dot M\xi$. Substituting: $\dot V \le -2\lambda V + 2\xi^T Mw$. Cauchy-Schwarz in M-inner product + (M0) gives the affine-contraction bound on $W = \sqrt{V}$; conversion to Euclidean norm via (M0) yields (CT-D). □

Result (contraction-template-S), conditional on (M0), (CT1)–(CT3-S), Itô-compatible metric

Ultimate bound — Model S (with Itô correction).

Under stochastic disturbance $d\xi = -F(\xi,t)\,dt + \sigma_\xi\,dW_t$ and a metric M satisfying the Itô-correction bound $\frac{1}{2}\sigma_\xi^2$ tr(M+ Hessian correction in drift direction) $\le c_{\text{Itô}}\,m_2$ locally (automatic for state-independent M; bounded by M's Hessian otherwise), the process, counted only while it is still inside the contraction region, satisfies

$$\mathbb{E}\big[V(\xi(t),t);\ \tau_R > t\big] \le V(\xi(0),0)\,e^{-2\lambda t} + \frac{n\sigma_\xi^2 c_{\text{Itô}} m_2}{2\lambda}\big(1 - e^{-2\lambda t}\big), \tag{AK.1}$$

where $\tau_R = \inf\{t : \|\xi(t)\| > R\}$. This is the killed moment, not the stopped one: the mass frozen on $\partial\mathcal{B}_R$ at exit does not contract, so a bound of this shape on $\mathbb{E}[V(\xi(t\wedge\tau_R))]$ is false whenever the stationary scale fits inside the region (A Prop A.1S(i) and Corollary A.1S.1). When (CT2) holds globally the unstopped moment obeys the same bound and the stationary scale is $n\sigma_\xi^2 c_{\text{Itô}} m_2/(2\lambda)$. Mean-square structural persistence requires

$$\lambda R^2 > \frac{n\sigma_\xi^2 c_{\text{Itô}}\, m_2^2}{2m_1}. \tag{CT-S-persist}$$

This is the natural extension of `#deriv-sector-condition`'s Prop A.1S region condition (Khasminskii 2012 ch. 5) to weighted metrics. When $M = I$ (Euclidean), $c_{\text{Itô}} = 1$, and the result reduces to Prop A.1S. What (CT-S-persist) certifies is the killed scale while certified; the exit-time comparison of Prop A.1S-τ is Euclidean (a radial comparison) and is not transferred to weighted metrics here.

Recovery of `#result-sector-persistence-template`. When $M \equiv I$ (Euclidean metric, time-independent), (CT2) reduces to

$$-\frac{\partial F}{\partial \xi} - \Big(\frac{\partial F}{\partial \xi}\Big)^T \preceq -2\lambda I, \tag{AK.2}$$

which is the **incremental Euclidean sector condition** (strong monotonicity of F in Euclidean norm, equivalently **DA2'-inc** in `#form-composition-closure`'s bridge-lemma language). This implies the one-point sector condition $\xi^T F(\xi) \ge \lambda\|\xi\|^2$ by integration along the ray $s\xi$, so $\alpha = \lambda$ when $M = I$. The template's ultimate bound ρ_ξ/α is reproduced with $m_1 = m_2 = 1$.

The structural observation: (CT2) at $M = I$ is strictly stronger than (T2) — it requires the Jacobian's symmetric part to be uniformly positive definite everywhere, matching DA2'-inc rather than (T2). The contraction formulation therefore **collapses (T2) and DA2'-inc into one condition** at the cost of requiring differential (Jacobian-level) rather than integral (inner-product-at-a-point) information. This is one of the template's structural contributions: the same condition that runs the template runs the bridge-lemma that `#form-composition-closure` depends on, so the gap between single-agent persistence and composite tracking faithfulness closes when (CT2) holds.

Sub-scope metric-α_1 / metric-α_2 / metric-β partition. The contraction formulation refines `#deriv-sector-condition`'s A2' sub-scope partition into three tiers:

Sub-scope metric-α_1 (Euclidean metric, AAT-internally derived via DA2'-inc ≡ (CT2) with M=I). Scalar Kalman, Euclidean strongly-convex gradient, L2-regularized convex, linear-PD-symmetric. These cases carry (CT2) with $M = I$ which is exactly DA2'-inc — a condition `#form-composition-closure` has required at the composite level all along. AAT's existing commitment already forces (CT2)-Euclidean at these cases; the template surfaces it.

Sub-scope metric-α_2 (non-Euclidean metric, derived under explicit conditions). Five cases lift here:

- *Matrix Kalman under information metric* $M = (P^-)^{-1}$: contraction rate $\lambda = \lambda_{\min}(H^T R^{-1} H)/2$ on the observable subspace; Euclidean $\kappa(P^-)$ degradation removed. Under (PI)/Čencov (see `#der-gain-sector-bridge` "Fisher-metric cases under parameterization-invariance"), the information-metric form is *uniquely forced* — AAT-internally derived rather than theorem-imported.
- *Exponential-family natural parameters under Fisher metric* $M = \mathbf{I}(\theta)$: contraction rate $\lambda = \eta$ globally on the interior of the natural-parameter domain (Fisher-conditioning degradation removed). Also AAT-internally forced under (PI)/Čencov.
- *Hessian-metric strongly-convex* $M = \nabla^2 L$: contraction rate $\lambda = \eta$ under bounded metric-derivative in the drift direction. Theorem-imported (Hessian metric chosen to match loss curvature; no AAT-internal axiom forces the specific coordinate).
- *Linear-Hurwitz-non-symmetric under Lyapunov metric*: M solves $MA + A^T M = -Q$ for $Q \succ 0$. Contraction rate $\lambda = \lambda_{\min}(Q)/(2\lambda_{\max}(M))$ in the M-metric. **New coverage:** asymmetric-stable linear corrections where Euclidean A2' fails. Theorem-imported (Lyapunov equation construction is standard; no AAT-internal axiom forces Q).
- *PID with bounded plant nonlinearity under Lyapunov metric*: derived under plant-Lipschitz bound $L_p < \lambda_{\min}(Q)/(2\lambda_{\max}(M))$. **Promotion from sub-scope β.** Theorem-imported for the Lyapunov-metric construction; specializations include SPR-tuned PID (phase margin as sector constant; see `#der-gain-sector-bridge` Verified Instances).

Sub-scope metric-β (contraction-metric formulation fails). Four cases:

- *Variational / amortized / approximate posteriors*: contraction to the *projected* target (best-in-class q^*) is derivable, but the projection error $\mathrm{KL}[q^* \| p_t]$ to the true posterior is a residual disturbance that contraction machinery cannot address.
- *Rule-based / symbolic / discontinuous updates*: non-smooth F; (CT2) requires C^1. Piecewise-smooth extensions (Di Bernardo et al. 2014) cover switched systems but not general rule-based reasoning.
- *Severely misspecified agents*: contraction to a wrong target is still wrong. Metric choice is silent on target validity.
- *Per-step SGD / human judgment*: noise-is-disturbance treatment identical to the Euclidean formulation; no improvement from metric choice.

This is **the seventh ladder** in `#disc-separability-pattern` (A2'-scope): metric-α_1 separable-core / metric-α_2 structured-repair / metric-β general-open.

Result (compositional-contraction), conditional on Slotine 2003 Thm 1–3 + per-sub-agent (CT1)–(CT3)

Compositional contraction.

When sub-agents individually satisfy the preconditions with metrics M_i and rates λ_i, the composite satisfies (CT1)–(CT3) under specific topologies:

(CC-parallel) Parallel composition. $\dot{x} = (\dot{x}_1, \dot{x}_2)^T$ with independent $(\dot{x}_i = -F_i(x_i) + w_i)$. The composite contracts under $M_c = \text{blockdiag}(M_1, M_2)$ at rate $\lambda_c = \min(\lambda_1, \lambda_2)$. Recovers `#der-team-persistence`'s weakest-link bound; cooperative coupling improves effective disturbance via the existing signed-coupling mechanism.

(CC-cascade) Hierarchical / lower-triangular cascade. If $\dot{x}_1 = f_1(x_1)$ contracts at λ_1 and $\dot{x}_2 = f_2(x_1, x_2)$ contracts in x_2 at λ_2 uniformly in x_1 with coupling gain $\|\partial f_2/\partial x_1\| \leq k$, the cascade contracts. Rate bounded below by $\min(\lambda_1, \lambda_2)$ up to coupling-gain-dependent adjustment.

(CC-feedback) Negative feedback with bounded loop gain (Slotine 2003 Thm 3). Two systems with rates λ_1, λ_2 connected by negative feedback with loop gains k_{12}, k_{21} — the closed loop contracts iff

$$k_{12}k_{21} < 4\lambda_1\lambda_2. \tag{CC-feedback}$$

Derived (CM2-M), from (CC-feedback) + '#deriv-critical-mass-composition' signed-coupling structure

Heterogeneous critical-mass — (CM2-M).

Specializing (CC-feedback) to the signed-coupling structure of `#deriv-critical-mass-composition` for heterogeneous Tier-1M sub-agents with metric-contraction rates λ_1, λ_2, coordination costs C_1, C_2, and feedback loop gains k_{12}, k_{21}:

$$(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4. \tag{CM2-M}$$

Specializing further to the matched-symmetric case ($\lambda_1 = \lambda_2 = \lambda, C_1 = C_2 = C, k_{12} = k_{21} = k$): $(\lambda - C)^2 > k^2/4 \iff \lambda - C > k/2$, which matches `#deriv-critical-mass-composition`'s (CM2) with $k = 2\gamma\mathcal{T}/R$ (up to normalization). **Heterogeneous-architecture composites now have a closed-form critical-mass inequality** where the matched-symmetric case was the only closed form previously available; `#deriv-critical-mass-composition`'s §6.1 obstruction on heterogeneous composites is thereby closed for the Tier-1M case.

Epistemic Status.

Conditional. Conditional on (M0) + (CT1)–(CT3) and the specific metric M chosen for each sub-scope α entry. The (CT-D) and (CT-S) ultimate-bound results are standard Lohmiller-Slotine theory (Lohmiller & Slotine 1998; Slotine 2003) specialized to AAT's per-instance state-variable choices; the proof technique is integration along the ray $s\xi$ followed by affine contraction. The contraction-metric primitive (Riemannian M) is standard in nonlinear control theory (Lohmiller-Slotine 1998; Forni & Sepulchre 2014 for the Finsler extension); AAT's contribution is the AAT-internal mapping of M-choices to sub-scope α_1/α_2 and the recognition that the compositional theorems (CC-parallel / CC-cascade / CC-feedback / CM2-M) directly generalize Part III's existing closure machinery.

The Jacobian-level observation. `#form-composition-closure`'s DA2'-inc condition is equivalent to (CT2) with $M = I$ for C^1 F on convex domains. AAT has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along under the name DA2'-inc; the contraction template's Euclidean specialization surfaces this at the single-agent level.

The (PI)/Čencov observation. Two metric-α_2 cases (information-metric Kalman, Fisher-metric exp-family) are AAT-internally forced under the (PI) axiom in `#scope-agent-identity` combined with Čencov's 1982 uniqueness theorem — the fourth primary instance of the meta-pattern in `#disc-additive-coordinate-forcing`. The Euclidean metric-α_1 case is **AAT-internally derived via DA2'-inc equivalence** rather than theorem-imported, structurally connecting it to the bridge lemma in `#form-composition-closure`. The remaining metric-α_2 cases (Hessian, Lyapunov-linear-Hurwitz,

Lyapunov-PID) are theorem-imported: the metric choice is problem-specific (matched to loss curvature or plant structure) rather than forced by an AAT-internal axiom.

Max attainable: *conditional*. The condition (the metric specification must be consistent with the system structure; the Slotine 2003 compositional theorems must apply to the topology) is inherent to the contraction-metric framework, not removable. Non-smooth systems (rule-based), strategic/game-theoretic equilibria, and systems without regular metric-state coupling remain honestly outside the template's scope (see §Honest failure modes).

Relationship to monotone-operator theory. The contraction-metric condition (CT2) is a specialization of monotone-operator theory's strong-monotonicity condition in a Hilbert space with a specific inner product structure (Rockafellar 1970; Bauschke & Combettes 2017 §§4, 22–28). AAT's `#deriv-sector-condition` reading of A2' α/β as operator-family classification — proximal / firmly nonexpansive / cocoercive / strongly-monotone-gradient / linear-PD — identifies the same mathematical content in operator-theoretic language; see that segment's Grounding paragraph for the correspondence. The present template does not claim the contraction-metric machinery as distinctively AAT — it claims the specific instantiation to AAT's state-variable choices and the composition with identifiability-floor + composition-scope-condition content as AAT-distinctive.

Honest failure modes.

The contraction-metric formulation does **not** lift AAT's coverage in five cases:

1. **Non-smooth dynamics.** Rule-based agents, state-machine controllers, threshold-triggered switches, and discontinuous policies fail the C^1 requirement. Piecewise-contraction extensions exist (Di Bernardo, Liuzza & Russo 2014, *SIAM J. Control Optim.* 52) but apply to specific switched structures, not general rule-based reasoning. This barrier is structural, not a gap in the metric machinery.

2. **Strategic / adversarial equilibria — three independent convergent obstructions.** The adversarial / strategic half of Part III is structurally outside contraction-metric scope for three distinct reasons that converge on the same conclusion; the convergence matters because it shows the limit is not an accident of one particular framework choice:

   (i) *Compositional-contraction applicability* (Slotine 2003 §III): compositional contraction theorems require each sub-system to be individually contracting *and* for the composition topology to preserve contraction. Two-agent strategic dynamics with opposing objectives yield **saddle-point equilibria of the joint best-response dynamics** — attracting in some directions and repelling in others. Contraction requires attracting fixed points; saddle points break the framework directly.

   (ii) *Passivity universality* (Khalil 2002 ch. 6 Thm 6.4): the dissipation inequality $\dot{S} \leq s(u, y)$ holds for *all* inputs u by the passivity property's universal quantifier. A strategic opponent can select u to maximize the recipient's storage-function growth rather than respecting the port structure. Passivity delivers cooperative/neutral robustness, not adversarial.

   (iii) *Last-iterate convergence failure* (Daskalakis, Ilyas, Syrgkanis & Zeng 2018 "Training GANs with optimism" *ICLR*): adversarial gradient-ascent-descent dynamics (fictitious play, no-regret dynamics) converge to Nash equilibria *in average-iterate* sense (Cesàro-mean) but generically **fail last-iterate convergence** for non-zero-sum games. No contraction metric exists for systems whose last iterates do not converge to an attractor.

   The three obstructions come from different theorem families (compositional-contraction / dissipativity-universality / differential-Lyapunov-attractor-theory) but converge on the same verdict: **contraction metrics cover the cooperative half of Part III** (`#der-team-persistence`, cooperative `#deriv-critical-mass-composition`, cooperative `#form-composition-closure`). The adversarial / strategic half belongs to `#deriv-strategic-composition` (equilibrium convergence via Monderer-Shapley / Rosen) and `#der-adversarial-destabilization` (asymmetric attacker-target dynamics). This is a scope boundary that AAT correctly honors, not a gap to be closed — the apparatus

that handles adversarial regimes is equilibrium-theoretic, structurally different from Lyapunov-contraction machinery.

3. **State-dependent metric stochastic coupling.** Pham & Slotine 2007 extends contraction to stochastic systems under small-noise bounds; the Itô-correction $c_{\text{Itô}}$ in (CT-S) is manageable for state-independent or slow-varying metrics. Strongly state-dependent M (adaptive-metric algorithms where M is learned alongside ξ) introduces cross-coupling that can destabilize contraction — the same structural issue `#deriv-adaptive-gain-dynamics` handles with its (MG-4) meta-gain coupling-boundedness condition.

4. **Multi-basin dynamics.** Lohmiller-Slotine contraction is local-on-a-region. Agents with multiple basins of attraction admit only basin-local metrics, giving a basin-chart structure rather than a single global metric. The basin-crossing boundary is exactly the `#result-structural-adaptation-necessity` trigger; the metric formulation makes the basin structure explicit but does not escape it.

5. **Identifiability-floor intersection.** The metric formulation operates downstream of identification. It assumes the correction function F points in a valid direction. Under the floors in `#disc-identifiability-floor` (on-policy L0-insufficiency no-go; L1' unobservable-C Cramér-Rao floor; composition-layer Instance 3 Liberzon no-go), the correction direction is structurally unavailable. Contraction metrics cannot recover identification — the two architectural moves are orthogonal.

Discussion.

Why factor this out. The Euclidean sector inequality in `#result-sector-persistence-template` (T2) matches a quadratic Lyapunov in Euclidean norm. For several AAT-relevant agent classes, the natural Lyapunov lives in a non-Euclidean metric (Fisher for statistical-manifold learning; Mahalanobis for Kalman; Hessian-induced for strongly-convex optimization; Lyapunov-equation-determined for asymmetric-stable linear systems). Stating AAT's persistence machinery once in parameter-free metric-formulation form makes visible that the sector condition and the contraction-metric condition are two states of the same apparatus — Euclidean and non-Euclidean — with the specific metric instantiated per sub-scope entry.

This is the *interior facet* of the stability certificate (`#disc-stability-certificate`) in the non-Euclidean metric: the contraction-metric M is the certificate, and which metric is available per sub-scope is the certificate-strength ladder (R0-strict widest contraction / R1 cocoercive / R2 Čencov-forced; R0-loss is the wider Lyapunov-stable rung below where the certificate exists at $\kappa = 0$) of H. The five non-Euclidean metric cases are five certificates of one object, not five separate conditions.

The compositional payoff. The topology-indexed closures (CC-parallel / CC-cascade / CC-feedback) extend to heterogeneous-architecture composites via (CM2-M), closing the §6.1 heterogeneous-architecture obstruction in `#deriv-critical-mass-composition`. This is Part III's single largest coverage expansion: from "closed form for symmetric-matched-Tier-1" to "closed form under any of four topology classes with per-sub-agent (CT1)–(CT3) verification."

Relationship to `#result-sector-persistence-template`. The two segments coexist. `#result-sector-persistence-template`'s Euclidean formulation remains the default for agents whose natural coordinate is already Euclidean; `#result-contraction-template` is invoked when a non-Euclidean metric is natural (matrix Kalman, exp-family, ill-conditioned gradient, asymmetric-stable linear, PID-with-bounded-plant-nonlinearity). Every `#result-sector-persistence-template` invocation can be re-read as a `#result-contraction-template` invocation at $M = I$; the inverse mapping goes from metric to Euclidean with a $\kappa(M)$ degradation. Which direction is natural depends on the instance.

Relationship to `#form-composition-closure`'s bridge lemma. The bridge lemma's DA2'-inc condition is equivalent to (CT2) with $M = I$ on convex C^1 domains. `#result-contraction-template` therefore sharpens the bridge-lemma understanding: Tier 1 (DA2'-inc globally) ≡ metric-contracting-globally at $M = I$; Tier 2 (DA2'-inc locally with $\kappa(D\hat{o})^2$ degradation) ≡ metric-contracting-locally with a specific local metric; Tier 3 (per-domain verification) ≡ no

globally valid metric exists. The tier structure of the bridge lemma carries over to `#result-contraction-template` as Tier 1M / 2M / 3M respectively.

Relationship to `#disc-identifiability-floor` Instance 3 (composition-layer no-go). Instance 3's §3.3 construction uses matched-symmetric-Tier-1 scalar systems; the coupling-sign bit is unidentifiable from component marginals. Escape route (E-d) in that Instance names "shared metric structure (common contraction metric; Lohmiller-Slotine 1998)" as an adjacent machinery the no-go motivates; `#result-contraction-template` is the segment that machinery lives in. The Instance 3 escape is therefore operationalized by a `#result-contraction-template` invocation with a composite metric constructed from sub-agent metrics per the topology (parallel / cascade / feedback).

Relationship to meta-patterns.

- `#disc-separability-pattern`*:* metric-α_1 / metric-α_2 / metric-β is the seventh ladder (A2'-scope), closing the "binary special case" gap that segment's Discussion flagged. Separable-core (metric-α_1, Euclidean, AAT-internally derived via DA2'-inc ≡ (CT2)-M=I) / structured-repair (metric-α_2, non-Euclidean, derived under explicit conditions including (PI)/Čencov for Fisher-metric cases) / general-open (metric-β, metric formulation fails).
- `#disc-identifiability-floor`*:* orthogonal axis. Contraction metrics lift the *geometry-of-correction* half of sub-scope α; identifiability-floor operates on the *what-correction-to-make* half. The two architectural moves compose (Instance 3 at composition layer + metric-Tier structure at single-agent layer) but do not interact.
- `#disc-additive-coordinate-forcing`*:* the (PI)/Čencov axiom that grounds two of five metric-α_2 cases is the fourth primary instance of the meta-pattern, adding a Čencov-invariance machinery alongside the three Cauchy-FE instances. See that segment's Four Instances tables.

Findings.

Topology-Indexed Compositional Closures via Contraction-Metric Generalization. **Brief:** AAT's central machinery for proving an adaptive system holds its mismatch bounded was originally written in the simplest geometric setting (the equivalent of measuring distance with a ruler in flat space). Many real systems have a more natural geometry — a Kalman filter "thinks" in terms of an information ellipsoid, a learning agent thinks in Fisher-information geometry, a controller designed for an asymmetric plant thinks in a different metric still. This result restates the persistence machinery in a way that uses each system's natural geometry, lifting five additional agent classes from sub-scope β (assumed) to sub-scope α_1/α_2 (derived under explicit conditions), and inheriting Slotine 2003's topology-indexed compositional closures (parallel / cascade / negative-feedback) at the composition layer.

Impact: Part III's largest single coverage expansion: from "closed-form composite persistence under matched-symmetric Tier-1" to "closed-form composite persistence under any of four topology classes with per-sub-agent contraction verification, including heterogeneous-architecture mixes via (CM2-M)." For Part II, the structural-transparency observation that DA2'-inc ≡ (CT2)-at-$M = I$ shows AAT's existing Euclidean machinery was already carrying the Jacobian-level contraction commitment under another name; what changes is making that commitment legible and lifting non-Euclidean cases (Fisher-metric Kalman, exp-family natural-parameter, ill-conditioned strongly-convex gradient, asymmetric-stable linear, PID with bounded plant) from theorem-imported to AAT-internally derived under named axioms. Three independent obstructions (Slotine 2003 saddle-point; passivity-universality; Daskalakis 2018 last-iterate non-convergence) bound the template's scope honestly — adversarial composition is structurally outside contraction analysis and properly handled by `#deriv-strategic-composition`'s equilibrium-convergence machinery.

Novelty Claim: *Claim synthesis* on contraction-metric machinery + AAT's sub-scope partition + (PI)/Čencov axiom. The contraction-metric primitive itself is standard nonlinear control theory; the topology-indexed closures are Slotine 2003. The synthesis is the AAT-internal mapping of metric choices to per-instance state-variable choices, the recognition that the Euclidean specialization is structurally the bridge-lemma's DA2'-inc, and the (PI)/Čencov axiom forcing two of the metric-α_2 cases to be AAT-internally derived rather than theorem-imported.

Related Work:

Table AK.1

ASF concern	*Prior-art language*	*Relationship / Positioning*
Differential-contraction primitive on Riemannian metrics	Lohmiller & Slotine 1998, "On Contraction Analysis for Non-linear Systems" *Automatica* 34:683–696 (published 1998, found pre-2026)	*formal antecedent* — supplies the (CT1)–(CT3) preconditions and the (CT-D) ultimate-bound argument adopted directly
Compositional contraction under topology	Slotine 2003, "Modular stability tools for distributed computation and control" *Int. J. Adapt. Control Signal Process.* 17:397–416 (published 2003, found pre-2026) — Theorems 1–3 (parallel; cascade; negative-feedback small-gain)	*formal antecedent* — supplies (CC-parallel) / (CC-cascade) / (CC-feedback) and the heterogeneous (CM2-M) extension. AAT adopts these directly, applied to the composition-validity question of `#form-composition-closure`
Finsler (direction-dependent) generalization	Forni & Sepulchre 2014, "A Differential Lyapunov Framework for Contraction Analysis" *IEEE TAC* 59 (published 2014, found 2026-04)	*adjacent literature* — Finsler-metric extension; flagged in Working Notes for highly anisotropic agents; not currently load-bearing
Strong monotonicity / firmly-nonexpansive operators	Rockafellar 1970, "On the maximality of sums of nonlinear monotone operators" *Trans. AMS*; Bauschke & Combettes 2017, *Convex Analysis and Monotone Operator Theory in Hilbert Spaces* §§4, 22–28	*formal antecedent* — (CT2) at $M = I$ is strong monotonicity in Hilbert space; the Rockafellar-Wets / Nesterov equivalence (DA2'-inc ⇔ Jacobian-symmetric-part PD ⇔ (CT2)-at-$M = I$) is the standard monotone-operator identity AAT uses to make the Euclidean metric-α_1 case AAT-internally derived rather than theorem-imported
Fisher-metric / information-geometry uniqueness	Čencov 1982, *Statistical Decision Rules and Optimal Inference* AMS Translations 53	*formal antecedent* — under the (PI) axiom in `#scope-agent-identity`, Čencov's theorem forces the Fisher-metric Kalman and exp-family natural-parameter cases AAT-internally
Bounded-loss composition under predictive compression	Subramanian, Sinha, Seraj & Mahajan 2020, "Approximate information state for approximate planning and reinforcement learning in partially observed systems" *arXiv:2010.08843* (published 2020, found 2026-04)	*adjacent literature* — closest mathematical neighbor to the bridge-lemma side of `#form-composition-closure`. The contraction-metric machinery in this segment supplies the (CT2) condition that, at $M = I$, *is* the bridge-lemma's DA2'-inc; Sub20's predictive-loss-to-control-error result lives in the same shape but at a coarser granularity (single-agent compressed control, not composite agent boundary)
Influence-based abstraction loss bounds	Congeduti, Mey & Oliehoek 2020, "Loss Bounds for Approximate Influence-Based Abstraction" *arXiv:2011.01788* (published 2020, found 2026-04)	*adjacent literature* — bounds value loss from approximate cross-agent influence; the (CC-feedback) topology with bounded loop gain is the contraction-metric counterpart at the dynamics layer
Saddle-point / adversarial limits of contraction	Daskalakis, Ilyas, Syrgkanis & Zeng 2018, "Training GANs with optimism" *ICLR* (published 2018, found 2026-04); Khalil 2002 *Nonlinear Systems* 3rd ed. ch. 6 Thm 6.4 (passivity-universality)	*contradicted by this finding* (in the sense of correctly bounding scope) — three independent obstructions to handling adversarial regimes converge: Slotine 2003 §III saddle-point limitation; passivity-universality's universal quantifier; Daskalakis 2018 last-iterate non-convergence. The convergence demonstrates the scope boundary is not an artifact of one framework; adversarial composition is structurally outside contraction analysis

ASF concern	*Prior-art language*	*Relationship / Positioning*
Switched-systems common-Lyapunov nonexistence	Liberzon 2003, *Switching in Systems and Control* Theorem 2.1	*formal antecedent* — supplies `#disc-identifiability-floor` Instance 3's no-go; the contraction-metric construction operationalizes escape route (d) (common contraction metric) for that Instance

Search Log:

- 2026-04 (*nominally comprehensive*): The composition-layer extension of this segment (the (CC-*) closures and (CM2-M)) was reviewed under the same Undermind pass that covered `#form-composition-closure`. Verdict: the contraction-metric machinery itself is standard adopted prior art (formal antecedent); the AAT-internal mapping of metric choices to per-instance sub-scope α_1/α_2 entries and the use of (PI)/Čencov to force two metric-α_2 cases AAT-internally are the synthesis-level contributions. The Sub20 / Con20 positioning at the composition-validity layer is recorded both here and at `#form-composition-closure` since the contraction template is what supplies the formal mechanism the bridge lemma uses.
- 2026-04 (*targeted*): Lohmiller-Slotine 1998 / Slotine 2003 / Forni-Sepulchre 2014 known and cited from the contraction-analysis literature. The three adversarial obstructions (Slotine 2003 §III; Khalil 2002 ch. 6; Daskalakis et al. 2018) are convergent scope-boundary evidence.
- 2026-04 (*intuition-only*, prior): expected the contraction-metric primitive to be standard control-theory; expected the composition theorems to be the more proprietary content. Comprehensive search confirmed both: the primitive is well-established, and the composition theorems are Slotine 2003's, with AAT's contribution lying in the application to `#form-composition-closure`'s admissibility-and-bridge-lemma framework.

Appendix AL

Derivation: Variational Approximate-A2' (ε-Fidelity)

Variational / approximate-posterior agents (VI, amortized VI, active-inference-style variational free energy) currently sit in A2' sub-scope β per `#form-sector-condition`: their correction functions target the *best-in-class* variational posterior q^* rather than the true posterior p, and the approximation gap can rotate the correction direction enough to break B1 directional fidelity (`#der-gain-sector-bridge`). Under a KL bound $\mathrm{KL}(q_\phi \| p) \leq \varepsilon$ on the variational approximation, directional fidelity recovers in a quantifiable form: **ε-fidelity B1**, with sector-constant degradation scaling as $O(\sqrt{\varepsilon})$ (Pinsker-tight). The sector-persistence template applies under a Regime-A / Regime-B decomposition — clean sector bound on an annulus away from the projection-error floor, approximation-dominated on a ball of radius $\delta_0 = O(\sqrt{\varepsilon})$ around the target. This promotes controlled-KL VI from sub-scope β to a new intermediate tier α' within the A2' partition (cf. the α / α_1 / α_2 / β refinements from `#deriv-adaptive-gain-dynamics` and `#deriv-fisher-whitened-update-rule`).

The result has a striking operational consequence: **mean-field VI is structurally sub-optimal as an adaptive system by a factor $\sqrt{1 - r^2}$ in the sector constant, regardless of compute or training data**, because the mean-field family does not match the Fisher metric. **Natural-gradient VI recovers the full sector parameter** (Khan-Lin 2017 conjugate-computation equivalence places it in full sub-scope α rather than α') because the natural-gradient direction matches the Fisher metric — no degradation. Where persistence-optimality matters, the variational-family choice is not interchangeable; the long-running Bayesian-deep-learning practical preference for natural-gradient over mean-field has a principled persistence-theoretic reading inside AAT.

Formal Expression.

ε-fidelity B1.

Derived (eps-fidelity-B1, from Pinsker + Cauchy-Schwarz)

Let the true posterior be $p(z \mid x)$ and the variational approximation $q_\phi(z \mid x)$ with $\mathrm{KL}(q_\phi \| p) \leq \varepsilon$. Under standard Lipschitz assumptions on the observation model and nested-support on the variational family, the total-variation distance bounds as $\|q_\phi - p\|_{TV} \leq \sqrt{\varepsilon/2}$ (Pinsker's inequality). Propagating this bound through the correction function via Cauchy-Schwarz:

$$(K\hat{P} - P^*)^T(\hat{P} - P^*) \geq (c_{\min} - C_H\sqrt{2\varepsilon}/\|\delta\|) \cdot \|\delta\|^2 \tag{AL.1}$$

where C_H is a constant depending on the observation-model Lipschitz constant. The effective sector constant

$$c_\varepsilon(\|\delta\|) = c_{\min} - C_H\sqrt{2\varepsilon}/\|\delta\| \tag{AL.2}$$

is state-dependent: large-mismatch regions see near-full sector constant; near-target regions see degraded sector constant due to approximation error dominating.

Formulation (regime-decomposition-variational)

Regime-A / Regime-B decomposition.

Define $\delta_0 = 2C_H\sqrt{2\varepsilon}/c_{\min}$ (the approximation-dominated radius around target). Then:

- **Regime A — clean sector bound** ($\|\delta\| > 2\delta_0$). On the annulus $\mathcal{B}_R \setminus \mathcal{B}_{2\delta_0}$, $c_\varepsilon \geq c_{\min}/2$: sector condition holds with constant $c_{\min}/2$. The template's ultimate bound $\rho_\xi/(c_{\min}/2)$ applies.
- **Regime B — approximation-dominated floor** ($\|\delta\| \leq 2\delta_0$). Near target, c_ε can be arbitrarily small; the correction may not contract. The ultimate bound gains an additive $O(\sqrt{\varepsilon})$ term: $R^*_\varepsilon = \rho_\xi/(c_{\min}/2) + \delta_0 = \rho_\xi/(c_{\min}/2) + O(\sqrt{\varepsilon})$.

Sector-persistence template instantiates with: - State variable $\xi = \hat{P} - P^*$ (variational mismatch to target posterior). - Effective ultimate bound R^*_ε on Euclidean norm. - Persistence requires $R^*_\varepsilon < R$, i.e., $\rho_\xi/(c_{\min}/2) + O(\sqrt{\varepsilon}) < R$.

Khasminskii stopping-time localization (same technique as `#deriv-sector-condition` Prop A.1S and the A2' strengthening spike) applies to the annulus Regime A; Regime B is handled by accepting δ_0 as a projection-error floor.

Derived (per-variational-case)

Per-case verdicts.

The α'-membership depends on the specific variational scheme:

Natural-gradient VI with exponential-family q_ϕ. Khan & Lin 2017 (*Conjugate-computation variational inference*) showed natural-gradient VI is equivalent to closed-form conjugate-Bayesian updates for exponential-family variational distributions. This recovers *full* sub-scope α membership (not merely α'), with A2' derived rather than ε-degraded, by converting the variational update into a Bayesian update in a reparameterized family. The $\varepsilon = 0$ limit is exact.

Mean-field VI. When the true posterior is approximately factorized ($p(z) \approx \prod_i p_i(z_i)$), mean-field VI achieves small ε and is the workhorse α' case. Ultimate bound degrades by additive $O(\sqrt{\varepsilon})$; sector persistence holds.

Amortized VI (VAE-style). Amortization adds a second approximation error (the variational network's function-class limit). KL bounds compose *additively*: δ_0 grows as $\sqrt{\varepsilon_{\text{family}} + \varepsilon_{\text{amort}} + \varepsilon_{\text{generalization}}}$. Under controlled-$\varepsilon$ amortized VI, α' membership holds with a larger floor.

Diffusion-posterior / energy-based with uncontrolled MCMC. No controlled ε bound; ε grows with mixing time. Stays firmly in sub-scope β.

Active inference (variational free energy). Conditional α' under exponential-family q + natural-gradient; ε-degraded α' otherwise. This does **not** force V-strong G-BP2 (presentation of AAT as control-theoretic specialization of active inference) — V-medium (KL-divergence-based cognitive cost) remains the appropriate scope commitment; the comparison trail is in the spike-routing cycle (CHANGELOG 2026-05-17).

Sub-scope α' in the A2' partition.

Derived (sub-scope-alpha-prime-partition)

The A2' sub-scope partition is structured as:

- **Sub-scope α** (derived under B1 directional fidelity; `#der-gain-sector-bridge`): Kalman, conjugate Bayesian, exponential-family natural parameters, strongly-convex gradient, L2-regularized, linear-PD-symmetric.
- **Sub-scope α_1/α_2** (`#result-contraction-template`): metric-formulation generalization of α.
- **Sub-scope α_3** (`#deriv-fisher-whitened-update-rule`): correlated evidence + Fisher-whitened under (PI)/Čencov.
- **Sub-scope α'** (this segment): controlled-KL VI under Pinsker's inequality with $O(\sqrt{\varepsilon})$ sector-constant degradation + Regime-A/B decomposition.
- **Sub-scope β**: uncontrolled-ε agents; non-smooth rule-based; severely misspecified; per-step SGD; human judgment.

This gives the full current picture: $\{\alpha, \alpha_1, \alpha_2, \alpha_3, \alpha', \beta\}$.

Epistemic Status.

Conditional. Max attainable: *exact* for the Pinsker-bound derivation (standard inequality). The $O(\sqrt{\varepsilon})$ rate is tight in general — improving to $O(\varepsilon)$ requires stronger inequalities (log-Sobolev, Talagrand T2) holding for the specific variational family and target; not chased in this segment but is a potential refinement for concentration-preserving families.

Load-bearing: - Pinsker + Cauchy-Schwarz gives the $O(\sqrt{\varepsilon})$ rate rigorously. - Regime-A annulus sector bound with $c_{\min}/2$ is derived (straightforward bookkeeping on the state-dependent c_ε). - Natural-gradient VI promotion to full α is standard (Khan & Lin 2017). - Mean-field VI as workhorse α' is standard variational analysis.

Not established: - $O(\varepsilon)$ rate (would require stronger inequalities). - Quantitative ε bounds for specific variational families in practice (amortization gaps are typically not estimable tightly). - α' membership under mode-missing VI (multimodal posteriors where q concentrates on a subset of modes). These may require separate analysis.

Honest Failure Modes.

Scope honesty — variational-alpha-prime-limits

- **Uncontrolled-ε agents**: diffusion posteriors with finite MCMC; VAEs trained without KL control; any VI without a bounded-ε certificate. Stays in β.
- **Mode-missing on strongly multimodal posteriors**: q misses modes; ε is irreducible regardless of optimization. α' fails structurally.
- **Non-Lipschitz observation models**: the C_H constant diverges; the Pinsker propagation bound doesn't hold. α' fails.
- **Support mismatch**: q's support is not a subset of p's support (or vice versa); KL is $+\infty$. α' fails.
- **Non-exponential-family + non-amortized + uncontrolled + multimodal**: the worst case, strictly β.

Discussion.

Relationship to `#form-strategy-complexity-cost`. The G-BP2 V-medium variational form landed in `#form-strategy-complexity-cost` uses KL divergence in the cognitive-cost term. This segment provides the complementary story on the persistence side: KL-bounded VI has ε-fidelity B1 and α' sector condition. The two together characterize the cognitive/persistence tradeoff for variational agents: ε controls both the cognitive cost (how far from the target) and the persistence degradation (how much sector-constant penalty). Large ε means cheap-but-persistently-weak; small ε means expensive-but-persistently-sharp.

Relationship to `#disc-compression-operations`. Variational compression is the first of the four AAT compression operations; its α' sector structure gives a concrete persistence guarantee for variational representation of M_t. The other three operations (strategy Σ_t, shared-intent, coarse-graining Λ) may admit similar ε-fidelity analyses where a KL-bound on the compression is available.

Meta-pattern positioning. - `#disc-separability-pattern`: α' sits on the structured-repair tier of the A2'-scope ladder (7th ladder from `#result-contraction-template`), alongside metric-α_2 and α_3. Each represents "sector condition recovered under explicit additional structure." - `#disc-additive-coordinate-forcing`: variational persistence sits *outside* this meta-pattern — no logarithmic coordinate is forced; Pinsker is a bound, not a Cauchy-FE argument. α' is an adjacent family member (shape: controlled-approximation-with-quantified-degradation), not a primary instance.

Appendix AM

Derivation: L1' Update-Bias Formula

Under L1' correlated-evidence regimes with unobservable common cause, the default log-odds edge-update (V) converges to a **biased fixed point** — edges settle at log-odds values that match the L1' root-probability rather than the marginal-edge truth. This segment derives a closed-form bias formula for the two-edge OR-root case, verifies it via Monte Carlo, and composes it with `#disc-identifiability-floor` Instance 2 and `#schema-strategy-persistence`'s forgetting prerequisite to produce a **dual forgetting-rate requirement** on strategic persistence.

The bias is the quantitative companion to F13's structural-identifiability-floor result: F13 shows identification is *structurally impossible* from single-channel observation; this segment shows the resulting *magnitude drift* is bounded, Lipschitz in correlation strength ρ, and Cramér-Rao-floored under forgetting. The dual forgetting-rate requirement reads operationally: forget *fast enough* to track non-stationarity (the plasticity lower bound from `#schema-strategy-persistence`) *and* fast enough to keep the L1' bias bounded (this segment's upper-bound-driving lower bound on $1 - \lambda$). When the two compose with `#form-consolidation-dynamics`'s stability cadence into an empty admissibility window, the agent must augment its observability or accept biased-fixed-point operation — the operational signature of a forgetting-window infeasibility is which constraint is binding.

Formal Expression.

Closed-form bias at matched-marginal initial conditions (two-edge OR-root). *Derived (L1-bias-formula-OR-root)*

Let root node G have two direct causal parents e_1, e_2 via OR-aggregation under L1' with latent common cause C producing correlation ρ between the parent edges. Assume matched-marginal initial conditions: $\hat{\mu}_j^{\text{init}} = \mu_j^*$ for each edge's marginal probability. Under the default log-odds update with per-edge gain $\eta_{\text{edge}} = 1/(n_k + 1)$ and identifiability coefficient ι_k, the per-cycle bias on edge k's log-odds update satisfies

$$B_k(\rho) = -\frac{\iota_k \cdot (1 - \mu_{\bar{k}}) \cdot \rho}{(n_k + 1) \cdot [(1 - \mu_1)^2 + (1 - \mu_2)^2]} \tag{AM.1}$$

where $\bar{k} \in \{1, 2\} \setminus \{k\}$ is the sibling edge. The bias is linear in ρ, decays as $1/(n_k + 1)$ with experience, topologically asymmetric via the Jacobian factor $(1 - \mu_{\bar{k}})/\|\mathbf{J}\|^2$.

Admissible region (precise). The denominator is the squared OR-aggregation Jacobian norm $\|\mathbf{J}\|^2 = (1 - \mu_1)^2 + (1 - \mu_2)^2$. The first-order perturbation is well-posed precisely when $\|\mathbf{J}\|$ is bounded below by some fixed $\epsilon_J > 0$ — equivalently, when at least one edge probability is bounded away from 1. This is the **Jacobian-norm admissibility condition** $\|\mathbf{J}\| \geq \epsilon_J$; it is *weaker* than requiring both μ_1, μ_2 to be bounded away from 1, and it is *sharper* than the qualitative "bounded away from determinism" reading: only the joint deterministic-success corner $\mu_1, \mu_2 \to 1$ together is excluded. As either edge alone approaches deterministic success the bias formula remains valid (the surviving sibling

edge carries the Jacobian norm); only the diagonal corner where both edges saturate is excluded, and there the per-cycle update itself vanishes (observations carry no residual information), so "no learning, no bias" is the operationally correct behaviour and the formula's apparent singularity is degenerate rather than physical. Inside the admissible region $\|\mathbf{J}\| \geq \epsilon_J$, the closed-form predictions are quantitatively verified by Monte Carlo; at the diagonal corner the segment defers to `#deriv-edge-credence-dynamics`'s degenerate-update analysis. The Jacobian-norm condition is the natural admissibility statement because the bias formula is the projection of the update onto $\mathbf{J}/\|\mathbf{J}\|^2$, so well-posedness requires only that this projection be defined.

For AND-root aggregation, the bias has the same structural form with opposite sign. For mixed OR-AND sub-plans, the bias decomposes sub-plan-wise.

Derived (biased-fixed-point-exists)

L1'-induced biased fixed point.

Under continued L1' evidence, the log-odds fixed point satisfies

$$\lambda_k^{\text{L1'-FP}} = \lambda_k^* + \int_0^\infty e^{-\eta_{\text{edge}} \iota_k t} B_k(\rho)\, dt = \lambda_k^* - \rho \cdot (1-\mu_{\bar{k}}) / \left[(1-\mu_1)^2 + (1-\mu_2)^2\right] \tag{AM.2}$$

(for constant-ι approximation). Each edge's log-odds converges to a **wrong value** that matches the L1' root probability rather than the marginal truth. **Plan-level correctness is purchased at edge-level miscalibration**: the agent predicts plan outcomes correctly on-policy (the L1' mixture on the sibling matches the L1' marginal), but its per-edge credences are systematically biased.

This is the quantitative companion to F1's on-policy undetectability (`#disc-identifiability-floor` Instance 1) at the marginal level: the bias accumulates at rate $1/(n_k+1)$, converges to a bounded non-zero value, and is *not detectable from on-policy data* because the mixture's marginal is matched to the on-policy observations.

Derived (observable-C-zero-bias)

Observable-C zero-bias result.

Under Prop B.7's five-way-gating with C observable, conditioning on C decomposes the L1' mixture into two single-child-per-component problems:

$$B_k(\rho \mid C = c) = 0 \quad \text{in expectation at conditional truth, for each } c \in \{0,1\}. \tag{AM.3}$$

Prop B.7's decomposition *exactly eliminates the bias* — the five gating conditions (facilitator monotonicity; C-observability; $\iota > 0$ for common-cause edge; marginal sector condition; identifiability of per-component rates) are precisely the conditions under which conditional-on-C updates restore exactness. This makes Prop B.7's derivation the quantitative escape from the L1' bias.

Derived (bias-floor-under-forgetting)

Unobservable-C Cramér-Rao bias floor under forgetting.

When C is unobservable and the agent uses experience discounting at rate $\lambda \in [0,1)$ to maintain plasticity (`#schema-strategy-persistence`'s forgetting prerequisite), the steady-state bias is

$$B_k^{SS}(\rho, \lambda) = -\frac{\iota_k \cdot \rho \cdot (1-\mu_{\bar{k}})}{(1-\lambda) \cdot [(1-\mu_1)^2 + (1-\mu_2)^2]}. \tag{AM.4}$$

The Cramér-Rao floor from `#disc-identifiability-floor` Instance 2's rank-1 Fisher matrix translates directly: no unbiased estimator can reduce the bias below $\rho/(1-\lambda) \cdot [\text{constant}]$. The bias is structurally present in any online estimator operating on single-channel observation of one child edge, not merely the specific default-signal estimator.

Dual forgetting-rate requirement.

Derived (dual-forgetting-requirement)

`#schema-strategy-persistence` requires forgetting rate $(1-\lambda) > \rho_\Sigma / R_\Sigma$ for asymptotic sector-persistence. The L1' bias floor above imposes a **second** constraint: to keep the bias bounded relative to the sector reserve,

$$(1-\lambda) > c_B \cdot \rho / R_\Sigma^{\text{bias}} \tag{AM.5}$$

where c_B is a topology-dependent constant from the Jacobian factor and R_Σ^{bias} is the tolerable bias-reserve within the sector region. The forgetting rate must satisfy **both**:

$$(1-\lambda) > \max\left(\frac{\rho_\Sigma}{R_\Sigma},\ \frac{c_B \cdot \rho}{R_\Sigma^{\text{bias}}}\right). \tag{AM.6}$$

When the admissibility window is empty (no forgetting rate satisfies both constraints simultaneously), no standard persistence regime works — the agent must augment (observable-C via instrumented regime indicators; multi-child joint observation; plan-level fallback) or accept biased-fixed-point operation.

Monte Carlo verification.

Empirical claim (monte-carlo-confirmation)

Two independent Monte Carlo implementations confirm the closed-form predictions — the original 2026-04-23 record (four scenarios: symmetric strong common-cause, asymmetric mild, degenerate $\rho = 0$, θ_C sweep; 400 trials × 5000 cycles, OR-root) and a fresh 2026-07-16 implementation that reproduces the original's scenario-A cumulative drift to 4 significant figures (-0.3996 vs -0.3997) and extends verification to the AND-root case: - Initial-cycle rate matches the closed form within $< 1\%$ for **both OR and AND roots** across the feasible covariance range $\rho \in \{0.04, 0.09, 0.16, 0.20, 0.24\}$ (note $\rho = \text{Cov}(Y_1, Y_2) \leq 1/4$ by construction) and in the asymmetric-edge case. - Sign matches prediction everywhere; **OR and AND roots produce opposite-sign biases of matching magnitude** (AND-root now simulated, not derivation-only). - Vanishing at $\rho = 0$ verified (degenerate L1'); no false positive under L0-matched truth (drift $\approx 3 \times 10^{-4}$ over 5000 cycles). - Cumulative drift is log-order (far from linear) and decelerates slightly faster than the frozen-state $\sum_t 1/(n+t+1)$ prediction, consistent with the bias-induced $\mathbf{p}(t)$ trajectory effect; the exact transient shape has no closed form (see Epistemic Status).

Epistemic Status.

Conditional. Max attainable: *exact* for the two-edge OR-root / AND-root matched-marginal case. The closed-form bias formula is derived via first-order perturbation of the log-odds update around matched-marginal truth; the Jacobian factor is specific to the DAG topology (OR vs AND vs mixed). *Robust qualitative* for deeper DAG structures where the bias Decompose sub-plan-wise but closed-form coefficients depend on per-sub-plan Jacobians.

What is load-bearing: - The linear-in-ρ scaling is exact (first-order in small correlation). - The $1/(n_k+1)$ experience scaling is exact (follows from `#deriv-edge-update-natural-parameter`'s Beta-Bernoulli gain). - The observable-C zero-bias result under Prop B.7 gating is exact. - The dual forgetting-rate requirement is a first-order composition of two separate constraints.

What is not established: - Closed-form bias for non-matched-marginal initial conditions (transient bias during convergence). Numerical simulation works; analytic form is messier. - Higher-than-second-order DAG structures (L2-general). Scope boundary. - Full coupled bias formula under `#deriv-adaptive-gain-dynamics` meta-gain machinery. Open.

Discussion.

Composition with `#disc-identifiability-floor` Instance 2. F13 establishes the structural no-go for single-channel L1' mixture identification (Cramér-Rao rank-1); this segment gives the **quantitative numerical-floor** companion — the bias is bounded, continuous in ρ, and vanishes at $\rho = 0$. Instance 2 answers "is it identifiable?" (no, structurally); this segment answers "if we update anyway, how wrong is the result?" (bounded drift to a biased fixed point).

Composition with `#deriv-fisher-whitened-update-rule`. The Fisher-whitened update recovers sharp *directional* fidelity but does not reduce the *magnitude* bias — Fisher whitening operates downstream of identification, and Instance 2's obstruction is an identification obstruction. The full picture for the default signal function under L1': **direction preserved (angle ≤ 45° per `#deriv-fisher-whitened-update-rule`)**, **magnitude biased by $B_k(\rho)$ with Cramér-Rao floor under forgetting (this segment)**, **observable-C restores exactness (Prop B.7)**, **undetectable on-policy (Instance 1 + Instance 2 composition)**. This is the honest quantitative account of what the default signal function does under Gemini's Gap A (validation under correlated failures).

Relationship to `#schema-strategy-persistence`'s forgetting. The segment's existing forgetting prerequisite (for asymptotic sector-persistence) becomes dual under L1': forgetting must *both* outpace sector disturbance *and* prevent bias accumulation. When the two constraints are compatible (admissibility window non-empty), the existing forgetting-rate requirement suffices with appropriate ρ-dependent tuning. When incompatible, augmentation is required — a structural scope narrowing that elevates observable-C instrumentation from convenience to prerequisite.

Connection to Z. The detection-latency theorem ($\mathbb{E}[T_{\text{detect}}] = \Omega((n_{\min}+1)/\varepsilon)$) assumes unbiased marginal tracking; under L1' bias, the pre-accumulated drift degrades effective ε for subsequent regime-change detection. High-operating-point agents with pre-accumulated L1' bias face a compounded myopia floor. Follow-on refinement possible in Z.

Appendix AN

Derived: Architecture No-Identifiability from On-Policy Summary Data

The framework's identifiability-floor at the *agent-internal architecture* layer. Two AAT agents whose linearized closed-loop residual dynamics are minimal state-space realizations related by an invertible similarity transformation produce *identical* on-policy summary statistics: identical innovation spectrum, identical sector-condition summary (α, R), identical similarity-invariant deviation summaries — and therefore identical observation law under on-policy, in-regime, summary-only access. An observer with that access cannot distinguish the two agents, even though they implement the correction in structurally distinct internal coordinates. The architectural degree of freedom AAT cares about — `#der-agent-opacity`'s "which internal mechanism" — lives entirely in the $GL(n)$ similarity fiber and is annihilated by the on-policy observation map.

The no-go is *exact* in the linear-Gaussian / Kalman sub-scope, anchored by Kalman 1963 and Ho-Kalman 1966 canonical-form non-uniqueness; *robust-qualitative* in the general sub-scope via the Causal Hierarchy Theorem applied at the agent-as-SCM layer (Bareinboim, Correa, Ibeling & Icard 2022). The mechanism is **not** Sylvester's law of inertia inherited for free — the *generating* group action that produces the indistinguishable pair is state-space similarity $T(\cdot)T^{-1}$, not the metric congruence $S^{\top}(\cdot)S$ that Sylvester governs. It *reduces*, however, to the same shape as the Instance-2 rank-collapse floor (S Prop B.7): the on-policy Fisher information of the realization manifold is identically zero along the similarity-fiber tangent distribution, so the observed-Fisher is rank-deficient on a structurally-forced indeterminacy manifold — Instance-2's mechanism with the manifold being a Lie-group fiber rather than a single indeterminacy submanifold. Escape-irreducibility under the agent's *metric/coordinate* freedom (distinct from the similarity freedom that *generates* the pair) is then Sylvester at one remove, identically to Instance 2.

Three structurally distinct escapes: (a) loop-interventional access (`#der-loop-interventional-access`) perturbs along similarity-fiber directions that on-policy data never excites — the intervention generates Level-2 data, the no-go is Level-1-only; (b) higher-moment / out-of-regime observation distinguishes the pair in the nonlinear sub-scope (AAT's β) — *provably void* in the linear-Gaussian sub-scope, since a Gaussian innovation carries no information beyond second order; (c) architecture instrumentation (direct read of the update rule or internal state) breaks the black-box scope, structurally the same kind as Instance 2's "instrument the latent" escape. A fourth proposed escape — horizon extension under the same policy — collapses into (a): more samples of an annihilated observation law cannot escape the annihilation.

The strengthened consequence is that `#der-loop-interventional-access` acquires a *third semantically-distinct deployment mode* — at the agent-internal layer — alongside Instance 1's agent-self-intervention (breaking causal-sufficiency degeneracy) and Instance 3's observer-on-sub-agent intervention (breaking coupling-sign degeneracy). The shared content across the three modes is "Level-2 data breaks a Level-1 degeneracy"; the distinct content is *which* degeneracy.

Formal Expression.

Definition (architecture-noidentifiability-setting)

The Setting.

Two AAT agents A and A' are *architecturally distinct, behaviorally identical from on-policy summary data* in a regime Ω when:

- both operate on the same regime-restricted trajectory under their respective on-policy execution;
- both share the same sector-condition summary (α, R) in Ω (A), equivalently the same ultimate-deviation statistics on the similarity-invariant summaries of $\|\delta\|$ in that regime;
- the observer has *on-policy, in-regime, summary-only* access — no off-policy intervention, no out-of-regime samples, no white-box read of the update rule or internal state.

Per `#scope-agent-identity`, A and A' are agents on singular trajectories with action-marginal invariance under change of internal-coordinate parameterization. The question is whether the architectural content — the basis in which the correction is implemented, internal to the agent — can be recovered from the observer's accessible data.

Derived (architecture-noidentifiability-Kalman, from Kalman-Ho canonical-form non-uniqueness applied to AAT-agent realizations), exact in the linear-Gaussian sub-scope

The No-Go Theorem (exact in the linear-Gaussian / Kalman sub-scope).

Let the linearized closed-loop residual dynamics of A and A' be minimal state-space realizations

$$d\delta_t = -F\,\delta_t\,dt + \sigma_w\,dW_t, \qquad d\delta'_t = -F'\,\delta'_t\,dt + \sigma'_w\,dW_t \tag{AN.1}$$

with F, F' Hurwitz (sector-condition-satisfying per `#deriv-sector-condition`) and the two realizations related by an invertible similarity $T \in GL(n)$:

$$F' = TFT^{-1}, \qquad \sigma'_w\,\sigma'^{\top}_w = T\,\sigma_w\,\sigma_w^{\top}\,T^{\top}. \tag{AN.2}$$

Claim. The stationary innovation process — hence the (α, R)-summary, the innovation spectrum, and every *similarity-invariant* summary of the on-policy observation law — is **identical** for A and A'. No on-policy, in-regime, summary-restricted statistic distinguishes them.

Derivation. The innovation's stationary law is determined by the pair $(F, \sigma_w\sigma_w^{\top})$ only through similarity-invariant quantities. Specifically:

- The eigenvalues of F (the α-spectrum entering the sector summary) are similarity-invariant.
- The stationary covariance Π solves the Lyapunov equation

$$F\Pi + \Pi F^{\top} = \sigma_w\,\sigma_w^{\top} \tag{AN.3}$$

(standard form for the OU drift $-F$ with F Hurwitz, sign convention matching `#deriv-sector-condition`), and transforms under similarity as $\Pi' = T\Pi T^{\top}$.

- The observed quantities are the *innovation spectrum* $\det(i\omega I - F)$ and its similarity-invariant moments — for example, the *deviation magnitude under any similarity-invariant norm* (a norm derived from the dynamics, such as the F-induced metric, rather than a fixed external basis). Each is annihilated by the similarity action.

The architectural content — the basis T, that is, *which* internal coordinates implement the correction — lives entirely in the similarity orbit and is annihilated by the on-policy observation map. □

Scope of "identical." The claim restricts to *similarity-invariant* summaries of the on-policy observation law. Fixed-external-basis quadratic forms in the raw residual are *not* in general similarity-invariant — for instance,

$$\mathbb{E}\,\|\delta\|^2 = \operatorname{tr}\Pi, \qquad \operatorname{tr}(T\,\Pi\,T^{\top}) \neq \operatorname{tr}\Pi \text{ in general} \tag{AN.4}$$

(equality only when T is orthogonal). The on-policy observer who consumes (α, R) summaries and the innovation spectrum sees only similarity-invariant content; an observer with access to a fixed external basis can in principle break the orbit by observing the raw covariance, but that is a *different* observation regime than the on-policy summary access the no-go is stated under.

The General Sub-Scope (robust qualitative).

Derived (architecture-noidentifiability-general, from causal hierarchy theorem applied at agent-as-SCM layer), robust qualitative

Outside the linear-Gaussian sub-scope, the no-go is robust-qualitative rather than exact. Model the agent as a structural causal model over its internal state space (per `#scope-agent-identity`'s token-level commitment to singular-trajectory agents). The Bareinboim, Correa, Ibeling & Icard (2022) Causal Hierarchy Theorem then applies at the agent-as-SCM layer: two SCMs over the agent's state space agreeing on Level-1 observation data — the on-policy summary distribution — cannot in general be distinguished on Level-2 questions about $do(\cdot)$ on the agent's internal coordinates.

The dual-anchor structure (sharp Kalman-Ho for the linear-Gaussian sub-scope; CHT-at-agent-as-SCM for the general case) is the same shape Instance 1 carries — explicit construction sharp / CHT general — and the tier separation is load-bearing: collapsing to "exact" overclaims; collapsing to "robust-qualitative" understates the sub-scope's sharpness.

Mechanism: Instance-2 Fisher Null on a Lie-Group Fiber.

Derived (mechanism-reduction-to-instance-2, from Fisher-information factorization through similarity-invariants)

The no-go's *generating* group action is state-space similarity $F \mapsto TFT^{-1}$ on the realization manifold. Sylvester's law of inertia (`#disc-identifiability-floor` §Discussion) governs metric *congruence* $\mathcal{G} \mapsto S^{\top}\mathcal{G}S$, a different group action; the no-go therefore does **not** inherit Sylvester at the generating-action layer.

The reduction to Instance-2 mechanism proceeds via Fisher geometry. Parameterize the agent by

$$\theta = (\text{similarity-invariants}, \text{orbit-coordinates}) \tag{AN.5}$$

— formally, the realization manifold fibered over the transfer-function manifold by the $GL(n)$ similarity group. By the §"No-Go Theorem" derivation, the on-policy observation law depends on θ only through the similarity-invariant base. The Fisher information of the on-policy observation channel, as a quadratic form on the full realization manifold, is therefore *identically zero* on the entire tangent distribution of the similarity fiber. That is the Instance-2 structure (Fisher rank-deficiency along a structurally-forced indeterminacy manifold; S Prop B.7), with the manifold being a $GL(n)$ Lie-group fiber rather than the mixture-indeterminacy manifold of Instance 2.

The agent's *escape-irreducibility under its metric/coordinate freedom* is then Sylvester at one remove — identically to Instance 2. Once the rank-deficient observed-Fisher is recognized, no reparameterization of the observation model refills the rank-deficient direction; congruence preserves inertia.

The two group actions and their roles, made precise:

- The **similarity action** $T(\cdot)T^{-1}$ on the realization manifold *generates* the indistinguishable architectural pair (A, A'). This is what makes the floor a floor at all, and it is *not* Sylvester — it is the fiber of the realization-to-transfer-function bundle.

- The **congruence action** $S^{\top}(\cdot)S$ on the observed-Fisher governs the agent's potential escape by reparameterizing its observation model. This *is* Sylvester, and it forbids the escape, identically to Instance 2.

This no-go is therefore a member of the rank-collapse subclass of `#disc-identifiability-floor` ({Instance 1, Instance 2, Instance 4} via Sylvester at one remove) rather than Instance 3's projection / Schur-complement obstruction. The additional structure beyond Instance 2 is that the rank-deficient direction is a Lie-group fiber.

Derived (escape-routes, from no-go scope conditions)

Boundary Characterization: Three Structurally Distinct Escapes.

The no-go's scope is (i) on-policy, (ii) in-regime, (iii) summary-only. Each scope-condition violation corresponds to an AAT capability that admits identification:

Table AN.1

Route	*Scope violated*	*AAT capability*	*Detection strength*
(a) Loop-interventional access	(i) on-policy	`#der-loop-interventional-access` — $do(\cdot)$ on the agent's input	Sharp in linear-Gaussian sub-scope; robust qualitative general
(b) Higher-moment / out-of-regime	(ii) in-regime + (iii) summary-only	Beyond sub-scope α: higher-moment statistics of the innovation; out-of-regime sampling	Provably void in linear-Gaussian sub-scope; informative in nonlinear sub-scope β
(c) Architecture instrumentation	(iii) summary-only (white-box)	Direct read of the update rule or internal state; same kind as Instance 2's "instrument the latent"	Strongest when available

Route (a) — loop-interventional access (sharp). A $do(\cdot)$ intervention on the agent's input perturbs the system along *similarity-fiber directions that on-policy data never excites*. The interventional response (correction-function Jacobian probed off the on-policy manifold) is *not* similarity-invariant: TFT^{-1} and F respond differently to a probe in a fixed external basis. The intervention generates Level-2 data; the §"No-Go Theorem" derivation is a Level-1-only statement. Genuinely escapes the orbit degeneracy.

Route (b) — higher-moment / out-of-regime (sub-scope-bounded). In the nonlinear sub-scope β, two agents matched at (α, R) and second-order generally differ at moments of order ≥ 3. In the linear-Gaussian sub-scope this route is *provably void*: a Gaussian innovation carries no information beyond second order, so higher-moment observation adds nothing — the similarity orbit remains invisible. The "escape exists here, provably not there" boundary is exactly the kind of sharp scope-statement that makes a floor instance informative.

Route (c) — architecture instrumentation. Direct read of the update rule or internal state breaks the black-box scope: the observer now has access to the basis T itself, not just its annihilated projection onto the on-policy observation law. Structurally the same kind as Instance 2's escape (i) ("instrument the latent" — S Prop B.7); genuinely distinct from (a) and (b) in *what the observer must have* (white-box access versus interventional access versus moment-level access).

Route (d), horizon extension under the same policy — does not escape. Passive observation at longer horizons under the *same* on-policy execution generates more *samples* of the same observation law; the §"No-Go Theorem" derivation is a statement about the *law* (the architectural content is annihilated by the spectrum map), not about the sample size. Horizon extension that takes the agent *out of the current regime* is no longer the same policy in the same regime — that is a regime change, which is structurally an exogenous intervention-like event, collapsing into route (a)/(b). Route (d) is therefore either void (same regime) or a special case of (a)/(b); the count of structurally distinct escapes is **three**, not four.

Why Fano Is the Finite-Sample Refinement, Not the Exact Anchor.

Derived (fano-degeneracy-at-zero-information), exact

A Fano-style anchor (Fano 1961, *Transmission of Information*) gives a lower bound on prediction error probability *given* a bound on the channel mutual information $I(A; \text{obs})$ between the architectural variable and the observation. Under the similarity-orbit construction, the on-policy observation law is *identical* for A and A' — not merely close — so $I(A; \text{obs}) = 0$ exactly. Fano degenerates to the trivial bound "error probability ≥ prior," vacuous and strictly weaker than the *exact* Kalman-Ho indistinguishability statement.

Fano is therefore the right tool for the *approximate / finite-sample* version of the no-go — architectures that are close but not equal in innovation spectrum, $I > 0$ small — and the wrong tool for the *exact* population-level no-go. The exact anchor is Kalman-Ho; Fano is honest open follow-on work on the finite-sample refinement, distinct from the floor itself. (Cross-reference: `#der-agent-opacity` Working Notes record the parallel finding from the observer-side H_b task.)

Strengthened Consequence.

Derived (third-mode-of-interventional-access)

The no-go elevates `#der-loop-interventional-access` to a load-bearing role at the *agent-internal* layer — the third semantically distinct deployment mode after:

1. **Mode 1: agent-self-intervention** — Instance 1's escape, breaking a *causal-sufficiency* degeneracy among the agent's action propositions (`#der-causal-insufficiency-detection`).

2. **Mode 2: observer-on-sub-agent intervention** — Instance 3's escape, breaking a *coupling-sign* degeneracy across composite sub-agents (J).

3. **Mode 3: observer-on-agent-input intervention** — this segment's escape, breaking a *similarity-orbit* degeneracy on the agent's internal architecture.

The three modes are *different group-theoretic objects*: each breaks a Level-1 degeneracy of a distinct kind, yet shares the load-bearing content "Level-2 data is the unique broadly-available violation of the Level-1 no-go." This converts the Instance-4 floor from "yet another no-go" into structural confirmation that `#der-loop-interventional-access` carries the same load-bearing role at *three* distinct AAT layers (action, composite, architecture), each escape preserving the framework's existing scope honesty rather than introducing new machinery.

Tier. *Exact* in the linear-Gaussian / Kalman sub-scope (α_1), anchored by Kalman-Ho canonical-form non-uniqueness. *Robust qualitative* in the general sub-scope, anchored by CHT-at-agent-as-SCM. **The exact / robust-qualitative tier boundary is load-bearing and must be carried consistently** — collapsing to flat "exact" overclaims the general case; collapsing to flat "robust qualitative" understates the sub-scope's sharpness. The mechanism reduction (§"Mechanism") is exact: the Fisher-null factorization through similarity-invariants follows in elementary steps from the §"No-Go Theorem" derivation.

The escape-count statement ("three structurally distinct escapes, with the proposed fourth collapsing into the first") is *exact* — the collapse of route (d) into (a) and the void of route (b) inside the linear-Gaussian sub-scope are both elementary consequences of the no-go statement itself. The escape-machinery mappings are *robust qualitative*, inheriting the tier of the cited segments (`#der-loop-interventional-access`, `#der-agent-opacity`).

Epistemic Status.

Conditional. Max attainable: *exact* for the Kalman-Ho derivation in the linear-Gaussian sub-scope; *robust qualitative* for the general case via CHT-at-agent-as-SCM; *exact* for the mechanism reduction to Instance-2 Fisher-null on a Lie-group fiber; *exact* for the escape count (three, not four) and for the Fano-degeneracy-at-$I = 0$ observation; *robust qualitative* for the escape-machinery mappings.

Load-bearing: - The Kalman-Ho construction makes the architectural-identifiability no-go *exact in a precisely characterized sub-scope* — replacing the prior sketch-grade framing (`#der-agent-opacity` Working Notes' candidate-status for a generic observer-side floor). - The mechanism reduction to Instance-2 Fisher-null on a $GL(n)$ fiber places the no-go in the rank-collapse subclass {Instance 1, Instance 2, Instance 4} via Sylvester at one remove, repairing `#disc-identifiability-floor`'s Sylvester-mechanism taxonomy to three rank-collapse members plus one projection/closure member ({Instance 3}). - The three-mode classification of `#der-loop-interventional-access` (causal-sufficiency / coupling-sign / similarity-orbit) is a structural unification across three identifiability-floor instances.

Not established: - The general (non-linear-Gaussian) sub-scope at *exact* tier. CHT-at-agent-as-SCM is robust-qualitative; an explicit construction-grade no-go in general sub-scope β is honestly open. - The finite-sample Fano refinement (architectures close-but-not-equal in innovation spectrum, $I(A; \text{obs}) > 0$ small). Open follow-on direction, distinct from the population-level no-go derived here. - Sharp characterization of the boundary between similarity orbits and behavioral equivalence classes when the agent's nonlinear dynamics admit *partial* spectrum identifiability (the sub-scope-β moment-cascade question).

Honest Limits.

- **Linearization scope.** The Kalman-Ho construction operates on the *linearized* closed-loop residual dynamics. Sector-condition-satisfying agents (A) admit such linearization in a basin around the singular trajectory; the no-go's exact tier inherits that scope. Outside the basin, the general-case CHT anchor applies but the construction is not closed-form.
- **Action-marginal invariance vs. internal-coordinate freedom.** Per `#scope-agent-identity`, the action space is coordinate-free and observed quantities are parameterization-invariant. The similarity orbit is a *different* invariance — internal to the realization manifold, distinct from action-marginal parameterization-invariance — and the no-go is about identifying the internal architecture, not about action-level identifiability.
- **Closed-loop residual dynamics, not open-loop plant.** The no-go is stated for the closed-loop δ-dynamics (`#def-mismatch-signal`) — what the on-policy observer actually sees. Open-loop plant identification from interventional experiments is route (a) and is *not* subject to this no-go.

Discussion.

Position in the identifiability-floor pattern. This no-go is the fourth instance of `#disc-identifiability-floor` and the third member of the rank-collapse subclass via Sylvester at one remove. The Sylvester-mechanism Discussion of the meta-segment (`#disc-identifiability-floor` §Discussion) is repaired by this addition: the three-rank-collapse taxonomy is {Instance 1, Instance 2, Instance 4} unified by congruence-preserves-inertia at the escape-irreducibility layer; Instance 3 stands alone as the projection/closure obstruction. The plurality-of-mechanism that makes the floor pattern a *family* rather than a single theorem is preserved (Sylvester for rank-collapse, Schur/projection for composition); the Sylvester-side is strengthened from two to three members.

Why Instance-2's mechanism, not new. The Fisher-null direction here is a $GL(n)$ Lie-group fiber rather than the single mixture-indeterminacy manifold of Instance 2, but the *kind* of obstruction is identical: a rank-deficient observed-Fisher along a structurally-forced indeterminacy manifold. Naming it a new mechanism would be inflation — the AAT-distinctive recognition is precisely that the architectural d.o.f. AAT cares about (`#der-agent-opacity`'s "which internal mechanism") is the similarity orbit and that the orbit is annihilated by the on-policy observation map; the

obstruction theorem itself is the same shape as Instance 2. Naming it Sylvester-for-free would be the opposite inflation — similarity and congruence are different group actions. The honest position is "Instance-2 mechanism on a Lie-group fiber, Sylvester at one remove on the escape."

Relationship to `#der-agent-opacity`'s observer-indexed H_b. This segment's no-go is structurally distinct from a generic observer-prediction-of-action floor: the architectural d.o.f. is invisible *at the observation-law level* regardless of observer instrumentation, not because of a particular observer's filtration. An observer with arbitrarily fine $\mathcal{F}_B^t$ over the on-policy summary distribution still cannot escape the similarity orbit; the observer-indexing of H_b is orthogonal. The natural relationship is at the meta-pattern level (both name floors in the family of `#disc-identifiability-floor`); the *quantities* are distinct (H_b is observer-conditional action entropy; this segment's invariant is the similarity orbit).

Conservative integration with the Sylvester recognition. The 2026-05-14 Sylvester-recognition Finding in `#disc-identifiability-floor` taxonomized two rank-collapse floors (Instances 1, 2) and explicitly said *"the floors do not share one mechanism — Sylvester for rank-collapse, a projection/closure obstruction for composition."* This segment's addition does not contradict that recognition; it extends the rank-collapse subclass to three members and preserves the projection/closure distinction for Instance 3. The plurality-as-feature framing of the meta-segment's Discussion is intact.

Multi-agent / Hafez bridge. The architectural d.o.f. that this no-go identifies as un-recoverable from on-policy summary data is structurally adjacent to `#der-agent-opacity`'s H_b in this sense: both quantities concern the *world's view of the agent's internal architecture* through information-theoretic limits. The Hafez et al. 2026 IDT pattern (Information Digital Twin, bi-predictability P) — operationally a low-H_b-preserving observation channel — is a structurally-instrumented escape, and its 89% perturbation-detection performance is consistent with this segment's route (c) "architecture instrumentation" classification.

Findings.

Architecture No-Identifiability from On-Policy Summary Data — the Fourth Identifiability Floor. **Brief:** Imagine two agents that both keep a desk tidy by the same rules: same overall reaction speed, same tolerance for clutter, same long-run mess level. One agent reasons by tracking pile heights; the other tracks distances between piles. Watching them work — under their normal habits, summary statistics only — you literally *cannot tell which one is running*. The visible behavior is the same, the variability is the same, the rate of clean-up is the same, the cluttered ceiling is the same. Different inner mechanisms, identical outer signature. There are only three honest ways to tell the agents apart: poke one in a way it wouldn't normally do (intervention), look at fine-grained details its rules would not normally generate (out-of-regime / higher-moment observation), or open the hood and read the wiring (white-box access). And a fourth pretender — "just watch longer" — does *not* work, because watching longer accumulates more samples of the same signature, never the missing distinction. This is the floor: it is structurally impossible to recover the inner-mechanism information from the outer summary alone, and the only paths forward are exactly the three named escapes, each of them violating a clearly-stated scope condition.

Impact: Closes the long-contested fourth instance of `#disc-identifiability-floor` with a sharp, dual-anchored no-go (Kalman-Ho linear-Gaussian sub-scope, exact; CHT-at-agent-as-SCM general, robust qualitative). Extends the Sylvester-mechanism rank-collapse subclass to three members {Instance 1, Instance 2, Instance 4} via the at-one-remove escape-irreducibility, while preserving the projection/closure distinctness of Instance 3 — the plurality-of-mechanism that makes the floor pattern a presentational family rather than a single theorem stays intact, sharpened. Elevates `#der-loop-interventional-access` to a third semantically-distinct deployment mode (agent-internal architecture, alongside agent-self-intervention and observer-on-sub-agent), unifying the strengthened-consequence content of three identifiability-floor instances under one piece of machinery. The result mathematically settles the Regime-C confound of the rho-decomposition (`#result-mismatch-decomposition` + X) as the same object projected onto the disturbance-statistic coordinate — one decision, not two.

Novelty Claim: *Claim derivation* of the architecture-noidentifiability floor with named scope (linear-Gaussian sub-scope, on-policy summary access) and named dual anchor (Kalman 1963 / Ho-Kalman 1966 / Anderson-Moore 1979

§10.4 for the sharp sub-scope; Bareinboim, Correa, Ibeling & Icard 2022 for the general case). The Kalman-Ho canonical-form non-uniqueness theorem is classical (1963/1966); the AAT-distinctive content is *recognizing* that the architectural degree of freedom the framework cares about — `#der-agent-opacity`'s "which internal mechanism" — *is* the similarity orbit, and that the orbit is annihilated by the on-policy observation map. *Claim differentiation* on mechanism: the obstruction is Instance-2's Fisher-null along a structurally-forced indeterminacy manifold (the manifold being a $GL(n)$ Lie-group fiber) rather than a new mechanism, with Sylvester preserving inertia on the escape side identically to Instance 2 — a *member* of the rank-collapse subclass, not a new mechanism family. *Claim recognition* that the previously-proposed Fano anchor degenerates at $I(A; \text{obs}) = 0$ (vacuous lower bound on prediction error under a zero-information channel), so Kalman-Ho is the exact-population anchor and Fano is the finite-sample refinement — a demonstrated no-go on the proposed Fano anchoring, not an assertion.

Related Work:

Table AN.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
State-space realization non-uniqueness up to similarity (the sharp sub-scope anchor)	Kalman, R. E. (1963), "Mathematical description of linear dynamical systems," *J. SIAM Control* 1(2):152–192; Ho, B. L. & Kalman, R. E. (1966), "Effective construction of linear state-variable models from input/output functions," *Regelungstechnik* 14:545–548; Anderson, B. D. O. & Moore, J. B. (1979), *Optimal Filtering*, Prentice-Hall, §10.4 — minimal realizations sharing an innovation spectrum form a similarity-equivalence class	*formal antecedent* — adopted with citation; AAT-distinctive recognition is that the architectural d.o.f. of `#der-agent-opacity` *is* the similarity orbit
Causal Hierarchy Theorem (the general-case anchor)	Bareinboim, E., Correa, J. D., Ibeling, D. & Icard, T. (2022), "On Pearl's Hierarchy and the Foundations of Causal Inference," in *Probabilistic and Causal Inference: The Works of Judea Pearl*, ACM	*formal antecedent* — applied at the agent-as-SCM layer; same anchor Instance 1 uses, here in the agent-internal-architecture setting
Fano's inequality (finite-sample refinement, not the exact anchor)	Fano, R. M. (1961), *Transmission of Information*, MIT Press; Cover, T. M. & Thomas, J. A. (2006), *Elements of Information Theory*, Wiley, §2.10	*adjacent* — vacuous at $I = 0$ for the exact construction; the right tool for the finite-sample architectures-close-but-not-equal version of the no-go (honest open)
Sylvester's law of inertia (escape-irreducibility, at one remove)	Sylvester, J. J. (1852), *Phil. Mag.* 4(23):138–142; Horn, R. A. & Johnson, C. R. (2013), *Matrix Analysis* (2nd ed.), Thm 4.5.8	*formal antecedent* — applied here to the *escape* side (congruence on the observed-Fisher), distinct from the generating action (similarity on the realization manifold). Sylvester forbids the escape by reparameterization identically to Instance 2
Fisher-information reparameterization law	Lehmann, E. L. & Casella, G. (1998), *Theory of Point Estimation* (2nd ed.), §2.5 — $\mathcal{G}_\varphi = S^\top \mathcal{G}_\theta S$ as the standard congruence law	*formal antecedent* — the law that makes the escape-side reparameterization a congruence action
Hafez et al. 2026 / IDT bi-predictability	Hafez, A. M., Khan, F. M. & Iqbal, M. (2026), *A Mathematical Theory of Agency and Intelligence* — IDT pattern, bi-predictability P	*adjacent* — IDT-as-architecture-instrumentation is route (c) of this segment's escape structure; the 89% IDT detection rate is consistent with white-box access escaping the orbit

Search Log: - 2026-05 (*targeted*): Kalman 1963 / Ho-Kalman 1966 / Anderson-Moore 1979 §10.4 verified as the canonical-form-non-uniqueness anchors; CHT-at-agent-as-SCM application of Bareinboim et al. 2022 verified con-

sistent with Instance 1's general-sub-scope use. Fano 1961 ↔ Cover-Thomas §2.10 verified for the finite-sample refinement framing. Sylvester 1852 + Horn-Johnson Thm 4.5.8 + Lehmann-Casella §2.5 inherited from the 2026-05-14 Sylvester-recognition Finding in `#disc-identifiability-floor`. - 2026-04 / 2026-05 (*intuition-only*) on whether the cross-instance recognition "the architectural d.o.f. AAT cares about *is* the similarity orbit, annihilated by the on-policy observation map" has been articulated in the system-identification or causal-inference literatures as the AAT-internal-agent-theory framing applied to multi-agent identifiability. The constituent pieces (Kalman-Ho canonical-form non-uniqueness; CHT-at-agent-as-SCM; Fisher rank-deficiency along Lie-group fibers) are textbook; the cross-instance unification as the fourth member of an integrated agent-theoretic floor pattern appears to be a presentational contribution under cursory search. Targeted future search candidates: system-identification literature on the agent-theoretic interpretation of canonical-form non-uniqueness; observability / identifiability literature for nonlinear agents; mechanism-grounded identifiability in multi-agent reinforcement learning.

Appendix AO

Derivation: Fisher-Whitened Edge Update Under Correlated Evidence

Under L1'/L2 correlated-evidence regimes, the default log-odds edge-update (from `#deriv-edge-update-natural-parameter`) retains correct *direction* — the angle between log-odds gradient and natural gradient never exceeds 45° at finite correlation ρ, so B1 directional fidelity (`#der-gain-sector-bridge`) is never actively violated — but its *magnitude alignment* degrades by a factor $\sqrt{1-r^2}$ in the sector constant. The Fisher-whitened correction restores sharp B1 on the Fisher-weighted inner product. Under the (PI) parameterization-invariance axiom named in `#scope-agent-identity` and promoted to a primary instance of `#disc-additive-coordinate-forcing` via Čencov 1982, Fisher whitening is **AAT-internally derivable** rather than externally imported. The result adds sub-scope α_3 (correlated evidence + Fisher-whitened update + Bayesian coherence → A2' derived) to the A2' partition and composes cleanly with the meta-gain machinery of `#deriv-adaptive-gain-dynamics` (Fisher whitening is a special case of meta-gain with $K_t = \mathbf{I}^{-1}(\lambda_t)$).

The framework's posture splits cleanly. *Without* the (PI) commitment, the Fisher-whitened correction is an externally imported choice from information geometry. *With* (PI), Čencov 1982's uniqueness theorem *forces* the Fisher metric as the unique Markov-invariant metric on the statistical manifold, and the whitened-update form is then the AAT-internally canonical correction for directional-fidelity preservation across reparameterizations. Operational prescription: where correlated evidence is the dominant regime, Fisher whitening recovers sharp persistence guarantees at the cost of meta-gain machinery; where it is mild or rare, the default log-odds update with $\sqrt{1-r^2}$ magnitude-alignment degradation suffices. Sibling at the model-parameter-update layer is AT (update *magnitude* under Fisher-local regime); together they make the Fisher-local invariance regime AAT-internally complete.

Formal Expression.

Angle characterization under L1' correlation. *Derived (angle-bound-finite-correlation)*

For a block-correlated two-edge evidential model (soft-facilitator Prop B.7 parameterization), the Fisher information matrix has off-diagonal entry

$$r = \frac{\theta_C(1-\theta_C)\Delta_1\Delta_2}{\sigma_1\sigma_2} \in [-1, 1], \tag{AO.1}$$

where θ_C is the latent-cause probability, $\Delta_j = p_{j|C} - p_{j|\neg C}$ is the separability gap, and σ_j is the marginal-edge standard deviation. The Euclidean angle between the log-odds update direction and the natural-gradient direction (at single-edge plan residual) satisfies

$$\theta_{\text{LO-NG}} = \arccos \frac{1}{\sqrt{1+r^2}} \leq 45° \quad \text{for all } r \in [-1,1]. \tag{AO.2}$$

The log-odds update **never flips sign** under any finite correlation — B1 is never actively violated, only degraded in magnitude. Under the operational claim that "the default signal function needs validation under correlated failures," the finding is: **the default direction is preserved; only magnitude alignment degrades**. The sector constant in B1 degrades by a factor $\sqrt{1-r^2}$ under Euclidean B1; the Fisher-weighted B1 recovers the unperturbed sector constant.

Definition (Fisher-whitened-update)

Fisher-whitened update rule.

In log-odds coordinates $\lambda \in \mathbb{R}^{|E|}$ with Fisher information $\mathbf{I}(\lambda)$, the Fisher-whitened edge update is

$$T_{\text{FW}}(\lambda) = \lambda - \eta_{\text{edge}} \cdot \mathbf{I}^{-1}(\lambda) \cdot \mathbf{J} \cdot (\hat{R}_\Sigma(\sigma(\lambda)) - y_G) \tag{AO.3}$$

(compared to the Euclidean log-odds update $T_{\text{edge}}(\lambda) = \lambda + \eta_{\text{edge}} \cdot \text{diag}(\iota) \cdot \mathbf{J}(y_G - \hat{R}_\Sigma)/\|\mathbf{J}\|^2$). The Fisher-weighted inner product $\langle a, b \rangle_{\mathbf{I}} = a^T \mathbf{I}^{-1} b$ makes the update's directional fidelity invariant under reparameterization of the natural-parameter coordinate.

Derived (Fisher-whitening-from-B1-parameterization-invariance)

Two AAT-internal axiom paths.

Path A (B1-parameterization-invariance). Require B1 directional fidelity to be *parameterization-invariant* in the sense of `#scope-agent-identity`'s (PI) axiom: the theorems about sub-scope α derivation in `#der-gain-sector-bridge` should not depend on arbitrary coordinate choices for M_t's natural parameters. Under (PI), B1 sub-scope α partition is coordinate-invariant iff the inner product defining the directional-fidelity condition is the Fisher metric (Čencov 1982 uniqueness theorem under (PI); extended by Ay-Jost-Lê-Schwachhöfer 2017). The Fisher-weighted inner product is therefore *forced* by (PI), and the Fisher-whitened update is the AAT-internally derived correction direction for directional-fidelity preservation across parameterizations.

Path B (Lyapunov-coordinate-matching via adjacent-family classification). In the adjacent-family framing of `#disc-additive-coordinate-forcing`, the Lyapunov coordinate is *matched* (not forced) to the sector condition. For natural-gradient updates, the canonical Lyapunov is Fisher-weighted (Amari 1998, "Natural gradient works efficiently in learning," *Neural Computation* 10); this matches the geometry of the update operator. The two paths converge on the same Fisher-weighted result; Path A forces it via axiomatics, Path B confirms it via adjacent-family coordinate-matching.

Under L0 (no correlation), $r = 0$ and $\mathbf{I}$ is diagonal — Fisher whitening is vacuous (reduces to the existing Euclidean log-odds update). The axioms pick out Fisher whitening *uniquely* only under L1'/L2 (correlated evidence) regimes; they are vacuously satisfied under L0.

Derived (sub-scope-alpha-3)

New sub-scope α_3.

Correlated-evidence + Fisher-whitened update + Bayesian coherence yields A2' *derived*:

$$(T_{\text{FW}}(\lambda) - \lambda^*)^T \mathbf{I}(\lambda^*)^{-1} (\lambda - \lambda^*) \geq \beta \|\lambda - \lambda^*\|_{\mathbf{I}}^2 \quad \text{with } \beta = \eta_{\text{edge}} \mu \tag{AO.4}$$

on the Fisher-metric sector region around the fixed point $\lambda^* = \text{logit}(\theta^*)$. The directional-fidelity proof of `#der-gain-sector-bridge` carries over with $\kappa(\mathbf{I})$ as Euclidean-transfer penalty — structurally identical to the Kalman case's $(P^-)^{-1}$-norm weighted-norm treatment. The Kalman case in `#der-gain-sector-bridge`'s Verified Instances is a special case of Fisher-weighted sector.

Sub-scope α_3 extends the refinement introduced by `#deriv-adaptive-gain-dynamics`: - **Sub-scope α_1**: fixed-gain, independent evidence (Euclidean B1 applies; the existing A2' partition). - **Sub-scope α_2**: adaptive-gain, independent evidence (meta-gain conditions (MG-1)–(MG-4) per `#deriv-adaptive-gain-dynamics`). - **Sub-scope α_3**: fixed-gain, correlated evidence (Fisher-whitened update; this segment).

Additional composition: adaptive-gain + correlated-evidence (sub-scope α_4) would compose `#deriv-adaptive-gain-dynamics` meta-gain machinery with Fisher whitening at the primary state; open.

Connection to meta-gain as special case.

Derived (Fisher-whitening-as-special-meta-gain)

The Fisher-whitened update is a meta-gain law in the sense of `#deriv-adaptive-gain-dynamics` with $K_t = \mathbf{I}^{-1}(\lambda_t)$ — a *degenerate* special case where the meta-gain is a deterministic function of the primary state rather than an independently learned variable. All four meta-gain conditions (MG-1)–(MG-4) are satisfied trivially: - (MG-1) Meta-gain sector: $\mathbf{I}^{-1}$ is symmetric-positive-definite on the interior of the natural-parameter domain. - (MG-2) Meta-gain bounded: $\|\mathbf{I}^{-1}\|$ bounded on compact natural-parameter subsets. - (MG-3) Smoothness: $\mathbf{I}^{-1}$ depends smoothly on λ for exponential families. - (MG-4) Primary-meta coupling bounded: $\mathbf{I}^{-1}$'s state-derivative in the drift direction is bounded for exponential families.

This hands `#deriv-adaptive-gain-dynamics` a **concrete second instance** of derivable meta-gain alongside adaptive-Kalman-with-Mehra-estimator (its primary instance). The machinery of meta-gain composition via `#result-sector-persistence-template` (augmented-state Lyapunov) applies directly.

L2 regime: candidate third `#disc-identifiability-floor` instance or strengthening of Instance 2.

Hypothesis (L2-latent-floor)

Under L2-latent regimes (unobservable correlation structure beyond what Fisher information can resolve), Fisher whitening fails: the correlation sub-block of Fisher is rank-deficient (Cramér-Rao floor on unobservable parameters). This parallels `#disc-identifiability-floor` Instance 2's L1'-unobservable-C Fisher-rank-1 obstruction — potentially a new L2 instance, or potentially a generalization of Instance 2 to higher correlation orders. Open whether these are distinct floors or a single unified obstruction.

L2-degenerate (perfect correlation, $r \to 1$) is a *structural* collapse — it requires DAG repair (edge merging) rather than update repair. This sits outside the Fisher-whitening framework.

Epistemic Status.

Conditional. Max attainable: *exact* under Path A ((PI) + Čencov combined with B1 directional fidelity) + L1'/L2 regimes + Bayesian coherence. The derivation is textbook information-geometry (Amari 1998; Amari-Nagaoka 2000) plus the AAT-internal (PI) axiom from `#scope-agent-identity`. What is AAT-distinctive is (a) the identification of (PI)/Čencov as the forcing machinery rather than merely a preferred choice; (b) the sub-scope α_3 labeling within the A2' partition; (c) the composition with `#deriv-adaptive-gain-dynamics`' meta-gain framework as a degenerate special case.

The angle bound is ≤ 45° for all finite r. This is a sharp load-bearing observation: AAT's existing default log-odds signal function is *directionally robust* under correlated failures. The bias formula in `#deriv-l1-update-bias` gives the quantitative magnitude-bias; this segment gives the qualitative direction-preservation. Together they characterize the default signal function's behavior under L1' fully: **direction preserved, magnitude biased, observable-C restores exactness (Prop B.7), unobservable-C floor per F13**.

Without (PI) adoption. If (PI) is not adopted as an AAT axiom, Fisher whitening remains a theorem-imported technique from information geometry. The sub-scope α_3 labeling still applies (correlated evidence + Fisher-whitened + Bayesian coherence → A2' derived) but without the AAT-internal forcing. This is the adjacent-family status.

Discussion.

Relationship to the default signal function. `#disc-credit-assignment-boundary`'s default log-odds signal was landed in the 2026-04-22/23 strengthening cycle via the Cauchy-FE uniqueness theorem in `#deriv-edge-update-natural-parameter`. Under L0 evidence independence, the default signal is exact. Under L1' correlated evidence, the default signal's direction is preserved (≤ 45° angle to natural-gradient direction) but magnitude degrades by $\sqrt{1 - r^2}$. This segment's Fisher-whitened update is the correction under (PI)/Čencov; it recovers sharp B1 under L1' by lifting the inner product from Euclidean to Fisher.

Relationship to `#disc-identifiability-floor` Instance 2. Instance 2 (L1' unobservable-C Cramér-Rao floor) refutes single-channel identification of mixture parameters. Under observable C, Prop B.7 gives the L1' transfer; under unobservable C, no update — including Fisher-whitened — recovers identification. Fisher whitening operates *downstream* of identification: it corrects the update direction given that the Fisher information is known; it does not manufacture Fisher information that the data cannot yield.

Composition with the (PI)/Čencov 4th primary instance. The Fisher-metric cases in `#der-gain-sector-bridge`'s Verified Instances table (Matrix Kalman, Exponential family in natural parameters) are also AAT-internally forced under (PI)/Čencov. Fisher whitening at the edge-update layer of Part II therefore shares the same AAT-internal axiom as the Kalman / exp-family cases in Part I's update machinery. This is a cross-layer consistency: the (PI) axiom operates uniformly across Parts I and II whenever the natural parameters are parameters on a statistical manifold.

Appendix AP

Derivation: Regime-Marginal Indistinguishability — Cross-Regime Witness for `#disc-identifiability-floor` Instance 3 (Broadened Scope)

This segment supplies the parameter-by-parameter witness construction for `#disc-identifiability-floor` Instance 3's broadened-scope claim (2026-05-21): from sub-agent marginal data alone, the coupling-topology bit distinguishing among R0 / R1 / R2 dynamic regimes (per `#disc-dynamic-regime-axis`) is unidentifiable. The witnesses are exact for symmetric-matched-Tier-1-scalar sub-agents under standard coupling models (Model D for disturbance; the same models `#deriv-critical-mass-composition` uses, where the Instance-3 coupling-sign-bit witness is established). The mechanism is rank-collapse / Sylvester-at-one-remove on the marginal-Fisher operator for the topology coordinate, parallel to the Instance-3 coupling-sign-bit mechanism but acting on a higher-dimensional topology coordinate.

The relationship to `#disc-identifiability-floor` Instance 3 is supporting-derivation — Instance 3 carries the no-go statement, the escape menu, and the strengthened consequence; this segment carries the explicit construction backing the broadened no-go.

Formal Expression.

Setup. Consider $N = 2$ scalar sub-agents indexed $i \in \{1, 2\}$, each with state $q_i \in \mathbb{R}$, base correction rate $\alpha > 0$, observation noise $w_i \sim \mathcal{N}(0, \sigma^2)$, base disturbance bound ρ, tempo $\mathcal{T}$. The composite state is $X^c = (q_1, q_2) \in \mathbb{R}^2$. An external observer B accesses *only* per-sub-agent marginal data: for each i, the marginal stationary distribution $\mu_\infty^{(i)}$, the local mismatch process, and the per-sub-agent update rule $f_M^{(i)}$ as a structural object (white-box on each sub-agent, observed in isolation). The observer does *not* see: the composite-level coupling topology (sign + structure of cross-agent influence), the joint trajectory $\{X_t^c\}$ (only the projection onto per-sub-agent marginals), the game-structure object $\{O^{(j)}\}_{j \neq i}$ from sub-agent i's frame, or composite-level convergence-rate-class diagnostics.

The inferential question: from per-sub-agent marginal data alone, can B identify the composite's dynamic regime $\mathcal{R}(X^c) \in \{\text{R0}, \text{R1}, \text{R2}\}$ per `#disc-dynamic-regime-axis` §B?

Derived (cross-regime-witness, from Cournot equilibrium structure + shared-target marginal-matching)

Witness 1: R0 vs R1 indistinguishability.

Composite A — R0 shared-objective contraction. Sub-agent objective $O_A^{(i)} = -\frac{1}{2}(q_1 + q_2 - T)^2$ for shared target T. Per-agent gradient-flow best-response:

$$\dot{q}_i \;=\; -\alpha\,(q_i + q_j - T) + w_i. \tag{AP.1}$$

The composite is R0 per `#disc-dynamic-regime-axis` §B.0: unique attracting fixed point at $(T/2, T/2)$, joint Lyapunov $V = \frac{1}{2}\|q - (T/2, T/2)\|^2$, exponential convergence at rate α. The stationary marginal distribution of each sub-agent's state, computed from the linearized Ornstein-Uhlenbeck dynamics, is

$$\mu^{(i)}_{\infty,A} \;=\; \mathcal{N}\big(\, T/2,\; \sigma^2/(2\alpha)\,\big). \tag{AP.2}$$

Composite B — R1 strategic potential-game equilibrium (Cournot). Sub-agent objective $O_B^{(i)} = q_i\,(a_0 - b(q_1 + q_2) - c)$ — quadratic-profit Cournot in canonical form. The game is a potential game (Monderer-Shapley 1996) with potential Φ_B satisfying $\partial\Phi_B/\partial q_i \;=\; \partial O_B^{(i)}/\partial q_i$ for each i. The unique Nash equilibrium $q^*_{B,\text{Nash}}$ is locally attracting under gradient flow, with local-curvature parameter $\lambda_{\min}(\nabla^2\Phi_B(q^*_{B,\text{Nash}}))$ controlling the rate. Linearizing the best-response gradient flow around the Nash and adding the same observation noise, the stationary marginal distribution of each sub-agent's state is

$$\mu^{(i)}_{\infty,B} \;=\; \mathcal{N}\big(\, q^*_{B,\text{Nash}},\; \sigma^2/(2\lambda_{\min}(\nabla^2\Phi_B))\,\big). \tag{AP.3}$$

Parameter-matching. Two free parameters on each side: (T, α) for Composite A, (a_0, b, c) for Composite B (with α implicit in the gradient-flow rate). Two matching conditions:

- **Mean-matching:** $T_A/2 = q^*_{B,\text{Nash}}$. Choose $T_A = 2\,q^*_{B,\text{Nash}}$.
- **Variance-matching:** $\sigma^2/(2\alpha) = \sigma^2/(2\,\lambda_{\min}(\nabla^2\Phi_B))$. Choose b in Composite B such that $\lambda_{\min}(\nabla^2\Phi_B) = \alpha$.

Under both conditions: $\mu^{(i)}_{\infty,A} = \mu^{(i)}_{\infty,B} = \mathcal{N}(q^*, \sigma^2/(2\alpha))$ for both i, where $q^* = T_A/2 = q^*_{B,\text{Nash}}$. The marginal *transient* second moments also match under the matched local-curvature condition — both are exponentially-converging Ornstein-Uhlenbeck-type processes per sub-agent in the linearization.

The R0/R1 witness. Composite A satisfies $\mathcal{R}(\Sigma_A) =$ R0 (unique global attractor + global Lyapunov); Composite B satisfies $\mathcal{R}(\Sigma_B) =$ R1 (unique-Nash equilibrium with local Lyapunov-on-deviation, but Nash structure rather than shared-target structure). The marginal-data observer sees identical $\mu^{(i)}$ in both; cannot distinguish $\mathcal{R} =$ R0 from $\mathcal{R} =$ R1. ∎

Derived (cross-regime-witness, from matching-pennies aggregation + shared-target marginal-matching)

Witness 2: R0 vs R2 indistinguishability.

Composite C — R2 cyclic-distributional regime (matching-pennies under continuous-action smoothing). Take the 2×2 matching-pennies payoff structure, smoothed to continuous actions via softmax-style best-response with temperature parameter τ. The composite has no pure-strategy Nash; the mixed-Nash is at $(1/2, 1/2)$; under no-regret dynamics (Hart-Mas-Colell 2000) the empirical joint distribution converges to the CCE set at rate $O(1/\sqrt{T})$. The per-sub-agent marginal stationary distribution under this regime is a continuous distribution with mean $1/2$ and bounded variance determined by τ and the noise level σ^2.

Parameter-matching to Composite A. Choose Composite A's parameters (T_A, α_A) to align with Composite C's per-sub-agent marginal mean ($T_A/2 = 1/2 \Rightarrow T_A = 1$) and variance (choose α_A such that $\sigma^2/(2\alpha_A)$ equals Composite C's marginal variance — solvable for appropriate τ).

The R0/R2 witness. Composite A is R0 (state-variable macro-state, joint Lyapunov). Composite C is R2 (distributional macro-state, no joint Lyapunov on $\mathcal{X}^c$; CCE convergence only). The marginal-data observer sees identical $\mu^{(i)}$ in both; cannot distinguish $\mathcal{R} = \text{R0}$ from $\mathcal{R} = \text{R2}$. This sharpening is qualitatively cleaner than the R0/R1 witness because the *joint* dynamic regimes differ qualitatively — convergent vs cyclic-in-joint-distribution — while the per-sub-agent marginals can be made stationary in both cases by appropriate observation-aggregation: cyclic in the *joint* distribution projects to *stationary* in the marginal distribution when the cycle is over the joint coordinate that the marginal averages out. ∎

Mechanism (Sylvester at one remove, same as Instances 1 / 2 / 4). *Discussion*

The composite-level topology coordinate — the bit (σ, p) distinguishing R0 / R1 / R2, where σ is the coupling-sign-bit (cooperative vs adversarial) and p is the potential-existence bit (potential/monotone game-structure vs none) — enters the per-sub-agent marginal observation law $p(q_i \mid \theta_{\text{topology}})$ only through the *fixed-point* and *local-curvature* parameters that the witness constructions above match across composites. Computing the Fisher information of the marginal observation law with respect to θ_{topology} yields a rank-deficient matrix: the topology coordinate has a null direction along the manifold

$$\left\{\theta : (\text{mean}, \text{variance})(\theta) = (q^*, \sigma^2/(2\alpha))\right\}. \tag{AP.4}$$

By Sylvester's law of inertia (`#disc-identifiability-floor` Sylvester-recognition Finding), no reparameterization of the marginal-observation coordinate system removes this null direction — the irreducibility is invariant under the observer's representational freedom. The escape requires *rank-augmentation*: observation of a coordinate the marginal projection has annihilated. This is exactly the same mechanism as `#disc-identifiability-floor` Instances 1, 2, and 4; the present derivation occupies the *same* rank-collapse subclass as Instance 3's original coupling-sign-bit derivation, with the rank-collapse occurring on a topology coordinate of higher dimension than Instance 3's single sign-bit.

The mechanism's *one-remove* status (per `#disc-identifiability-floor` Instance 4's Mechanism discussion) is preserved: the *generating* action producing the indistinguishable pair is the swap between cooperative-shared-objective coupling and strategic-potential-game coupling (a structural game-theoretic operation), not metric congruence; but the *escape-side* irreducibility under the observer's representational freedom is identical to Instance 2's Fisher-null-on-a-fiber. The architecture of the proof is parallel to Instance 4's, applied to a different topology coordinate.

Epistemic Status.

Conditional on the standard coupling models (Model D for disturbance; the same models `#deriv-critical-mass-composition` uses for Instance 3's original coupling-sign-bit construction). The two witness constructions are at parameter-by-parameter explicit level for the symmetric-matched-Tier-1-scalar case (R0 vs R1 with $T_A = 2q^*_{B,\text{Nash}}$ and $\lambda_{\min}(\nabla^2\Phi_B) = \alpha$ as the matching conditions; R0 vs R2 with mean-matching and variance-matching against the matching-pennies-with-continuous-smoothing marginal).

Max attainable: *exact* in the symmetric-matched-Tier-1-scalar sub-scope, anchored by the explicit parameter matching above; *robust qualitative* in the general heterogeneous case, inheriting the Liberzon 2003 / Dayawansa-Martin 1999 / Shorten et al. 2007 common-Lyapunov-nonexistence structure that `#disc-identifiability-floor` Instance 3 (Boundary Characterization) already invokes. The tier statement is the same one Instance 3 carries; the broader topology coordinate does not change the tier.

Discussion.

Why this is a supporting derivation rather than a sixth identifiability-floor instance. The witness constructions in §"Witness 1" and §"Witness 2" *do* derive a cross-regime no-go from per-sub-agent marginal data, but the integrity test on M1 (the five-element distinctness test) finds three of four distinctness criteria failing against the existing Instance 3:

- Same external theorem (Liberzon 2003 / Dayawansa-Martin 1999 / Shorten et al. 2007 — the same anchor Instance 3 invokes; no new external theorem required).
- Same mechanism family (rank-collapse / Sylvester-at-one-remove on the marginal-Fisher operator).
- Four of five escapes inherit verbatim from Instance 3 (a, b, c, d unchanged; only (e) is new).
- Two of three strengthened consequences inherit-with-broadening from Instance 3; only the `#deriv-strategic-composition` sub-scope α'/β' elevation is genuinely new.

Per the strengthen-first landing posture (`~/.claude/memory/epistemic-discipline/strengthen-before-soften.md` and its landing-half `~/.claude/memory/epistemic-discipline/integration-is-replacement.md`), this segment lands the *math* in canon (math-lives-in-segments discipline) while the *Instance 3 broadening* in `#disc-identifiability-floor` lands the *no-go statement, escape menu, and strengthened consequence* (integration-is-replacement at the meta-segment scope level). The candidate "Instance 6" framing is deleted; the broadening of Instance 3 carries the strengthened content; this segment carries the explicit witness backing.

The split — math segment plus meta-segment broadening rather than a new instance — preserves M1's instance-count integrity (the instances are *plural in mechanism*, not just plural in surface no-go) while ensuring the derived content lands in canon at full visibility.

Relationship to `#disc-dynamic-regime-axis`. This segment's witnesses are what concretely back `#disc-dynamic-regime-axis`'s §H "Identifiability floor for regime" — the regime-axis meta-segment names the negative-half complementarity in the abstract; this segment supplies the constructive witness showing the no-go is real and parameter-explicit. The relationship is parallel to how `#disc-identifiability-floor` Instance 3's coupling-sign-bit witness in `#deriv-critical-mass-composition` (asymmetric-limit treatment) backs the meta-segment-level no-go statement.

When the dynamic-regime axis advances to meta-segment status, this segment is the canonical backing for the regime-axis floor; the complementarity structure (`#disc-separability-pattern` ↔ `#disc-identifiability-floor` parallel for the regime axis) is then:

- *Positive half:* `#disc-dynamic-regime-axis` four-tier ladder (R0 / R1 / R2 / R3 with their respective Lyapunov machinery).
- *Negative half:* `#disc-identifiability-floor` Instance 3 (broadened) — backed by this segment's witnesses.

The double-duty Instance 3 serves both `#disc-separability-pattern` Contraction ladder *and* `#disc-dynamic-regime-axis` four-tier ladder; this segment's witnesses extend that double-duty to the regime-axis level explicitly.

Open extensions.

- **Composite-of-composites case.** The witness constructions handle finite-N matched-symmetric-Tier-1 sub-agents. The composite-of-composites question (regime-identification of meta-composites from sub-composite marginal data) plausibly inherits the Liberzon machinery one level up but is not derived here.
- **Finite-sample refinements.** The §"Witness 1" / §"Witness 2" constructions are exact-population statements (stationary distributions identical). The finite-sample minimax error of regime classification under observation budget T is the Fano-style refinement that the 2026-05-20 spike correctly identified as the *adjacent* tool. Should be derivable from `#der-agent-opacity` H_b machinery applied at the composite layer; not done here.

- **Mean-field R3 lift.** The §"Witness 1" / §"Witness 2" constructions handle R0 / R1 / R2 finite-*N*. The R3 mean-field tier (per `#disc-dynamic-regime-axis` §B.3) has its own identifiability question — from finite-*N* marginal data, can the limit MFG regime be identified? Plausibly governed by the Lasry-Lions-monotonicity-vs-not bit (the MFG-analog of the potential-existence bit handled here). Tied to the R3 lift in `#deriv-strategic-composition` Working Notes mean-field extension.

Findings.

Cross-Regime Marginal Indistinguishability (Witness Backing for Broadened Instance 3). **Brief:** When you watch only one player at a time in a multi-agent system, you cannot tell what game they are playing — and "what game they are playing" determines whether the system as a whole will settle to a target, settle to a strategic equilibrium, or cycle in distribution forever. Two concrete worked constructions show this: two firms in Cournot competition that produce identical per-firm production-quantity distributions to a rowing team optimizing for a shared target (different game, same marginal); two players in matching pennies whose mixed strategy produces identical per-player action distributions to the same rowing team (different *kind* of equilibrium, same marginal). The mechanism is the same as for the simpler coupling-sign-bit case: the topology bit lives in a direction the marginal projection annihilates, and Sylvester's law of inertia says you cannot recover an annihilated direction by changing the units on the ruler — only by looking from a new vantage point (intervening on the topology, or observing the joint trajectory's convergence-rate class).

Impact: Backs `#disc-identifiability-floor` Instance 3's 2026-05-21 broadening from coupling-sign-bit to coupling-topology-bit with explicit parameter-by-parameter witness constructions. Preserves the math-lives-in-segments discipline (the derivation lives in canon, not only in the spike). Provides the canonical worked instantiation for the broadened no-go, parallel to how `#deriv-critical-mass-composition`'s asymmetric-limit treatment carries Instance 3's original coupling-sign-bit witness. Operationalizes the "regime is unidentifiable from per-sub-agent marginals" claim that the dynamic-regime axis meta-segment (`#disc-dynamic-regime-axis`) references as the negative-half of its complementarity structure.

Novelty Claim: *Claim integration* — the witness constructions extend Instance 3's coupling-sign-bit construction to broader topology coordinates (R0 vs R1; R0 vs R2) using the same Liberzon 2003 / Dayawansa-Martin 1999 / Shorten et al. 2007 anchor and the same Sylvester-at-one-remove mechanism. The Cournot construction (Composite B in §"Witness 1") is standard Monderer-Shapley 1996 potential-game-as-Cournot; the matching-pennies construction (Composite C in §"Witness 2") is standard rock-paper-scissors-style cyclic-game smoothing. The contribution is the parameter-matching against R0 shared-target dynamics that produces marginal-indistinguishability across regimes — and the mechanism reduction to Instance 3's rank-collapse-on-topology-coordinate framework.

Related Work:

Table AP.1

ASF concern	*Prior-art language*	*Relationship / Positioning*
Common-Lyapunov nonexistence for switched systems	Liberzon 2003 *Switching in Systems and Control* §2.1; Dayawansa-Martin 1999 *IEEE TAC* 44:751; Shorten et al. 2007 *SIAM Review* 49:545	*formal antecedent* — same anchor `#disc-identifiability-floor` Instance 3 invokes; this segment extends the application from coupling-sign-bit to coupling-topology-bit
Cournot duopoly as potential game	Cournot 1838 *Recherches*; Monderer-Shapley 1996 *Games and Economic Behavior* 14:124 §3	*formal antecedent* — standard potential-game instantiation used as Composite B in Witness 1
Matching pennies / cyclic-game smoothing	Standard textbook (Fudenberg-Tirole 1991 *Game Theory* §1; Hart-Mas-Colell 2000 *Econometrica* 68:1127)	*formal antecedent* — standard cyclic-game template used as Composite C in Witness 2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Sylvester's law of inertia as escape-irreducibility	Sylvester 1852 *Phil. Mag.* 4(23):138; cf. `#disc-identifiability-floor` Sylvester-recognition Finding	*formal antecedent* — same congruence-invariance machinery the rank-collapse subclass relies on
Multivariate Markov chains with prescribed component behavior but different joint dependence	Bielecki-Jakubowski-Niewkeglowski 2011, *Intricacies of dependence between components of multivariate Markov chains: weak Markov consistency and weak Markov copulae* (DOI 10.1214/EJP.V18-2238)[^bg2-2026-05-21]	*closest structural analogue* — same-marginals-different-coupling-law construction at the multivariate Markov-chain level; the hidden distinction is filtration-level dependence rather than potential-vs-cyclic game structure, but the formal move (matched per-component behavior, distinct joint law) is the same shape as Witness 1 / Witness 2 at the game-topology level
Marginal-non-determining ergodic measures (stochastic-process side)	Courbage-Hamdan 2001, *A family of stationary processes with infinite memory having the same p-marginals* (DOI 10.4064/CM90-2-2)[^bg2-2026-05-21]; Courbage-Hamdan 1998, *An ergodic Markov chain is not determined by its two-dimensional marginal laws* (DOI 10.1016/S0167-7152(97)00096-5)[^bg2-2026-05-21]; Hamdan 2000, *Markov Chains with Positive Transitions Are Not Determined by Any p-Marginals* (DOI 10.1007/S006050070034 — journal-name unverified at landing time)[^bg2-2026-05-21]; Hamdan 2002, *An ergodic Markov chain is not determined by any p-marginals* (DOI 10.1016/S0019-3577(02)80028-3)[^bg2-2026-05-21]	*adjacent literature* — strong precedent for "finite-order marginals do not determine the generating law" at the stochastic-process level. Different object (Markov chains, not coupled best-response dynamics) but identical formal shape (matched marginals, distinct joint law)
Network-reconstruction limits from temporal data	Angulo-Moreno-Barabási-Liu 2015, *Fundamental limitations of network reconstruction* (arXiv:1508.03559)[^bg2-2026-05-21]	*adjacent literature* — graph topology is generically as hard to recover as the full interaction matrix without persistent excitation; underdetermines coupling from per-node observation. Closest network-level impossibility neighbor on the topology-from-component-data axis. Different setting (general dynamical, not strategic), but the underdetermination-from-reduced-observation pattern is shared
Delay-reconstruction recovers only transitive closure	Cummins-Gedeon-Spendlove 2015, *On the Efficacy of State Space Reconstruction Methods in Determining Causality* (SIAM J. Appl. Dyn. Syst. 14:335)[^bg2-2026-05-21]	*adjacent literature* — componentwise trajectory data recovers at most the transitive closure of the interaction graph, not direct edges or feedback orientation. Explicit negative result on what trajectory data cannot tell an observer; not a game-structure witness but operationally similar shape
Game-side observational equivalence (quantal-response rationalization; consistent-game-set characterization)	Haile-Hortaçsu-Kosenok 2008, *On the Empirical Content of Quantal Response Equilibrium* (DOI 10.1257/AER.98.1.180)[^bg2-2026-05-21] (any interior action distribution rationalizable as QRE); Ziani-Chandrasekaran-Ligett 2016, *Efficiently Characterizing Games Consistent with Perturbed Equilibrium Observations* (DOI 10.7907/Z91Z42CF)[^bg2-2026-05-21] (characterizing the whole consistent set of games rather than point-identifying)	*adjacent literature* — stronger observational-equivalence statements on the game side than mere nonidentification, but at the equilibrium-rationalization layer rather than the per-component-marginal-trajectory layer this segment treats. Game-topology distinction (potential vs non-potential; cooperative vs cyclic) not the target of these results

ASF concern	*Prior-art language*	*Relationship / Positioning*
Information-structure observational compatibility	Penalva-Ryall 2003, *Causal assessment in finite extensive-form games*[^bg2-2026-05-21]; Penalva-Zuasti-Fabra-Ryall 2004, *Empirical Implications of Information Structure in Finite-Length Extensive-Form Games*[^bg2-2026-05-21]; Lehrer-Rosenberg-Shmaya 2006, *Signaling and mediation in Bayesian games*[^bg2-2026-05-21]	*adjacent literature* — different information structures observationally compatible with respect to outcome / equilibrium distributions; another form of hidden structure surviving unchanged at the observable level. Different concern (information structure, signaling, mediation) but same shape (hidden distinction not visible at observable layer)

Novelty assessment. The cross-topology game witness with hidden distinction in potential-vs-non-potential or cooperative-vs-cyclic structure under matched per-component marginals appears uncodified relative to the prior-art search[^bg2-2026-05-21]; Witness 1 (R0-vs-R1 Cournot under matched marginal mean and variance) and Witness 2 (R0-vs-R2 matching-pennies under aggregation) are the specific instantiations the verdict identifies as uncodified. The closest structural analogues are Bielecki et al. 2011 (multivariate Markov-chain same-marginals-different-coupling) and the Courbage-Hamdan / Hamdan sequence (marginal-non-determining stochastic processes) — both at the stochastic-process layer rather than the game-topology layer. AAT's contribution: the *naming claim* on the game-topology-bit witness as the cross-regime indistinguishability instantiation of the established marginal-non-determining shape.

Search Log:

- 2026-05-21 (*targeted*): The core citations in §"Witness 1" / §"Witness 2" (Liberzon 2003, Dayawansa-Martin 1999, Shorten et al. 2007, Monderer-Shapley 1996, Hart-Mas-Colell 2000) are inherited from existing AAT machinery's citation set; the witness construction is a parameter-matched application of standard game-theoretic templates against R0 shared-objective dynamics — the constructions are not novel in themselves, and the contribution is the cross-regime marginal-indistinguishability *recognition* and the mechanism-reduction to Instance 3's rank-collapse framework.
- 2026-05-21 (*targeted second-pass*): A second-pass comprehensive-search pass searched specifically for prior art on (Q1) broader-topology-bit unidentifiability beyond the coupling-sign-bit witness, and (Q2) convergence-rate-class as a passive regime classifier. Verdict on both: *no exact hit found in this set*. The Related Work table above incorporates the verdict's strongest neighbors at adjacent layers; see footnote `[^bg2-2026-05-21]` for the verification-deferral notice on every neighbor citation introduced from this pass.

Appendix AQ

Derivation: Strategy-Proofness Impossibility (Gibbard-Satterthwaite)

The first charter instance of `#disc-implementation-impossibility`: Gibbard 1973 and Satterthwaite 1975 forbid an external designer from constructing a dominant-strategy incentive-compatible (DSIC), non-dictatorial, surjective social-choice function over ≥ 3 alternatives under unrestricted preferences. The AAT-side translation locates the designer's task in the composite-agent setting (`#scope-composite-agent`), recognizes the agents as composite sub-agents reporting preferences derived from their objectives $O_t^{(i)}$, and maps the four standard escapes against AAT machinery — with the preference-domain-restriction escape (single-peaked / single-crossing) named precisely as *adjacent to but not identical with* the sub-scope α' machinery of `#deriv-strategic-composition`. The adjacency is not identity: sub-scope α' secures best-response *convergence* under fixed reports; GS forbids *strategy-proof revelation*. The honest map of where AAT meets mechanism design is the contribution; the imported theorem is classical.

Formal Expression.

Setting. Let A be a finite alternative set with $|A| \geq 3$, and let L denote the set of strict linear orderings on A. A **social-choice function** is a map $f : L^n \to A$ that aggregates n agents' reported preferences $R = (R_1, \dots, R_n) \in L^n$ into a single outcome $f(R) \in A$.

Three constraints define the designer's regime:

Definition (dsic, strategy-proofness / dominant-strategy incentive compatibility)

$$\forall i,\ \forall R_{-i} \in L^{n-1},\ \forall R_i, R_i' \in L :\quad f(R_i, R_{-i})\, R_i\, f(R_i', R_{-i}). \tag{AQ.1}$$

Truth-telling is a dominant strategy: no agent i can profit from misreporting given any reports by the others.

Definition (non-dictatorial)

$$\neg\, \exists i^* :\ \forall R \in L^n,\ f(R) \text{ is } i^*\text{'s top-ranked alternative in } R_{i^*}. \tag{AQ.2}$$

No agent fully determines the outcome.

[Definition (surjective)] $f(L^n) = A$. Every alternative is reachable from some report profile.

Postulate (gibbard-satterthwaite-1973-75, imported), exact

External theorem.

> **Theorem (Gibbard 1973; Satterthwaite 1975).** If $|A| \geq 3$ and preferences are unrestricted (each R_i ranges over all of L), then no social-choice function $f : L^n \to A$ is simultaneously DSIC, non-dictatorial, and surjective.

References. Gibbard, A. (1973), "Manipulation of voting schemes: a general result," *Econometrica* 41(4):587–601. Satterthwaite, M. A. (1975), "Strategy-proofness and Arrow's conditions: existence and correspondence theorems for voting procedures and social welfare functions," *J. Econ. Theory* 10(2):187–217. Modern textbook synthesis: Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, §23.C; geometric/topological proof structure shared with Arrow's theorem in Reny, P. J. (2001), "Arrow's theorem and the Gibbard-Satterthwaite theorem: a unified approach," *Econ. Letters* 70:99–105.

No-go. A designer who insists simultaneously on DSIC, non-dictatorial, surjective, and unrestricted-preferences cannot construct the social-choice function. Any candidate f satisfying three of the four conditions violates the fourth.

Derived (aat-mechanism-translation, from '#scope-composite-agent' + '#deriv-strategic-composition'), conditional on the composite-revelation-mechanism reading

AAT-side translation.

Within AAT's composite-agent setting, the GS no-go translates as follows:

- The "agents" are *sub-agents* in a composite per `#scope-composite-agent`, each carrying its own objective $O_t^{(i)}$ inducing a preference ordering $R_i \in L$ over composite-level alternatives A (the joint outcomes reachable through the composite's strategic-composition machinery).
- A "social-choice function" $f : L^n \to A$ is a *mechanism* the designer (external to the composite agent) imposes — a rule that takes the sub-agents' *reports* $R = (R_1, \ldots, R_n)$ and produces a composite-level outcome. The designer's task is to choose f.
- "Strategy-proofness" requires that each sub-agent's truth-telling about its own $O_t^{(i)}$-induced ordering is its best-response strategy. The DSIC condition is stronger than Nash equilibrium: dominance against *all* other-agent reports, not just equilibrium-consistent ones.

The GS theorem then states: under the four stated conditions, no such mechanism exists. The designer faces a structural impossibility on what can be implemented over the composite of strategically-reporting sub-agents under unrestricted preference domain.

Boundary characterization.

Table AQ.1

Escape	*Mechanism-design route*	*AAT machinery?*
Preference-domain restriction	Single-peaked preferences (Black 1948 *J. Polit. Econ.* 56:23–34) admit the median-voter rule; single-crossing extends this. The domain restriction circumvents the unrestricted-preferences condition.	**Adjacent to AAT machinery.** Sub-scope α' of `#deriv-strategic-composition` (potential / monotone game structure) is *adjacent* to single-peaked / single-crossing — both restrict the preference-or-utility structure to admit convergence/aggregation guarantees — but **not identical** (see *Sub-scope α' vs preference-domain restriction* below).

Escape	*Mechanism-design route*	*AAT machinery?*
Weakened solution concept	Bayes-Nash incentive compatibility (BIC) instead of dominant-strategy IC — d'Aspremont & Gérard-Varet (1979) "Incentives and incomplete information," *J. Public Econ.* 11:25–45. The designer relaxes from dominance to expected-utility best-response under a common prior on others' types.	**Outside AAT machinery.** BIC requires a shared prior over types and Bayesian reasoning over expected reports; AAT's strategic-composition machinery does not currently formalize the mechanism-induced report game in BIC terms.
Randomization	Gibbard (1977) "Manipulation of schemes that mix voting with chance," *Econometrica* 45:665–681 — random-dictator and convex-combination mechanisms admit strategy-proofness in expectation.	**Outside AAT machinery.** AAT's machinery treats deterministic strategies Σ_t; randomized-mechanism design is not in canon.
Restricted strategy space (enforced truth-telling)	The revelation-principle-companion: if the designer can verify reports against ground truth (e.g., observable types), strategy-proofness becomes definitional rather than incentivized.	**Outside AAT machinery.** AAT's agent-spectrum (`#def-agent-spectrum`) does not currently include verification-of-self-reports machinery.

Strengthened consequence. The boundary-precise contribution: the AAT/mechanism-design intersection is at *one* of the four GS escapes, and that intersection is *adjacent without being identical*. The honest map looks like this.

Sub-scope α' vs preference-domain restriction. Three reframings under which sub-scope α' would *be* the GS escape all fail: sub-scope α' secures best-response *convergence* of the composite-dynamics given a fixed mechanism, not *strategy-proofness* of the mechanism itself. Concretely: agents in a potential game (the canonical sub-scope α' instance) can have profitable unilateral deviations under non-truthful preference revelation even when the joint best-response dynamics under the current revelation converge. The convergence-under-current-reports guarantee that sub-scope α' supplies is *not* the dominance-over-all-reports guarantee DSIC requires.

The honest read: sub-scope α' is *adjacent to* the preference-domain-restriction escape because both restrict the preference-or-payoff structure in similar ways (single-peaked preferences exhibit a potential-game-like ordinal structure on single-dimensional issue spaces; monotone games on lattices share order-theoretic features with single-crossing preference domains), but it is *not the same restriction*. Sub-scope α''s contribution to the composite-agent layer is convergence-given-mechanism, not mechanism-strategy-proofness. The boundary is named precisely; AAT does not claim to escape GS, but recognizes where the framework's strategic-composition machinery sits *next to* one of GS's classical escapes.

The other three escapes belong entirely to mechanism design and social-choice theory. The framework honestly documents that, and locates its current contribution as *recognition of the boundary* rather than *provision of the escapes*.

Epistemic Status.

Exact for the imported theorem (Gibbard 1973, Satterthwaite 1975 — classical theorems with closed-form proofs in the source literature, primary-source verification scheduled for promotion past `draft`).

Conditional for the AAT-side translation — the mapping from social-choice-function to composite-mechanism-over-sub-agent-reports is mechanical under the *composite-revelation-mechanism reading* (sub-agents per `#scope-composite-agent`; reports as inputs to a designer-chosen f; composite outcomes as elements of A), but the reading itself is a representational choice; alternative AAT readings (e.g., treating sub-agents as auctioning under VCG-like settings; treating f as a Σ-level commitment device) exist and would map GS to different AAT-side vocabulary.

Robust qualitative for the boundary characterization (sub-scope α' is adjacent-but-not-identical to preference-domain restriction; the other three escapes are outside AAT canon) — survives across instance details, with the specific functional adjacency (single-peaked ↔ potential-game-like structure on issue spaces) noted but not derived as identity.

Max attainable: *Exact* for the imported theorem; *derived conditional* for the AAT translation; *robust qualitative* for the boundary characterization. The AAT translation's max ceiling stays *conditional* unless an alternative reading is forced by some AAT-internal uniqueness argument (no such argument currently in hand).

Discussion.

Why this is not a `#disc-identifiability-floor` instance. GS fails the identifiability-floor's five-element shape on actor positioning: in Instances 1–4 of the floor the actor frustrated by the no-go is the *agent itself* (cannot detect, identify, certify, or recover an architectural d.o.f. from limited data), and the escape elevates *agent-side* information-augmentation machinery (loop-interventional access, latent observability, matched-Tier composite-extended observation, similarity-fiber resolution). GS's actor is the *designer*; the escapes are *design-constraint relaxations*. The cluster is structurally different and belongs in the sister meta-segment `#disc-implementation-impossibility`. This segment is GS's home there.

Sub-scope α' adjacency, named precisely. The Discussion above states sub-scope α' is *adjacent to* preference-domain restriction but *not identical*; this is load-bearing for the honest scope-marking discipline of `#disc-implementation-impossibility`. The framework's contribution here is recognition of where the strategic-composition machinery sits next to one of GS's classical escapes — not a claim that AAT supplies the escape. Mechanism-design proper supplies the escape; AAT documents the proximity.

Peer voice with the mechanism-design literature. The imported theorem is classical; the proof structure (Gibbard 1973 via the revelation-principle-style argument; Satterthwaite 1975 via direct manipulation construction; Reny 2001 via geometric/topological unification with Arrow) is mature. AAT does not re-prove the theorem; AAT does not displace the standard escape routes; AAT does not claim that the strategic-composition machinery solves mechanism design. The segment's contribution is honest recognition of the boundary, with the imported theorem cited under its original names and full bibliographic references.

Relationship to `#deriv-strategic-composition`. The placement of "mechanism-design impossibility" as a candidate-fourth-instance of `#disc-identifiability-floor` is wrong on actor-positioning grounds. GS's home is here in the sister meta-segment; sub-scope α' retains its load-bearing role in strategic-composition for best-response convergence, and is *additionally* recognized as adjacent to GS's preference-domain-restriction escape under the boundary characterization above.

Findings.

The Gibbard-Satterthwaite Translation and Its Sub-Scope α' Adjacency. **Brief:** The Gibbard-Satterthwaite theorem says that no voting rule can be at once strategy-proof (no one ever benefits from misreporting their preferences), non-dictatorial (no single voter always decides), surjective (every alternative is reachable), and broadly applicable (works on unrestricted preferences) when there are at least three alternatives. The classical escapes have been studied for decades — restrict the preference domain (require single-peaked preferences; the median voter rule works), allow randomization (random dictator is strategy-proof in expectation), accept a weaker incentive concept (Bayes-Nash IC under a shared prior), or constrain the strategy space (enforce truth-telling by verifying reports). The contribution here is the honest map: when the framework recognizes a composite agent whose sub-agents have their own objectives, the strategic-composition machinery's sub-scope α' (potential or monotone games — the regime where best-response dynamics converge) is *adjacent to* the preference-domain-restriction escape (single-peaked / single-crossing) — both restrict the preference structure to admit aggregation/convergence guarantees — but is *not the same restriction*: convergence-under-current-reports is not dominance-over-all-reports. The other three classical escapes belong entirely to mechanism design and social-choice theory; the framework documents them as outside the current canon. Mapping that boundary honestly is the contribution; the impossibility is classical.

Impact: Closes the candidate-fourth-instance flag in `#deriv-strategic-composition` Discussion by landing GS in its proper home — the designer-side `#disc-implementation-impossibility` meta-segment, with sub-scope α''s relationship to the preference-domain-restriction escape named precisely. The named adjacency-without-identity is the kind of

scope-honesty discipline the framework runs throughout: AAT recognizes where its machinery sits next to classical mechanism-design escape routes, and refuses the temptation to overclaim that sub-scope α' "supplies" the escape (it does not — it supplies a different guarantee at a different layer of the composite-agent dynamics). The result is a charter instance of the designer-side cluster with the AAT/mechanism-design boundary mapped with precision rather than enthusiasm.

Novelty Claim: *Claim recognition* of the AAT-side translation of the Gibbard-Satterthwaite theorem into the composite-agent setting (sub-agents per `#scope-composite-agent`; reports as inputs to a designer-chosen mechanism; composite outcomes as the alternative set); *claim differentiation* on the sub-scope $\alpha' \leftrightarrow$ preference-domain-restriction adjacency: sub-scope α' is *adjacent to but not identical with* the GS preference-domain escape (the three-reframing strengthen-first check is the documented argument). The imported theorem is classical (Gibbard 1973; Satterthwaite 1975); the AAT-side translation, the boundary-characterization annotations, and the honest adjacency-without-identity recognition are the contributions.

Related Work:

Table AQ.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
The imported theorem	Gibbard, A. (1973), "Manipulation of voting schemes: a general result," *Econometrica* 41(4):587–601 (found pre-2026); Satterthwaite, M. A. (1975), "Strategy-proofness and Arrow's conditions: existence and correspondence theorems for voting procedures and social welfare functions," *J. Econ. Theory* 10(2):187–217 (found pre-2026)	*formal antecedent* — primary-source verification scheduled for promotion past `draft`
Modern textbook synthesis	Mas-Colell, A., Whinston, M. D. & Green, J. R. (1995), *Microeconomic Theory*, Oxford University Press, §23.C (found pre-2026)	*adjacent literature* — used for the constraint-regime formulation in the Setting above
Geometric/topological proof unification with Arrow	Reny, P. J. (2001), "Arrow's theorem and the Gibbard-Satterthwaite theorem: a unified approach," *Econ. Letters* 70:99–105 (found 2026-05-20)	*adjacent literature* — names the topological-combinatorial mechanism shared with Arrow's theorem (Instance 3 of `#disc-implementation-impossibility`); informs the mechanism-family discussion in the meta-segment
Single-peaked preferences as the canonical domain-restriction escape	Black, D. (1948), "On the rationale of group decision-making," *J. Polit. Econ.* 56:23–34 (found pre-2026)	*adjacent literature* — the canonical escape; the sub-scope α' adjacency is named relative to this
Random-dictator and randomized-mechanism escapes	Gibbard, A. (1977), "Manipulation of schemes that mix voting with chance," *Econometrica* 45:665–681 (found pre-2026)	*adjacent literature* — outside AAT machinery; documented as honest scope-mark
Bayes-Nash IC weakening	d'Aspremont, C. & Gérard-Varet, L.-A. (1979), "Incentives and incomplete information," *J. Public Econ.* 11:25–45 (found pre-2026)	*adjacent literature* — outside AAT machinery; documented as honest scope-mark

Search Log:

- 2026-05-22 (*intuition-only* on the AAT-side composite-revelation-mechanism reading): The mapping from social-choice function to composite-mechanism-over-sub-agent-reports (sub-agents per `#scope-composite-agent`; reports as inputs; composite outcomes as A) is the natural AAT-side reading; alternative readings exist (treating f

as a Σ-level commitment device; treating sub-agents as VCG-auction bidders) and would map GS to different AAT-side vocabulary. The composite-revelation reading is chosen here for parsimony with `#deriv-strategic-composition`; targeted future search candidates would be alternative-reading derivations if a downstream use of this segment requires them.

Appendix AR

Derivation: Bilateral Trade Impossibility (Myerson-Satterthwaite)

The second charter instance of `#disc-implementation-impossibility`: Myerson & Satterthwaite 1983 forbid an external designer from constructing an ex-post efficient, Bayesian incentive compatible, individually rational, budget-balanced bilateral-trade mechanism when buyer and seller have private valuations drawn from overlapping continuous distributions. The AAT-side translation locates the buyer and seller as a two-sub-agent composite per `#scope-composite-agent`; recognizes that the two-agent strategic-composition machinery of `#deriv-strategic-composition` applies cleanly to the agents' best-response dynamics *given* a fixed mechanism but does not cover the designer's mechanism-choice problem; and honestly maps the four classical escapes as almost entirely outside AAT's current canon. The contribution here is *boundary precision*: bilateral-trade impossibility is *adjacent to* AAT (the strategic-composition layer engages the agents-given-mechanism dynamics) but the designer's task lives in mechanism-design proper. The honest scope-mark is the contribution.

Formal Expression.

Setting. Two parties: a buyer B with private valuation $v_b \in [0, \bar{v}_b]$ drawn from distribution F_b with density f_b, and a seller S with private valuation $v_s \in [0, \bar{v}_s]$ drawn from distribution F_s with density f_s. The supports overlap: there exists a non-empty interval $[a, b] \subseteq [0, \bar{v}_b] \cap [0, \bar{v}_s]$ such that f_b and f_s are both strictly positive on $[a, b]$. A **direct mechanism** is a pair (p, x) where $p : [0, \bar{v}_b] \times [0, \bar{v}_s] \to [0, 1]$ is the trade probability and $x : [0, \bar{v}_b] \times [0, \bar{v}_s] \to \mathbb{R}$ is the buyer's expected payment to the seller (with the seller receiving x). Equivalently with interim allocation rules $\bar{p}_b(v_b)$, $\bar{p}_s(v_s)$ and interim transfer rules $\bar{x}_b(v_b)$, $\bar{x}_s(v_s)$ for buyer-and-seller respectively.

Four constraints define the designer's regime:

[Definition (ex-post-efficient)] $p(v_b, v_s) = 1 \iff v_b \geq v_s$. Trade occurs exactly when the buyer values the good at least as much as the seller.

[Definition (bayesian-ic, BIC)] Each agent's truthful reporting is a Bayesian best response given the other's truthful strategy:

$$v_b \bar{p}_b(v_b) - \bar{x}_b(v_b) \;\geq\; v_b \bar{p}_b(\hat{v}_b) - \bar{x}_b(\hat{v}_b) \quad \forall v_b, \hat{v}_b, \tag{AR.1}$$

and symmetrically for the seller (with the sign reversed since the seller is paid).

[Definition (individual-rationality, IR)] Both parties participate voluntarily: expected interim payoff is nonnegative for every type:

$$v_b \bar{p}_b(v_b) - \bar{x}_b(v_b) \;\geq\; 0 \quad \forall v_b, \quad \text{and} \quad \bar{x}_s(v_s) - v_s \bar{p}_s(v_s) \;\geq\; 0 \quad \forall v_s. \tag{AR.2}$$

[Definition (budget-balance)] No external subsidy: the buyer's expected payment exactly equals the seller's expected receipt:

$$\mathbb{E}[x(v_b, v_s)] \;=\; 0 \text{ (net designer-side).} \tag{AR.3}$$

Postulate (myerson-satterthwaite-1983, imported), exact

External theorem.

> **Theorem (Myerson & Satterthwaite 1983).** If f_b and f_s are strictly positive on a common open interval and the supports overlap, no direct mechanism (p, x) is simultaneously ex-post efficient, BIC, IR, and budget-balanced.

Reference. Myerson, R. B. & Satterthwaite, M. A. (1983), "Efficient mechanisms for bilateral trading," *J. Econ. Theory* 29(2):265–281. Modern textbook synthesis: Krishna, V. (2002), *Auction Theory*, Academic Press, §5.3; Mas-Colell, A., Whinston, M. D. & Green, J. R. (1995), *Microeconomic Theory*, Oxford University Press, §23.E.

No-go. A designer demanding ex-post efficiency + BIC + IR + budget-balance cannot construct the bilateral-trade mechanism. Any candidate mechanism satisfying three of the four conditions violates the fourth. Equivalently: under private values with overlapping supports, the designer must accept a deficit (relax budget-balance), accept inefficiency (relax ex-post efficiency), accept that participation is not always individually rational, or weaken the incentive concept.

Derived (aat-bilateral-trade-translation, from '#scope-composite-agent'), conditional on the buyer-seller-as-two-sub-agent reading, partial

AAT-side translation.

Within AAT's composite-agent setting, the MS no-go translates partially:

- The "buyer" and "seller" are *sub-agents* in a two-sub-agent composite per `#scope-composite-agent`, each carrying its own objective $O_t^{(i)}$ — the buyer's $O_t^{(B)}$ rewarding acquisition at favorable price; the seller's $O_t^{(S)}$ rewarding relinquishing at favorable price.
- The "mechanism" (p, x) is a designer-imposed structure on the composite's joint action space and reward redistribution — outside AAT's agent-level machinery; AAT's strategic-composition machinery in `#deriv-strategic-composition` applies to the agents' best-response dynamics *given* the mechanism (p, x), not to the designer's choice of (p, x).

The honest translation: the *agents-given-mechanism layer* is internal to AAT (strategic-composition machinery applies); the *mechanism-choice layer* is external. The MS no-go applies at the mechanism-choice layer; AAT does not currently cover it.

Boundary characterization.

Table AR.1

Escape	*Mechanism-design route*	*AAT machinery?*
Subsidies (relax budget-balance)	VCG mechanism with deficit; pivotal-mechanism transfers. The designer covers any shortfall from external resources.	**Outside AAT machinery.** AAT does not currently formalize designer-side transfer pools or subsidy budgets.

Escape	*Mechanism-design route*	*AAT machinery?*
Second-best mechanisms (relax ex-post efficiency)	Myerson 1981 "Optimal auction design," *Math. Oper. Res.* 6:58–73 — characterizes revenue-optimal mechanisms with inefficient withholding in the marginal-cost-curve region; bilateral-trade analog accepts trade-failure when $v_b - v_s$ is small relative to the inefficiency cost.	**Outside AAT machinery.** Optimal-mechanism design under welfare-relaxation is the constructive side of mechanism design; deliberately excluded by the discipline filter (see `#disc-implementation-impossibility`'s "constructive mechanism-design" exclusion).
Common values (relax private values)	Information aggregation under shared-value components — Wilson 1977 "A bidding model of perfect competition," *Rev. Econ. Stud.* 44:511–518; the no-go softens because the agents' reports carry shared signal.	**Outside AAT machinery; adjacent** to `#scope-edge-update-causal-validity`'s regime-indexed identification structure. The shared-signal regime is structurally similar to AAT's partial-intervention Regime B (`#scope-edge-update-causal-validity`'s middle tier), but the formal connection has not been derived.
Long-run reputation / repeated interaction	Single-shot MS does not bind in indefinitely-repeated bilateral-trade games — Folk-theorem-style mechanisms with reputation backlash. Fudenberg-Maskin 1986 *Econometrica* 54:533–554 for the canonical Folk-theorem setup.	**Outside AAT machinery; adjacent** to the strategic-composition machinery in repeated-game form. The repeated-game extension of `#deriv-strategic-composition` is open canonical work; the MS-escape connection would land there if pursued.

Strengthened consequence. The MS boundary lies *almost entirely outside* AAT's current machinery — and the *honest delineation* is the contribution. AAT does not claim to escape MS; AAT does not provide subsidies, optimal-mechanism design, common-value aggregation, or repeated-game reputation machinery in its current canon. The framework's contribution at this charter instance is precisely the boundary recognition: the two-agent strategic-composition machinery in `#deriv-strategic-composition` applies cleanly to the buyer-seller best-response dynamics *given* a fixed mechanism (the agents-given-mechanism layer is internal), but the designer's mechanism-choice problem (the mechanism-choice layer) is external. The two escapes flagged as *adjacent* (common-values to Regime B; repeated-interaction to repeated-strategic-composition) are open avenues if AAT extends in those directions; the two flagged as *outside* (subsidies, second-best mechanisms) belong to mechanism-design proper.

The contribution is the same kind of scope-honesty discipline `#disc-identifiability-floor` runs on the agent side and `#deriv-strategy-proofness-impossibility` runs at a different AAT-machinery-intersection: name the boundary precisely; refuse to overclaim machinery the framework does not currently supply.

Epistemic Status.

Exact for the imported theorem (Myerson & Satterthwaite 1983 — classical theorem with closed-form proof in the source literature; primary-source verification scheduled for promotion past `draft`).

Conditional, partial for the AAT-side translation — the buyer-seller-as-two-sub-agent reading is the natural composite-agent translation, but the *agents-given-mechanism layer* is the only part of the no-go internal to AAT canon; the *mechanism-choice layer* (where the no-go binds) is external. The partial-translation is honest — AAT recognizes the bilateral-trade designer-task as adjacent but does not formalize the designer-side mechanism-choice problem.

Robust qualitative for the boundary characterization — four escapes documented with honest scope-marks; two adjacencies named (common-values ↔ Regime B; repeated-interaction ↔ repeated-strategic-composition); no precise functional intersection derived.

Max attainable: *Exact* for the imported theorem. *Derived conditional* for the AAT translation, with the *conditional* gating on the buyer-seller-two-sub-agent reading; an alternative reading (e.g., buyer and seller as *single-agent-with-two-roles* under intertemporal preference modeling) would change the translation but is not currently in canon. *Robust*

qualitative for the boundary characterization; lifting to *exact* on the common-values ↔ Regime B intersection would require formalizing the regime-correspondence, which is open.

Discussion.

Why this is not a `#disc-identifiability-floor` instance. Same actor-positioning argument as for the GS instance (`#deriv-strategy-proofness-impossibility`): the actor frustrated by the MS no-go is the *designer* (cannot construct a mechanism satisfying all four conditions), not an agent inferring from data. The escapes are *design-constraint relaxations* (relax budget-balance / efficiency / private-values / single-shot interaction), not information augmentations of an inferential regime. The cluster belongs in `#disc-implementation-impossibility`, not in the identifiability-floor.

Why the AAT translation is honestly partial. Unlike the GS instance (where sub-scope α' of `#deriv-strategic-composition` has a named adjacency to the preference-domain-restriction escape), MS has *no* AAT-internal escape adjacency. The two *adjacent* escapes (common-values ↔ Regime B; repeated-interaction ↔ repeated-strategic-composition) are open extensions, not current canon. The two *outside* escapes (subsidies, second-best mechanisms) belong to mechanism-design proper. The framework's contribution at this charter instance is therefore *purely the boundary recognition*: AAT covers the agents-given-mechanism layer, not the mechanism-choice layer; the MS no-go binds at the latter.

Peer voice with the mechanism-design literature. The imported theorem is classical; the proof structure (Myerson-Satterthwaite via Myerson's virtual-valuations machinery) is mature. AAT does not re-prove the theorem; AAT does not displace the standard escape routes; AAT does not claim that the strategic-composition machinery extends to mechanism-choice. The segment's contribution is honest recognition of the boundary, with the imported theorem cited under its original name and full bibliographic references.

Two open AAT-side extensions flagged. The common-values ↔ Regime B adjacency and the repeated-interaction ↔ repeated-strategic-composition adjacency are open avenues. If AAT extends in either direction — formalizing shared-signal information regimes alongside the current intervention-regime classification, or extending strategic-composition into repeated-game form — the MS instance's boundary characterization would gain AAT-machinery content where it currently has honest scope-marks. The flags here document the avenues without committing to either extension.

Findings.

The Bilateral Trade Boundary — Honest Scope-Marking as Contribution. **Brief:** When a buyer and seller each know their own valuation but not the other's, and the designer wants a trading mechanism that is at once fair (trade happens whenever the buyer values the good more than the seller does), incentive-honest (neither party can profitably misreport their valuation), voluntary (both parties want to participate), and self-funded (no outside subsidies), the Myerson-Satterthwaite theorem says all four cannot hold simultaneously. The classical escapes have been studied for decades — accept a budget deficit, accept inefficient withholding, exploit common-value information, or rely on long-run reputation in repeated trade. The honest contribution here is that *none* of these escapes is currently provided by the framework's machinery. The framework covers what the buyer and seller do *given* a mechanism (the two-sub-agent strategic-composition machinery applies cleanly to their best-response dynamics); the *designer's choice of mechanism* lives in mechanism-design proper, and the framework documents that boundary precisely rather than overclaiming reach. Two of the four classical escapes (common-values and repeated-interaction) are flagged as adjacent — open avenues if the framework extends in those directions — and two (subsidies and second-best mechanisms) belong entirely to mechanism design. The contribution is the map of where the framework meets the bilateral-trade boundary, not the provision of an escape.

Impact: The cleanest example of the `#disc-implementation-impossibility` cluster's discipline filter doing its honest work. Where the GS charter instance has a named sub-scope α' adjacency to make precise, the MS charter instance has *none* — the AAT-machinery contribution is purely boundary-recognition. That contrast within the cluster is itself epistemic content: the framework's contribution is precision about *where its machinery contributes* (GS: adjacent to preference-domain restriction) and *where it does not* (MS: outside almost entirely). Two open avenues flagged

(common-values ↔ Regime B; repeated-interaction ↔ repeated-strategic-composition) document where future framework extensions could intersect MS without committing to either.

Novelty Claim: *Claim recognition* of the AAT-side translation of the Myerson-Satterthwaite theorem under the buyer-seller-as-two-sub-agent reading; *claim differentiation* on the agents-given-mechanism layer being internal to AAT and the mechanism-choice layer being external — the no-go binds at the latter, which is the honest scope-mark. The imported theorem is classical (Myerson & Satterthwaite 1983); the AAT-side translation, the boundary-characterization annotations, the honest *almost-entirely-outside* scope-mark, and the two open adjacencies (common-values, repeated-interaction) are the contributions.

Related Work:

Table AR.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
The imported theorem	Myerson, R. B. & Satterthwaite, M. A. (1983), "Efficient mechanisms for bilateral trading," *J. Econ. Theory* 29(2):265–281 (found pre-2026)	*formal antecedent* — primary-source verification scheduled for promotion past `draft`
Modern textbook synthesis	Krishna, V. (2002), *Auction Theory*, Academic Press, §5.3 (found pre-2026); Mas-Colell, Whinston & Green (1995), §23.E	*adjacent literature* — used for the constraint-regime formulation in the Setting above
Optimal mechanism design (second-best escape)	Myerson, R. B. (1981), "Optimal auction design," *Math. Oper. Res.* 6(1):58–73 (found pre-2026)	*adjacent literature* — the canonical second-best-escape construction; outside AAT canon per the discipline filter exclusion
Common-values bidding (adjacent escape)	Wilson, R. (1977), "A bidding model of perfect competition," *Rev. Econ. Stud.* 44(3):511–518 (found pre-2026)	*adjacent literature* — flagged as adjacent to `#scope-edge-update-causal-validity` Regime B; formal correspondence is open
Folk theorem (repeated-interaction escape)	Fudenberg, D. & Maskin, E. (1986), "The folk theorem in repeated games with discounting or with incomplete information," *Econometrica* 54(3):533–554 (found pre-2026)	*adjacent literature* — flagged as adjacent to repeated-strategic-composition extension of `#deriv-strategic-composition`; formal correspondence is open

Search Log:

- 2026-05-22 (*intuition-only* on the buyer-seller-two-sub-agent reading): The natural composite-agent translation maps buyer and seller to two sub-agents per `#scope-composite-agent` with their own $O_t^{(i)}$. Alternative readings (single-agent-with-two-roles under intertemporal preference; large-economy aggregation limit) exist; not pursued here. Targeted future search candidates if alternative readings become relevant: Wilson 1977 large-economy bidding limit; behavioral / dual-self models of intertemporal trade.
- 2026-05-22 (*intuition-only* on the common-values ↔ Regime B adjacency): The common-values escape's structural feature (agents' reports carry shared signal about a partially-observable joint state) is reminiscent of `#scope-edge-update-causal-validity`'s Regime B (partial intervention; confounder-adjustment under side-channel observation). Whether the correspondence is formal or merely metaphorical is open. Targeted future search would compare the information-aggregation structure of common-values mechanisms (Wilson 1977; Pesendorfer-Swinkels 1997 *Econometrica* 65:1247–1281) against AAT's Regime B partial-intervention machinery.

- 2026-05-22 (*intuition-only* on the repeated-interaction ↔ repeated-strategic-composition adjacency): The repeated-bilateral-trade escape's structure (reputation backlash sustaining cooperation through grim-trigger or carrot-and-stick strategies) is the natural extension of strategic-composition to repeated games. Whether the formal correspondence holds is open. Targeted future search would compare the Folk-theorem machinery (Fudenberg-Maskin 1986; Fudenberg-Levine-Maskin 1994 *Econometrica* 62:997–1039) against a repeated-game extension of `#deriv-strategic-composition`.

Appendix AS

Derivation: Social Welfare Aggregation Impossibility (Arrow)

The third charter instance of `#disc-implementation-impossibility`: Arrow 1951 (expanded 1963) forbids an external designer from constructing a social welfare function aggregating n agents' strict preference orderings into a single social preference ordering over ≥ 3 alternatives under unrestricted domain (U), Pareto efficiency (P), Independence of Irrelevant Alternatives (IIA), and non-dictatorial (ND). The AAT-side translation locates the agents as composite sub-agents whose individual preference orderings are derived from their objectives $O_t^{(i)}$, and maps the four classical escapes with *two* AAT-machinery adjacencies named: the preference-domain-restriction escape (single-peaked / single-crossing → Condorcet-consistent rules) is *adjacent to* sub-scope α' of `#deriv-strategic-composition` (the same adjacency `#deriv-strategy-proofness-impossibility` names), and the cardinal-preference escape (utilitarianism, range voting) is *adjacent to* `#form-objective-functional`'s value-functional machinery — both adjacencies open avenues, not elevations to load-bearing. The contribution: two AAT-side adjacencies precisely mapped, with the cardinal-preference adjacency unique to Arrow (it does not appear in GS or MS) because Arrow's ordinal-only constraint is what AAT's value-functional formulation could in principle relax while GS's strategy-proofness regime is orthogonal to the ordinal/cardinal distinction.

Formal Expression.

Setting. Let A be a finite alternative set with $|A| \geq 3$, and let L denote the set of strict linear orderings on A. A **social welfare function** is a map $F : L^n \to L$ that aggregates n agents' strict preference orderings $R = (R_1, \ldots, R_n) \in L^n$ into a single social ordering $F(R) \in L$ on the same alternative set.

Four constraints define the designer's regime:

[Definition (unrestricted-domain, U)] The map F is defined on all of L^n — every possible profile of strict preference orderings is a valid input.

[Definition (pareto-efficient, P)] If every agent strictly prefers a over b (i.e., $a \, R_i \, b$ for all i), then the social ordering puts a strictly above b ($a \, F(R) \, b$).

[Definition (independence-of-irrelevant-alternatives, IIA)] The social ordering's comparison of any pair (a, b) depends only on the individuals' comparisons of (a, b) — not on their comparisons involving any third alternative c. Formally: for any two profiles R, R' such that each agent ranks a and b the same way in R and R', the social orderings $F(R)$ and $F(R')$ rank a, b the same way.

[Definition (non-dictatorial, ND)] No agent i^* such that for every profile R and every pair a, b with $a \, R_{i^*} \, b$ strictly, the social ordering $F(R)$ also puts a above b — i.e., no agent's strict preferences are imposed on society regardless of others.

Postulate (arrow-1951-1963, imported), exact

External theorem.

> **Theorem (Arrow 1951, 1963).** If $|A| \geq 3$, then no social welfare function $F : L^n \to L$ is simultaneously U, P, IIA, and ND.

References. Arrow, K. J. (1951, 2nd ed. 1963), *Social Choice and Individual Values*, Yale University Press. The expanded 1963 edition adds the simplified proof and discusses extensions. Modern proof structures: Geanakoplos, J. (2005), "Three brief proofs of Arrow's impossibility theorem," *Econ. Theory* 26:211–215; Reny, P. J. (2001), "Arrow's theorem and the Gibbard-Satterthwaite theorem: a unified approach," *Econ. Letters* 70:99–105 (geometric/topological unification with GS).

No-go. A designer demanding U + P + IIA + ND cannot construct the social welfare function. Any candidate F satisfying three of the four conditions violates the fourth.

Derived (aat-social-welfare-translation, from '#scope-composite-agent' + '#form-objective-functional'), conditional on the composite-aggregation-mechanism reading

AAT-side translation.

Within AAT's composite-agent setting, the Arrow no-go translates as follows:

- The "agents" are *sub-agents* in a composite per `#scope-composite-agent`, each carrying its own objective $O_t^{(i)}$ inducing a preference ordering $R_i \in L$ over composite-level alternatives A (the joint outcomes reachable through the composite's strategic-composition machinery). The ordinal preference R_i is the ordering induced by sub-agent i's value functional $V_{O_t^{(i)}}(\tau)$ on the alternative set A.
- A "social welfare function" $F : L^n \to L$ is a *meta-aggregation mechanism* the designer imposes — a rule that takes the sub-agents' ordinal preferences (or reports thereof) and produces a single social ordering on A. The designer's task is to choose F.
- The IIA condition is the load-bearing one structurally: it requires that the social comparison of a, b depend *only* on individuals' ordinal comparisons of a, b, not on their comparisons involving any other alternative — an *ordinal-only* informational restriction.

The Arrow theorem then states: under U + P + IIA + ND, no such F exists. The designer faces a structural impossibility on what can be aggregated over the composite of sub-agents under ordinal-only information.

Boundary characterization.

Table AS.1

Escape	*Mechanism-design / social-choice route*	*AAT machinery?*
Preference-domain restriction	Single-peaked preferences (Black 1948) admit Condorcet-consistent rules including the median voter; single-crossing extends this. Domain restriction circumvents U.	**Adjacent to AAT machinery.** Sub-scope α' of `#deriv-strategic-composition` (potential / monotone game structure) is *adjacent to* single-peaked / single-crossing — the same adjacency `#deriv-strategy-proofness-impossibility` names. *Not identical*: sub-scope α' secures best-response convergence in the composite-dynamics, not Condorcet-consistent aggregation in the welfare function.
Weakened IIA	Positional methods — Borda count (1781), range voting, Coombs method. Each uses ranks-and-positions across the full ordering, not just pairwise comparisons.	**Outside AAT machinery.** Positional aggregation is not in AAT canon.

Escape	*Mechanism-design / social-choice route*	*AAT machinery?*
Supermajority requirements	Replace simple-majority with k/n thresholds for $k > n/2$; admits constitutional / veto-respecting aggregators (e.g., the unanimity rule, two-thirds-majority rules).	**Outside AAT machinery.** Supermajority-thresholding mechanisms not in AAT canon.
Cardinal preferences	Relax ordinal-only IIA; allow social aggregation over cardinal utilities — utilitarian summation, weighted utilitarianism, range voting (treating ratings as cardinal). Sen 1970, *Collective Choice and Social Welfare*, Holden-Day, §3 gives the canonical treatment of cardinal-extension escapes.	**Adjacent to AAT machinery.** `#form-objective-functional`'s value-functional $V_{O_t}(\tau)$ carries cardinal value content — the value functional is real-valued, not ordinal-only. *Not identical*: AAT's value-functional formalism is per-sub-agent and per-trajectory, not a designer-side aggregation rule across sub-agents; lifting to aggregator-form requires a representational choice (utilitarian summation, weighted utilitarianism, etc.) that AAT does not currently make canonical.

Strengthened consequence. Arrow is the cluster instance with *two* AAT-side adjacencies:

Adjacency 1: sub-scope α' ↔ preference-domain restriction. Shared with `#deriv-strategy-proofness-impossibility`. Sub-scope α' of `#deriv-strategic-composition` (potential / monotone game structure) is adjacent to the single-peaked / single-crossing preference-domain restriction that admits Condorcet-consistent rules. The adjacency-without-identity is the same as in GS: sub-scope α' guarantees best-response *convergence* under fixed preferences, not Condorcet-consistent *aggregation* in the welfare function. The two are structurally related (both restrict preference structure to admit aggregation/convergence guarantees on a one-dimensional issue space) but operate at different layers of the composite-agent dynamics.

Adjacency 2: cardinal preferences ↔ `#form-objective-functional`. This adjacency is *unique to Arrow* among the three charter instances — GS's strategy-proofness regime is orthogonal to the ordinal/cardinal distinction (the manipulation no-go applies to any social-choice function on preference orderings regardless of whether utilities are ordinal or cardinal); MS's bilateral-trade setting already operates over cardinal valuations v_b, v_s so the Arrow IIA escape doesn't apply. For Arrow specifically, the ordinal-only IIA restriction is what AAT's value-functional formulation could *in principle* relax — the value functional V_{O_t} is real-valued and carries cardinal information per sub-agent. The adjacency is *open*: lifting AAT's per-sub-agent value-functional to a designer-side aggregator (utilitarian summation, weighted utilitarianism, range voting) requires a representational choice AAT does not currently make canonical. The open avenue is whether AAT extends to a designer-side welfare-aggregation formalism via the value-functional's cardinal content; if it does, Arrow's cardinal-preference escape would gain AAT-machinery internal content.

The other two escapes (weakened IIA via positional methods; supermajority requirements) belong entirely to social-choice theory and are honestly documented as outside AAT canon.

Epistemic Status.

Exact for the imported theorem (Arrow 1951, 1963 — classical theorem with closed-form proof in the source literature; multiple modern proofs available, e.g., Geanakoplos 2005, Reny 2001; primary-source verification scheduled for promotion past `draft`).

Conditional for the AAT-side translation — the composite-aggregation-mechanism reading is the natural composite-agent translation; alternative readings (treating sub-agents as factor-coalitions in a richer game; treating the social ordering as a Σ-level commitment device) exist and would map Arrow into different AAT-side vocabulary.

Robust qualitative for the boundary characterization — two named adjacencies (sub-scope α' ↔ preference-domain-restriction shared with GS; `#form-objective-functional` ↔ cardinal-preferences unique to Arrow); two outside AAT canon (positional methods, supermajority).

Max attainable: *Exact* for the imported theorem. *Derived conditional* for the AAT translation. *Robust qualitative* for the boundary characterization. The cardinal-preference adjacency could lift to *derived conditional* if AAT canonicalizes a designer-side welfare-aggregator construction over the per-sub-agent value functionals; that extension is open and would be its own framework-scope decision.

Discussion.

Why this is not a `#disc-identifiability-floor` instance. Same actor-positioning argument as GS and MS: the actor frustrated by the Arrow no-go is the *designer* (cannot construct a welfare function satisfying all four conditions), not an agent inferring from data. The escapes are *design-constraint relaxations* (relax U via domain restriction; relax IIA via positional methods or cardinal preferences; relax majority via supermajority), not information augmentations of an inferential regime. The cluster belongs in `#disc-implementation-impossibility`, not in the identifiability-floor.

The cardinal-preference adjacency is structurally unique to Arrow. Among the three charter instances of `#disc-implementation-impossibility`, only Arrow's IIA constraint produces a cardinal/ordinal distinction that AAT's value-functional formalism could engage. GS's strategy-proofness no-go applies regardless of utility cardinality; MS's bilateral-trade setting already operates over cardinal valuations. Arrow alone forbids aggregation under *ordinal-only* information, which is the constraint AAT's real-valued V_{O_t} could relax. The adjacency is *open* — AAT does not currently formalize a designer-side aggregator that consumes the per-sub-agent value functionals — but it is the open avenue most specifically illuminating about how the framework might engage social-choice theory in the future.

Peer voice with the social-choice literature. The imported theorem is classical; the proof structure (Arrow 1951 via the decisive-set / dictator-pivotal-position argument; Geanakoplos 2005's three brief proofs; Reny 2001's geometric/topological unification with GS) is mature. AAT does not re-prove the theorem; AAT does not displace the standard escape routes; AAT does not claim that the value-functional formalism solves social-choice theory. The segment's contribution is honest recognition of the two adjacencies — sub-scope α' (shared with GS) and `#form-objective-functional`'s cardinal value content (unique to Arrow) — with the imported theorem cited under its original name and full bibliographic references.

Relationship to `#form-objective-functional`. The cardinal-preference adjacency raises an open question about the value-functional's downstream extension. `#form-objective-functional` formalizes $V_{O_t}(\tau)$ as the agent's value functional over trajectories; the cardinal content is *within-agent* (used for the agent's own choice). Arrow's cardinal-preference escape would lift this to *cross-agent* (used by a designer to aggregate across sub-agents). Whether such a lift is a natural extension of AAT or requires new framework machinery (a designer-side aggregator separate from the per-sub-agent value functional) is the open question. `#deriv-reward-channel-learning-no-go` (sister to `#deriv-self-actuation-grounding`) recognizes the *single-interface narrowness* of V_{O_t} at the within-agent layer; the parallel cross-agent narrowness question (whether V_{O_t} aggregates honestly across sub-agents) is structurally adjacent but not yet posed in canon.

Findings.

The Arrow Translation and Its Two AAT-Side Adjacencies. **Brief:** Arrow's theorem says that no social welfare function — no rule for aggregating individuals' preferences into a single social preference ordering — can simultaneously satisfy four reasonable demands: be defined for every possible profile (unrestricted domain), respect unanimous preferences (Pareto efficiency), depend on each pair only through how individuals rank that pair (independence of irrelevant alternatives), and not be dictatorial (no single individual always decides). The classical escapes have been studied for decades — restrict the preference domain (single-peaked preferences admit the median voter / Condorcet rules), use a positional method (Borda count, range voting), require supermajorities, or allow cardinal preferences (weighted utilitarianism rather than ordinal-only aggregation). The contribution here is two named adjacencies on the AAT side:

the preference-domain-restriction escape is *adjacent to* sub-scope α' of the strategic-composition machinery — same adjacency as GS — and the cardinal-preference escape is *adjacent to* the framework's value-functional formalism (which carries cardinal value content per sub-agent). The cardinal adjacency is structurally unique to Arrow: GS's strategy-proofness is orthogonal to the cardinal/ordinal distinction; MS already operates over cardinal valuations. Both adjacencies are *adjacent without being identical* — open avenues if the framework extends in those directions, not current elevations of AAT machinery to load-bearing. The other two escapes (positional methods, supermajority) belong entirely to social-choice theory and are honestly documented as outside AAT canon. The map of where the framework meets Arrow is the contribution; the theorem is classical.

Impact: Closes the third charter-instance slot of `#disc-implementation-impossibility` with the cluster's *richest* AAT-side adjacency map — two named adjacencies, one shared with GS (sub-scope $\alpha' \leftrightarrow$ preference-domain restriction) and one unique to Arrow (`#form-objective-functional` $\leftrightarrow$ cardinal preferences). The Arrow-unique cardinal-preference adjacency is the open avenue most specifically illuminating about how the framework might engage social-choice theory in the future: AAT's per-sub-agent value functional V_{O_t} carries cardinal value content, and lifting it to a designer-side cross-agent aggregator (utilitarian summation, weighted utilitarianism) is the natural extension that would give the Arrow cardinal-preference escape AAT-machinery internal content. Whether such an extension is a natural growth of AAT or requires new framework machinery is the open question raised by the adjacency. Together with the GS instance (named adjacency, no open avenue) and MS instance (no current adjacency, two open avenues), the cluster's three charter instances span the full range of how AAT meets the designer-side impossibility cluster.

Novelty Claim: *Claim recognition* of the AAT-side translation of Arrow's theorem under the composite-aggregation-mechanism reading (sub-agents per `#scope-composite-agent`; preference orderings induced by per-sub-agent $V_{O_t^{(i)}}$; designer-chosen aggregator F producing social ordering); *claim differentiation* on the cardinal-preference adjacency being structurally unique to Arrow within the cluster — GS's strategy-proofness is orthogonal to the cardinal/ordinal distinction; MS's bilateral-trade setting already operates over cardinal valuations; only Arrow's ordinal-only IIA constraint produces a cardinal/ordinal distinction that AAT's value-functional formalism could engage. The imported theorem is classical (Arrow 1951, 1963); the AAT-side translation, the two-adjacency boundary characterization, and the cardinal-preference adjacency as the Arrow-unique open avenue are the contributions.

Related Work:

Table AS.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
The imported theorem	Arrow, K. J. (1951, 2nd ed. 1963), *Social Choice and Individual Values*, Yale University Press (found pre-2026)	*formal antecedent* — primary-source verification scheduled for promotion past `draft`
Modern proof structure	Geanakoplos, J. (2005), "Three brief proofs of Arrow's impossibility theorem," *Econ. Theory* 26:211–215 (found pre-2026); Reny, P. J. (2001), "Arrow's theorem and the Gibbard-Satterthwaite theorem: a unified approach," *Econ. Letters* 70:99–105 (found 2026-05-20)	*adjacent literature* — Geanakoplos for the constraint-regime formulation; Reny for the geometric-topological unification with GS that informs `#disc-implementation-impossibility`'s mechanism-family discussion
Single-peaked preferences as the domain-restriction escape	Black, D. (1948), "On the rationale of group decision-making," *J. Polit. Econ.* 56:23–34 (found pre-2026)	*adjacent literature* — the canonical escape; the sub-scope α' adjacency is named relative to this (shared with `#deriv-strategy-proofness-impossibility`)

ASF concern	*Prior-art language*	*Relationship / Positioning*
Cardinal-preference escapes via weighted utilitarianism	Sen, A. (1970), *Collective Choice and Social Welfare*, Holden-Day, §3 (found pre-2026)	*adjacent literature* — the canonical treatment of cardinal-extension escapes; the `#form-objective-functional` adjacency is named relative to this (unique to Arrow within the cluster)
Borda count (positional method escape)	Borda, J.-C. (1781), "Mémoire sur les élections au scrutin," *Histoire de l'Académie Royale des Sciences* (found pre-2026)	*adjacent literature* — outside AAT machinery; documented as honest scope-mark

Search Log:

- 2026-05-22 (*intuition-only* on the cardinal-preference adjacency): The cardinal-preference escape's connection to `#form-objective-functional` is named here for the first time; the natural correspondence (AAT's real-valued V_{O_t} carries cardinal value content per sub-agent; lifting to a designer-side aggregator is a representational choice) is intuitive but not derived. Targeted future search candidates: utilitarian and weighted-utilitarian social welfare functions (Harsanyi 1955, *J. Polit. Econ.* 63:309–321; Sen 1970 §3 cardinal-comparability spectrum); range voting and ratings-based methods (Smith 2000 and follow-ups); whether the per-sub-agent value functional's cardinal content can be made canonically aggregable without introducing new framework machinery.
- 2026-05-22 (*intuition-only* on the composite-aggregation-mechanism reading): Same composite-revelation-mechanism reading as `#deriv-strategy-proofness-impossibility`. Alternative readings exist (treating F as a Σ-level commitment device; treating sub-agents as factor-coalitions in a richer cooperative game); not pursued here.

Appendix AT

Derivation: Fisher-Local Update Gain

Under the **Fisher-local invariance regime** — smooth log-likelihood admitting non-degenerate local quadratic expansion at the current model parameter, with single-step update at first order in the step size — the natural-gradient Bayesian posterior mean shift has the form $\Delta\theta = K \cdot \tilde{\nabla}$ with **gain operator** $K = (H_M + H_L)^{-1} H_L$ and **scalar collapse** $\eta^* = U_M/(U_M + U_o)$ along the natural-gradient direction in the commuting / shared-eigenbasis case (always in 1-D; under (PI)/Čencov in the natural-gradient direction in higher dimensions). The result derives the form `#emp-update-gain` carries as an empirical claim, places it in the Fisher-local regime as exact, and recovers linear-Gaussian (Kalman) and conjugate-Bayesian instances as cases where the local quadratic expansion is globally exact. Companion at the model-parameter-update layer to `#deriv-fisher-whitened-update-rule`'s edge-update derivation: that segment derives update *direction* under correlated evidence; this segment derives update *magnitude* under the Fisher-local regime. Both share the (PI) parameterization-invariance axiom from `#scope-agent-identity` and Čencov 1982 uniqueness as AAT-internal forcing for the Fisher metric.

The same scalar $\eta^* = U_M/(U_M + U_o)$ falls out of three independent structural derivations — local-Gaussian Laplace expansion, Bregman / KL projection onto the local tangent plane (Pythagorean projection in the Fisher metric), and Cramér-Rao / inverse-Fisher precision-additive composition. The agreement is not coincidence: the Fisher-local regime *is* the regime where prior and likelihood compose as a Pythagorean projection in the Fisher metric. The matrix-valued K is the natural object — its eigenvalues per coordinate are $\eta_i^* = u_{M,i}/(u_{M,i} + u_{o,i})$ — and it serves as the **per-coordinate primitive** for the tensor adaptive tempo of `#def-adaptive-tempo` (scalar tempo recovered in the shared-eigenbasis collapse; tensor form preserves per-direction information the scalar averages away). The downstream consumer at the persistence layer is `#deriv-matrix-persistence-condition`, where cross-dimensional correction makes the matrix-Loewner condition strictly sharper than per-coordinate.

Formal Expression.

Setup — Fisher-local quadratic regime.

Definition (Fisher-local regime, conditions (R1)–(R3))

Let $\theta \in \mathbb{R}^d$ parameterize the agent's model and θ_t be the current point estimate (mode of the prior π_0, or center of expansion). Three regime conditions:

- **(R1) Smooth log-likelihood admitting non-degenerate local quadratic expansion.** $\log \pi_0$ and $\log p(o \mid \cdot)$ are C^3 in a neighborhood of θ_t, with prior precision $H_M := -\nabla^2 \log \pi_0(\theta_t)$ and observed information $H_L := -\nabla^2 \log p(o \mid \theta)\big|_{\theta_t}$ satisfying $H_M + H_L \succ 0$ (joint positive-definiteness; see §"Boundary admissibility" for why this is weaker than $H_M \succ 0$ and $H_L \succ 0$ both).
- **(R2) First-order-in-step-size regime.** The posterior update $\Delta\theta = \theta_{t+1} - \theta_t$ is small enough that $O(\|\Delta\theta\|^3)$ terms in the quadratic expansion are negligible compared to the quadratic terms.

- **(R3) Bayesian-coherent update.** θ_{t+1} is taken to be a coordinate of the posterior $p(\theta \mid o) \propto \pi_0(\theta) \cdot p(o \mid \theta)$ — mean, mode, or natural-parameter (they coincide for Gaussian posteriors, which is what the quadratic expansion produces).

Under (R1)–(R3), the local log-posterior is

$$\log p(\theta \mid o) = \text{const} + s^T(\theta - \theta_t) - \frac{1}{2}(\theta - \theta_t)^T(H_M + H_L)(\theta - \theta_t) + O(\|\theta - \theta_t\|^3), \tag{AT.1}$$

where $s := \nabla_\theta \log p(o \mid \theta)\big|_{\theta_t}$ is the score. The posterior is approximately $\mathcal{N}(\theta^\star, (H_M + H_L)^{-1})$ with mean shift

Derived (posterior-mean-shift)

$$\theta^\star - \theta_t = (H_M + H_L)^{-1} s. \tag{AT.2}$$

The AAT-vocabulary correspondences are

$$U_M := H_M^{-1}, \qquad U_o := H_L^{-1} \tag{AT.3}$$

— model uncertainty is the inverse prior precision, observation uncertainty is the inverse Fisher / inverse observed information (Cramér-Rao floor on what the observation pins down). The matrix forms generalize `#emp-update-gain`'s scalar U_M, U_o in the natural way.

Definition (natural gradient at the current point)

Gain decomposition on the natural-gradient direction.

The **natural gradient** of the log-likelihood at θ_t in the Fisher metric is

$$\tilde{\nabla} \log p(o \mid \theta_t) := \mathcal{I}(\theta_t)^{-1} s. \tag{AT.4}$$

For a single observation in the Fisher-local regime, $\mathcal{I}(\theta_t) = H_L$ up to the observed-vs-expected information distinction (standard under (R1)+regularity); the derivations below use H_L directly as the Fisher.

Derived (gain-decomposition)

Rewriting the posterior mean shift in the natural-gradient direction:

$$\Delta\theta = (H_M + H_L)^{-1} s = (H_M + H_L)^{-1} H_L \cdot (H_L^{-1} s) = K \cdot \tilde{\nabla}, \tag{AT.5}$$

with **gain operator**

$$\boxed{K := (H_M + H_L)^{-1} H_L.} \tag{AT.6}$$

The matrix-valued K is the natural object. The scalar collapse follows in the commuting / shared-eigenbasis case (always in 1-D; under (PI)/Čencov along the natural-gradient direction in higher dimensions): eigenvalues of K are

$$\eta_i^* = \frac{h_{L,i}}{h_{M,i} + h_{L,i}} = \frac{1/u_{o,i}}{1/u_{M,i} + 1/u_{o,i}} = \frac{u_{M,i}}{u_{M,i} + u_{o,i}} \tag{AT.7}$$

per coordinate, collapsing to the scalar

$$\boxed{\eta^* = \frac{U_M}{U_M + U_o}} \tag{AT.8}$$

— the form `#emp-update-gain` carries, derived from the Fisher-local quadratic expansion of prior and likelihood and applied to the natural-gradient direction.

Boundary admissibility — improper-prior and degenerate-likelihood directions.

Derived (admissibility-of-degenerate-curvature)

The gain operator $K = (H_M + H_L)^{-1} H_L$ is well-defined whenever $H_M + H_L \succ 0$ — not necessarily when both H_M and H_L are individually positive-definite. This admits **degenerate-likelihood directions** (observation uninformative along some coordinates, H_L rank-deficient): K's eigenvalue along those directions is 0, $u_{o,i} \to \infty$, $\eta_i^* \to 0$, no update along that direction — *trust the prior where the observation says nothing*. It also admits **improper-prior directions** (degenerate prior, $H_M = 0$ along some coordinate) when H_L pins the direction: $\eta_i^* \to 1$ along uninformative-prior directions where the observation is informative — *trust the observation where the prior says nothing*, the maximum-likelihood limit.

The minimal regime condition is $H_M + H_L \succ 0$, with $U_M/(U_M + U_o)$ defined per-eigendirection by the appropriate limit. This corresponds to standard improper-prior admissibility in Bayesian inference and to standard observation-non-identifiability admissibility in maximum-likelihood estimation.

Recovery of linear-Gaussian and conjugate-Bayesian cases.

Derived (global-exactness-special-cases)

When the quadratic expansion in (R1) holds *globally* — not just locally — the derivation gives η^* without any expansion error. Two canonical cases:

- **Linear-Gaussian (Kalman).** Prior $\mathcal{N}(\theta_t, U_M)$, likelihood $\mathcal{N}(o; H\theta, U_o)$ with linear observation operator H: the log-posterior is exactly quadratic. The scalar Kalman gain $K = U_M H^T (H U_M H^T + U_o)^{-1}$ collapses to $\eta^* = U_M/(U_M + U_o)$ in the scalar case (Kalman 1960; Bishop 2006 §3.3). Exact globally.
- **Conjugate Bayesian (Beta-Bernoulli, Dirichlet-multinomial, Gaussian-Gaussian, exponential-family natural-parameter conjugacy).** Natural-gradient VI is the exact posterior update at step size 1 for conjugate families (Khan-Lin 2017 conjugate-computation variational inference). The Fisher-local expansion is globally exact in natural-parameter coordinates; $\eta^* = U_M/(U_M + U_o)$ holds exactly with U_M, U_o read from the conjugate sufficient-statistic precision counts.

For general smooth models, the natural-gradient invariance theorem of Amari 1998 guarantees that the natural gradient is parameterization-invariant; the Fisher-local quadratic expansion is the order at which the natural gradient and the exact posterior agree to first order in step size.

Epistemic Status.

Conditional. Max attainable: *exact* under (R1)–(R3) + (PI) parameterization-invariance from `#scope-agent-identity`. Under (PI) and Čencov 1982 uniqueness, the Fisher metric is the AAT-internally canonical metric on the statistical manifold (the 4th primary instance of `#disc-additive-coordinate-forcing`); the natural-gradient direction is therefore the AAT-internally canonical reference direction along which η^* reads as $U_M/(U_M + U_o)$. The textbook information-geometry result (Amari 1998 natural-gradient invariance theorem) is imported; the (PI)/Čencov forcing in `#deriv-fisher-whitened-update-rule`'s Path A makes the import AAT-internal.

Linear-Gaussian (Kalman) and conjugate-Bayesian instances are cases where the local quadratic expansion is *globally* exact, so $\eta^* = U_M/(U_M + U_o)$ holds without truncation error.

Outside the Fisher-local regime — heavy-tailed posteriors (no second moment), structurally non-smooth likelihoods, multimodal uncertainty where the local quadratic misrepresents global structure — the quantitative form is *not* derived here. The *qualitative* direction (gain rises with model uncertainty, falls with observation noise) is preserved as *robust qualitative* from `#emp-update-gain`'s broader empirical scope; the first-order form is recovered locally; what need not hold is global quantitative fidelity.

This segment depends on (PI) from `#scope-agent-identity` only. The triple (PI)+(R)+(K) of the Markov-morphism layer (carried by the Fisher-Rao bias-bound machinery; the (R) Riemannian-structure and (K) KL-second-order-matching axioms are stronger) is *not* invoked here. If (R) and (K) are asserted at the scope-level in a future cycle, this segment's status remains conditional on (PI) alone unless explicitly extended.

Discussion.

Why read η^* off the natural-gradient direction. The Bayesian-coherent posterior mean shift $\Delta\theta = (H_M + H_L)^{-1}s$ is a fixed vector; its decomposition into "gain times reference direction" depends on the choice of reference direction. Two natural choices give two faces of the same shift:

- **Natural-gradient direction $\tilde{\nabla} = H_L^{-1}s$.** Coefficient: $\eta^*_{\text{NG}} = H_L/(H_M + H_L) = U_M/(U_M + U_o)$. *AAT-internally canonical* under (PI)/Čencov — the Fisher metric is the unique Markov-invariant metric, so the natural gradient is the unique coordinate-invariant gradient direction (Čencov 1982; extended by Ay-Jost-Lê-Schwachhöfer 2017).
- **Prior-curvature-rescaled direction $H_M^{-1}s$.** Coefficient: $\eta^\star_{\text{prior}} = H_M/(H_M + H_L) = U_o/(U_M + U_o)$. The *complement*; the same algebraic decomposition arises in Kalman literature as the prior weight in the convex combination "posterior = (1-K) prior + K observation."

The two coefficients are duals on the same posterior shift; their product reduces to the posterior covariance $(H_M + H_L)^{-1}$. AAT names the natural-gradient coefficient as the canonical η^* because (PI)/Čencov picks out the natural-gradient direction as the unique coordinate-invariant choice.

Three derivation routes converge. The same scalar $\eta^* = U_M/(U_M + U_o)$ falls out of three independent structural derivations:

Table AT.1

Route	*Lever*	*Surface meaning of η^**
Local-Gaussian Laplace expansion (§Setup–Gain decomposition above)	Algebraic completion of squares on additive log-posterior quadratics	Posterior mean shift coefficient on the natural-gradient direction
Bregman / KL projection onto the local tangent plane	First-order condition on the variational free energy along natural-gradient tilt	Projection coefficient of the posterior onto the local tangent plane (Pythagorean projection in Fisher metric per Amari-Nagaoka 2000 §3.2)
Cramér-Rao / inverse-Fisher	Precision-additive composition $1/U_M + 1/U_o$, inverted	Fractional observation contribution to posterior precision

The agreement is not coincidence: the Fisher-local regime *is* the regime where prior and likelihood compose as a Pythagorean projection in the Fisher metric — the same exponential-family / Bregman / Pythagorean structure underlying `#deriv-strategy-cost-regret-bound`'s ρ-decomposition. The three routes are three faces of one geometric object: precision-additive composition of two Gaussian information sources, read off the natural-gradient direction.

Sibling positioning vs `#deriv-fisher-whitened-update-rule`. That segment lives at the **edge-update layer** of Part II's strategy DAG: it derives the Fisher-whitening correction for the *direction* of edge updates under correlated evidence (L1'/L2). This segment lives at the **model-parameter-update layer** of Part I: it derives the *magnitude* of the natural-gradient Bayesian update via the Fisher-local invariance regime. Both depend on (PI) + Čencov as the AAT-internal axiom forcing the Fisher metric. Together they make the Fisher-local invariance regime AAT-internally complete: direction (whitening) at the edge layer and magnitude (gain) at the model-parameter layer are both derived from the same axiomatic chain.

Special case of `#deriv-adaptive-gain-dynamics`' meta-gain framework. The gain operator $K = (H_M + H_L)^{-1} H_L$ is a *deterministic function of the current state* — it depends on the prior covariance H_M^{-1} (part of the agent's M_t state) and on the observation Fisher H_L (depends on θ_t and o). It is therefore the **deterministic-meta-gain** special case of `#deriv-adaptive-gain-dynamics`' (MG-1)–(MG-4) machinery: the meta-gain is determined by the primary state rather than independently learned. All four meta-gain conditions are satisfied trivially (symmetric-positive-definite K on the interior; smoothness for exponential families; bounded primary-meta coupling). Resolving the epistemic-opacity question that `#def-observation-function` axiomatizes — i.e., the agent does not know U_o a priori and must estimate it from its own mismatch sequence — is one level up: it adds the Mehra-style meta-channel that `#deriv-adaptive-gain-dynamics` treats as its primary instance.

Downstream tempo and persistence machinery. `#emp-update-gain`'s Discussion §"Connection to adaptive tempo" names $\mathcal{T} = \nu \cdot \eta^*$. The derivation here promotes η^*'s tier from `empirical / robust-qualitative` (over the cross-domain validity tail — RL, PID, software-developer) to `derived (conditional on Fisher-local regime)` for the Kalman / conjugate / natural-gradient core. The tempo product $\mathcal{T} = \nu \cdot \eta^*$ inherits the regime distinction directly; the persistence-condition machinery (`#result-persistence-condition`, `#result-sector-condition-stability`) and the adaptive-tempo / aporia diagnostics (`#def-adaptive-tempo`, `#def-aporia`) all gain the tighter η^* exactness statement in their Fisher-local instances. The qualitative direction claim and the failure-mode framing (gain-collapse → epistrophe failure) survive unchanged outside the Fisher-local regime.

Tensor adaptive tempo connection. The matrix gain operator K is the per-coordinate primitive for tensor-valued adaptive tempo: $\mathcal{T} = \nu \cdot K$ as a matrix product, with the existing scalar $\mathcal{T} = \nu \cdot \eta^*$ recovered in the shared-eigenbasis limit. See `#def-adaptive-tempo`'s Tensor extension.

Appendix AU

Derivation: Scope of Tempo Additivity — Redundancy, Synergy, and the Echo-Chamber Bound

The additive tempo formula $\mathcal{T} = \sum_k \nu^{(k)}\eta^{(k)*}$ (`#def-adaptive-tempo`) is exact when the observation channels carry cross-channel-independent noise, and its deviation from the true joint capacity under correlated noise is **signed**: correlated channels can deliver *less* joint information than the additive accounting claims (redundancy — the additive form overcounts) or *more* (synergy — correlated noise can be cancelled, and the additive form undercounts). This segment derives the exact deviation in the Fisher-local regime, proves the additive form is an exact upper bound with an explicit, closed-form redundancy penalty in the **common-source noise regime** (the "echo chamber": a shared persistent bias contaminating several channels), proves an information-**saturation** theorem for persistent shared bias (no number of correlated channels buys information past the shared-bias floor), and establishes two no-gos: no sign-blind dependence measure (conditional mutual information, total correlation) can be the exact correction term, and no convention-free channel-local attribution of the deviation exists for three or more channels (the partial-information-decomposition obstruction). The joint tempo and the *total* deviation are canonical; per-channel corrected contributions are not.

Formal Expression.

Setup — the information reading of tempo.

Definition (effective tempo, Fisher-local regime)

Work in the Fisher-local / Gaussian location regime of AT: environment parameter θ, model uncertainty U_M (posterior variance), channels $k = 1, \dots, n$ with independent Poisson event rates $\nu^{(k)}$ and per-event observations $o^{(k)} = \theta + n^{(k)}$, noise variance $U_o^{(k)} = \mathrm{Var}(n^{(k)})$. Let $\mathfrak{J}$ denote the posterior-precision inflow rate delivered by the *optimal joint filter* (Fisher information about θ acquired per unit time). The **effective tempo** is the fractional uncertainty-contraction rate

$$\mathcal{T}_{\text{eff}} := -\frac{d}{dt}\log U_M(t) = U_M \cdot \mathfrak{J}, \tag{AU.1}$$

the second equality since $\frac{d}{dt}(1/U_M) = \mathfrak{J}$. Single-channel consistency: $\mathfrak{J} = \nu/U_o$ gives $\mathcal{T}_{\text{eff}} = \nu\, U_M/U_o$, which agrees with $\nu\eta^* = \nu\, U_M/(U_M + U_o)$ to first order in the per-event information ratio U_M/U_o — the same first-order-in-step-size regime (R2) under which AT operates and under which tempo is a rate at all. All statements below are exact at the Fisher-information level and transfer to gain-form tempo at that order.

Derived (additivity-under-independence)

Additivity under noise independence.

If the channel noises are independent across channels and events (conditional independence of the pooled event stream given θ), scores add and score cross-covariances vanish, so Fisher information is exactly additive:

$$\mathfrak{J}_{\text{joint}} = \sum_k \nu^{(k)} \frac{1}{U_o^{(k)}} = \mathfrak{J}_{\text{add}} \quad \Longrightarrow \quad \mathcal{J}_{\text{eff}} = \sum_k \nu^{(k)} \eta^{(k)*} \text{ (first order).} \tag{AU.2}$$

Independence is **sufficient but not necessary** for equality (see the harmonic-mean hypersurface below).

Derived (asynchrony refinement)

Independent Poisson streams almost surely have no simultaneous events. If noise is *fresh per event* (i.i.d. across events), instantaneous cross-channel noise correlation is never sampled and additivity remains exact regardless of it. What breaks additivity is noise dependence **across events** — a persistent shared noise component. The redundancy question is governed by the ratio of the noise-correlation timescale to event interarrival times.

Derived (signed-deviation)

The exact deviation, and its indefinite sign.

For a synchronized batch (one event per channel, jointly Gaussian noise Σ_n, $(\Sigma_n)_{kk} = U_o^{(k)}$), the joint Fisher information and the deviation from additivity are

$$J_{\text{joint}} = \mathbf{1}^\top \Sigma_n^{-1} \mathbf{1}, \qquad \Delta := \sum_k \frac{1}{U_o^{(k)}} - J_{\text{joint}}, \tag{AU.3}$$

and for two channels with noise covariance c (writing U_1, U_2 for the variances):

$$J_{\text{joint}} = \frac{U_1 + U_2 - 2c}{U_1 U_2 - c^2}, \qquad \Delta = \frac{c\left(2U_1 U_2 - c\,(U_1 + U_2)\right)}{U_1 U_2\,(U_1 U_2 - c^2)}. \tag{AU.4}$$

$\Delta = 0$ iff $c = 0$ or $c = 2U_1U_2/(U_1 + U_2)$ (the harmonic mean of the variances — admissible whenever $U_1 \neq U_2$): equality holds on a strictly larger set than independence, with redundancy and synergy cancelling exactly. Δ has **no definite sign**:

- **Anti-correlated noise (synergy).** $U_1 = U_2 = \sigma^2, \rho < 0$: $J_{\text{joint}} = 2/(\sigma^2(1+\rho)) > 2/\sigma^2$. Averaging cancels the noise; the additive form *under*counts.
- **Positively correlated, heterogeneous variances (synergy).** $U_1 = 1, U_2 = 100, c = 9.9$: $J_{\text{joint}} = 81.2/1.99 \approx 40.8 \gg 1.01$. The noisy channel serves as a noise *reference*; the combination $o^{(1)} - (c/U_2)\,o^{(2)}$ nearly cancels channel 1's noise.
- **Nonnegative equal-variance correlation, $n = 3$ (synergy).** Correlations $\rho_{12} = \rho_{23} = 0.7, \rho_{13} = 0$ (positive-definite): $\mathbf{1}^\top R^{-1} \mathbf{1} = 10 > 3$.

So "correlated channels ⇒ the additive formula overcounts" is false as a general claim; overcounting is the *redundant* half of a two-sided phenomenon whose other half is noise-cancelling synergy.

The echo-chamber regime — redundancy penalty and saturation.

Derived (common-source-redundancy-penalty)

Common-source noise model: $n^{(k)} = s + w^{(k)}$ with shared bias $s \sim \mathcal{N}(0, \sigma_s^2)$ and independent $w^{(k)} \sim \mathcal{N}(0, \sigma_{w,k}^2)$, so $\Sigma_n = \sigma_s^2 \mathbf{1}\mathbf{1}^\top + \mathrm{diag}(\sigma_{w,k}^2)$. By Sherman-Morrison, with $q_k := 1/\sigma_{w,k}^2$, $q := \sum_k q_k$, and $f(x) := x/(1 + \sigma_s^2 x)$:

$$J_{\text{joint}} = f(q) \;\le\; \sum_k f(q_k) = \sum_k \frac{1}{\sigma_s^2 + \sigma_{w,k}^2} = \sum_k J^{(k)}, \tag{AU.5}$$

by strict concavity of f with $f(0) = 0$ (strict subadditivity), with equality iff $\sigma_s^2 = 0$ or at most one channel is active. The **redundancy penalty** is explicit and closed-form: $R_s = \sum_k f(q_k) - f(q) > 0$. In this regime the additive tempo *is* an exact upper bound on effective tempo, for arbitrarily heterogeneous channels.

Derived (saturation-under-persistent-bias)

If the shared bias s is **persistent across events**, every observation measures $\theta + s$ plus fresh noise: the pooled data identify $\theta + s$ at additive rate, but the posterior precision about θ itself is bounded for all time by $1/U_M^{(0)} + 1/\sigma_s^2$, since $\mathrm{Var}(\theta \mid \theta + s) = (1/U_M^{(0)} + 1/\sigma_s^2)^{-1}$ and conditioning on the data is coarser than conditioning on $\theta + s$. Additive accounting claims precision growing without bound; the true joint information **saturates** at the shared-bias floor. No number of common-source channels, at any event rate, buys information past $1/\sigma_s^2$.

Two no-gos.

Derived (no-sign-blind-penalty)

No exact correction of the form $\mathcal{T}_{\text{eff}} = \mathcal{T}_{\text{add}} - g(D)$ exists for any dependence measure D invariant under noise-correlation sign flip — conditional mutual information $I(e^{(1)}; e^{(2)} \mid M_{\tau-})$, total correlation (Watanabe 1960), or any such functional. Proof: ρ and $-\rho$ in the equal-variance two-channel family give identical Gaussian conditional mutual information $-\frac{1}{2}\log(1 - \rho^2)$ but different joint informations $2/(\sigma^2(1 \pm \rho))$. The deviation depends on the dependence *geometry*, not the dependence *strength*; the exact correction is the Fisher deficit Δ, which requires Σ_n, not a scalar dependence summary. Conditional mutual information survives as a **witness**: in the Gaussian location model $\Delta \neq 0 \Rightarrow I > 0$ (converse false on the harmonic-mean hypersurface), and in the common-source regime both I and R_s increase with σ_s^2.

Derived (no-canonical-channel-local-decomposition — imported)

A channel-local, convention-free additive decomposition of the joint information rate into per-channel redundancy-corrected contributions is the demand of the **partial information decomposition** program, and it is known not to exist in general: Williams & Beer 2010 define the redundancy lattice; Bertschinger, Rauh, Olbrich, Jost & Ay 2014 give the two-source unique-information measure; Rauh, Bertschinger, Olbrich & Jost 2014 prove no nonnegative decomposition on the Williams-Beer lattice satisfying the identity axiom exists for three or more sources; Barrett 2015 shows the Gaussian collapse of the main two-source candidates. Consequence: $\mathcal{T}_{\text{joint}}$ and the total deviation Δ are canonical and exact; "channel k's corrected tempo contribution" is not a well-defined quantity for $n \geq 3$, and any AAT use of one must declare its attribution convention explicitly.

Epistemic Status.

Conditional. The derivations are exact within the Fisher-local / Gaussian location regime of AT (conditions (R1)–(R3)) plus the stated noise models (jointly Gaussian batch noise for the deviation formula; the common-source structure for the penalty and saturation theorems); the transfer from Fisher-information statements to gain-form tempo is first-order in the per-event information ratio, the same regime in which tempo is a rate. Additivity-under-independence is general (standard Fisher additivity, not merely Gaussian); the counterexamples are exact and finite, computed in-segment. The two-source PID landscape and the $n \geq 3$ impossibility are imported results, cited by their real names; the mapping of the PID obstruction onto channel-local tempo attribution is AAT's reading and is stated at that grade. Citation details for the imported PID results are flagged for primary-source verification before promotion past `draft` (see Working Notes). This segment supplies the derivation behind the channel-independence condition named in `#def-adaptive-tempo`'s Epistemic Status; it *refutes* the previously asserted general upper bound (the deviation is signed) and *establishes*

the bound on its true scope (common-source regime), so downstream uses of "additive tempo is conservative" are unsafe unless the common-source (or general nonpositive-Δ) structure is verified for the application.

Discussion.

What the correction to the prior assertion changes. The earlier claim — additive tempo is always an upper bound, equality iff channels informationally independent — read channel correlation as pure overlap. The derivation shows correlation is two-sided: overlapping *information* (common source) makes the additive form overcount, but correlated *noise* is also a resource — a correlated reference channel lets the filter cancel noise it could never average away, and then the additive form undercounts, potentially by large factors. Both directions matter for agent design: the redundant direction is the echo-chamber warning (diversity of observation *sources*, not count of observers), and the synergistic direction says a channel that is nearly useless in isolation (huge U_o, negligible η^*) can be extremely valuable jointly — a reason to retain "bad" sensors that are correlated with good ones' failure modes.

The echo chamber, quantitatively. The common-source saturation theorem is the formal content of the organizational intuition: three VPs reading one broken dashboard are one channel with extra steps, and past the shared-bias floor their combined reports carry zero marginal information about the world. A filter that *models* the channels as independent additionally manufactures false confidence — its internal U_M shrinks at the additive rate while true error stalls at the bias floor, driving $\eta^* \to 0$ while genuinely wrong: the gain-collapse pathology of `#emp-update-gain` arrived at through mis-modeled channel structure rather than miscalibrated uncertainty *[Discussion]*.

Why no simple penalty formula exists. The no-gos localize the obstruction precisely. A scalar dependence summary cannot carry the correction (sign-blindness), and a per-channel attribution cannot be made canonical for $n \geq 3$ (PID impossibility). What *is* available exactly: the joint object $\mathbf{1}^\top \Sigma_n^{-1} \mathbf{1}$ (tensor case: $A^\top \Sigma_n^{-1} A$) whenever the noise covariance structure is known or estimable, and the closed-form penalty on the common-source scope. This is a scope-precision result in the CS sense: the boundary of the additive form is now characterized rather than gestured at.

Anisotropy is orthogonal. With multivariate θ and channel maps A_k, cross-channel noise independence gives exact *matrix* additivity $\mathcal{J} = \sum_k \nu^{(k)} A_k^\top (\Sigma_n^{(k)})^{-1} A_k$ — channels "measuring different aspects" is handled by the tensor form of `#def-adaptive-tempo`, not by redundancy corrections. Redundancy/synergy is an off-block-diagonal-noise phenomenon; anisotropy is an eigenbasis/observation-map phenomenon; the common-source subadditivity holds in the Loewner order by the same Sherman-Morrison-Woodbury argument.

Composition consequence. `#der-team-persistence` inherits both directions: allied reports drawing on one upstream source are the common-source regime (additive communication tempo overcounts, saturating at the source's bias floor), while allies with anti-correlated errors triangulate — a genuinely super-additive composition effect the additive accounting misses in the favorable direction.

Appendix AV

Derivation: Matrix-Loewner Persistence Condition

Under matrix adaptive tempo $\mathcal{T}$ from `#def-adaptive-tempo`'s Tensor extension (per-channel primitive $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$ from `#deriv-fisher-local-update-gain`) and matrix disturbance covariance Σ_w, the operational persistence condition lifts from scalar to matrix-Loewner form: the agent persists iff $\mathcal{T}$ is Hurwitz (structural persistence) and the stationary covariance Σ_∞ — solving the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty \mathcal{T}^T = \Sigma_w$ — is strictly Loewner-dominated by the diagonal of squared per-direction critical thresholds $D_\delta := \mathrm{diag}(\delta^2_{\mathrm{critical},k})$. The matrix-Loewner condition $\Sigma_\infty \prec D_\delta$ recovers `#result-persistence-condition`'s scalar Model S form as the isotropic special case and `#result-per-dimension-persistence`'s per-coordinate form as the diagonal special case. Crucially, when $\mathcal{T}$ has off-diagonal entries that misalign with the coordinate basis — the generic situation under cross-dimensional correction — the per-coordinate condition is *unsafe*: it can declare persistence on systems whose stationary mismatch will exceed task-adequacy along the diagonal direction. The matrix-Loewner form is the canonical anisotropic persistence condition; per-coordinate is its diagonal-$\mathcal{T}$-axis-aligned-D_δ special case, sharp where the coordinate basis happens to be the correction-machinery's eigenbasis and unsafe outside it.

The framework's posture is sharp: **scalar and per-coordinate forms are special cases; the matrix form is the load-bearing object the others project from.** A constructive 2D counterexample (symmetric $\mathcal{T}$ with cross-coordinate correction, isotropic disturbance and threshold) makes the unsafety concrete — per-coordinate evaluates the wrong directions, returning PASS where matrix-Loewner correctly returns FAIL with the failing direction running 45° off-axis. The result composes with the per-direction information-rate floor: no Shannon-type aggregation across dimensions compensates for one direction being capacity-starved (*bandwidth-per-direction is the binding quantity, not total bandwidth*). The cleanest current application is multi-modal AI pipelines with asymmetric per-modality channel budgets, where aggregate-rate metrics systematically miss directional failures the matrix-Loewner reading catches.

Formal Expression.

Definition (matrix-Model-S-mismatch-dynamics)

Setup — Model S with matrix correction.

Consider linear stochastic mismatch dynamics

$$d\delta_t \;=\; -\mathcal{T}\,\delta_t\,dt \;+\; \Sigma_w^{1/2}\,dW_t \tag{AV.1}$$

with $\delta_t \in \mathbb{R}^d$, matrix adaptive tempo $\mathcal{T} \in \mathbb{R}^{d\times d}$, disturbance covariance $\Sigma_w \succ 0$, and standard d-dimensional Brownian W_t. Per `#def-adaptive-tempo`'s Tensor extension, $\mathcal{T} = \sum_k \nu^{(k)} K^{(k)}$ aggregates the per-channel matrix gain operators from `#deriv-fisher-local-update-gain`.

Derived (stationary-covariance, exact under Hurwitz $\mathcal{J}$ and $\Sigma_w \succ 0$)

Stationary covariance — continuous Lyapunov equation.

The stationary distribution of the linear stochastic dynamics is Gaussian $\mathcal{N}(0, \Sigma_\infty)$, with Σ_∞ the unique positive-definite solution of the **continuous Lyapunov equation**

$$\boxed{\mathcal{J}\,\Sigma_\infty \;+\; \Sigma_\infty\,\mathcal{J}^T \;=\; \Sigma_w.} \tag{AV.2}$$

Existence and uniqueness of $\Sigma_\infty \succ 0$ are equivalent to $\mathcal{J}$ being **Hurwitz** — all eigenvalues having strictly positive real part. The closed-form integral representation is

$$\Sigma_\infty \;=\; \int_0^\infty e^{-\mathcal{J}t}\,\Sigma_w\,e^{-\mathcal{J}^T t}\,dt. \tag{AV.3}$$

When $\mathcal{J}$ is symmetric and commutes with Σ_w, the solution simplifies to $\Sigma_\infty = \frac{1}{2}\mathcal{J}^{-1}\Sigma_w$. Otherwise the integral form (or any standard Lyapunov-equation solver) gives the stationary covariance.

Derived (matrix-persistence-loewner-Model-S)

The matrix-Loewner persistence condition.

The agent persists operationally iff:

(MP-1) Structural persistence. $\mathcal{J}$ is Hurwitz — equivalently, Σ_∞ exists as the unique positive-definite solution of the Lyapunov equation. This is the matrix analog of `#result-persistence-condition`'s structural condition $\alpha > 0$ (Hurwitz reduces to $\mathcal{J}_0 > 0$ in the scalar case).

(MP-2) Matrix-Loewner task adequacy.

$$\boxed{\Sigma_\infty \;\prec\; D_\delta \;:=\; \mathrm{diag}\big(\delta^2_{\mathrm{critical},k}\big)} \tag{AV.4}$$

with $\prec$ the strict Loewner (positive-definite) order: $D_\delta - \Sigma_\infty \succ 0$. The stationary mismatch ellipsoid is contained strictly inside the per-direction critical-threshold ellipsoid.

The two conditions are independent in the same sense as `#result-persistence-condition`'s structural-vs-task-adequacy split: (MP-1) is a property of the correction machinery alone; (MP-2) adds the domain-specific bound on per-direction tolerable mismatch.

Three equivalent restatements. The Loewner condition (MP-2) admits three equivalent forms:

(MP-2a) Per-direction. For every unit direction $\hat{v} \in \mathbb{R}^d$:

$$\hat{v}^T\Sigma_\infty\hat{v} \;<\; \hat{v}^T D_\delta \hat{v} \;=\; \sum_k v_k^2\,\delta^2_{\mathrm{critical},k}. \tag{AV.5}$$

The projected stationary variance along $\hat{v}$ must be less than the direction-projected squared threshold along the same direction.

(MP-2b) Spectral. The generalized eigenvalue problem $\Sigma_\infty x = \lambda D_\delta x$ satisfies $\lambda_{\max} < 1$, equivalently

$$\lambda_{\max}\big(D_\delta^{-1/2}\Sigma_\infty D_\delta^{-1/2}\big) \;<\; 1. \tag{AV.6}$$

The principal eigenvector identifies the *worst direction*; the eigenvalue's distance from 1 is the worst-direction safety margin.

(MP-2c) Ellipsoid containment. The stationary noise ellipsoid $\{x : x^T\Sigma_\infty^{-1}x \le 1\}$ is contained strictly inside the threshold ellipsoid $\{x : x^T D_\delta^{-1}x \le 1\}$.

The three are mathematically equivalent under standard matrix-analysis (Horn & Johnson 2013, §4.6 and §7). (MP-2a) is cleanest for hand reasoning, (MP-2b) for computation (single generalized-eigenvalue solve), (MP-2c) for geometric intuition.

Recovery of existing AAT persistence forms.

Table AV.1

Special case	*(MP-2) reduces to*	*Matches*
Isotropic ($\mathcal{T} = \mathcal{T}_0 I, \Sigma_w = \sigma_w^2 I$, $\delta_{\text{critical}} = \delta_0 \mathbf{1}$)	$\sigma_w^2/(2\mathcal{T}_0) < \delta_0^2$, i.e., $\mathcal{T}_0 > \sigma_w^2/(2\delta_0^2)$	Scalar Model S form, `#result-persistence-condition`
Diagonal $\mathcal{T}, \Sigma_w$ in the coordinate basis of D_δ	$\sigma_{w,k}^2/(2\mathcal{T}_{kk}) < \delta_{\text{critical},k}^2$ per coordinate k	Per-coordinate Model S RMS, `#result-per-dimension-persistence`
Symmetric $\mathcal{T}$ commuting with Σ_w, general D_δ	Per-eigendirection inequality in $\mathcal{T}$'s eigenbasis, with direction-projected thresholds	New content recovered from the matrix form — extends per-coordinate to non-axis-aligned $\mathcal{T}$

The third row is the content `#result-per-dimension-persistence` does not currently carry: when $\mathcal{T}$ and Σ_w share an eigenbasis that is *not* the coordinate basis of D_δ, the per-coordinate form misses the right direction-projected thresholds and the matrix-Loewner form handles them correctly.

Where per-coordinate is unsafe — a constructive counterexample. The deeper structural question: is the matrix-Loewner form merely a generalization that agrees with per-coordinate in all real cases, or is it *strictly sharper* — does there exist a regime where per-coordinate says PASS but matrix-Loewner correctly says FAIL? The construction below establishes the latter: the per-coordinate form is **unsafe** under cross-dimensional correction.

Construction. Take $d = 2, \Sigma_w = I$, and

$$\mathcal{T} = \begin{pmatrix} 1 & -0.9 \\ -0.9 & 1 \end{pmatrix}. \tag{AV.7}$$

This $\mathcal{T}$ is symmetric positive-definite with eigenvalues 1.9 (along $(1,-1)/\sqrt{2}$) and 0.1 (along $(1,1)/\sqrt{2}$) — strongly anisotropic correction, with the $(1,1)$ direction the weak axis. Per-coordinate naive reading of "diagonal of $\mathcal{T}$" sees uniform $\mathcal{T}_{11} = \mathcal{T}_{22} = 1$ and concludes correction is uniform at rate 1 on each coordinate.

Stationary covariance. Since $\mathcal{T}$ is symmetric, $\Sigma_\infty = \frac{1}{2}\mathcal{T}^{-1}\Sigma_w = \frac{1}{2}\mathcal{T}^{-1}$. Direct computation:

$$\Sigma_\infty = \frac{1}{0.38}\begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix} \approx \begin{pmatrix} 2.63 & 2.37 \\ 2.37 & 2.63 \end{pmatrix}, \tag{AV.8}$$

with eigenvalues 5.00 along $(1,1)/\sqrt{2}$ and 0.26 along $(1,-1)/\sqrt{2}$.

Set $\delta_{\text{critical}} = (1.7, 1.7)$, so $D_\delta = 2.89\, I$.

- *Per-coordinate check (*`#result-per-dimension-persistence` *form):* per-coordinate stationary variance is $\Sigma_{\infty,kk} = 2.63 < 2.89 = \delta^2_{\text{critical},k}$ for each k. **Per-coordinate says PASS.** ✓
- *Matrix-Loewner check (MP-2 here):* worst-direction projected variance is $\lambda_{\max}(\Sigma_\infty) = 5.00$ along $(1,1)/\sqrt{2}$. Direction-projected squared threshold along the same direction is $\hat{v}^T D_\delta \hat{v} = 2.89$. The condition $5.00 < 2.89$ fails. Equivalently, $\lambda_{\max}(D_\delta^{-1/2}\Sigma_\infty D_\delta^{-1/2}) = 5.00/2.89 \approx 1.73 > 1$. **Matrix-Loewner says FAIL. ✗**

Conclusion. Per-coordinate gives a false-pass. The agent will routinely produce mismatch vectors with RMS magnitude $\sqrt{5.00} = 2.24$ along the $(1,1)/\sqrt{2}$ direction, exceeding the direction-projected threshold of $\sqrt{2.89} = 1.70$ on that direction. The per-coordinate analysis missed the diagonal direction because each coordinate alone stays within $\delta_{\text{critical},k} = 1.7$ — the failure is *only* visible at the matrix-Loewner level. Cross-dimensional correction (the off-diagonal entries of $\mathcal{T}$) is what creates the diagonal-direction concentration that per-coordinate cannot detect.

Epistemic Status.

Conditional. Max attainable: *exact* under (i) Model S linear stochastic dynamics, (ii) Hurwitz matrix tempo $\mathcal{T}$, (iii) positive-definite disturbance covariance Σ_w, (iv) diagonal critical-threshold structure $D_\delta = \text{diag}(\delta^2_{\text{critical},k})$. The Lyapunov equation, the existence-iff-Hurwitz fact, and the matrix-Loewner equivalence forms are all standard from Bellman 1960 / Khalil 2002 / Horn-Johnson 2013; the AAT-distinctive content is the application to the persistence condition and the recognition that per-coordinate is strictly sharper-or-unsafe rather than uniformly sharper.

What is load-bearing:

- The matrix-Loewner form as the canonical anisotropic persistence condition, with scalar and per-coordinate forms as special cases recovered exactly.
- The strict-sharpness result via the §4 counterexample: per-coordinate gives a false-pass when $\mathcal{T}$ has off-diagonal entries that misalign with the coordinate basis. The per-coordinate form is not merely incomplete in this regime — it is unsafe.
- The three equivalent restatements (per-direction, spectral, ellipsoid-containment) with their different operational readings.
- The structural-vs-task-adequacy decomposition from `#result-persistence-condition` transfers directly: (MP-1) is the matrix structural condition (Hurwitz); (MP-2) is the matrix task adequacy (Loewner $\Sigma_\infty \prec D_\delta$). The two conditions remain independent — Hurwitz $\mathcal{T}$ does not imply $\Sigma_\infty \prec D_\delta$ for any particular D_δ, and conversely.

What is not established here:

- The nonlinear matrix-sector extension. Under sector-bounded $F(\mathcal{T}, \delta)$ with $\delta^T F \succeq \alpha I$ in matrix-Loewner form, the matrix-Lyapunov machinery extends to give an ultimate-bound matrix; the linear case here is the template instantiation. Full matrix-sector treatment is follow-on work in `#deriv-sector-condition`.
- The Model D matrix lift. Under deterministic bounded disturbance $\|w_t\|_M \leq \rho$ in a quadratic norm, the ultimate-bound becomes an ellipsoid containment problem (a linear matrix inequality). Mechanical lift; not derived here.
- The non-Hurwitz boundary behaviour (eigenvalue exactly on the imaginary axis). Outside Hurwitz, the linear analysis fails — no stationary distribution. This matches the scalar case's $\alpha > 0$ requirement.
- The promotion of `#result-adversarial-tempo-advantage` to matrix form. The matrix-Loewner form sharpens the adversarial result via worst-direction targeting (an adversary controlling Σ_w^{adv} maximizes $\lambda_{\max}(D_\delta^{-1/2}\Sigma_w^{\text{adv}}\mathcal{T}_{\text{eff}}^{-1}D_\delta^{-1/2})$), but the full adversarial promotion is a separate cycle.

- The matrix extension of `#deriv-persistence-cost`'s information-rate floor. The natural lift is $\dot{R} \geq \frac{1}{2}\mathrm{Tr}(\mathcal{T})$ — per-eigendirection application of the scalar derivation. Worth a cross-reference Working Note in that segment if it becomes load-bearing; not derived here.

Discussion.

Why the Loewner form is the natural matrix lift. The Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty \mathcal{T}^T = \Sigma_w$ is itself a matrix-Loewner statement: it says Σ_∞ balances correction (via $\mathcal{T}$) against disturbance (via Σ_w) in the PSD ordering. The task-adequacy condition $\Sigma_\infty \prec D_\delta$ reads this balance against the threshold ellipsoid. The whole structure lives in the cone of positive-semidefinite matrices ordered by Loewner dominance; scalar persistence is its 1D shadow, per-coordinate persistence is its axis-aligned-diagonal shadow, and the matrix form is the load-bearing object the shadows project from.

The structural-vs-task-adequacy decomposition transfers. `#result-persistence-condition`'s split — structural persistence (machinery works) vs task adequacy (machinery works well enough) — survives intact in the matrix lift: Hurwitz $\mathcal{T}$ is the matrix structural condition; Loewner $\Sigma_\infty \prec D_\delta$ is the matrix task-adequacy condition. The two are independent; an agent can have Hurwitz $\mathcal{T}$ (correction works in every direction) but fail Loewner on the worst direction (correction is not fast enough on the weak axis for the per-direction threshold). The remedies are different: structural failure requires changing the correction architecture (`#result-structural-adaptation-necessity`); task-adequacy failure requires either increasing $\mathcal{T}$ on the weak direction, decreasing Σ_w on the weak direction, or relaxing the threshold along that direction.

The weak-direction-bottleneck argument transfers and sharpens. `#result-per-dimension-persistence`'s key insight — the weak dimension is the bottleneck — generalizes to the weak *direction* in the matrix lift: the worst eigenvector of Σ_∞ relative to D_δ is what gates persistence. When this eigenvector aligns with a coordinate axis, the per-coordinate analysis catches it. When it points off-axis — the generic situation under cross-dimensional correction — the per-coordinate analysis misses it. The §4 counterexample is the canonical illustration: the worst direction is $(1, 1)/\sqrt{2}$, off-axis from both coordinate directions; per-coordinate evaluates the wrong directions; matrix-Loewner evaluates the right ones.

Cross-coordinate correction is the generic case under Fisher-local invariance. The matrix gain operator $K = (H_M + H_L)^{-1} H_L$ from `#deriv-fisher-local-update-gain` is diagonal in the coordinate basis only when prior precision H_M and observation Fisher H_L share an eigenbasis aligned with the coordinate axes. In general — and certainly in the natural-gradient regime where the agent's parameterization is statistical-manifold-natural rather than user-chosen — they do not, and K has off-diagonal entries. The matrix-Loewner condition becomes the canonical form for any Fisher-local-derived persistence analysis; the per-coordinate form is the exceptional case where prior, likelihood, and threshold all happen to share the coordinate axes.

Cross-reference to existing AAT machinery. The matrix-Loewner form composes with:

- `#result-sector-persistence-template` (T1)–(T3) as the linear-case template instantiation; the matrix sector inequality $\delta^T F + F^T \delta \succeq 2\alpha I$ would extend the matrix-Lyapunov result to nonlinear F via the template's (T2) condition lifted to matrix form.
- `#deriv-sector-condition` Prop A.1S (the Lyapunov-stochastic derivation underlying `#result-persistence-condition`'s scalar form) — the matrix derivation here is the analog at the matrix-Lyapunov level.
- `#deriv-fisher-local-update-gain` (matrix gain K as the per-channel primitive) and `#def-adaptive-tempo` (Tensor extension aggregating $\mathcal{T} = \sum_k \nu^{(k)} K^{(k)}$).

On the scope tag downstream. Until the broader promotion of `#result-persistence-condition`, `#result-adversarial-tempo-advantage`, `#form-composition-closure`, `#der-team-persistence`, `#deriv-critical-mass-composition` to invoke the matrix form directly, those segments continue to read scalar and inherit the "scalar / isotropic / nonredundant-channel

scope" tag from `#def-adaptive-tempo`'s Epistemic Status. The per-direction primitive is in place; the downstream promotion is a follow-on cycle in `TODO.md`.

Appendix AW

Derivation: The Survival-Imperative Exploration Drive

The unified policy objective in `#disc-ciy-unified-objective` combines pragmatic value Q_O with an epistemic exploration term (Causal Information Yield, CIY) weighted by $\lambda_{\text{info}} \propto U_M$. The logic is intuitive: explore when uncertain.

However, AAT's underlying Lyapunov dynamics (`#result-persistence-condition`) force a *second, structurally distinct* exploration drive: the **Survival Imperative**. This segment derives that an agent must actively seek low-noise (high CIY) observations not just to learn, but to physically survive environmental drift when its model is *confident*.

AAT therefore carries **two parallel exploration drives at opposite ends of the uncertainty spectrum**: the standard epistemic drive scaling as $\lambda_{\text{info}} \propto U_M$ (explore when uncertain — dominant when model uncertainty is high), and the survival drive scaling as $\lambda_{\text{surv}} \propto 1/U_M$ (explore when confident in a drifting world — dominant when model uncertainty is low). The result *bypasses the dark-room critique of active inference* structurally rather than parametrically: an AAT agent in a drifting world physically cannot stop exploring because exploration is forced by the same Lyapunov machinery that grounds the rest of Part II, not by a tunable preference parameter. The operational prescription — *confident agents are not safe agents in drifting environments* — is the formal grounding for the "productive paranoia" heuristic about not resting on past success. The scalar form here admits a "blank-wall" attack resolved in AX by lifting to a Linear Matrix Inequality on the Fisher Information Matrix with a positive-semidefinite matrix Lagrange multiplier; the dual-drive picture survives the lift in matrix form.

The Persistence-Bounded Objective.

The agent's primary goal is to maximize pragmatic value:

$$\max_{\pi} \mathbb{E}_{a\sim\pi}[Q_O(a)] \tag{AW.1}$$

The agent is subject to the structural persistence condition ($\alpha R > \rho^{\text{eff}}$), meaning it only survives if its steady-state mismatch $R^* < R$. Under Model D dynamics, $R^* = \rho^{\text{eff}}/(\eta^* c_{\min})$. Using the optimal update gain $\eta^* = \frac{U_M}{U_M+U_o}$, we can rewrite the survival constraint exactly:

$$\frac{U_M}{U_M + U_o} > \frac{\rho^{\text{eff}}}{R c_{\min}} \tag{AW.2}$$

Solving for U_o, we find the maximum observation noise the agent can tolerate without destabilizing:

$$U_o^{\max} = U_M \left(\frac{Rc_{\min}}{\rho^{\text{eff}}} - 1 \right) \tag{AW.3}$$

This creates a strict constraint on the agent's action selection:

$$\mathbb{E}_{a\sim\pi}[U_o(a)] \leq U_o^{\max}(M_t) \tag{AW.4}$$

The Tragedy of the Confident Agent.

Notice the proportionality: $U_o^{\max} \propto U_M$.

As $U_M \to 0$ (the agent becomes highly confident), $U_o^{\max} \to 0$. The survival constraint becomes infinitely tight.

If the environment is drifting ($\rho^{\text{eff}} > 0$), a confident agent inherently ignores noisy observations ($\eta^* \to 0$). To survive the drift, the agent *must* force an update. The only way to penetrate its own stubbornness is to generate an observation so pristine (low U_o) that the agent is forced to correct its course. Thus, a confident agent in a drifting world is *forced* to explore (seek low U_o) simply to survive.

Conversely, an uncertain agent (high U_M) has a loose constraint (large $U_o^{\max}$). It updates easily ($\eta^* \to 1$), so it can safely exploit noisy states without fear of structural collapse.

Deriving the Survival Lagrange Multiplier λ_{surv}.

Derivation (survival-exploration)

Formulating the KKT Lagrangian for this constrained optimization:

$$\mathcal{L} = \mathbb{E}[Q_O(a)] - \lambda' \cdot (\mathbb{E}[U_o(a)] - U_o^{\max}) \tag{AW.5}$$

The shadow price λ' scales as $1/(U_o^{\max} - \mathbb{E}[U_o])$. At the boundary, $\lambda' \propto 1/U_o^{\max}$, yielding:

$$\lambda' \propto \frac{\rho^{\text{eff}}}{(Rc_{\min} - \rho^{\text{eff}}) \cdot U_M} \tag{AW.6}$$

To align this with the CIY objective, we invoke the structural assumption that Causal Information Yield is inversely proportional to observation ambiguity: $\text{CIY}(a) \propto 1/U_o(a)$. This scalar reciprocal mapping is a heuristic placeholder: the mathematically clean correspondence between the survival-imperative Lagrangian and the CIY/EIG objective is the LMI / matrix-valued lift in `#deriv-causal-ib-lmi`, where the Fisher Information Matrix and a positive-semidefinite Lagrange multiplier Λ replace the scalar substitution below. Readers tracking exactness rather than directional fidelity should treat the rest of this section as the scalar shadow of that LMI derivation; the *honest limit* sub-section below and `#deriv-causal-ib-lmi` together carry the formal load.

Transforming the Lagrangian penalty from U_o to $1/U_o$ (which introduces a non-linear mapping of the multiplier, operating-point dependent), the equivalent survival weight on CIY becomes:

$$\lambda_{\text{surv}}(M_t) \propto \frac{\rho^{\text{eff}}}{(Rc_{\min} - \rho^{\text{eff}}) \cdot U_M} \tag{AW.7}$$

Epistemic Status.

Derived (conditional). The derivation of the $U_o^{\max}$ bound and the KKT multiplier λ' is exact, conditional on the continuous-time Lyapunov bounds from `#result-sector-condition-stability`. The mapping to $\lambda_{\text{surv}} \cdot \text{CIY}$ is *heuristic*, conditional on the assumed structural relationship $\text{CIY}(a) \propto 1/U_o(a)$ and the non-linear transformation of the shadow price.

Discussion.

Two Parallel Exploration Drives. AAT dictates two correlated but distinct motivations for exploration, acting at opposite ends of the uncertainty spectrum: 1. **Epistemic Information Gain ($\lambda_{\text{info}} \propto U_M$):** Explored in `#disc-ciy-unified-objective`. The agent explores to reduce its model uncertainty. Dominates when U_M is high. 2. **Lyapunov Survival Imperative ($\lambda_{\text{surv}} \propto 1/U_M$):** Derived here. The agent explores to force high-fidelity corrections against environmental drift. Dominates when U_M is low.

Resolving the IB β vs ρ conflation. This derivation clarifies a structural tension in the standard IB formulation (`#form-information-bottleneck`). Standard IB natively discards old information as ρ rises because the joint distribution $P(\mathcal{C}_t, O_{t+1})$ decorrelates. The agent's *representational preference* (β) does not need to change. However, the agent's *action policy* is tightly coupled to ρ. As ρ increases, the survival constraint tightens ($U_o^{\max} \downarrow$), forcing the Lagrange multiplier λ_{surv} to spike. The environment dictates the representation natively; survival dictates the policy.

Connection to Active Inference. This derivation completely sidesteps the Active Inference "preferences-as-priors" assumption (the dark room problem). The survival exploration drive exists purely because failing to bound U_o mathematically guarantees structural collapse ($\alpha R < \rho^{\text{eff}}$). Exploration is derived as a physical survival imperative, not a psychological preference.

Honest limit: the scalar formulation admits a "blank wall" attack. The survival constraint as derived bounds the *scalar* observation-noise expectation $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ — it cares only about the magnitude of U_o, not its alignment with the directions in which ρ^{eff} is acting. In multidimensional environments where some coordinates of the drift are inactive, an action that selects an observation channel orthogonal to the drift (a constant signal — the "blank wall") drives U_o to near-zero in that subspace and satisfies the scalar survival condition, while the agent's prediction error in the drifting subspace grows without bound because the observations carry no information about the drifting coordinates. The scalar reduction collapses the dimensionality that matters.

The scalar derivation here remains correct in its scope (1D environments and isotropic-disturbance multidimensional environments where any low-noise observation suffices), and the qualitative dual-drive picture survives the lift to higher dimensions. The full multidimensional resolution — upgrading the scalar inequality to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction rather than magnitude — is in `#deriv-causal-ib-lmi`.

Findings.

Survival-Imperative Exploration as Lyapunov-Forced Drive. **Brief:** A confident agent in a drifting world is mathematically forced to seek high-fidelity (low-noise) observations not because exploration is informationally valuable but because failing to do so violates the structural persistence condition and guarantees collapse. The maximum tolerable observation noise $U_o^{\max}$ scales with U_M — meaning the more confident the agent becomes, the tighter the survival-noise constraint becomes, and the more the agent is forced to actively seek low-ambiguity observations to penetrate its own stubbornness. The result identifies a second exploration drive ($\lambda_{\text{surv}} \propto 1/U_M$) structurally distinct from and complementary to the standard epistemic information-gain drive ($\lambda_{\text{info}} \propto U_M$), with the two acting at opposite ends of the uncertainty spectrum.

Impact: Provides AAT with a derivation of *why exploration is required* that does not route through epistemic-value-of-information arguments and therefore does not inherit the dark-room-problem objection that has dogged active-inference accounts. The survival imperative falls out of the same Lyapunov persistence condition that grounds the rest of Part II — exploration is forced by the same mathematics that forces the agent's adaptive responsiveness, not added as a separate motivational layer. The dual-drive picture also resolves a structural tension in standard Information Bottleneck formulations between the representational β and the environmental drift rate ρ: the environment dictates representation natively (IB's joint distribution decorrelates as ρ rises); survival dictates policy (the constraint tightens as ρ rises). The scalar form admits a "blank wall" attack, resolved in `#deriv-causal-ib-lmi` by lifting the constraint to an LMI on the Fisher Information Matrix; the dual-drive picture survives the lift in matrix form.

Novelty Claim: *Claim differentiation* on the structural source of agentic exploration. Active-inference work derives exploration from epistemic value inside expected free energy; bounded-rationality and information-cost frameworks derive exploration from utility-shaped information processing. This finding derives exploration from Lyapunov-persistence collapse — a structural-stability argument rather than a value-of-information argument — and identifies a second exploration multiplier inversely proportional to model uncertainty, complementary to the epistemic drive that scales directly with it.

Related Work:

- Friston, Rigoli, Ognibene, Mathys, FitzGerald & Pezzulo 2015, "Active inference and epistemic value," *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04 via Undermind Pillar 1 report) — *adjacent literature with structural contrast* — derives exploration from epistemic value inside expected free energy; the present finding's exploration drive is forced by Lyapunov bounds rather than preferences-as-priors, sidestepping the dark-room objection.
- Friston, Samothrakis & Montague 2012, "Active inference and agency: optimal control without cost functions," *Biological Cybernetics* 106:523–541 (published 2012, found 2026-04) — *adjacent literature* — broader active-inference framing within which the EFE epistemic-value account sits; same structural-vs-preference contrast.
- Ortega & Braun 2012, "Thermodynamics as a theory of decision-making with information-processing costs," doi:10.1098/rspa.2012.0683 (published 2012, found 2026-04) — *adjacent literature* — bounded-rationality framework deriving exploration-like behavior from information-processing costs; the present finding derives a structurally distinct second drive (Lyapunov survival) without inheriting the bounded-rationality objective form.
- Genewein, Leibfried, Grau-Moya & Braun 2015, "Bounded Rationality, Abstraction, and Hierarchical Decision-Making," *Frontiers Robotics AI* 2:27 (published 2015, found 2026-04) — *adjacent literature* — hierarchical extension of Ort12c; same structurally-different motivational source.
- Tishby, Pereira & Bialek 2000, "The information bottleneck method" (published 2000, found 2024) — *formal antecedent* — the standard IB formulation against which the β-vs-ρ tension is resolved; the survival imperative provides the missing policy-side coupling that standard IB does not natively carry.

Search Log:

- 2026-04 (*nominally comprehensive on the active-inference / bounded-rationality contrast*, via `ref/Novelty_defense_and_integration.md` Pillar 1): The Undermind report identified Friston 2015 as the canonical EFE-epistemic-value account against which this finding's structural-stability source for exploration differentiates, and the Ortega/Genewein bounded-rationality line as the closest information-cost-shapes-policy precursor. The verdict on Pillar 1 as a whole was *Conceptual Precursor* (High confidence) for the no-go half (`#der-causal-insufficiency-detection`); the survival-imperative half is the constructive complement, framed as differentiation from epistemic-value exploration rather than as a no-go contribution.

- 2025 (*intuition-only*): Pre-Undermind expectation was that the closest priors would be in active-inference (Friston) and Information Bottleneck (Tishby) literatures; this was confirmed and refined by the comprehensive search.

Appendix AX

Derivation: Linear Matrix Inequality Form of the Causal-IB Survival Bound

In `#deriv-causal-ib-exploration`, the survival-imperative exploration drive was derived in scalar form: bound $\mathbb{E}_\pi[U_o(a)] \leq U_o^{\max}$ to keep the agent's mismatch within survivable bounds. The scalar bound admits a "blank wall" attack — actions that minimize U_o in a subspace orthogonal to the drift (e.g., observing a constant signal uncorrelated with the environmental drift) satisfy the survival math without probing the drifting coordinates, and the agent's prediction in those subspaces diverges unboundedly. This segment lifts the survival constraint to a Linear Matrix Inequality on the Fisher Information Matrix, replacing the scalar shadow price with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction, not just by magnitude.

Under the matrix form, the action's exploration bonus is the trace $\mathrm{Tr}(\Lambda \cdot \mathcal{J}_o(a))$, and complementary slackness on the directional constraint *mathematically zeros out the bonus for blank-wall actions*: the bonus structurally requires the agent to probe the drifting directions. The result converts the unified policy objective from heuristic to conditional-theorem-grade under the matrix-LMI form; the structural form is grounded in AAT's physical survival bounds and standard semidefinite programming rather than treated as an ad-hoc heuristic. The same Fisher Information Matrix appears here in *three* structurally distinct framework loads — credit assignment (AO), coordinate forcing (`#disc-additive-coordinate-forcing` 4th instance), and now survival — because the framework's coordinates are forced rather than chosen; this segment is the first explicit cross-instance composition where coordinate-forcing and survival interact within the same machinery. Composition with the persistence-information-rate floor sharpens: the floor names *what bandwidth must be met*; the LMI names *in what directions to spend it*. Without the LMI the bandwidth could be supplied in irrelevant directions; with it, complementary slackness routes the bandwidth toward the drifting modes — a directional version of the central inequality.

Formal Expression.

Multidimensional mismatch dynamics.

Definition (multidim-mismatch)

The mismatch state $\delta_t \in \mathbb{R}^n$ evolves under linear-Gaussian dynamics with action-dependent observation noise:

$$\delta_{t+1} = (I - K_t H)\delta_t + w_t - K_t v_t(a_t) \tag{AX.1}$$

where $w_t \sim \mathcal{N}(0, Q_\rho)$ is the environmental drift, $v_t \sim \mathcal{N}(0, R_o(a_t))$ is the observation noise for action a_t, and the optimal Kalman gain is $K_t = P_t H^T (H P_t H^T + R_o(a_t))^{-1}$ with P_t the agent's prior uncertainty covariance.

The steady-state mismatch covariance Σ_δ satisfies the discrete-time algebraic Riccati equation. *Survival* requires the spectral bound

$$\lambda_{\max}(\Sigma_\delta) < R^2 \tag{AX.2}$$

— the matrix lift of the scalar bound $R^* < R$ from `#result-persistence-condition`.

Definition (matrix-ciy) **Fisher Information Matrix as matrix CIY.**

For the linear-Gaussian observation model, the action-conditional Fisher Information Matrix is

$$\mathcal{J}_o(a) = H^T R_o(a)^{-1} H. \tag{AX.3}$$

Where the scalar formulation in `#disc-ciy-unified-objective` defined CIY as a surrogate satisfying $\mathrm{CIY}(a) \propto 1/U_o(a)$, the multidimensional formulation identifies CIY directly with the action-conditional FIM. The CIY-U_o surrogate-mapping layer of approximation in the scalar derivation is removed at this layer; the residual CIY-vs-EIG concern (FIM equals EIG-rate only under Gaussian-linear conditions) persists as before.

The information-form Kalman update is additive in the FIM:

$$P_{t+1|t+1}^{-1} = P_{t+1|t}^{-1} + \mathcal{J}_o(a_t), \tag{AX.4}$$

where $P_{t+1|t} = AP_tA^T + Q_w$ is the prediction step that absorbs process noise. The simplification $P_{t+1}^{-1} = P_t^{-1} + \mathcal{J}_o(a_t)$ used in some passes refers to the information-update step alone; the full step adds the prediction contribution. The two coincide once the analysis is in steady state.

Derived (lmi-survival) **LMI survival constraint.**

The discrete-time algebraic Riccati equation admits a stabilizing steady-state solution with $\lambda_{\max}(\Sigma_\delta) < R^2$ if and only if the action policy π provides Fisher information that dominates the unstabilizable modes of the system drift. Specifically, the policy must satisfy the Linear Matrix Inequality

$$\mathbb{E}_{a\sim\pi}[\mathcal{J}_o(a)] \succeq \mathcal{J}_{\min}(Q_\rho, A, R^2) \tag{AX.5}$$

where $\mathcal{J}_{\min}$ is the symmetric positive-definite lower bound on the FIM required to keep the steady-state covariance's largest eigenvalue below R^2. The closed form for $\mathcal{J}_{\min}$ as a function of (Q_ρ, A, R^2) is given by the steady-state DARE condition — standard control-theory machinery (Boyd, Ghaoui, Feron & Balakrishnan 1994 Ch. 3–5; Anderson & Moore 1979). AAT's contribution at this layer is the framing: $\mathcal{J}_{\min}$ enters as the *survival-imperative* Fisher-information floor, not as a controller-design parameter.

Tensor Lagrangian.

Derived (tensor-lagrangian)

The agent maximizes pragmatic value under the LMI constraint:

$$\max_{\pi} \mathbb{E}_{\pi}[Q_O(a)] \quad \text{s.t.} \quad \mathbb{E}_{\pi}[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}. \tag{AX.6}$$

The Lagrangian for a matrix-inequality-constrained problem uses a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$:

$$\mathcal{L}(\pi, \Lambda) = \mathbb{E}_{\pi}[Q_O(a)] + \mathrm{Tr}(\Lambda \cdot (\mathbb{E}_{\pi}[\mathcal{I}_o(a)] - \mathcal{I}_{\min})) \,. \tag{AX.7}$$

KKT conditions are primal feasibility ($\mathbb{E}_{\pi}[\mathcal{I}_o(a)] \succeq \mathcal{I}_{\min}$), dual feasibility ($\Lambda \succeq 0$), and complementary slackness ($\mathrm{Tr}(\Lambda \cdot (\mathbb{E}_{\pi}[\mathcal{I}_o(a)] - \mathcal{I}_{\min})) = 0$). The optimal action selection rule is

$$a_t \in \arg\max_{a} [\, Q_O(a) + \mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a)) \,] \,. \tag{AX.8}$$

The trace inner-product $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ is the **matrix exploration bonus** — the multidimensional analog of $\lambda_{\text{surv}} \cdot \mathrm{CIY}(a)$ from the scalar derivation.

Resolution of the blank-wall attack.

Derived (blank-wall-resolution)

By complementary slackness, Λ has support only in eigendirections where the LMI constraint binds — i.e., directions where the agent's accumulated FIM sits at the threshold $\mathcal{I}_{\min}$. In non-drifting eigendirections of Q_ρ, $\mathcal{I}_{\min}$ contributes zero (no information needed there to survive), so Λ has zero weight in those directions.

For a blank-wall action — high $\mathcal{I}_o(a)$ in a non-drifting eigendirection — the trace product $\mathrm{Tr}(\Lambda \cdot \mathcal{I}_o(a))$ evaluates to zero: the action's FIM eigenvalues lie outside Λ's support. The agent receives no exploration bonus for the blank-wall action, even though the action would satisfy the scalar magnitude bound. **The matrix Lagrangian distinguishes by direction, not just by magnitude**, mathematically forbidding the trivial-exploration solutions that the scalar form admitted.

Scalar reduction.

Derived (scalar-recovery)

In the 1D case, H and $R_o(a)$ are scalars, $\mathcal{I}_o(a) = R_o(a)^{-1}$, and the LMI degenerates to the scalar inequality $\mathbb{E}_{\pi}[R_o(a)^{-1}] \geq \mathcal{I}_{\min}$ which, with $\mathcal{I}_{\min}^{-1} = U_o^{\max}$, recovers the scalar survival constraint of `#deriv-causal-ib-exploration`. The matrix multiplier Λ collapses to a scalar shadow price coinciding with λ_{surv}, and the matrix exploration bonus reduces to $\lambda_{\text{surv}} \cdot \mathrm{CIY}(a)$. The multidimensional derivation is structurally consistent with — and properly subsumes — the scalar one.

Tragedy of the Confident Agent — matrix form.

Derived (matrix-confidence-tragedy)

As the agent's prior covariance P_t shrinks in a particular eigendirection v, the agent's confidence in that direction grows. If v also has support in Q_ρ (a drifting eigendirection), the survival LMI requires $\mathcal{I}_{\min}$ to grow along v to compensate — a confident agent in a drifting direction must source information specifically along that direction, because its low P_t in v means its update gain is small there and slow to absorb new information. By complementary slackness, the matrix shadow price Λ develops a large principal eigenvalue along v, forcing the agent to choose actions whose $\mathcal{I}_o(a)$ has large support on v. The qualitative tragedy survives the lift: confident agents in drifting worlds are mathematically forced to probe the drifting eigendirections specifically, not just to minimize scalar noise.

Epistemic Status.

Derived (conditional). The structural derivation is exact under named conditions:

1. **Riccati stabilizability.** The discrete-time algebraic Riccati equation admits a stabilizing steady-state solution. Standard machinery: Boyd, Ghaoui, Feron & Balakrishnan 1994 §3–5 and Anderson & Moore 1979.
2. **Linear-Gaussian observation model.** The observation $o = Hx + v$ with $v \sim \mathcal{N}(0, R_o(a))$. Non-Gaussian or nonlinear models require local linearization or extended-Kalman treatment.
3. **Information-form Kalman in steady state.** The $\mathcal{J}_o$-additivity used in the FIM update reflects the information-update step alone; the prediction step's $AP_tA^T + Q_w$ contribution is absorbed into the steady-state Riccati analysis. Transient regimes admit the constraint shape but with $\mathcal{J}_{\min}$ a function of the transient covariance.
4. **DARE-imported $\mathcal{J}_{\min}$.** The closed-form expression for $\mathcal{J}_{\min}(Q_\rho, A, R^2)$ via the DARE is theorem-imported standard control theory. AAT's contribution is the framing — $\mathcal{J}_{\min}$ enters as the survival-imperative FIM floor, with Λ as the directional exploration-bonus shadow price — not the DARE solution itself.

The matrix-CIY mapping ($\mathcal{J}_o(a)$ as multidimensional CIY) is more rigorous than the scalar version: where the scalar formulation needed CIY $\propto 1/U_o$ as an assumed structural mapping, the matrix formulation directly identifies CIY with the action-conditional FIM, removing one of the two layers of approximation in `#disc-ciy-unified-objective`. The CIY-vs-EIG surrogate concern remains: FIM is the EIG-rate for Gaussian linear models but diverges from EIG under non-Gaussian or non-local conditions.

Discussion.

Connection to AAT's existing matrix-form machinery. The LMI repair sits naturally within several existing AAT components rather than being imported from outside:

- `#deriv-fisher-whitened-update-rule` derives Fisher-information-based credit assignment as AAT-internal under the (PI) parameterization-invariance axiom + Čencov 1982 uniqueness. The FIM appearing in the LMI is the same Fisher metric, used here for survival rather than for credit assignment — two roles for one geometric object.
- `#disc-additive-coordinate-forcing` lists Čencov-invariance as the 4th primary instance of the meta-pattern. The LMI's reliance on the FIM sits cleanly within that meta-pattern: the Fisher metric is forced by AAT-internal axioms, not adopted ad hoc.
- `#result-contraction-template` carries matrix-form sector machinery (Lohmiller-Slotine metric formulation) for Part III composition. The LMI's discrete-time Riccati is the control-theoretic analog of the contraction-metric framework — both are matrix-form sector conditions on multivariate dynamics.
- `#def-adaptive-tempo` will need a tensor extension to $\mathcal{T}$ for full alignment with this segment's directional structure; the LMI gives an explicit demand for tensor-valued adaptive rates.

Two parallel exploration drives — matrix lift. The scalar dual-drive distinction in `#disc-ciy-unified-objective` ($\lambda_{\text{info}} \propto U_M$ epistemic vs $\lambda_{\text{surv}} \propto 1/U_M$ Lyapunov-survival) generalizes to two matrix-valued drives that compose additively:

- Λ_{info}: epistemic information-gain matrix scaled by P_t. Dominates eigendirections where P_t is large (high uncertainty in some direction) — agent explores to learn that direction.
- Λ_{surv}: Lyapunov survival-imperative matrix derived above, with eigenvalue support where the LMI constraint binds. Dominates when P_t is small along a drifting eigendirection — agent explores to track the drift it's about to lose grip on.

The agent's total exploration shadow price is $\Lambda_{\text{info}} + \Lambda_{\text{surv}}$, with the relative direction-by-direction weights set by where the agent sits in (P_t, Q_ρ) space.

Connection to Active Inference. Same dark-room sidestep as the scalar version: the matrix-form survival imperative is forced by Lyapunov bounds, not preferences-as-priors. The matrix lift sharpens the argument by ruling out the non-exploratory "blank wall" trivial solutions that the scalar form admitted.

Empirical signature. A 2D drift environment with separable drifting/non-drifting subspaces is the simplest test: a scalar-Lagrangian agent succumbs to blank-wall traps; an LMI agent — with Λ developing weight specifically in the drifting direction — must select actions whose $\mathcal{I}_o$ aligns with the drift. The directional discrimination is empirically visible in the agent's action-selection histogram across the two subspaces.

Findings.

Matrix Lift of the Survival-Imperative Constraint via Fisher-Information LMI. **Brief:** Imagine an agent that's supposed to track something changing but only checks for new information by staring at a blank wall. A scalar version of the survival argument would let this work — the agent satisfies "I gathered enough information" without gathering information about the thing that's actually moving. This result fixes the math by making the argument directional: the agent has to gather information in the *directions* where the world is changing, not just gather some-information-anywhere. The scalar Causal-IB survival-imperative drive lifts to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier Λ that distinguishes by direction. Complementary slackness mathematically forbids "blank wall" actions that satisfy the scalar bound by sourcing information in non-drifting subspaces. The fix uses Fisher-geometric machinery already AAT-internal via `#deriv-fisher-whitened-update-rule` and `#disc-additive-coordinate-forcing`'s 4th instance.

Impact: Closes a structural failure of the scalar derivation that admitted trivially-satisfying-the-math actions (constant signals, walls, etc.) without actually probing the drifting modes the agent must track to survive. The matrix lift demonstrates that the same Fisher geometric object plays multiple structurally-distinct roles — credit assignment via `#deriv-fisher-whitened-update-rule`, coordinate-forcing via `#disc-additive-coordinate-forcing`, and now survival via the LMI here. The "Tragedy of the Confident Agent" insight survives the lift in sharper form: the matrix shadow price's eigenstructure picks out the directions where confidence has outpaced incoming information, mathematically forcing the agent to probe specifically those directions. The scalar reduction recovers `#deriv-causal-ib-exploration`'s constraint as the 1D special case, establishing structural consistency.

Novelty Claim: *Claim differentiation* on the directional discrimination of the survival-imperative exploration drive. The scalar finding in `#deriv-causal-ib-exploration` differentiated AAT's structural-stability source for exploration from active-inference's epistemic-value source; this finding sharpens that differentiation by ruling out the trivial-exploration solutions that the scalar form admitted, demonstrating that the structural-stability source is genuinely directional rather than scalar. The DARE-imported $\mathcal{I}_{\text{min}}$ and the matrix-Lagrangian KKT machinery are standard control-theoretic and convex-optimization apparatus; AAT's contribution at this layer is the framing that places $\mathcal{I}_{\text{min}}$ as a survival-imperative Fisher-information floor and Λ as the directional exploration-bonus shadow price.

Related Work:

Table AX.1

ASF concern	*Prior-art language*	*Relationship / Positioning*
LMI / matrix-inequality formulation of stability constraints	Boyd, El Ghaoui, Feron & Balakrishnan 1994, *Linear Matrix Inequalities in System and Control Theory*, SIAM (published 1994, found 2025)	*formal antecedent* — adopted machinery; the LMI form, KKT for matrix-inequality-constrained optimization, and the DARE-imported stabilizability condition are standard control-theoretic apparatus
Discrete-time algebraic Riccati equation for steady-state Kalman survival	Anderson & Moore 1979, *Optimal Filtering*, Prentice-Hall (published 1979, found 2025)	*formal antecedent* — DARE provides the closed form for $\mathcal{J}_{\min}(Q_\rho, A, R^2)$
Fisher Information Matrix as the natural geometric object for inference	Čencov 1982, *Statistical Decision Rules and Optimal Inference* (published 1982, found 2024)	*formal antecedent* — Čencov uniqueness forces the Fisher-Rao metric under (PI) parameterization-invariance, integrated AAT-internally via `#disc-additive-coordinate-forcing`'s 4th instance
Exploration as epistemic-value drive (alternative motivational source)	Friston, Rigoli, Ognibene et al. 2015, "Active inference and epistemic value" *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04 via Undermind Pillar 1 report)	*adjacent literature with structural contrast* — the matrix-form survival imperative is forced by Lyapunov bounds, not preferences-as-priors. The matrix lift sharpens the contrast by ruling out the non-exploratory blank-wall solutions that the scalar form admitted
Information-form Kalman filter / additive FIM update	Standard Kalman filter literature; Anderson & Moore 1979 (published 1979, found 2025)	*formal antecedent* — the additivity $P^{-1}_{t+1\|t+1} = P^{-1}_{t+1\|t} + \mathcal{J}_o(a_t)$ used in the LMI derivation is the information-form update step
Scalar survival-imperative drive being lifted	`#deriv-causal-ib-exploration` (AAT-internal, 2025)	*upstream segment being matrix-lifted* — the LMI repair preserves the dual-drive structure in matrix form ($\Lambda_{\text{info}} + \Lambda_{\text{surv}}$) and resolves the blank-wall structural failure flagged in the scalar segment's Discussion

Search Log:

- 2026-04 (*nominally comprehensive on the exploration-source contrast*, via `ref/Novelty_defense_and_integration.md` Pillar 1): The Undermind report on Pillar 1 ("Causal insufficiency and forced exploration") established the active-inference EFE framing as the canonical alternative source for exploration drives. This finding's matrix-Lagrangian LMI is the directional sharpening of the scalar derivation in `#deriv-causal-ib-exploration` and inherits that segment's differentiation posture against EFE-style epistemic-value accounts.
- 2025 (*targeted*): Boyd et al. 1994 and Anderson & Moore 1979 confirmed as the formal antecedents for the LMI / DARE machinery; Čencov 1982 confirmed as the formal antecedent for the FIM-as-natural-geometric-object move via the (PI) axiom. AAT's contribution at this layer is framing rather than apparatus.
- 2025 (*intuition-only*): Pre-search expectation was that the LMI form and DARE machinery were standard and would not be novel apparatus; this was confirmed. The novel content is the placement — using the FIM as a survival floor rather than a controller-design parameter, and integrating directional exploration with the broader Fisher-geometric meta-pattern.

Appendix AY

Observation: The Authored-World Laboratory — Identifiability by Construction

AAT has one calibration laboratory it *found* — software, whose epistemic privileges are catalogued in `#obs-software-epistemic-properties` — and one it is *building*: a deterministic authored world (working name **vivarium**) in which the environment's law-content, state, noise, and disturbance rate are not merely observable but authored, so that every quantity the framework's results quantify over (ρ, U_o, δ_t, $\mathcal{T}$, per-dimension mismatch, $\|\delta_{\text{critical}}\|$) has ground truth at scale inside dynamics rich enough that toy-model simplifications are structurally unavailable. This segment is the canonical introduction of the authored-world laboratory, placed ahead of the framework's validation simulations because the standing intent is to re-run those simulations in it; later segments and simulation reports cite this segment rather than re-describing the system.

Formal Expression.

Empirical Claim (system properties, by construction; build-state caveat in Epistemic Status)

The authored world provides four structural properties, each the *built* counterpart of a *found* software property in `#obs-software-epistemic-properties`:

V1. Ground truth for every framework quantity. Law-content is an explicit versioned registry of law-articles (keyed, memoizable invocation units); state is canonical and content-addressed; the disturbance rate ρ is dialable per region (weather volatility, water-regime schedules); δ_{critical} has literal in-world meaning. Where software's inspectability is limited by the agent-side comprehension bandwidth (P1 there), the authored world's is limited only by the author's choice of instrumentation: identifiability holds **by construction** rather than by fortunate configuration.

V2. Authored coordinatization of the exogenous space. All stochasticity is a stateless pure function of a world seed and a coordinate/identity key — fated noise: genuine chance to any inside agent, deterministic lookup to the author. The aleatoric boundary's frame-relativity is thereby operational rather than philosophical, and the system realizes the central object of `#deriv-mechanism-counterfactual-separation` — an explicit, held coordinatization of U — which makes **latent-anchored mechanism counterfactuals executable**: swap a law-article version, keep seed and keys, rerun. The separation theorem's demonstration experiment (a keyed law versus its key-permuted twin: indistinguishable at every hierarchy level, divergent under same-seed law-swap, with measurable disagreement mass) is runnable here and, at present, nowhere else available to the framework.

V3. Exact citation semantics. A claim validated in the authored world cites a specific world artifact — seed, generator versions, phase memo, intervention script — and content-addressed storage makes the citation exact and replayable

to bit-identity. This is the conjunction (α) deterministic outcomes, (β) replay at commensurate cost, (γ) content-addressed immutable environment state, which `#obs-software-epistemic-properties` P2 identifies as jointly held, in current standard practice, by software alone — and names as a configurational uniqueness with exactly this falsifier: a simulator brought under content-addressed recording with commensurate replay cost. The authored world is that falsifier under deliberate construction; when its fidelity program completes, the "unique" in P2 becomes "one of two," by design.

V4. A disturbance-rate schedule with a persistence reading. The world is built in phases, each running to convergence with its converged output promoted into fixed law for everything faster above it. The phase ladder is therefore a ρ-schedule: successive phases deliver environments of decreasing effective disturbance until agents of achievable tempo satisfy the persistence condition $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (`#result-persistence-condition`). An agent-bearing world is, in this reading, the terminal rung of a persistence ladder the authors descend deliberately.

Epistemic Status.

Discussion-grade as a whole, with the property claims empirical-by-construction and honestly dated. V1-V4 describe the system's design and its verified build state as of 2026-07-16: the world is mid-Phase-3 (geology and hydrology converged with absolute-tolerance conservation tests; climate next; **agents not yet embodied**), so V1-V3 are live properties of the running system while V4's agent-bearing terminus and everything downstream of the agent seam are design intent. Claims here about what the laboratory *will* offer AAT's agent-level results are forward-looking and marked so; claims about determinism, keyed noise, content-addressing, and conservation-honesty describe running code. The Level-4 executability claim (V2) inherits *exact* semantics from `#deriv-mechanism-counterfactual-separation`; that the system realizes the required coordinatization is a fact of its architecture. This segment deliberately does not depend on the world's agent layer existing: it introduces the laboratory, not results obtained in it.

Discussion.

Found versus built, and why the framework wants both. Software is the domain where the epistemic privileges happen to hold and a large practice already generates data inside them; the authored world is the domain where they hold because someone chose them, which buys three things software cannot: (a) *dialable* environment parameters — ρ can be scheduled, not merely measured; (b) *author-side access to the exogenous coordinatization* — software's version control gives literal Level-3 access (P2 there), while fated noise gives literal Level-**4** access, one full rung more; (c) freedom from the confound that software environments are themselves made of agent behavior. Conversely the authored world lacks software's ecological validity — nothing in it ships to production — so results transfer with stated construction assumptions, exactly the transfer-assumption discipline TST names for cross-domain claims. The two laboratories are complements, not competitors: found breadth against built depth.

The standing simulation intent. The framework's Part I validation simulations (AZ) were built as standalone numerical models; the standing intent is to re-run them in-vivo in the authored world as its phases reach the relevant dynamics, converting each from a bespoke script into a cited, replayable world artifact per V3. This segment exists partly so that migration has a canonical anchor: simulation reports should cite the laboratory, not re-describe it.

Near-term experiments the laboratory makes cheap. Beyond the Level-4 demonstration (V2), two are flagged in adjacent segments: the GA-1 housing experiment (transition-law versus observation-law ignorance — the experimenter chooses whether to perturb T or h, directly testing the verified asymmetry from the epistemic-target work), and the behavioral $\hat{\kappa}_{\text{processing}}$ estimator at scale (same event under different goal states across thousands of agent-events with controlled priors) once the agent seam lands.

Ethics gate. Everything agent-seam-ward is gated by the authored world's own ethics front door, which is deliberately more conservative than this framework's volume-04 gates and binds program-wide; the laboratory's agent-level offers are sequenced behind it, not around it.

Appendix AZ

Detail: Observation: Part I Validation Simulations

Six simulation variants validated, refined, and extended the analytical predictions from Part I, forcing regime splits and discovering scaling laws that the analytical derivations left ambiguous.

The simulations are **theory-shaping, not merely confirmatory** — three findings emerged from simulation that *forced refinements* to the analytical treatment: (1) the adversarial tempo exponent turned out to be regime-dependent (Model D vs Model S; coupling-dominant vs non-coupling-dominant), forcing the $b = 2$ / $b = 3/2$ / $b \to 1$ regime classification the linear ODE left ambiguous; (2) observation noise was found to gate adversarial advantage with optimal-gain partial recovery, a result that had no analytical treatment before the simulations; (3) per-dimension persistence emerged as load-bearing — anisotropic systems with gain varying across dimensions produced mismatch concentration on the weak direction that the scalar persistence condition systematically overestimated by up to 72%. The simulation program is treated as part of AAT's methodology rather than as external validation: where simulation surfaces something the analytical treatment misses, the analytical treatment is refined; where simulation confirms what the analytical treatment predicts, the result lifts in confidence.

Overview.

The track-b simulation program tested Part I's analytical predictions about mismatch dynamics, adversarial coupling, observation noise, correction function robustness, and anisotropic persistence. The simulations modeled discrete-time mismatch dynamics as AR(1) processes with five correction functions (linear, saturating, threshold/dead-zone, sigmoid, structural breakdown), sweeping parameter spaces across gain, disturbance rate, coupling strength, observation noise, and dimensional structure.

The simulations were theory-shaping, not merely confirmatory. In particular:

- The adversarial tempo exponent turned out to be regime-dependent (deterministic drift vs. stochastic noise, coupling-dominant vs. non-coupling-dominant), forcing a regime split that the mismatch ODE left ambiguous.
- Observation noise was found to gate the adversarial advantage, partially recoverable by optimal gain – a finding that had no analytical treatment before the simulations.
- The scalar persistence condition was shown to overestimate adaptive capacity by 72% in anisotropic systems, motivating the per-dimension persistence condition.

Variant Summary.

Table AZ.1

Variant	*Disturbance Model*	*Tested*	*Key Finding*	*Theory Impact*	*Promoted Segment*
A	**D** (deterministic drift)	Deterministic drift coupling; coupling-dominance sweep	Exponent $b \to 2.0$ in coupling-dominant limit (confirmed at 1.999)	Validates Model D derivation ($b = 2$)	`#result-adversarial-exponent-regimes`
B	**D+S** (interpolation)	Drift-noise interpolation across correction functions	Smooth transition between drift ($b = 2.0$) and noise ($b = 1.5$) regimes; coupling dominance is the key qualifier, not drift vs. noise per se	Confirmed that the coupling-dominant condition is quantitatively load-bearing	`#result-adversarial-exponent-regimes`
C	**S** (stochastic AR(1))	Exponent vs. gain (η) in stochastic model	Exponent drops toward 0.5 as $\eta \to 0$ (away from coupling dominance at fixed q_{base}); discrete AR(1) exponent never exceeds 1.5	Validates Model S derivation ($b = 3/2$)	`#result-adversarial-exponent-regimes`
D	**D vs. S** (separated)	Exponent vs. base noise (q_{base}) at fixed η	Continuous ODE exponent → 2.0 (Model D), discrete AR(1) exponent → 1.5 (Model S), as $q_{\text{base}} \to 0$	Definitively separated the two asymptotes; validates both derivations	`#result-adversarial-exponent-regimes`
E	**S** (stochastic AR(1))	Observation noise; optimal gain validation	Observation noise collapses adversarial exponent from ~ 1.0 to ~ 0.2; Riccati-optimal gain restores it to ~ 0.4; 52% mismatch reduction at moderate noise	Observation quality gates tempo advantage; optimal gain (`#emp-update-gain`) empirically validated	`#obs-gated-tempo-advantage`
F	**S** (stochastic AR(1))	Multi-dimensional anisotropic correction; targeted adversarial attack	Per-dimension theory exact to 4 significant figures; scalar tempo overestimates by 72%; targeted attack amplifies advantage by 17%	Validates Model S per-dimension steady state	`#result-per-dimension-persistence`
Hafez bridge	**S** (stochastic AR(1))	Bi-predictability P vs. AAT mismatch in adversarial and non-adversarial settings	P measures coupling architecture (scale-invariant); mismatch measures coupling performance; P is blind to adversarial dynamics	P and AAT mismatch are complementary diagnostics; H_b (agent opacity) has no direct AAT analog – potential gap for multi-agent work	–

Methodology.

Discrete mismatch dynamics. The environment follows a random walk $x_{t+1} = x_t + w_t$ with $w_t \sim N(0, q^2)$. The agent corrects via $\hat{x}_{t+1} = \hat{x}_t + \eta \cdot g(\delta_t)$, yielding the AR(1) mismatch process $\delta_{t+1} = (1 - \eta) \cdot \delta_t + w_t$ for linear g. This is the discrete-time analog of the mismatch ODE $d\|\delta\|/dt = -\mathcal{T} \cdot \|\delta\| + \rho$ from `#hyp-mismatch-dynamics`.

Parameter sweeps. Each variant swept its key parameter(s) across 7–20 values. Monte Carlo: 200 independent trials per parameter point, 10,000–20,000 timesteps per trial, with 2,000–5,000 step burn-in for steady-state convergence. Fixed random seeds for reproducibility.

Exponent fitting. Adversarial exponents were estimated by fitting $\log(\|\delta_B\|/\|\delta_A\|) = a + b \cdot \log(\mathcal{T}_A/\mathcal{T}_B)$ via weighted least squares, with weights inversely proportional to variance of each point's log-estimate. 95% confidence intervals via bootstrap (1,000 samples).

Correction functions. Five functions $g : \mathbb{R} \to \mathbb{R}$ were tested, all satisfying $g(0) = 0$ and $g'(0) = 1$: linear ($g(\delta) = \delta$), saturating ($g(\delta) = \delta/(1 + |\delta|/R)$), threshold ($g(\delta) = \delta \cdot \mathbf{1}[|\delta| > \epsilon]$), sigmoid ($g(\delta) = R \cdot \tanh(\delta/R)$), and structural breakdown ($g(\delta) = \delta \cdot \mathbf{1}[|\delta| < R_{\max}]$).

Simulation code (reproducibility). The full simulation suite — single-agent and two-agent initial simulations plus the four variant-extension drivers (A/B drift, C/D regimes, E/F extensions, Hafez bridge) and their per-variant result write-ups — is maintained as the project's reproducibility supplement and will be released as a citable code artifact at publication. The numerical outcomes that the body's findings rest on (steady-state formulas, the fitted exponents, the gating thresholds, the per-dimension errors) are stated self-contained in the tables and Key Findings below.

Key Findings.

The adversarial exponent is regime-dependent (now derived). The theory now distinguishes two disturbance models (Model D: bounded deterministic, GA-2; Model S: stochastic zero-mean, GA-2S), each producing a different steady-state scaling and therefore a different adversarial exponent:

- **Model D, coupling-dominant:** $\|\delta\|_{ss} = \rho/\mathcal{T}$. Adversarial exponent $b = 2$ (derived from Prop A.1; Variant A confirms at 1.999).
- **Model S, coupling-dominant:** $\|\delta\|_{\text{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$ (from Prop A.1S). Adversarial exponent $b = 3/2$ (derived; Variants C-D confirm at 1.481).
- **Non-coupling-dominant:** Exponent degrades smoothly toward 1.0 (Model D) or 0.5 (Model S) as base disturbance grows relative to adversarial coupling.

A measured exponent of $b \approx 1.05$ corresponds to the non-coupling-dominant regime (stochastic-noise coupling at moderate coupling dominance), consistent with the regime classification rather than contradicting it. Variants A–D systematically map the full regime space and validate the derived exponents. See `#result-adversarial-exponent-regimes` for the full treatment.

Observation noise gates adversarial advantage. Adding observation noise σ_{obs} to the mismatch signal collapsed the adversarial exponent from ~ 1.0 to ~ 0.2 at $\sigma_{\text{obs}} = 10 \times q_{\text{env}}$ (fixed gain). The Riccati-optimal gain (`#emp-update-gain`) partially restored the advantage to ~ 0.4, more than doubling it but not recovering the noise-free level. The optimal gain helps most in the moderate-noise regime, where it achieved a 52% mismatch reduction over fixed gain. See `#obs-gated-tempo-advantage`.

Per-dimension persistence is exact; scalar overestimates. In a 3-dimensional system with 5:1 gain variation across dimensions, the per-dimension AR(1) steady-state prediction matched simulation to 4 significant figures. The scalar persistence condition overestimated aggregate mismatch by 72%, because it averages across dimensions while the weak dimension dominates the L_2 norm. Isotropic gain allocation reduced overall mismatch by 13%; targeted adversarial attack on the weak dimension amplified the mismatch ratio by 17%. See `#result-per-dimension-persistence`.

Nonlinear correction creates thresholds, not lower exponents. Under deterministic drift, saturating, sigmoid, and breakdown correction functions did not simply reduce the adversarial exponent. Instead, they produced catastrophic divergence when disturbance exceeded the correction capacity ($\rho > \mathcal{T} \cdot R$). This is the persistence threshold failure from `#result-persistence-condition`, observed directly in simulation. The measured "exponents" above 3.0 for these functions were divergence artifacts, not meaningful scaling laws.

Hafez bridge: architecture vs. performance. Bi-predictability P (Hafez et al.) measures the informational architecture of agent-environment coupling; AAT mismatch measures the operational performance. In adversarial settings, P remained nearly constant (~ 0.268) across a 10:1 range of opponent tempo, while the mismatch ratio varied from 0.54 to 5.95. P is scale-invariant after z-score normalization, making it insensitive to adversarial dynamics. The two metrics are complementary: P diagnoses whether the coupling structure is sound; mismatch predicts how well it performs.

Epistemic Status.

Empirical. Simulation results are reproducible under fixed random seeds; the driver code and per-variant write-ups are maintained as a reproducibility supplement to be released as a citable code artifact at publication. The key results – regime-dependent exponents, observation noise gating, per-dimension exactness – have been promoted to first-class segments (`#result-adversarial-exponent-regimes`, `#obs-gated-tempo-advantage`, `#result-per-dimension-persistence`) with their own epistemic assessments. This appendix serves as reference for the simulation program as a whole.

Discussion.

Internal validation, not external. The simulations operate in the model's own terms: AR(1) processes with parameterized correction functions, Gaussian noise, and discrete-time dynamics. They validate the mathematical predictions within the model – confirming that the analytical steady-state formulas, the exponent regimes, and the per-dimension theory are correct as statements about these stochastic processes. The gap between "the math predicts X within its own model" and "real agents exhibit X" remains an empirical question for each domain.

What the simulations did not test. The simulations did not test whether the AR(1) mismatch dynamics are a good model for any particular real-world adaptive system. They also did not test cross-dimensional coupling (off-diagonal correction), non-Gaussian disturbances, non-stationary parameters, or multi-agent composition dynamics. The interaction between observation noise and the deterministic-drift exponent regime ($b = 2.0$) was not tested – Variant E used stochastic coupling only.

Theory-shaping role. The most important contribution of the simulations was not confirming predictions but forcing the theory to be more precise. The regime split in the adversarial exponent, the observation-noise gating mechanism, and the per-dimension bottleneck effect were all findings that the analytical derivations initially left ambiguous or unaddressed. The simulations forced the disturbance model split (Model D vs. Model S), which in turn enabled the analytical derivation of both exponents. The simulations now serve as validation of the derived results: Variant A validates $b = 2$ (Model D), Variants C-D validate $b = 3/2$ (Model S), Variant F validates the Model S per-dimension steady state.

Part II

Appendices: Operational Domains

Operational-specific appendices and end-to-end domain instantiations validating the theory chain.

Appendix BA

Detail: Operationalization — Estimation Procedures

Estimation recipes for core AAT quantities, bridging the measurement gap between formal objects and practical deployment.

The segment is **domain-honest** about which quantities are easily measured and which are operationally challenging: model uncertainty admits predictive-posterior / ensemble-spread / Bayesian-NN estimators; observation uncertainty admits channel calibration or residual variance; mismatch magnitude is directly measurable from prediction-versus-observation residuals; tempo from event rate times effective gain; the sector parameter via either the gain-sector bridge (where applicable — Bayesian on exponential families; gradient on strongly convex) or empirical correction-rate-vs-mismatch regression; disturbance parameters split per Model D ($\hat{\rho}_{\text{det}}$) versus Model S ($\hat{\sigma}_w$) with sequential dependence on $\hat{\mathcal{J}}$; critical-mismatch tolerance is a *domain parameter* not an estimated quantity (it encodes "how wrong can the model be before actions become harmful or ineffective?"). Model-class fitness is named as particularly challenging — not directly computable, requiring search across the model class; the observable signature is persistent mismatch despite adequate learning. Satisfaction gap and control regret require evaluating attainability, generally intractable in exact form. AAT's commitment is that the abstract machinery is operationalizable with stated quality trade-offs at each estimation step, not analytical idealization with no path to practice.

Measurement Targets.

Table BA.1

Quantity	*Role in AAT*	*Typical unit*	*Minimum data needed*
U_M	Model uncertainty (`#emp-update-gain`)	domain-specific variance/entropy	Predictive posterior or ensemble spread
U_o	Observation uncertainty (`#emp-update-gain`)	sensor variance/noise scale	Channel calibration or residual variance
$\rho_{\text{det}}(t)$	Mismatch injection rate bound, Model D (`#hyp-mismatch-dynamics`)	surprise per time	Time-series of mismatch magnitudes; worst-case bounds
σ_w	Disturbance noise scale, Model S (`#hyp-mismatch-dynamics`)	surprise per $\sqrt{\text{time}}$	Mismatch innovation variance (residual after correction)
ρ_{delib}	Local mismatch drift during pauses (`#der-deliberation-cost`)	surprise per time	Deliberation windows with no corrective action
α	Lower correction efficiency bound (`#result-sector-condition-stability`)	inverse time	Vector mismatch trajectories + correction term

Quantity	*Role in AAT*	*Typical unit*	*Minimum data needed*
R	Radius where local sector condition holds (`#result-sector-condition-stability`)	surprise magnitude	Same as α, plus breakdown detection
$\|\delta_{\text{critical}}\|$	Functional adequacy threshold (`#result-persistence-condition`)	surprise magnitude	Task-level performance curve vs mismatch

Estimator Cookbook.

Estimating U_M and U_o. Use the most native uncertainty representation available in the domain:

Table BA.2

Domain	U_M *estimator*	U_o *estimator*
Kalman / linear Gaussian	Prior predictive variance $P_{t\|t-1}$	Measurement-noise variance R_t (known or EM-estimated)
Conjugate Bayes	Posterior variance / inverse effective sample size	Likelihood variance (or precision inverse)
RL with ensembles	Across-head predictive variance $\text{Var}_i[Q_i(s,a)]$	TD-target noise variance over replay batches
Neural net regression	Ensemble or Laplace posterior variance	Aleatoric head output $\sigma^2(x)$
PID / classical control	State-estimation covariance from observer	Sensor noise from calibration + residual PSD

Operational Definition

For the scalar gain heuristic (`#emp-update-gain`), normalize to common units and compute:

$$\hat{\eta}_t^* = \frac{\hat{U}_{M,t}}{\hat{U}_{M,t} + \hat{U}_{o,t}} \tag{BA.1}$$

Estimating Disturbance Parameters ($\rho_{\text{det}}, \sigma_w, \rho_{\text{delib}}$). The disturbance model split (Model D vs. Model S; see `#hyp-mismatch-dynamics`, `#result-sector-condition-stability`) requires different estimation procedures depending on the disturbance character.

Identifying the disturbance model. The choice between Model D and Model S is domain-dependent: - **Model D** (bounded deterministic, GA-2): appropriate when the environment changes persistently and directionally. Estimate ρ_{det} from worst-case bounds on mismatch injection rate. Examples: adversary who maneuvers systematically, API that drifts, climate that shifts. - **Model S** (stochastic zero-mean, GA-2S): appropriate when the environment fluctuates unpredictably around a stable mean. Estimate σ_w from residual variance of mismatch innovations. Examples: market noise, sensor noise, random perturbations. - **Mixed environments** use both: estimate ρ_{det} from the persistent component (trend) and σ_w from the residual variance after detrending. The adversarial exponent interpolates smoothly between $b = 2$ (pure drift) and $b = 3/2$ (pure noise); see `#result-adversarial-exponent-regimes`.

Let $s_t = \|\delta_t\|$ in surprise units (e.g., negative log-likelihood residual scale).

Model D: Global mismatch injection rate ($\hat{\rho}$_det):

Operational Definition

$$\hat{\rho}_{\text{det}}(t) = \left[\frac{s_{t+\Delta t} - s_t}{\Delta t} + \hat{\mathcal{I}}_t s_t \right]_+ \tag{BA.2}$$

where $[x]_+ = \max(x, 0)$ and $\hat{\mathcal{I}}$ evaluated at time t is estimated adaptive tempo. Take the supremum (or a high quantile) over the estimation window.

Model S: Noise scale ($\hat{\sigma}$_w):

Operational Definition

$$\hat{\sigma}_w^2 = \text{Var}\left[\frac{s_{t+\Delta t} - (1 - \hat{\eta}^*) s_t}{\sqrt{\Delta t}} \right] \tag{BA.3}$$

i.e., the variance of the mismatch innovations (residuals after subtracting the expected correction). For AR(1) processes, this is the innovation variance $\hat{\rho}^2$ directly.

Note on estimation sequencing. Both estimators require $\hat{\mathcal{I}}_t$ (or $\hat{\eta}^*$), estimated from $\hat{\nu}$ and $\hat{\eta}^*$. Estimate the gain and event rate first (from the agent's internal statistics and observation timing), then use these to extract the disturbance parameters from the mismatch trajectory. This sequential structure avoids circularity but introduces sensitivity: errors in $\hat{\mathcal{I}}$ propagate into $\hat{\rho}_{\text{det}}$ and $\hat{\sigma}_w$.

Local pause-window drift for Derived 4.2 ($\hat{\rho}$_delib):

Operational Definition

$$\hat{\rho}_{\text{delib}} = \underset{w \in \mathcal{W}_{\text{pause}}}{\text{median}} \frac{s_{w,\text{end}} - s_{w,\text{start}}}{\Delta \tau_w} \tag{BA.4}$$

using windows where corrective action is suspended or effectively delayed.

Estimating α (sector lower bound). A uses $\delta^T F(\mathcal{T}, \delta) \geq \alpha \|\delta\|^2$ for $\|\delta\| \leq R$. Operationally:

1. Estimate $\dot{\delta}_t$ (finite differences or filtered derivative).
2. Compute $\widehat{F}_t = -\dot{\delta}_t + w_t$ where disturbance proxy w_t is estimated from exogenous perturbation channels or residual balancing.
3. Form ratios $r_t = (\delta_t^T \widehat{F}_t)/\|\delta_t\|^2$ on bins of $\|\delta_t\|$.
4. Set conservative lower bound $\hat{\alpha}$ as a low quantile (e.g., 10th percentile) of r_t in the valid region.

Estimating R (valid-region radius). Estimate R as the largest radius for which sector inequality violations remain below tolerance:

Operational Criterion

$$\hat{R} = \sup \left\{ r > 0 \,:\, \Pr\left(\delta^T \widehat{F} < \hat{\alpha} \|\delta\|^2 \,\middle|\, \|\delta\| \leq r \right) \leq \epsilon \right\} \tag{BA.5}$$

with a chosen violation tolerance ϵ (e.g., 5%).

Estimating $\|\delta_{\text{critical}}\|$. Define a mission-level performance metric J (reward rate, tracking error, service SLA, etc.). Set the critical mismatch threshold where performance crosses a minimum acceptable level $J_{\min}$:

Operational Definition

$$\|\hat{\delta}_{\text{critical}}\| = \inf\{d : \mathbb{E}[J \mid \|\delta\| = d] < J_{\min}\} \tag{BA.6}$$

This anchors `#result-persistence-condition` to real task outcomes.

Recommended Estimation Sequence.

1. Fix mismatch representation δ in one consistent unit system (prefer surprise-scale).
2. Estimate U_o from channel physics/calibration; estimate U_M from model uncertainty.
3. Validate gain behavior against `#emp-update-gain` ($\hat{\eta}^*$ trend checks).
4. Estimate ρ_{delib} from pause windows (`#der-deliberation-cost`) and $\rho(t)$ from full traces.
5. Estimate α and R from local correction dynamics (A).
6. Estimate $\|\delta_{\text{critical}}\|$ from task-performance degradation.
7. Compute derived diagnostics:
   - Tempo margin (Model D): $\hat{\mathcal{T}} - \hat{\rho}_{\text{det}}/\|\hat{\delta}_{\text{critical}}\|$
   - Tempo margin (Model S): $\hat{\mathcal{T}} - n\hat{\sigma}_w^2/(2\|\hat{\delta}_{\text{critical}}\|^2)$
   - Reserve: $\widehat{\Delta\rho^*} = \hat{\alpha}\hat{R} - \hat{\rho}_{\text{det}}$
   - Deliberation feasibility: $\Delta\eta^*(\Delta\tau)\|\delta_{\text{post}}\| - \hat{\rho}_{\text{delib}}\Delta\tau$

Decision-Theoretic Procedures.

Estimating λ (Exploration Price). The exploration weight $\lambda(M_t)$ in the unified policy objective (`#disc-ciy-unified-objective`) prices information in value-equivalent terms:

Table BA.3

Context	*λ estimator*	*Source*
Finite bandits	Gittins index from dynamic programming	Exact (Gittins 1979)
Linear-Gaussian	Probing cost in quadratic objective	Exact (dual control)
Discrete MDP	$(\text{VoI})^2/\text{info gain}$	Information-directed sampling (Russo & Van Roy)
General	$\hat{\lambda} = c \cdot \hat{U}_M/\hat{U}_o$	Heuristic: scale CIY weight by relative uncertainty

For the heuristic: when $U_M \gg U_o$ (highly uncertain model), exploration is cheap relative to exploitation risk, so λ should be large. When $U_M \ll U_o$ (confident model, noisy observations), exploitation dominates. The constant c is domain-specific.

Deliberation Stopping Policy. From `#der-deliberation-cost`, deliberation of duration $\Delta\tau$ is warranted when $\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau$. Operationally:

1. Estimate ρ_{delib} from prior pause windows.
2. Before each deliberation episode, estimate $\|\delta_{\text{post}}\|$ as current mismatch + $\rho_{\text{delib}} \cdot \Delta\tau_{\text{planned}}$.
3. Estimate $\Delta\eta^*(\Delta\tau)$ from the diminishing-returns profile of past deliberation episodes.
4. Stop deliberating when the marginal improvement rate $\partial\Delta\eta^*/\partial\Delta\tau$ drops below $\rho_{\text{delib}}/\|\delta_{\text{post}}\|$.

Structural-Switch Trigger. From `#result-structural-adaptation-necessity`, structural adaptation is indicated when parametric convergence leaves a mismatch floor. Operationally:

1. Estimate the current mismatch floor $\|\delta\|_{\text{floor}}$ from converged residual statistics.
2. Estimate post-switch expected mismatch as $\|\delta\|_{\text{new}} \approx \rho/\alpha'$ where α' is the sector bound under the candidate new model class.
3. Estimate transition cost C_{switch}: knowledge loss, retraining time ($\Delta\tau_{\text{switch}}$), and accumulated mismatch during transition ($\rho \cdot \Delta\tau_{\text{switch}}$).
4. Switch when: $(\|\delta\|_{\text{floor}} - \|\delta\|_{\text{new}}) \cdot T_{\text{horizon}} > C_{\text{switch}}$.

In a team or an ensemble, read a mismatch floor as structural only after the cheaper multi-agent remedies have been tried and the residual pattern has survived them: recalibrating trust in sources, routing queries away from failing members, improving channels. Residuals correlated across members point to a limitation they share; residuals confined to some members point to those members (`#der-team-persistence`).

Multi-Agent Quantities.

Agent i tracks, for each source j, the inputs to the communication gain of `#hyp-communication-gain` and the team quantities of `#der-team-persistence`:

Table BA.4

Quantity	*Role in AAT*	*Estimator*
$U_{o,ji}$	Channel noise in the communication gain	Message latency and the variance the channel adds to j's report, from channel calibration
$U_{\text{src},j} + U_{\text{align},ji}$	Source-quality and teleological-unity uncertainty	Variance of j's past reports against later ground truth, minus $\hat{U}_{o,ji}$, floored at zero; for probabilistic reports, a running Brier or log-loss score
V_{disagree}	Disagreement energy	Members' predictions on a common set of queries
$\Delta\mathcal{T}_i^{\text{cost}}(j)$	Coordination cost	i's direct-adaptation tempo with and without the channel to j

Residuals alone do not separate U_{src} from U_{align}: miscalibration and strategic distortion can leave the same residual variance. Separating them needs evidence about the direction of j's errors relative to j's own interests. Where that is unavailable, the sum is what enters the gain.

Two procedures follow from the same segments. An update that would be costly if wrong, from a source whose reliability is low or poorly known, waits for agreement from other sources: the risk-asymmetric trust of `#hyp-communication-gain` applied as a quorum. Agreement counts only between sources that do not share an upstream source; otherwise

the quorum counts one source several times (AU). And query actions are logged apart from actions on the world, since their information yield and cost differ (`#def-causal-information-yield`).

Estimator Uncertainty Guidance.

$\hat{\eta}^*$ **(gain estimate).** Approximate variance via the delta method:

$$\mathrm{Var}(\hat{\eta}^*) \approx \hat{\eta}^{*2}(1-\hat{\eta}^*)^2 \left[\frac{\mathrm{Var}(\hat{U}_M)}{\hat{U}_M^2} + \frac{\mathrm{Var}(\hat{U}_o)}{\hat{U}_o^2} \right] \tag{BA.7}$$

Near 0 or 1, stable (dominated by one source). Near 0.5, most volatile.

$\hat{\rho}$ **(mismatch injection rate).** Finite-difference estimation amplifies noise. Recommend smoothing before differencing. Minimum ~20 observations for stable trend; ~50 for reliable variance. The $[\cdot]_+$ clipping introduces positive bias at small ρ.

$\hat{\alpha}$ **(sector lower bound).** The 10th-percentile approach gives approximately 90% confidence under stationarity. Report quantile level, bin count, and bin width.

$\hat{R}$ **(sector-condition radius).** A sharp drop-off in sector-condition satisfaction is a strong signal; gradual decay suggests the sector condition may not hold cleanly.

Nonstationarity caveat. All estimators assume approximate stationarity over the estimation window. When environment regime changes are suspected (`#result-structural-adaptation-necessity`), re-estimate from post-change data only.

Epistemic Status.

This is a *procedures document* — estimation recipes, not theoretical claims. Each estimator inherits the epistemic status of the quantity it targets: the gain estimator's validity depends on `#emp-update-gain`'s structural claim; the sector-bound estimator's validity depends on `#result-sector-condition-stability`'s assumptions. The estimation procedures themselves are *conditional* on these theoretical foundations and on the stationarity and data-sufficiency assumptions noted above.

Appendix BB

Worked Example: 1D Active Tracking (Kalman Domain)

Every AAT quantity has an exact Kalman-filter counterpart. This is a *validation* of the formal chain — all quantities are computable in closed form.

This is AAT's **calibration on a problem with known exact answers**. AAT's contribution in the Kalman instance is not adding new results to Kalman theory; it is *embedding Kalman in a more general framework that handles the same primitives across non-linear, non-Gaussian, and non-fully-observable agent classes*, with the Kalman case as the special case where everything reduces to closed-form. The mismatch is the innovation; the optimal gain coincides with the Kalman gain (the uncertainty-ratio formula equals the Kalman gain in the linear-Gaussian case); the sector parameter equals tempo via `#der-gain-sector-bridge`; the persistence condition reduces to observability plus detectability. Where AAT predicts what Kalman theory says, the framework is minimally correct; where AAT predicts beyond Kalman scope (non-linear, non-Gaussian, partial observability, multi-agent), the framework's reach extends past Kalman's. The Kalman instance is the anchor point that grounds the abstract machinery in a familiar concrete case.

System.

The agent tracks scalar state x_t and chooses sensor mode $a_t \in \{L, H\}$:

- Dynamics: $x_{t+1} = x_t + v_t, v_t \sim \mathcal{N}(0, q), q = 0.25$
- Observation: $y_t = x_t + n_t^{(a_t)}$
- Low mode noise: $r_L = 9$; High mode noise: $r_H = 1$
- Event rate: $\nu = 5$ Hz
- High mode has higher energy cost

Chain Instantiation.

Scope (`#scope-adaptive-system`). *Mapping: exact.*

$\Omega_t = x_t$ is partially observed via a noisy channel, and the law (q, r_L, r_H, the random-walk form) is known, so the typed target is $S_t = x_t$: uncertainty of the state type. Because the state is a random walk, what is seen later still bears on it, so the future stream carries positive information about x_t at every t under either sensor mode (item (b) of `#scope-adaptive-system` with $A = C = 1$). The posterior variance converges to a positive steady state, so the warrant is standing. $\mathcal{A} = \{L, H\}$ is non-empty and causally affects observation quality.

Causal Structure (`#post-causal-structure`) + CIY (`#def-causal-information-yield`). *Mapping: exact.*

Action precedes observation and changes $P(y_t \mid do(a_t))$ through r_{a_t}. Using low mode as comparator action, the CIY is the interventional KL divergence:

Worked Quantity

$$\mathrm{CIY}(H) = D_{\mathrm{KL}}\big(P(y \mid do(H)) \,\|\, P(y \mid do(L))\big) \tag{BB.1}$$

$$= \frac{1}{2}\left[\log\left(\frac{P^- + r_L}{P^- + r_H}\right) + \frac{P^- + r_H}{P^- + r_L} - 1\right] \tag{BB.2}$$

With $P^- = 4.25$: $\mathrm{CIY}(H) \approx 0.161$ nats.

Model (`#form-agent-model`). *Mapping: exact.*

Model state $M_t = (\hat{x}_{t|t}, P_{t|t})$ — a compression of interaction history with recursive update.

Mismatch (`#def-mismatch-signal`). *Mapping: exact.*

The mismatch signal is the Kalman innovation:

$$\delta_t = y_t - \hat{x}_{t|t-1} \tag{BB.3}$$

The estimation error is $e_t = x_t - \hat{x}_{t|t-1}$, and the innovation is the estimation error plus the channel noise, $\delta_t = e_t + n_t^{(a_t)}$, so $\mathbb{E}[\delta_t^2] = P_t^- + r_{a_t}$. The agent sees δ_t and never sees e_t. The persistence bounds below are about e_t.

Update Gain (`#emp-update-gain`). *Mapping: exact.*

Scalar Kalman gain:

$$K_t = \frac{P_t^-}{P_t^- + r_{a_t}} \tag{BB.4}$$

With $P^- = 4.25$: $K(H) \approx 0.810, K(L) \approx 0.321$.

The exact uncertainty ratio mapping: $U_M = P_t^-, U_o = r_{a_t}, \eta^* = K_t$.

Exploration (`#def-causal-information-yield`).

Worked Objective

$$a_t^* = \arg\max_a \left[\mathbb{E}[\mathrm{value}(a) \mid M_t] + \lambda_t \, \mathbb{E}[\mathrm{CIY}(a) \mid M_t]\right] \tag{BB.5}$$

When uncertainty is high (P^- large), CIY term favors high mode. As uncertainty falls, policy shifts toward low-cost L.

Deliberation Threshold (`#der-deliberation-cost`). Suppose a planning pause of $\Delta\tau = 0.5$ s, with measured $\rho_{\text{delib}} = 0.40$ surprise/s.

Cost during pause: 0.20 surprise units. If the estimation error the agent expects to face on resuming is $\|\bar{\delta}_{\text{post}}\| = 0.70$ (an estimate; the agent observes only the innovation, whose RMS exceeds it by the channel noise, and the gain improvement must move toward the Kalman gain for the benefit term to read as a reduction, `#der-deliberation-cost`), deliberation is worthwhile when:

$$\Delta\eta^*(0.5) \cdot 0.70 > 0.20 \implies \Delta\eta^*(0.5) > 0.286 \tag{BB.6}$$

Structural Adaptation (`#result-structural-adaptation-necessity`). Assume a maneuvering regime change introduces sustained residual autocorrelation and a mismatch floor. The disturbance here is stochastic, so the trigger is the Model S one: the persistence condition is satisfiable at some gain iff the optimal filter's own error fits inside the region (`#result-persistence-condition`, the gain window's feasibility condition). If the estimated valid radius drops to $R = 0.08$ while the Kalman prior error is $P^- = 0.640$ (RMS 0.80, from the Lyapunov Bounds section), $R^2 = 0.0064 < P^-$: no gain persists in mean square, parametric adaptation cannot be certified at any tuning, and model-class change is required (e.g., constant-velocity → constant-acceleration process model).

Tempo + Persistence (`#def-adaptive-tempo`, `#result-persistence-condition`). Using action mix 70%H, 30%L:

$$\bar{\eta}^* = 0.7(0.810) + 0.3(0.321) = 0.663 \tag{BB.7}$$

$$\mathcal{T} = \nu\bar{\eta}^* = 5 \cdot 0.663 = 3.315 \text{ s}^{-1} \tag{BB.8}$$

These gains are the early-transient values at $P^- = 4.25$. In steady state the gains are smaller: under high mode throughout, $K_{ss} = 0.390$ (Lyapunov Bounds below) and $\mathcal{T}_{ss} = \nu K_{ss} = 1.95$ s^{-1} (the exact per-step contraction rate is $-\nu \ln(1 - K_{ss}) = 2.47$ s^{-1}); under low mode throughout, $P^- = 1.630$ and $K_{ss} = 0.153$. Tempo is a state-dependent quantity here, highest while the filter is still uncertain.

This system's disturbance is stochastic (a random walk observed through Gaussian noise), so the persistence question is the Model S one, worked in closed form below rather than through the Model D ratio $\rho/\mathcal{T}$.

Lyapunov Bounds (`#result-sector-condition-stability`, `#der-gain-sector-bridge`). *Mapping: exact (derived, not estimated).*

By `#der-gain-sector-bridge`, the correction of the estimation error is exactly linear, $F(e) = Ke$, so the sector parameter is $\alpha = K$ per observation and $\alpha = \nu K = \mathcal{T}$ in continuous time at the same gain (1.95 s^{-1} at the high-mode steady state; the Tempo section's 3.315 s^{-1} is the early-transient value). The disturbance on e has two parts: the random walk adds variance q per step, and the update feeds in the share $K^2 r$ of the channel noise (A, Setup). Per step $e_{t+1} = (1 - K_t)e_t - K_t n_t + \upsilon_t$.

Steady state, high mode throughout ($r = r_H = 1$). The Riccati equation gives $P^- = 0.640$, $K_{ss} = 0.390$, $P = 0.390$:

Table BB.1

Object	*RMS*
Estimation error after update, $\sqrt{P}$	0.62
Estimation error before update, $\sqrt{P^-}$	0.80

Object	*RMS*
Innovation, $\sqrt{P^- + r_H}$	1.28

The lag/noise trade-off, in one line. For a gain K held fixed, the prior error variance is

$$V(K) = \frac{q + K^2 r}{1 - (1 - K)^2} \tag{BB.9}$$

At $K = 0.39$ (the Kalman gain) it is 0.640. At $K = 0.81$, a gain held at its early transient value, it is 0.940. At $K = 0.15$ it is 0.982. The error is smallest at the Kalman gain: a larger gain removes lag faster and injects more channel noise.

Task adequacy. With $\|\delta_{\text{critical}}\| = 1$ read on the estimation error, the agent is adequate ($0.62 < 1$ after update, 0.80 before). Read on the innovation, the condition could never hold, since the innovation's RMS is at least $\sqrt{r_H} = 1$ whatever the gain. That is the distinction `#def-mismatch-signal` draws.

The region of validity. With model-class capacity $R = 1.4$ (a domain parameter), the prior error is Gaussian with standard deviation 0.80, so at any given time it lies outside $\mathcal{B}_R$ with probability 0.08. Under Model S the region is a typical scale, not a wall (A, Prop A.1S and Corollary A.1S.1); the error will leave it, after an expected stay that grows exponentially in $\alpha R^2/\sigma_w^2$ (Prop A.1S-τ), and because this filter runs at the Kalman gain, that exponent is at its maximum for the channel.

Mapping Quality Summary.

Table BB.2

AAT Concept	*Kalman Mapping*	*Status*
Scope	Exact	Definitional
Causal structure + CIY	Exact	Closed-form KL
Model (M_t as sufficient statistic)	Exact	Kalman state + covariance
Mismatch (δ_t = innovation; estimation error e_t)	Exact	Standard Kalman innovation; $\delta_t = e_t + n_t$
Gain ($\eta^* = K_t$)	Exact	Kalman gain IS uncertainty ratio
Tempo ($\mathcal{T} = \nu\bar{\eta}^*$)	Exact	Closed-form
Persistence condition	Exact	Model S, closed form on the estimation error
Sector parameter ($\alpha = \nu K = \mathcal{T}$ at the same gain)	Exact	Derived from Kalman gain via `#der-gain-sector-bridge`; 1.95 s^{-1} at the high-mode steady state
Lyapunov bounds	Exact	Steady-state error variance from the Riccati equation; minimized at the Kalman gain

Epistemic Status.

This is a *worked instantiation*, not a theoretical claim. Every mapping is exact — the Kalman domain is the canonical case where AAT's formal chain has closed-form realizations. The example validates that the formal chain is internally consistent and instantiable.

Appendix BC

Worked Example: Nonstationary Bandit (RL Domain)

AAT's conceptual architecture applies beyond the conjugate-Bayesian regime. The mapping here is *approximate* — it shows that AAT's concepts organize RL phenomena, but the quantitative relationships are structural analogies, not derivations. Where the mapping is exact versus approximate is marked explicitly.

This is the framework's bridge to RL theory. AAT does not subsume RL theory; it claims that typical RL methods (ε-greedy, UCB, Thompson sampling) are *operational approximations* of AAT primitives under regime-specific scope conditions — none exactly implements the optimal-gain principle, but their gains are *related to* the uncertainty-ratio gain and the methods converge in steady state toward optimal-gain behaviour. The worked example concentrates on the nonstationary case where AAT's machinery is most distinctive from classical RL: the forgetting prerequisite becomes operational (effective sample size per arm must be bounded or the agent's gain converges to zero); the per-arm tempo decomposition surfaces persistence as the binding constraint when uniform exploration is mismatched to drift.

System.

A k-armed bandit ($k = 4$) with slowly drifting reward means:

Formulation

$$\mu_i(t+1) = \mu_i(t) + w_i(t), \quad w_i(t) \sim \mathcal{N}(0, q), \quad q = 0.01 \tag{BC.1}$$

- Observation: Pulling arm $a_t = i$ yields reward $r_t = \mu_i(t) + \varepsilon_t, \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \sigma^2 = 1.0$.
- Event rate: $\nu = 1$ pull per step.
- Agent: Q-learning with fixed learning rate α and UCB action selection.

Chain Instantiation.

Scope (`#scope-agency`). *Mapping: exact.*

$\Omega_t = (\mu_1(t), \ldots, \mu_4(t))$ — not directly observable. Because the means drift, they are state, not law, under the invariance cut of `#scope-adaptive-system`; the law is the drift structure (q, σ^2, the Gaussian random-walk form), which the example fixes. $\mathcal{A} = \{1, 2, 3, 4\}$. The adaptive scope holds at every t, with a standing warrant: each mean takes a fresh Gaussian step every round, which renews what pulling its arm can reveal about where it is now (item (g′) of `#scope-adaptive-system`). (In the stationary variant the means do not drift; they are then law, and the agent stays in at every finite t. Under a Gaussian prior on the means, the information one pull carries about an arm's mean decays like $1/(2n)$ after n pulls of that arm, and never reaches zero, while the information in unboundedly many further pulls is infinite.)

Causal Structure (`#post-causal-structure`) + CIY (`#def-causal-information-yield`). *Mapping: exact for causal ordering; approximate for CIY formula.*

Discussion — Bandit CIY, Gaussian Case

The action (arm choice) temporally precedes the reward observation. Under the Gaussian reward model, CIY for arm i:

$$\mathrm{CIY}(i;\, M_{t-1}) = \frac{1}{2\sigma^2}\mathbb{E}_{j\sim q}\big[(\hat{\mu}_i - \hat{\mu}_j)^2\big] \tag{BC.2}$$

This measures how *distinctive* arm i's predicted reward is relative to alternatives — distinctiveness of predictions, not uncertainty about predictions. A proper uncertainty-aware CIY would require maintaining posterior variances per arm, which the basic Q-learning model does not track.

Model (`#form-agent-model`) and Sufficiency (`#def-model-sufficiency`). *Mapping: exact structural mapping; the model is deliberately impoverished.*

$$M_t = (\hat{\mu}_1, \ldots, \hat{\mu}_4,\ n_1, \ldots, n_4) \tag{BC.3}$$

Model sufficiency $S(M_t) < 1$ for two reasons: (1) No drift tracking — the model treats reward means as stationary. (2) No uncertainty representation — unlike a Bayesian agent that would maintain posterior variances.

Mismatch (`#def-mismatch-signal`). *Mapping: exact.*

$$\delta_t = r_t - \hat{\mu}_{a_t} \tag{BC.4}$$

Decomposes exactly: $\mathbb{E}[\delta_t^2] = (\hat{\mu}_{a_t} - \mu_{a_t}(t))^2 + \sigma^2$.

Update Gain (`#emp-update-gain`). *Mapping: approximate. This is where the bandit agent is most deficient relative to AAT-optimal behavior.*

The Q-learning update

$$\hat{\mu}_{a_t} \leftarrow \hat{\mu}_{a_t} + \alpha \cdot \delta_t \tag{BC.5}$$

uses a **degenerate constant gain**. It does not adapt to the agent's current uncertainty state.

Optimal gain via effective window. A fixed α is equivalent to exponential discounting with effective window $W = (1-\alpha)/\alpha$. The per-arm model uncertainty balances estimation variance and drift-induced bias:

$$U_M \approx \frac{\sigma^2}{W} + q \cdot W \tag{BC.6}$$

The optimal window $W^* = \sigma/\sqrt{q} = 10$ gives $\alpha^* \approx 0.091$, $U_M \approx 0.2$, and:

$$\eta^* = \frac{U_M}{U_M + U_o} = \frac{0.2}{1.2} \approx 0.167 \tag{BC.7}$$

The discrepancy between α^* and η^* arises because α operates on raw prediction errors while η^* accounts for the full uncertainty structure. The two converge when the model properly tracks its own uncertainty.

Exploration (`#def-causal-information-yield`). *Mapping: approximate structural analogy.*

UCB: $a_t = \arg\max_i [\hat{\mu}_i + c\sqrt{\ln t / n_i}]$

Table BC.1

AAT component	*UCB analog*	*Quality*
$\mathbb{E}[\text{value}(a) \mid M_t]$	$\hat{\mu}_i$	Exact
$\lambda(M_t) \cdot \mathbb{E}[\text{CIY}(a)]$	$c\sqrt{\ln t / n_i}$	Approximate

UCB captures the CIY structure (rarely-visited arms are more informative) but depends on visit count, not observation content.

Tempo + Persistence (`#def-adaptive-tempo`, `#result-persistence-condition`). *Mapping: approximate.*

Per-arm tempo. With approximately uniform exploration, each arm sampled at rate $\nu/k = 1/4$:

$$\mathcal{T}_i = \frac{\nu}{k} \cdot \alpha = 0.25 \times 0.091 = 0.023 \text{ step}^{-1} \tag{BC.8}$$

Per-arm drift rate: $\rho_i = \sqrt{q} = 0.1$ reward units $\cdot$ step^{-1}.

Persistence check: $\mathcal{T}_i = 0.023$ vs $\rho_i = 0.1$ — **fails**. Per-arm correction tempo is too low relative to drift. The agent cannot keep all arms' estimates current under uniform exploration.

Interpretation. This is expected and informative. AAT diagnoses exactly why: with 4 arms and one pull per step, each arm is visited too infrequently to track its drifting mean. Two remedies:

1. **Increase η^*** (raise α): Extract more per observation, but increase steady-state noise.
2. **Concentrate ν**: Abandon uniform exploration. Focus pulls on a subset, increasing per-arm ν/k_{active}. This is exactly what a good UCB policy does.

With focused exploration ($k_{\text{active}} = 2, \alpha = 0.2$): $\mathcal{T}_i = 0.5 \times 0.2 = 0.1$ — barely meets the threshold. The fundamental tension: with limited pulls and significant drift, the agent must accept either failing to track some arms or using high α that introduces noise.

Aggregate failure. $\mathcal{T} = \nu \cdot \alpha = 0.091$ vs $\rho = k \cdot \sqrt{q} = 0.4$. The agent's total adaptive capacity is outpaced by total environmental drift — a regime where model-based approaches (Bayesian bandits, Kalman bandit filters) with higher effective η^* have a structural advantage.

AAT Concept	*Bandit Mapping*	*Status*

Mapping Quality Summary.

Table BC.2

AAT Concept	*Bandit Mapping*	*Status*
Scope	Exact	Definitional
Causal structure	Exact	Structural
CIY	Approximate	Model lacks uncertainty representation
Model (M_t)	Exact	Definitional
Model sufficiency ($S < 1$)	Exact	Q-learner demonstrably insufficient
Mismatch (δ_t)	Exact	Standard prediction error
Gain (α as degenerate η^*)	Approximate	Fixed, does not adapt
Exploration (UCB as approximate CIY)	Approximate	Structural analogy
Tempo ($\mathcal{T} = (\nu/k) \cdot \alpha$ per arm)	Approximate	Assumes uniform allocation
Persistence condition	Exact (qualitative)	Threshold structure is robust

Epistemic Status.

This is a *structural mapping*, not a derivation. The mapping is *exact* for scope, causal ordering, model compression, and mismatch. It is *approximate* for gain, exploration, and tempo — precisely because the Q-learner does not represent its own uncertainty, which is what `#emp-update-gain` requires for optimal behavior. The gap between the Q-learner's fixed α and the AAT-optimal adaptive η^* is precisely the gap between a fixed-gain PID controller and a Kalman filter.

The mapping status is *conditional* because the quantitative relationships depend on the Gaussian reward model and specific parameterization. The qualitative conclusions (persistence failure under uniform exploration, the concentration-vs-noise tradeoff) should be robust.

Appendix BD

Worked Example: Nonstationary Bandit with Explicit Strategy (Part II Validation)

AAT's Part II machinery — objectives, strategy DAGs, the orient cascade, satisfaction gap, control regret, chain confidence decay, observability dominance, and edge update via gain — is exercised on a concrete system simple enough for closed-form analysis. The key payoff: Part II diagnostics tell the agent things that Part I quantities alone cannot.

System.

A 3-armed Bernoulli bandit with drifting success probabilities and an agent that maintains an explicit strategy DAG for arm selection.

Formulation

Environment. Three arms with time-varying success probabilities:

$$\theta_k(t) \in [0, 1], \quad k \in \{1, 2, 3\} \tag{BD.1}$$

At each step, pulling arm k yields reward $r_t \in \{0, 1\}$ with $P(r_t = 1 \mid a_t = k) = \theta_k(t)$. The best arm changes slowly: at rate ρ_θ, a random arm's success probability shifts by a random perturbation clipped to $[0.1, 0.9]$.

Concrete parameterization. At $t = 0$: $\theta_1(0) = 0.7$, $\theta_2(0) = 0.5$, $\theta_3(0) = 0.3$. Drift rate: $\rho_\theta = 0.05$ per step (one arm shifts by ± 0.05 every 20 steps on average). Horizon: $N_h = 100$ steps.

Agent. Maintains Beta-Bernoulli beliefs per arm and an explicit strategy DAG encoding its plan for achieving a target reward rate.

Part I Chain Instantiation (Summary).

Part I quantities map as in BC. The brief summary here establishes the epistemic baseline; the novel content is Part II.

Scope (`#scope-agency`). $\Omega_t = (\theta_1(t), \theta_2(t), \theta_3(t))$; $\mathcal{A} = \{1, 2, 3\}$. The arm probabilities are state that keeps drifting and persists into what later pulls show, so the adaptive scope holds at every t (`#scope-adaptive-system`). *Exact.*

Model (`#form-agent-model`). $M_t = (\alpha_k, \beta_k)_{k=1}^3$ — Beta posteriors per arm, with exponential discounting (effective window $W = 20$):

$$\hat{\theta}_k = \frac{\alpha_k}{\alpha_k + \beta_k} \tag{BD.2}$$

Exact structural mapping.

Mismatch (`#def-mismatch-signal`). $\delta_t = r_t - \hat{\theta}_{a_t}$. *Exact.*

Update gain (`#emp-update-gain`). The Beta-Bernoulli conjugate update with discounted pseudo-counts gives:

$$\eta_k^* = \frac{1}{\alpha_k + \beta_k + 1} \tag{BD.3}$$

For a well-observed arm with $\alpha_k + \beta_k \approx W = 20$: $\eta_k^* \approx 0.048$. *Exact for the pulled arm; zero for unpulled arms.*

Tempo (`#def-adaptive-tempo`). Per-arm tempo: $\mathcal{T}_k = \nu_k \cdot \eta_k^*$ where ν_k is the pull rate for arm k. Under uniform exploration: $\nu_k = 1/3$, so $\mathcal{T}_k \approx 0.016$ per step. *Approximate.*

Part II Chain Instantiation.

Objective (`#form-objective-functional`). *Mapping: exact.*

The agent's objective: achieve a target average reward rate over the horizon.

Definition (bandit-objective)

$$V_{O_t}(\tau_{0:N_h}) = \frac{1}{N_h} \sum_{t=0}^{N_h-1} r_t \tag{BD.4}$$

$$V_{O_t}^{\min} = 0.6 \tag{BD.5}$$

This is a utility-type objective with a satisfaction threshold: the agent counts the mission as successful if its average reward rate meets or exceeds 0.6. The value functional evaluates trajectories on $[0, 1]$ (average reward).

Value Object (`#def-value-object`). *Mapping: exact.*

Under the one-step improvement convention ($\pi_{\text{cont}} = \pi_{\text{current}}$):

$$V_O(M_t, \pi; N_h) = \mathbb{E}\left[\frac{1}{N_h} \sum_{s=t}^{t+N_h-1} r_s \,\middle|\, M_t, \pi \right] \tag{BD.6}$$

For a greedy policy that always pulls the estimated-best arm $k^* = \arg\max_k \hat{\theta}_k$:

$$V_O(M_t, \pi_{\text{greedy}}; N_h) = \hat{\theta}_{k^*} \tag{BD.7}$$

This is tractable: the agent's expected average reward under greedy play equals its belief about the best arm's success probability. With M_0: $V_O = \hat{\theta}_1 = 0.7$.

Action-value form:

$$Q_O(M_t, k; \pi_{\text{greedy}}, N_h) \approx \frac{1}{N_h}\hat{\theta}_k + \frac{N_h - 1}{N_h}\hat{\theta}_{k^*} \tag{BD.8}$$

The first pull contributes $1/N_h$ of the horizon; the rest follows the greedy policy.

Objective Attainability (`#def-satisfaction-gap`). *Mapping: exact.*

$$A_O(M_t; \Pi, N_h) = \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h) \tag{BD.9}$$

For a bandit with Beta-Bernoulli beliefs, the optimal policy (Gittins index or Bayes-adaptive) achieves an attainability that is bounded:

$$\max_k \hat{\theta}_k \leq A_O \leq \max_k \hat{\theta}_k + \epsilon_{\text{info}} \tag{BD.10}$$

where ϵ_{info} accounts for the information value of exploration (pulling a suboptimal arm now to identify the true best arm later). The upper bound is tight when exploration can reveal a better arm. For our initial beliefs: $A_O \approx 0.70 + \epsilon_{\text{info}}$.

Satisfaction gap:

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O = 0.6 - 0.70 \approx -0.10 \tag{BD.11}$$

$\delta_{\text{sat}} < 0$: the objective is **attainable**. The agent believes it can achieve a 0.6 average reward rate. The negative gap is margin — the agent has room to spare.

Control Regret (`#def-control-regret`). *Mapping: exact.*

Under an explore-then-exploit policy that spends 30 steps exploring uniformly, then exploits the estimated-best arm:

$$V_O(M_t, \pi_{\text{explore-exploit}}; 100) = \frac{30}{100} \cdot \bar{\theta} + \frac{70}{100} \cdot \hat{\theta}_{k^*(30)} \tag{BD.12}$$

where $\bar{\theta} = (\hat{\theta}_1 + \hat{\theta}_2 + \hat{\theta}_3)/3$ is the average expected reward during exploration and $\hat{\theta}_{k^*(30)}$ is the estimated-best arm after 30 observations. With initial beliefs: $V_O \approx 0.3 \cdot 0.5 + 0.7 \cdot 0.7 = 0.64$.

$$\delta_{\text{regret}} = A_O - V_O(\pi_{\text{explore-exploit}}) \approx 0.70 - 0.64 = 0.06 \tag{BD.13}$$

The agent is leaving 0.06 reward-rate points on the table with its current explore-then-exploit strategy — it explores too much. This is the signal for strategy revision (`#der-orient-cascade` step 3).

The 2x2 diagnostic (`#def-control-regret` Discussion):

Table BD.1

	$\delta_{sat} \leq 0$	$\delta_{sat} > 0$
$\delta_{\text{regret}} \approx 0$	Success	Capability limit
$\delta_{\text{regret}} \gg 0$	**Strategy problem**	Both

Our agent is in the **Strategy problem** cell: the goal is achievable, but the current policy is suboptimal. Part I alone (which reports $\hat{\theta}_k$, η_k^*, and $\mathcal{I}_k$) cannot distinguish "good strategy, hard goal" from "bad strategy, easy goal." The 2x2 diagnostic is Part II's value-add.

Strategy DAG (`#def-strategy-dag`, `#scope-and-or`). *Mapping: exact for structure. The forward pass is not the plan's probability here, because the three Phase 1 leaves are mutually exclusive (see "What the forward pass gets wrong here" below).*

The agent's strategy for achieving $V_{O_t}^{\min} = 0.6$:

```
 [root: "achieve ≥0.6 avg reward"] (AND) / \ [Phase 1: "identify best arm"] (OR) [Phase 2: "exploit best arm"]
/ | \ | [arm 1 [arm 2 [arm 3 [pull identified is best] is best] is best] arm reliably]
```

Nodes: - v_{root}: AND-node. Both phases must succeed: identify the best arm AND exploit it. Terminal satisfaction condition: average reward $\geq$ 0.6. - v_{id}: OR-node (Phase 1). Identifying the best arm succeeds if ANY one arm is correctly identified as best. - v_1, v_2, v_3: leaf condition nodes. "Arm k is the best arm." Base credence: $p_k(M_t) = P(\theta_k > \theta_j \; \forall j \neq k \mid M_t)$. - v_{exploit}: leaf action node (Phase 2). "Pulling the identified arm yields reward $\geq$ 0.6." Base credence: $p_{\text{exploit}}(M_t) = P(\theta_{k^*} \geq 0.6 \mid M_t)$, where k^* is the arm the agent currently identifies as best: arm 1 at $t = 0$, arm 2 after the revision at $t = 40$.

Edge credences. The edges from leaf nodes to their parents carry causal credences: - $p_{v_k, v_{\text{id}}}$: "correctly identifying arm k as best enables Phase 1 success." These are structural (close to 1.0) — if the agent correctly knows which arm is best, Phase 1 succeeds by definition. Set $p_{v_k, v_{\text{id}}} = 1.0$ for all k. - $p_{v_{\text{id}}, v_{\text{root}}}$: "having identified the best arm advances the root goal." Set to 0.95 (slight uncertainty about whether identification translates to exploitation success). - $p_{v_{\text{exploit}}, v_{\text{root}}}$: "successful exploitation advances the root goal." Set to 0.90 (the identified arm might drift below threshold during exploitation).

Leaf base credences from M_t. At $t = 0$ with $(\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3) = (0.7, 0.5, 0.3)$ and moderate pseudo-counts ($\alpha_k + \beta_k \approx 5$) from prior experience:

$$p_1(M_0) = P(\theta_1 > \theta_2, \theta_1 > \theta_3 \mid M_0) \approx 0.75 \tag{BD.14}$$

$$p_2(M_0) \approx 0.20, \quad p_3(M_0) \approx 0.05 \tag{BD.15}$$

$$p_{\text{exploit}}(M_0) = P(\theta_{k^*} \geq 0.6 \mid M_0) \approx 0.70 \tag{BD.16}$$

Status propagation (`#def-strategy-dag`, `#scope-and-or`):

Phase 1 (OR-node):

$$s_{v_{\text{id}}} = 1 - \prod_{k=1}^{3} (1 - p_{v_k, v_{\text{id}}} \cdot p_k) = 1 - (1 - 0.75)(1 - 0.20)(1 - 0.05) \tag{BD.17}$$

$$= 1 - (0.25)(0.80)(0.95) = 1 - 0.19 = 0.81 \tag{BD.18}$$

Phase 2 (leaf): $s_{v_{\text{exploit}}} = 0.70$.

Root (AND-node):

$$s_{v_{\text{root}}} = (p_{v_{\text{id}}, v_{\text{root}}} \cdot s_{v_{\text{id}}}) \cdot (p_{v_{\text{exploit}}, v_{\text{root}}} \cdot s_{v_{\text{exploit}}}) \tag{BD.19}$$

$$= (0.95 \times 0.81) \times (0.90 \times 0.70) = 0.770 \times 0.630 = 0.485 \tag{BD.20}$$

Plan-confidence score: $\hat{P}_\Sigma(M_0) = 0.485$. The strategy assigns about 49% confidence to achieving the objective. This is distinct from A_O (which optimizes over all policies) — $\hat{P}_\Sigma$ is the DAG's self-assessment of *this specific plan*.

What the forward pass gets wrong here. The score 0.485 is what the agent's forward pass computes. It is not the probability of this DAG, because the three Phase 1 leaves are not independent. Exactly one arm is the best arm, so the events "arm k is best" are disjoint and exhaustive, and their credences sum to 1. Which arm is best is a function of θ, a common cause that is not a node. So the DAG is causally insufficient although it is a tree and nothing in it looks shared (R, §"Exactness of Status Propagation"). For disjoint children, the OR-node's exact status is the sum of their transmitted credences: $s_{v_{\text{id}}} = \sum_k p_{v_k, v_{\text{id}}}\, p_k = 1.00$, not 0.81. Phase 1 is then certain, so the root's two inputs are independent, and the exact root value is $(0.95 \times 1.00) \times (0.90 \times 0.70) = 0.599$. The forward pass understates this plan by 0.11. Exclusive alternatives correlate negatively, which reverses the sign that shared infrastructure gives: the OR-node is underestimated, not overestimated (`#def-strategy-dag`, Correlation Hierarchy).

The exact value also exposes a modeling error. As written, Phase 1 cannot fail, because some arm is certainly the best. The uncertainty the plan means, whether the agent settles on the best arm, belongs in the edges $p_{v_k, v_{\text{id}}}$ ("the agent identifies arm k, given that k is best"), which the example set to 1 "by definition". With those edges below 1, Phase 1's exact status is $\sum_k p_{v_k, v_{\text{id}}}\, p_k$. The rest of this example keeps the DAG as written and reports the forward-pass scores the agent would compute, so that the cascade steps can be followed. Read those scores as the agent's computations, not as the plan's probabilities.

Chain Confidence Decay (`#der-chain-confidence-decay`). *Mapping: exact (mathematical identity).*

The root node's confidence illustrates chain decay through an AND structure. The root requires two children to succeed, each with their own sub-chains:

$$\log \hat{P}_\Sigma = \log(p_{v_{\text{id}}, v_{\text{root}}} \cdot s_{v_{\text{id}}}) + \log(p_{v_{\text{exploit}}, v_{\text{root}}} \cdot s_{v_{\text{exploit}}}) \tag{BD.21}$$

$$= \log(0.770) + \log(0.630) = -0.261 + (-0.462) = -0.724 \tag{BD.22}$$

Each term is negative; the AND combination accumulates them additively in log-space. If the agent added a third AND-child (e.g., "maintain budget"), even with $s_{\text{budget}} = 0.95$, the strategy-plan confidence would drop to $0.485 \times 0.95 = 0.461$. The pressure toward shallow, narrow DAGs is structural.

Contrast with the OR-node. The forward pass gives Phase 1 $s_{v_{\text{id}}} = 0.81$ even though no single arm has credence above 0.75, which looks like redundancy. It is not redundancy here. The three alternatives exclude one another, so the OR adds them up, to 1.00. The forward pass's 0.81 is the right answer for a different plan: three *independent* routes that succeed with probabilities 0.75, 0.20 and 0.05. There the OR does provide resilience, and the asymmetry between AND (multiplicative decay) and OR (complementary resilience) of `#der-chain-confidence-decay` and `#scope-and-or` applies. The same three numbers describe redundancy in one plan and a partition in the other, and only the plan's structure tells them apart.

Orient Cascade (`#der-orient-cascade`). *Mapping: exact for ordering; the content of each step is tractable in this domain.*

Suppose at $t = 40$, the agent has been pulling arm 1 (the initially-best arm) and observes a run of failures: the last 8 of 10 pulls yielded $r_t = 0$. Meanwhile, arm 2's true probability has drifted to $\theta_2(40) = 0.8$ (unbeknownst to the agent) and arm 1 has drifted to $\theta_1(40) = 0.3$.

The orient cascade proceeds:

Step 1: Reduce $\delta_{\text{epistemic}}$. Update M_t from the recent observations. The Beta posterior for arm 1 shifts:

$$\hat{\theta}_1 : 0.70 \to 0.38 \quad \text{(after 8 failures in 10 trials with discounted prior)} \tag{BD.23}$$

Arms 2 and 3 are unobserved since the exploration phase ended. Their posteriors remain at approximately $\hat{\theta}_2 = 0.52$, $\hat{\theta}_3 = 0.28$ (slightly regressed toward the prior due to discounting). *The epistemic update proceeds without reference to O_t or Σ_t* — directed separation (`#der-directed-separation`) holds because the Beta update depends only on the observation and the prior, not on the agent's goals.

Step 2: Evaluate δ_{sat}. With the updated M_t:

$$A_O(M_{40}) \approx \max_k \hat{\theta}_k + \epsilon_{\text{info}} = 0.52 + \epsilon_{\text{info}} \tag{BD.24}$$

$$\delta_{\text{sat}} = 0.6 - (0.52 + \epsilon_{\text{info}}) \tag{BD.25}$$

If $\epsilon_{\text{info}} \approx 0.05$ (exploration can reveal a better arm): $\delta_{\text{sat}} \approx 0.6 - 0.57 = 0.03 > 0$. The objective is now **marginally unmet** — the agent believes it may not be able to achieve 0.6 average reward. This triggers the disambiguation procedure (`#def-satisfaction-gap`): is the goal truly infeasible, or is M_t wrong about what's achievable?

Step 3: Evaluate δ_{regret}. Under the current greedy policy (still pulling arm 1 despite updated beliefs):

$$V_O(M_{40}, \pi_{\text{current}}) = \hat{\theta}_1 = 0.38 \tag{BD.26}$$

$$\delta_{\text{regret}} = 0.57 - 0.38 = 0.19 \tag{BD.27}$$

Control regret is high. The agent is pulling arm 1 out of inertia, but M_t now says arm 2 is better. *This is the Part II payoff*: Part I would report the mismatch δ_t on each pull and the updated $\hat{\theta}_1$, but it cannot diagnose that the *strategy* is the problem. The high δ_{regret} identifies the corrective action: revise Σ_t, not O_t.

Step 4: Evaluate $\delta_{\text{strategic}}$. The edge residual for the root AND-node's exploitation branch:

$$r_{\text{exploit}} = \mathbb{E}[\Delta V_O \mid \text{exploit arm 1}] - \Delta V_O^{\text{observed}} \tag{BD.28}$$

The agent predicted reward ≈ 0.70 per pull from arm 1; it observed ≈ 0.20. The residual is large and positive, localizing the problem to the exploitation edge: the agent's causal belief that "pulling arm 1 yields high reward" is wrong.

Simultaneously, the leaf credences update: $p_1(M_{40}) \approx 0.15$ (arm 1 is probably not the best), $p_2(M_{40}) \approx 0.60$ (arm 2 is probably best, by default from the prior). The forward-pass status of the Phase 1 OR-node shifts. Its exact status does not, since some arm is still the best; what changes is which arm, and the exploitation leaf below carries that.

Step 5: Revise Σ_t. The agent restructures its strategy: switch exploitation target to arm 2 (the currently estimated-best arm). In DAG terms: $p_1(M_{40})$ has dropped to 0.15 while $p_2(M_{40})$ has risen to 0.60. The OR-node now routes through arm 2. The exploitation leaf updates: $p_{\text{exploit}}(M_{40}) = P(\theta_2 \geq 0.6 \mid M_{40}) \approx 0.40$.

After strategy revision: $\hat{P}_\Sigma(M_{40}) \approx (0.95 \times 0.73) \times (0.90 \times 0.40) = 0.693 \times 0.360 = 0.249$. Plan confidence has dropped from 0.485 to 0.249 — the agent correctly recognizes that its situation has deteriorated. Respecting the partition, the exact value drops from 0.599 to $0.95 \times 0.90 \times 0.40 = 0.342$. The deterioration is real, and all of it is in the exploitation leaf.

What Part I alone cannot provide. At $t = 40$, Part I reports: $\hat{\theta}_1 = 0.38$, $\hat{\theta}_2 = 0.52$, $\hat{\theta}_3 = 0.28$, and the per-arm mismatch signals. This tells the agent what it believes about each arm. It does NOT tell the agent: - Whether its *goal*

is still achievable (δ_{sat}) - Whether its *strategy* is the problem or the goal is (δ_{regret}) - Which *part* of the strategy is failing ($\delta_{\text{strategic}}$, edge residuals) - What corrective action to take (the cascade's ordering)

Observability Dominance (`#der-observability-dominance`). *Mapping: exact mechanism; approximate quantitative form.*

When the agent stops pulling arm 3 after the exploration phase (say at $t = 30$), the observation rate for arm 3 drops to $\nu_3 = 0$. By `#emp-update-gain`:

$$\eta_3^* = \frac{U_{\text{edge},3}}{U_{\text{edge},3} + U_{\text{obs},3}} \tag{BD.29}$$

With $U_{\text{obs},3} \to \infty$ (no observations): $\eta_3^* \to 0$. Arm 3's credence $p_3(M_t)$ is **frozen at its exploration-phase value**.

For the strategy DAG, the leaf node "arm 3 is best" has: - Nominal credence: $p_3(M_t) \approx 0.05$ (from $t = 30$ beliefs) - Path observability: $\sigma_3 = 0$ (arm 3 is not being pulled) - Observability-adjusted confidence: $\text{conf}_{\text{obs}}(\text{arm 3 path}) = 0.05 \times 0 = 0$

Arm 3's path through the strategy is **epistemically dead** — the agent cannot learn whether arm 3 has become the best arm. If θ_3 drifts to 0.9 (which it might, given nonstationarity), the agent will never discover this through the exploitation-only policy.

The observability-dominance diagnostic. Compare arm 2 and arm 3:

Table BD.2

	Arm 2	*Arm 3*
Nominal credence p_k	0.52	0.28 (frozen)
Observability σ_k	moderate (occasional pulls)	0 (never pulled)
Adjusted confidence	$0.52 \times \sigma_2$	0
Can be updated?	Yes	No

Arm 2 is epistemically alive; arm 3 is epistemically dead. The agent should prefer exploring arm 2 (where it can learn) over arm 3 — unless it invests in making arm 3 observable again (pulling it despite low expected reward). This is the strategic case for exploration: not just expected value, but *observability maintenance*.

Part I vs Part II diagnosis. Part I reports $\hat{\theta}_3 = 0.28$ with growing uncertainty (the posterior variance increases under discounting with no new data). Part II adds the diagnostic: this arm's strategy path is epistemically dead ($\sigma_3 = 0$), and restoring observability has value beyond the arm's nominal expected reward.

Edge Update via Gain (`#hyp-edge-update-via-gain`). *Mapping: exact for the Beta-Bernoulli case.*

When the agent pulls arm 1 at $t = 35$ and observes failure ($r_t = 0$):

Leaf credence update. The Beta posterior for arm 1 updates:

$$\beta_1 \to \beta_1 + 1, \quad \hat{\theta}_1 = \frac{\alpha_1}{\alpha_1 + \beta_1 + 1} \tag{BD.30}$$

This is the standard conjugate update — the gain principle reduces to Bayesian updating in the binary case (`#hyp-edge-update-via-gain`).

Strategy propagation. The updated $\hat{\theta}_1$ changes $p_1(M_t)$ (the credence that arm 1 is best). This propagates through the OR-node to the root:

$$s_{v_{\text{id}}}^{\text{new}} = 1 - (1 - p_1^{\text{new}})(1 - p_2)(1 - p_3) \tag{BD.31}$$

Under the forward pass, the OR-node's status falls when p_1 drops and recovers as p_2 rises, which looks like redundancy absorbing a single-edge failure. Its exact status does not move at all, because the p_k always sum to 1. What looks like redundancy here is the bookkeeping of a partition. Redundancy proper needs independent alternatives ("Contrast with the OR-node", above).

The exploitation edge. The edge $p_{v_{\text{exploit}}, v_{\text{root}}} = 0.90$ is a causal credence: "pulling the identified-best arm reliably yields reward ≥ 0.6." After observing repeated failures from arm 1:

$$\text{signal} = 0 \quad \text{(failure observed)} \tag{BD.32}$$

$$p_{\text{exploit}}^{\text{new}} = p_{\text{exploit}}^{\text{old}} + \eta_{\text{edge}} \cdot (0 - p_{\text{exploit}}^{\text{old}}) \tag{BD.33}$$

With $\eta_{\text{edge}} = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$, the exploitation edge's credence decreases. The rate of decrease is governed by the gain principle: well-established edges (many prior successes, U_{edge} small) are slow to revise; newly formed edges are fast.

Directed Separation (`#der-directed-separation`). *Mapping: exact. The agent is Class 1 (Separated) by construction.*

The Beta-Bernoulli update for M_t processes (a_t, r_t) without reference to (O_t, Σ_t):

$$(\alpha_{a_t}, \beta_{a_t}) \to (\alpha_{a_t} + r_t, \beta_{a_t} + 1 - r_t) \tag{BD.34}$$

The update function f_M has no G_t argument. The strategy DAG then updates its leaf credences from the new M_t — this is $f_G(G_t, M_{t+1}, e_t)$. The agent architecture is modular: a separate estimator (Beta posteriors) and planner (DAG evaluation), connected through the state-estimate interface.

$\kappa_{\text{processing}} = 0$ — no goal information flows into the epistemic update. Part II results apply exactly.

Satisfaction Gap Dynamics. *Mapping: exact definition; approximate computation.*

Tracking δ_{sat} over time reveals the agent's evolving assessment of goal feasibility:

Table BD.3

Time	$\hat{\theta}_{k^*}$	A_O *(approx)*	δ_{sat}	*Interpretation*
$t = 0$	0.70	0.72	-0.12	Comfortably attainable
$t = 20$	0.68	0.70	-0.10	Still attainable
$t = 40$	0.52	0.57	$+0.03$	Marginally unmet
$t = 50$ (after exploring arm 2)	0.75	0.77	-0.17	Attainable again — M_t improved

$t = 70$	0.78	0.79	−0.19	Confidently attainable

The key transition is $t = 40 \to 50$: δ_{sat} goes from positive (unmet) to negative (attainable) — not because the goal changed or the environment improved, but because M_t improved. The agent explored arm 2, discovered $\hat{\theta}_2 \approx 0.75$, and its attainability assessment corrected upward. This matches the `#def-satisfaction-gap` disambiguation: the positive δ_{sat} at $t = 40$ was caused by a bad model, not an infeasible goal. The orient cascade's ordering (fix M_t first, then reassess δ_{sat}) produced the right diagnosis.

The Part II Payoff.

The purpose of this worked example is to demonstrate that Part II adds genuine diagnostic value beyond Part I. Here is the explicit comparison:

Scenario at $t = 40$: arm 1 failing, arm 2 actually best, agent pulling arm 1.

Part I diagnosis. The mismatch signal δ_t is large. The posterior $\hat{\theta}_1$ has dropped. Per-arm tempo $\mathcal{T}_1 \approx 0.016$ is adequate for tracking arm 1's drift. The agent knows arm 1 looks bad. Part I prescribes: update M_t faster (increase η^*) or observe more (increase ν).

Part II diagnosis. The orient cascade runs: 1. M_t updates correctly — arm 1's posterior drops to 0.38. (Directed separation: this happens goal-blindly.) 2. $\delta_{\text{sat}} = +0.03$ — the objective is marginally unmet. The agent's best-believed policy barely misses the target. 3. $\delta_{\text{regret}} = 0.19$ — the current policy (pulling arm 1) is far from the best available (pulling arm 2). 4. The 2x2 diagnostic says: "goal may be achievable, strategy is definitely suboptimal." Corrective action: revise Σ_t, then reassess δ_{sat}. 5. Edge residuals localize the problem: the exploitation branch's predicted-vs-observed value gap is large. The leaf credences identify arm 2 as the best candidate. 6. Observability check: arm 3 is epistemically dead ($\sigma_3 = 0$). The agent should invest in observing arm 2 (or arm 3) before committing.

Part II tells the agent: *switch to arm 2, but first explore to confirm — the goal is probably still achievable, the current strategy is the problem, not the goal.* Part I tells the agent: *arm 1 looks bad.* The difference is the difference between a diagnosis and a symptom.

Mapping Quality Summary.

Table BD.4

AAT Concept	*Bandit-Strategy Mapping*	*Status*	*Notes*
Scope (`#scope-agency`)	Exact	Definitional	Same as BC
Model M_t (`#form-agent-model`)	Exact	Structural	Beta posteriors per arm
Mismatch δ_t (`#def-mismatch-signal`)	Exact	Standard	Prediction error
Update gain η^* (`#emp-update-gain`)	Exact	Conjugate	Beta-Bernoulli gain
Tempo $\mathcal{T}$ (`#def-adaptive-tempo`)	Approximate	Per-arm decomposition	Assumes known allocation
Directed separation (`#der-directed-separation`)	Exact	Class 1 (Separated) by construction	Beta update is goal-blind
Objective O_t (`#form-objective-functional`)	Exact	Utility with threshold	$V_{O_t}^{\min} = 0.6$
Value object V_O (`#def-value-object`)	Exact	Closed-form (greedy)	$V_O = \hat{\theta}_{k^*}$
Satisfaction gap δ_{sat} (`#def-satisfaction-gap`)	Exact	Definition applied	$V_{O_t}^{\min} - A_O$

AAT Concept	*Bandit-Strategy Mapping*	*Status*	*Notes*
Control regret δ_{regret} (`#def-control-regret`)	Exact	Definition applied	$A_O - V_O(\pi_{\text{current}})$
Strategy DAG Σ_t (`#def-strategy-dag`)	Exact	AND/OR with 6 nodes	Small enough for manual propagation
AND/OR scope (`#scope-and-or`)	Partial	Root fits AND	Phase 1 is a disjunction of exclusive alternatives, which sums; the noisy-OR form does not fit it
Status propagation	Not exact here	Exclusive leaves make the tree causally insufficient	Forward pass 0.485; exact 0.599
Plan confidence $\hat{P}_\Sigma$ (`#def-strategy-dag`)	Exact as the forward pass's output	Underestimates this DAG's probability: the exclusive leaves correlate negatively	0.485 forward, 0.599 exact, at $t = 0$
Chain confidence decay (`#der-chain-confidence-decay`)	Exact	Mathematical identity	AND-node multiplicative; the OR-node here is a partition, not redundancy
Orient cascade (`#der-orient-cascade`)	Exact (ordering)	All 5 steps exercised	Content is tractable in this domain
Observability dominance (`#der-observability-dominance`)	Exact (mechanism)	Unpulled arms frozen	$\sigma_3 = 0 \Rightarrow \eta_3^* = 0$
Edge update via gain (`#hyp-edge-update-via-gain`)	Exact (binary case)	Beta-Bernoulli conjugate	Gain principle = Bayesian update
Strategic calibration (`#def-strategic-calibration`)	Approximate	Credit assignment tractable here	Single-arm attribution is clean
Satisfaction gap dynamics	Exact	Time series of δ_{sat}	Model improvement resolves positive gap

Quantities that map cleanly. All Part II definitions (objective, value object, satisfaction gap, control regret, strategy DAG) have exact instantiations. Plan confidence $\hat{P}_\Sigma$ is exact as the output of the forward pass, but here it is not the plan's probability. The Phase 1 leaves are exclusive, so the forward pass understates the plan (0.485 against 0.599). Even this toy shows the point of R that a tree is not enough: the leaves must be independent too. With shared infrastructure the error would run the other way, as an overestimate (`#def-strategy-dag`). The orient cascade ordering is exact and all five steps are exercised.

Quantities that map approximately. Strategic calibration is approximate because even in this simple domain, attributing value changes to specific DAG edges requires assumptions about the agent's execution fidelity. Observability dominance is exact in mechanism but the multiplicative form $\text{conf}_{\text{obs}} = \text{conf} \cdot \text{obs}$ is a first-order approximation.

Quantities that don't map. No Part II quantity fails to map in this domain, though plan confidence maps only as the agent's computation, not as the plan's probability (above). The system is simple enough that every quantity is computable (at least approximately). This is the advantage of the bandit setting: it is the simplest system that exercises all Part II machinery.

Epistemic Status.

This is a *worked instantiation* — it demonstrates that the Part II formal chain is internally consistent and instantiable. The mapping quality is *conditional*: the quantitative relationships depend on the Beta-Bernoulli reward model, the specific DAG structure, and the parameterization. The qualitative conclusions — especially the diagnostic value of the 2x2 table and the orient cascade's ordering — should be robust across reward models and DAG topologies.

The status is *conditional* rather than *exact* because the attainability A_O is approximated (the exact Bayes-optimal policy for a nonstationary bandit is intractable), and the strategic calibration mapping inherits the discussion-grade status of `#def-strategic-calibration`. The core Part II definitions (satisfaction gap, control regret, strategy-plan confidence) are exact instantiations, strategy-plan confidence as the forward pass's output. The plan's exact probability differs from it, because the Phase 1 leaves are exclusive.

Appendix BE

Worked Example: L1 Augmented DAG with Common-Cause Node

A concrete instantiation of the Correlation Hierarchy (`#def-strategy-dag`) and Proposition B.6 (S). Two OR-alternatives sharing an infrastructure dependency are modeled at L0 (independence) and L1 (augmented DAG), with the sector condition verified for L1.

The example **exhibits AAT's central observation about causally-insufficient DAGs**. At L0 (independence), AND/OR propagation systematically *overestimates* plan success: the strategy-plan-confidence comes out higher than the actual rate because the model treats the two paths as independently failing rather than as both depending on shared infrastructure. The overestimate is *exactly* the covariance between the paths' outcomes — the direction-of-bias formula for OR-structures. At L1 (augmented DAG with the infrastructure node factored *above* the correlated OR-alternatives as a strict-prerequisite AND-node), propagation is correct: conditional on infrastructure up, the paths are independent and standard propagation gives the exact answer. The L1 construction principle (factor the common cause above the correlation it creates) recovers the correct probability. The sector parameter under L1 augmentation has a three-way gating form — condition-testing rate, evidence-starvation on each path under θ_C, exploration-gating on the OR-node — with each factor reflecting a different mechanism. When the common cause is *unobservable*, the L1 augmentation breaks and `#disc-identifiability-floor` Instance 2 kicks in: the per-conditional decomposition is Cramér-Rao-floored. The agent's three escape routes (augment observability; multi-child joint observation; fall back to L0-on-marginals) are exercised below with their respective trade-offs.

Setup.

Two alternative paths to a goal G, sharing a common infrastructure dependency C:

- Infrastructure up with probability $\theta_C = 0.8$
- Path 1 succeeds with probability $\theta_{1|C} = 0.9$ when infrastructure is up, fails when down
- Path 2 succeeds with probability $\theta_{2|C} = 0.7$ when infrastructure is up, fails when down
- Goal succeeds if at least one path succeeds (OR)

Actual plan success probability (from the generative model):

$$P(G) = \theta_C \cdot [1 - (1 - \theta_{1|C})(1 - \theta_{2|C})] = 0.8 \cdot 0.97 = 0.776 \tag{BE.1}$$

L0 Analysis: The Independence Model.

The L0 DAG has two leaf nodes with marginal success probabilities:

$$\theta_1 = \theta_C \cdot \theta_{1|C} = 0.72, \qquad \theta_2 = \theta_C \cdot \theta_{2|C} = 0.56 \tag{BE.2}$$

OR-propagation:

$$\hat{P}_\Sigma^{L0} = 1 - (1 - \theta_1)(1 - \theta_2) = 1 - (0.28)(0.44) = 0.877 \tag{BE.3}$$

Overestimation: $0.877 - 0.776 = 0.101$ (13% relative error).

The overestimation equals $\mathrm{Cov}(X_1, X_2)$ — the covariance induced by the shared infrastructure. The OR-node's redundancy is illusory: both paths fail together when infrastructure fails, so the "backup" provides less protection than the independence model predicts. This is the $-\rho$ bias from the Correlation Hierarchy's direction-of-bias formula.

Naive L1: Why Adding a Node Is Not Sufficient.

The obvious L1 construction adds C as a parent of both paths:

- $B_1 = \mathrm{AND}(C, A_1), B_2 = \mathrm{AND}(C, A_2)$
- $G = \mathrm{OR}(B_1, B_2)$

Status propagation: $s_{B_1} = 0.8 \cdot 0.9 = 0.72$, $s_{B_2} = 0.8 \cdot 0.7 = 0.56$, $s_G = 1 - (0.28)(0.44) = 0.877$.

Same answer as L0. The OR-propagation treats B_1 and B_2 as marginally independent, but they are correlated through their shared parent C. Adding the common-cause node does not fix the overestimation because the node is *inside* the correlated structure (below the OR-node), not factored above it. The augmented graph is causally sufficient, since C is now a node, but C has two children that reconverge at G. So the graph is not a tree, and the forward pass is exact only on trees (R, §"Exactness of Status Propagation").

Correct L1: Factor the Common Cause Above the Correlation.

The correct L1 restructuring places C as an AND-prerequisite *above* the OR-structure:

- $G_{\text{sub}} = \mathrm{OR}(A_1, A_2)$ — the sub-plan assuming infrastructure holds
- $G = \mathrm{AND}(C, G_{\text{sub}})$ — infrastructure must hold AND sub-plan must work

Status propagation:

$$s_{G_{\text{sub}}} = 1 - (1 - \theta_{1|C})(1 - \theta_{2|C}) = 1 - (0.1)(0.3) = 0.97 \tag{BE.4}$$

$$s_G = \theta_C \cdot s_{G_{\text{sub}}} = 0.8 \cdot 0.97 = 0.776 \tag{BE.5}$$

Correct. Matches the actual plan success probability exactly.

Why this works: C is above the OR-node, so its value is factored out before the OR-propagation occurs. Conditional on $C = 1$, the two paths A_1 and A_2 are genuinely independent (their exogenous noise terms are independent — the common cause has been conditioned away). The OR-propagation's independence assumption holds within G_{sub}

because the correlation source is above, not among the siblings. In graph terms, the restructured graph is a tree: every node has one child, and that is the condition under which propagation is exact for all credences.

The L1 construction principle: *factor the common cause above the correlation it creates.* When a common cause correlates siblings under an OR-node, restructure so the common cause gates the entire OR-structure via an AND-relationship. This ensures conditional independence of the OR-children and makes standard propagation correct. The principle is a rewrite into a tree, $\mathrm{OR}(C \wedge A_1, C \wedge A_2) = C \wedge \mathrm{OR}(A_1, A_2)$, valid because C enters both paths as an AND-parent (a strict prerequisite).

Sector Condition Verification (Proposition B.6).

Three leaf nodes with Beta-Bernoulli dynamics and ε-greedy path selection:

Table BE.1

Leaf	*True value*	*Tested when*	*Expected correction*
C (condition)	$\theta_C = 0.8$	Every trial	$-\delta_C/(n_C + 1)$
A_1 (greedy)	$\theta_{1\|C} = 0.9$	$C = 1$ AND path 1 selected	$-\theta_C(1-\varepsilon)\,\delta_{A_1}/(n_{A_1} + 1)$
A_2 (explore)	$\theta_{2\|C} = 0.7$	$C = 1$ AND path 2 selected	$-\theta_C\varepsilon\,\delta_{A_2}/(n_{A_2} + 1)$

The sector parameter (full proof in Prop B.6 of S):

$$\alpha_\Sigma = \min\left(\frac{1}{n_C + 1},\ \frac{\theta_C(1-\varepsilon)}{n_{A_1} + 1},\ \frac{\theta_C\varepsilon}{n_{A_2} + 1}\right) \tag{BE.6}$$

Three-way gating. This is the first verified mixed AND/OR topology, combining:

1. **Condition testing** — C is tested every trial, same rate as a single edge (B.1)
2. **Evidence starvation** — action edges are gated by θ_C (tested only when $C = 1$), same mechanism as B.2
3. **Exploration gating** — OR-alternatives compete for test opportunities (ε vs $1 - \varepsilon$), same mechanism as B.4

The B.5b bridge applies (non-negative Jacobian, componentwise corrections): $\alpha_s = \alpha_\Sigma$ with no penalty.

L0 vs L1 Comparison.

Table BE.2

Quantity	*L0*	*L1 (correct)*
Leaf edges	2 (A_1, A_2 at marginals)	3 ($C, A_1 \mid C, A_2 \mid C$)
Φ at truth	0.877 (biased)	0.776 (correct)
Φ – actual success	+0.101 (overestimates)	0 (unbiased)
Bottleneck α_Σ (at $\varepsilon = 0.3$)	$0.3/(n_2 + 1)$	$0.24/(n_{A_2} + 1)$
Persistence threshold	Lower (easier)	Higher (harder)

The tradeoff: L0 has a higher sector parameter (easier to maintain persistence) but its reference value Φ^{L0} is systematically wrong — it overestimates actual success by the covariance ρ. L1 has a lower sector parameter (the θ_C attenuation makes action calibration harder) but its reference value Φ^{L1} is correct.

Both models have $\delta_s = 0$ when credences are at truth. The difference is what $\delta_s = 0$ *means*:

- **L0**: well-calibrated within the independence model — which overstates plan success by ρ
- **L1**: well-calibrated to actual plan success

L1 persistence is harder but calibration is honest. The extra leaf (C) adds a degree of freedom, and the condition-gating slows action calibration. But the resulting target (Φ^{L1}) is the true plan success probability, not a biased surrogate.

When Correct L1 Construction Is Not Possible.

The factoring-above principle works under two conjoint conditions: (a) the common cause cleanly gates all correlated children through a single AND-relationship (*topological*), and (b) the common cause is a *strict prerequisite* — $\theta_{\text{child}|\neg C} \approx 0$, so children cannot succeed when the common cause is absent (*semantic*). Both conditions must hold. The example above satisfies both: infrastructure gates all alternatives, and absent infrastructure forces failure ($\theta_{i|\neg C} = 0$).

Topological failure: mixed gating. When some OR-alternatives share a common cause and others do not, the factoring-above construction cannot place C above the full sub-plan. Conditioning-based propagation:

$$P(G) = \sum_c P(C = c) \cdot R_\Sigma(G \mid C = c) \tag{BE.7}$$

where $R_\Sigma(G \mid C = c)$ is computed by standard AND/OR propagation with C fixed. Cost: $O(2^k \cdot (|V| + |E|))$ where k is the number of common-cause nodes. Tractable for small k; exponential in general.

Semantic failure: soft facilitators. When the common cause is a soft facilitator — $\theta_{\text{child}|\neg C} > 0$, so children can still succeed when C is absent, just less reliably (favorable market conditions, a supportive environment, an enabling but non-essential resource) — the AND-prerequisite construction mathematically forces $P(\text{sub-plan} \mid \neg C) = 0$, which strictly understates actual success probability. This is a real failure mode even when the topology admits factoring-above. The natural repair at $O(1)$ cost per soft-facilitator node is the **mixture form (L1')**:

$$\hat{P}_\Sigma^{L1'} = \theta_C \cdot R_\Sigma(G \mid C) + (1 - \theta_C) \cdot R_\Sigma(G \mid \neg C) \tag{BE.8}$$

where the two conditional sub-plans may share structure but carry separate edge credences $p_{ij|C}$ and $p_{ij|\neg C}$. The parametric cost per soft-facilitator-affected edge doubles; the propagation remains polynomial. This is the gap between L1 (strict prerequisites) and L2 (arbitrary joint): L1' handles soft facilitators without the $O(2^k)$ exponential blowup of L2's explicit conditioning. Sector-condition verification for L1' is derived as **Proposition B.7** in S *when C is observable per trial*: under Beta-Bernoulli updating, componentwise updates per conditional branch, and facilitator monotonicity ($R_\Sigma(G_{|C}) \geq R_\Sigma(G_{|\neg C})$), the sector condition holds globally with five-way gating

$$\alpha_{L1'} = \min\left(\frac{1}{n_C+1},\ \min_{j \in \mathcal{J}_C} \frac{\theta_C \pi_{j|C}}{n_{j|C}+1},\ \min_{j \in \mathcal{J}_{\neg C}} \frac{(1-\theta_C)\pi_{j|\neg C}}{n_{j|\neg C}+1}\right) \tag{BE.9}$$

(condition testing; evidence starvation and exploration gating on each of the $C = 1$ and $C = 0$ branches). The B.5b transfer to plan value is lossless by facilitator monotonicity plus AND/OR monotonicity on each conditional sub-DAG. In the strict-prerequisite limit $\theta_{j|\neg C} \to 0$, B.7 reduces to B.6's three-way gating. **When C is not observable on a single observation channel**, the mixture is refuted as a no-go: the per-trial Fisher information matrix on $(\theta_C, p_{j|C}, p_{j|\neg C})$ factors as rank 1 around the indeterminacy manifold $\{\hat{\theta}_C \hat{p}_{|C} + (1 - \hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$, so by the Cramér-Rao bound

no unbiased online estimator on the joint conditional vector admits $\alpha > 0$. Repair routes when C is unobservable: (i) augment C-observability (recovers B.7); (ii) run $K \geq 2$ children jointly under the same C-realization so the joint Fisher matrix reaches rank $2K + 1$; or (iii) fall back to plan-level tracking on the marginal $\hat{\mu}_j$ (recovers B.1 at the scalar marginal, losing the per-conditional decomposition). See `#disc-identifiability-floor` for the meta-pattern shared with the on-policy detection no-go in `#der-causal-insufficiency-detection`.

For strategies with many mixed or soft common causes, the agent faces a practical choice: model the most important common causes at L1 (for strict prerequisites) or L1' (for soft facilitators), accept residual L0 bias from unmodeled causes, or invest in richer propagation algorithms. See `#def-strategy-dag` Correlation Hierarchy for the higher-level treatment of L1 / L1' / L2.

Detecting Latent Common Causes.

The detection of causal insufficiency and interventional localization of latent common causes is treated as a standalone result in `#der-causal-insufficiency-detection` — a *no-go theorem*: under purely on-policy short-circuit execution, no statistic of the agent's observable history can detect the latent, and the L0 residual in particular is *identically zero* on-policy. The residual figures of this example are the **marginal-sampling** regime ($\varepsilon = 1$: every trial, both paths observed regardless of policy): there the credences converge to the marginals θ_1, θ_2 and the L0 residual $\Phi^{L0} - \bar{y}_G$ converges to $+\rho$ (our example: $0.877 - 0.776 = 0.101$). Between the two regimes the residual scales as $O(\varepsilon)$ and serves only as a *confirmatory* signal; the primary detector is the pairwise sibling covariance test under exploration (`#der-causal-insufficiency-detection`).

On-policy observational equivalence (the no-go's shallow-case verification). This example is also the shallow case for which the no-go is *exact*, and the matched-world construction can be verified directly. The on-policy world $\mathcal{W}^*_{L0}$ has independent edges at the regime conditionals

$$\theta^*_1 = \theta_C\, \theta_{1|C} = 0.72, \qquad \theta^*_2 = \frac{\theta_C\,(1 - \theta_{1|C})\,\theta_{2|C}}{1 - \theta_C\,\theta_{1|C}} = \frac{0.056}{0.28} = 0.2. \tag{BE.10}$$

Under sequential short-circuit OR execution (try path 1; try path 2 only on failure), the three on-policy observable events have identical probabilities in both worlds:

Table BE.3

Event	$\mathcal{W}_{L1}$ *(latent C)*	$\mathcal{W}^*_{L0}$ *(independent)*
A_1 succeeds	0.72	$\theta^*_1 = 0.72$
A_1 fails, A_2 succeeds	$0.8 \cdot 0.1 \cdot 0.7 = 0.056$	$0.28 \cdot 0.2 = 0.056$
A_1 fails, A_2 fails	$0.2 + 0.8 \cdot 0.1 \cdot 0.3 = 0.224$	$0.28 \cdot 0.8 = 0.224$

The agent's on-policy history is drawn from the same distribution in both worlds — no on-policy statistic can tell them apart, which is the no-go's content instantiated on this example's numbers.

Epistemic Status.

Conditional on Beta-Bernoulli model. The L0 overestimation, L1 correction, and sector condition verification are exact algebra. The direction-of-bias formula ($\pm\rho$) is exact for two binary siblings **under marginal sampling** (credences at the marginals θ_j; under on-policy short-circuit execution the credences converge to regime conditionals instead and the residual vanishes — see `#der-causal-insufficiency-detection`); for deeper DAGs with multiple common causes, the bias involves higher-order covariance terms. The on-policy observational-equivalence table is exact arithmetic on the matched-world construction of `#der-causal-insufficiency-detection`. The L1 construction principle (factor common cause above the correlation) is general for single common causes gating OR-siblings; the conditioning-based extension for more complex topologies is standard (cutset conditioning or variable elimination in Bayesian networks). That naive L1 fails is exact as well: it is an instance of the tree condition in R.

Appendix BF

Worked Example: `#worked-example-cam` *(missing)*

Segment `worked-example-cam` *not yet written.*

Claim: Coevolving automata (Miller 2022): AAT ↔ Moore machine mapping, meta-machine as $\varepsilon^* = 0$ composition, simplest adaptive agent

Bibliography

[1] Andrei N. Tikhonov. "Systems of differential equations containing a small parameter multiplying the derivative". In: *Matematicheskii Sbornik* 31(73).3 (1952). In Russian, pp. 575–586 (cit. on p. 111).

[2] Hassan K. Khalil. *Nonlinear Systems*. 3rd. Upper Saddle River, NJ: Prentice-Hall, 2002 (cit. on pp. 111, 114).

[3] Ali Saberi and Hassan K. Khalil. "Quadratic-type Lyapunov functions for singularly perturbed systems". In: *IEEE Transactions on Automatic Control* 29.6 (1984), pp. 542–550. DOI: 10.1109/TAC.1984.1103586 (cit. on p. 111).

[4] P. Kokotovic, H. Khalil, and J. O'Reilly. *Singular perturbation methods in control : analysis and design*. 1986. DOI: 10.1137/1.9781611971118. URL: https://doi.org/10.1137/1.9781611971118 (cit. on p. 111).

[5] Lin-Yuan Chen, Nigel Goldenfeld, and Yoshitsugu Oono. "Renormalization group and singular perturbations: Multiple scales, boundary layers, and reductive perturbation theory". In: *Physical Review E* 54.1 (1996), pp. 376–394. DOI: 10.1103/PhysRevE.54.376 (cit. on p. 115).

[6] Hermann Haken. *Synergetics: An Introduction. Nonequilibrium Phase Transitions and Self-Organization in Physics, Chemistry, and Biology*. 3rd ed. Berlin: Springer, 1983 (cit. on p. 115).

[7] Karl Friston et al. "From pixels to planning: scale-free active inference". In: *Frontiers in Network Physiology* 5 (2025), p. 1521963. DOI: 10.3389/fnetp.2025.1521963 (cit. on p. 115).